$$= \int_0^a ax^2(a-x) - x^3(a-x) - x^2\frac{(a-x)^2}{2} + \underbrace{\frac{(a-x)^4}{3} - \frac{(a-x)^4}{4} + \frac{(a-x)^4}{12}}\, dx$$

$$= \int_0^a a^2x^2 - 2ax^3 + x^4 - \frac{x^4 - 2ax^3 + a^2x^2}{2} + \frac{(a-x)^4}{6}\, dx$$

$$= \int_0^a \frac{1}{2}a^2x^2 - ax^3 + \frac{1}{2}x^4 + \frac{(a-x)^4}{6}\, dx$$

$$= \frac{a^2x^3}{6} - \frac{ax^4}{4} + \frac{x^5}{10} - \frac{(a-x)^5}{30}\Bigg|_{x=0}^{x=a}$$

$$= \frac{a^5}{6} - \frac{a^5}{4} + \frac{a^5}{10} + \frac{a^5}{30} = \frac{10a^5 - 15a^5 + 6a^5 + 2a^5}{60}$$

$$= \frac{a^5}{20}$$

18. **18.** $\iiint_W z\,dx\,dy\,dz$, W is the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, $z = 1$, and the cylinder $x^2 + y^2 = 1$, with $x \geq 0$, $y \geq 0$.

W is described by $0 \leq x \leq 1$, $0 \leq y \leq \sqrt{1-x^2}$, $0 \leq z \leq 1$

$$\therefore \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^1 z\, dz\, dy\, dx = \int_0^1 \int_0^{\sqrt{1-x^2}} \frac{1}{2}\, dy\, dx$$

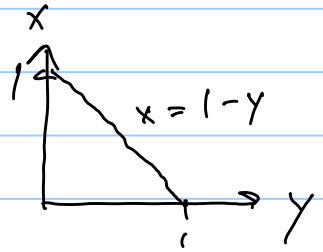
$$= \int_0^1 \frac{y}{2}\Big|_{y=0}^{y=\sqrt{1-x^2}}\, dx = \int_0^1 \frac{\sqrt{1-x^2}}{2}\, dx$$

$$= \frac{1}{2}\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\arcsin(x)\right]_{x=0}^{x=1}$$

$$= \frac{1}{4}\arcsin(1) = \underline{\frac{\pi}{8}}$$

19.

19. $\iiint_W x^2 \cos z\, dx\, dy\, dz$, W is the region bounded by $z = 0$, $z = \pi$, $y = 0$, $y = 1$, $x = 0$, and $x + y = 1$.



W: $0 \le z \le \pi$, $0 \le y \le 1$, $0 \le x \le 1-y$

$$\therefore \int_0^1 \int_0^{1-y} \int_0^{\pi} x^2 \cos z\, dz\, dx\, dy$$

$$= \int_0^1 \int_0^{1-y} \left(x^2 \sin z\Big|_{z=0}^{z=\pi}\right) dx\, dy = \int_0^1 \int_0^{1-y} 0\, dx\, dy$$

$$= \underline{0}$$

20.

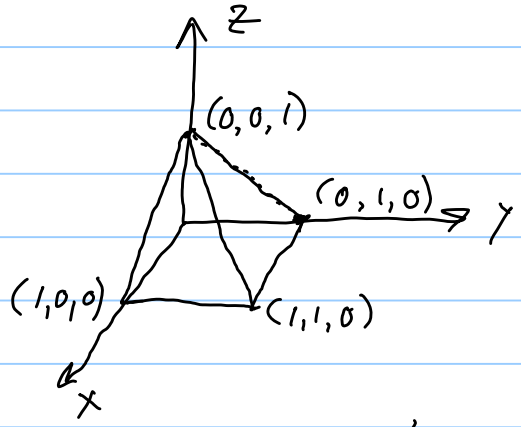
20. $\int_0^2 \int_0^x \int_0^{x+y} dz\, dy\, dx$

$$\int_0^2 \int_0^x \left(z\Big|_{z=0}^{z=x+y}\right) dy\, dx = \int_0^2 \int_0^x (x+y)\, dy\, dx$$

$$\int_0^2 \left(xy + \frac{y^2}{2} \Big|_{y=0}^{y=x} \right) dx = \int_0^2 x^2 + \frac{x^2}{2}\, dx = \int_0^2 \frac{3}{2} x^2\, dx$$

$$= \frac{3x^3}{6} \Big|_0^2 = \frac{1}{2}(8) = \underline{4}$$

21. **21.** $\iiint_W (1 - z^2)\, dx\, dy\, dz$, W is the pyramid with top vertex at (0, 0, 1) and base vertices at (0, 0, 0), (1, 0, 0), (0, 1, 0), and (1, 1, 0).



You could use the xy-plane as the base, break up the pyramid into 4 sections, using the 4-faces as tops to the $z=0$ plane, but that's a lot of work.

Better to use opposite faces as "top" and "bottom", and integrate the z component from $0 \le z \le 1$.

y-planes: $y = 0$ and plane for (0,0,1), (0,1,0), (1,1,0).

$\therefore Ax + By + Cz = D \Rightarrow$

$Ax + Dy + Dz = D$

$\therefore A(1) + D(1) + D(0) = D \Rightarrow A = 0$

$\therefore y + z = 1$

$\therefore 0 \le y \le 1 - z$

X-planes: $x = 0$ and $x + z = 1$

$\therefore 0 \le x \le 1 - z$

$$\therefore \int_0^1 \int_0^{1-z} \int_0^{1-z} (1-z^2)\, dx\, dy\, dz$$

$$= \int_0^1 \int_0^{1-z} \left(x - xz^2 \Big|_{x=0}^{x=1-z} \right) dy\, dz = \int_0^1 \int_0^{1-z} (1-z-(1-z)z^2\, dy\, dz$$

$$= \int_0^1 \int_0^{1-z} (1 - z - z^2 + z^3)\, dy\, dz = \int_0^1 \left(y - yz - yz^2 + yz^3 \Big|_{y=0}^{y=1-z} \right) dz$$

$$= \int_0^1 (1-z)(1-z-z^2+z^3)\, dz = \int_0^1 1 - 2z + 2z^3 - z^4\, dz$$

$$= z - z^2 + \frac{1}{2}z^4 - \frac{z^5}{5} \Big|_0^1 = \frac{1}{2} - \frac{1}{5} = \underline{\underline{\frac{3}{10}}}$$

22. **22.** $\iiint_W (x^2 + y^2)\, dx\, dy\, dz$; W is the same pyramid as in Exercise 21.

$$\int_0^1 \int_0^{1-z} \int_0^{1-z} (x^2+y^2)\, dx\, dy\, dz = \int_0^1 \int_0^{1-z} \frac{x^3}{3} + xy^2 \Big|_{x=0}^{x=1-z} dy\, dz$$

$$= \int_0^1 \int_0^{1-z} \frac{(1-z)^3}{3} + (1-z)y^2\, dy\, dz$$

$$= \int_0^1 \frac{(1-z)^3}{3} y + (1-z)\frac{y^3}{3} \Big|_{y=0}^{y=1-z} dz$$

$$= \int_0^1 \frac{(1-z)^4}{3} + \frac{(1-z)^4}{3} dz = \int_0^1 \frac{2}{3}(1-z)^4 dz$$

$$= \frac{2}{3}\left(-\frac{1}{5}\right)(1-z)^5 \Big|_0^1 = 0 - \left(-\frac{2}{15}(1)^5\right) = \frac{2}{15}$$

23.

23. $\int_0^1 \int_0^{2x} \int_{x^2+y^2}^{x+y} dz\, dy\, dx.$

$$\int_{x^2+y^2}^{x+y} dz = x+y-x^2-y^2$$

$$\int_0^{2x} x+y-x^2-y^2\, dy = xy + \frac{y^2}{2} - x^2 y - \frac{y^3}{3} \Big|_{y=0}^{y=2x}$$

$$= 2x^2 + \frac{4x^2}{2} - 2x^3 - \frac{8x^3}{3} = 4x^2 - \frac{14}{3}x^3$$

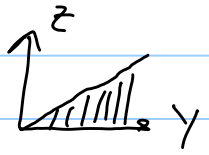
$$\int_0^1 4x^2 - \frac{14}{3}x^3\, dx = \frac{4}{3}x^3 - \frac{14}{12}x^4 \Big|_0^1 = \frac{4}{3} - \frac{7}{6} = \frac{1}{6}$$

24.

24. (a) Sketch the region for the integral
$\int_0^1 \int_0^x \int_0^y f(x, y, z)\, dz\, dy\, dx.$

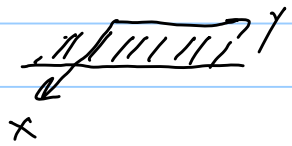
(b) Write the integral with the integration order $dx\, dy\, dz$.

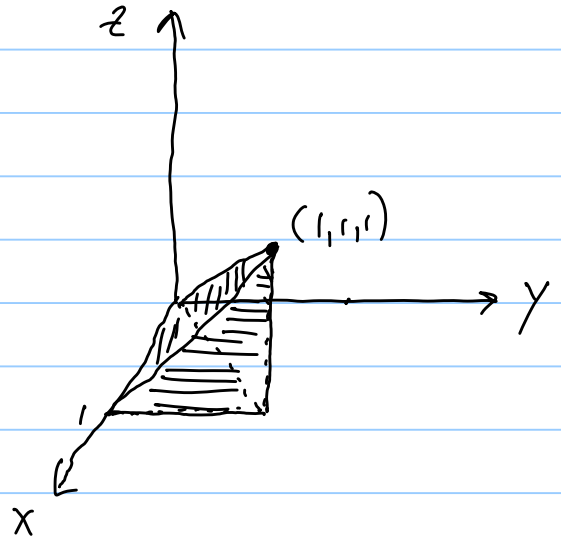
(a) $0 \le z \le y \Rightarrow$ z between $z=0$ and $z=y$.

$\therefore$ 

$0 \le y \le x \Rightarrow$ y between $y=0$ and $y=x$

$\therefore$

$0 \le x \le 1 \Rightarrow$ 

$\therefore$ 

(b) want z independent, so $\int_0^1 dz$

y a function of z: for each z', y goes from z' to 1. $\therefore z \le y \le 1$

$\therefore \int_z^1 dy$ $\therefore$ So far, $\int_0^1 \int_z^1 dy\,dz$

For each y', x goes from y' to 1. $\therefore \int_y^1 dx$

$$\therefore \int_0^1 \int_z^1 \int_y^1 f(x,y,z)\, dx\, dy\, dz$$

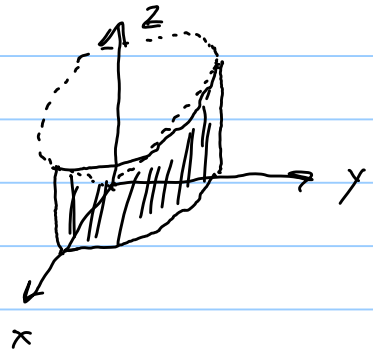
For the regions in Exercises 25 to 28, find the appropriate limits $\phi_1(x)$, $\phi_2(x)$, $\gamma_1(x, y)$, and $\gamma_2(x, y)$, and write the triple integral over the region W as an iterated integral in the form

$$\iiint_W f\, dV = \int_a^b \left\{ \int_{\phi_1(x)}^{\phi_2(x)} \left[\int_{\gamma_1(x,y)}^{\gamma_2(x,y)} f(x, y, z)\, dz \right] dy \right\} dx.$$

25. **25.** $W = \{(x, y, z) \mid \sqrt{x^2 + y^2} \le z \le 1\}$

From $z^2 = x^2$ and $z^2 = y^2$, $z^2 = x^2 + y^2$ is a cone, and since $0 \le \sqrt{x^2+y^2} \le z$, $z = \sqrt{x^2+y^2}$ is the "upper" portion of the cone.

$\therefore$ The solid $\sqrt{x^2+y^2} \le z \le 1$ is the volume "under" the upper portion of the cone, from $0 \le z \le 1$.



$\therefore -1 \le x \le 1$

$-\sqrt{1-x^2} \le y \le \sqrt{1-x^2}$

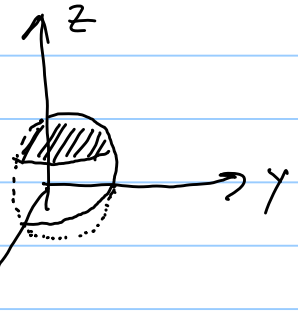
$\sqrt{x^2+y^2} \le z \le 1$

$$\therefore \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{\sqrt{x^2+y^2}}^{1} f(x,y,z)\, dz\, dy\, dx$$

26. **26.** $W = \{(x, y, z) \mid \frac{1}{2} \le z \le 1 \text{ and } x^2 + y^2 + z^2 \le 1\}$

$x^2 + y^2 + z^2 = 1$ is a sphere, ≤ 1 means inside of the

sphere, and $\frac{1}{2} \le z \le 1$ means the portion of the solid sphere above $z = \frac{1}{2}$.



When $z = \frac{1}{2}$, $x^2 + y^2 + \left(\frac{1}{2}\right)^2 = 1$

becomes $x^2 + y^2 = \frac{3}{4}$. For $y = 0$, x

The extremes of x come from $x^2 = \frac{3}{4}$, or $-\frac{\sqrt{3}}{2} \le x \le \frac{\sqrt{3}}{2}$

$\therefore y^2 = \frac{3}{4} - x^2$ becomes $-\sqrt{\frac{3}{4} - x^2} \le y \le \sqrt{\frac{3}{4} - x^2}$

For z, $x^2 + y^2 + z^2 = 1$ becomes $\frac{1}{2} \le z \le \sqrt{1 - x^2 - y^2}$

$$\therefore \int_{-\frac{\sqrt{3}}{2}}^{\sqrt{3}/2} \int_{-\sqrt{\frac{3}{4} - x^2}}^{\sqrt{\frac{3}{4} - x^2}} \int_{\frac{1}{2}}^{\sqrt{1 - x^2 - y^2}} f(x, y, z)\, dz\, dy\, dx$$

---

27. **27.** $W = \{(x, y, z) \mid x^2 + y^2 \le 1, z \ge 0 \text{ and } x^2 + y^2 + z^2 \le 4\}$

$x^2 + y^2 + z^2 \le 4 \Rightarrow$ inside sphere of radius 2.

$z \ge 0 \Rightarrow$ upper half of solid sphere.

$x^2 + y^2 \le 1 \Rightarrow$ cylinder of radius 1 inside upper half of solid sphere.

To get extremes of x, set $y = 0$ for $x^2 + y^2 = 1$ to get $-1 \le x \le 1$.

$\therefore$ Based on x, from $x^2+y^2=1$, $-\sqrt{1-x^2} \le y \le \sqrt{1-x^2}$

For z, from $x^2+y^2+z^2=4$, $0 \le z \le \sqrt{4-x^2-y^2}$

$$\therefore \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{0}^{\sqrt{4-x^2-y^2}} f(x,y,z)\, dz\, dy\, dx$$

28. **28.** $W = \{(x, y, z) \mid |x| \le 1, |y| \le 1, z \ge 0 \text{ and } x^2 + y^2 + z^2 \le 1\}$

$x^2+y^2+z^2 \le 1 \Rightarrow$ inside sphere of radius 1.

$z \ge 0 \Rightarrow$ upper half of sphere.

$|x| \le 1$, $|y| \le 1 \Rightarrow$ square of side 1, which "encompasses" the sphere, and so places no additional restrictions on W.

$\therefore$ For $z=0$, $y=0$, $x^2+y^2+z^2 \le 1 \Rightarrow -1 \le x \le 1$, and compatible with $|x| \le 1$.

For $z=0$, $x^2+y^2+z^2=1 \Rightarrow -\sqrt{1-x^2} \le y \le \sqrt{1-y^2}$

$$\therefore \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{0}^{\sqrt{1-x^2-y^2}} f(x,y,z)\, dz\, dy\, dx$$

29. **29.** Show that the formula using triple integrals for the volume under the graph of a positive function $f(x, y)$, on an elementary region D in the plane, reduces to the double integral of f over D.

$$\iiint_W dV \quad \text{becomes} \quad \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \int_0^{f(x,y)} dz\, dy\, dx$$

$$= \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} \left(z \Big|_0^{f(x,y)} \right) dy\, dx$$

$$= \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x,y)\, dy\, dx = \iint_D f(x,y)\, dA$$

assuming D is described by $a \le x \le b$, $\phi_1(x) \le y \le \phi_2(x)$.

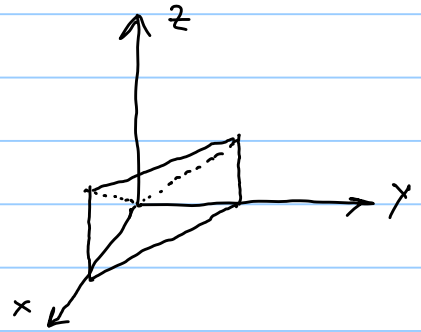
The same reasoning works for D defined by $a \le y \le b$, $\alpha_1(y) \le x \le \alpha_2(y)$, where the order of integration would be $dz\, dx\, dy$.

30. **30.** Let W be the region bounded by the planes $x = 0$, $y = 0$, $z = 0$, $x + y = 1$, and $z = x + y$.

(a) Find the volume of W.

(b) Evalute $\iiint_W x\,dx\,dy\,dz$.

(c) Evalute $\iiint_W y\,dx\,dy\,dz$.



(a) $0 \le x \le 1$, from $x + y = 1$, $0 \le y \le 1 - x$

and from $z = x+y$, $0 \le z \le x+y$.

$$\therefore V = \int_0^1 \int_0^{1-x} \int_0^{x+y} dz\,dy\,dx = \int_0^1 \int_0^{1-x} (x+y)\,dy\,dx$$

$$= \int_0^1 \left(xy + \frac{y^2}{2} \Big|_{y=0}^{y=1-x} \right) dx = \int_0^1 x(1-x) + \frac{(1-x)^2}{2}\,dx$$

$$= \int_0^1 x - x^2 + \frac{x^2}{2} - x + \frac{1}{2}\,dx = \int_0^1 -\frac{1}{2}x^2 + \frac{1}{2}\,dx$$

$$= -\frac{x^3}{6} + \frac{1}{2}x \Big|_0^1 = -\frac{1}{6} + \frac{1}{2} = \underline{\frac{1}{3}}$$

(5) $$\int_0^1 \int_0^{1-x} \int_0^{x+y} x\,dz\,dy\,dx = \int_0^1 \int_0^{1-x} x(x+y)\,dy\,dz$$

$$= \int_0^1 \left(x^2 y + \frac{xy^2}{2} \Big|_{y=0}^{y=1-x} \right) dz = \int_0^1 x^2(1-x) + \frac{x(1-x)^2}{2}\,dx$$

$$= \int_0^1 x^2 - x^3 + \frac{x^3 - 2x^2 + x}{2}\,dx = \int_0^1 -\frac{1}{2}x^3 + \frac{x}{2}\,dx$$

$$= -\frac{1}{8}x^4 + \frac{x^2}{4} \Big|_0^1 = -\frac{1}{8} + \frac{1}{4} = \underline{\frac{1}{8}}$$

(c) $\int_0^1 \int_0^{1-x} \int_0^{x+y} y \, dz\,dy\,dx = \int_0^1 \int_0^{1-x} y(x+y)\, dy\,dz$

$= \int_0^1 \left(\frac{xy^2}{2} + \frac{y^3}{3} \Big|_{y=0}^{y=1-x} \right) dx = \int_0^1 \frac{x(1-x)^2}{2} + \frac{(1-x)^3}{3}\, dx$

$= \int_0^1 \frac{x^3 - 2x^2 + x}{2} + \frac{-x^3 + 3x^2 - 3x + 1}{3}\, dx$

$= \int_0^1 \frac{x^3}{6} - \frac{x}{2} + \frac{1}{3}\, dx = \frac{x^4}{24} - x^2 + \frac{x}{3} \Big|_0^1$

$= \frac{1}{24} - 1 + \frac{1}{3} = -\frac{15}{24}$

31.

31. Let f be continuous and let B_ε be the ball of radius ε centered at the point (x_0, y_0, z_0). Let vol (B_ε) be the volume of B_ε. Prove that

$$\lim_{\varepsilon \to 0} \frac{1}{\text{vol}\,(B_\varepsilon)} \iiint_{B_\varepsilon} f(x, y, z)\, dV = f(x_0, y_0, z_0).$$

Use α instead of ϵ to avoid confusion with ϵ-δ in definition of limit.

∴ Trying to prove $\lim_{\alpha \to 0} \frac{1}{\text{vol}(B_\alpha)} \iiint_{B_\alpha} f(x,y,z)\, dV = f(x_0, y_0, z_0)$

Since f is continuous on closed B_α, it takes

on maximum and minimum values in B_α. Let those values be Max_α and Min_α.

From the definition of $\iiint_{B_\alpha} f(x,y,z)\,dV$,

$$(Min_\alpha)\,Vol(B_\alpha) \le \iiint_{B_\alpha} f(x,y,z)\,dV \le Max_\alpha\,Vol(B_\alpha)$$

$$\therefore\ Min_\alpha \le \frac{\iiint_{B_\alpha} f(x,y,z)\,dV}{Vol(B_\alpha)} \le Max_\alpha$$

Consider $\lim_{\alpha\to 0} Min_\alpha$ and $\lim_{\alpha\to 0} Max_\alpha$

Given $\epsilon > 0$, since f is continuous at (x_0,y_0,z_0),

$\exists\,\delta > 0$ s.t. if $\|(x,y,z)-(x_0,y_0,z_0)\| < \delta$ then

$|f(x,y,z)-f(x_0,y_0,z_0)| < \epsilon$.

f continuous $\Rightarrow$ f achieves a minimum in B_δ,

Say at (x',y',z'). $\therefore \|(x',y',z')-(x_0,y_0,z_0)\| < \delta$

so that $|f(x',y',z')-f(x_0,y_0,z_0)| < \epsilon$

and f achieves a maximum in B_δ,

say at (x'', y'', z''). $\therefore \|(x'', y'', z'') - (x_0, y_0, z_0)\| < \delta$

so that $|f(x'', y'', z'') - f(x_0, y_0, z_0)| < \epsilon$

Define this minimum, $f(x', y', z') = \text{Min}_\delta$

and this maximum $f(x'', y'', z'') = \text{Max}_\delta$

$\therefore |\text{Min}_\delta - f(x_0, y_0, z_0)| < \epsilon$ and $|\text{Max}_\delta - f(x_0, y_0, z_0)| < \epsilon$

$\therefore$ if $\alpha < \delta$ (same as $0 < |\alpha - 0| < \delta$), then

$B_\alpha \subset B_\delta$, both centred at (x_0, y_0, z_0),

and means that all $(x, y, z) \in B_\alpha \Rightarrow$

$(x, y, z) \in B_\delta$, so that $(x, y, z) \in B_\alpha \Rightarrow$

$|f(x, y, z) - f(x_0, y_0, z_0)| < \epsilon$, including

$|\text{Min}_\alpha - f(x_0, y_0, z_0)| < \epsilon$ and $|\text{Max}_\alpha - f(x_0, y_0, z_0)| < \epsilon$

$\therefore$ Given $\epsilon > 0$, $\exists\, \delta > 0$ s.t. if $0 < \alpha < \delta$, then

$|\text{Min}_\alpha - f(x_0, y_0, z_0)| < \epsilon$ and $|\text{Max}_\alpha - f(x_0, y_0, z_0)| < \epsilon$

$\therefore \lim_{\alpha \to 0} \text{Min}_\alpha = f(x_0, y_0, z_0)$ and $\lim_{\alpha \to 0} \text{Max}_\alpha = f(x_0, y_0, z_0)$

$$\lim_{\alpha \to 0} Min_\alpha \leq \lim_{\alpha \to 0} \frac{\iiint_{B_\alpha} f(x,y,z)\,dV}{vol(B_\alpha)} \leq \lim_{\alpha \to 0} Max_\alpha$$

and $\therefore \quad f(x_0,y_0,z_0) \leq \lim_{\alpha \to 0} \frac{\iiint_{B_\alpha} f(x,y,z)\,dV}{vol(B_\alpha)} \leq f(x_0,y_0,z_0)$

$\therefore \quad \lim_{\alpha \to 0} \frac{\iiint_{B_\alpha} f(x,y,z)\,dV}{vol(B_\alpha)} = f(x_0,y_0,z_0)$

Review Exercises for Chapter 5

Note Title 11/11/2016

Evaluate the integrals in Exercises 1 to 4.

1. **1.** $\int_0^3 \int_{-x^2+1}^{x^2+1} xy\,dy\,dx$

$$\int_0^3 \left(\frac{xy^2}{2} \Big|_{y=-x^2+1}^{y=x^2+1} \right) dx = \int_0^3 \frac{x(x^2+1)^2}{2} - \frac{x(-x^2+1)^2}{2}\,dx$$

$$= \int_0^3 \frac{2x(x^2+1)^2}{4}\,dx + \int_0^3 \frac{-2x(-x^2+1)^2}{4}\,dx$$

$$= \frac{(x^2+1)^3}{12}\Big|_0^3 + \frac{(-x^2+1)^3}{12}\Big|_0^3$$

$$= \frac{1000}{12} - \frac{1}{12} + \frac{-512}{12} - \frac{1}{12} = \frac{486}{12} = 40\tfrac{1}{2} = \frac{81}{2}$$

2. **2.** $\int_0^1 \int_{\sqrt{x}}^1 (x+y)^2\,dy\,dx$

Since $\frac{(x+y)^3}{3}\Big|_{y=\sqrt{x}}^{y=1}$ is messy, use change of integration limits.

$y=\sqrt{x}$ is increasing, $x=y^2$ is the inverse,

$\sqrt{0}=0$, $\sqrt{1}=1$

$$\therefore \int_0^1 \int_0^{y^2} (x+y)^2 \, dx \, dy = \int_0^1 \frac{(x+y)^3}{3} \Big|_{x=0}^{x=y^2} dy$$

$$= \int_0^1 \left[\frac{(y^2+y)^3}{3} - \frac{y^3}{3} \right] dy = \int_0^1 \frac{y^6+3y^5+3y^4+y^3}{3} - \frac{y^3}{3} \, dy$$

$$= \int_0^1 \frac{y^6}{3} + y^5 + y^4 \, dy = \frac{y^7}{21} + \frac{y^6}{6} + \frac{y^5}{5} \Big|_0^1$$

$$= \frac{1}{21} + \frac{1}{6} + \frac{1}{5} = \frac{10+35+42}{210} = \frac{87}{210} = \frac{29}{70}$$

3. **3.** $\int_0^1 \int_{e^x}^{e^{2x}} x \ln y \, dy \, dx$

$$\int_{e^x}^{e^{2x}} x \ln y \, dy = x \left[y \ln y - y \right]_{e^x}^{e^{2x}}$$

$$= x \left[e^{2x}(2x) - e^{2x} - (xe^x - e^x) \right]$$

$$= 2x^2 e^{2x} - xe^{2x} - x^2 e^x + xe^x$$

$\therefore$ Evaluate: $\int_0^1 2x^2 e^{2x} - xe^{2x} - x^2 e^x + xe^x \, dx$ [1]

$$\int_0^1 x^2 \, 2e^{2x} \, dx = x^2 e^{2x} \Big|_0^1 - \int_0^1 2xe^{2x} \, dx$$

$$= e^2 - 0 - \int_0^1 2x\, e^{2x}\, dx$$

$\therefore$ [1] becomes $e^2 + \int_0^1 -3xe^{2x} - x^2e^x + xe^x\, dx$ [2]

$$-\frac{3}{2}\int_0^1 2xe^{2x}\, dx = -\frac{3}{2}\left[xe^{2x}\Big|_0^1 - \int_0^1 e^{2x}\, dx \right]$$

$$= -\frac{3}{2}\left[e^2 - \frac{1}{2}e^{2x}\Big|_0^1 \right] = -\frac{3}{2}\left[e^2 - \frac{1}{2}e^2 + \frac{1}{2} \right]$$

$$= -\frac{3}{4}e^2 - \frac{3}{4}$$

$\therefore$ [2] becomes $\frac{1}{4}e^2 - \frac{3}{4} + \int_0^1 -x^2e^x + xe^x\, dx$ [3]

$$-\int_0^1 x^2e^x\, dx = -\left[x^2e^x\Big|_0^1 - 2\int_0^1 xe^x\, dx \right]$$

$$= -e + 2\int_0^1 xe^x\, dx$$

$\therefore$ [3] becomes $\frac{1}{4}e^2 - e - \frac{3}{4} + 3\int_0^1 xe^x\, dx$ [4]

$$3\int_0^1 xe^x\, dx = 3\left[xe^x\Big|_0^1 - \int_0^1 e^x\, dx \right]$$

$$= 3\left[e - 0 - e + 1\right] = 3$$

$$\therefore [4] \text{ becomes } \frac{1}{4}e^2 - e - \frac{3}{4} + 3 = \frac{1}{4}e^2 - e + \frac{9}{4}$$

4. **4.** $\int_0^1 \int_1^2 \int_2^3 \cos[\pi(x+y+z)]\,dx\,dy\,dz.$

$$\int_2^3 \cos[\pi(x+y+z)]\,dx = \frac{1}{\pi}\sin[\pi(x+y+z)]\Big|_{x=2}^{x=3}$$

$$= \frac{1}{\pi}\left[\sin(3\pi + \pi y + \pi z) - \sin(2\pi + \pi y + \pi z)\right]$$

now using $\sin(n\pi + b) = \cos(n\pi)\sin(b)$

$$= \frac{1}{\pi}\left[-\sin(\pi y + \pi z) - \sin(\pi y + \pi z)\right]$$

$$= -\frac{2}{\pi}\sin(\pi y + \pi z)$$

$$\therefore \int_1^2 -\frac{2}{\pi}\sin(\pi y + \pi z)\,dy = \frac{2}{\pi^2}\cos(\pi y + \pi z)\Big|_{y=1}^{y=2}$$

$$= \frac{2}{\pi^2}\left[\cos(2\pi + \pi z) - \cos(\pi + \pi z)\right]$$

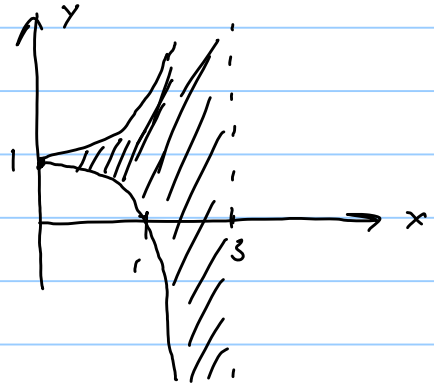
now using $\cos(n\pi + b) = \cos(n\pi)\cos(b)$

$$= \frac{2}{\pi^2}\left[\cos(\pi z) + \cos(\pi z)\right] = \frac{4}{\pi^2}\cos(\pi z)$$

$$\therefore \int_0^1 \frac{4}{\pi^2}\cos(\pi z)\,dz = \frac{4}{\pi^3}\sin(\pi z)\Big|_0^1 = 0$$

Reverse the order of integration of the integrals in Exercises 5 to 8 and evaluate.

5. **5.** The integral in Exercise 1 **1.** $\int_0^3 \int_{-x^2+1}^{x^2+1} xy\,dy\,dx$



$y = -x^2 + 1 \Rightarrow x = \sqrt{1-y}$

$x = 3 \Rightarrow y = -8$

$y = x^2 + 1 \Rightarrow x = \sqrt{y-1}$

$x = 3 \Rightarrow y = 10$

$\therefore \quad 1 \le y \le 10, \ \sqrt{y-1} \le x \le 3$

$-8 \le y \le 1, \ \sqrt{1-y} \le x \le 3$

$$\therefore \int_{-8}^{1}\int_{\sqrt{1-y}}^{3} xy\,dx\,dy + \int_{1}^{10}\int_{\sqrt{y-1}}^{3} xy\,dx\,dy$$

$$\int_{-8}^{1}\int_{\sqrt{1-y}}^{3} xy\,dx\,dy = \int_{-8}^{1} \frac{x^2 y}{2}\Big|_{x=\sqrt{1-y}}^{x=3} dy$$

$$= \int_{-8}^{1} \frac{9y}{2} - \frac{(1-y)y}{2}\,dy = \int_{-8}^{1} 4y + \frac{y^2}{2}\,dy$$

$$= 2y^2 + \frac{y^3}{6}\Big|_{-8}^{1} = 2 + \frac{1}{6} - \left(128 - \frac{512}{6}\right)$$

$$= -126 + \frac{513}{6}$$

$$\int_1^{10}\int_{\sqrt{y-1}}^{3} xy\,dx\,dy = \int_1^{10} \frac{x^2y}{2}\Big|_{x=\sqrt{y-1}}^{x=3} dy$$

$$= \int_1^{10} \frac{9y}{2} - \frac{(y-1)y}{2}\,dy = \int_1^{10} 5y - \frac{y^2}{2}\,dy$$

$$= \frac{5}{2}y^2 - \frac{y^3}{6}\Big|_1^{10} = 250 - \frac{1000}{6} - \frac{5}{2} + \frac{1}{6}$$

$$= 250 - \frac{5}{2} - \frac{999}{6}$$

$$\therefore -126 + \frac{513}{6} + 250 - \frac{5}{2} - \frac{999}{6} = 124 - \frac{5}{2} - \frac{486}{6}$$

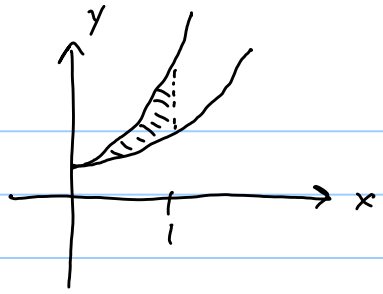
$$= 124 - \frac{5}{2} - \frac{162}{2} = \frac{248}{2} - \frac{167}{2} = \frac{81}{2}$$

6. **6.** The integral in Exercise 2 **2.** $\int_0^1 \int_{\sqrt{x}}^1 (x+y)^2\,dy\,dx$

This was done in #2 using $\int_0^1 \int_0^{y^2} (x+y)^2\,dx\,dy$

$$= \frac{29}{70}$$

7. **7.** The integral in Exercise 3 **3.** $\int_0^1 \int_{e^x}^{e^{2x}} x \ln y \, dy \, dx$



$y = e^x \Rightarrow x = \ln y$

$e^0 \Rightarrow 1, \quad e^1 \Rightarrow e$

$y = e^{2x} \Rightarrow x = \frac{1}{2}\ln y, \quad e^{2(0)} = 1, \quad e^{2(1)} = e^2$

$\therefore$ Break into 2 adjacent x-simple regions.

$1 \le y \le e : \frac{1}{2}\ln y \le x \le \ln y$

$e \le y \le e^2 : \frac{1}{2}\ln y \le x \le 1$

$$\therefore \int_1^e \int_{\frac{1}{2}\ln y}^{\ln y} x \ln y \, dx \, dy + \int_e^{e^2} \int_{\frac{1}{2}\ln y}^{1} x \ln y \, dx \, dy$$

[1]: $$\int_1^e \int_{\frac{1}{2}\ln y}^{\ln y} x \ln y \, dx \, dy = \int_1^e \frac{x^2}{2} \ln y \Big|_{x=\frac{1}{2}\ln y}^{x=\ln y} dy$$

$$= \int_1^e \frac{(\ln y)^3}{2} - \frac{1}{8}(\ln y)^3 \, dy = \frac{3}{8}\int_1^e (\ln y)^3 \, dy$$

Using $\int (\log x)^m dx = x(\log x)^m - m\int (\log x)^{m-1} dx$

$$= \frac{3}{8}\left[y(\ln y)^3 \Big|_1^e - 3\int_1^e (\ln y)^2 \, dy \right]$$

$$= \frac{3}{8}\left[e - 0 - 3\left(y(\ln y)^2 \Big|_1^e - 2\int_1^e (\ln y)\, dy \right) \right]$$

$$= \frac{3}{8}\left[e - 3(e) + 6\left(y\ln y - y \right)\Big|_1^e \right]$$

$$= \frac{3}{8}\left[-2e + 6\left(e - e - (0-1) \right) \right]$$

$$= \frac{3}{8}\left[-2e + 6 \right] = -\frac{3}{4}e + \frac{9}{4}$$

[2]: $$\int_e^{e^2}\int_{\frac{1}{2}\ln y}^{1} x \ln y \, dx\, dy = \int_e^{e^2} \frac{x^2}{2}\ln y \Big|_{x=\frac{1}{2}\ln y}^{x=1} dy$$

$$= \int_e^{e^2} \frac{1}{2}\ln y - \frac{1}{8}(\ln y)^3 \, dy$$

$$= \frac{1}{2}\left[y\ln y - y \right]_e^{e^2} - \frac{1}{8}\int_e^{e^2} (\ln y)^3 \, dy$$

$$= \frac{1}{2}\left[2e^2 - e^2 - (e-e) \right] - \frac{1}{8}\int_e^{e^2} (\ln y)^3 \, dy$$

$$= \frac{e^2}{2} - \frac{1}{8}\int_e^{e^2} (\ln y)^3 \, dy$$ [2]

Using $\int (\log x)^m dx = x(\log x)^m - m\int (\log x)^{m-1} dx$

$$\int_e^{e^2} (\ln y)^3 dy = y(\ln y)^3 \Big|_e^{e^2} - 3\int_e^{e^2} (\ln y)^2 dy$$

$$= (8e^2 - e) - 3\left[y(\ln y)^2 \Big|_e^{e^2} - 2\int_e^{e^2} \ln y\, dy \right]$$

$$= 8e^2 - e - 3\left[4e^2 - e - 2(y\ln y - y)\Big|_e^{e^2} \right]$$

$$= 8e^2 - e - 12e^2 + 3e + 6\left[2e^2 - e^2 - (e - e) \right]$$

$$= -4e^2 + 2e + 6e^2 = 2e^2 + 2e$$

$\therefore$ [2] becomes $\frac{e^2}{2} - \frac{1}{8}(2e^2 + 2e) = \frac{1}{4}e^2 - \frac{e}{4}$

$\therefore$ [1] + [2] $= -\frac{3}{4}e + \frac{9}{4} + \frac{1}{4}e^4 - \frac{e}{4} = \frac{1}{4}e^4 - e + \frac{9}{4}$

8. **8.** The integral in Exercise 4 **4.** $\int_0^1 \int_1^2 \int_2^3 \cos[\pi(x + y + z)]\, dx\, dy\, dz.$

$$\int_2^3 \int_1^2 \int_0^1 \cos[\pi x + \pi y + \pi z]\, dz\, dy\, dx$$

$$= \int_2^3 \int_1^2 \frac{1}{\pi} \sin[\pi x + \pi y + \pi z]\Big|_{z=0}^{z=1} dy\, dx$$

$$= \int_2^3 \int_1^2 \frac{1}{\pi} \left[\sin(\pi x + \pi y + \pi) - \sin(\pi x + \pi y) \right] dy\,dx$$

Now use $\sin(\alpha + \pi) = -\sin\alpha$

$$= \int_2^3 \int_1^2 -\frac{2}{\pi} \sin(\pi x + \pi y)\, dy\,dx$$

$$= \int_2^3 \frac{2}{\pi^2} \cos(\pi x + \pi y) \Big|_{y=1}^{y=2} dx$$

$$= \int_2^3 \frac{2}{\pi^2} \left[\cos(\pi x + 2\pi) - \cos(\pi x + \pi) \right] dx$$

Now use $\cos(\alpha + n\pi) = (-1)^n \cos\alpha$

$$= \int_2^3 \frac{4}{\pi^2} \cos(\pi x)\, dx = \frac{4}{\pi^3} \sin(\pi x) \Big|_{x=2}^{x=3} = \underline{0}$$

9. **9.** Evaluate the integral $\int_0^1 \int_0^x \int_0^y (y + xz)\, dz\, dy\, dx$.

$$\int_0^y (y + xz)\, dz = yz + \frac{xz^2}{2} \Big|_{z=0}^{z=y} = y^2 + \frac{xy^2}{2}$$

$$\int_0^x \left(y^2 + \frac{xy^2}{2}\right) dy = \frac{y^3}{3} + \frac{xy^3}{6} \Big|_{y=0}^{y=x} = \frac{x^3}{3} + \frac{x^4}{6}$$

$$\int_0^1 \frac{x^3}{3} + \frac{x^4}{6}\,dx = \frac{x^4}{12} + \frac{x^5}{30}\Big|_0^1 = \frac{1}{12} + \frac{1}{30} = \frac{5}{60} + \frac{2}{60}$$

$$= \frac{7}{60}$$

10. **10.** Evaluate $\int_0^1 \int_y^{y^2} e^{x/y}\,dx\,dy$.

$$\int_y^{y^2} e^{x/y}\,dx = y e^{\frac{x}{y}}\Big|_{x=y}^{x=y^2} = y e^{y} - ye$$

$$\therefore \int_0^1 (y e^y - ey)\,dy = y e^y\Big|_0^1 - \int_0^1 e^y - e\frac{y^2}{2}\Big|_0^1$$

$$= (e-0) - (e-1) - \left(\frac{e}{2} - 0\right)$$

$$= 1 - \frac{e}{2}$$

11. **11.** Evaluate $\int_0^1 \int_0^{(\arcsin y)/y} y\cos xy\,dx\,dy$.

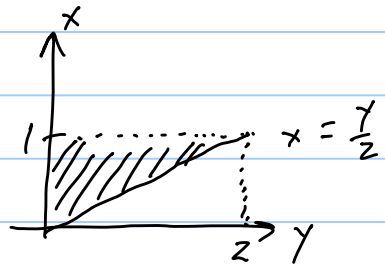
$$\int_0^{\frac{\arcsin y}{y}} y\cos(xy)\,dx = \sin(xy)\Big|_{x=0}^{x=\frac{\arcsin y}{y}}$$

$$= \sin\left[y \cdot \frac{\arcsin y}{y}\right] - 0 = \sin(\arcsin y) = y$$

$$\therefore \int_0^1 y\,dy = \frac{y^2}{2}\Big|_0^1 = \underline{\underline{\frac{1}{2}}}$$

12. **12.** Change the order of integration and evaluate

$$\int_0^2 \int_{y/2}^1 (x+y)^2\,dx\,dy.$$



$x = \frac{y}{2}$ is an increasing function of y.

Inverse function is $y = 2x$

$$\frac{y}{2}\Big|_{y=0} = 0 \qquad \frac{y}{2}\Big|_{y=2} = 1$$

$$\therefore \int_0^1 \int_0^{2x} (x+y)^2\,dy\,dx$$

$$= \int_0^1 \frac{(x+y)^3}{3}\Big|_{y=0}^{y=2x} dx = \int_0^1 \frac{27}{3}x^3 - \frac{x^3}{3}\,dx$$

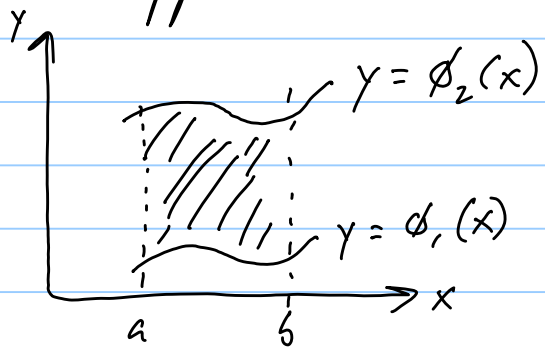
$$= \frac{26}{12}x^4\Big|_0^1 = \frac{26}{12} = \underline{\underline{\frac{13}{6}}}$$

13. **13.** Show that evaluating $\iint_D dx\,dy$, where D is a y-simple region, reproduces the formula from one-variable calculus for the area between two curves.

Since D is a y-simple region, let the lower bound be $y = \phi_1(x)$, upper bound $y = \phi_2(x)$.

$\therefore \phi_1(x) \le y \le \phi_2(x)$, and suppose $a \le x \le b$.

$$\therefore \iint_D dx\,dy = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} dy\,dx$$



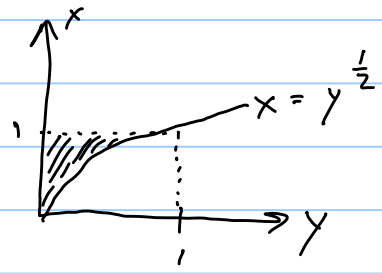
$$= \int_a^b y \Big|_{y=\phi_1(x)}^{y=\phi_2(x)} dx$$

$$= \int_a^b [\phi_2(x) - \phi_1(x)]\,dx = \text{area between between 2 curves: } \phi_1(x) \text{ and } \phi_2(x).$$

14.

14. Change the order of integration and evaluate

$$\int_0^1 \int_{y^{1/2}}^1 (x^2 + y^3 x)\,dx\,dy.$$



$x = \sqrt{y}$ is an increasing function
inverse is: $y = x^2$

$$\sqrt{y}\Big|_{y=0} = 0 \qquad \sqrt{y}\Big|_{y=1} = 1$$

$$\therefore \int_0^1 \int_0^{x^2} (x^2 + y^3 x)\,dy\,dx = \int_0^1 x^2 y + \frac{y^4 x}{4}\Big|_{y=0}^{y=x^2} dx$$

$$= \int_0^1 \left(x^4 + \frac{x^9}{4}\right) dx = \frac{x^5}{5} + \frac{x^{10}}{40}\Bigg|_0^1 = \frac{1}{5} + \frac{1}{40} = \frac{9}{40}$$

15.

15. Let D be the region in the xy plane inside the unit circle $x^2 + y^2 = 1$. Evaluate $\iint_D f(x, y)\, dx\, dy$ in each of the following cases:

$$-1 \le x \le 1$$
$$-\sqrt{1-x^2} \le y \le \sqrt{1-x^2}$$

(a) (a) (a)] $f(x, y) = xy$

$$\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} xy \, dy\, dx = \int_{-1}^{1} \frac{xy^2}{2}\Bigg|_{y=-\sqrt{1-x^2}}^{y=\sqrt{1-x^2}} dx$$

$$= \int_{-1}^{1} \frac{x(1-x^2)}{2} - \frac{x(1-x^2)}{2}\, dx = \int_{-1}^{1} 0\, dx = 0$$

(5) (b) $f(x, y) = x^2 y^2$

$$\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} x^2 y^2 \, dy\, dx = \int_{-1}^{1} \frac{x^2 y^3}{3}\Bigg|_{y=-\sqrt{1-x^2}}^{y=\sqrt{1-x^2}} dx$$

$$= \int_{-1}^{1} \frac{x^2(1-x^2)^{\frac{3}{2}}}{3} + \frac{x^2(1-x^2)^{3/2}}{3}\, dx = \frac{2}{3}\int_{-1}^{1} x^2(1-x^2)^{3/2}\, dx$$

Let $x = \sin\theta$ $\quad x = -1 \Rightarrow \theta = -\frac{\pi}{2}$, $\quad x = 1 \Rightarrow \theta = \frac{\pi}{2}$

$(1-x^2)^{3/2} = \cos^3\theta \quad dx = \cos\theta\, d\theta$

$$\therefore \frac{2}{3}\int_{-1}^{1} x^2(1-x^2)^{3/2}\,dx = \frac{2}{3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2\theta\cos^4\theta\,d\theta$$

Now use $\sin^2\theta = \frac{1-\cos 2\theta}{2}$, $\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$$\therefore \frac{2}{3}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1-\cos 2\theta}{2}\right)\left(\frac{1+\cos 2\theta}{4}\right)^2 d\theta$$

$$= \frac{1}{12}\int_{-\frac{\pi}{2}}^{\pi/2} \left(1+\cos 2\theta - \cos^2 2\theta - \cos^3 2\theta\right) d\theta$$

$$= \frac{1}{12}\int_{-\frac{\pi}{2}}^{\pi/2} \left[1+\cos 2\theta - \left(\frac{1+\cos 2\theta}{2}\right) - (1-\sin^2 2\theta)\cos 2\theta\right] d\theta$$

$$= \frac{1}{12}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left[\frac{1}{2} + \frac{\cos 2\theta}{2} - \cos 2\theta + \cos 2\theta \sin^2 2\theta\right] d\theta$$

$$= \frac{1}{12}\left[\frac{\theta}{2}\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} - \frac{\sin 2\theta}{4}\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \frac{1}{6}\sin^3 2\theta\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\right]$$

$$= \frac{1}{12}\left[\frac{\pi}{2} - 0 + \frac{1}{6}(0)\right] = \frac{\pi}{24}$$

(c) (c) $f(x, y) = x^3 y^3$

$$\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} x^3 y^3 \, dy \, dx = \int_{-1}^{1} \frac{x^3 y^4}{4} \Big|_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \, dx$$

$$= \int_{-1}^{1} \frac{x^3 (1-x^2)^2}{4} - \frac{x^3 (1-x^2)^2}{4} \, dx$$

$$= \int_{-1}^{1} 0 \, dx = \underline{0}$$

16.

16. Find $\iint_D y[1 - \cos(\pi x/4)]\,dx\,dy$, where D is the region in Figure 5.R.1.

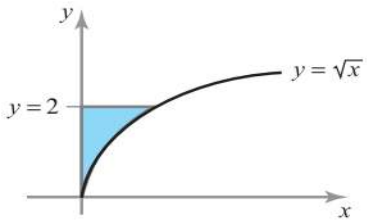


figure **5.R.1** The region of integration for Exercise 16.

For $0 \le y \le 2$,
$0 \le x \le 4$

$$\int_0^4 \int_{\sqrt{x}}^2 y \left[1 - \cos\left(\frac{\pi x}{4}\right)\right] dy \, dx$$

$$= \int_0^4 \frac{y^2}{2} \left[1 - \cos\left(\frac{\pi x}{4}\right)\right]_{y=\sqrt{x}}^{y=2} dx$$

$$= \int_0^4 \left[2 - 2\cos\left(\frac{\pi x}{4}\right) - \frac{x}{2} + \frac{x}{2}\cos\left(\frac{\pi x}{4}\right)\right] dx$$

$$= 2x\Big|_0^4 - \frac{8}{\pi}\sin\left(\frac{\pi x}{4}\right)\Big|_0^4 - x^2\Big|_0^4 + \int_0^4 \frac{x}{2}\cos\left(\frac{\pi}{4}x\right)dx$$

$$= (8 - 0 - 16) + \int_0^4 \frac{x}{2}\cos\left(\frac{\pi}{4}x\right)dx$$

$$= -8 + \frac{1}{2}\int_0^4 x\cos\left(\frac{\pi x}{4}\right)dx$$

$$= -8 + \frac{1}{2}\left[\frac{4}{\pi}x\sin\left(\frac{\pi x}{4}\right)\Big|_0^4 - \frac{4}{\pi}\int_0^4 \sin\left(\frac{\pi x}{4}\right)dx\right]$$

$$= -8 + 0 - \frac{2}{\pi}\left[-\frac{4}{\pi}\cos\left(\frac{\pi x}{4}\right)\right]_0^4$$

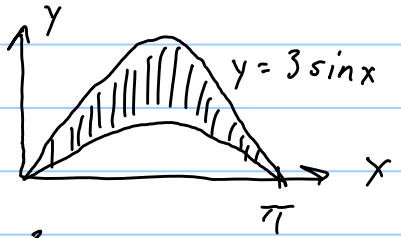
$$= -8 + \frac{8}{\pi^2}\left[\cos(\pi) - \cos(0)\right]$$

$$= -8 - \frac{16}{\pi^2}$$

Evaluate the integrals in Exercises 17 to 24. Sketch and identify the type of the region (corresponding to the way the integral is written).

17. **17.** $\int_0^{\pi}\int_{\sin x}^{3\sin x} x(1+y)\,dy\,dx$

y-simple:



$$\int_{\sin x}^{3\sin x} x(1+y)\,dy = xy + \frac{xy^2}{2}\Big|_{y=\sin x}^{y=3\sin x}$$

$$= 3x\sin x + \frac{9}{2}x\sin^2 x - x\sin x - \frac{x}{2}\sin^2 x$$

$$= 2x\sin x + 4x\sin^2 x$$

$$\therefore \int_0^{\pi} (2x\sin x + 4x\sin^2 x)\,dx =$$

$$-2x\cos x\Big|_0^{\pi} + 2\int_0^{\pi}\cos x\,dx + \int_0^{\pi} 4x\sin^2 x\,dx$$

$$= 2\pi + 2\sin x\Big|_0^{\pi} + \int_0^{\pi} 4x\left(\frac{1-\cos 2x}{2}\right)dx$$

using $\sin^2 x = \frac{1-\cos 2x}{2}$

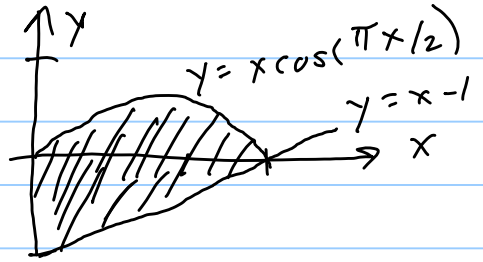
$$= 2\pi + 0 + \int_0^{\pi} 2x\,dx - \int_0^{\pi} 2x\cos 2x\,dx$$

$$= 2\pi + \pi^2 - \left[x\sin 2x\Big|_0^{\pi} - \int_0^{\pi}\sin 2x\,dx\right]$$

$$= 2\pi + \pi^2 - \left[0 + \frac{1}{2}\cos 2x\Big|_0^{\pi}\right]$$

$$= 2\pi + \pi^2 - \left\{\frac{1}{2} - \frac{1}{2}\right\} = \underline{2\pi + \pi^2}$$

18. **18.** $\displaystyle\int_0^1 \int_{x-1}^{x\cos(\pi x/2)} (x^2 + xy + 1)\,dy\,dx$



y-simple

$$\int_{x-1}^{x\cos\left(\frac{\pi x}{2}\right)} (x^2 + xy + 1)\,dy = x^2y + \frac{xy^2}{2} + y \Bigg|_{y=x-1}^{y=x\cos\left(\frac{\pi x}{2}\right)}$$

$$= x^2\left[x\cos\left(\frac{\pi x}{2}\right)\right] + \frac{x}{2}\left[x^2\cos^2\left(\frac{\pi x}{2}\right)\right] + x\cos\left(\frac{\pi x}{2}\right)$$
$$- \left[x^2(x-1) + x\frac{(x-1)^2}{2} + (x-1)\right]$$

$$= x^3\cos\left(\frac{\pi x}{2}\right) + \frac{x^3}{2}\cos^2\left(\frac{\pi x}{2}\right) + x\cos\left(\frac{\pi x}{2}\right)$$

$$- x^3 + x^2 - \frac{x^3}{2} + x^2 - \frac{x}{2} - x + 1$$

$$= x^3\cos\left(\frac{\pi x}{2}\right) + \frac{x^3}{2}\cos^2\left(\frac{\pi x}{2}\right) + x\cos\left(\frac{\pi x}{2}\right)$$

$$-\frac{3}{2}x^3 + 2x^2 - \frac{3}{2}x + 1$$

[1] $$\int_0^1 x^3\cos\left(\frac{\pi x}{2}\right)dx = \frac{2}{\pi}x^3\sin\left(\frac{\pi x}{2}\right)\Big|_0^1 - \frac{3\cdot 2}{\pi}\int_0^1 x^2\sin\left(\frac{\pi x}{2}\right)dx$$

$$= \frac{2}{\pi} - 0 - \frac{6}{\pi}\left[-\frac{2}{\pi}x^2\cos\left(\frac{\pi}{2}x\right)\Big|_0^1 + \frac{2\cdot 2}{\pi}\int_0^1 x\cos\left(\frac{\pi}{2}x\right)dx\right]$$

$$= \frac{2}{\pi} - \frac{6}{\pi}\left[0 + \frac{4}{\pi}\int_0^1 x\cos\left(\frac{\pi}{2}x\right)dx\right]$$

$$= \frac{2}{\pi} - \frac{24}{\pi^2}\left[\frac{2}{\pi}x\sin\left(\frac{\pi}{2}x\right)\Big|_0^1 - \frac{2}{\pi}\int_0^1 \sin\left(\frac{\pi x}{2}\right)dx\right]$$

$$= \frac{2}{\pi} - \frac{24}{\pi^2}\left[\frac{2}{\pi} + \frac{4}{\pi^2}\cos\left(\frac{\pi x}{2}\right)\Big|_0^1\right]$$

$$= \frac{2}{\pi} - \frac{24}{\pi^2}\left[\frac{2}{\pi} + \left(0 - \frac{4}{\pi^2}\right)\right]$$

$$= \frac{2}{\pi} - \frac{48}{\pi^3} + \frac{96}{\pi^4}$$

[2] $$\int_0^1 \frac{x^3}{2}\cos^2\left(\frac{\pi x}{2}\right)dx = \frac{1}{2}\int_0^1 x^3\left(\frac{1+\cos(\pi x)}{2}\right)dx$$

using $\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$$= \frac{1}{4}\int_0^1 x^3\,dx + \frac{1}{4}\int_0^1 x^3\cos(\pi x)\,dx$$

$$\frac{1}{4}\int_0^1 x^3\,dx = \frac{1}{16}x^4\Big|_0^1 = \frac{1}{16}$$

$$\frac{1}{4}\int_0^1 x^3\cos(\pi x)\,dx = \frac{1}{4\pi}x^3\sin(\pi x)\Big|_0^1 - \frac{3}{4\pi}\int_0^1 x^2\sin(\pi x)\,dx$$

$$= 0 - \frac{3}{4\pi}\left[-\frac{1}{\pi}x^2\cos(\pi x)\Big|_0^1 + \frac{2}{\pi}\int_0^1 x\cos(\pi x)\,dx\right]$$

$$= -\frac{3}{4\pi}\left[\frac{1}{\pi} + \frac{1}{\pi}x\sin(\pi x)\Big|_0^1 - \frac{1}{\pi}\int_0^1 \sin(\pi x)\,dx\right]$$

$$= -\frac{3}{4\pi}\left[\frac{1}{\pi} + 0 + \frac{1}{\pi^2}\cos(\pi x)\Big|_0^1\right]$$

$$= -\frac{3}{4\pi}\left\{\frac{1}{\pi} + \frac{1}{\pi^2}(-1-1)\right\} = -\frac{3}{4\pi}\left\{\frac{1}{\pi} - \frac{2}{\pi^2}\right\}$$

$$= -\frac{3}{4\pi^2} + \frac{3}{2\pi^3}$$

$$\therefore \frac{1}{16} - \frac{3}{4\pi^2} + \frac{3}{2\pi^3}$$

[3] $$\int_0^1 x\cos\left(\frac{\pi x}{2}\right)dx = \frac{2}{\pi}x\sin\left(\frac{\pi x}{2}\right)\Big|_0^1 - \frac{2}{\pi}\int_0^1 \sin\left(\frac{\pi x}{2}\right)dx$$

$$= \frac{2}{\pi} + \frac{2}{\pi}\left[\frac{2}{\pi}\cos\left(\frac{\pi x}{2}\right)\Big|_0^1\right]$$

$$= \frac{2}{\pi} + \frac{2}{\pi}\left[0 - \frac{2}{\pi}\right] = \frac{2}{\pi} - \frac{4}{\pi^2}$$

$$\therefore [1] + [2] + [3] =$$

$$\left(\frac{2}{\pi} - \frac{48}{\pi^3} + \frac{96}{\pi^4}\right) + \left(\frac{1}{16} - \frac{3}{4\pi^2} + \frac{3}{2\pi^3}\right) + \left(\frac{2}{\pi} - \frac{4}{\pi^2}\right)$$

$$= \frac{1}{16} + \frac{4}{\pi} - \frac{19}{4\pi^2} - \frac{93}{2\pi^3} + \frac{96}{\pi^4}$$

$$[4] \int_0^1 -\frac{3}{2}x^3 + 2x^2 - \frac{3}{2}x + 1 \, dx$$

$$= -\frac{3}{8}x^4 + \frac{2}{3}x^3 - \frac{3}{4}x^2 + x \Big|_0^1$$

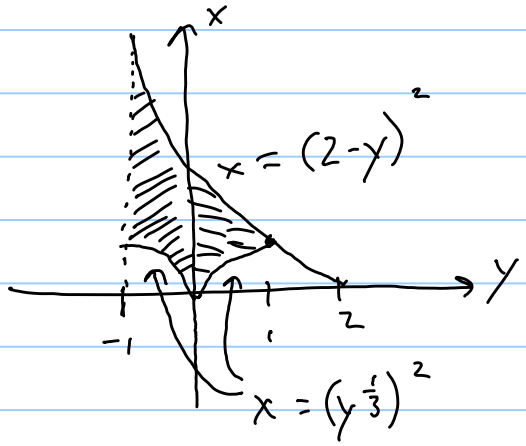
$$= -\frac{3}{8} + \frac{2}{3} - \frac{3}{4} + 1 = \frac{-9 + 16 - 18 + 24}{24} = \frac{13}{24}$$

$$\therefore [1] + [2] + [3] + [4] =$$

$$\frac{29}{48} + \frac{4}{\pi} - \frac{19}{4\pi^2} - \frac{93}{2\pi^3} + \frac{96}{\pi^4}$$

19.

19. $\displaystyle\int_{-1}^{1}\int_{y^{2/3}}^{(2-y)^2} \left(\frac{3}{2}\sqrt{x} - 2y\right) dx\,dy$



2 adjacent x-simple regions

$$\int_{y^{2/3}}^{(2-y)^2} \left(\frac{3}{2}\sqrt{x} - 2y\right) dx =$$

$$\frac{3}{2}\left(\frac{2}{3}\right)x^{\frac{3}{2}} - 2xy \Big|_{x=y^{2/3}}^{x=(2-y)^2}$$

$$= (2-y)^3 - 2(2-y)^2 y - \left[y - 2y^{\frac{5}{3}}\right]$$

$$= (2-y)^3 - 2y^3 + 8y^2 - 9y + 2y^{5/3}$$

$$\therefore \int_{-1}^{1} (2-y)^3 - 2y^3 + 8y^2 - 9y + 2y^{5/3}\, dy$$

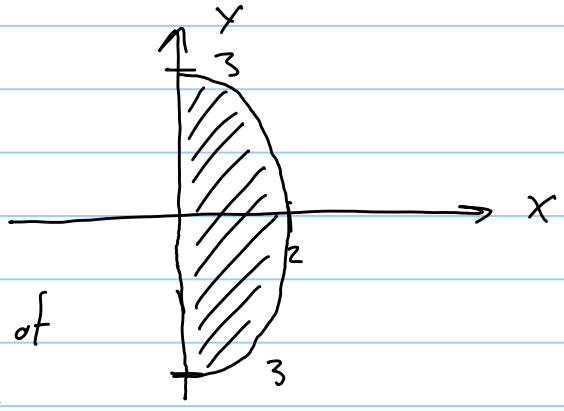
$$= -\frac{(2-y)^4}{4} - \frac{y^4}{2} + \frac{8}{3}y^3 - \frac{9}{2}y^2 + \frac{3}{4}y^{8/3}\Big|_{-1}^{1}$$

$$= -\frac{1}{4} - \frac{1}{2} + \frac{8}{3} - \frac{9}{2} + \frac{3}{4} - \left[-\frac{81}{4} - \frac{1}{2} - \frac{8}{3} - \frac{9}{2} + \frac{3}{4}\right]$$

$$= 20 + \frac{16}{3} = \frac{76}{3}$$

20.

20. $\displaystyle\int_0^2 \int_{-3(\sqrt{4-x^2})/2}^{3(\sqrt{4-x^2})/2} \left(\frac{5}{\sqrt{2+x}} + y^3\right) dy\, dx$



$y = \frac{3}{2}\sqrt{4-x^2} \Rightarrow \frac{x^2}{4} + \frac{y^2}{9} = 1$

$\therefore$ y-simple (opposite sides of ellipse).

$$\int_{-\frac{3}{2}\sqrt{4-x^2}}^{\frac{3}{2}\sqrt{4-x^2}} \left(\frac{5}{\sqrt{2+x}} + y^3\right) dy\, dx$$

$$= \frac{5y}{\sqrt{x+2}} + \frac{y^4}{4}\Bigg|_{y=-\frac{3}{2}\sqrt{4-x^2}}^{y=\frac{3}{2}\sqrt{4-x^2}}$$

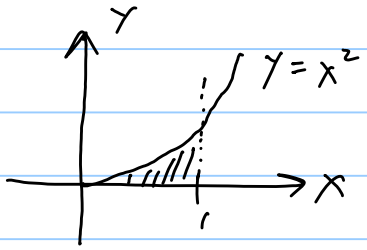
$$= \frac{15\sqrt{4-x^2}}{\sqrt{x+2}} = 15\sqrt{2-x}$$ since, for $0 \le x \le 2$, $\sqrt{x+2} \neq 0$

$$\therefore \int_0^2 15\sqrt{2-x}\,dx = -15\left(\tfrac{2}{3}\right)(2-x)^{3/2}\Big|_0^2$$

$$= 0 - \left(-10(2)^{3/2}\right) = 10(2\sqrt{2}) = \underline{\underline{20\sqrt{2}}}$$

21. **21.** $\int_0^1 \int_0^{x^2} (x^2 + xy - y^2)\,dy\,dx$

y-simple region



$$\int_0^{x^2} (x^2 + xy - y^2)\,dy = x^2y + \frac{xy^2}{2} - \frac{y^3}{3}\Big|_{y=0}^{y=x^2}$$

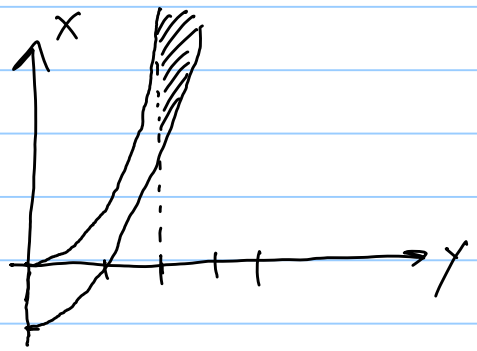
$$= x^4 + \frac{x^5}{2} - \frac{x^6}{3}$$

$$\therefore \int_0^1 x^4 + \frac{x^5}{2} - \frac{x^6}{3}\,dx = \frac{x^5}{5} + \frac{x^6}{12} - \frac{x^7}{21}\Big|_0^1$$

$$= \frac{1}{5} + \frac{1}{12} - \frac{1}{21} = \frac{84 + 35 - 20}{420} = \frac{99}{420} = \underline{\underline{\frac{33}{140}}}$$

22. **22.** $\int_2^4 \int_{y^2-1}^{y^3} 3\,dx\,dy$

x-simple regions



$$\int_2^4 \int_{y^2-1}^{y^3} 3\,dx\,dy = \int_2^4 3x\Big|_{x=y^2-1}^{x=y^3}\,dy$$

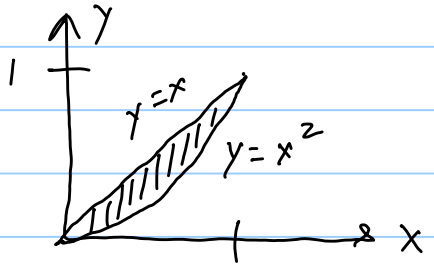
$$= \int_2^4 3y^3 - 3y^2 + 3\,dy = \frac{3y^4}{4} - y^3 + 3y \Big|_2^4$$

$$= \frac{3}{4}(256) - 64 + 12 - \left(\frac{3}{4}(16) - 8 + 6\right)$$

$$= \frac{3}{4}(240) - 56 + 6 = 180 - 50 = \underline{130}$$

23. **23.** $\int_0^1 \int_{x^2}^{x} (x+y)^2\,dy\,dx$

y - Simple region



$$\int_0^1 \int_{x^2}^{x} (x+y)^2\,dy\,dx =$$

$$\int_0^1 \frac{(x+y)^3}{3}\Big|_{y=x^2}^{y=x}\,dx =$$

$$\int_0^1 \frac{8x^3}{3} - \frac{(x+x^2)^3}{3}\,dx = \int_0^1 \frac{8x^3 - (x^6 + 3x^5 + 3x^4 + x^3)}{3}\,dx$$

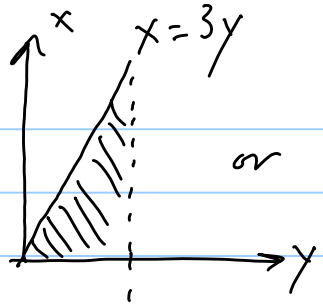
$$= \frac{1}{3}\int_0^1 -x^6 - 3x^5 - 3x^4 + 7x^3\,dx = \frac{1}{3}\left[-\frac{x^7}{7} - \frac{x^6}{2} - \frac{3}{5}x^5 + \frac{7x^4}{4}\right]_0^1$$

$$= \frac{1}{3}\left(-\frac{1}{7} - \frac{1}{2} - \frac{3}{5} + \frac{7}{4}\right) = \frac{1}{3}\left(\frac{-20 - 70 - 84 + 245}{140}\right)$$

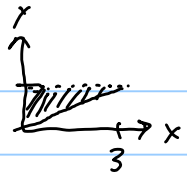
$$= \frac{1}{3}\left(\frac{71}{140}\right) = \underline{\frac{71}{420}}$$

24. **24.** $\int_0^1 \int_0^{3y} e^{x+y}\,dx\,dy$

x-simple region



or



$$\int_0^1 \int_0^{3y} e^{x+y}\,dx\,dy = \int_0^1 e^{x+y}\Big|_{x=0}^{x=3y}\,dy$$

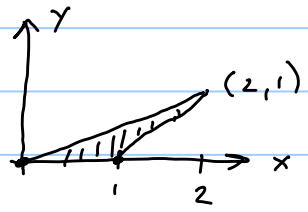
$$= \int_0^1 e^{4y} - e^{y}\,dy = \frac{1}{4}e^{4y} - e^{y}\Big|_0^1 =$$

$$\frac{1}{4}e^4 - e - \left(\frac{1}{4}e^0 - e^0\right) = \frac{1}{4}e^4 - e - \left(\frac{1}{4} - 1\right)$$

$$= \frac{e^4}{4} - e + \frac{3}{4}$$

In Exercises 25 to 27, integrate the given function f over the given region D.

25. **25.** $f(x, y) = x - y$; D is the triangle with vertices (0, 0), (1, 0), and (2, 1).



It's easiest to make D an x-simple region: $x = y + 1$ as the "top", and $x = 2y$ as the "bottom", for $0 \le y \le 1$.

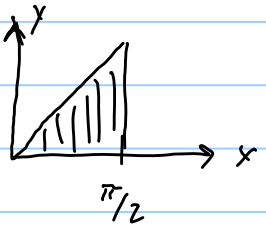
$$\therefore \int_0^1 \int_{y+1}^{2y} (x - y)\,dx\,dy = \int_0^1 \frac{x^2}{2} - xy\Big|_{x=y+1}^{x=2y}\,dy$$

$$= \int_0^1 \frac{(2y)^2}{2} - 2y^2 - \left[\frac{(y+1)^2}{2} - (y+1)y\right] dy$$

$$= \int_0^1 0 - \left[\frac{y^2 + 2y + 1}{2} - y^2 - y\right] dy$$

$$= \int_0^1 \frac{-y^2+1}{2}\, dy = -\frac{y^3}{6} + \frac{y}{2}\Big|_0^1 = -\frac{1}{6} + \frac{1}{2} = \frac{1}{3}$$

26. **26.** $f(x, y) = x^3 y + \cos x$; D is the triangle defined by $0 \le x \le \pi/2$, $0 \le y \le x$.



$$\int_0^{\frac{\pi}{2}} \int_0^x (x^3 y + \cos x)\, dy\, dx$$

$$= \int_0^{\frac{\pi}{2}} \int_y^{\frac{\pi}{2}} (x^3 y + \cos x)\, dx\, dy$$

$$= \int_0^{\frac{\pi}{2}} \frac{x^4}{4} y + \sin x \Big|_{x=y}^{x=\frac{\pi}{2}} dy$$

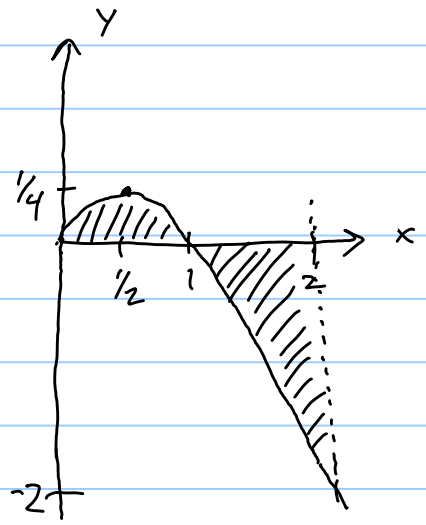
$$= \int_0^{\frac{\pi}{2}} \frac{\pi^4}{64} y + 1 - \frac{y^5}{4} - \sin y\, dy$$

$$= \frac{\pi^4}{128} y^2 + y - \frac{y^6}{24} + \cos y \Big|_0^{\frac{\pi}{2}}$$

$$= \frac{\pi^6}{2^9} + \frac{\pi}{2} - \frac{\pi^6}{2^6 \cdot 2^3 \cdot 3} + 0 - (0+0-0+1)$$

$$= \frac{\pi^6}{256} + \frac{\pi}{2} - 1$$

27. **27.** $f(x, y) = x^2 + 2xy^2 + 2$; D is the region bounded by the graph of $y = -x^2 + x$, the x axis, and the lines $x = 0$ and $x = 2$.



$$-x^2 + x = -(x^2 - x) = -\left(x - \tfrac{1}{2}\right)^2 + \tfrac{1}{4}$$

Assume D is two regions: $0 \le x \le 1$ and $1 \le x \le 2$, as description in problem is imprecise.

$$\therefore \int_0^1 \int_0^{-x^2+x} (x^2 + 2xy^2 + 2)\,dy\,dx + \int_1^2 \int_{-x^2+x}^0 (x^2 + 2xy^2 + 2)\,dy\,dx$$

$$(1) \int_0^1 \int_0^{-x^2+x} (x^2 + 2xy^2 + 2)\,dy\,dx = \int_0^1 x^2 y + \frac{2}{3} x y^3 + 2y \Big|_{y=0}^{y=-x^2+x} dx$$

$$= \int_0^1 x^2(-x^2+x) + \underbrace{\frac{2}{3} x(-x^2+x)^3}_{\frac{2}{3}(-x^7+3x^6-3x^5+x^4)} + 2(-x^2+x)\,dx$$

$$= \int_0^1 -x^4 + x^3 - 2x^2 + 2x - \frac{2}{3}x^7 + 2x^6 - 2x^5 + \frac{2}{3}x^4 \, dx$$

$$= -\frac{x^5}{3\cdot 5} + \frac{x^4}{4} - \frac{2}{3}x^3 + x^2 - \frac{2}{24}x^8 + \frac{2}{7}x^7 - \frac{2}{6}x^6 \Big|_0^1$$

$$= -\frac{1}{15} + \frac{1}{4} - \frac{2}{3} + 1 - \frac{1}{12} + \frac{2}{7} - \frac{1}{3} \qquad 2^2\cdot 3\cdot 5\cdot 7 = 420$$

$$= \frac{-28 + 105 - 280 + 420 - 35 + 120 - 140}{420}$$

$$= \frac{162}{420} = \frac{81}{210} = \underline{\frac{27}{70}}$$

(2) $$\int_1^2 \int_{-x^2+x}^{0} (x^2 + 2xy^2 + 2)\, dy\, dx = \int_1^2 x^2y + \frac{2}{3}xy^3 + 2y \Big|_{y=-x^2+x}^{y=0} dx$$

Now using same calculations as in (1), but using (-1)

$$= \int_1^2 x^4 - x^3 + 2x^2 - 2x + \frac{2}{3}x^7 - 2x^6 + 2x^5 - \frac{2}{3}x^4 \, dx$$

$$= \frac{x^5}{15} - \frac{x^4}{4} + \frac{2}{3}x^3 - x^2 + \frac{1}{12}x^8 - \frac{2}{7}x^7 + \frac{1}{3}x^6 \Big|_1^2$$

$$= \frac{32}{15} - \frac{16}{4} + \frac{2}{3}(8) - 4 + \frac{1}{12}(256) - \frac{2}{7}(128) + \frac{1}{3}(64) + \frac{27}{70}$$

$$= \frac{32(28) - 4(420) + 16(140) - 4(420) + 256(35) - 256(60) + 64(148)}{420} + \frac{27}{70}$$

$$= \frac{2336}{420} + \frac{27}{70}$$

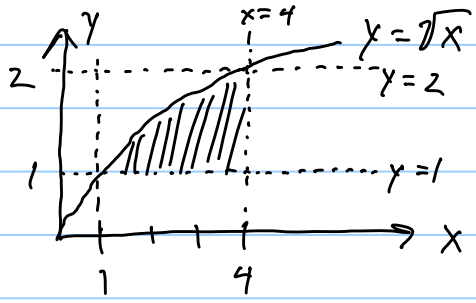
$$\therefore (1) + (2) = \left(\frac{2336}{420} + \frac{27}{70}\right) + \left(\frac{27}{70}\right) = \frac{2336 + 162 + 162}{420}$$

$$= \frac{2660}{420} = \frac{2^2 \cdot 5 \cdot 7 \cdot 19}{2^2 \cdot 3 \cdot 5 \cdot 7} = \frac{19}{3}$$

In Exercises 28 and 29, sketch the region of integration, interchange the order, and evaluate.

28.

28. $\int_1^4 \int_1^{\sqrt{x}} (x^2 + y^2)\, dy\, dx$



$y = \sqrt{x}$ is increasing on $1 \le x \le 4$

$\sqrt{1} = 1$, $\sqrt{4} = 2$ inverse of $y = \sqrt{x}$ is $x = y^2$

$$\therefore \int_1^2 \int_{y^2}^4 (x^2 + y^2)\, dx\, dy = \int_1^2 \frac{x^3}{3} + xy^2 \Big|_{x=y^2}^{x=4} dy$$

$$= \int_1^2 \frac{64}{3} + 4y^2 - \left(\frac{y^6}{3} + y^4\right) dy = \frac{64}{3}y + \frac{4}{3}y^3 - \frac{y^7}{21} - \frac{y^5}{5} \Big|_1^2$$

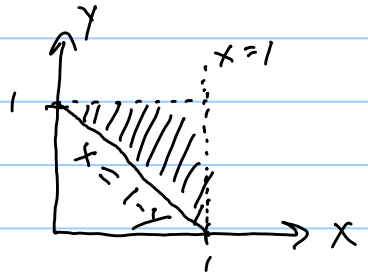
$$= \frac{128}{3} + \frac{32}{3} - \frac{128}{21} - \frac{32}{5} - \left(\frac{64}{3} + \frac{4}{3} - \frac{1}{21} - \frac{1}{5}\right)$$

$7 \cdot 3 \cdot 5 = 105$

$$= \frac{128(35) + 32(35) - 128(5) - 32(21) - 68(35) + 5 + 21}{105}$$

$= \frac{1934}{105} = \frac{2 \cdot 967}{3 \cdot 5 \cdot 7} = \frac{1934}{105}$

29. **29.** $\int_0^1 \int_{1-y}^1 (x + y^2)\, dx\, dy$



$X = f(y) = 1 - y$ is a decreasing function, inverse is $y = 1 - x$. $f(0) = 1$, $f(1) = 0$

$\therefore \int_{f(1)}^{f(0)} \int_{1-x}^{1} \Rightarrow \int_0^1 \int_{1-x}^1 (x + y^2)\, dy\, dx$

$= \int_0^1 xy + \frac{y^3}{3} \Big|_{y=1-x}^{y=1} dx$

$= \int_0^1 x + \frac{1}{3} - \left[x(1-x) + \frac{(1-x)^3}{3} \right] dx$

$-x^3 + 3x^2 - 3x + 1$

$= \int_0^1 \frac{1}{3} + x^2 + \left(\frac{x^3}{3} - x^2 + x - \frac{1}{3} \right) dx$

$= \int_0^1 \frac{x^3}{3} + x\, dx = \frac{x^4}{12} + \frac{x^2}{2} \Big|_0^1 = \frac{1}{12} + \frac{1}{2} = \frac{7}{12}$

30. **30.** Show that

$$4e^5 \le \iint_{[1,3]\times[2,4]} e^{x^2+y^2}\,dA \le 4e^{25}.$$

The box $[1,3]\times[2,4]$ has area $(3-1)(4-2)=4$

$e^{x^2+y^2}$ has a minimum when x is smallest ($x=1$) and y is smallest ($y=2$) over the box.

$\therefore$ Minimum $= e^{1^2+2^2} = e^5$

Similarly, a max is obtained when x and y have maximum values: $x=3, y=4$.

$\therefore$ Max $= e^{3^2+4^2} = e^{25}$

$\therefore\ e^5 \le e^{x^2+y^2} \le e^{25}$ for all $(x,y) \in D = [1,3]\times[2,4]$

$\therefore$

$$4e^5 \le \iint_{[1,3]\times[2,4]} e^{x^2+y^2}\,dA \le 4e^{25}$$

31. **31.** Show that

$$4\pi \le \iint_D (x^2+y^2+1)\,dx\,dy \le 20\pi,$$

where D is the disk of radius 2 centered at the origin.

A disk of radius 2 has area $\pi(2)^2 = 4\pi$

D is the region $[-2,2]\times[-2,2]$ s.t. $x^2+y^2 \le 4$

$f(x,y) = x^2+y^2+1$ has a minimum at $(x,y)=(0,0)$

so that $f(0,0) = 1$.

Over D, $f(x,y)$ has a maximum when $x^2+y^2 = 4$.

$$\therefore f(x,y)_{max} = (x^2+y^2)_{max} + 1 = 4+1 = 5$$

$$\therefore \quad 1 \le f(x,y) \le 5$$

$$\therefore$$

$$1\cdot(4\pi) = 4\pi \le \iint_D (x^2+y^2+1)\,dx\,dy \le 20\pi = 5\cdot(4\pi)$$

32.

32. Suppose W is a ***path-connected region***; that is, given any two points of W there is a continuous path joining them. If f is a continuous function on W, use the intermediate-value theorem to show that there is at least one point in W at which the value of f is equal to the average of f over W; that is, the integral of f over W divided by the volume of W. (Compare this with the mean-value theorem for double integrals.) What happens if W is not connected?

Assume W is closed or compact, and $\therefore$ has a finite volume.

f continuous on compact $W \Rightarrow f$ achieves a minimum and maximum on W.

Let $\vec{x} \in W$ s.t. $f(\vec{x}) = m$ is a minimum.
$\vec{y} \in W$ s.t. $f(y) = M$ is a maximum.

Let $\vec{p}(t)$, $t \in R$, $\vec{p}(t) \in W$ be the continuous path connecting $\vec{x}$ and $\vec{y}$. Let $\vec{p}(a) = \vec{x}$, $\vec{p}(b) = \vec{y}$

By the mean value inequality,

$$m \cdot Vol(W) \le \iiint_W f\,dV \le M \cdot Vol(W)$$

$$\therefore f(\vec{\rho}(a)) = f(\vec{x}) = m \le \frac{\iiint_W f\,dV}{Vol(W)} \le M = f(\vec{y}) = f(\vec{\rho}(b))$$

Since f and ρ are continuous, $f \circ \rho$ is continuous on $[a,b]$

$\therefore$ By Intermediate Value Theorem, there is a

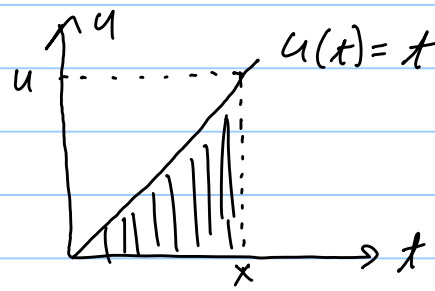
$$t_0 \in [a,b] \text{ s.t. } f(\vec{\rho}(t_0)) = \frac{\iiint_W f\,dV}{Vol(W)}$$

Let $\vec{z} = \vec{\rho}(t_0)$. $\vec{\rho}$ a continuous path in W means $\vec{z} \in W$. $\therefore f(\vec{z}) = \dfrac{\iiint_W f\,dV}{Vol(W)}$

If W is not connected, Then can't make the assertion.

33. **33.** Prove: $\int_0^x \left[\int_0^t F(u)\,du\right] dt = \int_0^x (x-u)\,F(u)\,du.$

Region D is $0 \le u \le t$ and $0 \le t \le x$



$u(t) = t$ is an increasing function, inverse is $t = u$ $\quad u(0) = 0$, $u(x) = x$.

$$\therefore \int_0^x \int_0^t du\,dt \Rightarrow \int_{u(0)=0}^{u(x)=x} \int_u^x dt\,du = \int_0^x \int_u^x dt\,du$$

$\therefore$ Interchanging order of integration,

$$\int_0^x \int_0^t F(u)\,du\,dt = \int_0^x \int_u^x F(u)\,dt\,du$$

But $\int_u^x F(u)\,dt = F(u)\int_u^x dt$ as $F(u)$ acts

like a constant relative to variable t.

and $$\int_u^x dt = t\Big|_u^x = x - u$$

$$\therefore \int_u^x F(u)\,dt = (x-u)\,F(u)$$

$$\therefore \int_0^x \int_0^t F(u)\,du\,dt = \int_0^x \int_u^x F(u)\,dt\,du = \int_0^x (x-u)\,F(u)\,du$$

Evaluate the integrals in Exercises 34 to 36.

34.

34. $\int_0^1 \int_0^z \int_0^y xy^2 z^3 \, dx \, dy \, dz$

$$\int_0^1 \int_0^z \left[\int_0^y xy^2 z^3 \, dx \right] dy \, dz = \int_0^1 \int_0^z \frac{x^2}{2} y^2 z^3 \Big|_{x=0}^{x=y} dy \, dz$$

$$= \int_0^1 \int_0^z \frac{y^4}{2} z^3 \, dy \, dz = \int_0^1 \frac{y^5 z^3}{10} \Big|_{y=0}^{y=z} dz$$

$$= \int_0^1 \frac{z^8}{10} \, dz = \frac{z^9}{90} \Big|_0^1 = \underline{\frac{1}{90}}$$

35.

35. $\int_0^1 \int_0^y \int_0^{x/\sqrt{3}} \frac{x}{x^2 + z^2} \, dz \, dx \, dy$

$$\int_0^{x/\sqrt{3}} \frac{x}{x^2 + z^2} \, dz = x \left[\frac{1}{x} \arctan \frac{z}{x} \right]_{z=0}^{z=\frac{x}{\sqrt{3}}}, \quad x \neq 0$$

$$= \arctan\left(\frac{\sqrt{3}}{3}\right) = \frac{\pi}{6}$$

$$\int_0^y \frac{\pi}{6} \, dx = \frac{\pi}{6} x \Big|_0^y = \frac{\pi}{6} y$$

$$\int_0^1 \frac{\pi}{6} y \, dy = \frac{\pi}{6} \frac{y^2}{2} \Big|_0^1 = \underline{\frac{\pi}{12}}$$

Note: The first integral assumes $x \neq 0$,

but the second states $0 \leq |x| \leq y$

To justify, let $0 < |x| \leq y$

and take limit as $x \to 0$

$$\therefore \lim_{x \to 0} x \left[\frac{1}{x} \arctan \frac{z}{x} \right]_{z=0}^{z=\frac{x}{\sqrt{3}}}$$

$$= \lim_{x \to 0} \left(\arctan \frac{\sqrt{3}}{3} - \arctan(0) \right)$$

$$= \lim_{x \to 0} \frac{\pi}{6} = \frac{\pi}{6}$$

$\therefore$ Define integral as its limit

$$\int_0^{\frac{\sqrt{x}}{3}} \frac{x}{x^2 + z^2} \, dz = \frac{\pi}{6}$$

Note, if $x < 0$, $\int \frac{dz}{x^2 + z^2} = \frac{1}{-x} \arctan\left(\frac{z}{-x}\right) + C$

$$= \frac{1}{x} \arctan\left(\frac{z}{x}\right) + C$$

36. **36.** $\int_1^2 \int_1^z \int_{1/y}^2 yz^2 \, dx \, dy \, dz$

$$\int_{\frac{1}{y}}^{2} yz^2 \, dx = xyz^2 \Big|_{x=\frac{1}{y}}^{x=2} = 2yz^2 - z^2$$

$$\int_1^z (2yz - z^2) \, dy = zy^2 - yz^2 \Big|_{y=1}^{y=z} = z^3 - z^3 - (z - z^2)$$

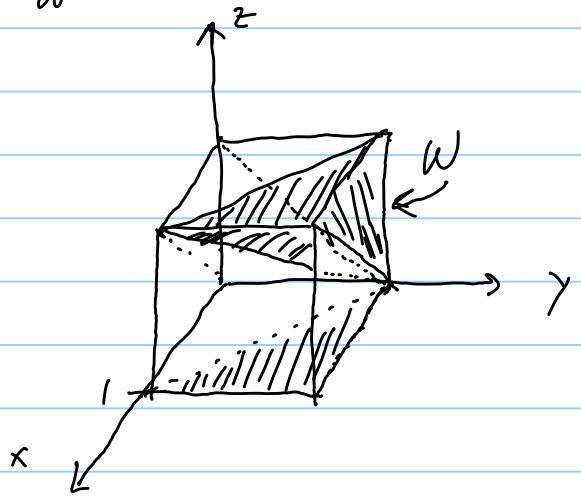
$$= z^2 - z$$

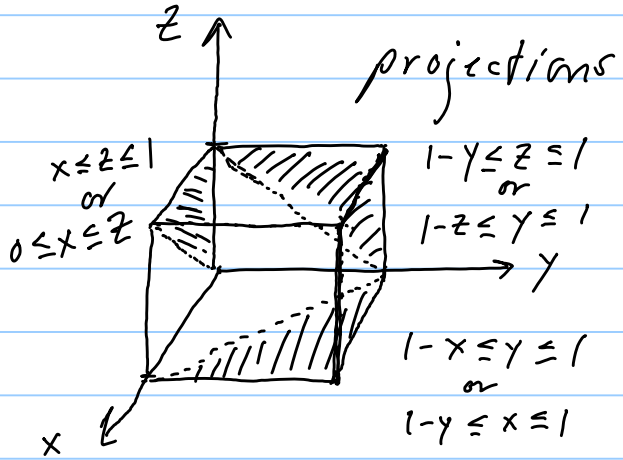
$$\therefore \int_1^2 z^2 - z \, dz = \frac{z^3}{3} - \frac{z^2}{2} \Big|_1^2 = \frac{8}{3} - 2 - \left(\frac{1}{3} - \frac{1}{2}\right)$$

$$= \frac{7}{3} - \frac{3}{2} = \frac{14 - 9}{6} = \frac{5}{6}$$

37. **37.** Write the iterated integral $\int_0^1 \int_{1-x}^1 \int_x^1 f(x, y, z) \, dz \, dy \, dx$ as an integral over a region in $\mathbb{R}^3$ and then rewrite it in five other possible orders of integration.

$$\iiint_W f(x,y,z) \, dV \text{, } W \text{ is } 0 \le x \le 1,\ 1-x \le y \le 1,\ x \le z \le 1$$





(1) Consider the projection of the region onto the xy-plane. This is described as:

$0 \le x \le 1$ and $1-x \le y \le 1$

or

$0 \le y \le 1$ and $1-y \le x \le 1$

The z extent in both cases is $x \le z \le 1$

$$\therefore \int_0^1 \int_{1-x}^1 \int_x^1 f \, dz \, dy \, dx = \int_0^1 \int_{1-y}^1 \int_x^1 f \, dz \, dx \, dy$$

(2) Consider the projection of the region onto the yz-plane. This is described as:

$0 \le y \le 1$ and $1-y \le z \le 1$

or

$0 \le z \le 1$ and $1-z \le y \le 1$

Now consider $x = \phi_1(y,z)$ and $x = \phi_2(y,z)$.

The "bottom" of this region is a plane containing $(1,0,1)$, $(0,1,0)$, $(0,1,1)$

Using $Ax + By + Cz = D$, $(0,1,0) \Rightarrow B = D$

$\therefore Ax + By + Cz = B$ $(0,1,1) \Rightarrow B + C = B \Rightarrow C = 0$

$\therefore Ax + By = B$ $(1,0,1) \Rightarrow A = B \Rightarrow Bx + By = B$,

or $x + y = 1$, or $x = 1 - y = \phi_1(y,z)$

The "top" of this region is a plane containing $(1,0,1)$, $(1,1,1)$, $(0,1,0)$.

Using $Ax+By+Cz=D$, $(0,1,0) \Rightarrow B=D$

$\therefore Ax+By+Cz=B$ $\quad (1,0,1) \Rightarrow A+C=B$ $\quad (1,1,1) \Rightarrow A+B+C=B$ $\Big\} \therefore B=0$

$\therefore A+C=0$, $C=-A$. $\therefore Ax-Az=0$, or $x=z$

$\therefore x=z=\phi_2(y,z)$

$$\therefore \int_0^1 \int_{1-y}^{1} \int_{1-y}^{z} f \, dx\,dz\,dy = \int_0^1 \int_{1-z}^{1} \int_{1-y}^{z} f \, dx\,dy\,dz$$

(3) Consider the projection of the region onto the xz-plane. This is described as:

$0 \le x \le 1$ and $x \le z \le 1$

or

$0 \le z \le 1$ and $0 \le x \le z$

Now consider $y=\phi_1(x,z)$ and $y=\phi_2(x,z)$

The "top" is $y=1=\phi_2(x,z)$

The "bottom" plane contains $(0,1,1)$, $(0,1,0)$, $(1,0,1)$

This plane was described in (2) above as $x+y=1$, or $y=1-x=\phi_1(x,z)$

$$\therefore \int_0^1 \int_x^1 \int_{1-x}^{1} f \, dy\,dz\,dx = \int_0^1 \int_0^z \int_{1-x}^{1} f \, dy\,dx\,dz$$

Note: The answer in the back of the text mistakenly has $\int_0^1 \int_z^1 \int_{1-x}^{1} f \, dy \, dx \, dz$

The range of x is $0 \le x \le z$, not $z \le x \le 1$

6.1 The Geometry of Maps from R^2 to R^2

Note Title 11/22/2016

1. **1.** Determine if the following functions $T: \mathbb{R}^2 \to \mathbb{R}^2$ are one-to-one and/or onto.

(a) (a) $T(x, y) = (2x, y)$

$T(x,y) = T(x',y') \Rightarrow (2x,y) = (2x',y')$

$\Rightarrow x = x',\ y = y' \Rightarrow (x,y) = (x',y')$

$\therefore$ One-to-one

Let $(u,v) \in \mathbb{R}^2$. $\therefore (2x,y) = (u,v) \Rightarrow x = \frac{u}{2},\ y = v$

$\therefore T(x,y) = T(\frac{1}{2}u, v) = (2(\frac{1}{2}u), v) = (u,v)$.

$\therefore$ Onto

(b) (b) $T(x, y) = (x^2, y)$

Not one-to-one since $T(x,y) = T(-x,y)$

Not onto: for $(-1, 0) \in \mathbb{R}^2$, there is no $(x,y) \in \mathbb{R}^2$ s.t. $T(x,y) = (-1,0)$.

(c) (c) $T(x, y) = (\sqrt[3]{x}, \sqrt[3]{y})$

$T(x,y) = T(x',y') \Rightarrow (\sqrt[3]{x}, \sqrt[3]{y}) = (\sqrt[3]{x'}, \sqrt[3]{y'})$

$\Rightarrow x = x',\ y = y'$.

$\therefore$ One-to-one

Let $(u,v) \in \mathbb{R}^2$. $\therefore (\sqrt[3]{x}, \sqrt[3]{y}) = (u,v) \Rightarrow$

$x = u^3, \; y = v^3$

$\therefore T(x,y) = T(u^3, v^3) = (u,v)$. $\therefore$ Onto

(d) (d) $T(x, y) = (\sin x, \cos y)$

Since $T(x,y) = T(x,-y)$ as $\cos(y) = \cos(-y)$,

Then not one-to-one

Not onto since there is no $(x,y) \in \mathbb{R}^2$

s.t. $T(x,y) = (a,b)$ for $|a|>1, |b|>1$

2. **2.** Determine if the following functions $T: \mathbb{R}^2 \to \mathbb{R}^2$ are one-to-one and/or onto.

Actually, $T: \mathbb{R}^3 \to \mathbb{R}^3$

(a) (a) $T(x, y, z) = (2x + y + 3z, 3y - 4z, 5x)$

If $T(x,y,z) = T(x',y',z')$ then

$2x + y + 3z = 2x' + y' + 3z' \quad [1]$

$3y - 4z = 3y' - 4z' \quad [2]$

$5x = 5x' \quad [3]$

$[3] \Rightarrow x = x'$ $\therefore [1]$ becomes

$$y + 3z = y' + 3z' \quad [4]$$

Multiply [2] by 3, [4] by 4:

$$9y - 12z = 9y' - 12z' \quad [3^*]$$

$$4y + 12z = 4y' + 12z' \quad [4^*]$$

Add to get: $13y = 13y' \Rightarrow y = y'$

$\therefore$ [2] becomes $-4z = -4z' \Rightarrow z = z'$

$\therefore (x, y, z) = (x', y', z')$

$\therefore$ One-to-one

Let $(u, v, w) \in \mathbb{R}^3$

$$\therefore \begin{bmatrix} 2 & 1 & 3 \\ 0 & 3 & -4 \\ 5 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 3 & -4 \\ 5 & 0 & 0 \end{bmatrix}$ Since $\det A = 5(-4-9) \neq 0$, then A^{-1} exists.

$$\therefore \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} u \\ v \\ w \end{bmatrix}$$

$\therefore T$ is onto

(b) $T(x, y, z) = (y \sin x, z \cos y, xy)$

(b) Since $T(x,y,z) = T(-x,-y,z)$, not one-to-one

Consider $(u, v, 0)$ where $u \neq 0$.

There is no (x,y,z) s.t. $(y\sin x, z\cos y, xy) = (u,v,0)$

for $xy = 0 \Rightarrow$ x or y must be 0.

$\therefore y \sin x = 0$.

$\therefore$ Not onto

(c) $T(x, y, z) = (xy, yz, xz)$

(c) Since $T(1,1,1) = T(-1,-1,-1)$, not one-to-one

Consider $(1,1,-1) \in \mathbb{R}^3$

Is there an (x,y,z) s.t. $(xy, yz, xz) = (1,1,-1)$?

$xz = -1 \Rightarrow x < 0$ and $z > 0$ or $x > 0$ and $z < 0$

(1) Suppose $x < 0$ and $z > 0$

$xy = 1 \Rightarrow y < 0$.

$\therefore yz < 0$ so $yz \neq 1$

(2) Suppose $x > 0$ and $z < 0$

$\therefore xy = 1 \Rightarrow y > 0$

$\therefore yz < 0$ so $yz \neq 1$

$\therefore$ There is no (x,y,z) s.t. $T(x,y,z) = (1,1,-1)$.

$\therefore$ Not onto

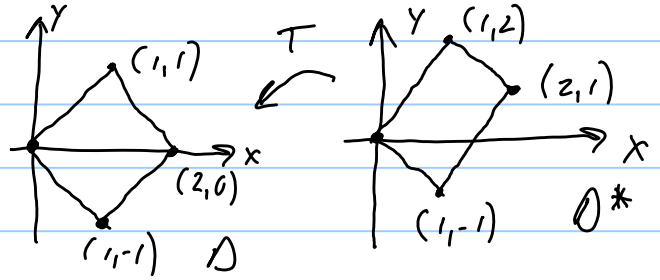
(d) (d) $T(x, y, z) = (e^x, e^y, e^z)$

Since $e^u = e^v \Rightarrow u = v$, then T is one-to-one

Since $e^x > 0$ for all x, there is no (x,y,z)

s.t. $T(x,y,z) = (-1,-1,-1)$. $\therefore$ Not onto

3. **3.** Let D be a square with vertices $(0, 0)$, $(1, 1)$, $(2, 0)$, $(1, -1)$ and D^* be a parallelogram with vertices $(0, 0)$, $(1, 2)$, $(2, 1)$, $(1, -1)$. Find a linear map T taking D^* onto D.



Let $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ Assign vertices in D^* to D and try to solve, using Theorem 1, p. 310 of text.

$\therefore \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \begin{matrix} a + 2b = 1 & [1] \\ c + 2d = 1 & [2] \end{matrix}$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} \Rightarrow \begin{matrix} 2a + b = 2 \\ 2c + d = 0 \end{matrix} \quad \begin{matrix}[3] \\ [4]\end{matrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \Rightarrow \begin{matrix} a - b = 1 \\ c - d = -1 \end{matrix} \quad \begin{matrix}[5] \\ [6]\end{matrix}$$

$[1], [3], [5]$:
$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad \begin{matrix}[1] \\ [3] \\ [5]\end{matrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & -3 \\ 0 & -3 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow a = 1,\ b = 0$$

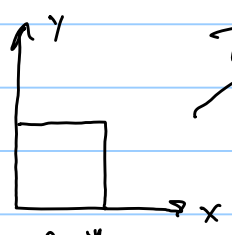
$[2], [4], [6]$
$$\begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad \begin{matrix}[2] \\ [4] \\ [6]\end{matrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 \\ 0 & -3 \\ 0 & -3 \end{bmatrix}\begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ -2 \end{bmatrix} \Rightarrow d = \frac{2}{3},\ c = -\frac{1}{3}$$

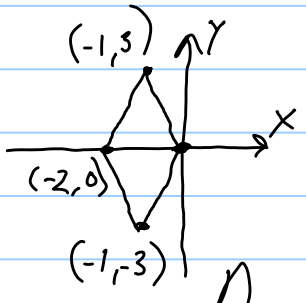
$$\therefore T = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -\frac{1}{3} & \frac{2}{3} \end{bmatrix}$$

4.

4. Let D be a parallelogram with vertices $(0, 0), (-1, 3), (-2, 0), (-1, -3)$. Let $D^* = [0, 1] \times [0, 1]$. Find a linear map T such that $T(D^*) = D$.



T



Use, Theorem 1, p. 310 of text. D^* D

You always want $(0,0) \to (0,0)$ for linear maps.

Here, imagine a counterclockwise rotation so that $(0,1) \to (-1,-3)$, $(1,1) \to (-2,0)$, $(1,0) \to (-1,3)$

$\therefore$ Let $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$\therefore \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & -2 & -1 \\ -3 & 0 & 3 \end{bmatrix}$$

Looking at the transpose of each side,

$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ -2 & 0 \\ -1 & 3 \end{bmatrix}$$ add (-1)(row 3) to row 2

$$\therefore \begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} -1 & -3 \\ -1 & -3 \\ -1 & 3 \end{bmatrix} \Rightarrow b = -1,\ d = -3,\ a = -1,\ c = 3$$

$$\therefore T = \begin{bmatrix} -1 & -1 \\ -3 & 3 \end{bmatrix}, \text{ or } T(x,y) = (-x-y, -3x+3y)$$

5.

5. Let $S^* = (0, 1] \times [0, 2\pi)$ and define $T(r, \theta) = (r\cos\theta, r\sin\theta)$. Determine the image set S. Show that T is one-to-one on S^*.

$T(r,\theta) = (x,y)$. $x^2+y^2 = r^2\cos^2\theta + r^2\sin^2\theta = 1$, a circle

$\therefore$ T appears to map to a disc of radius 1, minus $(0,0)$

as there is no (r,θ) s.t. $T(r,\theta)=(0,0)$.

$T(r,\theta)=T(r',\theta') \Rightarrow (r\cos\theta, r\sin\theta)=(r'\cos\theta', r'\sin\theta')$

(a) For $0\le\theta<\frac{\pi}{2}$: $\sin\theta$ and $\cos\theta$ are one-to-one.

$\therefore \sin\theta=\sin\theta' \Rightarrow \theta=\theta'$

$\cos\theta=\cos\theta' \Rightarrow \theta=\theta'$

$\therefore (r\cos\theta, r\sin\theta)=(r'\cos\theta', r'\sin\theta')$

$=(r'\cos\theta, r'\sin\theta)$

$\Rightarrow r'=r$ since $\cos\theta\neq 0$ on $0\le\theta<\frac{\pi}{2}$

$\therefore T(r,\theta)=T(r',\theta') \Rightarrow (r,\theta)=(r',\theta')$

(b) For $\theta=\frac{\pi}{2}$: $T(r,\frac{\pi}{2})=(0,r)$

$\therefore T(r,\frac{\pi}{2})=T(r',\frac{\pi}{2}) \Rightarrow (0,r)=(0,r') \Rightarrow r=r'$

(c) For $\frac{\pi}{2}<\theta\le\pi$: as in (a), $\sin\theta$ and $\cos\theta$ are one-to-one on this interval.

$\therefore (r\cos\theta, r\sin\theta)=(r'\cos\theta', r'\sin\theta')$

$=(r'\cos\theta, r'\sin\theta)$

$\Rightarrow r'=r$ since $\cos\theta\neq 0$ on $\pi/2<\theta\le\pi$

(d) For $\pi<\theta\le\frac{3}{2}\pi$: as in (a), $\sin\theta$ and $\cos\theta$ are one-to-one on this interval.

$\therefore (r\cos\theta, r\sin\theta)=(r'\cos\theta', r'\sin\theta')$

$=(r'\cos\theta, r'\sin\theta)$

$\Rightarrow r'=r$ since $\sin\theta\neq 0$ on this interval.

(e) $\frac{3}{2}\pi < \theta < 2\pi$: as in (a), $\sin\theta$ and $\cos\theta$ are one-to-one on this interval.

$\therefore (r\cos\theta, r\sin\theta) = (r'\cos\theta', r'\sin\theta')$

$= (r'\cos\theta, r'\sin\theta)$

$\Rightarrow r' = r$ since $\cos\theta \neq 0$ on this interval.

$\therefore$ (a) $\rightarrow$ (e) $\Rightarrow$ when $T(r,\theta) = T(r',\theta')$ for $0 < r \leq 1$, $0 \leq \theta < 2\pi$, then $(r,\theta) = (r',\theta')$

$\therefore$ T is one-to-one on S^*

6.

6. Define

$$T(x^*, y^*) = \left(\frac{x^* - y^*}{\sqrt{2}}, \frac{x^* + y^*}{\sqrt{2}} \right).$$

Show that T rotates the unit square, $D^* = [0, 1] \times [0, 1]$.

To show rotation, must show every point (x^*, y^*) of the original unit square maps to the same angle θ from the orrginal point, and is at the same distance from the origin as the original point.

(1) The origin of this rotation is (0,0) as

$$T(0,0) = \left(\frac{0-0}{\sqrt{2}}, \frac{0+0}{\sqrt{2}} \right) = (0,0).$$

(2) The distance from the origin after the mapping is:

$$\sqrt{\left(\frac{x^*-y^*}{\sqrt{2}}\right)^2+\left(\frac{x^*+y^*}{\sqrt{2}}\right)^2}=\sqrt{\frac{(x^*)^2-2x^*y^*+(y^*)^2+(x^*)^2+2x^*y^*+(y^*)^2}{2}}$$

$$=\sqrt{\frac{2(x^*)^2+2(y^*)^2}{2}}=\sqrt{(x^*)^2+(y^*)^2}$$

$\therefore$ Distance from the origin is preserved.

(3) Angle between (x^*, y^*) and $T(x^*, y^*)$:

Use dot product:

$$(x^*, y^*)\cdot\left(\frac{x^*-y^*}{\sqrt{2}}, \frac{x^*+y^*}{\sqrt{2}}\right)=\|(x^*, y^*)\|\,\|T(x^*, y^*)\|\cos\theta$$

$$\frac{(x^*)^2-x^*y^*}{\sqrt{2}}+\frac{x^*y^*+(y^*)^2}{\sqrt{2}}=\|(x^*, y^*)\|^2\cos\theta \quad [1]$$

Since, by (2) above, $\|(x^*, y^*)\|=\|T(x^*, y^*)\|=\sqrt{(x^*)^2+(y^*)^2}$

$\therefore$ [1] becomes:

$$\frac{(x^*)^2+(y^*)^2}{\sqrt{2}}=\left(\sqrt{(x^*)^2+(y^*)^2}\right)^2\cos\theta, \text{ or}$$

$$\frac{1}{\sqrt{2}}=\cos\theta, \quad \theta=\frac{\pi}{4}$$

$\therefore$ Every point (x^*, y^*) rotates $\frac{\pi}{4}$ radians counterclockwise under T.

7.

7. Let $D^* = [0, 1] \times [0, 1]$ and define T on D^* by $T(u, v) = (-u^2 + 4u, v)$. Find the image D. Is T one-to-one?

$-u^2+4u$ is a parabola, concave downward.

But on the interval $[0,1]$, it is an

increasing function: $f(u) = -u^2+4u$, $f'(u) = -2u+4$

$\therefore f'(u) > 0$ for $0 \le u \le 1$. $\therefore f(1) = 3$ is

the max, and $0 \le -u^2+4u \le 3$.

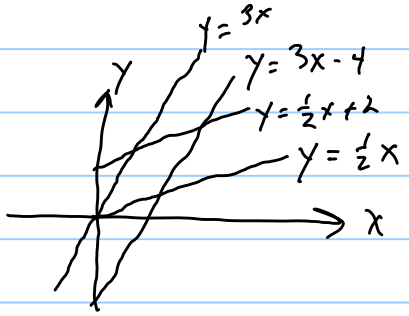
$\therefore D = [0,3] \times [0,1]$

$\therefore$ The image is a single-sheet parabolic cylinder

The image is one-to-one.

8.

8. Let D^* be the parallelogram bounded by the lines $y = 3x - 4$, $y = 3x$, $y = \frac{1}{2}x$, and $y = \frac{1}{2}(x + 4)$. Let $D = [0, 1] \times [0, 1]$. Find a T such that D is the image of D^* under T.



Use Theorem 1, p. 310 of text

Find the corners of D^*

$y = \frac{1}{2}x$ and $y = 3x - 4$: $3x - 4 = \frac{1}{2}x$, $5x = 8$, $x = \frac{8}{5}$

$\therefore y = \frac{1}{2}\left(\frac{8}{5}\right) = \frac{4}{5}$.

$\therefore \left(\frac{8}{5}, \frac{4}{5}\right)$

$y = 3x - 4$ and $y = \frac{1}{2}x + 2$: $3x - 4 = \frac{1}{2}x + 2$, $5x = 12$,

$\therefore x = \frac{12}{5}$ $\quad y = \frac{1}{2}\left(\frac{12}{5}\right) + 2 = \frac{16}{5}$

$\therefore \left(\frac{12}{5}, \frac{16}{5}\right)$

$y = \frac{1}{2}x + 2$ and $y = 3x$: $3x = \frac{1}{2}x + 2$, $5x = 4$, $x = \frac{4}{5}$

$\therefore y = \frac{1}{2}\left(\frac{4}{5}\right) + 2 = \frac{12}{5}$

$\therefore \left(\frac{4}{5}, \frac{12}{5}\right)$

$\therefore (0,0), \left(\frac{8}{5}, \frac{4}{5}\right), \left(\frac{12}{5}, \frac{16}{5}\right), \left(\frac{4}{5}, \frac{12}{5}\right)$

$\therefore (0,0) \quad \left(\frac{8}{5}, \frac{4}{5}\right) \quad \left(\frac{12}{5}, \frac{16}{5}\right) \quad \left(\frac{4}{5}, \frac{12}{5}\right)$

$\downarrow \qquad \downarrow \qquad \downarrow \qquad \downarrow$

$(0,0) \quad (1,0) \quad (1,1) \quad (0,1)$

$\therefore$ Let $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\therefore \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 8/5 & 12/5 & 4/5 \\ 0 & 4/5 & 16/5 & 12/5 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

Taking transposes,

$$\begin{bmatrix} 0 & 0 \\ 8/5 & 4/5 \\ 12/5 & 16/5 \\ 4/5 & 12/5 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} 0 & 0 \\ 8 & 4 \\ 12 & 16 \\ 4 & 12 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 5 & 0 \\ 5 & 5 \\ 0 & 5 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} 0 & 0 \\ 0 & -20 \\ 0 & -20 \\ 4 & 12 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 5 & -10 \\ 5 & -10 \\ 0 & 5 \end{bmatrix}$$

$\therefore 0(a) - 20(b) = 5 \Rightarrow b = -\frac{1}{4}$

$4(a) + 12(b) = 0 \Rightarrow 4a + 12(-\frac{1}{4}) = 0 \Rightarrow a = \frac{3}{4}$

$0(c) - 20(d) = -10 \Rightarrow d = \frac{1}{2}$

$4(c) + 12(d) = 5 \Rightarrow 4c + 12(\frac{1}{2}) = 5, \quad c = -\frac{1}{4}$

$$\therefore T = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 3/4 & -1/4 \\ -1/4 & \frac{1}{2} \end{bmatrix}$$

or $T(x^*, y^*) = \left(\frac{3}{4}x^* - \frac{1}{4}y^*, -\frac{1}{4}x^* + \frac{1}{2}y^*\right)$

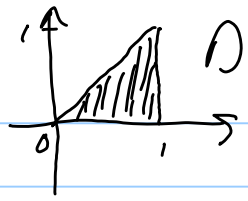
9.

9. Let $D^* = [0, 1] \times [0, 1]$ and define T on D^* by $T(x^*, y^*) = (x^* y^*, x^*)$. Determine the image set D. Is T one-to-one? If not, can we eliminate some subset of D^* so that on the remainder T is one-to-one?

(a) $x^* y^*$ varies from 0 to 1, so $x^* y^* \in [0, 1]$ and clearly $x^* \in [0, 1]$

$\therefore D = [0, 1] \times [0, 1]$

Note $x^* y^* \leq x^*$ $\therefore$ The first coordinate is always "underneath" the second coordinate.

∴ 

D is bounded by the triangle whose corners are (0,0), (1,0), (1,1), and contains all points inside the triangle.

(b) Not one-to-one as $T(0,1) = T(0,0) = (0,0)$ and $(0,1) \neq (0,0)$

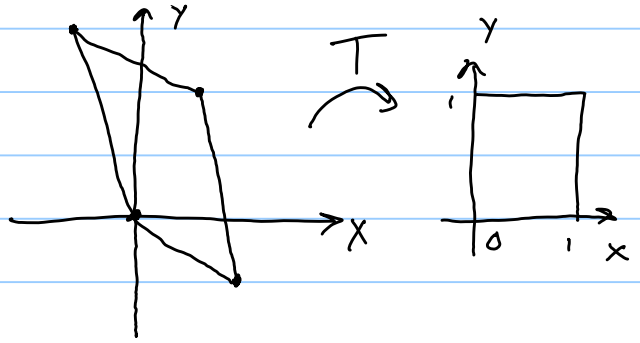
(c) If $D^* = (0,1] \times [0,1]$ Then $T(x,y) = T(a,b)$

$\Rightarrow (xy, x) = (ab, a) \Rightarrow x = a \Rightarrow xy = xb$

$\Rightarrow y = b$ as $x \neq 0$, so $\Rightarrow (x,y) = (a,b)$

and ∴ one-to-one.

10. **10.** Let D^* be the parallelogram with vertices at $(-1, 3)$, $(0, 0)$, $(2, -1)$, and $(1, 2)$, and D be the rectangle $D = [0, 1] \times [0, 1]$. Find a T such that D is the image set of D^* under T.



Let $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and make points

$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \to \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\begin{bmatrix} -1 \\ 3 \end{bmatrix} \to \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 2 \end{bmatrix} \to \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 2 \\ -1 \end{bmatrix} \to \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\therefore \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & -1 & 1 & 2 \\ 0 & 3 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$

taking transposes,

$$\begin{bmatrix} 0 & 0 \\ -1 & 3 \\ 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$

add row(3) to row(2)
subtract 2 times row(3) from row(4)

$$\Rightarrow \begin{bmatrix} 0 & 0 \\ 0 & 5 \\ 1 & 2 \\ 0 & -5 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 2 \\ 1 & 1 \\ -1 & -2 \end{bmatrix}$$

$\therefore 5b = 1, \quad b = \frac{1}{5}$

$5d = 2, \quad d = \frac{2}{5}$

$a + 2b = 1,$

$a + 2\left(\frac{1}{5}\right) = 1, \quad a = \frac{3}{5}$

$c + 2d = 1,$

$c + 2\left(\frac{2}{5}\right) = 1, \quad c = \frac{1}{5}$

$$\therefore T = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 3/5 & 1/5 \\ 1/5 & 2/5 \end{bmatrix}$$

11.

11. Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the spherical coordinate mapping defined by $(\rho, \phi, \theta) \mapsto (x, y, z)$, where

$$x = \rho \sin\phi \cos\theta, \qquad y = \rho \sin\phi \sin\theta, \qquad z = \rho \cos\phi.$$

Let D^* be the set of points (ρ, ϕ, θ) such that $\phi \in [0, \pi]$, $\theta \in [0, 2\pi]$, $\rho \in [0, 1]$. Find $D = T(D^*)$. Is T one-to-one? If not, can we eliminate some subset of D^* so that, on the remainder, T will be one-to-one?

(a) D is the unit sphere centred at $(0,0,0)$.

(b) T is <u>not</u> one-to-one as:

(1) For $\theta \in [0, 2\pi]$, $\cos(0) = \cos(2\pi)$

$\therefore (\rho, \phi, 0)$ is the same point as $(\rho, \phi, 2\pi)$

so $T(\rho, \phi, 0) = T(\rho, \phi, 2\pi)$

(2) For $\rho = 0$, $T(0, \phi, \theta) = T(0, \phi^*, \theta^*)$ for $\phi \neq \phi^*$, $\theta \neq \theta^*$.

(3) For $\phi = 0$ or $\phi = \pi$, $\sin\phi = 0$, so

$T(\rho, 0, \theta) = T(\rho, 0, \theta^*)$ for $\theta \neq \theta^*$

and $T(\rho, \pi, \theta) = T(\rho, \pi, \theta^*)$ for $\theta \neq \theta^*$

(c) From (b), restrict D^* to

$\rho \in (0, 1]$, $\theta \in [0, 2\pi)$, $\phi \in (0, \pi)$

In Exercises 12 and 13, let $T(\mathbf{x}) = A\mathbf{x}$, where A is a 2×2 matrix.

12. **12.** Show that T is one-to-one if and only if the determinant of A is not zero.

(1) Suppose $T(\vec{x}) = T(\vec{y})$ $\therefore$ $A\vec{x} = A\vec{y}$

Since $\det A \neq 0$, A^{-1} exists. $\therefore$ $A^{-1}A\vec{x} = A^{-1}A\vec{y}$

$\Rightarrow I\vec{x} = I\vec{y} \Rightarrow \vec{x} = \vec{y}$, where I = identity matrix

$\therefore$ $\det A \neq 0 \Rightarrow T$ is one-to-one

(2) Suppose $T(\vec{x}) = T(\vec{y}) \Rightarrow \vec{x} = \vec{y}$ (i.e., T one-to-one)

$\therefore A\vec{x} = A\vec{y} \Rightarrow \vec{x} = \vec{y}$

$\therefore A\vec{x} - A\vec{y} = \vec{0} \Rightarrow \vec{x} - \vec{y} = \vec{0}$

or $A(\vec{x} - \vec{y}) = \vec{0} \Rightarrow \vec{x} - \vec{y} = \vec{0}$

Let $\vec{z} = \vec{x} - \vec{y}$. $\therefore A\vec{z} = \vec{0} \Rightarrow \vec{z} = \vec{0}$.

i.e., columns of A are independent

$\Rightarrow$ columns form a basis for R^2 since A is square

$\Rightarrow A\vec{x}_1 = \vec{e}_1$ and $A\vec{x}_2 = \vec{e}_2$ is solvable

$\Rightarrow A[\vec{x}_1\ \vec{x}_2] = I$, so $[\vec{x}_1\ \vec{x}_2] = A^{-1}$

$\Rightarrow \det A \neq 0$

13. **13.** Show that $\det A \neq 0$ if and only if T is onto.

(1) if $\det A \neq 0$, then A^{-1} exists.

$\therefore$ For any $\vec{y} \in R^2$, let $\vec{x} = A^{-1}\vec{y}$

$\therefore A\vec{x} = AA^{-1}\vec{y} = I\vec{y} = \vec{y}$

$\therefore$ for any $\vec{y} \in R^2$, $\exists$ an $\vec{x} \in R^2$ s.t. $T\vec{x} = \vec{y}$. $\therefore$ onto

(2) Suppose T is onto.

$\therefore$ for every $\vec{y} \in \mathbb{R}^2$, $\exists\ \vec{x} \in \mathbb{R}^2$ s.t. $A\vec{x} = \vec{y}$

Let $\vec{y} = \vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{z} = \vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$\therefore \exists\ \vec{x}_1$ s.t. $A\vec{x}_1 = \vec{e}_1$ and $A\vec{x}_2 = \vec{e}_2$

$\therefore A\left[\vec{x}_1\ \vec{x}_2\right] = \left[\vec{e}_1\ \vec{e}_2\right] = I$

$\therefore \left[\vec{x}_1\ \vec{x}_2\right] = A^{-1}$

$\therefore AA^{-1} = I$, $\det(AA^{-1}) = \det(I) = 1$

$\therefore \det(A)\det(A^{-1}) = 1$, so $\det A \neq 0$

14.

14. Suppose $T: \mathbb{R}^2 \to \mathbb{R}^2$ is linear and is given by $T(\mathbf{x}) = A\mathbf{x}$, where A is a 2×2 matrix. Show that if $\det A \neq 0$, then T takes parallelograms onto parallelograms. [HINT: The general parallelogram in $\mathbb{R}^2$ can be described by the set of points $\mathbf{q} = \mathbf{p} + \lambda\mathbf{v} + \mu\mathbf{w}$ for $\lambda, \mu \in (0, 1)$ where $\mathbf{p}$, $\mathbf{v}$, $\mathbf{w}$ are vectors in $\mathbb{R}^2$ with $\mathbf{v}$ not a scalar multiple of $\mathbf{w}$.]

(1) Let $P = \{\vec{q} : \vec{q} = \vec{p} + \lambda\vec{v} + \mu\vec{w},\ \lambda, \mu \in [0,1],$

$\vec{p}, \vec{v}, \vec{w} \in \mathbb{R}^2$, and $\vec{v}, \vec{w}$ not scalar multiples$\}$

Since T is linear, then

$T(\vec{q}) = T(\vec{p} + \lambda\vec{v} + \mu\vec{w}) = T(\vec{p}) + \lambda T(\vec{v}) + \mu T(\vec{w})$

Let $\vec{p}^* = T(\vec{p}) = A\vec{p}$, $\vec{v}^* = T(\vec{v}) = A\vec{v}$, $\vec{w}^* = T(\vec{w}) = A\vec{w}$

$\therefore \vec{p}^*, \vec{v}^*, \vec{w}^* \in R^2$

$\therefore T(\vec{q}) = \vec{p}^* + \lambda \vec{v}^* + \mu \vec{w}^*$

Must now show $\vec{v}^*$ and $\vec{w}^*$ are not scalar multiples.

Suppose $\vec{v}^* = c\vec{w}^*$, $c \in R$, $c \neq 0$.

Since $\det A \neq 0$, A^{-1} exists, so

$\vec{v} = A^{-1}\vec{v}^* = A^{-1}(c\vec{w}^*) = cA^{-1}\vec{w}^* = c\vec{w}$,

But $\vec{v}, \vec{w}$ are not scalar multiples,

so there is no $c \in R$ s.t. $\vec{v}^* = c\vec{w}^*$

$\therefore \vec{p}^* + \lambda \vec{v}^* + \mu \vec{w}^*$ is a parallelogram

$\therefore T$ maps parallelograms to parallelograms.

(2) Suppose $\vec{q}$ is a parallelogram given by:

$\vec{q} = \vec{p} + \lambda \vec{v} + \mu \vec{w}$, $\vec{v}, \vec{w}$ not scalar multiples.

Since $\det A \neq 0$, then A^{-1} exists (and T is onto).

Let $\vec{q}^* = A^{-1}(\vec{p} + \lambda \vec{v} + \mu \vec{w})$

$= A^{-1}\vec{p} + \lambda A^{-1}\vec{v} + \mu A^{-1}\vec{w}$

Let $\vec{p}^{\,*} = A^{-1}\vec{p}$, $\vec{v}^{\,*} = A^{-1}\vec{v}$, $\vec{w}^{\,*} = A^{-1}\vec{w}$

$\therefore \vec{q}^{\,*} = \vec{p}^{\,*} + \lambda\vec{v}^{\,*} + \mu\vec{w}^{\,*}$

and $T(\vec{q}^{\,*}) = \vec{q}$

As in (1), $\vec{v}^{\,*}$ and $\vec{w}^{\,*}$ can't be scalar multiples.

$\therefore$ given any parallelogram $\vec{q} \in R^2$, there is a parallelogram $\vec{q}^{\,*} \in R^2$ s.t. $T(\vec{q}^{\,*}) = \vec{q}$.

$\therefore$ (1) + (2) $\Rightarrow$ T maps parallelograms <u>onto</u> parallelograms.

15.

15. A map $T : \mathbb{R}^2 \to \mathbb{R}^2$ is called **affine** if $T(\mathbf{x}) = A\mathbf{x} + \mathbf{v}$, where A is a 2×2 matrix, and $\mathbf{v}$ is a fixed vector in $\mathbb{R}^2$. Show that Exercises 12, 13, and 14 hold for T.

(a)

12. Show that T is one-to-one if and only if the determinant of A is not zero.

(1) Suppose $\det A \neq 0$. If $T(\vec{x}) = T(\vec{y})$, then

$A\vec{x} + \vec{v} = A\vec{y} + \vec{v} \Rightarrow A\vec{x} = A\vec{y}$.

$\det A \neq 0 \Rightarrow A^{-1}$ exists. $\therefore A^{-1}A\vec{x} = A^{-1}A\vec{y} \Rightarrow$

$\vec{x} = \vec{y}$. $\therefore$ T is one-to-one.

(2) Suppose T is one-to-one.

$\therefore T(\vec{x}) = T(\vec{y}) \Rightarrow \vec{x} = \vec{y}$.

$\therefore A\vec{x} + \vec{v} = A\vec{y} + \vec{v} \Rightarrow \vec{x} = \vec{y}$,

or $A\vec{x} = A\vec{y} \Rightarrow \vec{x} = \vec{y}$, or

$A(\vec{x} - \vec{y}) = \vec{0} \Rightarrow \vec{x} \cdot \vec{y} = \vec{0} \Rightarrow$

columns of A are independent $\Rightarrow$

A, which is 2×2 square, has an inverse

$\therefore A A^{-1} = I$, $\det(A)\det(A^{-1}) = \det(I)$

$\therefore \det(A)\det(A^{-1}) = 1 \Rightarrow \det(A) \neq 0$.

(6) **13.** Show that $\det A \neq 0$ if and only if T is onto.

(1) Assume $\det A \neq 0$. $\therefore A^{-1}$ exists.

Let $\vec{y} \in \mathbb{R}^2$, and $\vec{v}$ be the fixed vector.

Need to find $\vec{x} \in \mathbb{R}^2$ s.t. $T(\vec{x}) = \vec{y}$

But $T(\vec{x}) = A\vec{x} + \vec{v}$.

$\therefore$ Let $\vec{x} = A^{-1}(\vec{y} - \vec{v})$

$\therefore T(\vec{x}) = A\left(A^{-1}(\vec{y}-\vec{v})\right) + \vec{v}$

$= (\vec{y}-\vec{v}) + \vec{v} = \vec{y}.$

$\therefore T$ is onto

(2) Assume T is onto. $\therefore$ Given any $\vec{y} \in \mathbb{R}^2$, $\exists\ \vec{x} \in \mathbb{R}^2$ s.t. $T(\vec{x}) = \vec{y}$, or $A\vec{x} + \vec{v} = \vec{y}$, where $\vec{v}$ is the fixed vector for T.

$\therefore$ Let $\vec{y} = \vec{e_1} + \vec{v}$, $\vec{e_1} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\therefore \exists\ \vec{x_1}$ s.t. $A\vec{x_1} + \vec{v} = \vec{e_1} + \vec{v}$, or

$$A\vec{x_1} = \vec{e_1}$$

Similarly, for $\vec{y} = \vec{e_2} + \vec{v}$, $\vec{e_2} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$,

$$\exists\ \vec{x_2} \text{ s.t. } A\vec{x_2} = \vec{e_2}$$

$\therefore A\begin{bmatrix} \vec{x_1} & \vec{x_2} \end{bmatrix} = \begin{bmatrix} \vec{e_1} & \vec{e_2} \end{bmatrix}$, or

$$A\begin{bmatrix} \vec{x_1} & \vec{x_2} \end{bmatrix} = I.$$

$$\therefore \begin{bmatrix} \vec{x_1} & \vec{x_2} \end{bmatrix} = A^{-1}$$

$\therefore T$ onto $\Rightarrow A^{-1}$ exists $\Rightarrow \det(A) \neq 0$.

(c)

14. Suppose $T: \mathbb{R}^2 \to \mathbb{R}^2$ is linear and is given by $T(\mathbf{x}) = A\mathbf{x}$, where A is a 2×2 matrix. Show that if $\det A \neq 0$, then T takes parallelograms onto parallelograms. [HINT: The general parallelogram in $\mathbb{R}^2$ can be described by the set of points $\mathbf{q} = \mathbf{p} + \lambda\mathbf{v} + \mu\mathbf{w}$ for $\lambda, \mu \in (0, 1)$ where $\mathbf{p}$, $\mathbf{v}$, $\mathbf{w}$ are vectors in $\mathbb{R}^2$ with $\mathbf{v}$ not a scalar multiple of $\mathbf{w}$.]

(1) Suppose $\det A \neq 0$. $\therefore A^{-1}$ exists.

Let $\vec{q} = \vec{a} + \lambda\vec{b} + \mu\vec{c}$, $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^2$,

$\lambda, \mu \in [0, 1]$, $\vec{b}, \vec{c}$ not scalar multiples.

$\therefore T(\vec{q}) = T(\vec{a} + \lambda\vec{b} + \mu\vec{c})$

$= A(\vec{a} + \lambda\vec{b} + \mu\vec{c}) + \vec{v}$

$= A\vec{a} + \lambda A\vec{b} + \mu A\vec{c} + \vec{v}$

$= (A\vec{a} + \vec{v}) + \lambda A\vec{b} + \mu A\vec{c}$

Let $\vec{a}^* = A\vec{a} + \vec{v}$, $\vec{b}^* = A\vec{b}$, $\vec{c}^* = A\vec{c}$

$\therefore T(\vec{q}) = \vec{a}^* + \lambda\vec{b}^* + \mu\vec{c}^*$, which is the form for a parallelogram.

Must show $\vec{b}^*, \vec{c}^*$ are not scalar multiples.

Suppose they are scalar multiples,

i.e., $k \in \mathbb{R}$ s.t. $\vec{b}^* = k\vec{c}^*$.

$\vec{b} = A^{-1}\vec{b}^* = kA^{-1}\vec{c}^* = k\vec{c}$,

a contradiction since there is no

k s.t. $\vec{b} = k\vec{c}$.

$\therefore$ $\det A \neq 0 \Rightarrow T$ maps parallelograms to parallelograms.

(2) Suppose $\vec{q}^*$ is a parallelogram.

$\therefore \vec{q}^* = \vec{a}^* + \lambda\vec{b}^* + \mu\vec{c}^*$, $\vec{a}, \vec{b}, \vec{c}^* \in \mathbb{R}^2$,

$\lambda, \mu \in [0,1]$, $\vec{b}^*, \vec{c}$ not scalar multiples.

To prove T is onto, need to find a parallelogram $\vec{q}$ s.t. $T(\vec{q}) = \vec{q}^*$,

or $A\vec{q} + \vec{v} = \vec{q}^* = \vec{a}^* + \lambda\vec{b}^* + \mu\vec{c}^*$,

or $A\vec{q} = (\vec{a}^* - \vec{v}) + \lambda\vec{b}^* + \mu\vec{c}^*$

$\therefore$ Let $\vec{q} = A^{-1}(\vec{a}^* - \vec{v}) + \lambda A^{-1}\vec{b}^* + \mu A^{-1}\vec{c}^*$

This is the form of a parallelogram

And $A^{-1}\vec{b}^*$ and $A^{-1}\vec{c}^*$ are not scalar multiples, for if they were, then $A^{-1}\vec{b}^* = kA^{-1}\vec{c}^*$, some $k \in \mathbb{R}$

$\Rightarrow AA^{-1}\vec{b}^* = kAA^{-1}\vec{c}^*$

$\Rightarrow \vec{b} = k\vec{c}^*$, a contradiction

$\therefore$ (1) & (2) $\Rightarrow$ affine T maps parallelograms onto parallelograms.

16. **16.** Suppose $T: \mathbb{R}^2 \to \mathbb{R}^2$ is as in Exercise 14 and that $T(P^*) = P$ is a parallelogram. Show that P^* is a parallelogram.

This was done in 14 (2)

17. **17.** Show that T is not one-to-one.

Assume $T(\vec{x}) = A\vec{x}$. For T do not be one-to-one, $\det(A) = 0$ by #12.

$\therefore$ Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$. $\therefore \det A = 0$

Let $\vec{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

$\therefore A\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $A\vec{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\therefore A\vec{x} = A\vec{y}$ but $\vec{x} \neq \vec{y}$,

$\therefore T$ is not one-to-one.

6.2 The Change of Variables Theorem

Note Title 12/14/2016

1.

1. Suggest a substitution/transformation that will simplify the following integrands, and find their Jacobians.

(a) $\iint_R (3x + 2y)\sin(x - y)\,dA$

(b) $\iint_R e^{(-4x+7y)}\cos(7x - 2y)\,dA$

(a) Make $x-y$ one variable, and $3x+2y$ the other

$$\therefore \begin{bmatrix} 3 & 2 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}, \quad T = \begin{bmatrix} 3 & 2 \\ 1 & -1 \end{bmatrix}$$

$$\therefore T^{-1} = \frac{1}{-3-2}\begin{bmatrix} -1 & -2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1/5 & 2/5 \\ 1/5 & -3/5 \end{bmatrix}$$

$$\begin{bmatrix} 1/5 & 2/5 \\ 1/5 & -3/5 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}, \text{ or } \begin{aligned} x &= \tfrac{1}{5}(u+2v) \\ y &= 1/5(u-3v) \end{aligned}$$

$$|\det(T^{-1})| = \frac{1}{5} = \text{Jacobian}$$

$$\therefore \iint_{R^*} u\sin(v)\left(\tfrac{1}{5}\right)du\,dv$$

(b) Make $(-4x+7y) = u$, $(7x-2y) = v$

$$\therefore \begin{bmatrix} -4 & 7 \\ 7 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}, \quad T = \begin{bmatrix} -4 & 7 \\ 7 & -2 \end{bmatrix}$$

$$\therefore T^{-1} = -\frac{1}{41}\begin{bmatrix} -2 & -7 \\ -7 & -4 \end{bmatrix} \qquad \det(T^{-1}) = -\frac{1}{41}$$

$$\therefore \quad x = \tfrac{1}{41}(2u + 7v) \qquad \text{Jacobian} = \left|-\tfrac{1}{41}\right| = \tfrac{1}{41}$$
$$y = \tfrac{1}{41}(7u + 4v)$$

$$\therefore \iint_{R^*} e^u \cos(v)\left(\tfrac{1}{41}\right) du\,dv$$

2.

2. Suggest a substitution/transformation that will simplify the following integrands, and find their Jacobians.

(a) $\iint_R (5x + y)^3 (x + 9y)^4 \, dA$

(b) $\iint_R x\sin(6x + 7y) - 3y\sin(6x + 7y) \, dA$

(a) Let $u = 5x + y \quad v = x + 9y$

$$\therefore \begin{bmatrix} 5 & 1 \\ 1 & 9 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}, \quad T = \begin{bmatrix} 5 & 1 \\ 1 & 9 \end{bmatrix} \quad \det(T) = 46$$

$$\therefore T^{-1} = \frac{1}{46}\begin{bmatrix} 9 & -1 \\ -1 & 5 \end{bmatrix} \quad \det(T^{-1}) = \tfrac{1}{46}$$

$$\therefore \quad x = \tfrac{1}{46}(9u - v) \qquad y = \tfrac{1}{46}(-u + 5v)$$

$$\text{Jacobian} = \left|\det(T^{-1})\right| = \tfrac{1}{46}$$

$$\iint_{R^*} u^3 v^4 \left(\tfrac{1}{46}\right) du\,dv$$

(6) $\iint_R x\sin(6x+7y) - 3y\sin(6x+7y)\,dA =$

$\iint_R (x-3y)\sin(6x+7y)\,dA$

$\therefore$ Let $u = x-3y$, $v = 6x+7y$, $T = \begin{bmatrix} 1 & -3 \\ 6 & 7 \end{bmatrix}$

$\therefore \begin{bmatrix} 1 & -3 \\ 6 & 7 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$ $\det(T) = 25$

$\therefore T^{-1} = \frac{1}{25}\begin{bmatrix} 7 & 3 \\ -6 & 1 \end{bmatrix}$ $\therefore T^{-1}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$,

$x = \frac{1}{25}(7u+3v)$, $y = \frac{1}{25}(-6u+v)$

Jacobian $= |\det(T^{-1})| = \frac{1}{25}$

$\therefore \iint_{R^*} u\sin(v)\left(\frac{1}{25}\right) du\,dv$

3. **3.** Let D be the unit disk: $x^2 + y^2 \le 1$. Evaluate

$$\iint_D \exp(x^2 + y^2)\,dx\,dy$$

by making a change of variables to polar coordinates.

As $T(r,\theta) = (r\cos\theta, r\sin\theta)$, $\left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$

$= r$

$$\iint_D \exp(x^2+y^2)\,dx\,dy = \iint_{D^*} \exp(r^2)\, r\,dr\,d\theta$$

$$= \int_0^1 \int_0^{2\pi} \exp(r^2)\, r\,dr\,d\theta = 2\pi \int_0^1 e^{r^2} r\,dr$$

$$= \pi \int_0^1 e^{r^2}\, d(r^2) = \pi e^{r^2}\Big|_0^1 = \pi(e-1)$$

4.

4. Let D be the region $0 \le y \le x$ and $0 \le x \le 1$. Evaluate

$$\iint_D (x+y)\,dx\,dy$$

by making the change of variables $x = u + v$, $y = u - v$. Check your answer by evaluating the integral directly by using an iterated integral.

(a) $x + y = (u+v) + (u-v) = 2u$

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}, \quad T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$\therefore |\det(T)| = |-1-1| = 2$

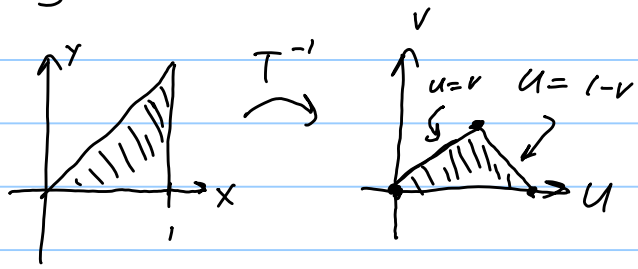
$\therefore \iint_{D^*} 2u\,(2)\,du\,dv$

What is D^*? $T^{-1} = \frac{1}{-2}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

$$\therefore \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

$\therefore T^{-1}(0,0) = (0,0)$
$T^{-1}(1,0) = (\frac{1}{2}, \frac{1}{2})$
$T^{-1}(1,1) = (1,0)$

$\therefore$



$\therefore$ a "flip" and rotation

This can be viewed as a "u-simple" region in uv-plane.

"Upper" boundary: $u = 1 - v$

"Lower" boundary: $u = v$

$$\therefore \iint_{D^*} (4u)\,du\,dv = \int_0^{\frac{1}{2}} \int_v^{1-v} (4u)\,du\,dv$$

$$= \int_0^{\frac{1}{2}} \left(2u^2 \Big|_{u=v}^{u=1-v} \right) dv = \int_0^{\frac{1}{2}} \left[2(1-v)^2 - 2v^2 \right] dv$$

$$= \int_0^{\frac{1}{2}} 2(v^2 - 2v + 1) - 2v^2 \, dv = \int_0^{\frac{1}{2}} -4v + 2 \, dv$$

$$= -2v^2 + 2v \Big|_0^{\frac{1}{2}} = -2\left(\tfrac{1}{2}\right)^2 + 2\left(\tfrac{1}{2}\right) = \underline{\underline{\tfrac{1}{2}}}$$

(6) $$\iint_D (x+y)\,dx\,dy = \int_0^1 \int_0^x (x+y)\,dy\,dx$$

$$= \int_0^1 \left[xy + \frac{y^2}{2} \Big|_{y=0}^{y=x} \right] dx = \int_0^1 \left(x^2 + \frac{x^2}{2} \right) dx$$

$$= \int_0^1 \frac{3x^2}{2}\,dx = \frac{x^3}{2}\Big|_0^1 = \frac{1}{2}$$

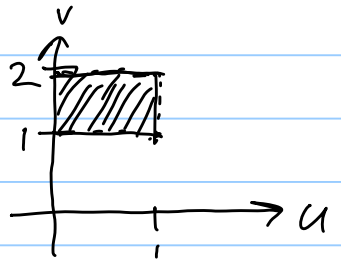
5.

5. Let $T(u, v) = (x(u, v), y(u, v))$ be the mapping defined by $T(u, v) = (4u, 2u + 3v)$. Let D^* be the rectangle $[0, 1] \times [1, 2]$. Find $D = T(D^*)$ and evaluate

(a) $\iint_D xy\,dx\,dy$

(b) $\iint_D (x - y)\,dx\,dy$

by making a change of variables to evaluate them as integrals over D^*.



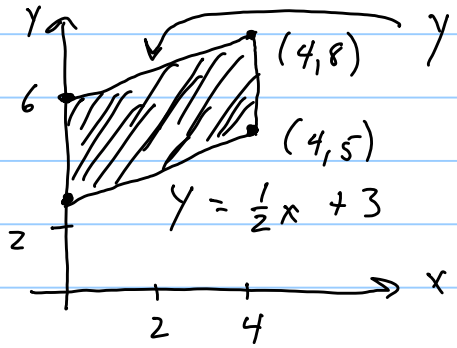
$$T = \begin{bmatrix} 4 & 0 \\ 2 & 3 \end{bmatrix} \text{ as } \begin{bmatrix} 4 & 0 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 4u \\ 2u+3v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$T(0,1) = (0,3)$ $\quad T(1,2) = (4,8)$

$T(1,1) = (4,5)$ $\quad T(0,2) = (0,6)$

$\therefore D:$

$0 \le x \le 4$



$\frac{1}{2}x + 3 \le y \le \frac{1}{2}x + 6$

$\therefore$ Rectange $D^* \xrightarrow{T}$ parallelogram D

$$|\det(T)| = 12$$

(a) $$\iint_D xy\,dx\,dy = \iint_{D^*} (4u)(2u+3v)(12)\,du\,dv$$

$$= \int_1^2 \int_0^1 (96u^2 + 144uv)\,du\,dv$$

$$= \int_1^2 \left(32u^3 + 72u^2v\Big|_{u=0}^{u=1}\right) dv$$

$$= \int_1^2 (32 + 72v)\,dv = 32v + 36v^2\Big|_1^2$$

$$= 64 + 144 - (32+36) = \underline{\underline{140}}$$

(b) $$\iint_D (x-y)\,dx\,dy = \iint_{D^*} [4u - (2u+3v)](12)\,du\,dv$$

$$= \int_1^2 \int_0^1 (24u - 36v)\,du\,dv = \int_1^2 12u^2 - 36uv\Big|_{u=0}^{u=1} dv$$

$$= \int_1^2 (12 - 36v)\,dv = 12v - 18v^2\Big|_1^2$$

$= 24 - 72 - (12 - 18) = \underline{-42}$

6. **6.** Repeat Exercise 5 for $T(u, v) = (u, v(1 + u))$.

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{bmatrix} 1 & 0 \\ v & 1 \end{bmatrix} \quad \therefore \left|\frac{\partial(x,y)}{\partial(u,v)}\right| = 1$$

(a) $$\iint_D xy\,dx\,dy = \iint_{D^*} (u)(v(1+u))(1)\,du\,dv$$

$$= \int_1^2 \int_0^1 (uv + u^2v)\,du\,dv$$

$$= \int_1^2 \left(\frac{u^2v}{2} + \frac{u^3v}{3}\Big|_{u=0}^{u=1}\right) dv = \int_1^2 \frac{v}{2} + \frac{v}{3}\,dv$$

$$= \frac{v}{4}^2 + \frac{v}{6}^2 \Big|_1^2 = 1 + \frac{2}{3} - \left(\frac{1}{4} + \frac{1}{6}\right) = \underline{\frac{5}{4}}$$

(b) $$\iint_D (x-y)\,dx\,dy = \iint_{D^*} [u - (v + uv)](1)\,du\,dv$$

$$= \int_1^2 \int_0^1 (u - v - uv)\,du\,dv$$

$$= \int_1^2 \left(\frac{u^2}{2} - uv - \frac{u^2 v}{2} \Big|_{u=0}^{u=1} \right) dv$$

$$= \int_1^2 \left(\frac{1}{2} - v - \frac{v}{2} \right) dv = \int_1^2 \left(\frac{1}{2} - \frac{3v}{2} \right) dv$$

$$= \frac{v}{2} - \frac{3v^2}{4} \Big|_1^2 = 1 - 3 - \left(\frac{1}{2} - \frac{3}{4} \right) = -\frac{7}{4}$$

7.

7. Evaluate

$$\iint_D \frac{dx\,dy}{\sqrt{1+x+2y}},$$

where $D = [0, 1] \times [0, 1]$, by setting $T(u, v) = (u, v/2)$ and evaluating an integral over D^*, where $T(D^*) = D$.

$T = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$, so $\begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} u \\ v/2 \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$

To find D^*, look at the 4 corners of D.

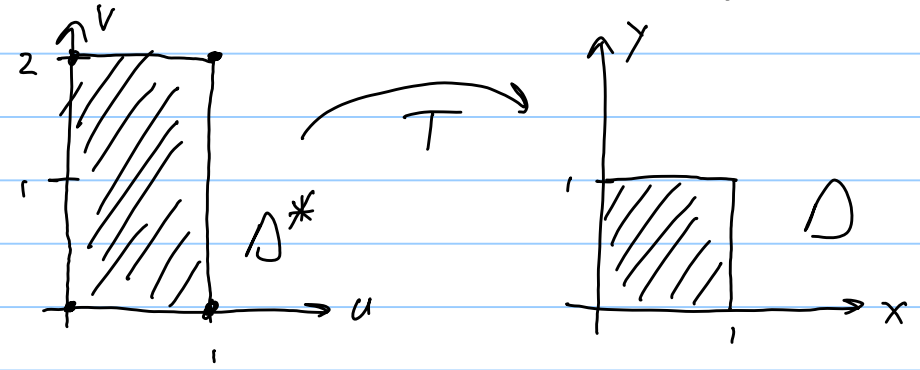
$$\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

$\therefore$



$D^* = [0,1] \times [0,2]$

$$|\det(T)| = \frac{1}{2}, \quad x = u, \quad y = \frac{v}{2}$$

$\therefore$

$$\iint_D \frac{dx\,dy}{\sqrt{1+x+2y}} = \iint_{D^*} \frac{\left(\frac{1}{2}\right) du\,dv}{\sqrt{1+u+v}}$$

$$= \int_0^2 \int_0^1 \frac{du\,dv}{2\sqrt{1+u+v}}$$

$$\frac{1}{2}\int_0^1 (1+u+v)^{-\frac{1}{2}}\,du = \frac{1}{2}(2)(1+u+v)^{\frac{1}{2}}\Big|_{u=0}^{u=1}$$

$$= (2+v)^{\frac{1}{2}} - (1+v)^{\frac{1}{2}}$$

$\therefore$

$$\int_0^2 (2+v)^{\frac{1}{2}} - (1+v)^{\frac{1}{2}}\,dv$$

$$= \frac{2}{3}(2+v)^{3/2} - \frac{2}{3}(1+v)^{3/2}\Big|_{v=0}^{v=2}$$

$$= \frac{2}{3}(8) - \frac{2}{3}\sqrt{27} - \left(\frac{2}{3}\sqrt{8} - \frac{2}{3}\right)$$

$$= \frac{18}{3} - 2\sqrt{3} - \frac{4}{3}\sqrt{2}$$

It would have been easier to evaluate directly without making the change of variables.

8. **8.** Define $T(u, v) = (u^2 - v^2, 2uv)$. Let D^* be the set of (u, v) with $u^2 + v^2 \le 1$, $u \ge 0$, $v \ge 0$. Find $T(D^*) = D$. Evaluate $\iint_D dx\,dy$.

$u^2 + v^2 \le 1$ is a disc of radius 1, and with $u \ge 0, v \ge 0$, this is the disc in quadrant I.

From $\iint_D dx\,dy$, $f(x,y) = 1 \quad \therefore f(x(u,v), y(u,v)) = 1$

From $T(u,v) = (u^2 - v^2, 2uv)$, $\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} 2u & -2v \\ 2v & 2u \end{vmatrix} = 4u^2 + 4v^2$

$4u^2 + 4v^2 \ge 0$ for $u \ge 0, v \ge 0$.

$$\therefore \iint_D dx\,dy = \iint_{D^*} (4u^2 + 4v^2)\,du\,dv$$

The problem is finding what D^* is.

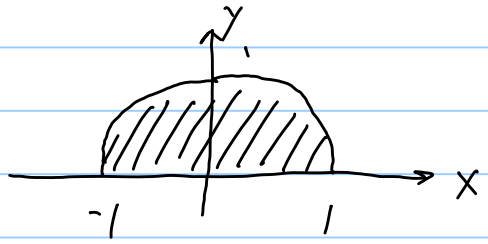
Consider $u = \cos\theta$, $v = \sin\theta$ in Quadrant I, for $0 \le \theta \le \pi/2$. This satisfies $u^2 + v^2 \le 1$, and covers the disc one-to-one in Quadrant I.

Then $u^2 - v^2 = \cos^2\theta - \sin^2\theta = \cos 2\theta$

$2uv = 2\cos\theta\sin\theta = \sin 2\theta$

And for $0 \le \theta \le \frac{\pi}{2}$, $(\cos 2\theta, \sin 2\theta)$ is the unit disk in Quadrants I & II.

$\therefore D$:



$$\therefore \iint_D dx\,dy = \int_0^1 \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} dx\,dy$$

$$= \int_0^1 2\sqrt{1-y^2}\,dy = y\sqrt{1-y^2} + \arcsin(y)\Big|_0^1$$

$$= \arcsin(1) = \frac{\pi}{2}$$

Perhaps a more elegant solution is to do another change of variables, this time on D^*.

From above, $$\iint_D dx\,dy = \iint_{D^*} 4u^2 + 4v^2\,du\,dv$$

Where D^* = unit disk in Quadrant I.

Consider $T^*(r,\theta) = (r\cos\theta, r\sin\theta)$

or $T^*(D^{**}) = D^*$, where

$D^{**} = [0,1] \times [0, \frac{\pi}{2}]$ for (r,θ).

$$\left| \frac{\partial T^*(u,v)}{\partial(r,\theta)} \right| = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

where $u = r\cos\theta$, $v = r\sin\theta$. This is one-to-one over D^{**}.

$$\therefore 4u^2 + 4v^2 = 4(r\cos\theta)^2 + 4(r\sin\theta)^2 = 4r^2$$

$$\therefore \iint_D dx\,dy = \iint_{D^*} 4u^2 + 4v^2 \, du\,dv = \iint_{D^{**}} (4r^2)(r)\,dr\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^1 4r^3 \, dr\,d\theta = \int_0^{\frac{\pi}{2}} r^4 \Big|_{r=0}^{r=1} d\theta$$

$$= \int_0^{\frac{\pi}{2}} d\theta = \frac{\pi}{2}$$

Here, you don't even need to know what D is.

9.

9. Let $T(u, v)$ be as in Exercise 8. By making a change of variables, "formally" evaluate the "improper" integral

$$\iint_D \frac{dx\,dy}{\sqrt{x^2+y^2}}.$$

[NOTE: This integral (and the one in the next exercise) is *improper*, because the integrand $1/\sqrt{x^2+y^2}$ is neither continuous nor bounded on the domain of integration. (The theory of improper integrals is discussed in Section 6.4.)]

For $T(u,v) = (u^2 - v^2, 2uv)$, $\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = 4u^2 + 4v^2$

as seen in #8.

Here, $f(x,y) = \frac{1}{\sqrt{x^2+y^2}}$, not 1 as in #8.

$\therefore f(x(u,v), y(u,v)) = f(u^2-v^2, 2uv) =$

$$\frac{1}{\sqrt{(u^2-v^2)^2 + (2uv)^2}} = \frac{1}{\sqrt{u^4 - 2u^2v^2 + v^4 + 4u^2v^2}}$$

$$= \frac{1}{\sqrt{u^4 + 2u^2v^2 + v^4}} = \frac{1}{\sqrt{(u^2+v^2)^2}}$$

$$= \frac{1}{u^2+v^2}, \text{ for } u \geq 0, v \geq 0, u^2+v^2 \leq 1$$

$$\therefore \iint_D \frac{dx\,dy}{\sqrt{x^2+y^2}} = \iint_{D^*} \frac{(4u^2+4v^2)}{u^2+v^2}\,du\,dv$$

$$= \iint_{D^*} 4\,du\,dv$$

where D^* = unit disk in Quadrant I from #8 above

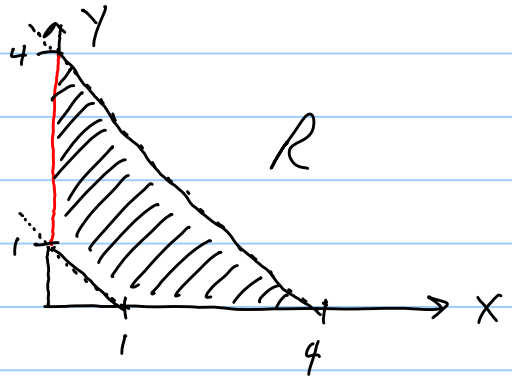
$$= \int_0^1 \int_0^{\sqrt{1-u^2}} 4 \, dv \, du = \int_0^1 4\sqrt{1-u^2} \, du$$

(see table of integrals #38 of text)

$$= 2u\sqrt{1-u^2} + 2\arcsin(u) \Big|_0^1$$

$$= 2\arcsin(1) = 2\left(\frac{\pi}{2}\right) = \underline{\pi}$$

10. **10.** Calculate $\iint_R \frac{1}{x+y}\,dy\,dx$, where R is the region bounded by $x = 0$, $y = 0$, $x + y = 1$, $x + y = 4$, by using the mapping $T(u, v) = (u - uv, uv)$.



(a) $\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} 1-v & -u \\ v & u \end{vmatrix}$

$= u - uv + uv = u$

Since $y = uv$, and $y \geq 0$ for $(x,y) \in R$,

then $u \geq 0$ and $v \geq 0$ or $u \leq 0$ and $v \leq 0$ [1]

For $x = u(1-v) \geq 0$ for $(x,y) \in R$,

then $u \geq 0$ and $1 - v \geq 0$ or $u \leq 0$ and $1 - v \leq 0$ [2]

[1] & [2] $\Rightarrow$ $u \geq 0$ and $0 \leq v \leq 1$, or

$u \leq 0$ and $1 \leq v$ and $v \leq 0$, a conflict

$\therefore u \geq 0$ and $0 \leq v \leq 1$

$\therefore \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = u$

(6) $f(x(u,v), y(u,v)) = \frac{1}{x(u,v) + y(u,v)} = \frac{1}{(u - uv) + (uv)}$

$= \frac{1}{u}$

$\therefore f(x(u,v), y(u,v)) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \frac{1}{u} \cdot u = 1$

$\therefore \iint_R \frac{1}{x+y} \, dx \, dy = \iint_{R^*} du \, dv = \text{Area}(R^*)$

(c) Look at $T(R^*) = R$

By definition of R^*, T is onto

Now consider $T(u,v) = T(u',v')$ for $(u,v) \in R^*$, $(u',v') \in R^*$

$\therefore (u - uv, uv) = (u' - u'v', u'v')$

$\Rightarrow uv = u'v'$ (2nd coordinates equal)

$\therefore u - uv = u' - u'v'$, so $u - u'v' = u' - u'v'$

$\Rightarrow u = u'$. $\therefore uv = u'v' \Rightarrow v = v'$ for $u \neq 0$

But $u \neq 0$ for $(u,v) \in R^*$, because if $u = 0$,

$T(0,v) = (0,0)$, and $(0,0) \notin R$.

$\therefore T(u,v) = T(u',v') \Rightarrow u = u', v = v'$

$\therefore T$ is one-to-one and onto so T^{-1} exists.

$\therefore$ Look at corners: $(1,0), (4,0), (0,4), (0,1)$

$T(u,v) = (1,0) \Rightarrow (u - uv, uv) = (1,0)$

$\Rightarrow (u,v) = (1,0)$

$T(u,v) = (4,0) \Rightarrow (u - uv, uv) = (4,0)$

$\Rightarrow (u,v) = (4,0)$

$T(u,v) = (0,4) \Rightarrow (u - uv, uv) = (0,4)$

$\Rightarrow u = 4, v = 1$, so $(u,v) = (4,1)$

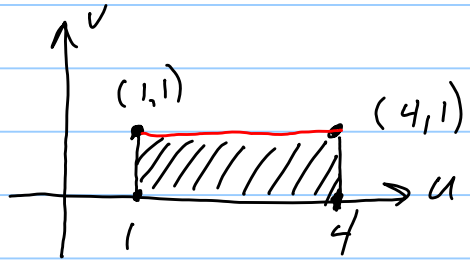
$T(u,v) = (0,1) \Rightarrow (u - uv, uv) = (0,1)$

$\Rightarrow u = 1, v = 1$, so $(u,v) = (1,1)$

Note $T(u,1) = (0,u)$, so the top horizontal

border of R^* corresponds to left vertical border of R.

$\therefore$ R^* :



$$\therefore \iint_{R^*} du\,dv = \int_0^1 \int_1^4 du\,dv = 3$$

$$\therefore \iint_R \frac{1}{x+y}\,dy\,dx = \iint_{R^*} du\,dv = \underline{3}$$

11. **11.** Evaluate $\iint_D (x^2 + y^2)^{3/2}\,dx\,dy$, where D is the disk $x^2 + y^2 \le 4$.

Try a polar coordinate transfer:

$$T(r,\theta) = (r\cos\theta, r\sin\theta)$$

$\therefore$ $0 \le r \le 2$, $0 \le \theta \le 2\pi$

$$\left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| = r, \quad (x^2+y^2)^{3/2} = (r^2)^{3/2} = r^3$$

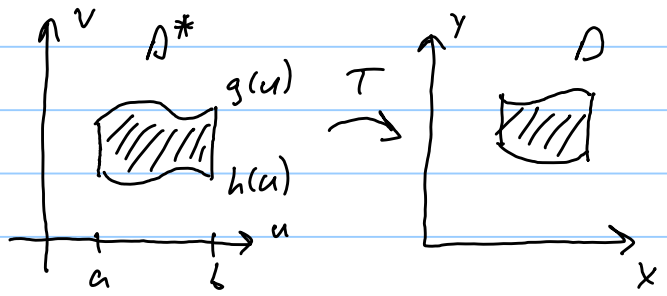
$$\therefore \iint_D (x^2+y^2)^{3/2}\,dx\,dy = \int_0^{2\pi}\int_0^2 (r^3)\, r\,dr\,d\theta$$

$$= \int_0^{2\pi} \frac{r^5}{5}\Big|_{r=0}^{r=2} d\theta = \int_0^{2\pi} \frac{32}{5}\,d\theta = \frac{64}{5}\pi$$

12.

12. Let D^* be a v-simple region in the uv plane bounded by $v = g(u)$ and $v = h(u) \le g(u)$ for $a \le u \le b$. Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the transformation given by $x = u$ and $y = \psi(u, v)$, where ψ is of class C^1 and $\partial\psi/\partial v$ is never zero. Assume that $T(D^*) = D$ is a y-simple region; show that if $f\colon D \to \mathbb{R}$ is continuous, then

$$\iint_D f(x, y)\,dx\,dy = \iint_{D^*} f(u, \psi(u, v))\left|\frac{\partial\psi}{\partial v}\right| du\,dv.$$



From chapter 5 of text, f is integrable over D.

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ \partial\psi/\partial u & \frac{\partial\psi}{\partial v} \end{vmatrix}$$

$$= \left|\frac{\partial\psi}{\partial v}\right| = \text{absolute value of } \partial\psi/\partial v \neq 0 \text{ by assumption.}$$

$\therefore$ By Theorem 2, p. 319 of text,

$$\iint_D f(x,y)\,dx\,dy = \iint_{D^*} f(x(u,v), y(u,v))\left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$$

$$= \iint_{D^*} f(u, \psi(u,v))\left|\frac{\partial\psi}{\partial v}\right| du\,dv$$

Note: $T(u,v) = (u, \psi(u,v))$ is one-to-one.

$T(u,v) = T(u',v') \Rightarrow (u, \psi(u,v)) = (u', \psi(u',v'))$.

$\Rightarrow u = u'$. $\therefore \psi(u,v) = \psi(u,v')$. But $\frac{\partial \psi}{\partial v} \neq 0$

$\Rightarrow g(v) = \psi(u,v)$, for fixed u, is either increasing or decreasing on any v-axis of D^*.

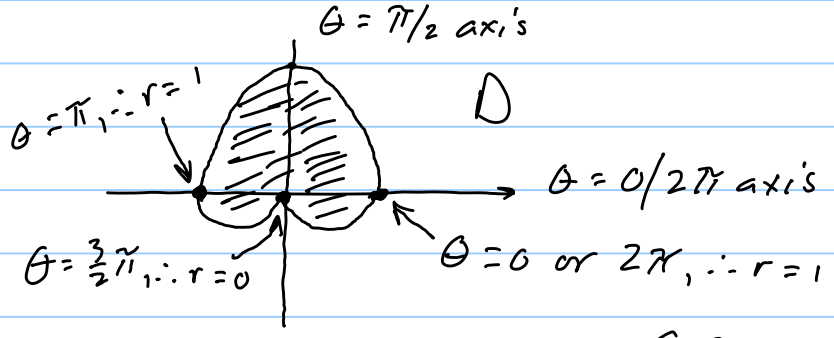
$\therefore g^{-1}$ exists, so $g(v) = g(v') \Rightarrow$

$g^{-1}(g(v)) = g^{-1}(g(v')) \Rightarrow v = v'$.

$\therefore T(u,v) = T(u',v') \Rightarrow u = u'$ and $v = v'$.

13. **13.** Use double integrals to find the area inside the curve $r = 1 + \sin\theta$.

Assume $0 \leq \theta \leq 2\pi$. $\therefore 0 \leq r \leq 2$ as $-1 \leq \sin\theta \leq 1$



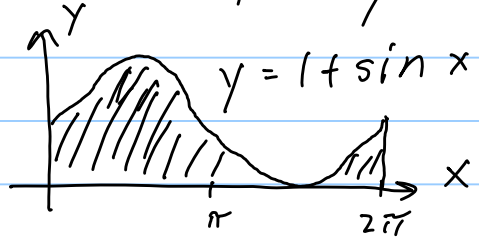
Curve is $r(\theta) = 1 + \sin\theta$

Let $f(r,\theta) = 1$. $\therefore$ Area $= \iint_D f(r,\theta)\,dr\,d\theta = \iint_0 dr\,d\theta$

Now need a region D^* so that $T(D^*) = D$, making the integration easier. Use a polar transformation: $(r,\theta) = T(x,y) = (y\cos x, y\sin x)$,

where $0 \le x \le 2\pi$, $0 \le y \le 1+\sin x$

$\therefore D^*$:



Although D^* doesn't look simple, the integration will be familiar.

$$\frac{\partial(r,\theta)}{\partial(x,y)} = \begin{bmatrix} -y\sin x & \cos x \\ y\cos x & \sin x \end{bmatrix}$$

$$\therefore \left| \det \frac{\partial(r,\theta)}{\partial(x,y)} \right| = \left| -y\sin^2 x - y\cos^2 x \right| = y$$

$$\therefore \iint_D dr\,d\theta = \iint_{D^*} y\,dy\,dx = \int_0^{2\pi}\int_0^{1+\sin x} y\,dy\,dx$$

$$= \int_0^{2\pi} \frac{y^2}{2}\Big|_{y=0}^{y=1+\sin x} dx = \int_0^{2\pi} \frac{(1+\sin x)^2}{2}\,dx$$

$$= \frac{1}{2}\int_0^{2\pi} 1 + 2\sin x + \sin^2 x\,dx$$

using $\cos^2 x - \sin^2 x = \cos 2x$,
$\therefore (1-\sin^2 x) - \sin^2 x = \cos 2x$
$\therefore \frac{1-\cos 2x}{2} = \sin^2 x$

$$= \int_0^{2\pi} \frac{1}{2} + \sin x + \frac{1-\cos 2x}{4}\,dx$$

$$= \int_0^{2\pi} \frac{3}{4} + \sin x - \frac{\cos 2x}{2}\, dx$$

$$= \frac{3}{4}x - \cos x - \frac{\sin 2x}{4}\Bigg|_0^{2\pi} = \frac{3}{2}\pi - 1 - 0 - (0 - 1 - 0)$$

$$= \frac{3}{2}\pi$$

Note: Could have originally set variables as $y = 1 + \sin x$, and then change using $(x,y) = T(r,\theta)$. However, the process is the same.

14.

14. (a) Express $\int_0^1 \int_0^{x^2} xy\, dy\, dx$ as an integral over the triangle D^*, which is the set of (u, v) where $0 \le u \le 1$, $0 \le v \le u$. (HINT: Find a one-to-one mapping T of D^* onto the given region of integration.)

(b) Evaluate this integral directly and as an integral over D^*.

(a) Let $T(u,v) = (u, v^2) = (x,y)$. For $0 \le u \le 1$, $0 \le v \le u$,

$$T(u,v) = T(u',v') \Rightarrow (u, v^2) = (u', v'^2)$$

$$\Rightarrow u = v' \text{ and } v = v'. \quad \therefore T \text{ is one-to-one.}$$

$$\therefore \frac{\partial(x,y)}{\partial(u,v)} = \begin{bmatrix} 1 & 0 \\ 0 & 2v \end{bmatrix} \quad \therefore \left|\frac{\partial(x,y)}{\partial(u,v)}\right| = 2v \ge 0$$

$$\therefore \int_0^1 \int_0^{x^2} xy\, dy\, dx = \iint_{D^*} (u)(v^2)(2v)\, du\, dv$$

$$= \int_0^1 \int_0^u 2uv^3 \, dv \, du$$

(6) (1) $$\int_0^1 \int_0^u 2uv^3 \, dv \, du = \int_0^1 \left(\frac{1}{2} uv^4 \Big|_{v=0}^{v=u} \right) du$$

$$= \int_0^1 \frac{1}{2} u^5 \, du = \frac{1}{12} u^6 \Big|_0^1 = \frac{1}{12}$$

(2) $$\int_0^1 \int_0^{x^2} xy \, dy \, dx = \int_0^1 \left(x \frac{y^2}{2} \Big|_{y=0}^{y=x^2} \right) dx$$

$$= \int_0^1 \frac{x^5}{2} \, dx = \frac{x^6}{12} \Big|_0^1 = \frac{1}{12}$$

15. **15.** Integrate $ze^{x^2+y^2}$ over the cylinder $x^2 + y^2 \le 4, 2 \le z \le 3$.

$$\int_2^3 \int_0^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} z e^{x^2+y^2} \, dy \, dx \, dz$$

Use cylindrical transformation:

$$T(r, \theta, z) = (r\cos\theta, r\sin\theta, z) = (x, y, z)$$

$$\therefore r^2 = x^2 + y^2 \qquad \left| \frac{\partial(x, y, z)}{\partial(r, \theta, z)} \right| = r$$

where $0 \le r \le 2$, $0 \le \theta \le 2\pi$, $2 \le z \le 3$

$$\therefore \int_2^3 \int_0^{2\pi} \int_0^2 z e^{r^2}(r)\, dr\, d\theta\, dz$$

$$= \int_2^3 \int_0^{2\pi} \frac{1}{2} z e^{r^2} \Big|_{r=0}^{r=2} d\theta\, dz$$

$$= \int_2^3 \int_0^{2\pi} \frac{1}{2} z (e^4 - 1)\, d\theta\, dz = \int_2^3 \frac{1}{2} z (e^4-1)\theta \Big|_{\theta=0}^{\theta=2\pi} dz$$

$$\int_2^3 \pi (e^4-1) z\, dz = \pi (e^4-1) \frac{z^2}{2} \Big|_{z=2}^{z=3} = \frac{5}{2}\pi (e^4-1)$$

16. **16.** Let D be the unit disk. Express $\iint_D (1 + x^2 + y^2)^{3/2}\, dx\, dy$ as an integral over $[0, 1] \times [0, 2\pi]$ and evaluate.

Use the polar transformation: $T(r, \theta) = (r\cos\theta, r\sin\theta)$

$\left| \frac{\partial(x,y)}{\partial(r,\theta)} \right| = r$, and $x^2 + y^2 = r^2$

$$\therefore \iint_D (1 + x^2 + y^2)^{3/2}\, dx\, dy = \iint_{D^*} (1 + r^2)^{3/2}(r)\, dr\, d\theta$$

$$= \int_0^{2\pi} \int_0^1 (1+r^2)^{3/2} \, r \, dr \, d\theta$$

$$= \int_0^{2\pi} \frac{1}{5}(1+r^2)^{5/2} \Big|_{r=0}^{r=1} \, d\theta$$

$$= \int_0^{2\pi} \frac{1}{5}(2)^{5/2} - \frac{1}{5} \, d\theta$$

$$= \int_0^{2\pi} \frac{4}{5}\sqrt{2} - \frac{1}{5} \, d\theta = \frac{2\pi}{5}(4\sqrt{2}-1)$$

17. **17.** Using polar coordinates, find the area bounded by the *lemniscate* $(x^2+y^2)^2 = 2a^2(x^2-y^2)$.

Note: xy-graph has symmetry with respect to x-axis and y-axis. Assume $a > 0$.

Let $T(r,\theta) = (r\cos\theta, r\sin\theta) = (x,y)$ $\therefore \left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| = r$

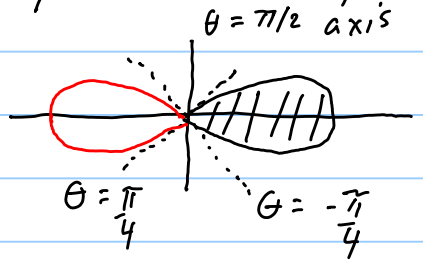
and $x^2+y^2 = r^2$, $x^2-y^2 = r^2(\cos^2\theta - \sin^2\theta) = r^2\cos 2\theta$

When $y=0$, $(x^2+y^2)^2 = 2a^2(x^2-y^2) \Rightarrow x^4 = 2a^2x^2$, $x = \pm\sqrt{2}\,a$

When $x=0$, $(x^2+y^2)^2 = 2a^2(x^2-y^2) \Rightarrow y^4 = 2a^2(-y^2) \Rightarrow y=0$.

Since $(x^2+y^2)^2 \geq 0$, $x^2-y^2 \geq 0$, $\therefore |y| \leq |x|$

$(x^2+y^2)^2 = 2a^2(x^2-y^2) \Rightarrow r^4 = 2a^2r^2\cos 2\theta \Leftrightarrow r^2 = 2a^2\cos 2\theta$



As $-\frac{\pi}{4} \le \theta \le \frac{\pi}{4}$, curve traces out loop shown at left, for $r \ge 0$.

For $r \le 0$, symmetric curve in — is traced out. Areas are equal.

Let D^* be just the loop for $r \ge 0$.

$\therefore D^* = \left\{(r,\theta) : 0 \le r \le \sqrt{2}a,\ -\frac{\pi}{4} \le \theta \le \frac{\pi}{4}\right\}$

If $\theta > \frac{\pi}{4}$, $2a^2\cos 2\theta < 0$, and $r^2 \ge 0$.

$$\therefore \iint_D dx\,dy = 2\iint_{D^*} r\,dr\,d\theta = 2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\int_0^{f(\theta)} r\,dr\,d\theta$$

Where $r = f(\theta) = \sqrt{2a^2\cos 2\theta}$

$$\therefore 2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\int_0^{\sqrt{2a^2\cos 2\theta}} r\,dr\,d\theta = 2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{r^2}{2}\Big|_{r=0}^{r=\sqrt{2a^2\cos 2\theta}} d\theta$$

$$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} 2a^2\cos 2\theta\,d\theta = a^2\sin 2\theta\Big|_{-\frac{\pi}{4}}^{\pi/4}$$

$= a^2 - (-a^2) = \underline{2a^2}$

18. **18.** Redo Exercise 15 of Section 5.3 using a change of variables and compare the effort involved in each method.

15. Find the volume of the region inside the surface $z = x^2 + y^2$ and between $z = 0$ and $z = 10$.

Use a cylindrical coordinate conversion.

$$\text{Volume} = \iiint_{W^*} r \, dr \, d\theta \, dz$$

where $0 \le z \le 10$, $0 \le \theta \le 2\pi$, $0 \le r \le \sqrt{z}$ (from $z = x^2 + y^2 = r^2$)

$$\therefore V = \int_0^{10} \int_0^{2\pi} \int_0^{\sqrt{z}} r \, dr \, d\theta \, dz$$

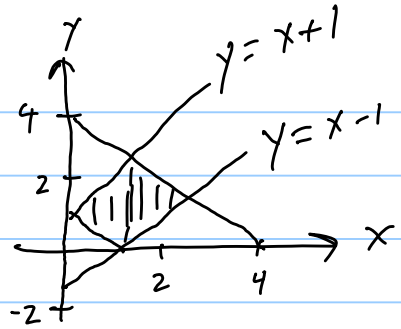
$$= \int_0^{10} \int_0^{2\pi} \frac{1}{2} r^2 \Big|_{r=0}^{r=\sqrt{z}} \, d\theta \, dz$$

$$= \int_0^{10} \int_0^{2\pi} \frac{z}{2} \, d\theta \, dz = \int_0^{10} \pi z \, dz$$

$$= \frac{\pi z^2}{2} \Big|_0^{10} = \underline{50\pi}$$

19.

19. Calculate $\iint_R (x+y)^2 e^{x-y}\,dx\,dy$, where R is the region bounded by $x+y=1$, $x+y=4$, $x-y=-1$, and $x-y=1$.



Let $u = x+y$, $v = x-y$

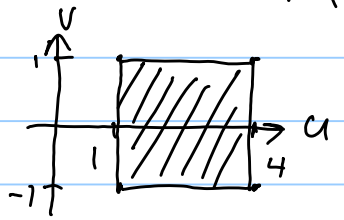
$\therefore \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$, $T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. $\det(T) = -2 \neq 0$

$\therefore T^{-1} = \frac{1}{-2}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$ $\therefore T^{-1}(u,v) = (x,y)$

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = |\det(T^{-1})| = \left|-\tfrac{1}{2}\right| = \tfrac{1}{2}$$

Corners for R: $(0,1)$, $(1,0)$, $(\frac{5}{2},\frac{3}{2})$, $(\frac{3}{2},\frac{5}{2})$

$\therefore$ Corners for R^*: $T(0,1) = (1,-1)$ $T(\frac{5}{2},\frac{3}{2}) = (4,1)$
$T(1,0) = (1,1)$ $T(\frac{3}{2},5/2) = (4,-1)$

$\therefore R^*$ 

$1 \le u \le 4$
$-1 \le v \le 1$

$$\therefore \iint_R (x+y)^2 e^{x-y}\,dx\,dy = \iint_{R^*} u^2 e^v \left|\det(T^{-1})\right| du\,dv$$

$$= \int_{-1}^{1}\int_{1}^{4} u^2 e^v \left(\tfrac{1}{2}\right) du\,dv$$

$$= \int_{-1}^{1} \left(\frac{e^v}{2} \frac{u^3}{3} \Big|_{u=1}^{u=4} \right) dv = \int_{-1}^{1} \frac{64}{6} e^v - \frac{e^v}{6} \, dv$$

$$= \frac{63}{6} e^v \Big|_{v=-1}^{v=1} = \frac{21}{2}\left(e - \frac{1}{e}\right)$$

20.

20. Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be defined by

$$T(u, v, w) = (u \cos v \cos w, \, u \sin v \cos w, \, u \sin w).$$

(a) Show that T is onto the unit sphere; that is, every (x, y, z) with $x^2 + y^2 + z^2 = 1$ can be written as $(x, y, z) = T(u, v, w)$ for some (u, v, w).

(b) Show that T is not one-to-one.

(a) T is much like spherical coordinates.

Note $[u\cos(v)\cos(w)]^2 + [u\sin(v)\cos(w)]^2 + [u\sin(w)]^2$

$= u^2\cos^2(w)[\cos^2(v) + \sin^2(v)] + u^2\sin^2(w)$

$= u^2\cos^2(w) + u^2\sin^2(w) = u^2$

$\therefore \|T(u,v,w)\|^2 = u^2 = \|(x,y,z)\|^2 = x^2+y^2+z^2 = 1$

$\therefore$ Let $u = \sqrt{x^2+y^2+z^2}$ (analogous to ρ)

Let w = radian measure of point from the xy-plane (analogous to $\frac{\pi}{2} - \phi$).

$\therefore$ Let $\sin(w) = \frac{z}{\sqrt{x^2+y^2+z^2}}$, or $w = \text{Arcsin}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)$

so $-\frac{\pi}{2} \le w \le \frac{\pi}{2}$ as $-1 \le z \le 1$

Let v = radian measure from x-axis of projected point onto the xy-plane ($\equiv \theta$).

$\therefore$ Let $\cos(v) = \frac{x}{\sqrt{x^2+y^2}}$, $\therefore v = \text{Arccos}\left(\frac{x}{\sqrt{x^2+y^2}}\right)$

To cover the entire circle,

Let $v = \text{Arccos}\left(\frac{x}{\sqrt{x^2+y^2}}\right)$ for $y \geq 0$

$v = -\text{Arccos}\left(\frac{x}{\sqrt{x^2+y^2}}\right)$ for $y < 0$

$\therefore -\pi < v \leq \pi$

$\therefore$ Given any (x,y,z) s.t. $x^2+y^2+z^2=1$,

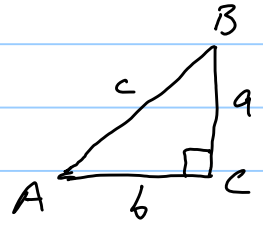
Choose $u = \sqrt{x^2+y^2+z^2}$

$v = \text{Arccos}\left(\frac{x}{\sqrt{x^2+y^2}}\right)$, $y \geq 0$

$= -\text{Arccos}\left(\frac{x}{\sqrt{x^2+y^2}}\right)$, $y < 0$

$w = \text{Arcsin}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)$

Note: Given any right triangle ABC with sides a, b, c labelled opposite the angles $\angle A, \angle B, \angle C$, respectively.



$c^2 = a^2 + b^2$, $\cos(\angle A) = \frac{b}{c}$, so $\angle A = \text{Arccos}\left(\frac{b}{\sqrt{a^2+b^2}}\right)$

$$\therefore \sin(\angle A) = \frac{a}{\sqrt{a^2+b^2}} = \sin\left(\operatorname{Arccos}\frac{b}{\sqrt{a^2+b^2}}\right)$$

i.e., you switch variables under the radical.

e.g., $\sin\left(\operatorname{Arccos}\frac{r}{\sqrt{r^2+s^2}}\right) = \frac{s}{\sqrt{r^2+s^2}}$, $r, s > 0$.

$\therefore$ Given $[x,y,z]$, with $y \geq 0$

$$T(u,v,w) = [u\cos(v)\cos(w), u\sin(v)\cos(w), u\sin(w)]$$

$$= \Bigg[\sqrt{x^2+y^2+z^2}\cos\left(\operatorname{Arccos}\left(\frac{x}{\sqrt{x^2+y^2}}\right)\right)\cos\left(\operatorname{Arcsin}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)\right),$$

$$\sqrt{x^2+y^2+z^2}\sin\left(\operatorname{Arccos}\left(\frac{x}{\sqrt{x^2+y^2}}\right)\right)\cos\left(\operatorname{Arcsin}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)\right),$$

$$\sqrt{x^2+y^2+z^2}\sin\left(\operatorname{Arcsin}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)\right) \Bigg]$$

$$= \Bigg[\sqrt{x^2+y^2+z^2}\ \frac{x}{\sqrt{x^2+y^2}}\ \frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}},$$

$$\sqrt{x^2+y^2+z^2}\ \frac{y}{\sqrt{x^2+y^2}}\ \frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}},$$

$$\sqrt{x^2+y^2+z^2}\ \frac{z}{\sqrt{x^2+y^2+z^2}} \Bigg]$$

$$= [x, y, z]$$

Note use of $\sin\left(\operatorname{Arccos}\left(\frac{x}{\sqrt{x^2+y^2}}\right)\right) = \frac{y}{\sqrt{x^2+y^2}}$ (assuming $y \geq 0$)

and $\cos\left(\operatorname{Arcsin}\left(\frac{z}{\sqrt{x^2+y^2+z^2}}\right)\right) = \cos\left(\operatorname{Arcsin}\left(\frac{z}{\sqrt{r^2+z^2}}\right)\right)$

$$= \frac{r}{\sqrt{r^2+z^2}} = \frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}} \quad (\text{letting } r^2 = x^2+y^2)$$

using the note above.

Also note $T(u,-v,w) = [u\cos(v)\cos(w), -u\sin(v)\cos(w), u\sin(w)]$
so if $y < 0$, the y-coordinate will have a negative contribution.

Thus, given a (x,y,z), $\exists\ (u,v,w)$ s.t.

$T(u,v,w) = (x,y,z)$, where $x^2+y^2+z^2=1$,

so T is <u>onto</u>.

(6) Since $u = 1 = \sqrt{x^2+y^2+z^2}$,

$T(u,v,w) = (\cos(v)\cos(w), \sin(v)\cos(w), \sin(w))$

Let $w = \frac{\pi}{2}$. $\therefore T(1, v, \frac{\pi}{2}) = (0,0,1)$.

$\therefore$ Even when $v_1 \neq v_2$, $T(1, v_1, \frac{\pi}{2}) = T(1, v_2, \frac{\pi}{2})$

$\therefore$ T is not one-to-one.

21. **21.** Integrate $x^2 + y^2 + z^2$ over the cylinder $x^2 + y^2 \le 2, -2 \le z \le 3$.

$$\iiint_R (x^2+y^2+z^2)\,dx\,dy\,dz$$ Use a cylindrical transformation

$x = r\cos\theta \quad y = r\sin\theta \quad z = z$

$$\therefore x^2+y^2+z^2 \Rightarrow r^2+z^2$$

$$\therefore \iiint_{R^*} (r^2+z^2)\, r\,dr\,d\theta\,dz = \int_{-2}^{3}\int_{0}^{2\pi}\int_{0}^{\sqrt{2}} (r^2+z^2)\, r\,dr\,d\theta\,dz$$

$$= \int_{-2}^{3}\int_{0}^{2\pi}\int_{0}^{\sqrt{2}} r^3+rz^2\,dr\,d\theta\,dz = \int_{-2}^{3}\int_{0}^{2\pi} \frac{r^4}{4}+\frac{r^2}{2}z^2\Big|_{r=0}^{r=\sqrt{2}}\,d\theta\,dz$$

$$= \int_{-2}^{3}\int_{0}^{2\pi} (1+z^2)\,d\theta\,dz = \int_{-2}^{3} 2\pi+2\pi z^2\,dz$$

$$= 2\pi z + \frac{2\pi z^3}{3}\Big|_{z=-2}^{z=3} = 6\pi+18\pi-\left(-4\pi-\frac{16}{3}\pi\right)$$

$$= 28\pi+\frac{16}{3}\pi = 33\tfrac{1}{3}\pi = \underline{\underline{\frac{100}{3}\pi}}$$

22. **22.** Evaluate $\int_0^\infty e^{-4x^2}\,dx$.

$$\int_0^\infty e^{-4x^2}dx = \lim_{u\to\infty}\int_0^u e^{-4x^2}dx = \lim_{u\to\infty}\int_0^u e^{-(2x)^2}dx$$

Let $y=2x$ $\therefore$ when $x=u$, $y=2u$ $dy=2dx$

$$\therefore \lim_{u\to\infty}\int_0^u e^{-(2x)^2}dx = \lim_{y\to\infty}\int_0^{2y} e^{-(y)^2}\left(\tfrac{1}{2}\right)dy$$

$$= \frac{1}{2}\lim_{y\to\infty}\int_0^{2y} e^{-y^2}dy = \frac{1}{2}\left[\frac{1}{2}\lim_{y\to\infty}\int_{-2y}^{2y} e^{-y^2}dy\right]$$

using symmetry

$$= \frac{1}{4}\sqrt{\pi}$$

23. **23.** Let B be the unit ball. Evaluate

$$\iiint_B \frac{dx\,dy\,dz}{\sqrt{2+x^2+y^2+z^2}}$$

by making the appropriate change of variables.

Use spherical coordinates

$$\therefore f(x,y,z) = \frac{1}{\sqrt{2+x^2+y^2+z^2}} \Rightarrow \frac{1}{\sqrt{2+\rho^2}}$$

$$dx\,dy\,dz \Rightarrow \rho^2\sin\phi\,d\rho\,d\theta\,d\phi$$

$$\therefore \iiint_{B^*} \frac{\rho^2\sin\phi}{\sqrt{2+\rho^2}}\,d\rho\,d\theta\,d\phi$$

$$= \int_0^1\int_0^{\pi}\int_0^{2\pi} \frac{\rho^2\sin\phi}{\sqrt{2+\rho^2}}\,d\theta\,d\phi\,d\rho$$

$$= 2\pi\int_0^1\int_0^{\pi} \frac{\rho^2\sin\phi}{\sqrt{2+\rho^2}}\,d\phi\,d\rho$$

$$= 2\pi \int_0^1 \left(\frac{\rho^2}{\sqrt{2+\rho^2}} - \cos\phi \Big|_{\phi=0}^{\phi=\pi} \right) d\rho$$

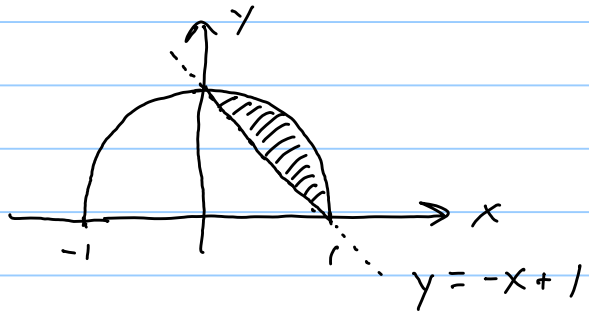
$$= 4\pi \int_0^1 \frac{\rho^2}{\sqrt{2+\rho^2}}\, d\rho$$ Use table of integrals #58 of text

$$= 4\pi \left[\frac{\rho\sqrt{2+\rho^2}}{2} - \log\left(\rho + \sqrt{2+\rho^2}\right) \right]_{\rho=0}^{\rho=1}$$

$$= 4\pi \left[\frac{\sqrt{3}}{2} - \log\left(1+\sqrt{3}\right) - \left(0 - \log\sqrt{2}\right) \right]$$

$$= 4\pi \left[\frac{\sqrt{3}}{2} + \log\sqrt{2} - \log\left(1+\sqrt{3}\right) \right]$$

24. **24.** Evaluate $\iint_A [1/(x^2+y^2)^2]\, dx\, dy$, where A is determined by the conditions $x^2+y^2 \le 1$ and $x+y \ge 1$.



Use a polar conversion

$0 \le \theta \le \frac{\pi}{2}$ $\quad r^2 = x^2+y^2$ $\quad x = r\cos\theta$ $\quad y = r\sin\theta$

$x+y \ge 1 \Rightarrow r(\cos\theta + \sin\theta) \ge 1$

$\therefore r \ge \frac{1}{\cos\theta+\sin\theta}$, and $\cos\theta+\sin\theta \ne 0$ for $0 \le \theta \le \frac{\pi}{2}$

$x^2+y^2 \le 1 \Rightarrow r^2 \le 1$, or $r \le 1$.

$$\therefore \iint_A \frac{1}{(x^2+y^2)^2}\,dx\,dy = \iint_{A^*} \frac{r\,dr\,d\theta}{r^4}$$

$$= \int_0^{\frac{\pi}{2}} \int_{\frac{1}{\cos\theta+\sin\theta}}^{1} \frac{1}{r^3}\,dr\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{r^{-2}}{-2} \Big|_{r=\frac{1}{\cos\theta+\sin\theta}}^{r=1} \right) d\theta$$

$$= -\frac{1}{2} \int_0^{\frac{\pi}{2}} 1 - (\cos\theta+\sin\theta)^2\,d\theta$$

$$= -\frac{1}{2} \int_0^{\frac{\pi}{2}} -2\cos\theta\sin\theta\,d\theta = \frac{1}{2}\int_0^{\frac{\pi}{2}} \sin 2\theta\,d\theta$$

$$= -\frac{1}{4}\cos 2\theta \Big|_0^{\frac{\pi}{2}} = \frac{1}{4} - \left(-\frac{1}{4}\right) = \underline{\frac{1}{2}}$$

25.

25. Evaluate $\iiint_W \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^{3/2}}$, where W is the solid bounded by the two spheres $x^2+y^2+z^2=a^2$ and $x^2+y^2+z^2=b^2$, where $0<b<a$.

Use spherical coordinates.

$x^2+y^2+z^2 = \rho^2$ $\qquad z = \rho\cos\phi$

$$x = \rho \sin\phi \cos\theta \qquad y = \rho \sin\phi \sin\theta$$

$$\iiint_W \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^{3/2}} = \iiint_{W^*} \frac{\rho^2 \sin\phi}{(\rho^2)^{3/2}}\, d\rho\, d\theta\, d\phi$$

$$= \int_0^{\pi} \int_0^{2\pi} \int_b^a \frac{\sin\phi}{\rho}\, d\rho\, d\theta\, d\phi$$

$$= \int_0^{\pi} \int_0^{2\pi} \left(\sin\phi \log\rho \Big|_{\rho=b}^{\rho=a} \right) d\theta\, d\phi$$

$$= \int_0^{\pi} \int_0^{2\pi} \sin\phi \log\left(\frac{a}{b}\right) d\theta\, d\phi = \int_0^{\pi} 2\pi \log\left(\frac{a}{b}\right) \sin\phi\, d\phi$$

$$= -2\pi \log\left(\frac{a}{b}\right) \cos\phi \Big|_{\phi=0}^{\phi=\pi} = 2\pi \log\left(\frac{a}{b}\right) \left[1 - (-1)\right]$$

$$= \underline{4\pi \log\left(\frac{a}{b}\right)}$$

26. **26.** Use spherical coordinates to evaluate:

$$\int_0^3 \int_0^{\sqrt{9-x^2}} \int_0^{\sqrt{9-x^2-y^2}} \frac{\sqrt{x^2+y^2+z^2}}{1+[x^2+y^2+z^2]^2}\, dz\, dy\, dx$$

$$x^2 + y^2 + z^2 = \rho^2$$

For the integration limits, the original region is a sphere, centered at $(0,0,0)$, of radius 3. But with $0 \le x \le 3$, $0 \le y$, and $0 \le z$, so sphere in quadrant I.

This is analogous to $0 \le \rho \le 3$, $0 \le \phi \le \frac{\pi}{2}$, and $0 \le \theta \le \pi/2$.

Jacobian is $\rho^2 \sin\phi$

$$\therefore \iiint_{R^*} \frac{\rho}{1+(\rho^2)^2} \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi =$$

$$\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \int_0^3 \frac{\rho^3 \sin\phi}{1+\rho^4} \, d\rho \, d\theta \, d\phi$$

$$= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \left(\sin\phi \, \frac{1}{4} \log(1+\rho^4) \Big|_{\rho=0}^{\rho=3} \right) d\theta \, d\phi$$

$$= \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \sin\phi \, \frac{\log(82)}{4} \, d\theta \, d\phi$$

$$= \int_0^{\frac{\pi}{2}} \frac{\pi \log(82)}{8} \sin\phi \, d\phi = \frac{\pi \log(82)}{8} (-\cos\phi) \Big|_{\phi=0}^{\phi=\frac{\pi}{2}}$$

$$= \frac{\pi \log(82)}{8}$$

27.

27. Let D be a triangle in the (x, y) plane with vertices $(0, 0)$, $(\frac{1}{2}, \frac{1}{2})$, $(1, 0)$. Evaluate:

$$\iint_D \cos \pi \left(\frac{x-y}{x+y} \right) dx\, dy$$

by making the appropriate change of variables.

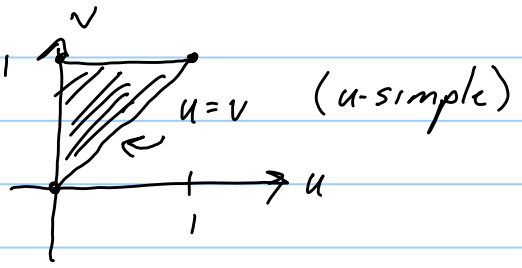
Let $u = x - y$, $v = x + y$ $\therefore \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$

$\therefore T = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$, $\det(T) = 2$, $T^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

$\therefore \begin{bmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$ $\det(T^{-1}) = \frac{1}{2}$

$\therefore \iint_{D^*} \cos \pi \left(\frac{u}{v} \right) \left(\frac{1}{2} \right) du\, dv$

D^* : $T(0,0) = (0,0)$
$T(\frac{1}{2}, \frac{1}{2}) = (0,1)$
$T(1,0) = (1,1)$



$\therefore 0 \le v \le 1 \quad 0 \le u \le v$

$\therefore \int_0^1 \int_0^v \cos \pi \left(\frac{u}{v} \right) \left(\frac{1}{2} \right) du\, dv$

$$\int_0^V \frac{1}{2}\cos\left(\frac{\pi u}{V}\right) du = \frac{V}{2\pi}\sin\left(\frac{\pi u}{V}\right)\Big|_{u=0}^{u=V}$$

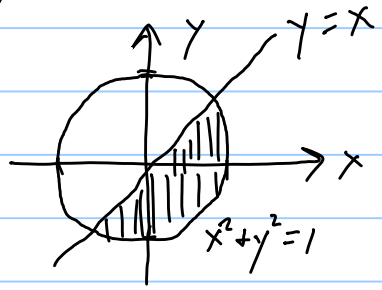
$$= 0 - 0 = 0$$

$$\therefore \int_0^1 0 \, dv = \underline{0}.$$

$$\therefore \iint_D \cos\pi\left(\frac{x-y}{x+y}\right) dx\,dy = \underline{\underline{0}}$$

Note: Integrating $\iint_{D^*}$ using u-simple regions is much easier (because of 0) than integrating using v-simple regions.

28. **28.** Evaluate $\iint_D x^2\,dx\,dy$, where D is determined by the two conditions $0 \le x \le y$ and $x^2 + y^2 \le 1$.



A polar conversion using $0 \le r \le 1$ and $-\frac{3}{4}\pi \le \theta \le \frac{\pi}{4}$ covers D.

$$\therefore \iint_D x^2\,dx\,dy = \iint_{D^*} (r\cos\theta)^2\, r\,dr\,d\theta$$

$$= \int_{-\frac{3}{4}\pi}^{\frac{\pi}{4}} \int_0^1 r^3 \cos^2\theta \, dr\, d\theta = \int_{-\frac{3}{4}\pi}^{\frac{\pi}{4}} \left(\frac{r^4}{4} \cos^2\theta \Big|_{r=0}^{r=1} \right) d\theta$$

$$= \int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}} \cos^2\theta \, d\theta = = \int_{-\frac{3\pi}{4}}^{\frac{\pi}{4}} \frac{1+\cos 2\theta}{2} \, d\theta$$

$$= \frac{\theta}{2} + \frac{\sin 2\theta}{4} \Big|_{\theta = -\frac{3\pi}{4}}^{\theta = \frac{\pi}{4}}$$

$$= \frac{\pi}{8} + \frac{1}{4} - \left(-\frac{3\pi}{8} + \frac{1}{4} \right) = \frac{\pi}{2}$$

29.

29. Integrate $\sqrt{x^2 + y^2 + z^2}\, e^{-(x^2+y^2+z^2)}$ over the region described in Exercise 25.

solid bounded by the two spheres $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 + z^2 = b^2$, where $0 < b < a$.

Use spherical coordinates: $\rho^2 = x^2 + y^2 + z^2$

Here, $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi$

$$\therefore \int_0^{2\pi} \int_0^{\pi} \int_b^a \rho\, e^{-\rho^2} (\rho^2 \sin\phi) \, d\rho \, d\phi \, d\theta$$

Look at $\int_b^a \rho^3 e^{-\rho^2} d\rho$ Let $u = \rho^2$. $\therefore du = 2\rho\, d\rho$

When $\rho = b$, $u = b^2$, when $\rho = a$, $u = a^2$

$$\therefore \int_b^a \rho^3 e^{-\rho^2} d\rho = \frac{1}{2}\int_{b^2}^{a^2} u e^{-u} du \qquad \text{integrate by parts}$$

$$= -\frac{1}{2} u e^{-u}\Big|_{b^2}^{a^2} + \frac{1}{2}\int_{b^2}^{a^2} e^{-u} du$$

$$= -\frac{1}{2} a^2 e^{-a^2} + \frac{1}{2} b^2 e^{-b^2} - \frac{1}{2} e^{-u}\Big|_{b^2}^{a^2}$$

$$= -\frac{1}{2} e^{-a^2}(a^2+1) + \frac{1}{2} e^{-b^2}(b^2+1)$$

$$\int_0^{\pi} \sin\phi \, d\phi = -\cos\phi\Big|_0^{\pi} = 2$$

$$\int_0^{2\pi} d\theta = 2\pi$$

$$\therefore \; -2\pi e^{-a^2}(a^2+1) + 2\pi e^{-b^2}(b^2+1)$$

30. **30.** Evaluate the following by using cylindrical coordinates.

(a) $\iiint_B z\,dx\,dy\,dz$, where B is the region within the cylinder $x^2 + y^2 = 1$ above the xy plane and below the cone $z = (x^2 + y^2)^{1/2}$

(a) since $r^2 = x^2 + y^2 = 1$, $0 \le r \le 1$

$0 \le \theta \le 2\pi$ $\quad 0 \le z \le (r^2)^{\frac{1}{2}} = r$

$$\therefore \iiint_{B^*} (z)\, r\,dr\,d\theta\,dz = \int_0^{2\pi}\int_0^1\int_0^r zr\,dz\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^1 \left(\frac{z^2}{2}\Big|_{z=0}^{z=r}\right) r\,dr\,d\theta = \int_0^{2\pi}\int_0^1 \frac{r^3}{2}\,dr\,d\theta$$

$$= \int_0^{2\pi} \frac{r^4}{8}\Big|_{r=0}^{r=1} d\theta = \int_0^{2\pi} \frac{1}{8}\,d\theta = \frac{2\pi}{8} = \frac{\pi}{4}$$

(6) (b) $\iiint_W (x^2 + y^2 + z^2)^{-1/2}\,dx\,dy\,dz$, where W is the region determined by the conditions $\frac{1}{2} \le z \le 1$ and $x^2 + y^2 + z^2 \le 1$

$x^2 + y^2 = r^2$, $\quad x^2 + y^2 + z^2 \le 1 \Rightarrow r^2 + z^2 \le 1$

$\Rightarrow r \le \sqrt{1 - z^2}$

$\therefore 0 \le r \le \sqrt{1 - z^2}$

remember: r is in a plane parallel to xy-plane.

$0 \le \theta \le 2\pi$

$$\therefore \iiint_{W^*} \frac{1}{\sqrt{r^2+z^2}}\, r\, dr\, d\theta\, dz = \int_0^{2\pi}\int_{\frac{1}{2}}^{1}\int_0^{\sqrt{1-z^2}} \frac{r}{\sqrt{r^2+z^2}}\, dr\, dz\, d\theta$$

$$= \int_0^{2\pi}\int_{\frac{1}{2}}^{1} \left(\sqrt{r^2+z^2}\,\Big|_{r=0}^{r=\sqrt{1-z^2}}\right) dz\, d\theta$$

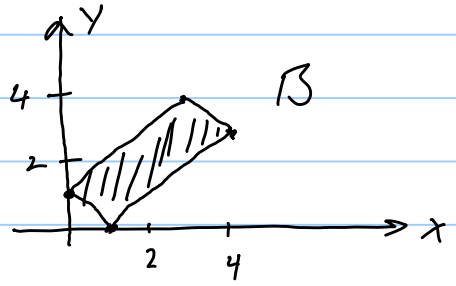
$$= \int_0^{2\pi}\int_{\frac{1}{2}}^{1} (1-z)\, dz\, d\theta = \int_0^{2\pi}\left(z-\frac{z^2}{2}\,\Big|_{z=\frac{1}{2}}^{z=1}\right) d\theta$$

$$= \int_0^{2\pi} 1-\frac{1}{2}-\left(\frac{1}{2}-\frac{1}{8}\right) d\theta = \int_0^{2\pi} \frac{1}{8}\, d\theta$$

$$= \frac{2\pi}{8} = \frac{\pi}{4}$$

31.

31. Evaluate $\iint_B (x+y)\, dx\, dy$, where B is the rectangle in the xy plane with vertices at (0, 1), (1, 0), (3, 4), and (4, 3).



Note the lines of two sides of B have form $y = x + k_1$, two sides have $y = -x + k_2$ or $x + y = k_2$, $x - y = -k_2$.

$x+y$ is in the integrand, $\therefore$ make a change of variable of form $u = x+y$, $v = x-y$.

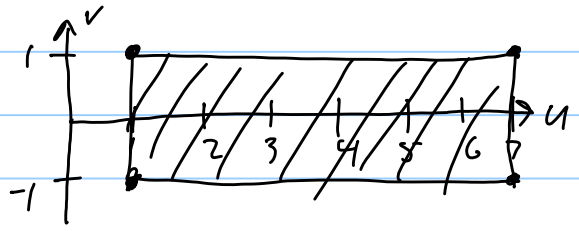
$\therefore T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, so $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+y \\ x-y \end{bmatrix}$

and $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$, where $\begin{bmatrix} u \\ v \end{bmatrix}$ = new corner for B^*

The new corners determine new limits of integration.

$$\therefore \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 4 & 3 \\ 1 & 0 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 7 & 7 \\ -1 & 1 & 1 & -1 \end{bmatrix}$$

$\therefore$



$$\det(T) = -2$$

$$\therefore \det(T^{-1}) = -\frac{1}{2}$$

If is $T^{-1} : (u,v) \to (x,y)$, or $T^{-1} : B^* \to B$

$$T^{-1} = -\frac{1}{2}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix}$$

$$\therefore \left| \frac{\partial(x,y)}{\partial(u,v)} \right| = \left| -\frac{1}{4} - \frac{1}{4} \right| = \frac{1}{2}$$

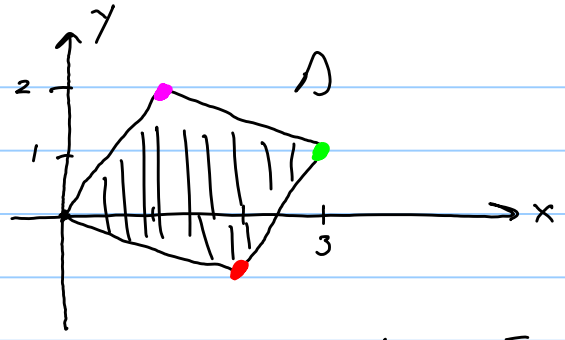
$$\therefore \iint_B (x+y)\,dx\,dy = \iint_{B^*} u\left(\frac{1}{2}\right) du\,dv$$

using $u = x+y$, $v = x-y$

$$= \int_{-1}^{1}\int_{1}^{7} \frac{u}{2}\,du\,dv = \int_{-1}^{1} \frac{u^2}{4}\Big|_{u=1}^{u=7} dv$$

$$= \int_{-1}^{1} \left(\frac{49}{4} - \frac{1}{4}\right) dv = 12v\Big|_{-1}^{1} = \underline{24}$$

32. **32.** Evaluate $\iint_D (x+y)\,dx\,dy$, where D is the square with vertices at $(0, 0)$, $(1, 2)$, $(3, 1)$, and $(2, -1)$.



Border lines are: $y = 2x$, $y = 2x - 5$

$y = -\frac{1}{2}x$, $y = -\frac{1}{2}x + \frac{5}{2}$

or $y - 2x = 0$, $y - 2x = -5$; $2y + x = 0$, $2y + x = 5$

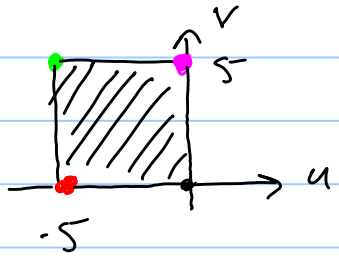
If $u = y - 2x$, $v = 2y + x$ Then $u + 2v = 5y$, $v - 2u = 5x$ $\Rightarrow$ $y = \frac{u+2v}{5}$ [1], $x = \frac{v-2u}{5}$ [2]

$\therefore \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$ $\therefore 3v - u = 5y + 5x$, $x + y = \frac{3v-u}{5}$

Look at corners for new limits of integration (uv-plane)

$$\begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 2 & 3 & 1 \\ 0 & -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & -5 & -5 & 0 \\ 0 & 0 & 5 & 5 \end{bmatrix}$$

$\therefore D^*$



From [1], [2], $\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} -2/5 & 1/5 \\ 1/5 & 2/5 \end{vmatrix} = \frac{1}{5}$

$$\therefore \iint_D (x+y)\,dx\,dy = \iint_{D^*} \left(\frac{3v-u}{5}\right)\left(\frac{1}{5}\right) du\,dv$$

$$= \int_0^5 \int_{-5}^{0} \frac{3v-u}{25}\, du\, dv = \int_0^5 \left(\frac{3vu}{25} - \frac{u^2}{50}\Big|_{u=-5}^{u=0} \right) dv$$

$$= \int_0^5 0 - \left(-\frac{3v}{5} - \frac{1}{2}\right) dv = \int_0^5 \frac{3v}{5} + \frac{1}{2}\, dv$$

$$= \frac{3v^2}{10} + \frac{v}{2}\Big|_0^5 = \frac{75}{10} + \frac{5}{2} - 0 = \underline{10}$$

Note: The strategy above is to look at the sides of D

(1) Assign u = side 1, v = side 2 ($ax+by$ form)

(2) Solve x, y in terms of u, v; i.e., get T^{-1}.

$$\therefore T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}, \quad T^{-1}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

(3) Convert $f(x,y)$ to $f(x(u,v), y(u,v))$

(4) Jacobian = $|\det(T^{-1})|$

(5) $T\begin{bmatrix} x \\ y \end{bmatrix}$ gives new corners $\begin{bmatrix} u \\ v \end{bmatrix}$ from old corners $\begin{bmatrix} x \\ y \end{bmatrix}$. New corners $\begin{bmatrix} u \\ v \end{bmatrix}$ yield new limits of integration for $\iint_{D^*}$

Method 2: Use given corners and assign new "rectangular" corners, respecting orientation.

(1) $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$, $\begin{bmatrix} x \\ y \end{bmatrix}$ = old corners,

$\begin{bmatrix} u \\ v \end{bmatrix}$ = new, assigned corners,

$T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, and solve for $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$

(2) Find T^{-1}, Jacobian is $|\det(T^{-1})|$

$= \left|\frac{1}{\det(T)}\right|$

(3) $T^{-1}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$, so now have

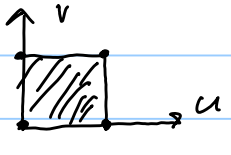
$x(u,v)$, $y(u,v)$ $\left[x = n_1 u + n_2 v,\ y = m_1 u + m_2 v\right]$

(4) Convert $f(x,y)$ into $f(x(u,v), y(u,v))$

(5) $\iint_D f(x,y)\,dx\,dy = \int_{u_1}^{u_2}\int_{v_1}^{v_2} f(u,v)\left|\det(T^{-1})\right| du\,dv$

In the problem above, make new corners

$(0,0)$, $(1,0)$, $(1,1)$, $(0,1)$



(1) $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 0 & 2 & 3 & 1 \\ 0 & -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

Taking transposes, $\begin{bmatrix} 0 & 0 \\ 2 & -1 \\ 3 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$

$$\therefore \begin{bmatrix} 0 & 0 \\ 0 & -5 \\ 0 & -5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & -2 \\ 1 & -2 \\ 0 & 1 \end{bmatrix} \quad \begin{array}{l} -5b = 1, \ b = -\frac{1}{5} \\ a + 2b = 0, \ a = \frac{2}{5} \\ -5d = -2, \ d = \frac{2}{5} \\ c + 2d = 1, \ c = \frac{1}{5} \end{array}$$

$$\therefore T = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} \frac{2}{5} & -\frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} \end{bmatrix} \quad \det(T) = \frac{1}{5}$$

(2) $$T^{-1} = 5 \begin{bmatrix} \frac{2}{5} & \frac{1}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \quad \det(T^{-1}) = 5$$

(3) $$T^{-1} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 2u + v \\ -u + 2v \end{bmatrix}$$

$$\therefore \begin{array}{l} x = 2u + v \\ y = -u + 2v \end{array}$$

(4) $$f(x,y) = x + y = (2u + v) + (-u + 2v) = u + 3v$$

(5) $$\therefore \iint_D x + y \, dx\,dy = \iint_{D^*} (u + 3v)(5) \, du\,dv$$

$$= \int_0^1 \int_0^1 (5u + 15v) \, du\,dv = \int_0^1 5\frac{u^2}{2} + 15uv \Big|_{u=0}^{u=1} dv$$

$$= \int_0^1 \left(\frac{5}{2} + 15v\right) dv = \frac{5}{2}v + \frac{15v^2}{2} \Big|_0^1 = \underline{10}$$

33.

33. Let E be the ellipsoid $(x^2/a^2) + (y^2/b^2) + (z^2/c^2) \le 1$, where a, b, and c are positive.

(a) Find the volume of E.

(b) Evaluate

$$\iiint_E [(x^2/a^2) + (y^2/b^2) + (z^2/c^2)]\, dx\, dy\, dz.$$

(HINT: Change variables and then use spherical coordinates.)

(a)

$$\text{Vol } E = \iiint_D dx\,dy\,dz, \quad D = \text{ellipsoid}$$

$$= \int_{-a}^{a} \int_{-b\sqrt{1-x^2/a^2}}^{b\sqrt{1-x^2/a^2}} \int_{-c\sqrt{1-x^2/a^2-y^2/b^2}}^{c\sqrt{1-x^2/a^2-y^2/b^2}} dz\,dy\,dx$$

Let $x = au$, $y = bv$, $z = cw$ (assume $a, b, c > 0$).

$$\therefore \text{Jacobian} = \left| \frac{\partial(x,y,z)}{\partial(u,v,w)} \right| = \begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix} = abc$$

$$\therefore \iiint_D dx\,dy\,dz = \iiint_{D^*} abc\, du\,dv\,dw$$

and D^* becomes a unit ball.

Now use a spherical coordinate transformation.

$$\therefore \iiint_{D^*} abc\, du\,dv\,dw = \iiint_{D^{**}} abc\, \rho^2 \sin\phi\, d\rho\, d\theta\, d\phi$$

As D^{**} is a unit ball, limits of integration are $0 \le \rho \le 1$, $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi$.

$$= abc \int_0^{\pi} \int_0^{2\pi} \int_0^1 \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi$$

$$= abc \int_0^{\pi} \int_0^{2\pi} \left(\frac{\rho^3}{3} \Big|_{\rho=0}^{\rho=1} \right) \sin\phi \, d\theta \, d\phi$$

$$= \frac{abc}{3} \int_0^{\pi} \int_0^{2\pi} \sin\phi \, d\theta \, d\phi$$

$$= \frac{2\pi abc}{3} \int_0^{\pi} \sin\phi \, d\phi = \frac{2\pi abc}{3} \left[-\cos\phi \right]_0^{\pi}$$

$$= \frac{4\pi}{3} abc$$

(b) Using the above substitution,

$$\iiint_D \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) dx\,dy\,dz =$$

$$\iiint_{D^*} (u^2 + v^2 + w^2)(abc) \, du\,dv\,dw =$$

$$abc\int_0^{\pi}\int_0^{2\pi}\int_0^{1} \rho^2\,(\rho^2\sin\phi)\,d\rho\,d\theta\,d\phi$$

$$= abc\int_0^{\pi}\int_0^{2\pi}\left(\sin\phi\,\frac{\rho^5}{5}\Big|_{\rho=0}^{\rho=1}\right)d\theta\,d\phi$$

$$= \frac{abc}{5}\int_0^{\pi}\int_0^{2\pi}\sin\phi\,d\theta\,d\phi = \frac{2\pi}{5}abc\int_0^{\pi}\sin\phi\,d\phi$$

$$= \frac{2\pi\,abc}{5}\Big[-\cos\phi\Big]\Big|_0^{\pi} = \frac{4\pi\,abc}{5}$$

34. **34.** Using spherical coordinates, compute the integral of $f(\rho, \phi, \theta) = 1/\rho$ over the region in the first octant of $\mathbb{R}^3$, which is bounded by the cones $\phi = \pi/4$, $\phi = \arctan 2$ and the sphere $\rho = \sqrt{6}$.

$0 \le \rho \le \sqrt{6}$, $0 \le \theta \le \frac{\pi}{2}$, $\frac{\pi}{4} \le \phi \le \arctan 2$

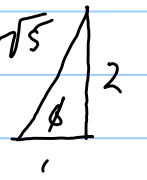
Note: we are changing variables here from xyz (problem states $\mathbb{R}^3$ 1st octant).

$$\therefore \int_{\frac{\pi}{4}}^{\arctan 2}\int_0^{\frac{\pi}{2}}\int_0^{\sqrt{6}} \left(\frac{1}{\rho}\right)(\rho^2\sin\phi)\,d\rho\,d\theta\,d\phi$$

$$= \int_{\frac{\pi}{4}}^{\arctan 2} \int_{0}^{\frac{\pi}{2}} \left(\sin\phi \, \frac{\rho^2}{2} \Big|_{\rho=0}^{\rho=\sqrt{6}} \right) d\theta \, d\phi$$

$$= \int_{\frac{\pi}{4}}^{\arctan 2} \int_{0}^{\frac{\pi}{2}} 3\sin\phi \, d\theta \, d\phi = \int_{\frac{\pi}{4}}^{\arctan 2} \frac{3\pi}{2} \sin\phi \, d\phi$$

$$= -\frac{3\pi}{2} \cos\phi \Big|_{\phi=\frac{\pi}{4}}^{\phi=\arctan 2}$$



$\phi = \arctan 2$

$\cos\phi = \frac{1}{\sqrt{5}}$

$$= -\frac{3\pi}{2} \cos(\arctan 2) - \left(-\frac{3\pi}{2} \frac{\sqrt{2}}{2} \right)$$

$$= -\frac{3\pi}{2} \left(\frac{\sqrt{5}}{5} \right) + \frac{3\sqrt{2}}{4}\pi = 3\pi \left(\frac{\sqrt{2}}{4} - \frac{\sqrt{5}}{10} \right)$$

35.

35. The mapping $T(u, v) = (u^2 - v^2, 2uv)$ transforms the rectangle $1 \le u \le 2$, $1 \le v \le 3$ of the uv plane into a region R of the xy plane.

(a) Show that T is one-to-one.

(b) Find the area of R using the change of variables formula.

(a) Let $T(u,v) = T(x,y) = (a,b)$ where $1 \le u \le 2, 1 \le v \le 3$
$1 \le x \le 2, 1 \le y \le 3$

$\therefore u^2 - v^2 = a$

$2uv = b$

(1) $\therefore v = \frac{6}{2u}$ as $u \neq 0$, $\therefore u^2 - \left(\frac{6}{2u}\right)^2 = a$

or $u^2 - \frac{6^2}{4u^2} = a$, $4u^4 - 4au^2 - 6^2 = 0$

$$\therefore u^2 = \frac{4a \pm \sqrt{16a^2 - 4(4)(-6^2)}}{8}$$

$$= \frac{4a \pm 4\sqrt{a^2+6^2}}{8} = \frac{a \pm \sqrt{a^2+6^2}}{2}$$

Since $u^2 > 0$ for $1 \leq u \leq 2$, and $\sqrt{a^2+6^2} > a^2$

reject the root $\frac{a - \sqrt{a^2+6^2}}{2}$

$$\therefore u^2 = \frac{a+\sqrt{a^2+6^2}}{2}, \quad u = \sqrt{\frac{a+\sqrt{a^2+6^2}}{2}}$$

(2) Similarly, $u = \frac{6}{2v}$ as $v \neq 0$ for $1 \leq v \leq 3$

$\therefore \left(\frac{6}{2v}\right)^2 - v^2 = a$, $6^2 - 4v^4 = 4av^2$

Or, $4v^4 + 4av^2 - 6^2 = 0$

$$\therefore v^2 = \frac{-4a \pm \sqrt{16a^2 - 4(4)(-6^2)}}{8}$$

$$= \frac{-a \pm \sqrt{a^2+6^2}}{2}$$

Since $\sqrt{a^2+b^2} > a > 0$, reject $\frac{-a-\sqrt{a^2+b^2}}{2}$

as $v^2 > 0$ for $1 \le v \le 3$

$\therefore v^2 = \frac{-a+\sqrt{a^2+b^2}}{2}$, $v = \sqrt{\frac{-a+\sqrt{a^2+b^2}}{2}}$

$\therefore T(u,v) = (a,b) \Rightarrow u = \sqrt{\frac{a+\sqrt{a^2+b^2}}{2}}$, $v = \sqrt{\frac{-a+\sqrt{a^2+b^2}}{2}}$

Similarly, $T(x,y) = (a,b) \Rightarrow x = \sqrt{\frac{a+\sqrt{a^2+b^2}}{2}}$, $y = \sqrt{\frac{-a+\sqrt{a^2+b^2}}{2}}$

$\therefore T(u,v) = T(x,y) \Rightarrow u = x$ and $v = y$.

$\therefore T$ is one-to-one

(6) (b) Find the area of R using the change of variables formula.

$$\iint_D dx\,dy = \iint_{D^*} \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv$$

where $D^* = \{(u,v) : 1 \le u \le 2,\ 1 \le v \le 3\}$

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} 2u & -2v \\ 2v & 2u \end{vmatrix} = 4u^2 + 4v^2$$

$$\therefore \iint_{D^*} \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv = \int_1^3 \int_1^2 (4u^2 + 4v^2)\, du\,dv$$

$$= \int_1^3 \left(\frac{4}{3}u^3 + 4uv^2 \Big|_{u=1}^{u=2} \right) dv$$

$$= \int_1^3 \frac{4}{3}(8) + 8v^2 - \frac{4}{3} - 4v^2\, dv$$

$$= \int_1^3 \left(4v^2 + \frac{28}{3} \right) dv = \frac{4}{3}v^3 + \frac{28}{3}v \Big|_1^3$$

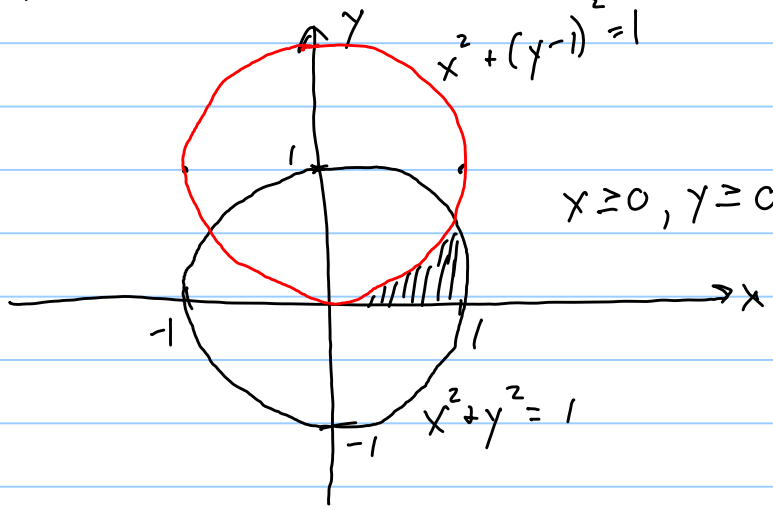
$$= 36 + 28 - \left(\frac{4}{3} + \frac{28}{3} \right) = 64 - \frac{32}{3} = \frac{192}{3} - \frac{32}{3} = \frac{160}{3}$$

36.

36. Let R denote the region inside $x^2 + y^2 = 1$, but outside $x^2 + y^2 = 2y$ with $x \geq 0$, $y \geq 0$.

(a) Sketch this region.

(a) $x^2 + y^2 = 2y \iff x^2 + (y-1)^2 = 1$



(6) (b) Let $u = x^2 + y^2$, $v = x^2 + y^2 - 2y$. Sketch the region D in the uv plane, which corresponds to R under this change of coordinates.

Since $0 \le x^2+y^2 \le 1$, $0 \le u \le 1$

For $x^2 + (y-1)^2 \ge 1$, $v+1 \ge 1$, or $v \ge 0$

$T(x,y) = (x^2+y^2,\ x^2+y^2-2y) = (u,v)$

What is T^{-1}?

$x^2+y^2 = u$ so $x^2+y^2-2y = u-2y = v \Rightarrow y = \frac{u-v}{2}$

$\therefore x^2 + \left(\frac{u-v}{2}\right)^2 = u$, $4x^2 = 4u - (u-v)^2$

$= 4u - u^2 + 2uv - v^2$

$\therefore x^2 = u - \frac{u^2}{4} + \frac{uv}{2} - \frac{v^2}{4}$

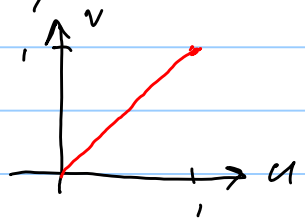
$\therefore T^{-1}(u,v) = \left(u + \frac{uv}{2} - \frac{u^2}{4} - \frac{v^2}{4},\ \frac{u-v}{2}\right)$

Not very helpful, except for $\left|\frac{\partial(x,y)}{\partial(u,v)}\right|$

$\therefore$ Look at borders in x-y plane and graph borders in uv-plane using $T(x,y)$

With $(x,y) = (x,0)$, $T(x,0) = (x^2, x^2)$, $0 \le x \le 1$

$\therefore$

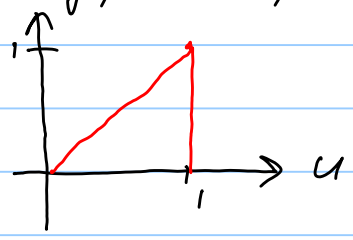


With $x^2+y^2=1$, $T(x,y)=(1,1-2y)$, $0\le y$

But since $x^2+y^2-2y\ge 0$, $1-2y\ge 0$, $y\le \frac{1}{2}$

$\therefore\ 0\le y\le \frac{1}{2}$, $0\ge -2y\ge -1$, $0\le 1-2y\le 1$

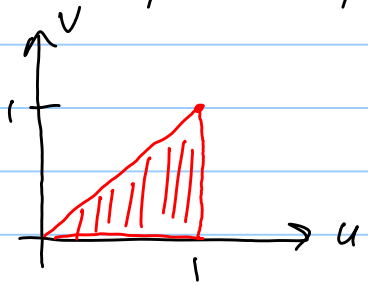
$\therefore$



The top border in xy-plane is $x^2+y^2-2y=0$, with $0\le y\le \frac{1}{2}$, so $x^2+y^2\le 1 \Rightarrow x^2\le \frac{3}{4}$, $0\le x\le \sqrt{3}/2$.

$\therefore\ T(x,y)=(2y,0)$, so $0\le 2y\le 1$.

$\therefore$



$\therefore\ 0\le u\le 1$, $0\le v\le u$

$$J=\left|\frac{\partial(x,y)}{\partial(u,v)}\right| \quad \text{with} \quad x=u+\frac{uv}{2}-\frac{u^2}{4}-\frac{v^2}{4}, \quad y=\frac{u-v}{2}$$

$$=\begin{vmatrix} 1+\frac{v}{2}-\frac{u}{2} & \frac{u}{2}-\frac{v}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{vmatrix}$$

$$=\left|-\frac{1}{2}-\frac{v}{4}+\frac{u}{4}-\frac{u}{4}+\frac{v}{4}\right|=\frac{1}{2}$$

$$\therefore \iint_R x e^y \, dx\,dy = \iint_D \left(u + \frac{uv}{2} - \frac{u^2}{4} - \frac{v^2}{4}\right) e^{\frac{u-v}{2}} \left(\frac{1}{2}\right) du\,dv$$

$$= \int_0^1 \int_0^u \left(\frac{u}{2} + \frac{uv}{4} - \frac{u^2}{8} - \frac{v^2}{8}\right) e^{\frac{u-v}{2}} \, dv\,du$$

$$\int_0^u \left(\frac{u}{2} + \frac{uv}{4} - \frac{u^2}{8} - \frac{v^2}{8}\right) e^{\frac{u}{2}} e^{-\frac{v}{2}} \, dv$$

$$= \underbrace{\int_0^u \frac{u}{2} e^{\frac{u}{2}} e^{-\frac{v}{2}} dv}_{(1)} + \underbrace{\int_0^u e^{\frac{u}{2}} \frac{uv}{4} e^{-\frac{v}{2}} dv}_{(2)} - \underbrace{\int_0^u \frac{u^2}{8} e^{\frac{u}{2}} e^{-\frac{v}{2}} dv}_{(3)}$$

$$\underbrace{- \int_0^u e^{\frac{u}{2}} \frac{v^2}{8} e^{-\frac{v}{2}} dv}_{(4)}$$

$$= (-2) \frac{u}{2} e^{\frac{u}{2}} e^{-\frac{v}{2}} \Big|_{v=0}^{v=u} \qquad (1)$$

$$+ \frac{u e^{\frac{u}{2}}}{4} \left[\frac{v e^{-\frac{v}{2}}}{\left(-\frac{1}{2}\right)} \Big|_{v=0}^{v=u} \right] - \frac{u e^{\frac{u}{2}}}{4} \frac{1}{\left(-\frac{1}{2}\right)} \int_0^u e^{-\frac{v}{2}} dv \qquad (2)$$

$$- \left[\frac{u^2 e^{\frac{u}{2}}}{8} (-2) e^{-\frac{v}{2}} \Big|_{v=0}^{v=u} \right] \qquad (3)$$

$$-\frac{e^{\frac{u}{2}}}{8}\left[\frac{v^2 e^{-\frac{v}{2}}}{\left(-\frac{1}{2}\right)}\Bigg|_{v=0}^{v=u} - \frac{2}{\left(-\frac{1}{2}\right)}\int_0^u v e^{-\frac{v}{2}}\,dv\right] \qquad (4)$$

$$= \quad -u - \left(-u e^{\frac{u}{2}}\right) \qquad (1)$$

$$+\frac{u e^{\frac{u}{2}}}{4}\left\{-2u e^{-\frac{u}{2}}\right\} + \frac{u e^{\frac{u}{2}}}{2}\left[(-2)e^{-\frac{v}{2}}\Big|_{v=0}^{v=u}\right] \qquad (2)$$

$$-\left[\frac{u^2 e^{\frac{u}{2}} e^{-\frac{u}{2}}}{-4} - \left(\frac{u^2 e^{\frac{u}{2}}}{-4}\right)\right] \qquad (3)$$

$$-\frac{e^{\frac{u}{2}}}{8}\left[-2u^2 e^{-\frac{u}{2}}\right] + 4\left[\frac{v e^{-\frac{v}{2}}}{\left(-\frac{1}{2}\right)}\Bigg|_{v=0}^{v=u} - \frac{1}{\left(-\frac{1}{2}\right)}\int_0^u e^{-\frac{v}{2}}dv\right] \qquad (4)$$

$$= -u + u e^{\frac{u}{2}} - \frac{u^2}{2} + \frac{u e^{\frac{u}{2}}}{2}\left[-2e^{-\frac{u}{2}} - (-2)\right] \qquad (1)+(2)$$

$$+\frac{u^2}{4} - \frac{u^2 e^{\frac{u}{2}}}{4} \qquad (3)$$

$$+\frac{u^2}{4} - 8u e^{-\frac{u}{2}} + 2\left[(-2)e^{-\frac{v}{2}}\Big|_{v=0}^{v=u}\right] \qquad (4)$$

$$= -u + u e^{\frac{u}{2}} - \frac{u^2}{2} - u + u e^{\frac{u}{2}} \qquad (1)+(2)$$

$$+\frac{u^2}{4} - \frac{u^2 e^{\frac{u}{2}}}{4} \qquad (3)$$

$$+\frac{u^2}{4} - 8ue^{-\frac{u}{2}} + 2\left[-2e^{-\frac{u}{2}} - (-2)\right] \qquad (4)$$

$$= -2u + 2ue^{\frac{u}{2}} - \frac{u^2}{4} - \frac{u^2 e^{\frac{u}{2}}}{4} \qquad (1),(2),(3)$$

$$+\frac{u^2}{4} - 8ue^{-\frac{u}{2}} - 4e^{-\frac{u}{2}} + 4 \qquad (4)$$

$$= -2u + 2ue^{\frac{u}{2}} - \frac{u^2 e^{\frac{u}{2}}}{4} - 8ue^{-\frac{u}{2}} - 4e^{-\frac{u}{2}} + 4$$

$$= 4 - 2u - 4e^{-\frac{u}{2}} + 2ue^{\frac{u}{2}} - 8ue^{-\frac{u}{2}} - \frac{u^2 e^{\frac{u}{2}}}{4} \qquad (5)$$

$$\therefore \int_0^1 \overbrace{4 - 2u - 4e^{-\frac{u}{2}}}^{[6]} + \underbrace{2ue^{\frac{u}{2}}}_{[7]} - \underbrace{8ue^{-\frac{u}{2}}}_{[8]} - \underbrace{\frac{u^2 e^{\frac{u}{2}}}{4}}_{[9]} \, du$$

$$= \left(4u - u^2 + 8e^{-\frac{u}{2}}\right)\Big|_0^1 \qquad [6]$$

$$+ 2\left[\frac{ue^{\frac{u}{2}}}{\left(\frac{1}{2}\right)}\Big|_0^1 - \frac{1}{\left(\frac{1}{2}\right)}\int_0^1 e^{\frac{u}{2}}\,du\right] \qquad [7]$$

$$- 8\left[\frac{ue^{-\frac{u}{2}}}{\left(-\frac{1}{2}\right)}\Big|_0^1 - \frac{1}{\left(-\frac{1}{2}\right)}\int_0^1 e^{-\frac{u}{2}}\,du\right] \qquad [8]$$

$$- \frac{1}{4}\left[\frac{u^2 e^{\frac{u}{2}}}{\left(\frac{1}{2}\right)}\Big|_0^1 - \frac{2}{\left(\frac{1}{2}\right)}\int_0^1 ue^{\frac{u}{2}}\,du\right] \qquad [9]$$

$$= 4 - 1 + 8e^{-\frac{1}{2}} - (8) \qquad [6]$$

$$+ 2\left[2e^{\frac{1}{2}} - 2\left(2e^{\frac{u}{2}}\Big|_0^1\right)\right] \qquad [7]$$

$$- 8\left[-2e^{-\frac{1}{2}} + 2\left(-2e^{-\frac{u}{2}}\Big|_0^1\right)\right] \qquad [8]$$

$$- \frac{1}{4}\left[2e^{\frac{1}{2}} - 4\left[\frac{ue^{\frac{u}{2}}}{\left(\frac{1}{2}\right)}\Big|_0^1 - \frac{1}{\left(\frac{1}{2}\right)}\int_0^1 e^{\frac{u}{2}}\,du\right]\right] \qquad [9]$$

$$= -5 + 8e^{-\frac{1}{2}} + 4e^{\frac{1}{2}} - 4\left[2e^{\frac{1}{2}} - 2\right] \qquad [6]+[7]$$

$$+ 16e^{-\frac{1}{2}} - 16\left[-2e^{-\frac{1}{2}} - (-2)\right] \qquad [8]$$

$$- \frac{1}{2}e^{\frac{1}{2}} + \left[2e^{\frac{1}{2}} - 2\left(2e^{\frac{u}{2}}\Big|_0^1\right)\right] \qquad [9]$$

$$= -5 + 8e^{-\frac{1}{2}} + 4e^{\frac{1}{2}} - 8e^{\frac{1}{2}} + 8 \qquad [6]+[7]$$

$$+ 16e^{-\frac{1}{2}} + 32e^{-\frac{1}{2}} - 32 \qquad [8]$$

$$- \frac{1}{2}e^{\frac{1}{2}} + 2e^{\frac{1}{2}} - 2\left(2e^{\frac{1}{2}} - 2\right) \qquad [9]$$

$$= -29 + 56e^{-\frac{1}{2}} - 4e^{\frac{1}{2}} \qquad [6],[7],[8]$$

$$- \frac{1}{2}e^{\frac{1}{2}} + 2e^{\frac{1}{2}} - 4e^{\frac{1}{2}} + 4 \qquad [9]$$

$$= -25 + 56e^{-\frac{1}{2}} - \frac{13}{2}e^{\frac{1}{2}}$$

37.

37. Let D be the region bounded by $x^{3/2} + y^{3/2} = a^{3/2}$, for $x \geq 0$, $y \geq 0$, and the coordinate axes $x = 0$, $y = 0$.

Express $\iint_D f(x, y)\, dx\, dy$ as an integral over the triangle D^*, which is the set of points $0 \leq u \leq a$, $0 \leq v \leq a - u$. (Do not attempt to evaluate.)

Note for $x^{3/2} + y^{3/2} = a^{3/2}$ and $x, y \geq 0$, then $0 \leq x \leq a$ and $0 \leq y \leq a$.

The problem wants the transformation to be a triangle with sides a, not $a^{3/2}$

$\therefore$ If $u = a^{-\frac{1}{2}} x^{3/2}$, then $a^{\frac{1}{2}} u = x^{3/2}$, so $x = a^{\frac{1}{3}} u^{2/3}$

$\therefore$ When $0 \leq x \leq a$, at $x = a$, $u = a^{-\frac{1}{2}}(a)^{3/2} = a$

$\therefore$ $u = a^{-\frac{1}{2}} x^{3/2}$ and $v = a^{-\frac{1}{2}} y^{3/2}$ is the correct transformation so D^* is a triangle with sides a.

$\therefore$ $x = a^{\frac{1}{3}} u^{2/3}$, $y = a^{\frac{1}{3}} v^{2/3}$

$\therefore$ $\dfrac{\partial(x,y)}{\partial(u,v)} = \begin{bmatrix} \frac{2}{3} a^{\frac{1}{3}} u^{-\frac{1}{3}} & 0 \\ 0 & \frac{2}{3} a^{\frac{1}{3}} v^{-\frac{1}{3}} \end{bmatrix}$

$\therefore$ $J = \frac{4}{9} a^{2/3} u^{-\frac{1}{3}} v^{-\frac{1}{3}}$

$\therefore$ $\iint_D f(x,y)\, dx\, dy = \iint_{D^*} f\left(a^{\frac{1}{3}} u^{2/3}, a^{\frac{1}{3}} v^{2/3}\right) \left(\frac{4}{9} a^{2/3} u^{-\frac{1}{3}} v^{-\frac{1}{3}}\right) du\, dv$

$$= \frac{4}{9} a^{2/3} \iint_{D^*} f\left(a^{\frac{1}{3}} u^{2/3}, a^{\frac{1}{3}} v^{2/3}\right) u^{-\frac{1}{3}} v^{-\frac{1}{3}} \, du\, dv$$

38.

38. Show that $S(\rho, \theta, \phi) = (\rho \sin\phi \cos\theta,\ \rho \sin\phi \sin\theta,\ \rho \cos\phi)$, the spherical change-of-coordinate mapping, is one-to-one except on a set that is a union of finitely many graphs of continuous functions.

Let $S(\rho_1, \theta_1, \phi_1) = S(\rho_2, \theta_2, \phi_2) = (x, y, z)$ [1]

and assume $\rho_1 > 0$, $\rho_2 > 0$.

(a) $\therefore x^2 + y^2 + z^2 = (\rho_1^2 \sin^2\phi_1 \cos^2\theta_1) + (\rho_1^2 \sin^2\phi_1 \sin^2\theta_1) + (\rho_1^2 \cos^2\phi_1)$

$= \rho_1^2 \sin^2\phi_1 \left[\cos^2\theta_1 + \sin^2\theta_1\right] + \rho_1^2 \cos^2\phi_1$

$= \rho_1^2 \sin^2\phi_1 + \rho_1^2 \cos^2\phi_1 = \rho_1^2$

Similarly, $x^2 + y^2 + z^2 = \rho_2^2$. $\therefore \rho_1^2 = \rho_2^2 \Rightarrow \rho_1 = \rho_2$

(b) $\therefore$ [1] $\Rightarrow (\rho_1 \sin\phi_1 \cos\theta_1, \rho_1 \sin\phi_1 \sin\theta_1, \rho_1 \cos\phi_1) =$

$(\rho_2 \sin\phi_2 \cos\theta_2, \rho_2 \sin\phi_2 \sin\theta_2, \rho_2 \cos\phi_2)$

$\Rightarrow (\sin\phi_1 \cos\theta_1, \sin\phi_1 \sin\theta_1, \cos\phi_1) =$

$(\sin\phi_2 \cos\theta_2, \sin\phi_2 \sin\theta_2, \cos\phi_2)$

$\therefore \cos\phi_1 = \cos\phi_2$ But $0 \le \phi_1 \le \pi$, $0 \le \phi_2 \le \pi$

and on $0 \le w \le \pi$, $\cos(w)$ is one-to-one

$\therefore \cos\phi_1 = \cos\phi_2 \Rightarrow \phi_1 = \phi_2$

(c) $\therefore$ [1] $\Rightarrow \sin\phi\sin\theta_1 = \sin\phi\sin\theta_2$ (y-coordinates)

If $\phi \neq 0, \pi$, then $\sin\phi \neq 0$.

$\therefore \sin\theta_1 = \sin\theta_2$

Also, $\cos\theta_1 = \cos\theta_2$ (x-coordinates).

Over $0 \le \theta < 2\pi$, $\sin\theta_1 = \sin\theta_2 \Rightarrow$

$\theta_1 = \theta_2$ or $\pi - \theta_1 = \theta_2$ (and $\theta_1 \neq \theta_2$)

But $\pi - \theta_1 = \theta_2$ is not possible since

then $\cos(\pi - \theta_1) = \cos(\theta_2)$, and

$$\cos(\pi - \theta_1) = \cos(\pi)\cos(\theta_1) + \sin(\pi)\sin(\theta_1)$$
$$= -\cos(\theta_1)$$

$\therefore \pi - \theta_1 = \theta_2 \Rightarrow -\cos(\theta_1) = \cos(\theta_2)$

contradicting $\cos(\theta_1) = \cos(\theta_2)$

$\therefore$ Over $0 \le \theta < 2\pi$, [1] $\Rightarrow \theta_1 = \theta_2$

$\therefore$ [1] $\Rightarrow$ (a) $\rho_1 = \rho_2$ (assuming $\rho \neq 0$)

(b) $\phi_1 = \phi_2$ (assuming $0 < \phi < \pi$)

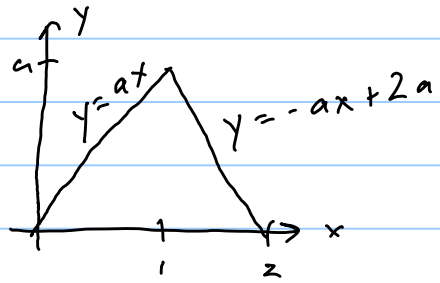
(c) $\theta_1 = \theta_2$ (assuming $0 \leq \theta < 2\pi$)

$\therefore$ $S(\rho, \theta, \phi)$ is one-to-one with the finite limitations mentioned above.

6.3 Applications

Note Title 1/2/2017

1. **1.** Find the coordinates of the center of mass of an isosceles triangle of uniform density bounded by the x axis, $y = ax$, and $y = -ax + 2a$.



$$\bar{x} = \frac{\iint_D x\,\delta\,dx\,dy}{\iint_D \delta\,dx\,dy} = \frac{\iint_D x\,dx\,dy}{\iint_D dx\,dy}$$

Similarly, $\bar{y} = \dfrac{\iint_D y\,dx\,dy}{\iint_D dx\,dy}$

Consider D to be x-simple.

$\therefore x = \frac{y}{a}$ and $x = \frac{2a-y}{a} = 2 - \frac{y}{a}$

$$\therefore \iint_D dx\,dy = \int_0^a \int_{\frac{y}{a}}^{2-\frac{y}{a}} dx\,dy = \int_0^a 2 - \frac{2y}{a}\,dy$$

$$= 2y - \frac{y^2}{a}\Big|_{y=0}^{a} = 2a - a = a$$

consistent with (area of triangle)(density) = $(\frac{1}{2}\,2a)\delta = \delta a$.

$$\therefore \bar{x} = \frac{1}{a}\int_0^a \int_{\frac{y}{a}}^{2-\frac{y}{a}} x\,dx\,dy = \frac{1}{a}\int_0^a \left(\frac{x^2}{2}\Big|_{x=\frac{y}{a}}^{x=2-\frac{y}{a}}\right) dy$$

$$= \frac{1}{a}\int_0^a \frac{1}{2}\left[\left(2-\frac{y}{a}\right)^2-\left(\frac{y}{a}\right)^2\right]dy$$

$$= \frac{1}{a}\int_0^a \frac{1}{2}\left[4-\frac{4y}{a}\right]dy = \frac{1}{2a}\left[4y-\frac{2y^2}{a}\right]_{y=0}^{y=a}$$

$$= \frac{1}{2a}\left[4a-2a\right] = \underline{1}$$

$$\bar{y} = \frac{1}{a}\int_0^a \int_{\frac{y}{a}}^{2-\frac{y}{a}} y\,dx\,dy = \frac{1}{a}\int_0^a y\left(2-\frac{2y}{a}\right)dy$$

$$= \frac{1}{a}\int_0^a 2y-\frac{2y^2}{a}\,dy = \frac{1}{a}\left[y^2-\frac{2}{3}\frac{y^3}{a}\right]_{y=0}^{y=a}$$

$$= \frac{1}{a}\left[a^2-\frac{2}{3}a^2\right] = \underline{\frac{a}{3}}$$

$\therefore$ Center of mass $= \underline{\underline{\left(1, \frac{a}{3}\right)}}$

2.

2. Assuming uniform density, find the coordinates of the center of mass of the semicircle $y = \sqrt{r^2 - x^2}$, with $y \geq 0$.

x-coordinate will be $x=0$ from symmetry.

$$\bar{y} = \frac{\iint_D y\,dx\,dy}{\text{Area of } D} = \frac{1}{\frac{1}{2}\pi r^2}\int_{-r}^{r}\int_0^{\sqrt{r^2-x^2}} y\,dy\,dx$$

$$= \frac{2}{\pi r^2} \int_{-r}^{r} \left[\frac{y^2}{2} \Big|_{y=0}^{y=\sqrt{r^2-x^2}} \right] dx = \frac{1}{\pi r^2} \int_{-r}^{r} (r^2 - x^2)\, dx$$

$$= \frac{1}{\pi r^2} \left[r^2 x - \frac{x^3}{3} \right]_{x=-r}^{x=r}$$

$$= \frac{1}{\pi r^2} \left[r^3 - \frac{r^3}{3} - \left(-r^3 + \frac{r^3}{3} \right) \right] = \frac{1}{\pi r^2} \left[\frac{4}{3} r^3 \right]$$

$$= \frac{4r}{3\pi}$$

$\therefore$ Centre of mass $= \left(0, \frac{4r}{3\pi}\right)$

3.

3. Find the average of $f(x, y) = y \sin xy$ over $D = [0, \pi] \times [0, \pi]$.

$$\iint_D f(x,y)\, dx\, dy = \int_0^{\pi} \int_0^{\pi} y \sin(xy)\, dx\, dy$$

$$= \int_0^{\pi} \left[-\cos(xy) \Big|_{x=0}^{x=\pi} \right] dy = \int_0^{\pi} -\cos(\pi y) + 1 \; dy$$

$$= -\frac{1}{\pi} \sin(\pi y) + y \Big|_0^{\pi} = \pi - \frac{\sin(\pi^2)}{\pi}$$

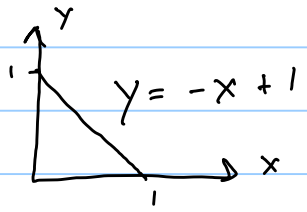
But $\sin(\pi^2) = \sin(\pi)\cos(\pi) + \cos(\pi)\sin(\pi) = 0 + 0 = 0$

$\therefore \iint_D f(x,y)\, dx\, dy = \pi$

Area of $D = \pi^2$. $\therefore$ Average $= \frac{\pi}{\pi^2} = \frac{1}{\pi}$

4.

4. Find the average of $f(x, y) = e^{x+y}$ over the triangle with vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$.



Area of triangle $= \frac{1}{2}$

$$\therefore \iint_D f(x,y)\,dx\,dy = \int_0^1 \int_0^{-x+1} e^{x+y}\,dy\,dx$$

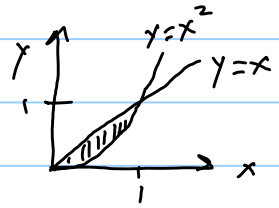
$$= \int_0^1 \left(e^{x+y} \Big|_{y=0}^{y=-x+1} \right) dx = \int_0^1 e - e^x\,dx$$

$$= ex - e^x \Big|_{x=0}^{x=1} = e - e - (0 - 1) = 1$$

$$\therefore \text{Average} = \frac{1}{1/2} = \underline{2}$$

5.

5. Find the center of mass of the region between $y = x^2$ and $y = x$ if the density is $x + y$.



$$\text{Mass of } D = \iint_D \delta(x,y)\,dx\,dy = \int_0^1 \int_{x^2}^{x} (x+y)\,dy\,dx$$

$$= \int_0^1 \left(xy + \frac{y^2}{2} \Big|_{y=x^2}^{y=x} \right) dx = \int_0^1 \left\{ x^2 + \frac{x^2}{2} - \left(x^3 + \frac{x^4}{2} \right) \right\} dx$$

$$= \int_0^1 -\frac{x^4}{2} - x^3 + \frac{3}{2}x^2\,dx = \frac{-x^5}{10} - \frac{x^4}{4} + \frac{x^3}{2}\Big|_{x=0}^{1}$$

$$= -\frac{1}{10} - \frac{1}{4} + \frac{1}{2} = \frac{-2-5+10}{20} = \frac{3}{20}$$

$$\therefore \bar{x} = \frac{20}{3}\int_0^1 \int_{x^2}^{x} x(x+y)\,dy\,dx = \frac{20}{3}\int_0^1 x^2y + \frac{xy^2}{2}\Big|_{y=x^2}^{y=x} dx$$

$$= \frac{20}{3}\int_0^1 \left[x^3 + \frac{x^3}{2} - \left(x^4 + \frac{x^5}{2}\right)\right] dx$$

$$= \frac{20}{3}\left(\frac{3}{8}x^4 - \frac{x^5}{5} - \frac{x^6}{12}\right)\Big|_{x=0}^{x=1}$$

$$= \frac{20}{3}\left(\frac{3}{8} - \frac{1}{5} - \frac{1}{12}\right) = \frac{20}{3}\left(\frac{45-24-10}{120}\right) = \frac{20}{3}\cdot\frac{11}{120}$$

$$= \frac{11}{18}$$

$$\bar{y} = \frac{20}{3}\int_0^1 \int_{x^2}^{x} y(x+y)\,dy\,dx = \frac{20}{3}\int_0^1 \frac{xy^2}{2} + \frac{y^3}{3}\Big|_{y=x^2}^{y=x} dx$$

$$= \frac{20}{3}\int_0^1 \frac{x^3}{2} + \frac{x^3}{3} - \left(\frac{x^5}{2} + \frac{x^6}{3}\right) dx$$

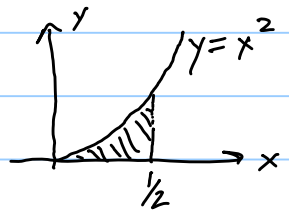
$$= \frac{20}{3}\left[\frac{5}{6}\cdot\frac{x^4}{4} - \frac{x^6}{12} - \frac{x^7}{21}\right]_{x=0}^{x=1}$$

$$= \frac{20}{3}\left[\frac{5}{24} - \frac{1}{12} - \frac{1}{21}\right] = \frac{20}{3}\left[\frac{35 - 14 - 8}{2^3 \cdot 3 \cdot 7}\right]$$

$$= \frac{5}{3}\left(\frac{13}{2 \cdot 3 \cdot 7}\right) = \frac{65}{126}$$

$\therefore$ Center of mass $= \left(\frac{11}{18}, \frac{65}{126}\right)$

6. **6.** Find the center of mass of the region between $y = 0$ and $y = x^2$, where $0 \le x \le \frac{1}{2}$.



Assume uniform density $= \delta$

Mass of area $= \delta \iint_D dx\,dy = \delta \int_0^{\frac{1}{2}} \int_0^{x^2} dy\,dx = \delta \int_0^{\frac{1}{2}} x^2\,dx$

$$= \delta \left.\frac{x^3}{3}\right|_0^{\frac{1}{2}} = \delta \frac{1}{24}$$

$\therefore \bar{x} = \frac{24}{\delta} \iint_D \delta x\,dx\,dy = 24 \int_0^{\frac{1}{2}} \int_0^{x^2} x\,dy\,dx$

$$= 24 \int_0^{1/2} x^3\,dx = 24 \left[\frac{x^4}{4}\right]_{x=0}^{x=\frac{1}{2}} = \frac{24}{64} = \frac{3}{8}$$

$\bar{y} = \frac{24}{\delta} \iint_D \delta y\,dx\,dy = 24 \int_0^{\frac{1}{2}} \int_0^{x^2} y\,dy\,dx$

$$= 24 \int_0^{\frac{1}{2}} \left.\frac{y^2}{2}\right|_{y=0}^{y=x^2} dx = 24 \int_0^{1/2} \frac{x^4}{2}\,dx = 12 \left[\frac{x^5}{5}\right]_0^{\frac{1}{2}}$$

$= \frac{12}{32(5)} = \frac{3}{40}$

$\therefore$ Centre of mass $= \left(\frac{3}{8}, \frac{3}{40}\right)$

7.

7. A sculptured gold plate D is defined by $0 \le x \le 2\pi$ and $0 \le y \le \pi$ (centimeters) and has mass density $\delta(x, y) = y^2 \sin^2 4x + 2$ (grams per square centimeter). If gold sells for \$7 per gram, how much is the gold in the plate worth?

$$\text{Mass} = \iint_D \delta(x,y)\,dx\,dy = \int_0^{2\pi}\int_0^{\pi} y^2\sin^2(4x) + 2 \; dy\,dx$$

$$= \int_0^{2\pi} \left[\frac{y^3}{3}\sin^2(4x) + 2y\right]_{y=0}^{y=\pi} dx$$

$$= \int_0^{2\pi} \frac{\pi^3}{3}\sin^2(4x) + 2\pi \; dx \qquad \text{use } \sin^2\theta = \frac{1-\cos 2\theta}{2}$$

$$= \int_0^{2\pi} \frac{\pi^3}{3}\left(\frac{1-\cos 8x}{2}\right) + 2\pi \; dx$$

$$= \frac{\pi^3}{6}x - \frac{\pi^3}{48}\sin 8x + 2\pi x \Big|_0^{2\pi}$$

$$= \frac{\pi^4}{3} - 0 + 4\pi^2 = \frac{\pi^4}{3} + 4\pi^2 \text{ grams}$$

$\therefore$ Cost $= 7\left(\frac{\pi^4}{3} + 4\pi^2\right) = \$503.64$

8. **8.** In Exercise 7, what is the average mass density in grams per square centimeter?

From #7, total mass $= \frac{\pi^4}{3} + 4\pi^2$ grams

Total area of plate $D = (2\pi)(\pi) = 2\pi^2$

$\therefore$ Average density $= \dfrac{\frac{\pi^4}{3} + 4\pi^2}{2\pi^2} = \dfrac{\pi^2}{6} + 2$

9. **9.** (a) Find the mass of the box $[0, \frac{1}{2}] \times [0, 1] \times [0, 2]$, assuming the density to be uniform.

(b) Same as part (a), but with a mass density $\delta(x, y, z) = x^2 + 3y^2 + z + 1$.

(a) Mass = (Density)(Volume) $= \delta\left(\frac{1}{2} \times 1 \times 2\right) = \delta$

where δ = uniform density of the mass.

(b) Mass $= \iiint_V \delta(x,y,z)\, dx\,dy\,dz$

$$= \int_0^{1/2}\int_0^1\int_0^2 (x^2 + 3y^2 + z + 1)\, dz\,dy\,dx$$

assuming mass orientation is $0 \le x \le \frac{1}{2}$, $0 \le y \le 1$, $0 \le z \le 2$

$$= \int_0^{\frac{1}{2}} \int_0^{1} \left(x^2 z + 3y^2 z + \frac{z^2}{2} + z \Big|_{z=0}^{2} \right) dy\, dx$$

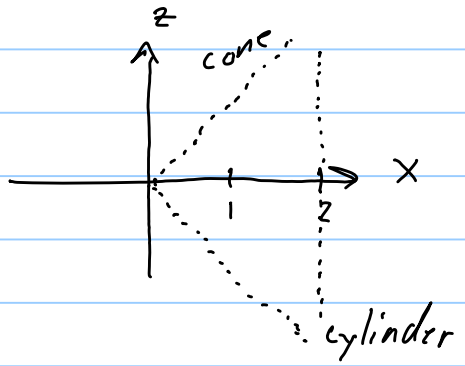
$$= \int_0^{1/2} \int_0^{1} 2x^2 + 6y^2 + 4 \; dy\, dx$$

$$= \int_0^{\frac{1}{2}} \left(2x^2 y + 2y^3 + 4y \Big|_{y=0}^{1} \right) dx$$

$$= \int_0^{\frac{1}{2}} 2x^2 + 6 \, dx = \frac{2}{3}x^3 + 6x \Big|_{x=0}^{\frac{1}{2}}$$

$$= \frac{1}{12} + 3 = \underline{\underline{3\frac{1}{12}}}$$

10. **10.** Find the mass of the solid bounded by the cylinder $x^2 + y^2 = 2x$ and the cone $z^2 = x^2 + y^2$ if the density is $\delta = \sqrt{x^2 + y^2}$.

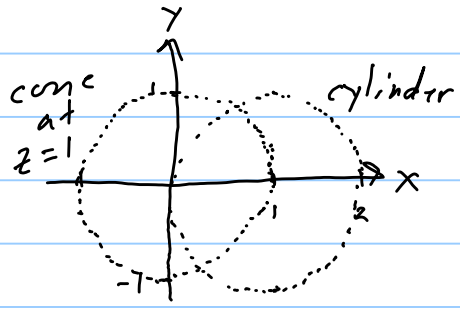


Cylinder $x^2 + y^2 = 2x \Leftrightarrow (x-1)^2 + y^2 = 1$

V is described as $0 \le x \le 2$,

$$-\sqrt{2x - x^2} \le y \le \sqrt{2x - x^2},$$

$$-\sqrt{x^2+y^2} \le z \le \sqrt{x^2+y^2}$$



$$\therefore \text{Mass} = \iiint_V \delta(x,y,z)\,dx\,dy\,dz$$

$$= \int_0^2 \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} \int_{-\sqrt{x^2+y^2}}^{\sqrt{x^2+y^2}} \sqrt{x^2+y^2}\,dz\,dy\,dx$$

It likely will be easier to use cylindrical coordinates

$$\text{Mass} = \iiint_V \delta(r,\theta,z)\, r\,dr\,d\theta\,dz$$

For $\delta(x,y,z) = \sqrt{x^2+y^2}$, $\delta(r,\theta,z) = r$

For V, $(x-1)^2+y^2=1$ translates to $r = 2\cos\theta$:

$$(r\cos\theta - 1)^2 + r^2\sin^2\theta = 1 \iff r^2\cos^2\theta - 2r\cos\theta + 1 + r^2\sin^2\theta = 1$$
$$\iff r^2 - 2r\cos\theta = 0$$
$$\iff r = 2\cos\theta$$

$$\therefore V: -\frac{\pi}{2} \le \theta \le \frac{\pi}{2},\ 0 \le r \le 2\cos\theta,\ -r \le z \le r$$

$$\therefore \text{Mass} = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\cos\theta} \int_{-r}^{r} r^2\,dz\,dr\,d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\cos\theta} \left(r^2 z \Big|_{z=-r}^{z=r}\right) dr\,d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{2\cos\theta} 2r^3\,dr\,d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{r^4}{2} \Big|_{r=0}^{r=2\cos\theta} \right) d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 8\cos^4\theta \, d\theta$$

use $\cos^2 x = \frac{1+\cos 2x}{2}$

$$= 8\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \left(\frac{1+\cos 2\theta}{2} \right)^2 d\theta = 2\int_{-\pi/2}^{\pi/2} 1 + 2\cos 2\theta + \cos^2 2\theta \, d\theta$$

$$= \int_{-\pi/2}^{\pi/2} 2\,d\theta + 4\int_{-\pi/2}^{\pi/2} \cos 2\theta \, d\theta + 2\int_{-\pi/2}^{\pi/2} \cos^2 2\theta \, d\theta$$

$$= 2\theta \Big|_{-\pi/2}^{\pi/2} + 2\sin 2\theta \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + 2\int_{-\frac{\pi}{2}}^{\pi/2} \frac{1+\cos 4\theta}{2} \, d\theta$$

$$= (2\pi) + (0) + \theta \Big|_{-\pi/2}^{\pi/2} + \frac{\sin 4\theta}{4} \Big|_{-\pi/2}^{\pi/2}$$

$$= 2\pi + \pi + 0 = 3\pi$$

$\therefore$ Mass $= 3\pi$

11. **11.** Find the mass of the solid ball of radius 5 with density given by

$$\delta(x, y, z) = 2x^2 + 2y^2 + 2z^2 + 1$$

assuming the center of the ball is at the origin.

Use spherical coordinates. $\delta(\rho, \theta, \phi) = 2\rho^2 + 1$

$$\text{Mass} = \iiint_V \delta(\rho, \theta, \phi)\, \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi$$

$$= \int_0^{2\pi}\int_0^{\pi}\int_0^{5} (2\rho^2+1)\rho^2\sin\phi \, d\rho\, d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi}\int_0^{5} \sin\phi\,(2\rho^4+\rho^2)\, d\rho\, d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \sin\phi \left[\frac{2}{5}\rho^5 + \frac{\rho^3}{3}\right]_{\rho=0}^{5} d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \sin\phi \left(1250 + \frac{125}{3}\right) d\phi\, d\theta$$

$$= \int_0^{2\pi} \frac{3875}{3}\left[-\cos\phi\right]_{\phi=0}^{\pi} d\theta = \int_0^{2\pi} \frac{7750}{3}\, d\theta$$

$$= \frac{15500\pi}{3}$$

12. **12.** A solid disk of radius 9 and height 2 is placed at the origin, so that it can be expressed by $x^2 + y^2 = 81$ and $0 \le z \le 2$. If the disk has a density given by

$$\delta(x, y, z) = 2x^2 + 2y^2 + 2z^2 + 1,$$

find its mass.

Use cylindrical coordinates. $\delta(r,\theta,z) = 2r^2 + 2z^2 + 1$

$$\text{Mass} = \iiint_V \delta(r,\theta,z)\, r\, dr\, d\theta\, dz$$

$$= \int_0^2 \int_0^{2\pi} \int_0^9 (2r^2 + 2z^2 + 1)\, r\, dr\, d\theta\, dz$$

$$= \int_0^2 \int_0^{2\pi} \int_0^9 2r^3 + 2rz^2 + r\, dr\, d\theta\, dz$$

$$= \int_0^2 \int_0^{2\pi} \frac{r^4}{2} + r^2 z^2 + \frac{r^2}{2} \Big|_{r=0}^{9} d\theta\, dz$$

$$= \int_0^2 \int_0^{2\pi} \left(\frac{6561}{2} + 81z^2 + \frac{81}{2}\right) d\theta\, dz = \int_0^2 (3321 + 81z^2)\, 2\pi\, dz$$

$$= 6642\pi z + 27z^3 \Big|_{z=0}^{z=2}$$

$$= 13{,}284\pi + 216$$

13. **13.** Find the center of mass of the region bounded by $x + y + z = 2$, $x = 0$, $y = 0$, and $z = 0$, assuming the density to be uniform.

$z = 2 - x - y$. At $z = 0$, $x + y = 2$. x-intercept: $(2, 0, 0)$

$\therefore$ Volume $= \int_0^2 \int_0^{2-x} \int_0^{2-x-y} dz\, dy\, dx$

$$= \int_0^2 \int_0^{2-x} (2-x-y)\,dy\,dx = \int_0^2 2y - xy - \frac{y^2}{2}\Big|_{y=0}^{y=2-x} dx$$

$$= \int_0^2 2(2-x) - x(2-x) - \frac{(2-x)^2}{2}\,dx$$

$$= \int_0^2 \left(2 - 2x + \frac{x^2}{2}\right) dx = 2x - x^2 + \frac{x^3}{6}\Big|_0^2 = \frac{4}{3}$$

$$\bar{x} = \frac{1}{4/3} \int_0^2 \int_0^{2-x} \int_0^{2-x-y} x\,dz\,dy\,dx$$

$$= \frac{3}{4} \int_0^2 \int_0^{2-x} (2x - x^2 - xy)\,dy\,dx = \frac{3}{4} \int_0^2 2xy - x^2y - \frac{xy^2}{2}\Big|_{y=0}^{y=2-x} dx$$

$$= \frac{3}{4} \int_0^2 2x(2-x) - x^2(2-x) - x\frac{(2-x)^2}{2}\,dx$$

$x^2 - 4x + 4$

$$= \frac{3}{4} \int_0^2 4x - 2x^2 - 2x^2 + x^3 - \frac{(x^3 - 4x^2 + 4x)}{2}\,dx$$

$$= \frac{3}{4} \int_0^2 \frac{x^3}{2} - 2x^2 + 2x\,dx = \frac{3}{4}\left[\frac{x^4}{8} - \frac{2}{3}x^3 + x^2\right]_0^2$$

$$= \frac{3}{4}\left(2 - \frac{16}{3} + 4\right) = \frac{1}{2}$$

$$\bar{y} = \frac{3}{4}\int_0^2\int_0^{2-x}\int_0^{2-x-y} y\,dz\,dy\,dx = \frac{3}{4}\int_0^2\int_0^{2-x} y(2-x-y)\,dy\,dx$$

$$= \frac{3}{4}\int_0^2\int_0^{2-x}(2y - xy - y^2)\,dy\,dx = \frac{3}{4}\int_0^2 \left(y^2 - \frac{xy^2}{2} - \frac{y^3}{3}\Big|_{y=0}^{y=2-x}\right)dx$$

$$= \frac{3}{4}\int_0^2 (2-x)^2 - \frac{x(2-x)^2}{2} - \frac{(2-x)^3}{3}\,dx$$

$$= \frac{3}{4}\int_0^2 x^2 - 4x + 4 - \frac{(x^3 - 4x^2 + 4x)}{2} + \frac{(x^3 - 6x^2 + 12x - 8)}{3}\,dx$$

$$6x^2 - 24x + 24 - 3x^3 + 12x^2 - 12x + 2x^3 - 12x^2 + 24x - 16$$

$$= \frac{3}{4}\int_0^2 \frac{-x^3 + 6x^2 - 12x + 8}{6}\,dx = \frac{1}{8}\left[-\frac{x^4}{4} + 2x^3 - 6x^2 + 8x\right]_{x=0}^{2}$$

$$= \frac{1}{8}(-4 + 16 - 24 + 16) = \frac{1}{8}(4) = \underline{\frac{1}{2}}$$

$$\bar{z} = \frac{3}{4}\int_0^2\int_0^{2-x}\int_0^{2-x-y} z\,dz\,dy\,dx = \frac{3}{4}\int_0^2\int_0^{2-x} \frac{z^2}{2}\Big|_{z=0}^{z=2-x-y}\,dy\,dx$$

$$= \frac{3}{4}\int_0^2\int_0^{2-x} \frac{4 - 4x - 4y + x^2 + 2xy + y^2}{2}\,dy\,dx$$

$$= \frac{3}{8}\int_0^2 4y - 4xy - 2y^2 + x^2y + xy^2 + \frac{y^3}{3}\Big|_{y=0}^{y=(2-x)}\,dx$$

$$= \frac{3}{8}\int_0^2 4(2-x) - 4x(2-x) - 2(2-x)^2 + x^2(2-x) + x(2-x)^2 + \frac{(2-x)^3}{3}\,dx$$

$$8 - 4x - 8x + 4x^2 - 2x^2 + 8x - 8 + 2x^2 - x^3 + x^3 - 4x^2 + 4x + \frac{(2-x)^3}{3}$$

$$= \frac{3}{8}\int_0^2 \frac{(2-x)^3}{3}\,dx = \frac{1}{8}(-1)\frac{(2-x)^4}{4}\Big|_{x=0}^{x=2}$$

$$= -\frac{1}{32}(0-16) = \frac{16}{32} = \frac{1}{2}$$

$$\therefore \text{Center of Mass} = (\bar{x}, \bar{y}, \bar{z}) = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$$

14. **14.** Find the center of mass of the cylinder $x^2 + y^2 \le 1,\ 1 \le z \le 2$ if the density is $\delta = (x^2 + y^2)z^2$.

$$\text{Mass} = \iiint_V \delta\,dV, \text{ use cylindrical coordinates}$$

$$\therefore \delta = (r^2\cos^2\theta + r^2\sin^2\theta)z^2 = r^2z^2$$

$$\therefore \text{Mass} = \int_1^2\int_0^1\int_0^{2\pi} (r^2z^2)\,r\,d\theta\,dr\,dz$$

$$= \int_1^2\int_0^1 2\pi r^3z^2\,dr\,dz = \int_1^2 \pi z^2\frac{r^4}{2}\Big|_{r=0}^{1}\,dz$$

$$= \int_1^2 \frac{\pi}{2}z^2\,dz = \frac{\pi}{2}\left[\frac{z^3}{3}\right]_{z=1}^{2}$$

$$= \frac{\pi}{2}\left[\frac{8}{3} - \frac{1}{3}\right] = \frac{7\pi}{6}$$

$$\bar{x} = \frac{\iiint_V x\,\delta\,dV}{\text{Mass}}$$ Again, use cylindrical coordinates $\delta = r^2z^2$ $x = r\cos\theta$

$$= \frac{6}{7\pi}\int_1^2\int_0^1\int_0^{2\pi} (r\cos\theta)(r^2z^2)\,r\,d\theta\,dr\,dz$$

$$= \frac{6}{7\pi}\int_1^2\int_0^1 r^4z^2\sin\theta\Big|_{\theta=0}^{\theta=2\pi} = 0$$ $\sin(2\pi) = \sin(0) = 0$

which makes sense since δ is symmetric with respect to z-axis: $\delta(x,y,z) = \delta(-x,y,z) = \delta(x,-y,z)$

$$\bar{y} = \frac{\iiint_V y\,\delta\,dV}{\text{Mass}}$$ Use cylindrical coordinates with $\delta = r^2z^2$ $y = r\sin\theta$

$$= \frac{6}{7\pi}\int_1^2\int_0^1\int_0^{2\pi} (r\sin\theta)(r^2z^2)\,r\,d\theta\,dr\,dz$$

$$= \frac{6}{7\pi}\int_1^2\int_0^1 r^4z^2\left[-\cos\theta\right]_{\theta=0}^{\theta=2\pi} dr\,dz$$

$$= \frac{6}{7\pi}\int_1^2\int_0^1 r^4z^2\left[-1-(-1)\right] dr\,dz = 0$$

$$\bar{z} = \frac{\iiint_V z\,\delta\,dV}{\text{mass}}$$ Use cylindrical coordinates $\delta = r^2 z^2$

$$= \frac{6}{7\pi}\int_1^2\int_0^1\int_0^{2\pi} (z)(r^2z^2)\,r\,d\theta\,dr\,dz$$

$$= \frac{6}{7\pi}(2\pi)\int_1^2\int_0^1 z^3 r^3\,dr\,dz = \frac{12}{7}\int_1^2 z^3\,\frac{r^4}{4}\Big|_0^1\,dz$$

$$= \frac{12}{7}\cdot\frac{1}{4}\left[\frac{z^4}{4}\right]_{z=1}^{z=2} = \frac{3}{7}\left(\frac{16}{4}-\frac{1}{4}\right) = \frac{45}{28}$$

$\therefore$ Centre of mass $= \left(0, 0, \frac{45}{28}\right)$

15. **15.** Find the average value of $\sin^2 \pi z \cos^2 \pi x$ over the cube $[0, 2] \times [0, 4] \times [0, 6]$.

Volume of cube $= (2)(4)(6) = 48$

Assume $0 \le x \le 2,\ 0 \le y \le 4,\ 0 \le z \le 6$

$$\therefore \text{Ave} = \frac{1}{48}\int_0^2\int_0^6\int_0^4 \sin^2(\pi z)\cos^2(\pi x)\,dy\,dz\,dx$$

$$= \frac{4}{48}\int_0^2 \cos^2(\pi x)\,dx \int_0^6 \sin^2(\pi z)\,dz$$

$\cos^2\theta - \sin^2\theta = \cos 2\theta,\ 2\cos^2\theta - 1 = \cos 2\theta,\ 1 - 2\sin^2\theta = \cos 2\theta$

$$= \frac{1}{12} \int_0^2 \frac{1+\cos 2\pi x}{2}\,dx \int_0^6 \frac{1-\cos 2\pi z}{2}\,dz$$

$$= \frac{1}{48}\left[x + \frac{\sin 2\pi x}{2\pi}\right]_0^2 \left[z - \frac{\sin 2\pi z}{2\pi}\right]_0^6$$

$$= \frac{1}{48}\left[2+0-(0+0)\right]\left[6-0-(0-0)\right]$$

$$= \frac{12}{48} = \frac{1}{4}$$

16. **16.** Find the average value of e^{-z} over the ball $x^2 + y^2 + z^2 \le 1$.

Volume of ball = 1. Use spherical coordinates

$z = \rho\cos\phi$

$$\therefore \text{Ave} = \int_0^1 \int_0^{\pi} \int_0^{2\pi} e^{-\rho\cos\phi}\,\rho^2 \sin\phi\, d\theta\, d\phi\, d\rho$$

$$= 2\pi \int_0^1 \int_0^{\pi} e^{-\rho\cos\phi}\,\rho^2 \sin\phi\, d\phi\, d\rho$$

$$= 2\pi \int_0^1 \rho^2 \left[\frac{e^{-\rho\cos\phi}}{\rho}\right]_{\phi=0}^{\phi=\pi} d\rho$$

$$= 2\pi \int_0^1 \rho\left[e^{\rho} - e^{-\rho}\right] d\rho$$

$$= 2\pi \int_0^1 \rho e^{\rho}\, d\rho - 2\pi \int_0^1 \rho e^{-\rho}\, d\rho$$

$$= 2\pi \left[\rho e^{\rho} - e^{\rho}\right]_{\rho=0}^{\rho=1} - 2\pi \left[-\rho e^{-\rho} - e^{-\rho}\right]_{\rho=0}^{\rho=1}$$

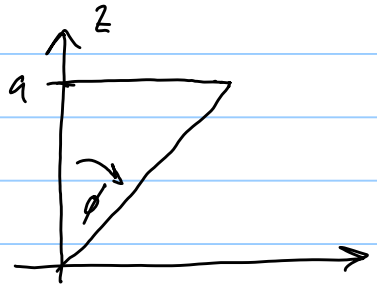
$$= 2\pi \left[e - e - (0-1)\right] - 2\pi \left[-e^{-1} - e^{-1} - (0-1)\right]$$

$$= 2\pi - 2\pi(-2e^{-1} + 1)$$

$$= \frac{4\pi}{e}$$

17.

17. A solid with constant density is bounded above by the plane $z = a$ and below by the cone described in spherical coordinates by $\phi = k$, where k is a constant $0 < k < \pi/2$. Set up an integral for its moment of inertia about the z axis.



Let δ = density, a constant.

Use cylindrical coordinates. An elemental mass is $\delta\, dV = \delta\, r\, dr\, d\theta\, dz$, $0 \le z \le a$, $0 \le \theta \le 2\pi$, $0 \le r \le z\tan(k)$

The distance^2 from the z-axis of the elemental mass is $(= x^2 + y^2)$ $(r\cos\theta)^2 + (r\sin\theta)^2 = r^2$

$$\therefore \quad I_z = \int_0^a \int_0^{2\pi} \int_0^{z\tan(k)} \delta\, r^3\, dr\, d\theta\, dz$$

With spherical coordinates, an elemental mass is
$\delta\, dV = \delta \rho^2 \sin\phi \, d\rho\, d\theta\, d\phi$, $0 \le \theta \le 2\pi$, $0 \le \phi \le K$,
$0 \le \rho \le \frac{a}{\cos\phi} = a \sec\phi$

The distance^2 from the z-axis of the elemental mass is $(\equiv x^2 + y^2)$ $(\rho \sin\phi \cos\theta)^2 + (\rho \sin\phi \sin\theta)^2$
$= \rho^2 \sin^2\phi$

$$\therefore I_z = \int_0^K \int_0^{2\pi} \int_0^{a\sec\phi} (\rho^2 \sin^2\phi)\, \delta \rho^2 \sin\phi \, d\rho\, d\theta\, d\phi$$

$$= \int_0^K \int_0^{2\pi} \int_0^{a\sec\phi} \delta \rho^4 \sin^3\phi \, d\rho\, d\theta\, d\phi$$

18.

18. Find the moment of inertia around the y axis for the ball $x^2 + y^2 + z^2 \le R^2$ if the mass density is a constant δ.

$$I_y = \iiint_V (x^2 + z^2)\, \delta\, dV$$ Use spherical coordinates.

$dV = \rho^2 \sin\phi \, d\rho\, d\theta\, d\phi$ $x^2 + z^2 = (\rho \sin\phi \cos\theta)^2 + (\rho \cos\phi)^2$
$= \rho^2 \sin^2\phi \cos^2\theta + \rho^2 \cos^2\phi$

$$\therefore I_y = \int_0^{\pi} \int_0^{2\pi} \int_0^{R} \delta \left[\rho^2 \sin^2\phi \cos^2\theta + \rho^2 \cos^2\phi\right] \rho^2 \sin\phi \, d\rho\, d\theta\, d\phi$$

$$= \delta \int_0^{\pi} \int_0^{2\pi} \int_0^{R} \left[\sin^3\phi \cos^2\theta + \cos^2\phi \sin\phi \right] \rho^4 \, d\rho \, d\theta \, d\phi$$

$$= \delta \int_0^{\pi} \int_0^{2\pi} \left[\sin^3\phi \cos^2\theta + \cos^2\phi \sin\phi \right] \frac{\rho^5}{5} \Big|_{\rho=0}^{\rho=R} \, d\theta \, d\phi$$

$$= \frac{\delta R^5}{5} \int_0^{\pi} \int_0^{2\pi} \sin^3\phi \left(\frac{1+\cos 2\theta}{2} \right) + \cos^2\phi \sin\phi \, d\theta \, d\phi$$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$$= \frac{\delta R^5}{5} \int_0^{\pi} \sin^3\phi \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_{\theta=0}^{\theta=2\pi} + \cos^2\phi \sin\phi \left[\theta \right]_{\theta=0}^{\theta=2\pi} \, d\phi$$

$$= \frac{\delta R^5}{5} \int_0^{\pi} \pi \sin^3\phi + 2\pi \cos^2\phi \sin\phi \, d\phi$$

$$= \frac{\delta \pi R^5}{5} \int_0^{\pi} \sin\phi \left(1 - \cos^2\phi\right) + 2\cos^2\phi \sin\phi \, d\phi$$

$$= \frac{\delta \pi R^5}{5} \int_0^{\pi} \left(\sin\phi + \cos^2\phi \sin\phi \right) d\phi$$

$$= \frac{\delta \pi R^5}{5} \left[-\cos\phi - \frac{\cos^3\phi}{3} \right]_{\phi=0}^{\phi=\pi}$$

$$= \frac{\delta \pi R^5}{5}\left[1 + \frac{1}{3} - \left(-1 - \frac{1}{3}\right)\right] = \frac{\delta \pi R^5}{5}\left[\frac{8}{3}\right]$$

$$= \frac{8\delta \pi R^5}{15}$$

19. **19.** Find the gravitational potential on a mass m of a spherical planet with mass $M = 3 \times 10^{26}$ kg, at a distance of 2×10^8 m from its center.

$$V = -\frac{GMm}{r} = -\frac{(6.67 \times 10^{-11})(3 \times 10^{26})m}{2 \times 10^8}$$

$$= -10 \times 10^7 m = -1.0 \times 10^8 m \ \text{N·m}$$

20. **20.** Find the gravitational force exerted on a 70-kg object at the position in Exercise 19.

$$F = \frac{GMm}{r^2} = \frac{(6.67 \times 10^{-11})(3 \times 10^{26})(70)}{(2 \times 10^8)^2}$$

$$= \frac{(1.0 \times 10^8)(70)}{(2 \times 10^8)} = 35 \text{ Newtons}$$

21. **21.** A body W in xyz coordinates is called *symmetric with respect to a given plane* if for every particle on one side of the plane there is a particle of equal mass located at its mirror image through the plane.

(a) Discuss the planes of symmetry for an automobile shell.

(a) The vertical plane down the middle dividing the

car in half (left from right) is the only plane of symmetry.

(6) (b) Let the plane of symmetry be the xy plane, and denote by W^+ and W^- the portions of W above and below the plane, respectively. By our assumption, the mass density $\delta(x, y, z)$ satisfies $\delta(x, y, -z) = \delta(x, y, z)$. Justify the following steps:

$$\begin{aligned}\bar{z} \cdot \iiint_W \delta(x, y, z)\, dx\, dy\, dz &= \iiint_W z\delta(x, y, z)\, dx\, dy\, dz \\ &= \iiint_{W^+} z\delta(x, y, z)\, dx\, dy\, dz + \iiint_{W^-} z\delta(x, y, z)\, dx\, dy\, dz \\ &= \iiint_{W^+} z\delta(x, y, z)\, dx\, dy\, dz + \iiint_{W^+} -w\delta(u, v, -w)\, du\, dv\, dw \\ &= 0.\end{aligned}$$

(1) By definition, $\bar{z} = \dfrac{\iiint_W z\,\delta(x,y,z)\,dx\,dy\,dz}{\iiint_W \delta(x,y,z)\,dx\,dy\,dz}$

as the Mass $= \iiint_W \delta(x,y,z)\,dx\,dy\,dz$

$\therefore \bar{z} \cdot \iiint_W \delta(x,y,z)\,dx\,dy\,dz = \iiint_W z\,\delta(x,y,z)\,dx\,dy\,dz$

(2) $\iiint_W = \iiint_{W^+} + \iiint_{W^-}$ is just additivity of integrals (property (iv) on p. 275 of text).

(3) By symmetry, $\delta(x,y,z) = \delta(x,y,-z)$

and again by symmetry, integrating

$-a \le z \le -b \le 0$ on W^- is the same as integrating $0 \le b \le -z \le a$ on W^+

$$\therefore \int_{-a}^{-b} z\,dz = \int_{a}^{b} -z\,dz, \text{ or } \int_{W^-} z = \int_{W^+} -z$$

In the formula, the u, v, w are dummy variables

$$\therefore \iiint_{W^-} z\,\delta(x,y,z)\,dx\,dy\,dz = \iiint_{W^+} -z\,\delta(x,y,-z)\,dx\,dy\,dz$$

(4) $$\iiint_{W^+} -z\,\delta(x,y,-z) = \iiint_{W^+} -w\,\delta(u,v,-w)\,du\,dv\,dw$$

as the x,y,z and u,v,w are dummy variables.

$\therefore$ Last step is:

$$\iiint_{W^+} z\,\delta(x,y,z)\,dx\,dy\,dz - \iiint_{W^+} z\,\delta(x,y,z)\,dx\,dy\,dz$$

and this is $\underline{0}$.

(c) (c) Explain why part (b) proves that if a body is symmetrical with respect to a plane, then its center of mass lies in that plane.

(6) proves $\bar{z} = 0$, as the mass in not zero.

$\therefore$ z-coordinate of center of mass is in the

plane of symmetry (the xy-plane).

(d) (d) Derive this law of mechanics: *If a body is symmetric with respect to two planes, then its center of mass lies on their line of intersection.*

From (c), one coordinate lies on one plane, another lies on the other plane. To lie on both planes, the center of mass must lie on the common aspect of each plane — i.e., the line of intersection.

22. **22.** A uniform rectangular steel plate of sides a and b rotates about its center of mass with constant angular velocity ω.

(a) The kinetic energy equals $\frac{1}{2}$(mass)(velocity)^2. Argue that the kinetic energy of any element of mass $\delta\, dx\, dy$(δ = constant) is given by $\delta(\omega^2/2)(x^2 + y^2)\, dx\, dy$, provided the origin (0, 0) is placed at the center of mass of the plate.

(a) Given constant density σ, an element of mass is $\sigma \Delta\, dx dy$, where Δ = the thickness of the plate, so $\Delta\, dx dy$ is an element of volume. Let $\delta = \sigma\Delta$, a constant. ∴ An element of mass = $\delta\, dx dy$ (dx = width, dy = length).

If the position of the element of mass is (x, y), its distance from the origin (center of mass) is $\sqrt{x^2+y^2}$.

A unit of mass rotating about an axis at angular velocity ω moves at speed $r\omega$, where r = distance from axis of rotation.

∴ $r\omega = \sqrt{x^2+y^2}\,\omega$.

Kinetic energy of a unit mass m is $\frac{1}{2}mv^2$.

$$\therefore \frac{1}{2}mv^2 = \frac{1}{2}(\delta\, dx\, dy)\left(\sqrt{x^2+y^2}\,\omega\right)^2$$

$$= \delta\,\frac{\omega^2}{2}(x^2+y^2)\,dx\,dy$$

(5) (b) Justify the formula for kinetic energy:

$$\text{K.E.} = \iint_{\text{plate}} \delta\frac{\omega^2}{2}(x^2+y^2)\,dx\,dy.$$

A Riemann sum of the kinetic energy of all the elemental masses yields the total kinetic energy, summing over the region of "plate"

$$\therefore \text{K.E.} = \iint_{\text{plate}} \delta\,\frac{\omega^2}{2}(x^2+y^2)\,dx\,dy$$

(c) (c) Evaluate the integral, assuming that the plate is described by the inequalities $-a/2 \le x \le a/2$, $-b/2 \le y \le b/2$.

$$\int_{-\frac{a}{2}}^{\frac{a}{2}}\int_{-\frac{b}{2}}^{\frac{b}{2}} \delta\,\frac{\omega^2}{2}(x^2+y^2)\,dy\,dx = \frac{\delta\omega^2}{2}\int_{-a/2}^{a/2}\int_{-b/2}^{b/2}(x^2+y^2)\,dy\,dx$$

$$= \frac{\delta\omega^2}{2}\int_{-a/2}^{a/2} x^2y + \frac{y^3}{3}\Big|_{y=-b/2}^{y=b/2}\,dx$$

$$= \frac{\delta w^2}{2} \int_{-a/2}^{a/2} \frac{6}{2}x^2 + \frac{6^3}{24} - \left(-\frac{6}{2}x^2 - \frac{6^3}{24}\right) dx$$

$$= \frac{\delta w^2}{2} \int_{-a/2}^{a/2} 6x^2 + \frac{6^3}{12}\, dx = \frac{\delta w^2}{2}\left[\frac{6x^3}{3} + \frac{6^3}{12}x\right]_{x=-\frac{a}{2}}^{x=a/2}$$

$$= \frac{\delta w^2}{2}\left[\frac{6a^3}{24} + \frac{6^3}{12}\cdot\frac{a}{2} - \left(-\frac{6a^3}{24} - \frac{6^3}{12}\frac{a}{2}\right)\right]$$

$$= \frac{\delta w^2}{2}\left\{\frac{6a^3}{12} + \frac{6^3 a}{12}\right\} = \frac{\delta w^2}{24}\left(a^3 6 + a 6^3\right)$$

23.

23. As is well known, the density of a typical planet is not constant throughout the planet. Assume that planet C.M.W. has a radius of 5×10^8 cm and a mass density (in grams per cubic centimeter)

$$\rho(x, y, z) = \begin{cases} \dfrac{3 \times 10^4}{r}, & r \geq 10^4 \text{ cm}, \\ 3, & r \leq 10^4 \text{ cm}, \end{cases}$$

where $r = \sqrt{x^2 + y^2 + z^2}$. Find a formula for the gravitational potential outside C.M.W.

(a) Mass of the constant density portion is:

$$\rho \frac{4}{3}\pi (10^4 \text{ cm})^3 = (3)\frac{4}{3}\pi 10^{12} = 4\pi 10^{12} \text{ g}$$

(b) Need to find the mass of the "shell" between 10^4 cm and 5×10^8 cm.

Use spherical coordinates. A unit of volume of

the shell is $\rho^2 \sin\phi \, d\rho \, d\theta \, d\phi$.

$\therefore$ Mass of the unit volume is $\frac{3\times10^4}{\rho} \rho^2 \sin\phi \, d\rho \, d\theta \, d\phi$

$= (3\times10^4 \text{ cm}) \rho \sin\phi \, d\rho \, d\theta \, d\phi$.

$\therefore$ Total mass $= (3\times10^4) \int_0^{\pi} \int_0^{2\pi} \int_{10^2}^{5\times10^6} \rho \sin\phi \, d\rho \, d\theta \, d\phi$

$= 3\times10^4 \int_0^{\pi} \int_0^{2\pi} \sin\phi \left. \frac{\rho^2}{2} \right|_{\rho=10^4}^{\rho=5\times10^8} d\theta \, d\phi$

$= 3\times10^4 \int_0^{\pi} \int_0^{2\pi} (12.5\times10^{16}) \sin\phi \, d\theta \, d\phi$

$= 3.75\times10^{21} \int_0^{\pi} \int_0^{2\pi} \sin\phi \, d\theta \, d\phi = 7.5\times10^{21}\pi \int_0^{\pi} \sin\phi \, d\phi$

$= (7.5\times10^{21}\pi) \left[-\cos\phi \right]_{\phi=0}^{\phi=\pi}$

$= 15\pi\times10^{21} \text{ g}$

(c) Total mass of planet, as seen from a point "outside" c.m.w. is:

$4\pi\times10^{12} \text{ g} + 15\pi\times10^{21} \text{ g} \simeq 15\pi\times10^{21} \text{ g}$

$= 15\pi\times10^{18} \text{ kg}$

(d) ∴ Potential "outside" C.M.W is $-\frac{GMm}{R}$

$= -\left(6.67\times10^{-11}\ \frac{N\cdot m^2}{Kg^2}\right)\left(15\pi\times10^{18}\ Kg\right)\frac{m}{R}$, $R > 5\times10^6\ m$

$= -3.14\times10^{9}\ \frac{m}{R}\ N\cdot m$ $\left(N\cdot m \equiv Kg\left(\frac{m}{sec}\right)^2\right)$

24. **24.** Let D be a region in the part of the xy plane with $x > 0$. Assume D has uniform density, area $A(D)$, and center of mass $(\bar{x}, \bar{y})$. Let W be the solid obtained by rotating D about the y axis. Show that the volume of W is given by

$$\text{vol}(W) = 2\pi\bar{x}A(D).$$

Consider an element of A(D), call it dA. The solid swept out by dA around the y-axis is $2\pi x\,dA$, where x is the distance of dA from the y-axis. That is, the circumference, $2\pi x$, time the area of the element, dA.

∴ Total volume swept out by D is:

$$Vol(W) = \iint_D 2\pi x\,dA$$

By definition, $\bar{x} = \frac{\iint_D x\,dxdy}{\iint_D dxdy} = \frac{\iint_D x\,dxdy}{A(D)}$

$$\therefore\ \bar{x}A(D) = \iint_D x\,dxdy$$

$\therefore 2\pi \bar{x} A(D) = 2\pi \iint_D x\,dx\,dy = \iint_D 2\pi x\,dx\,dy$

But $dA = dx\,dy$ (an element of $A(D)$).

$\therefore 2\pi \bar{x} A(D) = \iint_D 2\pi x\,dx\,dy = \iint_D 2\pi x\,dA = Vol(w)$

$\therefore Vol(w) = 2\pi \bar{x} A(D)$

25. **25.** Use the previous exercise to show that if a doughnut is obtained by rotating the circle $(x - a)^2 + y^2 = r^2$ about the y axis, then the volume of the doughnut is $2\pi^2 ar^2$.

Assume $a > r$

The center of mass of the circle is at $(a, 0)$, or a units from the y-axis. $\therefore \bar{x} = a$.

The area of the circle of radius r is πr^2

$\therefore Vol(w) = 2\pi \bar{x} A(D) = 2\pi (a)(\pi r^2) = 2\pi^2 a r^2$

6.4 Improper Integrals

Note Title 1/9/2017

In Exercises 1 to 4, evaluate the following integrals if they exist (discuss how you define the integral if it was not given in the text).

1.

1. $\iint_D \frac{1}{\sqrt{xy}}\, dA$, where $D = [0, 1] \times [0, 1]$

Choose $0 < \delta$, $0 < \epsilon$ s.t. $D_{\delta,\epsilon} \subset D$

By Fubini's Theorem,

$$\iint_{D_{\delta,\epsilon}} \frac{1}{\sqrt{xy}}\, dx\,dy = \int_{\delta}^{1-\delta}\int_{\epsilon}^{1-\epsilon} \frac{1}{\sqrt{xy}}\, dx\,dy$$

$$= \int_{\delta}^{1-\delta} \frac{1}{\sqrt{y}}\, dy \int_{\epsilon}^{1-\epsilon} \frac{1}{\sqrt{x}}\, dx = \left(2\sqrt{y}\Big|_{y=\delta}^{y=1-\delta}\right)\left(2\sqrt{x}\Big|_{x=\epsilon}^{x=1-\epsilon}\right)$$

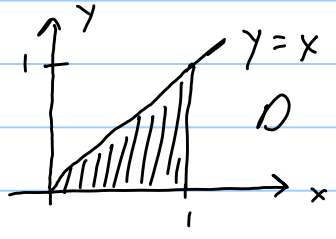
$$= 4\left(\sqrt{1-\delta} - \sqrt{\delta}\right)\left(\sqrt{1-\epsilon} - \sqrt{\epsilon}\right)$$

$$\therefore \lim_{(\delta,\epsilon)\to(0,0)} \iint_{D_{\delta,\epsilon}} \frac{1}{\sqrt{xy}}\, dA = \lim_{(\delta,\epsilon)\to(0,0)} 4\left(\sqrt{1-\delta} - \sqrt{\delta}\right)\left(\sqrt{1-\epsilon} - \sqrt{\epsilon}\right)$$

$$= \underline{4}$$

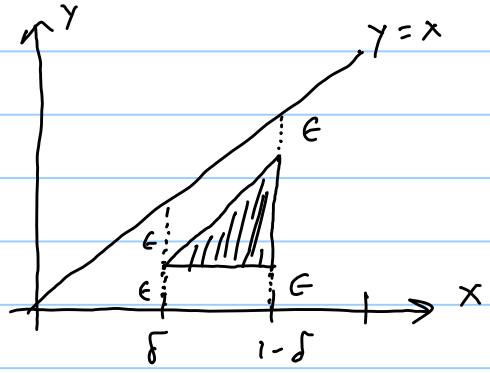
2.

2. $\iint_D \frac{1}{\sqrt{|x-y|}}\, dx\,dy$, where $D = \{(x, y) \mid 0 \le x \le 1, 0 \le y \le 1, y \le x\}$



For $y \le x$, $x - y \ge 0$ so $|x-y| = x-y$.

Let $\delta > 0$, $\epsilon > 0$ and consider $\delta \le x \le 1-\delta$ and $\epsilon \le y \le 1-\epsilon$



The segment from δ to $1-\delta$ must be > 0, so $1-\delta-\delta > 0 \Rightarrow 2\delta < 1$, or $\delta < \frac{1}{2}$. $\therefore$ choose δ s.t. $0 < \delta < \frac{1}{2}$.

Similarly, the height of the triangle must be > 0. The smallest height is at $x = \delta$ (height $= 0$). So $2\epsilon < \delta$, so choose ϵ s.t. $\epsilon < \frac{\delta}{2}$.

$\therefore D_{\delta,\epsilon} \subset D$ and on $D_{\delta,\epsilon}$ $x - y > 0$.

$\therefore$
$$\iint_{D_{\delta,\epsilon}} \frac{1}{\sqrt{|x-y|}}\,dx\,dy = \iint_{D_{\delta,\epsilon}} \frac{1}{\sqrt{x-y}}\,dx\,dy$$

$$= \int_{\delta}^{1-\delta} \int_{\epsilon}^{x-\epsilon} \frac{1}{\sqrt{x-y}}\,dy\,dx \qquad \text{by Fubini's Theorem}$$

$$= \int_{\delta}^{1-\delta} \left(-2\sqrt{x-y}\,\Big|_{y=\epsilon}^{y=x-\epsilon} \right) dx$$

$$= \int_{\delta}^{1-\delta} \left(2\sqrt{x-\epsilon} - 2\sqrt{\epsilon} \right) dx$$

$$= (2)\left(\frac{2}{3}\right)(x-\epsilon)^{3/2}\Big|_{x=\delta}^{x=1-\delta} - 2\sqrt{\epsilon}\,x\Big|_{x=\delta}^{x=1-\delta}$$

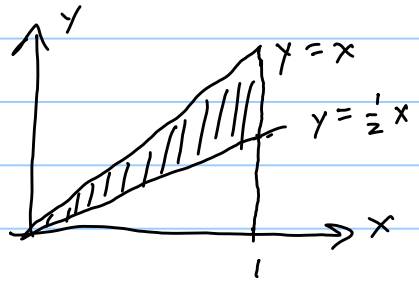
$$= \frac{4}{3}(1-\delta-\epsilon)^{3/2} - \frac{4}{3}(\delta-\epsilon)^{3/2} - 2\sqrt{\epsilon}(1-\delta) + 2\sqrt{\epsilon}\,\delta$$

$$\therefore \lim_{(\delta,\epsilon)\to(0,0)} \iint_{D_{\delta,\epsilon}} \frac{1}{\sqrt{|x-y|}}\,dx\,dy = \frac{4}{3} - 0 - 0 + 0 = \frac{4}{3}$$

$$\therefore \iint_D \frac{1}{\sqrt{|x-y|}}\,dx\,dy = \underline{\frac{4}{3}}$$

3.

3. $\iint_D (y/x)\,dx\,dy$, where D is bounded by $x = 1$, $x = y$, and $x = 2y$



Consider D_δ : $0 < \delta < 1$, $\delta \le x \le 1$

$\therefore \frac{y}{x}$ is defined over all of D_δ.

$$\therefore \iint_{D_\delta} \frac{y}{x}\,dx\,dy = \int_\delta^1 \int_{\frac{x}{2}}^{x} \frac{y}{x}\,dy\,dx$$

$$= \int_\delta^1 \left(\frac{y^2}{2x}\Big|_{y=\frac{x}{2}}^{y=x}\right) dx = \int_\delta^1 \left(\frac{x}{2} - \frac{x}{8}\right) dx$$

$$= \frac{3}{8}\,\frac{x^2}{2}\Big|_{x=\delta}^{x=1} = \frac{3}{16}\left[1-\delta^2\right]$$

$$\therefore \iint_D \frac{y}{x}\,dx\,dy = \lim_{\delta\to 0}\iint_{D_\delta}\frac{y}{x}\,dy\,dx = \lim_{\delta\to 0}\frac{3}{16}[1-\delta^2]$$

$$= \frac{3}{16}$$

4. **4.** $\int_0^1\int_0^{e^y}\log x\,dx\,dy$

Let $D_\delta : 0\le y\le 1,\ 0<\delta\le x\le e^y$

$$\therefore \int_0^1\int_0^{e^y}\log x\,dx\,dy = \lim_{\delta\to 0^+}\int_0^1\int_\delta^{e^y}\log x\,dx\,dy$$

$$= \lim_{\delta\to 0^+}\int_\delta^{e^y}\log x\,dx = \lim_{\delta\to 0^+}\Big[x\log x - x\Big]_{x=\delta}^{x=e^y}$$

$$= \lim_{\delta\to 0^+}\left[e^y\log e^y - e^y - \delta\log\delta + \delta\right]$$

$$= ye^y - e^y - \lim_{\delta\to 0^+}\delta\log\delta = ye^y - e^y - \lim_{\delta\to 0^+}\frac{\log\delta}{\frac{1}{\delta}}$$

$$= ye^y - e^y - \lim_{\delta\to 0^+}\frac{\frac{1}{\delta}}{-\frac{1}{\delta^2}} \quad (\text{L'Hopital's})$$

$$= ve^v - e^v - \lim_{\delta \to 0^+} (-\delta) = \underline{ve^v - e^v}$$

5.

5. Let $D = [0, 1] \times [0, 1]$. Let $0 < \alpha < 1$ and $0 < \beta < 1$. Evaluate:

$$\iint_D \frac{dx\,dy}{x^\alpha y^\beta}.$$

Let $0 < \delta < 1$, $0 < \epsilon < 1$, $D_{\delta,\epsilon}$: $\delta \le x \le 1$, $\epsilon \le y \le 1$

$$\therefore \iint_D \frac{dx\,dy}{x^\alpha y^\beta} = \lim_{(\delta,\epsilon)\to(0,0)} \iint_{D_{\delta,\epsilon}} \frac{dx\,dy}{x^\alpha y^\beta}$$

$$= \lim_{(\delta,\epsilon)\to(0,0)} \int_\delta^1 \int_\epsilon^1 \frac{1}{x^\alpha y^\beta}\, dy\, dx$$

$$= \lim_{(\delta,\epsilon)\to(0,0)} \left[\frac{x^{1-\alpha}}{1-\alpha} \right]_{x=\delta}^{x=1} \left[\frac{y^{1-\beta}}{1-\beta} \right]_{y=\epsilon}^{y=1}$$

$$= \lim_{(\delta,\epsilon)\to(0,0)} \left[\frac{1}{1-\alpha} - \frac{\delta^{1-\alpha}}{1-\alpha} \right] \left[\frac{1}{1-\beta} - \frac{\epsilon^{1-\beta}}{1-\beta} \right]$$

$$= \left(\frac{1}{1-\alpha} \right)\left(\frac{1}{1-\beta} \right) \quad \text{as} \quad \lim_{x\to 0} x^r = 0 \text{ for } r > 0.$$

C.

6. Let $D = [1, \infty) \times [1, \infty]$. Let $1 < \gamma$ and $1 < \rho$. Evaluate:

$$\iint_D \frac{dx\,dy}{x^\gamma y^\rho}.$$

Let $\delta > 1$, $\epsilon > 1$. Define $D_{\delta,\epsilon}$: $1 \le x \le \delta$, $1 \le y \le \epsilon$

$$\therefore \iint_D \frac{dx\,dy}{x^\gamma y^\rho} = \lim_{\substack{\delta \to \infty^+ \\ \epsilon \to \infty^+}} \iint_{D_{\delta,\epsilon}} \frac{dx\,dy}{x^\gamma y^\rho}$$

$$= \lim_{\substack{\delta \to \infty^+ \\ \epsilon \to \infty^+}} \int_1^\delta \int_1^\epsilon \frac{dy\,dx}{x^\gamma y^\rho} = \lim_{\substack{\delta \to \infty^+ \\ \epsilon \to \infty^+}} \left[\frac{x^{1-\gamma}}{1-\gamma}\right]_{x=1}^{x=\delta} \left[\frac{y^{1-\rho}}{1-\rho}\right]_{y=1}^{y=\epsilon}$$

$$= \lim_{\substack{\delta \to \infty^+ \\ \epsilon \to \infty^+}} \left[\frac{\delta^{1-\gamma}}{1-\gamma} - \frac{1}{1-\gamma}\right]\left[\frac{\epsilon^{1-\rho}}{1-\rho} - \frac{1}{1-\rho}\right]$$

$$= \lim_{\substack{\delta \to \infty^+ \\ \epsilon \to \infty^+}} \left[\frac{1}{(1-\gamma)\,\delta^{\gamma-1}} + \frac{1}{\gamma-1}\right]\left[\frac{1}{(1-\rho)\,\epsilon^{\rho-1}} + \frac{1}{\rho-1}\right]$$

$$= \left[0 + \frac{1}{\gamma-1}\right]\left[0 + \frac{1}{\rho-1}\right] = \frac{1}{(\gamma-1)(\rho-1)}$$

7.

7. (a) Evaluate

$$\iint_D \frac{dA}{(x^2+y^2)^{2/3}},$$

where D is the unit disk in $\mathbb{R}^2$.

(a) Use polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, $0 \le \theta \le 2\pi$ and $0 < \delta \le r \le 1$ $\quad dA = r\,dr\,d\theta$

$$\therefore \iint_D \frac{dA}{(x^2+y^2)^{2/3}} = \lim_{\delta\to 0} \iint_{D_\delta} \frac{r\,dr\,d\theta}{(r^2)^{2/3}}$$

$$= \lim_{\delta\to 0} \int_\delta^1 \int_0^{2\pi} r^{-\frac{1}{3}}\,dr\,d\theta = 2\pi \lim_{\delta\to 0} \int_\delta^1 r^{-\frac{1}{3}}\,dr$$

$$= 2\pi \lim_{\delta\to 0} \left(\frac{3}{2} r^{\frac{2}{3}} \Big|_{r=\delta}^{r=1} \right) = 2\pi \lim_{\delta\to 0} \left(\frac{3}{2} - \frac{3}{2}\delta^{2/3} \right)$$

$$= 2\pi \left(\frac{3}{2}\right) = \underline{3\pi}$$

(b)

(b) Determine the real numbers λ for which the integral

$$\iint_D \frac{dA}{(x^2+y^2)^\lambda}$$

is convergent, where again D is the unit disk.

Using polar coordinates as in (a),

$$\iint_D \frac{dA}{(x^2+y^2)^\lambda} = \lim_{\delta\to 0} \iint_{D_\delta} \frac{r\,dr\,d\theta}{(r^2)^\lambda}, \text{ where } D_\delta \text{ is } 0 \le \theta \le 2\pi,\ 0 < \delta \le r \le 1$$

$$= \lim_{\delta \to 0} \int_{\delta}^{1} \int_{0}^{2\pi} r^{1-2\lambda} \, dr d\theta = 2\pi \lim_{\delta \to 0} \int_{\delta}^{1} r^{1-2\lambda} \, dr$$

$$= 2\pi \lim_{\delta \to 0} \left(\frac{r^{2-2\lambda}}{2-2\lambda} \Big|_{r=\delta}^{r=1} \right) , \quad 2-2\lambda \neq 0$$

$$= 2\pi \lim_{\delta \to 0} \left(\frac{1}{2-2\lambda} - \frac{\delta^{2-2\lambda}}{2-2\lambda} \right) \qquad \lim_{\delta \to 0} \delta^n = 0 \text{ only if } n > 0$$

$$= 2\pi \left(\frac{1}{2-2\lambda} - 0 \right) , \text{ for } 2-2\lambda > 0$$

$$= \frac{2\pi}{2-2\lambda} , \text{ for } 2-2\lambda > 0, \text{ or } \underline{\lambda < 1}$$

8.

8. (a) Discuss how you would define $\int\int_D f \, dA$ if D is an unbounded region—for example, the set of (x, y) such that $a \leq x < \infty$ and $\phi_1(x) \leq y \leq \phi_2(x)$, where $\phi_1 \leq \phi_2$ are given (Figure 6.4.5).

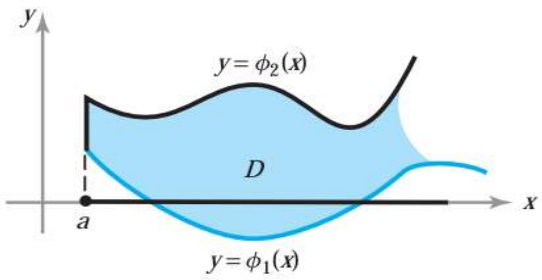


figure **6.4.5** An unbounded region D.

(a) Let D_δ be $a \leq x \leq \delta$, $\phi_1(x) \leq y \leq \phi_2(x)$

$$\iint_D f \, dA = \lim_{\delta \to +\infty} \iint_{D_\delta} f \, dA \text{ , if limit exists.}$$

$$= \lim_{\delta \to +\infty} \int_a^{\delta} \int_{\phi_1(x)}^{\phi_2(x)} f(x,y) \, dy dx$$

(6)

(b) Evaluate $\int\int_D xye^{-(x^2+y^2)} \, dx \, dy$ if $x \geq 0$, $0 \leq y \leq 1$.

Let D_δ be as in (a)

$$\lim_{\delta\to\infty}\int_0^{\delta}\int_0^{1} xy\,e^{-(x^2+y^2)}\,dy\,dx$$

$$=\lim_{\delta\to\infty}\int_0^{1} y e^{-y^2}\,dy\int_0^{\delta} x e^{-x^2}\,dx$$

$$=\left(\frac{e^{-y^2}}{-2}\Big|_{y=0}^{y=1}\right)\left(\lim_{\delta\to\infty}\frac{e^{-x^2}}{-2}\Big|_{x=0}^{x=\delta}\right)$$

$$=\left(-\frac{1}{2e}+\frac{1}{2}\right)\left(\lim_{\delta\to\infty}-\frac{1}{2e^{\delta^2}}+\frac{1}{2}\right)$$

$$=\left(\frac{1}{2}-\frac{1}{2e}\right)\left(\frac{1}{2}\right)=\frac{1}{4}\left(1-\frac{1}{e}\right)$$

9.

9. Using Exercise 8, integrate e^{-xy} for $x \geq 0$, $1 \leq y \leq 2$ in two ways. Assuming Fubini's theorem can be used, show that

$$\int_0^{\infty} \frac{e^{-x}-e^{-2x}}{x}\,dx = \log 2.$$

(a) Let D_δ be $1\le y\le 2$, $0<a\le x\le b$

$$\therefore \lim_{\substack{a\to 0\\ b\to\infty}}\int_a^{b}\int_1^{2} e^{-xy}\,dy\,dx = \lim_{\substack{a\to 0\\ b\to\infty}}\int_1^{2}\int_a^{b} e^{-xy}\,dx\,dy$$

as $e^{-xy}>0$

$$=\lim_{\substack{a\to 0\\ b\to\infty}}\int_1^{2}\left(\frac{e^{-xy}}{-y}\Big|_{x=a}^{x=b}\right)dy$$

$$= \lim_{\substack{a\to 0\\ b\to\infty}} \int_1^2 \left(\frac{1}{ye^{ay}} - \frac{1}{ye^{by}}\right) dy = \int_1^2 \left(\frac{1}{y} - 0\right) dy$$

$$= \int_1^2 \frac{dy}{y} = \log y \Big|_{y=1}^{y=2} = \underline{\log 2}$$

(6) Integrating by dy first in (a),

$$\lim_{\substack{a\to 0\\ b\to\infty}} \int_a^b \int_1^2 e^{-xy}\, dy\, dx = \lim_{\substack{a\to 0\\ b\to\infty}} \int_a^b \left(\frac{e^{-xy}}{-x}\Big|_{y=1}^{y=2}\right) dx$$

$$= \lim_{\substack{a\to 0\\ b\to\infty}} \int_a^b \left(\frac{e^{-x}}{x} - \frac{e^{-2x}}{x}\right) dx$$

$$= \int_0^\infty \frac{e^{-x} - e^{-2x}}{x}\, dx = \log 2 \quad \text{from (a)}$$

10.

10. Show that the integral

$$\int_0^1 \int_0^a (x/\sqrt{a^2 - y^2})\, dy\, dx$$

exists, and compute its value.

Define D_δ to be $0 \le x \le 1,\ 0 \le y \le \delta < a$

$$\therefore \int_0^1 \int_0^a \frac{x}{\sqrt{a^2-y^2}}\, dy\, dx = \lim_{\delta\to a} \int_0^1 \int_0^\delta \frac{x}{\sqrt{a^2-y^2}}\, dy\, dx$$

$$= \lim_{\delta \to a} \int_0^1 x\,dx \int_0^\delta \frac{1}{\sqrt{a^2-y^2}}\,dy$$

$$= \frac{1}{2} \lim_{\delta \to a} \left(\operatorname{Arcsin} \frac{y}{a} \Big|_{y=0}^{y=\delta} \right)$$

$$= \frac{1}{2} \lim_{\delta \to a} \left(\operatorname{Arcsin} \frac{\delta}{a} - 0 \right) = \frac{1}{2} \operatorname{Arcsin}(1)$$

$$= \frac{\pi}{4}$$

11.

11. Discuss whether the integral

$$\iint_D \frac{x+y}{x^2+2xy+y^2}\,dx\,dy$$

exists where $D = [0, 1] \times [0, 1]$. If it exists, compute its value.

$$\frac{x+y}{x^2+2xy+y^2} = \frac{x+y}{(x+y)^2} = \frac{1}{x+y}, \quad x+y \neq 0.$$

For this D, $x+y=0$ at $(0,0)$.

$$\therefore \iint_D \frac{x+y}{x^2+2xy+y^2}\,dx\,dy = \int_0^1 \int_0^1 \frac{1}{x+y}\,dx\,dy$$

$$= \int_0^1 \log(x+y) \Big|_{x=0}^{x=1} dy$$

$$= \int_0^1 \log(1+y) - \log(y)\, dy \qquad \int \ln x = x\ln x - x$$

$$= (1+y)\ln(1+y)\Big|_{y=0}^{y=1} - (1+y)\Big|_{y=0}^{y=1} - \int_0^1 \ln(y)\, dy$$

$$= 2\ln(2) - (2-1) - \lim_{a\to 0^+} \int_a^1 \ln(y)\, dy$$

$$= 2\ln(2) - 1 - \lim_{a\to 0^+} \Big\{ y\ln(y) - y \Big\}_{y=a}^{y=1}$$

$$= 2\ln(2) - 1 - \lim_{a\to 0^+} y\ln(y)\Big|_{y=a}^{y=1} + \lim_{a\to 0^+}(1-a)$$

$$= 2\ln(2) - \lim_{a\to 0^+}(0 - a\ln(a))$$

$$= 2\ln(2) + \lim_{a\to 0^+} \frac{\ln(a)}{\frac{1}{a}} = 2\ln(2) + \lim_{a\to 0^+} \frac{\frac{1}{a}}{-\frac{1}{a^2}}$$

$$= 2\ln(2) + \lim_{a\to 0^+}(-a) = \underline{2\ln(2)}$$

12. **12.** We can also consider improper integrals of functions that fail to be continuous on entire curves lying in some region D. For example, by breaking $D = [0, 1] \times [0, 1]$ into two regions, define and then discuss the convergence of the integral

$$\iint_D \frac{1}{\sqrt{|x-y|}}\, dx\, dy.$$

$|x-y| = 0$ when $y = x$. $\therefore$ Break D up into 2

regions, one above $y=x$, one below $y=x$.

$$D^+ = \{(x,y) \mid 0 \le x \le 1,\ 0 \le y \le 1,\ y \ge x\}$$

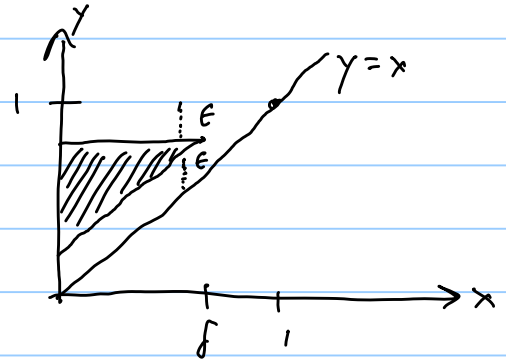
$$D^- = \{(x,y) \mid 0 \le x \le 1,\ 0 \le y \le 1,\ y \le x\}$$

$D = D^+ \cup D^-$ $\therefore$ $\displaystyle\iint_D \frac{dx\,dy}{\sqrt{|x-y|}} = \iint_{D^+} \frac{dx\,dy}{\sqrt{|x-y|}} + \iint_{D^-} \frac{dx\,dy}{\sqrt{|x-y|}}$

From problem #2 above, $\displaystyle\iint_{D^-} \frac{dx\,dy}{\sqrt{|x-y|}} = \frac{4}{3}$

For D^+, $\sqrt{|x-y|} = \sqrt{y-x}$, and similar to #2,

$$\iint_{D^+} \frac{dx\,dy}{\sqrt{|x-y|}} =$$



$$\lim_{\substack{\delta \to 1^- \\ \epsilon \to 0^+}} \int_0^{\delta} \int_{x+\epsilon}^{1-\epsilon} \frac{dy\,dx}{\sqrt{y-x}}$$

$$= \lim_{\substack{\delta \to 1^- \\ \epsilon \to 0^+}} \int_0^{\delta} \left(2\sqrt{y-x} \,\Big|_{y=x+\epsilon}^{y=1-\epsilon} \right) dx$$

$$= \lim_{\substack{\delta \to 1^- \\ \epsilon \to 0^+}} \int_0^{\delta} \left(2\sqrt{1-x-\epsilon} - 2\sqrt{\epsilon}\right) dx = \lim_{\delta \to 1^-} \int_0^{\delta} 2\sqrt{1-x}\, dx$$

$$= \lim_{\delta \to 1^-} 2\left(-\frac{2}{3}\right)(1-x)^{3/2} \Big|_0^{\delta} = -\frac{4}{3} \lim_{\delta \to 1^-} \left[(1-\delta)^{3/2} - 1\right]$$

$$= \frac{4}{3}.$$

$$\therefore \iiint_0 \frac{1}{\sqrt{|x-y|}}\, dx\, dy = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$$

13.

13. Let W be the first octant of the ball $x^2 + y^2 + z^2 \le a^2$, where $x \ge 0$, $y \ge 0$, $z \ge 0$. Evaluate the improper integral

$$\iiint_W \frac{(x^2 + y^2 + z^2)^{1/4}}{\sqrt{z + (x^2 + y^2 + z^2)^2}}\, dx\, dy\, dz$$

by changing variables.

Use spherical coordinates: $x^2+y^2+z^2 = \rho^2$, $z = \rho\cos\phi$, $0 \le \rho \le a$, $0 \le \theta \le \frac{\pi}{2}$, $0 \le \phi \le \frac{\pi}{2}$

$$\therefore \int_0^a \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \frac{(\rho^2)^{\frac{1}{4}}}{\sqrt{\rho\cos\phi + \rho^4}} \rho^2 \sin\phi\, d\theta\, d\phi\, d\rho$$

$$= \frac{\pi}{2} \int_0^a \int_0^{\frac{\pi}{2}} \rho^{5/2} \sin\phi \left(\rho\cos\phi + \rho^4\right)^{-\frac{1}{2}} d\phi\, d\rho$$

$$= \frac{\pi}{2} \int_0^a \rho^{5/2} \left(-\frac{2}{\rho}\right)\left(\rho\cos\phi + \rho^4\right)^{\frac{1}{2}} \Big|_{\phi=0}^{\phi=\frac{\pi}{2}} d\rho$$

$$= -\pi \int_0^a \rho^{3/2} \left[(\rho^4)^{\frac{1}{2}} - (\rho + \rho^4)^{\frac{1}{2}} \right] d\rho$$

$$= -\pi \int_0^a \rho^{7/2} - \rho^2 (1+\rho^3)^{\frac{1}{2}} \, d\rho$$

$$= -\pi \left[\frac{2}{9} \rho^{9/2} \Big|_0^a - \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)(1+\rho^3)^{\frac{3}{2}} \Big|_0^a \right]$$

$$= -\pi \left[\frac{2}{9} a^{9/2} - \frac{2}{9}(1+a^3)^{3/2} + \frac{2}{9} \right]$$

$$= -\frac{2\pi}{9} \left[a^{9/2} - (1+a^3)^{3/2} + 1 \right]$$

14. **14.** Let f be a nonnegative function that may be unbounded and discontinuous on the boundary of an elementary region D. Let g be a similar function such that $f(x, y) \le g(x, y)$ whenever both are defined. Suppose $\iint_D g(x, y)\, dA$ exists. Argue informally that this implies the existence of $\iint_D f(x, y)\, dA$.

This is somewhat analogous to a bounded series composed of positive terms: a monotonically increasing series that is bounded above has a limit. The upper bound in this case is $\iint_D g(x,y)\, dA$. However, "somewhat" is used because increasing the graniness of a partition doesn't guarantee that $S_{n+1} > S_n$, i.e., the Riemann sum of a finer partition may not be larger than a coarser partition.

$f(x,y)$ is nonnegative.

$\therefore 0 \le \lim_{n\to\infty} \sum_{j,k}^{n-1} f(\vec{c}_{jk})\,\Delta x\,\Delta y \le \iint_D g(x,y)\,dA$

15. **15.** Use Exercise 14 to show that

$$\iint_D \frac{\sin^2(x-y)}{\sqrt{1-x^2-y^2}}\,dy\,dx$$

exists where D is the unit disk $x^2 + y^2 \le 1$.

$\frac{\sin^2(x-y)}{\sqrt{1-x^2-y^2}}$ is nonnegative on D.

$\frac{\sin^2(x-y)}{\sqrt{1-x^2-y^2}} \le \frac{1}{\sqrt{1-x^2-y^2}}$ as $|\sin\theta| \le 1$ for all θ.

From Example #2 in text (p. 343), $\iint_D \frac{1}{\sqrt{1-x^2-y^2}}\,dA = 2\pi$

$\therefore$ From #14 above, $\iint_D \frac{\sin^2(x-y)}{\sqrt{1-x^2-y^2}}\,dA$ exists.

16. **16.** Let f be as in Exercise 14 and let g be a function such that $0 \le g(x, y) \le f(x, y)$ whenever both are defined. Suppose that $\iint_D g(x, y)\,dA$ does not exist. Argue informally that $\iint_D f(x, y)\,dA$ cannot exist.

By #14, if $\iint_D f\,dA$ did exist, then $\iint_D g\,dA$ would exist.

More directly, since $g(x,y)$ is nonnegative,

the Riemann sum of $g(x,y)$ over D is like an unbounded nonnegative sequence.

17. **17.** Use Exercise 16 to show that

$$\iint_D \frac{e^{x^2+y^2}}{x-y}\,dy\,dx$$

does not exist, where D is the set of (x, y) with $0 \le x \le 1$ and $0 \le y \le x$.

For all $0 \le x^2$, $e^0 \le e^{x^2} \Rightarrow 1 \le e^{x^2}$. Similarl for $0 \le y^2$, $1 \le e^{y^2}$. $\therefore$ $1 \le e^{x^2} \cdot e^{y^2} = e^{x^2+y^2}$

By Example #3 of text, p. 343, $\iint_D \frac{1}{x-y}\,dA$ does not exist, where D is $0 \le x \le 1$, $0 \le y \le x$.

$\therefore$ By #16 above, $\iint_D \frac{1}{x-y}\,dA \le \iint_D \frac{e^{x^2+y^2}}{x-y}\,dA$

does not exist.

18. **18.** Let D be the unbounded region defined as the set of (x, y, z) with $x^2 + y^2 + z^2 \ge 1$. By making a change of variables, evaluate the improper integral

$$\iiint_D \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^2}.$$

Use spherical coordinates

$$\iiint_D \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^2} = \iiint_D \frac{\rho^2 \sin\phi}{\rho^4}\,d\rho\,d\phi\,d\theta$$

$$= \int_1^{\infty} \int_0^{\pi} \int_0^{2\pi} \frac{\sin\phi}{\rho^2} \, d\theta \, d\phi \, d\rho$$

$$= 2\pi \int_1^{\infty} \int_0^{\pi} \frac{\sin\phi}{\rho^2} \, d\phi \, d\rho = -2\pi \int_1^{\infty} \frac{\cos\phi}{\rho^2} \Big|_{\phi=0}^{\phi=\pi} d\rho$$

$$= 4\pi \int_1^{\infty} \frac{d\rho}{\rho^2} = 4\pi \lim_{a \to \infty} (-\rho^{-1}) \Big|_{\rho=1}^{\rho=a}$$

$$= 4\pi \lim_{a \to \infty} \left[-\frac{1}{a} + 1 \right] = 4\pi$$

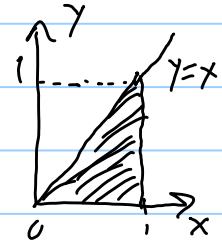
19.

19. Evaluate

$$\int_0^1 \int_0^y \frac{x}{y} \, dx \, dy \quad \text{and} \quad \int_0^1 \int_x^1 \frac{x}{y} \, dy \, dx.$$

Does Fubini's theorem apply?

(a) $$\int_0^1 \int_0^y \frac{x}{y} \, dx \, dy = \lim_{a \to 0} \int_a^1 \int_0^y \frac{x}{y} \, dx \, dy$$



$$= \lim_{a \to 0} \int_a^1 \left(\frac{x^2}{2y} \Big|_{x=0}^{x=y} \right) dy = \lim_{a \to 0} \int_a^1 \frac{y^2}{2y} \, dy$$

$$= \lim_{a \to 0} \frac{y^2}{4} \Big|_{y=a}^{y=1} = \lim_{a \to 0} \left\{ \frac{1}{4} - \frac{a^2}{4} \right\} = \underline{\frac{1}{4}}$$

Theorem 3 (Fubini's Theorem) on p. 343 of text does apply, since x/y is nonnegative on D, continuous on D except at $(x,0)$, a boundary of D.

$$\therefore \iint_D \frac{x}{y}\,dA = \int_0^1\int_0^y \frac{x}{y}\,dx\,dy = \int_0^1\int_x^1 \frac{x}{y}\,dy\,dx$$

(6)
$$\int_0^1\int_x^1 \frac{x}{y}\,dy\,dx = \int_0^1 \left(x \ln y \Big|_{y=x}^{y=1}\right) dx$$

$$= \int_0^1 -x\ln x\,dx = -x^2\left\{\frac{\ln x}{2} - \frac{1}{4}\right\}\Big|_{x=0}^{x=1}$$

$$= \frac{1}{4} + \frac{1}{2}\lim_{a\to 0^+} x^2\ln x = \frac{1}{4} + \frac{1}{2}\lim_{a\to 0^+} \frac{\ln x}{\frac{1}{x^2}}$$

$$= \frac{1}{4} + \frac{1}{2}\lim_{a\to 0^+} \frac{\frac{1}{x}}{\frac{-2}{x^3}} = \frac{1}{4} + \frac{1}{2}\lim_{a\to 0^+} -\frac{x^2}{2} = \frac{1}{4} + 0$$

$$= \underline{\frac{1}{4}}$$

20.

20. In Exercise 17 of Section 5.2 we showed that

$$\int_0^1\int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\,dy\,dx \neq \int_0^1\int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\,dx\,dy.$$

Thus, Fubini's theorem does not hold here, even though the iterated improper integrals both exist. What went wrong?

First, $\frac{x^2-y^2}{(x^2+y^2)^2}$ is not defined at $(0,0)$. This in itself does not violate Theorem 3, p. 343 of the text (Fubini's Theorem).

However, because of the term x^2-y^2, the function is not nonnegative in every neighborhood around $(0,0)$: if $r>0$, $D_r(0,0)$ contains points (x,y) for which $x^2-y^2>0$ (i.e., $x>y$), and points (x,y) for which $x^2-y^2<0$ (i.e., $x<y$). This does violate the premise of Fubini's Theorem.

21.

21. If $0 \le f(x, y) \le g(x, y)$ for all $(x, y) \in D$, and the improper integral of g

$$\iint_D g(x, y)\, dx\, dy$$

exists, then $\iint_D f(x, y)\, dx\, dy$ also exists. Use this fact and exercises 5 and 6 to argue that if $0 < \alpha, \beta < 1$ and $1 < \gamma, \rho$, then

$$\iint_D \frac{dx\, dy}{x^\alpha y^\beta + x^\gamma y^\rho}$$

exists, where $D = [0, \infty) \times [0, \infty)$.
[HINT: Write $D = D_1 \cup D_2$ and apply Exercise 14 to each D_i separately.]

It really should state $0 \le f(x,y) \le g(x,y)$ for all $(x,y) \in D$ except at boundary points of D for which f and g may not be defined. Better worded in exercise #14.

First consider $D_1 = [0,1] \times [0,1]$ for $(x,y) \in D_1$.

For $(x,y) \neq 0$, $\frac{1}{x^\alpha y^\beta + x^\gamma y^\rho} < \frac{1}{x^\alpha y^\beta}$

since $0 < x^\alpha y^\beta < x^\alpha y^\beta + x^\gamma y^\rho$ and $0 < x^\gamma y^\rho$

Since, by exercise #5, $\iint_{D_1} \frac{1}{x^\alpha y^\beta}\,dxdy$ exists,

Then by the premise $\iint_{D_1} \frac{1}{x^\alpha y^\beta + x^\gamma y^\rho}\,dxdy$ exists.

Now consider $D_2 = [1,\infty) \times [1,\infty)$ for $(x,y) \in D_2$

For every $(x,y) \in D_2$, $\frac{1}{x^\alpha y^\beta + x^\gamma y^\rho} < \frac{1}{x^\gamma y^\rho}$

since $x^\gamma y^\rho < x^\alpha y^\beta + x^\gamma y^\rho$ and $0 < x^\alpha y^\beta$

Since, by exercise #6, $\iint_{D_2} \frac{1}{x^\gamma y^\rho}\,dxdy$ exists,

then by the premise $\iint_{D_2} \frac{1}{x^\alpha y^\beta + x^\gamma y^\rho}\,dxdy$ exists.

$\therefore \iint_{D_1} \frac{1}{x^\alpha y^\beta + x^\gamma y^\rho}\,dxdy + \iint_{D_2} \frac{1}{x^\alpha y^\beta + x^\gamma y^\rho}\,dxdy =$

$\iint_{D} \frac{1}{x^\alpha y^\beta + x^\gamma y^\rho}\,dxdy$ exists since $D = D_1 \cup D_2$.

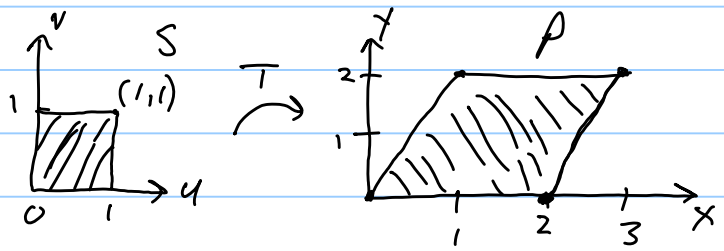
Review Exercises for Chapter 6

Note Title 1/15/2017

1.

1. (a) Find a linear transformation taking the square $S = [0, 1] \times [0, 1]$ to the parallelogram P with vertices $(0, 0)$, $(2, 0)$, $(1, 2)$, $(3, 2)$.

(b) Write down a change of variables formula appropriate to the transformation you found in part (a).



(a) Let $T = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $\therefore \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 1 \\ 0 & 2 & 2 \end{bmatrix}$

Taking transposes: $\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 3 & 2 \\ 1 & 2 \end{bmatrix}$

$\Rightarrow$ $a(1) + b(0) = 2,\ a = 2$ $\quad c(1) + d(0) = 0,\ c = 0$

$a(0) + b(1) = 1,\ b = 1$ $\quad c(0) + d(1) = 2,\ d = 2$

$\Rightarrow T = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

(b) $\det(T) = 4$. Note the $T: (u,v) \to (x,y)$.

So, $\iint_S f \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv = \iint_P f\,dx\,dy$

Here, $T(u,v) = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 2u+v \\ 2v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$

$\therefore \left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} = \det(T) = 4$

$$\therefore \iint_{D^*} f(x(u,v), y(u,v)) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du\,dv = \iint_D f(x,y)\,dx\,dy$$

$$\text{or, } \iint_{D^*} f(T(u,v))\,\det(T)\,du\,dv = \iint_{T(D^*)} f(x,y)\,dx\,dy$$

∴ In this example, $T(S) = P$

$$\iint_S f(x(u,v), y(u,v))\,\det(T)\,du\,dv = \iint_P f(x,y)\,dx\,dy$$

$$\text{or } \int_0^1 \int_0^1 f(2u+v, 2v)\,(4)\,du\,dv = \iint_P f(x,y)\,dx\,dy$$

(simple region, complex integrand → complex region, simple integrand)

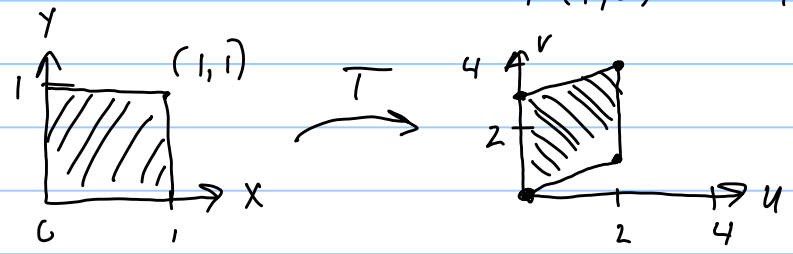
2. **2.** (a) Find the image of the square $[0, 1] \times [0, 1]$ under the transformation $T(x, y) = (2x, x + 3y)$.

   (b) Write down a change of variables formula appropriate to the transformation and the region you found in part (a).

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ x+3y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

(a) Look at corners: $T(0,0)=(0,0)$ $T(1,1)=(2,4)$

$T(1,0)=(2,1)$ $T(0,1)=(0,3)$



parallelogram

$$\begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

(b) S = square P = parallelogram

$$\iint_S f(u(x,y),v(x,y))\left|\frac{\partial(u,v)}{\partial(x,y)}\right| dx\,dy = \iint_P f(u,v)\,du\,dv$$

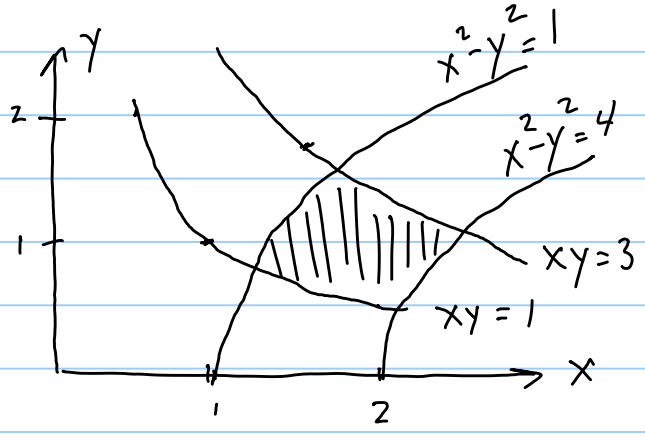
$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \det(T) = \det\begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = 6$$

So,

$$6\iint_S f(2x, x+3y)\,dx\,dy = \iint_P f(u,v)\,du\,dv$$

---

3. **3.** Let B be the region in the first quadrant bounded by the curves $xy=1$, $xy=3$, $x^2-y^2=1$, and $x^2-y^2=4$. Evaluate $\iint_B (x^2+y^2)\,dx\,dy$ using the change of variables $u=x^2-y^2$, $v=xy$.



$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x^2-y^2 \\ xy \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

$$\therefore \left|\frac{\partial(u,v)}{\partial(x,y)}\right| = \begin{vmatrix} 2x & -2y \\ y & x \end{vmatrix} = 2x^2+2y^2 = 2(x^2+y^2)$$

Note The given integrand already contains a form of the Jacobian determinant.

$$\therefore \frac{1}{2}\iint_B 2(x^2+y^2)\,dx\,dy = \iint_B \frac{1}{2}\left|\frac{\partial(u,v)}{\partial(x,y)}\right| dx\,dy = \iint_{B^*} \frac{1}{2}\,du\,dv$$

where $f(x,y) = \frac{1}{2} = f(u,v)$

[using Theorem 2, p. 319 of text, setting $B = D^*$, $B^* = D$, and swapping the dummy variables x,y and u,v. Also assuming T is one-to-one and onto].

So what is $B^* = T(B)$?

Note That the borders of B are $xy = 1$, $xy = 3$, (so $v = 1$, $v = 3$) and $x^2 - y^2 = 1$, $x^2 - y^2 = 4$ (so $u = 1$, $u = 4$)

$$\therefore B^* = \{(u,v) : 1 \le u \le 4,\ 1 \le v \le 3\}$$

$$\therefore \iint_{B^*} \frac{1}{2}\,du\,dv = \frac{1}{2}\int_1^4 \int_1^3 dv\,du = \frac{1}{2}\int_1^4 \left(v\Big|_1^3\right) du$$

$$= \frac{1}{2}(2)\int_1^4 du = \frac{1}{2}(2)\left(u\Big|_1^4\right) = \frac{1}{2}(2)(3) = \underline{3}$$

Note: T is one-to-one: Suppose $T\begin{bmatrix} x \\ y \end{bmatrix} = T\begin{bmatrix} 9 \\ 6 \end{bmatrix}$

Then $x^2-y^2 = a^2-b^2$ and $x\neq 0, y\neq 0$

$xy = ab$ $\quad a\neq 0, b\neq 0$

$\therefore x = \frac{ab}{y} \Rightarrow \left(\frac{ab}{y}\right)^2 + b^2 = a^2 + y^2$

$\therefore a^2b^2 + y^2b^2 = a^2y^2 + y^4$

$\Rightarrow (a^2+y^2)b^2 = (a^2+y^2)y^2$

$\Rightarrow b^2 = y^2 \Rightarrow y = b$

and from $xy = a(y)$, $x = a$

T is onto: Given any $(u,v) \in \{1\le u\le 4, 1\le v\le 3\}$

Need to find an $(x,y) \in B$ s.t. $T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$

i.e., $x^2-y^2 = u$, $xy = v$

Let $y = \frac{v}{x}$ $\therefore x^2 - \left(\frac{v^2}{x^2}\right) = u$

$\therefore x^4 - ux^2 - v^2 = 0$, $x^2 = \frac{u \pm \sqrt{u^2 - 4(-v^2)}}{2}$

$x^2 = \frac{u + \sqrt{u^2+4v^2}}{2}$ since $x^2 > 0$

$\therefore x = \sqrt{\frac{u + \sqrt{u^2+4v^2}}{2}}$, and $y = \frac{v}{x}$

and these satisfy $1\le x^2-y^2\le 4$, $1\le xy\le 3$ under the above construction.

4.

4. In parts (a) to (d), make the indicated change of variables. (Do not evaluate.)

(a) $\int_0^1 \int_{-1}^1 \int_{-\sqrt{(1-y^2)}}^{\sqrt{(1-y^2)}} (x^2+y^2)^{1/2}\,dx\,dy\,dz$, cylindrical coordinates

$x = r\cos\theta$
$y = r\sin\theta$
$z = z$
$\frac{\partial(x,y,z)}{\partial(r,\theta,z)} = r$

Here, $0 \le r \le 1$, $0 \le \theta \le 2\pi$, $(x^2+y^2)^{\frac{1}{2}} = r$

$\therefore \int_0^1 \int_0^{2\pi} \int_0^1 r^2\,dr\,d\theta\,dz$

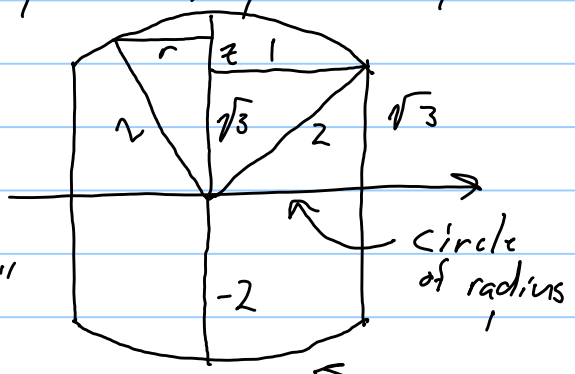
(b) $\int_{-1}^1 \int_{-\sqrt{(1-y^2)}}^{\sqrt{(1-y^2)}} \int_{-\sqrt{(4-x^2-y^2)}}^{\sqrt{(4-x^2-y^2)}} xyz\,dz\,dx\,dy$, cylindrical coordinates

$x = r\cos\theta$ $\quad J = r$
$y = r\sin\theta$
$z = z$

A sphere of radius 2: $-\sqrt{4-x^2-y^2} \le z \le \sqrt{4-x^2-y^2}$
$\Rightarrow z^2 \le 4-x^2-y^2 \Rightarrow x^2+y^2+z^2 \le 2$

"caps" top & bottom of a cylinder of radius 1: $-1 \le y \le 1$, $-\sqrt{1-y^2} \le x \le \sqrt{1-y^2} \Rightarrow x^2+y^2 \le 1$.

$\therefore 0 \le r \le 1$
$0 \le \theta \le 2\pi$
$-\sqrt{3} \le z \le \sqrt{3}$ for cylinder minus top & bottom "caps"



For caps: $r^2+z^2 = 4$, $z = \pm\sqrt{4-r^2}$

$\therefore \int_0^{2\pi} \int_0^1 \int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} (r\cos\theta)(r\sin\theta)z\,(r)\,dz\,dr\,d\theta$

$$= \int_0^{2\pi} \int_0^{1} \int_{-\sqrt{4-r^2}}^{\sqrt{4-r^2}} r^3 \cos\theta \sin\theta \; z \, dz \, dr \, d\theta$$

(C) (c) $\int_{-\sqrt{2}}^{\sqrt{2}} \int_{-\sqrt{(2-y^2)}}^{\sqrt{(2-y^2)}} \int_{\sqrt{(x^2+y^2)}}^{\sqrt{(4-x^2-y^2)}} z^2 \, dz \, dx \, dy$, spherical coordinates

$z = \rho\cos\phi \qquad J = \rho^2 \sin\phi$

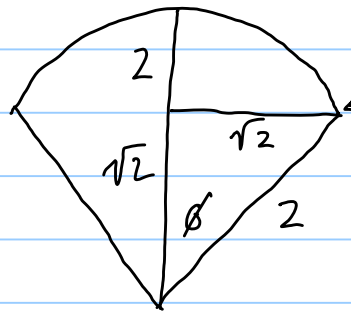
$y = \rho \sin\phi \sin\theta$

$x = \rho \sin\phi \cos\theta$

$-\sqrt{2} \le y \le \sqrt{2}$, $-\sqrt{2-y^2} \le x \le \sqrt{2-y^2}$ is a circle of radius $\sqrt{2}$

$z^2 = x^2 + y^2$ is a cone, $z \ge \sqrt{x^2+y^2}$ a cone above xy-plane.

$z = \sqrt{4 - x^2 - y^2}$ is a sphere of radius 2.



$\sqrt{x^2+y^2} = \sqrt{4-x^2-y^2}$

$\cos\phi = \frac{\sqrt{2}}{2}$, $\phi = \frac{\pi}{4}$

$\therefore \; 0 \le \rho \le 2$

$0 \le \theta \le 2\pi$

$0 \le \phi \le \pi/4$

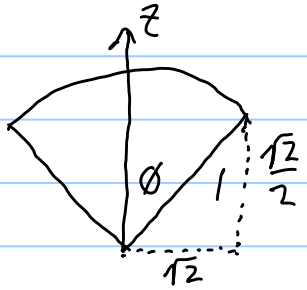
$$\therefore \int_0^{\pi/4} \int_0^{2\pi} \int_0^{2} (\rho\cos\phi)^2 (\rho^2 \sin\phi) \, d\rho \, d\theta \, d\phi$$

$$= \int_0^{\frac{\pi}{4}} \int_0^{2\pi} \int_0^{2} \rho^4 \cos^2\phi \sin\phi \, d\rho \, d\theta \, d\phi$$

(d) (d) $\int_0^1 \int_0^{\pi/4} \int_0^{2\pi} \rho^3 \sin 2\phi \, d\theta \, d\phi \, d\rho$, rectangular coordinates

Note $\sin 2\phi = 2\sin\phi\cos\phi$, so the Jacobian of $T\begin{bmatrix}\rho\\ \theta\\ \phi\end{bmatrix} = \begin{bmatrix}x\\ y\\ z\end{bmatrix}$ seems to be embedded in the integrand. This seems easier than finding the Jacobian of $T\begin{bmatrix}x\\ y\\ z\end{bmatrix} = \begin{bmatrix}\rho\\ \theta\\ \phi\end{bmatrix} = \begin{bmatrix}\sqrt{x^2+y^2+z^2}\\ \text{Arctan}\frac{y}{x}\\ \text{Arccos}\frac{z}{\sqrt{x^2+y^2+z^2}}\end{bmatrix}$

$$\therefore \int_0^1 \int_0^{\frac{\pi}{4}} \int_0^{2\pi} \rho^3 \, 2\sin\phi\cos\phi \, d\theta \, d\phi \, d\rho$$



$$= \int_0^1 \int_0^{\frac{\pi}{4}} \int_0^{2\pi} 2\rho\cos\phi \, (\rho^2\sin\phi) \, d\theta \, d\phi \, d\rho \quad [1]$$

This cone projects to the xy-plane as a disc of radius $\sqrt{\frac{1}{2}} = \frac{\sqrt{2}}{2}$ $\left(\sqrt{\frac{1}{2}}^2 + \sqrt{\frac{1}{2}}^2 = 1^2\right)$

$\therefore$ For x: $-\frac{\sqrt{2}}{2} \le x \le \frac{\sqrt{2}}{2}$

For y: $x^2+y^2 = \frac{1}{2}$, $-\sqrt{\frac{1}{2}-x^2} \le y \le \sqrt{\frac{1}{2}-x^2}$

For z, a sphere of radius 1, $x^2+y^2+z^2 = 1$,

The upper limit is $z \le \sqrt{1-x^2-y^2}$

The lower bound for z is a cone: $z^2 = x^2+y^2$,

or $\sqrt{x^2+y^2} \le z$ (upper half of cone).

From [1],

$$\int_0^1 \int_0^{\frac{\pi}{4}} \int_0^{2\pi} 2\rho\cos\phi \left|\frac{\partial(x,y,z)}{\partial(\rho,\theta,\phi)}\right| d\theta\, d\phi\, d\rho$$

$$= \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \int_{-\sqrt{\frac{1}{2}-x^2}}^{\sqrt{\frac{1}{2}-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{1-x^2-y^2}} 2\sqrt{x^2+y^2+z^2}\,\frac{z}{\sqrt{x^2+y^2+z^2}}\, dz\, dy\, dx$$

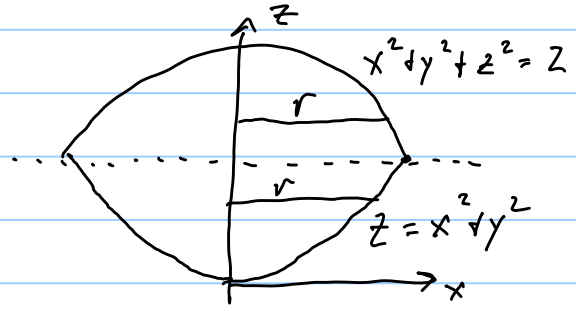
$$= \int_{-\frac{\sqrt{2}}{2}}^{\frac{\sqrt{2}}{2}} \int_{-\sqrt{\frac{1}{2}-x^2}}^{\sqrt{\frac{1}{2}-x^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{1-x^2-y^2}} 2z\, dz\, dy\, dx$$

---

5. **5.** Find the volume inside the surfaces $x^2 + y^2 = z$ and $x^2 + y^2 + z^2 = 2$.

$z = x^2+y^2$ is a paraboloid

$x^2+y^2+z^2 = 2$ is a sphere of radius $\sqrt{2}$



The point of intersection is: $x^2+y^2 = z \Rightarrow (z)+z^2 = 2$,

$(z-1)(z+2) = 0$, $z = 1$.

$\therefore$ projection onto xy-plane is a unit disc as $z = 1 = x^2+y^2$.

$\therefore$ Using cylindrical coordinates, $0 \le \theta \le 2\pi$,

$0 \le r \le 1$.

z extends from $z = x^2+y^2$ to $z = \sqrt{2-x^2-y^2}$

But $x^2+y^2 = r^2$, $\therefore$ $r^2 \le z \le \sqrt{2-r^2}$

$$\therefore \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{x^2+y^2}^{\sqrt{2-x^2-y^2}} dz\,dy\,dx = \iiint_{V^*} r\,dr\,d\theta\,dz$$

$$= \int_0^{2\pi} \int_0^1 \int_{r^2}^{\sqrt{2-r^2}} r\,dz\,dr\,d\theta$$

$$= \int_0^{2\pi} \int_0^1 \left(r z \Big|_{z=r^2}^{z=\sqrt{2-r^2}} \right) dr\,d\theta = \int_0^{2\pi} \int_0^1 r(2-r^2)^{\frac{1}{2}} - r^3\,dr\,d\theta$$

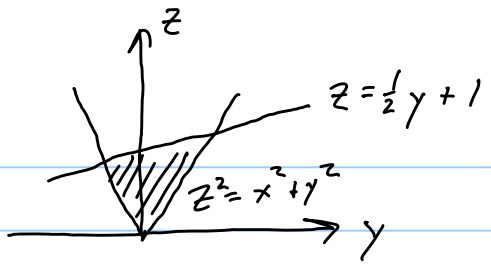
$$= \int_0^{2\pi} -\tfrac{1}{2}\left(\tfrac{2}{3}\right)(2-r^2)^{\frac{3}{2}} \Big|_{r=0}^{1} - \frac{r^4}{4}\Big|_{r=0}^{1} \, d\theta$$

$$= \int_0^{2\pi} \left\{ -\frac{1}{3} - \left(-\frac{2\sqrt{2}}{3}\right) - \frac{1}{4} \right\} d\theta = \left(\frac{2\sqrt{2}}{3} - \frac{7}{12}\right) 2\pi$$

$$= \pi \left(\frac{4\sqrt{2}}{3} - \frac{7}{6}\right)$$

C.

6. Find the volume enclosed by the cone $x^2 + y^2 = z^2$ and the plane $2z - y - 2 = 0$.



The object projects onto the xy-plane as an ellipse.

Find the equation of the projection.

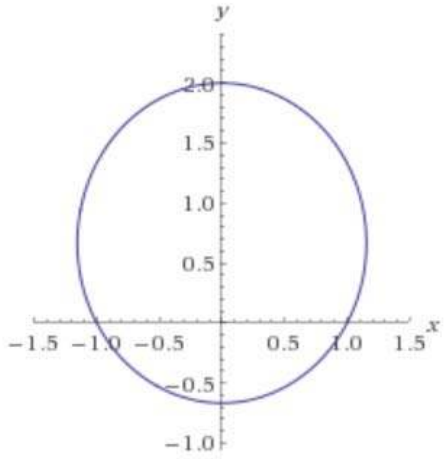
$$x^2 + y^2 = z^2 = \left(\tfrac{1}{2}y + 1\right)^2 = \tfrac{1}{4}y^2 + y + 1$$

$$\therefore\ x^2 + \tfrac{3}{4}y^2 - y = 1 \Rightarrow \frac{x^2}{\frac{3}{4}} + y^2 - \tfrac{4}{3}y = \tfrac{4}{3}$$

$$\Rightarrow \frac{x^2}{\frac{3}{4}} + \left(y - \tfrac{2}{3}\right)^2 = \tfrac{4}{3} + \tfrac{4}{9} = \tfrac{16}{9}$$

$$\Rightarrow \frac{x^2}{\frac{4}{3}} + \frac{\left(y - \frac{2}{3}\right)^2}{16/9} = 1, \text{ or } \frac{x^2}{\left(\sqrt{4/3}\right)^2} + \frac{\left(y - \frac{2}{3}\right)^2}{(4/3)^2} = 1$$

This is an ellipse centered at $(0, \frac{2}{3})$
major axis of $4/3$ along y-axis,
minor axis of $\sqrt{4/3}$ along x-axis.



From analytic geometry,

$$\frac{x^2}{a^2(1-e^2)} + \frac{y^2}{a^2} = 1$$

with focus at $(0, \pm ae)$
e = eccentricity, assuming no translation.

$$\therefore\ a^2(1-e^2) = \frac{16}{9}\left(1-e^2\right) = \frac{4}{3},\ 1-e^2 = \frac{3}{4},\ e = \frac{1}{2}$$

$$f = \pm ae = \pm \frac{4}{3}\left(\frac{1}{2}\right) = \pm \frac{2}{3}$$

The equation of the ellipse above, if a focus is at the pole, and directrix parallel and below the polar axis (the x-axis) is:

$$r = \frac{ed}{1 - e\sin\theta}\ .\ d = \frac{a}{e} - ae = \frac{a(1-e^2)}{e}$$

In this case, $d = \dfrac{\frac{4}{3}\left(1-\frac{1}{4}\right)}{\frac{1}{2}} = 2$

$$\therefore\ r = \frac{\frac{1}{2}(2)}{1-\frac{1}{2}\sin\theta} = \frac{2}{2-\sin\theta}$$

That is, $r=0$ corresponds to the lower focus at $(0, -2/3)$.

But the above ellipse has the lower focus translated upward by $2/3$, and $\therefore$ is at the origin

$\therefore\ r = \dfrac{2}{2-\sin\theta}$ describes the ellipse in the xy-plane. in polar coordinates

(a) Cylindrical coordinates:

The solid can be described as (r, θ, z) such that

$$0 \le \theta \le 2\pi,\ 0 \le r \le \frac{2}{2-\sin\theta}$$

$\sqrt{x^2+y^2} \le z \le \frac{1}{2}y + 1$. Since $\sqrt{x^2+y^2} = r$, $y = r\sin\theta$

$\therefore \; r \le z \le \frac{r\sin\theta}{2} + 1$

$$\therefore \; V = \int_0^{2\pi} \int_0^{\frac{2}{2-\sin\theta}} \int_r^{\frac{r\sin\theta}{2}+1} r \, dz\, dr\, d\theta$$

From Wolfram:

Definite integral:

$$\int_0^{2\pi} \int_0^{\frac{2}{2-\sin(x)}} \int_r^{\frac{1}{2}r\sin(x)+1} r \, dz\, dr\, dx = \frac{8\pi}{9\sqrt{3}} \approx 1.61227$$

The integration can be performed up to a point:

$$= \int_0^{2\pi} \int_0^{\frac{2}{2-\sin\theta}} \left(r\, z \Big|_{z=r}^{z=\frac{r\sin\theta}{2}+1} \right) dr\, d\theta$$

$$= \int_0^{2\pi} \int_0^{\frac{2}{2-\sin\theta}} \left(\frac{r^2\sin\theta}{2} + r - r^2 \right) dr\, d\theta$$

$$= \int_0^{2\pi} \frac{r^3}{3}\left(\frac{\sin\theta}{2} - 1\right) + \frac{r^2}{2} \Big|_{r=0}^{r=\frac{2}{2-\sin\theta}} d\theta$$

$$= \int_0^{2\pi} \frac{8}{3(2-\sin\theta)^3} \frac{(\sin\theta - 2)}{2} + \frac{2}{(2-\sin\theta)^2} \, d\theta$$

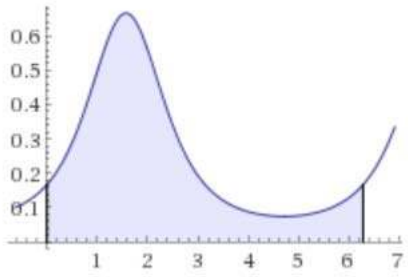
$$= \int_0^{2\pi} -\frac{4}{3}\frac{1}{(2-\sin\theta)^2} + \frac{2}{(2-\sin\theta)^2}\, d\theta$$

$$= \frac{2}{3}\int_0^{2\pi} \frac{1}{(2-\sin\theta)^2}\, d\theta$$

Definite integral:

$$\int_0^{2\pi} \frac{2}{3\,(2-\sin(x))^2}\, dx = \frac{8\pi}{9\sqrt{3}} \approx 1.6123$$

Visual representation of the integral:



From Wolfram:

which agrees with the above. The indefinite integral is:

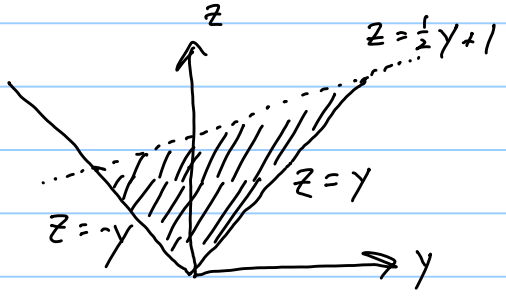
$$\int \frac{2}{3\,(2-\sin(x))^2}\, dx = \frac{2\left(3\cos(x) - 4\sqrt{3}\,(\sin(x)-2)\tan^{-1}\left(\frac{1-2\tan\left(\frac{x}{2}\right)}{\sqrt{3}}\right)\right)}{27\,(\sin(x)-2)} + \text{constant}$$

Such rational functions of sine can be evaluated using the substitution $u = \tan\frac{\theta}{2}$ so that $\sin\theta = \frac{2u}{1+u^2}$, then using partial fractions. Too much work.

$\therefore$ Volume $= \underline{\frac{8\pi}{9\sqrt{3}}}$

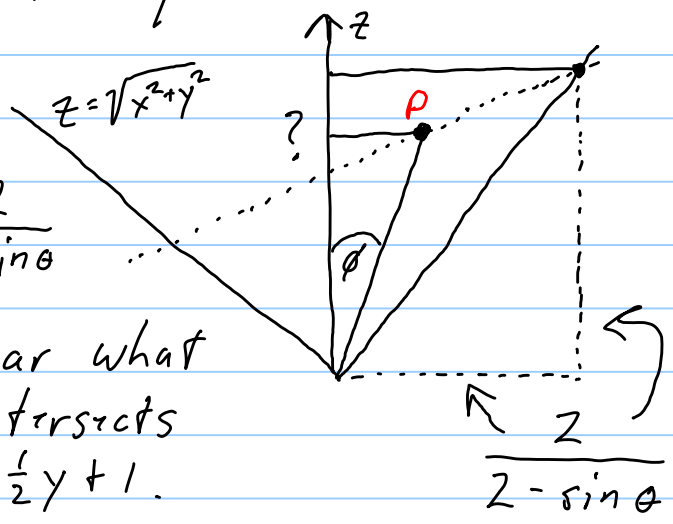
(b) Spherical coordinates:

The solid can be described as $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi/4$



But for a fixed θ and ϕ, it is not immediately clear what the extent of ρ is.

From above, the outer edge of the cone has height $= \frac{2}{2-\sin\theta}$



However, it is not clear what height the ray ϕ intersects the angled plane $z = \frac{1}{2}y + 1$.

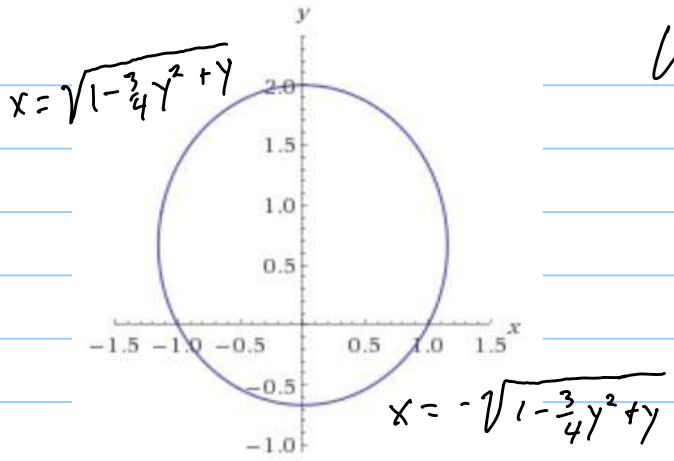
Viewed in profile, the plane $z = \frac{1}{2}y + 1$ will intersect the z-axis at different angles depending on θ. Thus, the extent of ρ from 0 to ρ is not clear.

A height of $\frac{2}{\cos\phi\,(2-\sin\theta)}$ works fine for $\phi = \frac{\pi}{4}$, but this formula doesn't work for other values of ϕ.

As a result, this problem does not lend itself easily to computation via spherical coordinates.

(c) Rectangular coordinates

From $x^2 + \frac{3}{4}y^2 - y = 1$, $x = \pm\sqrt{1 - \frac{3}{4}y^2 + y}$



Use x-simple curves.

Limits for y: $-\frac{2}{3} \le y \le 2$

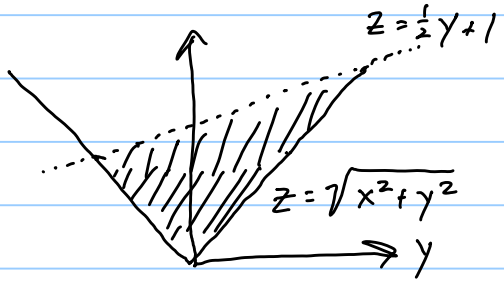
obtained by setting $x = 0$

so that

$\frac{3}{4}y^2 - y = 1 \Rightarrow y^2 - \frac{4}{3}y = \frac{4}{3}$

$\Rightarrow (y-\frac{2}{3})^2 = \frac{16}{9} \Rightarrow y = \frac{2}{3} \pm \frac{4}{3}$

$\Rightarrow y = -2/3, 2$



$\therefore$

$$V = \int_{-\frac{2}{3}}^{2} \int_{-\sqrt{1-\frac{3}{4}y^2+y}}^{\sqrt{1-\frac{3}{4}y^2+y}} \int_{\sqrt{x^2+y^2}}^{\frac{1}{2}y+1} dz\,dx\,dy$$

From Wolfram:

Definite integral:

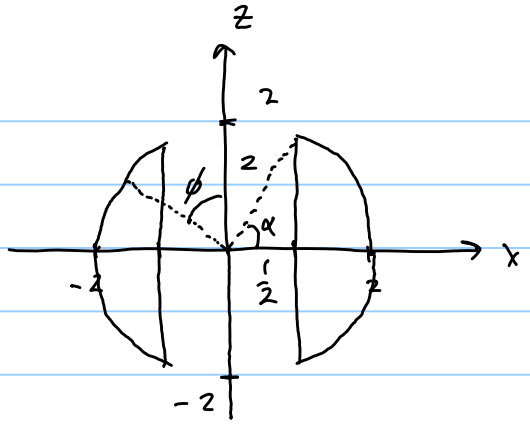
$$\int_{-\frac{2}{3}}^{2} \int_{-\sqrt{1-\frac{3y^2}{4}+y}}^{\sqrt{1-\frac{3y^2}{4}+y}} \int_{\sqrt{x^2+y^2}}^{\frac{y}{2}+1} 1\,dz\,dx\,dy = 1.61227$$

This agrees with cylindrical coordinates of

$$\frac{8\pi}{9\sqrt{3}} \approx 1.61$$

7.

7. A cylindrical hole of diameter 1 is bored through a sphere of radius 2. Assuming that the axis of the cylinder passes through the center of the sphere, find the volume of the solid that remains.

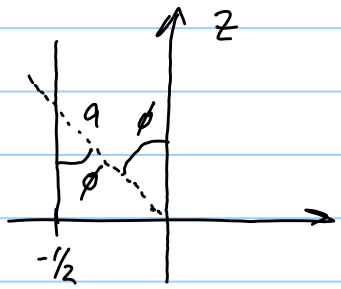


Use spherical coordinates

$\cos\alpha = \frac{1}{4}$, $\sin\phi = \frac{1}{4}$

$\therefore \text{Arcsin}\left(\frac{1}{4}\right) \le \phi \le \pi - \text{Arcsin}\left(\frac{1}{4}\right)$, $0 \le \theta \le 2\pi$

ρ is a function of ϕ, not θ.



Note that $|a\sin\phi| = \frac{1}{2}$

where a = hypotenuse of triangle

$\frac{1}{2\sin\phi}$ = minimum of ρ given ϕ.

$\therefore \frac{1}{2\sin\phi} \le \rho \le 2$.

$\therefore$

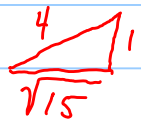
$$\text{Volume} = \int_0^{2\pi} \int_{\text{Arcsin}\left(\frac{1}{4}\right)}^{\pi - \text{Arcsin}\left(\frac{1}{4}\right)} \int_{\frac{1}{2\sin\phi}}^{2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_{\sin^{-1}\left(\frac{1}{4}\right)}^{\pi - \sin^{-1}\left(\frac{1}{4}\right)} \sin\phi \, \frac{\rho^3}{3} \Big|_{\rho = \frac{1}{2\sin\phi}}^{\rho = 2} \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_{\sin^{-1}\left(\frac{1}{4}\right)}^{\pi - \sin^{-1}\left(\frac{1}{4}\right)} \frac{8}{3}\sin\phi - \frac{1}{24}\frac{1}{\sin^2\phi} \, d\phi \, d\theta$$

$\frac{1}{\sin^2\phi} = \csc^2\phi$

$$= \int_0^{2\pi} \frac{8}{3}\left[-\cos\phi\right]_{\phi=\sin^{-1}\left(\frac{1}{4}\right)}^{\pi-\sin^{-1}\left(\frac{1}{4}\right)} - \frac{1}{24}\left[-\cot\phi\right]_{\sin^{-1}\left(\frac{1}{4}\right)}^{\pi-\sin^{-1}\left(\frac{1}{4}\right)} d\theta$$



If $\sin\phi = \frac{1}{4}$, $\cos\phi = \frac{\sqrt{15}}{4}$, $\tan\phi = \frac{1}{\sqrt{15}}$

$$\therefore \sin^{-1}\left(\frac{1}{4}\right) \Leftrightarrow \cos^{-1}\left(\frac{\sqrt{15}}{4}\right) \Leftrightarrow \tan^{-1}\left(\frac{1}{\sqrt{15}}\right) \Leftrightarrow \cot^{-1}\left(\sqrt{15}\right)$$

$$= \int_0^{2\pi} \frac{8}{3}\left[-\cos\phi\right]_{\phi=\cos^{-1}\left(\frac{\sqrt{15}}{4}\right)}^{\pi-\cos^{-1}\left(\frac{\sqrt{15}}{4}\right)} + \frac{1}{24}\left[\cot\phi\right]_{\cot^{-1}\left(\sqrt{15}\right)}^{\pi-\tan^{-1}\left(\frac{1}{\sqrt{15}}\right)} d\theta$$

$\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$ $\quad$ $\cot(\alpha-\beta) = \frac{1+\tan\alpha\tan\beta}{\tan\alpha-\tan\beta}$

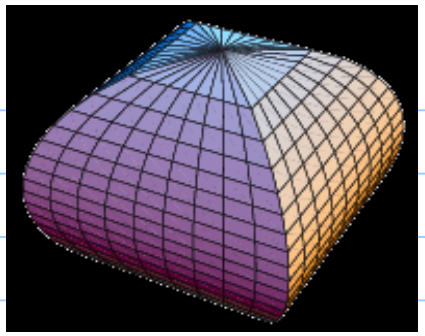
$$= \int_0^{2\pi} \frac{8}{3}\left[\frac{\sqrt{15}}{4} - \left(-\frac{\sqrt{15}}{4}\right)\right] + \frac{1}{24}\left[\frac{1}{-\frac{1}{\sqrt{15}}} - \sqrt{15}\right] d\theta$$

$$= \int_0^{2\pi} \frac{4}{3}\sqrt{15} - \frac{1}{12}\sqrt{15}\, d\theta = \int_0^{2\pi} \frac{5}{4}\sqrt{15}\, d\theta$$

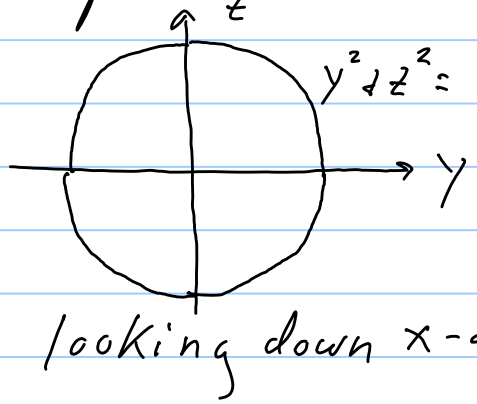
$$= \frac{5\sqrt{15}}{2}\pi$$

8.

8. Let C_1 and C_2 be two cylinders of infinite extent, of diameter 2, and with axes on the x and y axes, respectively. Find the volume of their intersection, $C_1 \cap C_2$.



Using above figure, place the two "points" on the z-axis. ∴ One cylinder (purple) is parallel to the y-axis, the other (yellow) parallel to the x-axis.



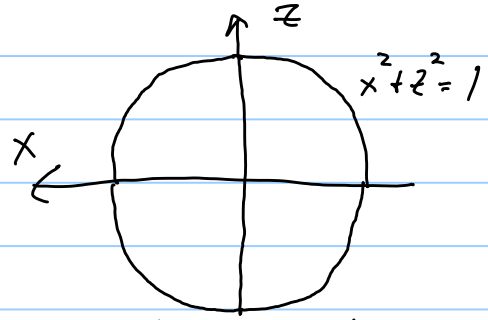
looking down x-axis

The solid projects onto the yz-plane as a circle of radius 1: $y^2+z^2=1$

$\therefore -1 \le z \le 1, \quad -\sqrt{1-z^2} \le y \le \sqrt{1-z^2}$

The extent of x depends on z

$$-\sqrt{1-z^2} \le x \le \sqrt{1-z^2}$$



looking down y-axis

$\therefore$

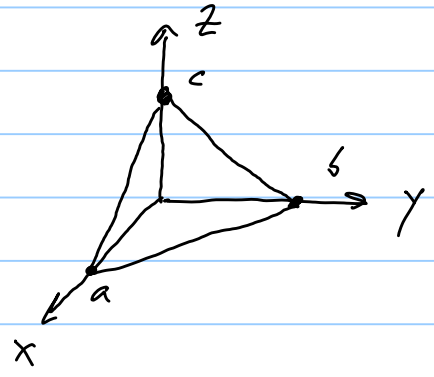
$$\text{Volume} = \int_{-1}^{1} \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} dx\,dy\,dz$$

$$= \int_{-1}^{1} \int_{-\sqrt{1-z^2}}^{\sqrt{1-z^2}} 2\sqrt{1-z^2}\,dy\,dz$$

$$= \int_{-1}^{1} 4(1-z^2)\,dz = \int_{-1}^{1} 4-4z^2\,dz = 4z - \frac{4}{3}z^3 \Big|_{-1}^{1}$$

$$= 4\cdot\frac{4}{3} - \left(-4 + \frac{4}{3}\right) = 8 - \frac{8}{3} = \underline{\underline{\frac{16}{3}}}$$

9. **9.** Find the volume bounded by $x/a + y/b + z/c = 1$ and the coordinate planes.



A plane with axis intercepts of $(a,0,0)$, $(0,b,0)$, $(0,0,c)$

Unknown if a, b, c are positive or negative.

Note that $\int_0^n = -\int_n^0$, so will take absolute value of final formula as "volume" ≥ 0.

$\therefore$ Assume $0 \leq x \leq a$ (if would be $a \leq x \leq 0$ if $a \leq 0$).

Note plane intersects xy-plane in a line,

$y = -\frac{b}{a}x + b$ $\therefore$ $0 \leq y \leq -\frac{b}{a}x + b$

$\frac{z}{c} = 1 - \frac{x}{a} - \frac{y}{b}$, $\therefore$ $z = c - \frac{c}{a}x - \frac{c}{b}y$

$\therefore$ $0 \leq z \leq c - \frac{c}{a}x - \frac{c}{b}y$

Assuming $a, b, c \neq 0$. If one is 0, volume = 0.

$$\therefore \text{Volume} = \int_0^a \int_0^{-\frac{b}{a}x+b} \int_0^{c-\frac{c}{a}x-\frac{c}{b}y} dz\,dy\,dx$$

$$= \int_0^a \int_0^{-\frac{b}{a}x+b} \left(c - \frac{c}{a}x - \frac{c}{b}y\right) dy\,dx$$

$$= \int_0^a cy - \frac{c}{a}xy - \frac{c}{2b}y^2 \Big|_{y=0}^{y=-\frac{b}{a}x+b} dx$$

$$= \int_0^a c\left(-\frac{b}{a}x+b\right) - \frac{c}{a}x\left(-\frac{b}{a}x+b\right) - \frac{c}{2b}\left(-\frac{b}{a}x+b\right)^2 dx$$

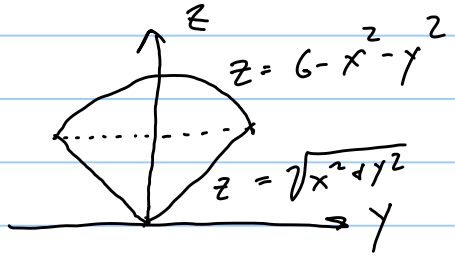
$$-\frac{bc}{a}x + bc + \frac{bc}{a^2}x^2 - \frac{bc}{a}x - \frac{bc}{2a^2}x^2 + \frac{bc}{a}x - \frac{bc}{2}$$

$$= \int_0^a \frac{bc}{2} - \frac{bc}{a}x + \frac{bc}{2a^2}x^2\,dx = \frac{bcx}{2} - \frac{bcx^2}{2a} + \frac{bcx^3}{6a^2}\Big|_{x=0}^{x=a}$$

$$= \frac{abc}{2} - \frac{abc}{2} + \frac{abc}{6} = \frac{abc}{6} \quad \therefore \text{Volume} = \frac{|abc|}{6}$$

10. **10.** Find the volume determined by $z \leq 6 - x^2 - y^2$ and $z \geq \sqrt{x^2 + y^2}$.

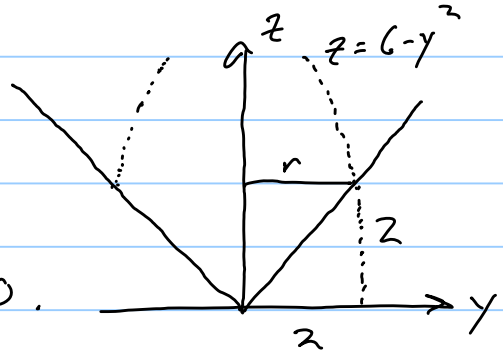
Basically, an ice cream cone.



Use cylindrical coordinates: $0 \le \theta \le 2\pi$

Since $r^2 = x^2 + y^2$, $\therefore z = 6 - r^2$ at top.

$\therefore \quad r \le z \le 6 - r^2$.



What is extent of r?

From symmetry, let $x = 0$.

From paraboloid, $z = 6 - y^2$

From cone, $z = y$

$\therefore 6 - y^2 = y$, or $y^2 + y - 6 = 0, (y+3)(y-2) = 0$

$\therefore y = 2$, since $y \ge 0$.

$\therefore \quad 0 \le r \le 2$

$\therefore$

$$\text{Volume} = \int_0^{2\pi} \int_0^2 \int_r^{6-r^2} r \, dz \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^2 rz \Big|_{z=r}^{z=6-r^2} dr \, d\theta = \int_0^{2\pi} \int_0^2 6r - r^3 - r^2 \, dr \, d\theta$$

$$= \int_0^{2\pi} 3r^2 - \frac{r^4}{4} - \frac{r^3}{3} \Big|_{r=0}^{r=2} d\theta = \int_0^{2\pi} \left(12 - 4 - \frac{8}{3}\right) d\theta$$

$$= \int_0^{2\pi} \frac{16}{3}\, d\theta = \frac{32}{3}\pi$$

11.

11. The *tetrahedron* defined by $x \geq 0,\ y \geq 0,\ z \geq 0,\ x + y + z \leq 1$ is to be sliced into n segments of equal volume by planes parallel to the plane $x + y + z = 1$. Where should the slices be made?

From problem # 9, Volume of tetrahedron bounded by $\frac{x}{a} + \frac{y}{a} + \frac{z}{a} = 1$ is $\frac{a^3}{6}$.

Volume of $x+y+z=1$ is $\therefore \frac{1}{6}$

$\therefore$ Each slice is to have volume $\frac{1}{6n}$

Let the slices be made at:

$x = y = z = a_1$,

$x = y = z = a_2$

$\vdots$

$x = y = z = a_n = 1$

$$\therefore \frac{a_1^3}{6} + \left[\frac{a_2^3}{6} - \frac{a_1^3}{6}\right] + \left[\frac{a_3^3}{6} - \frac{a_2^3}{6}\right] + \cdots + \left[\frac{a_n^3}{6} - \frac{a_{n-1}^3}{6}\right] = \frac{1}{6}$$

$$\frac{a_1^3}{6} = \frac{1}{6n},\ \frac{a_2^3 - a_1^3}{6} = \frac{1}{6n},\ \ldots\ \frac{a_n^3 - a_{n-1}^3}{6} = \frac{1}{6n}$$

$$a_1^3 = \frac{1}{n},\ a_2^3 = \frac{2}{n},\ a_3^3 = \frac{3}{n},\ \ldots,\ a_n^3 = \frac{n}{n}$$

$\therefore a_1 = \sqrt[3]{\frac{1}{n}}, \; a_2 = \sqrt[3]{\frac{2}{n}}, \; a_3 = \sqrt[3]{\frac{3}{n}}, \; \ldots, \; a_n = 1$

$\therefore$ Each plane is of form: $\frac{x}{a_k} + \frac{y}{a_k} + \frac{z}{a_k} = 1$

where $a_k = \sqrt[3]{\frac{k}{n}}$, $k = 1, 2, \ldots, n$

---

12. **12.** Let E be the solid ellipsoid $E = \{(x, y, z) \mid (x^2/a^2) + (y^2/b^2) + (z^2/c^2) \leq 1\}$, where $a > 0$, $b > 0$, and $c > 0$. Evaluate

$$\iiint xyz \, dx \, dy \, dz$$

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(a) (a) over the whole ellipsoid; and

Projection of ellipsoid on xy-plane yields:

$-a \leq x \leq a, \; -b\sqrt{1 - \frac{x^2}{a^2}} \leq y \leq b\sqrt{1 - x^2/a^2}$

$$\therefore \int_{-a}^{a} \int_{-b\sqrt{1 - x^2/a^2}}^{b\sqrt{1 - x^2/a^2}} \int_{-c\sqrt{1 - x^2/a^2 - y^2/b^2}}^{c\sqrt{1 - x^2/a^2 - y^2/b^2}} xyz \, dz \, dy \, dx$$

$$= \int_{-a}^{a} \int_{-b\sqrt{1 - x^2/a^2}}^{b\sqrt{1 - x^2/a^2}} xy \frac{z^2}{2} \Bigg|_{z = -c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}}^{z = c\sqrt{1 - x^2/a^2 - y^2/b^2}} dy \, dx$$

$$= \int_{-a}^{a} \int_{-b\sqrt{1-x^2/a^2}}^{b\sqrt{1-x^2/a^2}} \frac{xy}{2}\left[c^2\left(1-\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)-(-c)^2\left(1-\frac{x^2}{a^2}-\frac{y^2}{b^2}\right)\right] dy\,dx$$

$$= \int_{-a}^{a} \int_{-b\sqrt{1-x^2/a^2}}^{b\sqrt{1-x^2/a^2}} \frac{xy}{2}(0)\,dy\,dx = \underline{0}$$

Same result if integrate $dz\,dx\,dy$, $dx\,dy\,dz$, ...etc. because of symmetry of xyz.

(5) (b) over that part of it in the first quadrant:

$x \geq 0$, $y \geq 0$, and $z \geq 0$, $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2} \leq 1$.

$$\int_0^a \int_0^{b\sqrt{1-x^2/a^2}} \int_0^{c\sqrt{1-x^2/a^2-y^2/b^2}} xyz\,dz\,dy\,dx$$

$$= \int_0^a \int_0^{b\sqrt{1-x^2/a^2}} xy\frac{z^2}{2}\Bigg|_{z=0}^{z=c\sqrt{1-x^2/a^2-y^2/b^2}} dy\,dx$$

$$= \int_0^a \int_0^{b\sqrt{1-x^2/a^2}} \frac{c^2}{2}\left(xy-\frac{x^3y}{a^2}-\frac{xy^3}{b^2}\right) dy\,dx$$

$$= \frac{c^2}{2}\int_0^a \frac{xy^2}{2}-\frac{x^3y^2}{2a^2}-\frac{xy^4}{4b^2}\Bigg|_{y=0}^{y=b\sqrt{1-x^2/a^2}} dx$$

$$= \frac{c^2}{2}\int_0^a \frac{b^2}{2}\left(x - \frac{x^3}{a^2}\right) - \frac{b^2}{2a^2}\left(x^3 - \frac{x^5}{a^2}\right) - \frac{b^2}{4}x\left(1 - \frac{x^2}{a^2}\right)^2 dx$$

$$-\frac{x}{2}\left(1 - \frac{2x^2}{a^2} + \frac{x^4}{a^4}\right)$$

$$= \frac{b^2c^2}{4}\int_0^a x - \frac{x^3}{a^2} - \frac{x^3}{a^2} + \frac{x^5}{a^4} - \frac{x}{2} + \frac{x^3}{a^2} - \frac{x^5}{2a^4}\, dx$$

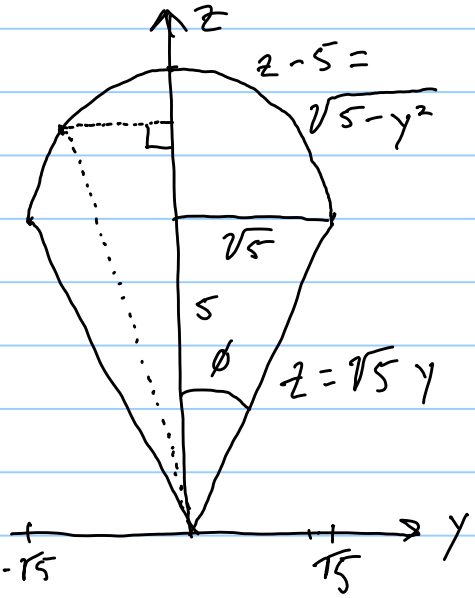
$$= \frac{b^2c^2}{4}\int_0^a \frac{x}{2} - \frac{x^3}{a^2} + \frac{x^5}{2a^4}\, dx$$

$$= \frac{b^2c^2}{4}\left[\frac{x^2}{4} - \frac{x^4}{4a^2} + \frac{x^6}{12a^4}\right]_{x=0}^{x=a}$$

$$= \frac{b^2c^2}{4}\left[\frac{a^2}{4} - \frac{a^2}{4} + \frac{a^2}{12}\right] = \frac{a^2b^2c^2}{48}$$

13.

13. Find the volume of the "ice cream cone" defined by the inequalities $x^2 + y^2 \le \frac{1}{5}z^2$, and $0 \le z \le 5 + \sqrt{5 - x^2 - y^2}$.



$x^2 + y^2 = \frac{1}{5}z^2 \iff z = \sqrt{5}\sqrt{x^2+y^2}$

For $x = 0$ (or $y = 0$) this is equivalent to $z = \pm\sqrt{5}\,y$

$z = 5 + \sqrt{5 - x^2 - y^2} \iff z - 5 = \sqrt{5 - x^2 - y^2}$

For $x = 0$ (or $y = 0$) this is equivalent to $(z-5)^2 + y^2 = 5$, a circle of radius $\sqrt{5}$, centered at $(y, z) = (0, 5)$

Try spherical coordinates.

For this problem, $0 \le \theta \le 2\pi$, $0 \le \phi \le \arctan\left(\frac{\sqrt{5}}{5}\right)$

as the "cone" meets the sphere at $z = 5$, $y = \sqrt{5}$ $(x = 0)$.

For a given θ and ϕ, ρ ranges from 0 to the edge of the sphere (top of ice cream cone). ρ is independent of θ, but depends on ϕ.

$$\rho\cos\phi = z = 5 + \sqrt{5 - x^2 - y^2}$$

$$= 5 + \sqrt{5 - (\rho\sin\phi\cos\theta)^2 - (\rho\sin\phi\sin\theta)^2}$$

$$= 5 + \sqrt{5 - \rho^2\sin^2\phi}$$

$$\therefore (\rho\cos\phi - 5)^2 = 5 - \rho^2\sin^2\phi$$

$$\rho^2\cos^2\phi - 10\rho\cos\phi + 25 = 5 - \rho^2\sin^2\phi$$

$$\rho^2 - 10\rho\cos\phi + 20 = 0$$

This looks messy.

Try cylindrical coordinates

$0 \le \theta \le 2\pi$, $0 \le r \le \sqrt{5}$ (from the above figure)

$\sqrt{5(x^2+y^2)} \le z \le 5 + \sqrt{5 - x^2 - y^2}$, or

$$\sqrt{5}\,r \le z \le 5 + \sqrt{5 - r^2}$$

$$\therefore \text{Volume} = \int_0^{2\pi} \int_0^{\sqrt{5}} \int_{\sqrt{5}\,r}^{5+\sqrt{5-r^2}} r \, dz \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\sqrt{5}} r\left(5 + \sqrt{5 - r^2}\right) - \sqrt{5}\,r^2 \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\sqrt{5}} 5r - \sqrt{5}\,r^2 + r\sqrt{5 - r^2} \, dr \, d\theta$$

$$= \int_0^{2\pi} \frac{5}{2}r^2 - \frac{\sqrt{5}}{3}r^3 - \left(\frac{1}{2}\right)\left(\frac{2}{3}\right)\left(5 - r^2\right)^{3/2} \Big|_{r=0}^{r=\sqrt{5}} \, d\theta$$

$$= \int_0^{2\pi} \frac{25}{2} - \frac{25}{3} + \frac{1}{3}5^{\frac{3}{2}} \, d\theta = 2\pi\left(\frac{25}{6} + \frac{5\sqrt{5}}{3}\right)$$

$$= \frac{\pi}{3}\left(25 + 10\sqrt{5}\right)$$

14. **14.** Let ρ, θ, ϕ be spherical coordinates in $\mathbb{R}^3$ and suppose that a surface surrounding the origin is described by a continuous positive function $\rho = f(\theta, \phi)$. Show that the volume enclosed by the surface is

$$V = \frac{1}{3}\int_0^{2\pi}\int_0^{\pi} [f(\theta, \phi)]^3 \sin\phi \, d\phi \, d\theta.$$

A surface surrounding the origin must have points in all 8 quadrants.

$\therefore 0 \le \theta \le 2\pi, \; 0 \le \phi \le \pi, \; 0 \le \rho \le f(\theta, \phi)$ describes the solid. The Jacobian $= \rho^2 \sin\phi$

$$\therefore \quad V = \int_0^{2\pi}\int_0^{\pi}\int_0^{f(\theta,\phi)} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \frac{\rho^3}{3}\sin\phi \Big|_{\rho=0}^{\rho=f(\theta,\phi)} d\phi \, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \frac{[f(\theta,\phi)]^3}{3}\sin\phi \, d\phi \, d\theta$$

15. **15.** Using an appropriate change of variables, evaluate

$$\iint_B \exp[(y-x)/(y+x)]\, dx\, dy,$$

where B is the interior of the triangle with vertices at $(0, 0)$, $(0, 1)$, and $(1, 0)$.

The structure of exp( ) suggests letting

$u = y - x$, $v = y + x$, or $T\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$

or $T^{-1}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} y - x \\ y + x \end{bmatrix} \therefore T^{-1} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}$

$\therefore T = -\frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$

$$\left|\frac{\partial(x,y)}{\partial(u,v)}\right| = \left|\det(T)\right| = \left|-\frac{1}{4} - \frac{1}{4}\right| = \frac{1}{2}$$

$$\therefore \iint_{B^*} f(x(u,v), y(u,v)) \left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv = \iint_B f(x,y)\,dx\,dy$$

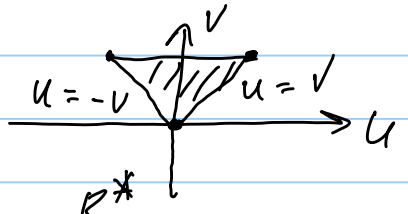
$$T\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{-u+v}{2} \\ \frac{u+v}{2} \end{bmatrix}$$

or $x = \frac{v-u}{2}$, $y = \frac{u+v}{2}$

$T^{-1}(0,0) = (0,0)$, $T^{-1}(0,1) = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$T^{-1}(1,0) = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$\therefore B^*$ has corners $(0,0)$, $(1,1)$, $(-1,1) = (u,v)$



A u-simple region

$\therefore 0 \le v \le 1$, from $u = -v$ to $u = v$

$$\therefore \iint_{B^*} f(x(u,v), y(u,v)) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du\,dv$$

$$= \int_0^1 \int_{-v}^{v} e^{\frac{u}{v}} \left(\frac{1}{2}\right) du\,dv = \frac{1}{2} \int_0^1 v e^{\frac{u}{v}} \Big|_{u=-v}^{u=v} dv$$

$$= \frac{1}{2} \int_0^1 \left(ve - ve^{-1}\right) dv = \frac{1}{2}\left(e - e^{-1}\right) \frac{v^2}{2} \Big|_{v=0}^{v=1}$$

$$= \frac{1}{4}\left(e - \frac{1}{e}\right)$$

16. **16.** Suppose the density of a solid of radius R is given by $(1 + d^3)^{-1}$, where d is the distance to the center of the sphere. Find the total mass of the sphere.

(Density)(Volume) = mass.

Use spherical coordinates on a sphere.

Volume unit $= \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$

$\therefore$ Mass of unit $= \left(\frac{1}{1+\rho^3}\right)\left(\rho^2 \sin\phi \, d\rho \, d\phi \, d\theta\right)$

$$\therefore \text{Mass} = \int_0^{2\pi} \int_0^{\pi} \int_0^{R} \frac{1}{1+\rho^3} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi} \frac{1}{3} \sin\phi \log(1+\rho^3) \Big|_{\rho=0}^{\rho=R} \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi} \frac{\log(1+R^3)}{3} \sin\phi \, d\phi \, d\theta$$

$$= \frac{\log(1+R^3)}{3} \int_0^{2\pi} -\cos\phi \Big|_{\phi=0}^{\phi=\pi} \, d\theta$$

$$= \frac{\log(1+R^3)}{3} \int_0^{2\pi} 2 \, d\theta = \frac{4\pi}{3} \log(1+R^3)$$

17.

17. The density of the material of a spherical shell whose inner radius is 1 m and whose outer radius is 2 m is $0.4d^2$ g/cm^3, where d is the distance to the center of the sphere in meters. Find the total mass of the shell.

Use spherical coordinates on the spherical shell.

Mass = (Density)(Volume) = $(0.4\rho^2)(\rho^2 \sin\phi \, d\rho \, d\phi \, d\theta)$

$0.4\,g/cm^3 = 0.4\,g/(0.01m)^3 = 4\times10^5\,g/m^3$

$\therefore$ Mass = $\int_0^{2\pi} \int_0^{\pi} \int_1^2 (4\times10^5)\rho^4 \sin\phi \, d\rho \, d\phi \, d\theta$

$$= (4\times 10^5)\int_0^{2\pi}\int_0^{\pi} \sin\phi \,\frac{\rho^5}{5}\Big|_{\rho=1}^{\rho=2} \, d\phi\, d\theta$$

$$= (8\times 10^4)\int_0^{2\pi}\int_0^{\pi} \sin\phi \,[32-1]\, d\phi\, d\theta$$

$$= (8\times 10^4)(31)\int_0^{2\pi} -\cos\phi\Big|_{\phi=0}^{\phi=\pi} \, d\phi\, d\theta$$

$$= 2.48\times 10^6 \int_0^{2\pi} 2\, d\theta = (2.48\times 10^6)(4\pi)\ g$$

$$= (9.92\times 10^6)\pi \text{ grams}$$

18. **18.** If the shell in Exercise 17 were dropped into a large tank of pure water, would it float? What if the shell leaked? (Assume that the density of water is exactly 1 g/cm^3.)

(a) It's volume is: $\frac{4}{3}\pi(200\text{ cm})^3 = \left(\frac{32}{3}\times 10^6\right)\pi\text{ cm}^3$

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{(9.92\times 10^6)\pi \text{ grams}}{(32/3\times 10^6)\pi\text{ cm}^3}$$

$$= 0.93\ \text{g/cm}^3$$

As it is less dense than water, it will float.

(b) If the shell leaked, the inside would fill with water. The inside of the shell has

volume $= \frac{4}{3}\pi(100\,cm)^3 = \left(\frac{4}{3}\times 10^6\right)\pi\ cm^3$

$\therefore$ Mass of water inside shell is:

$\left[\left(\frac{4}{3}\times 10^6\right)\pi\ cm^3\right]\left(1\,g/cm^3\right) = \left(\frac{4}{3}\times 10^6\right)\pi$ grams.

$\therefore$ Mass shell + inside water is:

$(9.92\times 10^6)\pi + \left(\frac{4}{3}\times 10^6\right)\pi = (11.25\times 10^6)\pi\ g$

$\therefore$ Density of leaked shell is:

$$\frac{\text{Mass}}{\text{Volume}} = \frac{(11.25\times 10^6)\pi\ g}{\left(\frac{32}{3}\times 10^6\right)\pi\ cm^3} = 1.06\,g/cm^3$$

Density of leaked shell is greater than water, and so it will sink.

19.

19. The temperature at points in the cube $C = \{(x, y, z) \mid -1 \le x \le 1, -1 \le y \le 1, \text{ and } -1 \le z \le 1\}$ is $32d^2$, where d is the distance to the origin.

(a) What is the average temperature?

(b) At what points of the cube is the temperature equal to the average temperature?

(a) Sum the temperature in all the volume "elements" of the sphere, then divide by total volume.

Total volume $= 2^3 = 8$

$$\sum \text{Temp} = \int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} 32(x^2+y^2+z^2)\,dx\,dy\,dz$$

$$= 32\int_{-1}^{1}\int_{-1}^{1} \frac{x^3}{3} + y^2x + z^2x \Big|_{x=-1}^{x=1} dy\,dz$$

$$= 32\int_{-1}^{1}\int_{-1}^{1} \frac{2}{3} + 2y^2 + 2z^2\,dy\,dz$$

$$= 32\int_{-1}^{1} \frac{2}{3}y + \frac{2}{3}y^3 + 2z^2y \Big|_{y=-1}^{y=1} dz$$

$$= 32\int_{-1}^{1} \left(\frac{4}{3} + \frac{4}{3} + 4z^2\right) dz$$

$$= 32\left[\frac{8}{3}z + \frac{4}{3}z^3\right]_{z=-1}^{z=1} = 32\left[\frac{16}{3} + \frac{8}{3}\right]$$

$$= 32\left(\frac{24}{3}\right) = 32(8)$$

$$\therefore \text{Ave Temp} = \frac{32(8)}{8} = \underline{32}$$

(6) $T(D) = 32d^2$. $\therefore 32d^2 = 32$, or $d = 1$.

$\therefore$ Points where ave. temp $= 32$ are on the sphere of radius 1.

20.

20. Use cylindrical coordinates to find the center of mass of the region defined by

$$y^2 + z^2 \le \frac{1}{4}, \qquad (x-1)^2 + y^2 + z^2 \le 1, \qquad x \ge 1.$$

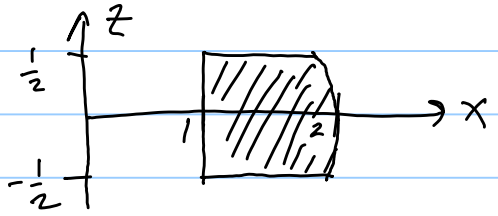
Assume uniform density.

$y^2 + z^2 \le \frac{1}{4}$ is a circular pipe extending to $\pm\infty$ along x-axis, radius $\frac{1}{2}$.

$(x-1)^2 + y^2 + z^2 \le 1$ is a unit sphere, center at $(1,0,0)$.

$x \ge 1$ restricts sphere to hemisphere, pipe to $+\infty$.

∴ A pipe of radius $\frac{1}{2}$, along x-axis from $x=1$ with a unit spherical cap at $x=2$



Make x-axis the traditional "z-axis" of cylindrical coordinates.

∴ $0 \le \theta \le 2\pi$, $0 \le r \le \frac{1}{2}$, where $r^2 = y^2 + z^2$

From $(x-1)^2 = 1 - y^2 - z^2 = 1 - r^2$,

$$x - 1 = \sqrt{1-r^2}, \quad x = 1 + \sqrt{1-r^2}$$

$$\therefore \; 1 \le x \le 1 + \sqrt{1-r^2}$$

$$\therefore \text{Volume} = \int_0^{2\pi}\int_0^{\frac{1}{2}}\int_1^{1+\sqrt{1-r^2}} r\,dx\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{1}{2}} rx\Big|_{x=1}^{x=1+\sqrt{1-r^2}}\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{1}{2}} r\sqrt{1-r^2}\,dr\,d\theta = \int_0^{2\pi}\left(-\tfrac{1}{2}\right)\left(\tfrac{2}{3}\right)(1-r^2)^{\frac{3}{2}}\Big|_{r=0}^{r=\frac{1}{2}} d\theta$$

$$= -\frac{1}{3}\int_0^{2\pi}\left[\left(\tfrac{3}{4}\right)^{\frac{3}{2}} - 1\right]d\theta = \int_0^{2\pi}\left(\frac{1}{3}-\frac{\sqrt{3}}{8}\right)d\theta = \left(\frac{1}{3}-\frac{\sqrt{3}}{8}\right)2\pi$$

From symmetry, center of mass is on the x-axis.
$\therefore$ Just compute $\bar{x}$.

$$\therefore \bar{x} = \frac{\displaystyle\int_0^{2\pi}\int_0^{\frac{1}{2}}\int_1^{1+\sqrt{1-r^2}} x\,r\,dx\,dr\,d\theta}{2\pi\left(\frac{1}{3}-\frac{\sqrt{3}}{8}\right)}$$

$$\int_0^{2\pi}\int_0^{\frac{1}{2}}\int_1^{1+\sqrt{1-r^2}} x\,r\,dx\,dr\,d\theta = \int_0^{2\pi}\int_0^{\frac{1}{2}} r\frac{x^2}{2}\Big|_{x=1}^{x=1+\sqrt{1-r^2}}\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{1}{2}} \frac{r\left(1+\sqrt{1-r^2}\right)^2}{2} - \frac{r}{2}\,dr\,d\theta$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^{\frac{1}{2}} r\left(1 + 2\sqrt{1-r^2} + 1 - r^2\right) - r \; dr\,d\theta$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^{\frac{1}{2}} 2r\sqrt{1-r^2} + r - r^3 \; dr\,d\theta$$

$$= \frac{1}{2}\int_0^{2\pi} \left(-\frac{2}{3}\right)\left(1-r^2\right)^{3/2} + \frac{r^2}{2} - \frac{r^4}{4}\Bigg|_{r=0}^{r=\frac{1}{2}} d\theta$$

$$= \frac{1}{2}\int_0^{2\pi} \left[-\frac{\sqrt{3}}{4} + \frac{1}{8} - \frac{1}{64} - \left(-\frac{2}{3}\right)\right] d\theta$$

$$= \frac{2\pi}{2}\left[\frac{7}{64} + \frac{2}{3} - \frac{\sqrt{3}}{4}\right] = \pi\left(\frac{7}{64} + \frac{2}{3} - \frac{\sqrt{3}}{4}\right)$$

$$\therefore \bar{x} = \frac{\pi\left(\frac{7}{64} + \frac{2}{3} - \frac{\sqrt{3}}{4}\right)}{2\pi\left(\frac{1}{3} - \frac{\sqrt{3}}{8}\right)} = \frac{\left(\frac{7}{64} + \frac{2}{3} - \frac{\sqrt{3}}{4}\right)}{2\left(\frac{1}{3} - \frac{\sqrt{3}}{8}\right)}$$

$$\approx 1.468$$

Note: solution manual states "shifted down $\frac{1}{2}$ unit", but actually was shifted down 1 unit. Using 1 unit, the answer is the same as above: 1.468.

21.

21. Find the center of mass of the solid hemisphere

$$V = \{(x, y, z) \mid x^2 + y^2 + z^2 \le a^2 \text{ and } z \ge 0\},$$

if the density is constant.

Volume $= \frac{1}{2}\left(\frac{4}{3}\pi a^3\right) = \frac{2}{3}\pi a^3$

By symmetry, center of mass on z-axis.

Use spherical coordinates. Distance from xy-plane is $\rho\cos\phi$.

$$\therefore \bar{z} = \frac{\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\int_0^{a} (\rho\cos\phi)\,\rho^2\sin\phi\, d\rho\, d\phi\, d\theta}{\frac{2}{3}\pi a^3}$$

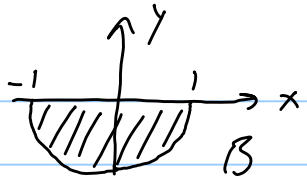
$$= 2\pi\left(\frac{3}{2\pi a^3}\right)\int_0^{\frac{\pi}{2}}\int_0^{a} \rho^3\left(\tfrac{1}{2}\right)(2\cos\phi\sin\phi)\, d\rho\, d\phi$$

$$= \frac{3}{2a^3}\int_0^{\frac{\pi}{2}}\int_0^{a} \sin 2\phi\,\rho^3\, d\rho\, d\phi$$

$$= \frac{3}{2a^3}\left[\frac{\rho^4}{4}\right]_{\rho=0}^{\rho=a}\left[-\frac{1}{2}\cos 2\phi\right]_{\phi=0}^{\phi=\frac{\pi}{2}}$$

$$= \frac{3}{2a^3}\left[\frac{a^4}{4}\right]\left[1\right] = \frac{3}{8}a \quad \therefore \left(0, 0, \frac{3}{8}a\right)$$

22. **22.** Evaluate $\iint_B e^{-x^2-y^2}\,dx\,dy$, where B consists of those (x, y) satisfying $x^2 + y^2 \le 1$ and $y \le 0$.



$$\iint_B e^{-x^2-y^2}\,dx\,dy = \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{0} e^{-x^2-y^2}\,dy\,dx$$

Use a polar coordinates conversion.

$r = x^2+y^2$, $\therefore\ 0 \le r \le 1$, $-\pi \le \theta \le 0$, Jacobian $= r$

$$\therefore \int_{-\pi}^{0}\int_{0}^{1} e^{-r^2} r\,dr\,d\theta = \int_{-\pi}^{0} \left(-\tfrac{1}{2}\right) e^{-r^2}\Big|_{r=0}^{r=1} d\theta$$

$$= -\frac{1}{2}\int_{-\pi}^{0}\left(e^{-1}-1\right)d\theta = \frac{1-e^{-1}}{2}\left[\theta\right]_{-\pi}^{0} = \frac{\pi}{2}\left(1-\frac{1}{e}\right)$$

23. **23.** Evaluate

$$\iiint_S \frac{dx\,dy\,dz}{(x^2+y^2+z^2)^{3/2}},$$

where S is the solid bounded by the spheres $x^2 + y^2 + z^2 = a^2$ and $x^2 + y^2 + z^2 = b^2$, where $a > b > 0$.

Use spherical coordinates: $x^2+y^2+z^2 = \rho^2$, $b \le \rho \le a$

$$\therefore \int_0^{2\pi}\int_0^{\pi}\int_b^a \frac{\rho^2 \sin\phi\,d\rho\,d\phi\,d\theta}{(\rho^2)^{3/2}} = \int_0^{2\pi}\int_0^{\pi}\int_b^a \frac{\sin\phi}{\rho}\,d\rho\,d\phi\,d\theta$$

$$= \left[\ln \rho\right]_{\rho=6}^{\rho=9} \left[-\cos\phi\right]_{\phi=0}^{\phi=\pi} \left[\theta\right]_{\theta=0}^{\theta=2\pi}$$

$$= \left[\ln \frac{9}{6}\right]\left[2\right]\left[2\pi\right] = \underline{4\pi \ln \frac{9}{6}}$$

24.

24. Evaluate $\iiint_D (x^2 + y^2 + z^2)\,xyz\,dx\,dy\,dz$ over each of the following regions.

(a)

(a) The sphere $D = \{(x, y, z) \mid x^2 + y^2 + z^2 \le R^2\}$

Use spherical coordinates: $x^2+y^2+z^2 = \rho^2$

$$xyz = (\rho\sin\phi\cos\theta)(\rho\sin\phi\sin\theta)(\rho\cos\phi)$$

$$= \rho^3(\sin^2\phi\cos\phi)(\sin\theta\cos\theta)$$

$$= \rho^3\left(\frac{\sin 2\theta}{2}\right)(\sin^2\phi\cos\phi)$$

Sphere: $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi$, $0 \le \rho \le R$

Jacobian $= \rho^2\sin\phi$

$$\therefore \int_0^{2\pi}\int_0^{\pi}\int_0^{R} (\rho^2)\left(\frac{\rho^3}{2}\sin 2\theta\,\sin^2\phi\cos\phi\right)(\rho^2\sin\phi)\,d\rho\,d\phi\,d\theta$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^{\pi}\int_0^{R} (\rho^7)(\sin 2\theta)(\sin^3\phi\cos\phi)\,d\rho\,d\phi\,d\theta$$

$$= \frac{1}{2}\left[\frac{\rho^8}{8}\right]_{\rho=0}^{\rho=R}\left[-\frac{\cos 2\theta}{2}\right]_{\theta=0}^{\theta=2\pi}\left[\frac{\sin^4\phi}{4}\right]_{\phi=0}^{\phi=\pi}$$

$$= \frac{1}{2}\left[\frac{R^8}{8}\right]\left[-\frac{1}{2}-\left(-\frac{1}{2}\right)\right]\left[0-0\right] = \underline{0}$$

(6) (b) The hemisphere $D = \{(x, y, z) \mid x^2 + y^2 + z^2 \le R^2 \text{ and } z \ge 0\}$

Use the above spherical coordinates, but here,

$$0 \le \phi \le \frac{\pi}{2}$$

$$\therefore \frac{1}{2}\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\int_0^{R}(\rho^7)(\sin 2\theta)(\sin^3\phi\cos\phi)\,d\rho\,d\phi\,d\theta$$

$$= \frac{1}{2}\left[\frac{\rho^8}{8}\right]_{\rho=0}^{\rho=R}\left[-\frac{\cos 2\theta}{2}\right]_{\theta=0}^{\theta=2\pi}\left[\frac{\sin^4\phi}{4}\right]_{\phi=0}^{\phi=\frac{\pi}{2}}$$

$$= \frac{1}{2}\left[\frac{R^8}{8}\right]\left[-\frac{1}{2}-\left(-\frac{1}{2}\right)\right]\left[\frac{1}{4}-0\right] = \underline{0}$$

$= 0$

(c) (c) The octant $D = \{(x, y, z) \mid x \ge 0, y \ge 0, z \ge 0, \text{ and } z^2 + y^2 + z^2 \le R^2\}$

Use the above spherical coordinates, but here,

$$0 \le \theta \le \frac{\pi}{2},\quad 0 \le \phi \le \frac{\pi}{2},\quad 0 \le \rho \le R$$

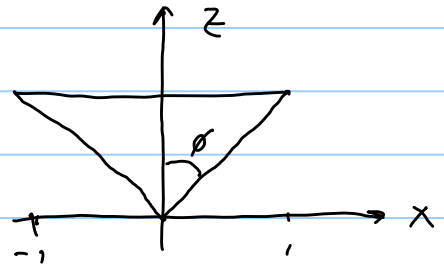
$$\therefore \frac{1}{2}\int_0^{\frac{\pi}{2}}\int_0^{\frac{\pi}{2}}\int_0^{R}(\rho^7)(\sin 2\theta)(\sin^3\phi\cos\phi)\,d\rho\,d\phi\,d\theta$$

$$= \frac{1}{2}\left[\frac{\rho^8}{8}\right]_{\rho=0}^{\rho=R}\left[-\frac{\cos 2\theta}{2}\right]_{\theta=0}^{\theta=\frac{\pi}{2}}\left[\frac{\sin^4\phi}{4}\right]_{\phi=0}^{\phi=\frac{\pi}{2}}$$

$$= \frac{1}{2}\left[\frac{R^8}{8}\right]\left[\frac{1}{2}-\left(-\frac{1}{2}\right)\right]\left[\frac{1}{4}-0\right]$$

$$= \frac{R^8}{64}$$

25. **25.** Let C be the cone-shaped region $\{(x, y, z) \mid \sqrt{x^2 + y^2} \le z \le 1\}$ in $\mathbb{R}^3$ and evaluate the integral $\iiint_C (1 + \sqrt{x^2 + y^2})\,dx\,dy\,dz$.



Convert to spherical coordinates. $0 \le \theta \le 2\pi$, $0 \le \phi \le \frac{\pi}{4}$

Note $z = \rho\cos\phi$, so $0 \le \rho \le \frac{1}{\cos\phi}$ as $0 \le z \le 1$.

$$x^2 + y^2 = (\rho\sin\phi\cos\theta)^2 + (\rho\sin\phi\sin\theta)^2$$

$$= \rho^2\sin^2\phi$$

$$\therefore 1 + \sqrt{x^2+y^2} = 1 + \rho\sin\phi$$

Jacobian $= \rho^2\sin\phi$

$$\therefore \int_0^{2\pi}\int_0^{\frac{\pi}{4}}\int_0^{\frac{1}{\cos\phi}} (1+\rho\sin\phi)\,\rho^2\sin\phi \; d\rho\, d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{\pi}{4}}\int_0^{\frac{1}{\cos\phi}} \rho^2\sin\phi + \rho^3\sin^2\phi \; d\rho\, d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{\pi}{4}} \sin\phi\,\frac{\rho^3}{3} + \sin^2\phi\,\frac{\rho^4}{4}\Big|_{\rho=0}^{\rho=\frac{1}{\cos\phi}} \; d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\frac{\pi}{4}} \frac{\sin\phi}{3\cos^3\phi} + \frac{\sin^2\phi}{4\cos^4\phi} \; d\phi\, d\theta$$

$$= \int_0^{2\pi} -\frac{1}{3}\left(-\frac{1}{2}\right)\cos^{-2}\phi\Big|_{\phi=0}^{\phi=\frac{\pi}{4}} d\theta + \int_0^{2\pi}\int_0^{\frac{\pi}{4}} \frac{1}{4}\tan^2\phi\sec^2\phi \; d\phi\, d\theta$$

$$= \int_0^{2\pi} \frac{1}{6}\left[\frac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} - 1\right] d\theta + \int_0^{2\pi} \frac{1}{4}\left[\frac{\tan^3\phi}{3}\right]_{\phi=0}^{\phi=\frac{\pi}{4}} d\theta$$

$$= \int_0^{2\pi} \frac{1}{6}\, d\theta + \int_0^{2\pi} \frac{1}{4}\left(\frac{1}{3}\right) d\theta = \frac{2\pi}{6} + \frac{2\pi}{12}$$

$$= \frac{\pi}{2}$$

26. **26.** Find $\iiint_{\mathbb{R}^3} f(x, y, z)\, dx\, dy\, dz$, where $f(x, y, z) = \exp\left[-(x^2 + y^2 + z^2)^{3/2}\right]$.

Use spherical coordinates. $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi$

$$x^2 + y^2 + z^2 = \rho^2, \quad 0 \le \rho < \infty$$

$$\therefore \lim_{a \to \infty} \int_0^{2\pi} \int_0^{\pi} \int_0^{a} e^{-\rho^3} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= \lim_{a \to \infty} \int_0^{2\pi} d\theta \int_0^{\pi} \sin\phi \, d\phi \int_0^{a} e^{-\rho^3} \rho^2 \, d\rho$$

$$= \lim_{a \to \infty} (2\pi) \left[-\cos\phi\right]_0^{\pi} \left[-\frac{1}{3} e^{-\rho^3}\right]_0^{a}$$

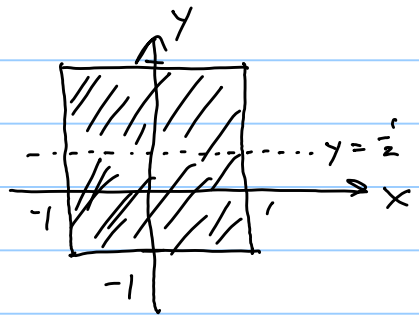
$$= 4\pi \lim_{a \to \infty} \left\{ -\frac{1}{3e^{a^3}} + \frac{1}{3} \right\} = \frac{4\pi}{3}$$

27. **27.** The *flexural rigidity* EI of a uniform beam is the product of its Young's modulus of elasticity E and the moment of inertia I of the cross section of the beam with respect to a horizontal line l passing through the center of gravity of this cross section. Here

$$I = \iint_R [d(x, y)]^2 \, dx\, dy,$$

where $d(x, y)$ = the distance from (x, y) to l and R = the cross section of the beam being considered.

(a) (a) Assume that the cross section R is the rectangle $-1 \le x \le 1$, $-1 \le y \le 2$, and l is the line $y = 1/2$. Find I.

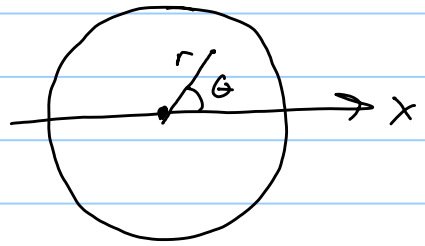


$$d(x,y) = \sqrt{(y-\tfrac{1}{2})^2} = |y-\tfrac{1}{2}|, \quad \therefore d(x,y) = (y-\tfrac{1}{2})^2$$

$$\therefore I = \int_{-1}^{1}\int_{-1}^{2} (y-\tfrac{1}{2})^2 \, dy\,dx = \int_{-1}^{1} \tfrac{1}{3}(y-\tfrac{1}{2})^3 \Big|_{y=-1}^{y=2} dx$$

$$\int_{-1}^{1} \tfrac{1}{3}\left[(\tfrac{3}{2})^3 - (-\tfrac{3}{2})^3\right] dx = \int_{-1}^{1} \tfrac{9}{4}\, dx = \underline{\underline{\tfrac{9}{2}}}$$

(1) (b) Assume the cross section R is a circle of radius 4 and l is the x axis. Find I, using polar coordinates.



$$d = r\sin\theta, \quad d^2 = r^2\sin^2\theta$$

$$\therefore I = \int_0^{2\pi}\int_0^4 (r^2\sin^2\theta)\, r\, dr\, d\theta$$

$$= \int_0^{2\pi} \frac{r^3}{3}\sin^2\theta \Big|_{r=0}^{r=4} d\theta = \frac{64}{3}\int_0^{2\pi} \sin^2\theta\, d\theta$$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta$

$$= \frac{64}{3}\int_0^{2\pi} \frac{1-\cos 2\theta}{2}\, d\theta = \frac{64}{3}\left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_{\theta=0}^{\theta=2\pi}$$

$$= \underline{\underline{\frac{64}{3}\pi}}$$

28. **28.** Find $\iiint_{\mathbb{R}^3} f(x, y, z)\, dx\, dy\, dz$, where

$$f(x, y, z) = \frac{1}{[1 + (x^2 + y^2 + z^2)^{3/2}]^{3/2}}.$$

Use spherical coordinates. $\rho^2 = x^2 + y^2 + z^2$

$$\therefore f(x(\rho,\theta,\phi), y(\rho,\theta,\phi), z(\rho,\theta,\phi)) = \frac{1}{(1+\rho^3)^{3/2}}$$

$0 \le \theta \le 2\pi$, $0 \le \phi \le \pi$, $0 \le \rho \le a$, $\lim_{a\to\infty}$

$$\therefore \iiint_{\mathbb{R}^3} f\, dx\,dy\,dz = \lim_{a\to\infty} \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \frac{\rho^2 \sin\phi}{(1+\rho^3)^{3/2}}\, d\rho\, d\phi\, d\theta$$

$$= \lim_{a\to\infty} \int_0^{2\pi}\int_0^{\pi} \left(\tfrac{1}{3}\right)(-2)(1+\rho^3)^{-\frac{1}{2}} \sin\phi \Big|_{\rho=0}^{\rho=a} d\phi\, d\theta$$

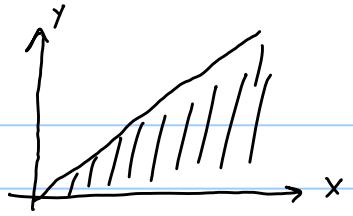
$$= -\frac{2}{3} \lim_{a\to\infty} \int_0^{2\pi}\int_0^{\pi} \left[\frac{1}{\sqrt{1+a^3}} - 1\right] \sin\phi\, d\phi\, d\theta$$

$$= -\frac{2}{3} \left[\lim_{a\to\infty}\left(\frac{1}{\sqrt{1+a^3}} - 1\right)\right]\left[\int_0^{2\pi} d\theta\right]\left[\int_0^{\pi} \sin\phi\, d\phi\right]$$

$$= -\frac{2}{3}\,[-1]\,[2\pi]\,[2] = \frac{8\pi}{3}$$

29.

29. Suppose D is the unbounded region of $\mathbb{R}^2$ given by the set of (x, y) with $0 \le x < \infty$, $0 \le y \le x$. Let $f(x, y) = x^{-3/2} e^{y-x}$. Does the improper integral $\iint_D f(x, y)\, dx\, dy$ exist?



$$\iint_D f(x,y)\,dx\,dy = \lim_{\substack{a\to\infty \\ b\to 0^+}} \int_b^a \int_0^x x^{-3/2} e^{y-x}\,dy\,dx$$

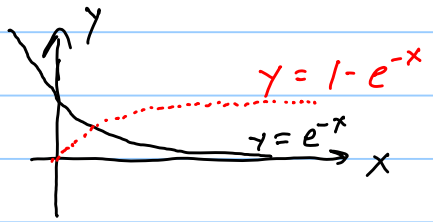
$$= \lim_{\substack{a\to\infty \\ b\to 0^+}} \int_b^a x^{-\frac{3}{2}} e^{y-x} \Big|_{y=0}^{y=x}\,dx$$

$$= \lim_{\substack{a\to\infty \\ b\to 0^+}} \int_b^a x^{-\frac{3}{2}} - x^{-\frac{3}{2}} e^{-x}\,dx$$

$$= \lim_{\substack{a\to\infty \\ b\to 0^+}} \int_b^a x^{-\frac{3}{2}}\left(1 - e^{-x}\right)dx$$

$$= \lim_{b\to 0^+} \int_b^1 x^{-\frac{3}{2}}\left(1-e^{-x}\right)dx + \lim_{a\to\infty} \int_1^a x^{-\frac{3}{2}}\left(1-e^{-x}\right)dx$$

(a) $1 - e^{-x} < 1$ for $1 \le x < \infty$



$\therefore\ x^{-\frac{3}{2}}\left(1-e^{-x}\right) < x^{-\frac{3}{2}}$ for $1 \le x < \infty$

$\therefore\ \int_1^a x^{-\frac{3}{2}}\left(1-e^{-x}\right)dx \le \int_1^a x^{-\frac{3}{2}}\,dx$

Since $\lim_{a\to\infty} \int_1^a x^{-\frac{3}{2}}\,dx = \lim_{a\to\infty} \left[-2x^{-\frac{1}{2}}\right]_1^a$

$= \lim_{a\to\infty} \left[\frac{-2}{\sqrt{a}} + 2\right] = 2$

Then since $x^{-\frac{3}{2}}(1-e^{-x}) > 0$ for $x \geq 1$,

and is bounded above by $x^{-3/2}$,

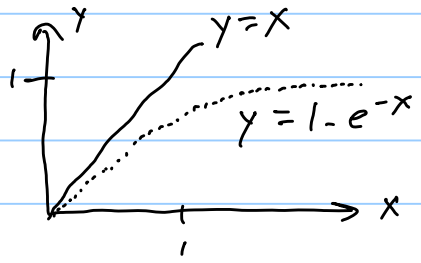
and $\int_1^\infty x^{-\frac{3}{2}}\,dx$ exists,

Then $\int_1^\infty x^{-\frac{3}{2}}(1-e^{-x})\,dx$ exists

(6) On $0 < x \leq 1$, $1 - e^{-x} < x$

$\frac{d}{dx}(1-e^{-x}) = e^{-x}$, so max slope is at $x = 0$,

$e^{-0} = 1$, so $1 - e^{-x} < x$



$\therefore x^{-\frac{3}{2}}(1-e^{-x}) < x^{-\frac{3}{2}} x = x^{-\frac{1}{2}}$ for $0 < x < 1$.

$\therefore \int_\delta^1 x^{-\frac{3}{2}}(1-e^{-x}) < \int_\delta^1 x^{-\frac{1}{2}}\,dx$ for $\delta > 0$

since $0 < x^{-\frac{3}{2}}(1-e^{-x})$ for $0 < x < 1$.

But $\lim_{b\to 0^+} \int_b^1 x^{-\frac{1}{2}}\,dx = \lim_{b\to 0^+} \left[2x^{\frac{1}{2}}\right]_b^1$

$= \lim_{b\to 0^+} \left[2 - 2\sqrt{b}\right] = 2.$

$\therefore \lim_{b\to 0^+} \int_b^1 x^{-\frac{3}{2}}(1-e^{-x})$ exists.

$\therefore \int_0^1 x^{-3/2}(1-e^{-x})$ exists.

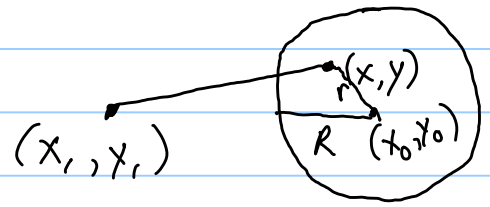
(a) & (b) $\Rightarrow \int_0^\infty x^{-3/2}(1-e^{-x})\,dx$ exists.

$\therefore \iint_D x^{-3/2} e^{y-x}\,dx\,dy$ exists.

30. **30.** If the world were two-dimensional, the laws of physics would predict that the gravitational potential of a mass point is proportional to the logarithm of the distance from the point. Using polar coordinates, write an integral giving the gravitational potential of a disk of constant density.

$V(x_1,y_1) = K\sigma\,dV \ln\sqrt{(x-x_1)^2+(y-y_1)^2}$, potential at (x_1,y_1)

$\sigma\,dV$ = mass of "point mass" of volume dV,
K = a constant, (x,y) = point in disk



Disk of density σ, radius R
center at (x_0,y_0).

$$\therefore \iint_U k\sigma \ln \sqrt{(x-x_1)^2+(y-y_1)^2}\, dx\, dy \;,\; U = \text{disk}$$

Let $(x_0, y_0) = (0, 0)$ for ease.

$$\therefore V(x_1, y_1) = k\sigma \int_0^{2\pi} \int_0^R \ln \sqrt{(r\cos\theta - x_1)^2 + (r\sin\theta - y_1)^2}\, r\, dr\, d\theta$$

31.

31. (a) Evaluate the improper integral

$$\int_0^\infty \int_0^y xe^{-y^3}\, dx\, dy.$$

(a) $$\int_0^\infty e^{-y^3} \frac{x^2}{2}\Big|_{x=0}^{x=y} dy = \int_0^\infty e^{-y^3} \frac{y^2}{2}\, dy$$

$$= \lim_{a\to\infty} -\frac{1}{6} e^{-y^3}\Big|_{y=0}^{y=a} = \lim_{a\to\infty} \left[-\frac{1}{6e^{a^3}} + \frac{1}{6} \right] = \frac{1}{6}$$

(6)

(b) Evaluate

$$\iint_B (x^4 + 2x^2y^2 + y^4)\, dx\, dy,$$

where B is the portion of the disk of radius 2 [centered at (0, 0) in the first quadrant].

Use polar coordinates. $x = r\cos\theta$, $y = r\sin\theta$

$0 \le r \le 2, \quad 0 \le \theta \le \frac{\pi}{2} \qquad \text{Jacobian} = r$

$x^4 + 2x^2y^2 + y^4 \Rightarrow (x^2+y^2)^2 \Rightarrow r^4$

$$\therefore \int_0^{\frac{\pi}{2}} \int_0^2 (r^4)\, r\, dr\, d\theta = \int_0^{\frac{\pi}{2}} \frac{r^6}{6}\Big|_{r=0}^{r=2} d\theta$$

$$= \frac{64}{6}\left(\frac{\pi}{2}\right) = \frac{16}{3}\pi$$

32.

32. Let f be a nonnegative function on an x-simple or a y-simple region $D \subset \mathbb{R}^2$ and that is continuous except for points on the boundary of D and at most finitely many points interior to D. Give a suitable definition of $\iint_D f\, dA$.

If D is y-simple, define $D_{\delta,\epsilon}$ as

$$D_{\delta,\epsilon} = \{(x,y) : a+\delta \le x \le b-\delta,\ \phi_1(x)+\epsilon \le y \le \phi_2(x)-\epsilon\}$$

If D is x-simple, define $D_{\delta,\epsilon}$ as

$$D_{\delta,\epsilon} = \{(x,y) : \phi_1(y)+\delta \le x \le \phi_2(y)-\delta,\ c+\epsilon \le y \le d-\epsilon\}$$

Suppose there are n points of discontinuity in D, whose coordinates are (x_i, y_i), $i = 1, \ldots, n$.

Let $B^i_{\delta,\epsilon} = \{(x,y) : x_i-\delta < x < x_i+\delta,\ y_i-\epsilon < y < y_i+\epsilon\}$

(so $B^i_{\delta,\epsilon}$ is open)

Let $D^*_{\delta,\epsilon} = D_{\delta,\epsilon} - B^1_{\delta,\epsilon} - B^2_{\delta,\epsilon} - \dots - B^n_{\delta,\epsilon}$

Choosing δ, ϵ small enough so that $B^i_{\delta,\epsilon} \subset D_{\delta,\epsilon}$

Define $\iint_D f\,dA = \lim_{\substack{\delta \to 0 \\ \epsilon \to 0}} \iint_{D^*_{\delta,\epsilon}} f\,dA$

33.

33. Evaluate $\int\int_{\mathbb{R}^2} f(x, y)\,dx\,dy$, where $f(x, y) = 1/(1 + x^2 + y^2)^{3/2}$. (HINT: You may assume that changing variables and Fubini's theorem are valid for improper integrals.)

Use a polar conversion: $x^2 + y^2 = r^2$

$\therefore \dfrac{1}{(1+x^2+y^2)^{3/2}} \Rightarrow \dfrac{1}{(1+r^2)^{3/2}}$ Jacobian $= r$

$0 \le \theta \le 2\pi, \quad 0 \le r \le a < \infty$

$$\therefore \lim_{a\to\infty} \int_0^{2\pi} \int_0^a \frac{r\,dr\,d\theta}{(1+r^2)^{3/2}} = \lim_{a\to\infty} \int_0^{2\pi} -(1+r^2)^{-\frac{1}{2}} \Big|_{r=0}^{r=a} d\theta$$

$$= \lim_{a\to\infty} \int_0^{2\pi} -\frac{1}{\sqrt{1+a^2}} + 1\; d\theta = \int_0^{2\pi} d\theta = \underline{2\pi}$$

as $\lim_{a\to\infty} \dfrac{1}{\sqrt{1+a^2}} = 0$

7.1 The Path Integral

Note Title 2/3/2017

In Exercises 1 to 4, find an appropriate parametrization for the given piecewise-smooth curve in $\mathbb{R}^2$, with the implied orientation.

1.

1. The curve C, which goes along the circle of radius 3, from the point $(3, 0)$ to the point $(-3, 0)$, and then in a straight line along the x-axis back to $(3, 0)$

Consider $(3\cos t, 3\sin t)$ for $0 \le t \le \pi$

and then for $\pi \le t \le 7\pi$, $\left(\frac{t}{\pi} - 4, 0\right)$

$$\therefore \quad c(t) = \begin{cases} (3\cos t, 3\sin t), & t \in [0, \pi] \\ \left(\frac{t}{\pi} - 4, 0\right), & t \in [\pi, 7\pi] \end{cases}$$

2.

2. The curve C, which goes along $y = x^2$ from the point $(0, 0)$ to the point $(2, 4)$, then in a straight line from $(2, 4)$ to $(0, 4)$, and then along the y-axis back to $(0, 0)$

For $y = x^2$: $c(t) = (t, t^2)$ for $t \in [0, 2]$

Line from $(2,4)$ to $(0,4)$: $(4-t, 4)$ for $t \in [2, 4]$

Line from $(0,4)$ to $(0,0)$: $(0, 8-t)$, $t \in [4, 8]$.

$$\therefore \quad c(t) = \begin{cases} (t, t^2), & t \in [0, 2] \\ (4-t, 4), & t \in [2, 4] \\ (0, 8-t), & t \in [4, 8] \end{cases}$$

3.

3. The curve C, which goes along $y = \sin x$ from the point $(0, 0)$ to the point $(\pi, 0)$, and then along the x-axis back to $(0, 0)$

$$c(t) = \begin{cases} (t, \sin t), & t \in [0, \pi] \\ (2\pi - t, 0), & t \in [\pi, 2\pi] \end{cases}$$

4.

4. The closed curve C described by the ellipse

$$\frac{(x-2)^2}{4} + \frac{(y-3)^2}{9} = 1$$

oriented counterclockwise

Centre of ellipse at $(2,3)$, minor axis $= 2$, major $= 3$

Let $x = 2 + 2\cos\theta$, $y = 3 + 3\sin\theta$

$\therefore (x-2)^2 = 4\cos^2\theta \quad (y-3)^2 = 9\sin^2\theta$

$\therefore \frac{(x-2)^2}{4} + \frac{(y-3)^2}{9} = \frac{4\cos^2\theta}{4} + \frac{9\sin^2\theta}{9} = 1.$

$\therefore c(\theta) = (2 + 2\cos\theta, 3 + 3\sin\theta), \quad \theta \in [0, 2\pi]$

In Exercises 5 to 8, find an appropriate parametrization for the given piecewise-smooth curve in $\mathbb{R}^3$.

5.

5. The intersection of the plane $z = 3$ with the elliptical cylinder

$$\frac{x^2}{9} + \frac{y^2}{16} = 1$$

$$c(\theta) = (3\cos\theta, 4\sin\theta, 3), \quad \theta \in [0, 2\pi]$$

6. **6.** The triangle formed by traveling from the point (1, 2, 3) to (0, −2, 1), to (6, 4, 2), and back to (1, 2, 3)

$(1,2,3)$ to $(0,-2,1)$: $(1,2,3) + t[(0,-2,1)-(1,2,3)]$

$= (1-t, 2-4t, 3-2t)$, $0 \le t \le 1$

$(0,-2,1)$ to $(6,4,2)$: $(0,-2,1) + s[(6,4,2)-(0,-2,1)]$

$= (0,-2,1) + s(6,6,1)$, $0 \le s \le 1$

Let $t-1=s$, so $t=s+1$, $\therefore 1 \le t \le 2$

$\therefore$ Segment $= (0,-2,1) + (t-1)(6,6,1)$

$= (0,-2,1) + (-6,-6,-1) + t(6,6,1)$

$= (-6,-8,0) + t(6,6,1)$, $1 \le t \le 2$

$(6,4,2)$ to $(1,2,3)$: $(6,4,2) + s[(1,2,3)-(6,4,2)]$

$= (6,4,2) + s(-5,-2,1)$, $0 \le s \le 1$

Let $t-2=s$, so $t=s+2$, $\therefore 2 \le t \le 3$

$\therefore$ segment $= (6,4,2) + (t-2)(-5,-2,1)$

$= (6,4,2) + (10,4,-2) + t(-5,-2,1)$

$= (16,8,0) + t(-5,-2,1)$, $2 \le t \le 3$

$\therefore$ Triangle $=$

$$\begin{cases} (1-t, 2-4t, 3-2t), & 0 \le t \le 1 \\ (-6+6t, -8+6t, t), & 1 \le t \le 2 \\ (16-5t, 8-2t, t), & 2 \le t \le 3 \end{cases}$$

7. **7.** The intersection of the surfaces $y = x$ and $z = x^3$, from the point $(-3, -3, 9)$ to $(2, 2, 4)$

Should be $(2,2,8)$.

If look at $-3 \to 2$: $-3 + t(2-(-3)) = -3+5t$, $0 \le t \le 1$.

$\therefore [-3+5t, -3+5t, (-3+5t)^3]$, $0 \le t \le 1$

8. **8.** The intersection of the cylinder $y^2 + z^2 = 1$ and the plane $z = x$

The circle $y^2+z^2=1$ can be parametrized as

$(\ , \cos\theta, \sin\theta)$, $0 \le \theta \le 2\pi$

Since $x = z$, $C(\theta) = (\sin\theta, \cos\theta, \sin\theta)$, $0 \le \theta \le 2\pi$

9. **9.** Let $f(x, y, z) = y$ and $\mathbf{c}(t) = (0, 0, t)$, $0 \le t \le 1$. Prove that $\int_{\mathbf{c}} f\,ds = 0$.

$$\int_C f\,ds = \int_0^1 f(\vec{c}(t))\,\|\vec{c}\,'(t)\|\,dt$$

$$= \int_0^1 f(0,0,t)\,\|(0,0,1)\|\,dt = \int_0^1 0(1)\,dt = 0$$

10. **10.** Evaluate the following path integrals $\int_{\mathbf{c}} f(x, y, z)\,ds$, where

(a) $f(x, y, z) = x + y + z$ and
$\mathbf{c}$: $t \mapsto (\sin t, \cos t, t)$, $t \in [0, 2\pi]$

(a) $f(x(t), y(t), z(t)) = f(\sin t, \cos t, t) = \sin t + \cos t + t$

$\vec{c}\,'(t) = (\cos t, -\sin t, 1)$

$$\therefore \int_C f\,ds = \int_0^{2\pi} [\sin t + \cos t + t]\, \|(\cos t, -\sin t, 1)\|\, dt$$

$$= \int_0^{2\pi} [\sin t + \cos t + t] \sqrt{\cos^2 t + \sin^2 t + 1}\, dt$$

$$= \int_0^{2\pi} [\sin t + \cos t + t] \sqrt{2}\, dt$$

$$= \sqrt{2} \left[-\cos t + \sin t + \frac{t^2}{2} \right]_{t=0}^{t=2\pi}$$

$$= \sqrt{2} \left[-1 + 0 + \frac{(2\pi)^2}{2} - (-1 + 0 + 0) \right]$$

$$= 2\sqrt{2}\,\pi^2$$

(6) (b) $f(x, y, z) = \cos z$, $\mathbf{c}$ as in part (a)

$$f(x(t), y(t), z(t)) = \cos(z(t)) = \cos(t)$$

since $z(t) = t$, from $\vec{c}(t) = (\sin t, \cos t, t)$.

$$\therefore \int_C f\,ds = \int_0^{2\pi} [\cos t] \sqrt{\cos^2 t + \sin^2 t + 1}\, dt$$

$$= \sqrt{2} \int_0^{2\pi} \cos(t)\, dt = \sqrt{2} \left[\sin t\right]_{t=0}^{t=2\pi} = \underline{0}$$

11.

11. Evaluate the following path integrals $\int_{\mathbf{c}} f(x, y, z)\, ds$, where

(a) $f(x, y, z) = \exp\sqrt{z}$, and $\mathbf{c}$: $t \mapsto (1, 2, t^2)$, $t \in [0, 1]$

(b) $f(x, y, z) = yz$, and $\mathbf{c}$: $t \mapsto (t, 3t, 2t)$, $t \in [1, 3]$

(a) $\vec{c}\,'(t) = (0, 0, 2t)$. $\therefore \|\vec{c}\,'(t)\| = \sqrt{4t^2} = 2t$, for $t \in [0,1]$

$$f(\vec{c}(t)) = \exp(\sqrt{t^2}) = e^t$$

$$\therefore \int_c f\, ds = \int_0^1 e^t (2t)\, dt = 2\left[te^t - e^t\right]_0^1$$

$$= 2\left[e - e - (0 - 1)\right] = \underline{2}$$

(b) $\vec{c}\,'(t) = (1, 3, 2)$. $\therefore \|\vec{c}\,'(t)\| = \sqrt{14}$

$$f(\vec{c}(t)) = y(t)\, z(t) = (3t)(2t) = 6t^2$$

$$\therefore \int_c f\, ds = \int_1^3 (6t^2)(\sqrt{14})\, dt = \sqrt{14}\left[2t^3\right]_1^3$$

$$= \sqrt{14}\,(54 - 2) = \underline{52\sqrt{14}}$$

12. **12.** Evaluate the integral of $f(x, y, z)$ along the path $\mathbf{c}$, where

(a) $f(x, y, z) = x \cos z$, $\mathbf{c}: t \mapsto t\mathbf{i} + t^2\mathbf{j}$, $t \in [0, 1]$

(b) $f(x, y, z) = (x + y)/(y + z)$, and $\mathbf{c}: t \mapsto \left(t, \frac{2}{3}t^{3/2}, t\right)$, $t \in [1, 2]$

(a) $\vec{c}(t) = (t, t^2, 0)$, $\vec{c}\,'(t) = (1, 2t, 0)$ $\|\vec{c}\,'(t)\| = \sqrt{1+4t^2}$

$f(\vec{c}(t)) = (t)(\cos(0)) = t$

$$\therefore \int_C f\,ds = \int_0^1 t\sqrt{1+4t^2}\,dt = \left(\tfrac{2}{3}\right)\left(\tfrac{1}{8}\right)(1+4t^2)^{\frac{3}{2}}\Big|_0^1$$

$$= \frac{1}{12}\left[1+4-(1+0)\right] = \underline{\frac{1}{3}}$$

(b) $\vec{c}\,'(t) = (1, t^{\frac{1}{2}}, 1)$, $\|\vec{c}\,'(t)\| = \sqrt{2+t}$

$$f(\vec{c}(t)) = \frac{t + \frac{2}{3}t^{3/2}}{\frac{2}{3}t^{3/2} + t} = 1 \quad \text{for } 1 \le t \le 2$$

$$\therefore \int_C f\,ds = \int_1^2 (1)\sqrt{2+t}\,dt = \left(\tfrac{2}{3}\right)(2+t)^{3/2}\Big|_1^2$$

$$= \frac{2}{3}\left[4^{3/2} - 3^{3/2}\right] = \frac{2}{3}\left[8 - 3\sqrt{3}\right] = \underline{\frac{16}{3} - 2\sqrt{3}}$$

13. **13.** Let $f: \mathbb{R}^3 \setminus \{xz \text{ plane}\} \to \mathbb{R}$ be defined by $f(x, y, z) = 1/y^3$. Evaluate $\int_{\mathbf{c}} f(x, y, z)\,ds$, where $\mathbf{c}: [1, e] \to \mathbb{R}^3$ is given by $\mathbf{c}(t) = (\log t)\mathbf{i} + t\mathbf{j} + 2\mathbf{k}$.

$\vec{c}(t) = (\log t, t, 2)$, $\vec{c}\,'(t) = \left(\frac{1}{t}, 1, 0\right)$, $1 \le t \le e$

$$f(\vec{c}(t)) = \frac{1}{t^3} \qquad \|\vec{c}\,'(t)\| = \sqrt{\frac{1}{t^2}+1} = \frac{\sqrt{1+t^2}}{t}$$

$$\therefore \int_c f\,ds = \int_1^e \left(\frac{1}{t^3}\right)\left(\frac{\sqrt{1+t^2}}{t}\right)dt = \int_1^e \frac{\sqrt{1+t^2}}{t^4}\,dt$$

$$= -\frac{\sqrt{(t^2+1)^3}}{3t^3}\Bigg|_1^e \qquad \text{using text table of integrals \# 70}$$

$$= \left[-\frac{\sqrt{(e^2+1)^3}}{3e^3} - \left(-\frac{\sqrt{8}}{3}\right)\right] = \frac{2\sqrt{2}}{3} - \frac{\sqrt{(e^2+1)^3}}{3e^3}$$

14.

14. (a) Show that the path integral of $f(x, y)$ along a path given in polar coordinates by $r = r(\theta)$, $\theta_1 \le \theta \le \theta_2$, is

$$\int_{\theta_1}^{\theta_2} f(r\cos\theta, r\sin\theta)\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\,d\theta.$$

(b) Compute the arc length of the path $r = 1 + \cos\theta$, $0 \le \theta \le 2\pi$.

Let $\vec{c}(\theta) = (r\cos\theta, r\sin\theta)$ for $\theta_1 \le \theta \le \theta_2$

$$= (r(\theta)\cos\theta, r(\theta)\sin\theta) = [x(\theta), y(\theta)]$$

$$\therefore f(\vec{c}(\theta)) = f(r(\theta)\cos\theta, r(\theta)\sin\theta)$$

$$\frac{d}{d\theta}\vec{c}(\theta) = \vec{c}\,'(\theta) = [-r(\theta)\sin\theta + r'(\theta)\cos\theta,\ r(\theta)\cos\theta + r'(\theta)\sin\theta]$$

$$\therefore \|\vec{c}\,'(\theta)\| = [r^2\sin^2\theta + (r')^2\cos^2\theta - 2rr'\sin\theta\cos\theta + r^2\cos^2\theta + (r')^2\sin^2\theta + 2rr'\cos\theta\sin\theta]^{\frac{1}{2}}$$

$$= \left[r^2 + (r')^2 \right]^{\frac{1}{2}} = \sqrt{r(\theta)^2 + \left(\frac{dr(\theta)}{d\theta}\right)^2}$$

$$\therefore \int_C f\,ds = \int_{\theta_1}^{\theta_2} f(r\cos\theta, r\sin\theta) \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

noting for simplicity that $r = r(\theta)$

(6) Here, $f(x,y) = 1$, so $f(r\cos\theta, r\sin\theta) = 1$

just like $\iiint_W f\,dV$ = volume of W when $f = 1$.

$$\therefore \text{Arc length} = \int_C ds = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + (r')^2}\, d\theta$$

$$r^2(\theta) = (1+\cos\theta)^2, \quad \frac{dr}{d\theta} = -\sin\theta, \quad \left(\frac{dr}{d\theta}\right)^2 = \sin^2\theta$$

$$\therefore \int_C ds = \int_0^{2\pi} \sqrt{(1+\cos\theta)^2 + \sin^2\theta}\, d\theta$$

$$= \int_0^{2\pi} \sqrt{2 + 2\cos\theta}\, d\theta = \sqrt{2} \int_0^{2\pi} \sqrt{1+\cos\theta}\, d\theta$$

$$= 2\int_0^{2\pi} \sqrt{\frac{1+\cos\theta}{2}}\, d\theta$$

Use $\cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}$

$\cos\frac{\theta}{2} \geq 0$ for $0 \leq \theta \leq \pi$

$-\cos\frac{\theta}{2} \geq 0$ for $\pi \leq \theta \leq 2\pi$

$$= 2\int_0^{\pi} \cos\frac{\theta}{2}\,d\theta + 2\int_{\pi}^{2\pi} -\cos\frac{\theta}{2}\,d\theta$$

$$= 2(2)\sin\frac{\theta}{2}\Big|_0^{\pi} + 2(-2)\sin\frac{\theta}{2}\Big|_{\pi}^{2\pi}$$

$$= 4(1-0) + (-4)(0-1) = \underline{\underline{8}}$$

15.

15. Let $f(x, y) = 2x - y$, and consider the path $x = t^4$, $y = t^4$, $-1 \leq t \leq 1$.

(a) Compute the integral of f along this path and interpret the answer geometrically.

(b) Evaluate the arc-length function $s(t)$ and redo part (a) in terms of s (you may wish to consult Exercise 2, Section 4.2).

(a) $\vec{c}(t) = (t^4, t^4)$, $\vec{c}\,'(t) = (4t^3, 4t^3)$

$f(\vec{c}(t)) = 2(t^4) - (t^4) = t^4$

$\|\vec{c}\,'(t)\| = \sqrt{16t^6 + 16t^6} = 4\sqrt{2t^6}$

Note that $\|\vec{c}\,'(t)\|$ is to be positive over the path.

The parametrization $\vec{c}(t) = (t^4, t^4)$ is

basically the line $y = x$. For $t \in [-1, 0]$, this from $(1,1)$ to $(0,0)$, and from $t \in [0,1]$, it is $(0,0)$ back to $(1,1)$, an equal path.

$$\therefore \int_C f\,ds = 2\int_0^1 f\,ds = 2\int_0^1 (t^4)\left(4\sqrt{2t^6}\right)dt$$

$$= 2\int_0^1 4\sqrt{2}\,t^7\,dt, \text{ since } \sqrt{2t^6} = \sqrt{2}t^3 \text{ for } t \in [0,1]$$

$$= \sqrt{2}\,t^8\Big|_0^1 = \underline{\sqrt{2}}$$

Since $f(\vec{c}(t)) = t^4$, the graph is $(x(t), y(t), z(t)) = (t^4, t^4, t^4)$, or a triangle from $(0,0,0)$ to $(1,1,1)$.
$\therefore$ Base is $\sqrt{2}$, height is 1, area is $\frac{\sqrt{2}}{2}$.

Going from $(0,0,0)$ to $(1,1,1)$, then back to $(0,0,0)$ gives twices the area of the triangle.

(6) From page 232 of text (section 4.2)

$$s(t) = \int_a^t \|\vec{c}\,'(u)\|\,du, \quad \|\vec{c}\,'(u)\| = 4\sqrt{2u^6} \text{ from above}$$

For $a=-1$, $-1 \le u \le 0$, $\|\vec{c}\,'(u)\| = -4\sqrt{2}\,u^3$

$$s(t) = \int_{-1}^{t} -4\sqrt{2}\,u^3\,du = -\sqrt{2}\,u^4\Big|_{-1}^{t}$$

$$= -\sqrt{2}\,t^4 - (-\sqrt{2}) = \sqrt{2} - \sqrt{2}\,t^4$$

when $t=0$, $s(0)=\sqrt{2}$

For $0 \le u \le 1$, $\|\vec{c}\,'(u)\| = 4\sqrt{2}\,u^3$

Since $s(t)$ is supposed to represent total length from the start ($t=-1$), need to add length from $t=-1$ to $t=0$ for $t \in [0,1]$

$\therefore$

$$s(t) = s(0) + \int_{0}^{t} 4\sqrt{2}\,u^3\,du$$

$$= \sqrt{2} + \sqrt{2}\,u^4\Big|_{0}^{t}$$

$$= \sqrt{2} + \sqrt{2}\,t^4$$

$$\therefore s(t) = \begin{cases} \sqrt{2} - \sqrt{2}\,t^4, & -1 \le t \le 0 \\ \sqrt{2} + \sqrt{2}\,t^4, & 0 \le t \le 1 \end{cases}$$

To find a parametrization of the path from $(1,1)$ to $(0,0)$, and then back to $(1,1)$; note the length of the path from $(1,1)$ to $(0,0)$ is $\sqrt{2}$.

s is 0 at $(1,1)$ and $\sqrt{2}$ at $(0,0)$.

$$\therefore \vec{c}(s) = \left(1-\frac{s}{\sqrt{2}}\right)(1,1) + \frac{s}{\sqrt{2}}(0,0)$$

$$= \left(1-\frac{s}{\sqrt{2}}\right)(1,1), \quad 0 \le s \le \sqrt{2}$$

Coordinate parametrization going back is:

$$\vec{c}(t) = \left(1-\frac{t}{\sqrt{2}}\right)(0,0) + \frac{t}{\sqrt{2}}(1,1), \quad 0 \le t \le \sqrt{2}$$

or, $\sqrt{2} \le t+\sqrt{2} \le 2\sqrt{2}$, let $s = t+\sqrt{2}$

$$\vec{c}(s) = \left(1-\frac{s-\sqrt{2}}{\sqrt{2}}\right)(0,0) + \frac{s-\sqrt{2}}{\sqrt{2}}(1,1)$$

$$= \left(\frac{s}{\sqrt{2}}-1\right)(1,1), \quad \sqrt{2} \le s \le 2\sqrt{2}$$

using $(1-t)\vec{a} + t\vec{b}$, $0 \le t \le 1$, for $\vec{a}$ to $\vec{b}$

$$\therefore \vec{c}(s) = \begin{cases} \left(1-\frac{s}{\sqrt{2}}\right)(1,1), & 0 \le s \le \sqrt{2} \\ \left(\frac{s}{\sqrt{2}}-1\right)(1,1), & \sqrt{2} \le s \le 2\sqrt{2} \end{cases}$$

$$\therefore \vec{c}\,'(s) = \begin{cases} \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right), & 0 \le s \le \sqrt{2} \\ \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right), & \sqrt{2} \le s \le 2\sqrt{2} \end{cases}$$

$$\|\vec{c}\,'(s)\| = \sqrt{\tfrac{1}{2}+\tfrac{1}{2}} = 1.$$

$$f(x(s),y(s)) = \begin{cases} 2\left(1-\frac{s}{\sqrt{2}}\right)-\left(1-\frac{s}{\sqrt{2}}\right), & 0 \le s \le \sqrt{2} \\ \quad = \left(1-\frac{s}{\sqrt{2}}\right) & \\ 2\left(\frac{s}{\sqrt{2}}-1\right)-\left(\frac{s}{\sqrt{2}}-1\right), & \sqrt{2} \le s \le 2\sqrt{2} \\ \quad = \left(\frac{s}{\sqrt{2}}-1\right) & \end{cases}$$

$$\therefore \int_C f\,ds = \int_0^{\sqrt{2}} \left(1-\frac{s}{\sqrt{2}}\right)(1)\,ds + \int_{\sqrt{2}}^{2\sqrt{2}} \left(\frac{s}{\sqrt{2}}-1\right)(1)\,ds$$

$$= s - \frac{s^2}{2\sqrt{2}}\Bigg|_0^{\sqrt{2}} + \frac{s^2}{2\sqrt{2}} - s\Bigg|_{\sqrt{2}}^{2\sqrt{2}}$$

$$= \sqrt{2} - \frac{1}{\sqrt{2}} + \left[\frac{4\cdot 2}{2\sqrt{2}} - 2\sqrt{2} - \left(\frac{2}{2\sqrt{2}} - \sqrt{2}\right)\right]$$

$$= \frac{1}{\sqrt{2}} + \left[\frac{3}{\sqrt{2}} - \sqrt{2}\right] = \frac{4}{\sqrt{2}} - \sqrt{2}$$

$$= 2\sqrt{2} - \sqrt{2} = \underline{\sqrt{2}}$$

Same answer as in (a) using length as a parametrization.

Exercises 16 to 19 are concerned with the application of the path integral to the problem of defining the average value of a scalar function along a path. Define the number

$$\frac{\int_{\mathbf{c}} f(x, y, z)\, ds}{l(\mathbf{c})}$$

to be the ***average value*** *of f along $\mathbf{c}$. Here $l(\mathbf{c})$ is the length of the path:*

$$l(\mathbf{c}) = \int_{\mathbf{c}} \|\mathbf{c}'(t)\|\, dt.$$

(This is analogous to the average of a function over a region defined in Section 6.3.)

16.

16. (a) Justify the formula $[\int_{\mathbf{c}} f(x, y, z)\, ds]/l(\mathbf{c})$ for the average value of f along $\mathbf{c}$ using Riemann sums.

(b) Show that the average value of f along $\mathbf{c}$ in Example 1 is $(1 + \frac{4}{3}\pi^2)$.

(c) In Exercise 10(a) and (b) above, find the average value of f over the given curves.

(a) Let $[a,b]$ be an interval, subdivide it by a partition $a = t_0 < t_1 < t_2 < \dots < t_n = b$

Let $\vec{c}(t)$, $t \in [a,b]$ be C^1, and thereby breaking path $\vec{c}(t)$ into paths $\vec{c}_i$ defined on $[t_i, t_{i+1}]$, for $0 \le i \le n-1$.

When n is large, $[t_i, t_{i+1}]$ is small, and so is

$$\Delta s_i = \int_{t_i}^{t_{i+1}} \|\vec{c}_i'(t)\|\, dt$$

$\therefore f(\vec{c}_i(t))$ is fairly constant on $[t_i, t_{i+1}]$

$\therefore$ For some $t \in [t_i, t_{i+1}]$,

consider the sum $S_n = \sum_{i=0}^{n-1} f(\vec{c}_i)\, \Delta s_i$

The average value of f on the path would be:

$$f_{ave} = \frac{\sum_{i=0}^{n-1} f(\vec{c}_i)\,\Delta s_i}{\text{length of path}}$$

since $\Delta s_0 + \Delta s_1 + \cdots + \Delta s_{n-1} = \text{length of path}$

But by mean value theorem, $\Delta s_i = \|\vec{c}_i'(t^*)\|\,\Delta t_i$,

for some $t^* \in [t_i, t_{i+1}]$

$$\therefore \sum_{i=0}^{n-1} f(\vec{c}_i)\,\Delta s_i = \sum_{i=0}^{n-1} f(\vec{c}_i)\,\|\vec{c}_i'(t^*)\|\,\Delta t_i$$

$$= \frac{S_n}{\text{length of path}} = \frac{S_n}{\ell(\vec{c})}$$

$$= \frac{S_n}{\int_C ds}$$

As $n \to \infty$, $S_n \to \int_I f(x,y,z)\,\|\vec{c}'(t)\|\,dt$

$$= \int_C f\,ds$$

$$\therefore f_{ave} = \frac{\int_c f\,ds}{\int_c ds}$$

(b) Show that the average value of f along $\mathbf{c}$ in Example 1 is $(1 + \frac{4}{3}\pi^2)$.

example 1 Let $\mathbf{c}$ be the helix $\mathbf{c}: [0, 2\pi] \to \mathbb{R}^3$, $t \mapsto (\cos t, \sin t, t)$ (see Figure 2.4.9), and let $f(x, y, z) = x^2 + y^2 + z^2$. Evaluate the integral $\int_{\mathbf{c}} f(x, y, z)\,ds$.

Here $\int_c ds = \int_0^{2\pi} \|\vec{c}'(t)\|\,dt$

and $\vec{c}'(t) = (-\sin t, \cos t, 1)$

$\therefore \|\vec{c}'(t)\| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}$

$\therefore \int_0^{2\pi} \sqrt{2}\,dt = 2\sqrt{2}\,\pi$

From the text, $\int_c f\,ds = \frac{2\sqrt{2}\,\pi}{3}(3 + 4\pi^2)$

$\therefore f_{ave} = \frac{\frac{2\sqrt{2}\,\pi\,(3 + 4\pi^2)}{3}}{2\sqrt{2}\,\pi} = \frac{(3 + 4\pi^2)}{3}$

$= 1 + \frac{4}{3}\pi^2$

(c) In Exercise 10(a) and (b) above, find the average value of f over the given curves.

10(a): $\int_C f\,ds = 2\sqrt{2}\,\pi^2$

$\vec{c}(t) = (\sin t, \cos t, t)$, for $t \in [0, 2\pi]$

as in (b) above, $\int_C ds = 2\sqrt{2}\,\pi$

$\therefore f_{ave} = \dfrac{2\sqrt{2}\,\pi^2}{2\sqrt{2}\,\pi} = \underline{\pi}$

10(b): $\int_C f\,ds = 0 \quad \therefore f_{ave} = 0$

17. **17.** Find the average y coordinate of the points on the semicircle parametrized by $\mathbf{c}$: $[0, \pi] \to \mathbb{R}^3$, $\theta \mapsto (0, a\sin\theta, a\cos\theta)$; $a > 0$.

Length of semicircle of radius $a = \pi a$

$f(x, y, z) = y = a\sin\theta$

$\therefore \int_C f\,ds = \int_0^{\pi} (a\sin\theta)\,\|\vec{c}\,'(\theta)\|\,d\theta$

$\vec{c}\,'(\theta) = (0, a\cos\theta, -a\sin\theta) \quad \therefore \|\vec{c}\,'(\theta)\| = a$

$\therefore \int_C f\,ds = \int_0^{\pi} (a\sin\theta)(a)\,d\theta = -a^2\cos\theta\,\Big|_0^{\pi}$

$= a^2 - (-a^2) = 2a^2$

$$\therefore \quad y_{ave} = \frac{2a^2}{\pi a} = \frac{2a}{\pi}$$

18. **18.** Suppose the semicircle in Exercise 17 is made of a wire with a uniform density of 2 grams per unit length.

(a) What is the total mass of the wire?

(b) Where is the center of mass of this configuration of wire? (Consult Section 6.3.)

(a) Mass = (density)(length) = $(2)(\pi a) = 2\pi a$ grams

(b) By symmetry, since wire is in yz-plane and a semicircle with center $(0,0,0)$, extending from $(0,0,a)$ to $(0,0,-a)$, the center of mass will have coordinates $(0,y,0)$.

The wire has uniform density $= \sigma = 2$

$$\therefore \text{ center of mass } = \frac{\int_C \sigma y \, ds}{\int_C \sigma \, ds} = \frac{\int_C y \, ds}{\int_C ds}$$

This computation was made in #17,

and so y-coordinate $= \frac{2a}{\pi}$.

$\therefore$ Center of mass $= \underline{(0, 2a/\pi, 0)}$.

19.

19. Let $\mathbf{c}$ be the path given by $\mathbf{c}(t) = (t^2, t, 3)$ for $t \in [0, 1]$.

(a) Find $l(\mathbf{c})$, the length of the path.

(b) Find the average y coordinate along the path $\mathbf{c}$.

(a) $\vec{c}\,'(t) = (2t, 1, 0)$. $\therefore \|\vec{c}\,'(t)\| = \sqrt{4t^2+1+0} = \sqrt{4t^2+1}$

$$l(\vec{c}) = \int_c ds = \int_0^1 \sqrt{4t^2+1}\, dt$$

letting $u = 2t$
$\therefore du = 2dt$

$$= \frac{1}{2}\int_0^2 \sqrt{u^2+1}\, du = \frac{1}{2}\left[\frac{u}{2}\sqrt{u^2+1} + \frac{1}{2}\ln\left(u+\sqrt{u^2+1}\right)\right]_0^2$$

$$= \frac{1}{2}\left[\sqrt{5} + \frac{1}{2}\ln(2+\sqrt{5})\right] = \frac{\sqrt{5}}{2} + \frac{1}{4}\ln(2+\sqrt{5})$$

(b) $$\int_c t\,\|\vec{c}\,'(t)\|\,dt = \int_0^1 t\sqrt{4t^2+1}\, dt$$

$$= \left(\frac{2}{3}\right)\left(\frac{1}{8}\right)(4t^2+1)^{3/2}\Big|_0^1 = \frac{1}{12}\left(5^{3/2} - 1\right)$$

$$= \frac{5\sqrt{5}-1}{12}$$

$\therefore$ Ave of y-coord $= \dfrac{\frac{5\sqrt{5}-1}{12}}{\frac{2\sqrt{5}+\ln(2+\sqrt{5})}{4}}$

$$= \frac{5\sqrt{5} - 1}{6\sqrt{5} - 3\ln(2+\sqrt{5})}$$

20.

20. Show that the path integral of a function $f(x, y)$ over a path C given by the graph of $y = g(x)$, $a \le x \le b$ is given by:

$$\int_C f\,ds = \int_a^b f(x, g(x))\sqrt{1 + [g'(x)]^2}\,dx$$

Conclude that if $g : [a, b] \to \mathbb{R}$ is piecewise continuously differentiable, then the length of the graph of g on $[a, b]$ is given by:

$$\int_C f\,ds = \int_a^b \sqrt{1 + g'(x)^2}\,dx.$$

(a) $\vec{c}(x) = (x, g(x))$, for $a \le x \le b$

$\therefore \frac{d}{dx}\vec{c}(x) = \vec{c}'(x) = (1, g'(x))$.

$\therefore \|\vec{c}'(x)\| = \sqrt{1 + (g'(x))^2}$

On $\vec{c}$, $f(x,y) = f(x, g(x))$.

$\therefore \int_c f\,ds = \int_a^b f(x, g(x))\sqrt{1 + [g'(x)]^2}\,dx$

(b) For $f(x,y) = 1$, the above formula gives the length of $g(x)$

$\therefore$ length $g(x)$ on $[a,b] = \int_a^b \sqrt{1 + g'(x)^2}\,dx$

21.

21. If g: $[a, b] \to \mathbb{R}$ is piecewise continuously differentiable, let the *length of the graph* of g on $[a, b]$ be defined as the length of the path $t \mapsto (t, g(t))$ for $t \in [a, b]$. Show that the length of the graph of g on $[a, b]$ is

$$\int_a^b \sqrt{1 + [g'(x)]^2}\, dx.$$

As shown in #20 above, given $\vec{c}(t) = (t, g(t))$,

$\vec{c}\,'(t) = (1, g'(t))$. $\therefore \|\vec{c}\,'(t)\| = \sqrt{1 + [g'(t)]^2}$

$\int_c f\,ds$ where $f = 1$ becomes $\int_c ds$ and is the length of the curve traced out by $\vec{c}(t)$.

$\therefore$ length of the graph (= length of curve from $\vec{c}(t)$) is

$$\int_c ds = \int_a^b \|\vec{c}\,'(t)\|\,dt = \int_a^b \sqrt{1 + [g'(t)]^2}\,dt$$

22.

22. Use Exercise 21 to find the length of the graph of $y = \log x$ from $x = 1$ to $x = 2$.

$$\int_1^2 \sqrt{1 + \left(\frac{1}{x}\right)^2}\,dx = \int_1^2 \frac{\sqrt{x^2+1}}{x}\,dx$$

use table of integrals #56 from text.

$$= \sqrt{x^2+1} - \log\left[\frac{1 + \sqrt{x^2+1}}{x}\right]\Bigg|_{x=1}^{x=2}$$

$$= \left[\sqrt{5} - \log\left(\frac{1+\sqrt{5}}{2}\right) - \left(\sqrt{2} - \log(1+\sqrt{2})\right) \right]$$

$$= \sqrt{5} - \sqrt{2} + \log\left(\frac{2+2\sqrt{2}}{1+\sqrt{5}}\right)$$

23. **23.** Use Exercise 20 to evaluate the path integral of $f(x, y) = y$ over the graph of the semicircle $y = \sqrt{1 - x^2}$, $-1 \le x \le 1$.

$$\int_{-1}^{1} f(x, g(x)) \sqrt{1 + g'(x)^2}\, dx$$

Here, $f(x, g(x)) = g(x) = \sqrt{1-x^2}$ as $g(x) = \sqrt{1-x^2}$ and $g'(x) = \frac{1}{2}(1-x^2)^{-\frac{1}{2}}(-2x) = -\frac{x}{\sqrt{1-x^2}}$

$$= \int_{-1}^{1} \sqrt{1-x^2}\sqrt{1+\frac{x^2}{1-x^2}}\, dx$$

$$= \lim_{a \to 1} \int_{-a}^{a} \sqrt{1-x^2}\sqrt{\frac{1}{1-x^2}}\, dx = \lim_{a \to 1} \int_{-a}^{a} (1)\, dx = 2$$

24. **24.** Compute the path integral of $f(x, y) = y^2$ over the graph $y = e^x$, $0 \le x \le 1$.

$$f(x, y) = y^2 = (e^x)^2 = e^{2x}$$

$$\therefore \int_0^1 (e^{2x}) \sqrt{1 + (e^x)^2}\, dx = \int_0^1 e^{2x} \sqrt{1 + e^{2x}}\, dx$$

$$= \left(\frac{2}{3}\right)\left(\frac{1}{2}\right)\left(1+e^{2x}\right)^{3/2}\Big|_{x=0}^{x=1} = \frac{1}{3}\left[1+e^2-(1+1)\right]$$

$$= \frac{e^2-1}{3}$$

25. **25.** Compute the path integral of $f(x, y, z) = xyz$ over the path $c(t) = (\cos t, \sin t, t)$, $0 \le t \le \frac{\pi}{2}$.

$$\vec{c}\,'(t) = (-\sin t, \cos t, 1) \qquad \|\vec{c}\,'(t)\| = \sqrt{\sin^2 t + \cos^2 t + 1} = \sqrt{2}$$

From $\vec{c}(t) = (x(t), y(t), z(t))$, $f(x,y,z) = t\cos t \sin t$

$$\therefore \int_c f\,ds = \int_0^{\frac{\pi}{2}} (t\cos t \sin t)(\sqrt{2})\,dt$$

$$= \frac{\sqrt{2}}{2}\int_0^{\frac{\pi}{2}} t\sin 2t\,dt$$

Let $u = t$ $dv = \sin 2t\,dt$
$du = dt$ $v = -\frac{1}{2}\cos 2t$
integration by parts

$$= \frac{\sqrt{2}}{2}\left[-\frac{t}{2}\cos 2t\Big|_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} -\frac{1}{2}\cos 2t\,dt\right]$$

$$= \frac{\sqrt{2}}{2}\left[\frac{\pi}{4} - 0 + \frac{1}{4}\sin 2t\Big|_0^{\frac{\pi}{2}}\right]$$

$$= \frac{\sqrt{2}\,\pi}{8}$$

26. **26.** Find the mass of a wire formed by the intersection of the sphere $x^2 + y^2 + z^2 = 1$ and the plane $x + y + z = 0$ if the density at (x, y, z) is given by $\rho(x, y, z) = x^2$ grams per unit length of wire.

Mass = (density)(length). $\therefore \Delta m = \rho(x,y,z)\,ds$

Total mass = $\sum \Delta m = \sum \rho\, ds = \int_C \rho\, ds$

Need to find a parameterization for the curve.

From $z = -x - y$ of the plane,

$$x^2 + y^2 + (-x-y)^2 = 1 \Rightarrow 2x^2 + 2y^2 + 2xy = 1$$

From analytic geometry, can do a rotation transformation to get rid of xy term. Resulting curve likely an ellipse, but then need to find a parameterization for the ellipse. A lot of messy work.

Other idea: The intersection is a unit circle in the $x+y+z=0$ plane, since the plane goes through $(0,0,0)$.

$\therefore$ if can find orthogonal unit vectors in that plane, can use $x' = \cos\theta$ and $y' = \sin\theta$ to parametrize the circle.

$(1,-1,0)$ is in the plane $(1+(-1)+0=0)$
$(1,1,-2)$ is in the plane $(1+1-2=0)$
and $(1,-1,0)\cdot(1,1,-2) = 0$, so the two

vectors are perpendicular.

Unit vectors are $\frac{1}{\sqrt{2}}(1,-1,0)$ and $\frac{1}{\sqrt{6}}(1,1,-2)$

$\therefore$ A parameterization of the circle is

$$\vec{c}(\theta) = \frac{1}{\sqrt{2}}(1,-1,0)\cos\theta + \frac{1}{\sqrt{6}}(1,1,-2)\sin\theta, \quad 0 \le \theta \le 2\pi$$

$$\therefore \vec{c}\,'(\theta) = -\frac{1}{\sqrt{2}}(1,-1,0)\sin\theta + \frac{1}{\sqrt{6}}(1,1,-2)\cos\theta$$

To find $\|\vec{c}\,'(\theta)\|$, note if $\vec{a} = \vec{b} + \vec{c}$,

$$\|\vec{a}\| = \sqrt{\vec{b}\cdot\vec{b} + 2\vec{b}\cdot\vec{c} + \vec{c}\cdot\vec{c}}$$

In this case, $\vec{b}\cdot\vec{c} = 0$. $\therefore \|\vec{a}\| = \sqrt{\vec{b}\cdot\vec{b} + \vec{c}\cdot\vec{c}}$

$$\therefore \|\vec{c}\,'(\theta)\| = \sqrt{\frac{\cos^2\theta}{2}(1+1+0) + \frac{\sin^2\theta}{6}(1+1+4)}$$

$$= \sqrt{\cos^2\theta + \sin^2\theta} = 1.$$

Now, $\vec{c}(\theta) = (x(\theta), y(\theta), z(\theta))$

$$\therefore x(\theta) = \frac{\cos\theta}{\sqrt{2}} + \frac{\sin\theta}{\sqrt{6}}$$

$$\rho(x(\theta), y(\theta), z(\theta)) = x(\theta)^2 = \left(\frac{\cos\theta}{\sqrt{2}} + \frac{\sin\theta}{\sqrt{6}}\right)^2$$

$$= \frac{\cos^2\theta}{2} + \frac{\sin^2\theta}{6} + \frac{\sin 2\theta}{\sqrt{12}}$$

Note $\cos 2\theta = \cos^2\theta - \sin^2\theta \Rightarrow \cos^2\theta = \frac{1+\cos 2\theta}{2}$, $\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$$\therefore \int_C f\,ds = \int_0^{2\pi} \left(\frac{\cos^2\theta}{2} + \frac{\sin^2\theta}{6} + \frac{\sin 2\theta}{\sqrt{12}} \right)(1)\,d\theta$$

$$= \int_0^{2\pi} \frac{1+\cos 2\theta}{4} + \frac{1-\cos 2\theta}{12} + \frac{\sin 2\theta}{\sqrt{12}}\,d\theta$$

$$= \left[\frac{1}{4}\theta + \frac{\sin 2\theta}{8} + \frac{1}{12}\theta - \frac{\sin 2\theta}{24} - \frac{\cos 2\theta}{2\sqrt{12}} \right]_{\theta=0}^{\theta=2\pi}$$

$$= \frac{1}{4}(2\pi) + 0 + \frac{1}{12}(2\pi) - 0 - 0 = \frac{2}{3}\pi \text{ grams}$$

27.

27. Evaluate $\int_C f\,ds$, where $f(x, y, z) = z$ and $\mathbf{c}(t) = (t\cos t, t\sin t, t)$ for $0 \le t \le t_0$.

$$\vec{c}\,'(t) = (\cos t - t\sin t, \sin t + t\cos t, 1)$$

$$\vec{c}\,' \cdot \vec{c}\,' = \cos^2 t - 2t\cos t\sin t + t^2\sin^2 t + \sin^2 t + 2t\cos t\sin t + t^2\cos^2 t + 1$$

$$= 1 + t^2 + 1 = t^2 + 2$$

$$\therefore \|\vec{c}\,'(t)\| = \sqrt{t^2+2}$$

$$f(x(t), y(t), z(t)) = z(t) = t$$

$$\therefore \int_C f\,ds = \int_0^{t_0} t\sqrt{t^2+2}\,dt = \left(\frac{2}{3}\right)\left(\frac{1}{2}\right)(t^2+2)^{3/2}\Big|_0^{t_0}$$

$$= \frac{1}{3}\left[(t_0^2+2)^{3/2} - 2^{3/2} \right] = \frac{(t_0^2+2)^{3/2} - 2\sqrt{2}}{3}$$

28.

28. Write the following limit as a path integral of $f(x, y, z) = xy$ over some path $\mathbf{c}$ on $[0, 1]$ and evaluate:

$$\lim_{N\to\infty} \sum_{i=1}^{N-1} t_i^2 \left(t_{i+1}^2 - t_i^2\right),$$

where $t_1, \ldots, t_N$ is a partition of $[0, 1]$.

$$\lim_{N\to\infty} \sum_{i=1}^{N-1} t_i^2 \left(t_{i+1}^2 - t_i^2\right) = \lim_{N\to\infty} \sum_{i=1}^{N-1} t_i^2 (t_{i+1} + t_i)(t_{i+1} - t_i)$$

$f(x(t), y(t), z(t)) = x(t)\,y(t)$.

$\therefore$ If $\vec{c}(t) = (t, t, 0)$, $f(\vec{c}(t)) = t^2$,
but $\vec{c}\,'(t) = (1, 1, 0)$ which doesn't help
with the $(t_{i+1} + t_i)(t_{i+1} - t_i)$.

If $\vec{c}(t) = (t^2, 1, 0)$, $f(\vec{c}(t)) = t^2$,
and $\vec{c}\,'(t) = (2t, 0, 0)$. $\therefore \|\vec{c}\,'(t)\| = 2t$
This can be $t_{i+1} + t_i \simeq 2t^*$, where
$t^* \in [t_i, t_{i+1}]$.

i.e., $\lim_{N\to\infty} (t_{i+1} + t_i) = 2t_i$

$$\therefore \lim_{N\to\infty} \sum_{i=1}^{N-1} t_i^2 (t_{i+1} + t_i)(t_{i+1} - t_i) = \lim_{N\to\infty} \sum_{i=1}^{N-1} t_i^2 (2t_i)\,\Delta t =$$

$$\lim_{N\to\infty} \sum_{i=1}^{N-1} f(\vec{c}(t_i)) \, \|\vec{c}\,'(t_i)\| \, \Delta t = \int_C f \, ds$$

where $f(x,y,z) = (x,y)$, and $\vec{c}(t) = (t^2, 1, 0)$

Actually, $\vec{c}(t)$ could be $(1, t^2, K)$ or $(t^2, 1, K)$,

where K = constant, since $\|\vec{c}\,'(t)\| = 2t$

and $f(\vec{c}(t)) = t^2$ for all these

parameterizations, $t \in [0,1]$.

$$\therefore \int_C f \, ds = \int_0^1 t^2 (2t)\, dt = \frac{1}{2} t^4 \Big|_0^1 = \frac{1}{2}$$

$$\therefore \lim_{N\to\infty} \sum_{i=1}^{N-1} t_i^2 (t_{i+1}^2 - t_i^2) = \frac{1}{2}$$, t_i a partition on $[0,1]$

Note: index of i started at $i = 1$

So $t_1 = 0 < t_2 < t_3 < \dots < t_N = 1$.

If chose $i = 0$ so $t_0 = 0 < t_1 < t_2 \cdots < t_N = 1$,

you would get same result as first term

of $\sum_{i=1}^{N-1} t_i^2 (t_{i+1}^2 - t_i^2)$ is 0.

29.

29. Consider paths that connect the points $A = (0, 1)$ and $B = (1, 0)$ in the xy plane, as in Figure 7.1.5.[2]

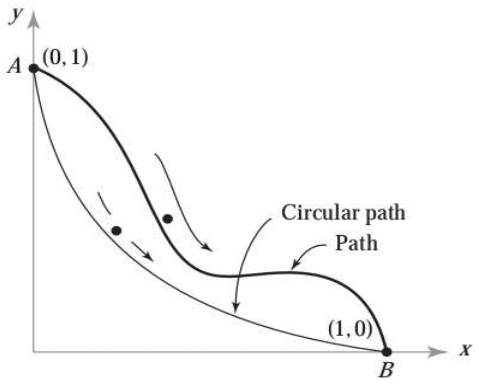


figure **7.1.5** A curve joining the points A and B.

[2] We thank Tanya Leise for suggesting this exercise.

Galileo contemplated the following question: Does a bead falling under the influence of gravity from a point A to a point B along a curve do so in *the least possible time* if that curve is a circular arc? For any given path, the time of transit T is a path integral

$$T = \int \frac{dt}{v},$$

where the bead's velocity is $v = \sqrt{2gy}$, where g is the gravitational constant. In 1697, Johann Bernoulli challenged the mathematical world to find the path in which the bead would roll from A to B in the least time. This solution would determine whether Galileo's considerations had been correct.

(a) Calculate T for the straight-line path $y = 1 - x$.

(b) Write a formula for T for Galileo's circular path, given by $(x - 1)^2 + (y - 1)^2 = 1$.

Incidentally, Newton was the first to send his solution [which turned out to be a cycloid—the same curve (inverted) that we studied in Section 2.4, Example 4], but he did so anonymously. Bernoulli was not fooled, however. When he received the solution, he immediately knew its author, exclaiming, "I know the Lion from his paw." While the solution of this problem is a cycloid, it is known in the literature as the *brachistrochrone*. This was the beginning of the important field called the *calculus of variations*.

From energy considerations,

starting at height h_0

$mgh_0 + \frac{1}{2}mv_0^2 = mgh_f + \frac{1}{2}mv_f^2$, f = final position.

$\therefore$ Assuming $v_0 = 0$, $\sqrt{-2g\Delta h} = v_f$

Note Δh is negative since $h_f < h_0$.

The velocity function is $\therefore v(y) = \sqrt{2g(1-y)}$

Since time = $\frac{\text{distance}}{\text{velocity}}$, for a given segment Δs, the time is $\frac{\Delta s}{v}$

If $\vec{c}(t)$ is the path, then $\Delta s = \|\vec{c}'(t^*)\|\Delta t$, $t^* \in \Delta t$.

As the partition increases,

$$\text{total time} = \lim_{N\to\infty} \sum_{i=0}^{N-1} \frac{\Delta s}{v} = \lim_{N\to\infty} \sum_{i=0}^{N-1} \frac{\|\vec{c}\,'(x^*)\|\,\Delta t}{v}$$

$$= \int_0^1 \frac{\|\vec{c}\,'(t)\|\,dt}{\sqrt{2g(1-y(t))}}, \quad \vec{c}(t) \text{ defined on } [0,1]$$

(a) For $y = 1 - x$, $\vec{c}(t) = (t, 1-t)$, $t \in [0,1]$

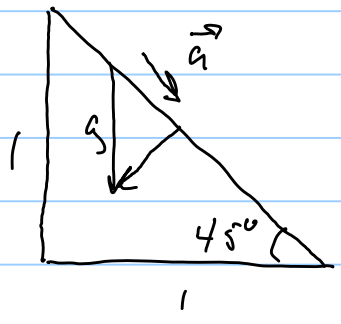
$\therefore \vec{c}\,'(t) = (1,-1)$, $\|\vec{c}\,'(t)\| = \sqrt{2}$

$$\sqrt{2g[1-y(t)]} = \sqrt{2g[1-(1-t)]} = \sqrt{2gt}$$

$$\therefore \int_0^1 \frac{\|\vec{c}\,'(t)\|}{\sqrt{2g[1-y(t)]}}\,dt = \int_0^1 \frac{\sqrt{2}}{\sqrt{2gt}}\,dt = \frac{1}{\sqrt{g}} \int_0^1 t^{-\frac{1}{2}}\,dt$$

$$= \frac{1}{\sqrt{g}}\, 2t^{\frac{1}{2}}\Big|_0^1 = \frac{2}{\sqrt{g}}$$

To check, from mechanics,



$d = \frac{1}{2} a t^2$, $d = \sqrt{2}$

and $\vec{a} = \vec{g}\cos 45^\circ = \frac{\sqrt{2}}{2} g$

$$\therefore \sqrt{2} = \frac{1}{2}\left(\frac{\sqrt{2}}{2}g\right)t^2, \quad \frac{4}{g} = t^2, \quad t = \frac{2}{\sqrt{g}}$$

(5) (b) Write a formula for T for Galileo's circular path, given by $(x-1)^2 + (y-1)^2 = 1$.

A parameterization for the circular path is

$$\vec{c}(\theta) = (1+\cos\theta, 1+\sin\theta), \quad \pi \le \theta \le \tfrac{3}{2}\pi$$

$$\therefore v = \sqrt{2g[1-y(\theta)]} = \sqrt{2g[1-(1+\sin\theta)]} = \sqrt{-2g\sin\theta}$$

$$\vec{c}\,'(\theta) = (-\sin\theta, \cos\theta), \quad \|\vec{c}\,'(\theta)\| = 1$$

$$\therefore T = \int_{\pi}^{3/2\,\pi} \frac{d\theta}{\sqrt{-2g\sin\theta}} = \frac{1}{\sqrt{2g}} \int_{0}^{\frac{\pi}{2}} \frac{d\theta}{\sqrt{\sin\theta}}$$

Alternatively, $(x-1)^2 + (y-1)^2 = 1 \Rightarrow (y-1)^2 = 1-(x-1)^2$

$$\therefore y-1 = \mp\sqrt{1-(x-1)^2}$$, choose the negative root since dealing with $0 \le y \le 1$

$$\therefore y = 1-\sqrt{1-(x-1)^2}$$

$$\therefore \vec{c}(x) = \left(x, 1-\sqrt{1-(x-1)^2}\right) = \left(x, 1-\sqrt{2x-x^2}\right)$$

$$\therefore v = \sqrt{2g\sqrt{2x-x^2}} = \sqrt{2g}\,(2x-x^2)^{\frac{1}{4}}$$

$$\vec{c}'(x) = \left[1, -\frac{1}{2}(2x-x^2)^{-\frac{1}{2}}(2-2x)\right]$$

$$= \left[1, \frac{x-1}{\sqrt{2x-x^2}}\right]$$

$$\therefore \|\vec{c}'(x)\| = \sqrt{1 + \frac{(x-1)^2}{(2x-x^2)}} = \sqrt{\frac{2x-x^2+x^2-2x+1}{2x-x^2}}$$

$$= \frac{1}{\sqrt{2x-x^2}} = \frac{1}{(2x-x^2)^{\frac{1}{2}}}$$

$$\therefore T = \int_0^1 \frac{\|\vec{c}'(x)\|\,dx}{\sqrt{2g[1-y(x)]}} = \int_0^1 \frac{1}{(2x-x^2)^{\frac{1}{2}}} \cdot \frac{1}{\sqrt{2g}\,(2x-x^2)^{1/4}}\,dx$$

$$= \frac{1}{\sqrt{2g}} \int_0^1 \frac{dx}{(2x-x^2)^{3/4}}$$

7.2 Line Integrals

Note Title 2/10/2017

1.

1. Evaluate the line integral

$$\int_C \mathbf{F} \cdot d\mathbf{s},$$

where $\mathbf{F}(x, y) = y^2\mathbf{i} - xy\mathbf{j}$ and C is the part of the circle $x^2 + y^2 = 1$ that starts at $(1, 0)$ and ends at $(0, 1)$, oriented counterclockwise.

Let $\vec{c}(\theta) : [0, \frac{\pi}{2}] \to [\cos\theta, \sin\theta]$

$\therefore \vec{c}'(\theta) = [-\sin\theta, \cos\theta]$

$\vec{F}(\vec{c}(\theta)) = [\sin^2\theta, -\cos\theta\sin\theta]$

$$\therefore \int_C \vec{F} \cdot d\vec{s} = \int_0^{\frac{\pi}{2}} (\sin^2\theta, -\cos\theta\sin\theta) \cdot (-\sin\theta, \cos\theta)\, d\theta$$

$$= \int_0^{\frac{\pi}{2}} -\sin^3\theta - \cos^2\theta\sin\theta \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} -\sin\theta(1-\cos^2\theta)\, d\theta + \frac{\cos^3\theta}{3}\Big|_0^{\frac{\pi}{2}}$$

$$= \cos\theta\Big|_0^{\frac{\pi}{2}} - \frac{\cos^3\theta}{3}\Big|_0^{\frac{\pi}{2}} + \frac{\cos^3\theta}{3}\Big|_0^{\frac{\pi}{2}}$$

$$= 0 - 1 = \underline{-1}$$

2.

2. Repeat Problem 1 for $\mathbf{F} = y^2\mathbf{i} + 2xy\mathbf{j}$, where C is the entire unit circle $x^2 + y^2 = 1$.

Let $\vec{c}(\theta) : [0, 2\pi] \to [\cos\theta, \sin\theta]$

$\therefore \vec{c}'(\theta) = (-\sin\theta, \cos\theta)$

$\vec{F}(\vec{c}(\theta)) = [\sin^2\theta, 2\cos\theta\sin\theta]$

$\vec{F}(\vec{c}(\theta)) \cdot \vec{c}'(\theta) = -\sin^3\theta + 2\cos^2\sin\theta$

$$\therefore \int_C \vec{F} \cdot d\vec{s} = \int_0^{2\pi} -\sin^3\theta + 2\cos^2\sin\theta \, d\theta$$

$$= \int_0^{2\pi} -\sin\theta(1-\cos^2\theta) - \frac{2}{3}\cos^3\theta \Big|_0^{2\pi}$$

$$= \cos\theta \Big|_0^{2\pi} - \frac{\cos^3\theta}{3}\Big|_0^{2\pi} - \frac{2\cos^3\theta}{3}\Big|_0^{2\pi}$$

$$= 1 - 1 - \cos^3\theta \Big|_0^{2\pi} = 0 - (1-1) = \underline{0}$$

3.

3. Let $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Evaluate the integral of $\mathbf{F}$ along each of the following paths:

(a) $\mathbf{c}(t) = (t, t, t), \quad 0 \le t \le 1$

(b) $\mathbf{c}(t) = (\cos t, \sin t, 0), \quad 0 \le t \le 2\pi$

(c) $\mathbf{c}(t) = (\sin t, 0, \cos t), \quad 0 \le t \le 2\pi$

(d) $\mathbf{c}(t) = (t^2, 3t, 2t^3), \quad -1 \le t \le 2$

(a) $\vec{c}'(t) = (1,1,1)$ $\vec{F}(\vec{c}(t)) = (t,t,t)$

$\therefore \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t) = 3t$

$\therefore \int_C \vec{F} \cdot ds = \int_0^1 3t\,dt = \frac{3}{2}t^2\Big|_0^1 = \underline{\frac{3}{2}}$

(b) $\vec{c}'(t) = (-\sin t, \cos t, 0)$ $\vec{F}(\vec{c}(t)) = (\cos t, \sin t, 0)$

$\therefore \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t) = -\sin t\cos t + \cos t \sin t + 0 = 0$

$\therefore \int_C \vec{F} \cdot ds = \underline{0}$

(c) $\vec{c}'(t) = (\cos t, 0, -\sin t)$ $\vec{F}(\vec{c}(t)) = (\sin t, 0, \cos t)$

$\therefore \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t) = \cos t \sin t + 0 - \sin t \cos t = 0$

$\therefore \int_C \vec{F} \cdot ds = 0$

(d) $\vec{c}'(t) = (2t, 3, 6t^2)$ $\vec{F}(\vec{c}(t)) = (t^2, 3t, 2t^3)$

$\therefore \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t) = 2t^3 + 9t + 12t^5$

$\therefore \int_C \vec{F} \cdot ds = \int_{-1}^2 2t^3 + 9t + 12t^5\,dt$

$$= \frac{1}{2}t^4 + \frac{9}{2}t^2 + 2t^6 \Big|_{-1}^{2}$$

$$= \left(8 - \frac{1}{2}\right) + \left(18 - \frac{9}{2}\right) + (128 - 2) = 154 - 7$$

$$= \underline{147}$$

4.

4. Evaluate each of the following line integrals:

(a) $\int_{\mathbf{c}} x\,dy - y\,dx$, $\mathbf{c}(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$

(b) $\int_{\mathbf{c}} x\,dx + y\,dy$, $\mathbf{c}(t) = (\cos \pi t, \sin \pi t)$, $0 \le t \le 2$

(c) $\int_{\mathbf{c}} yz\,dx + xz\,dy + xy\,dz$, where $\mathbf{c}$ consists of straight-line segments joining (1, 0, 0) to (0, 1, 0) to (0, 0, 1)

(d) $\int_{\mathbf{c}} x^2\,dx - xy\,dy + dz$, where $\mathbf{c}$ is the parabola $z = x^2$, $y = 0$ from $(-1, 0, 1)$ to $(1, 0, 1)$.

(a) $\vec{f}(x,y) = (-y, x)$ $\quad \vec{c}\,'(t) = (-\sin t, \cos t)$

$\vec{f}(\vec{c}(t)) = (-\sin t, \cos t)$. $\quad \therefore \vec{f} \cdot \vec{c}\,' = \sin^2 t + \cos^2 t = 1$

$$\therefore \int_C \vec{f} \cdot d\vec{s} = \int_0^{2\pi} 1\,dt = \underline{2\pi}$$

(b) $\vec{f}(x,y) = (x,y)$ $\quad \vec{c}\,'(t) = (-\pi \sin \pi t, \pi \cos \pi t)$

$\vec{f}(\vec{c}(t)) = (\cos \pi t, \sin \pi t)$

$\therefore \vec{f} \cdot \vec{c}\,' = -\pi \cos \pi \sin \pi t + \pi \sin \pi t \cos \pi t = 0$

$$\therefore \int_C \vec{f} \cdot d\vec{s} = \int_0^2 0\,dt = \underline{0}$$

(c) $\vec{f}(x,y,z) = (yz, xz, xy)$

Note $\vec{f} = \nabla f$, where $f(x,y,z) = xyz$

$$\therefore \int_C \vec{f} \cdot d\vec{s} = f(0,0,1) - f(1,0,0) = 0 - 0 = \underline{0}$$

(d) $\vec{f}(x,y,z) = (x^2, -xy, 1)$ $\quad \vec{c}(t) = (t, 0, t^2)$, $t \in [-1,1]$

$$\therefore \vec{f}(\vec{c}(t)) = (t^2, 0, 1) \quad \vec{c}\,'(t) = (1, 0, 2t)$$

$$\therefore \vec{f} \cdot \vec{c}\,' = t^2 + 0 + 2t = t^2 + 2t$$

$$\therefore \int_C \vec{f} \cdot d\vec{s} = \int_{-1}^{1} t^2 + 2t \, dt = \frac{t^3}{3} + t^2 \Big|_{-1}^{1}$$

$$= \frac{1}{3} + 1 - \left(-\frac{1}{3} + 1\right) = \underline{\frac{2}{3}}$$

5.

5. Consider the force field $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Compute the work done in moving a particle along the parabola $y = x^2$, $z = 0$, from $x = -1$ to $x = 2$.

$$\vec{c}(t) = (t, t^2, 0), \quad \vec{c}\,'(t) = (1, 2t, 0)$$

$$\vec{F}(\vec{c}(t)) = (t, t^2, 0) \quad \therefore \vec{F}(\vec{c}) \cdot \vec{c}\,' = t + 2t^3$$

$$\therefore \int_{\vec{C}} \vec{F} \cdot d\vec{s} = \int_{-1}^{2} t + 2t^3 = \frac{t^2}{2} + \frac{t^4}{2} \Big|_{-1}^{2} =$$

$2 + 8 - \left(\frac{1}{2} + \frac{1}{2}\right) = \underline{9}$

6. Let $\mathbf{c}$ be a smooth path.

(a) Suppose $\mathbf{F}$ is perpendicular to $\mathbf{c}'(t)$ at the point $\mathbf{c}(t)$. Show that

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = 0.$$

Let $\vec{c}(t) : [a,b] \to \mathbb{R}^n$

$\vec{F} : \mathbb{R}^n \to \mathbb{R}^n$

(a) $$\int_{\vec{c}} \vec{F} \cdot d\vec{s} = \int_a^b \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t)\, dt = \int_a^b 0\, dt = 0$$

(b) If $\mathbf{F}$ is parallel to $\mathbf{c}'(t)$ at $\mathbf{c}(t)$, show that

$$\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}} \|\mathbf{F}\|\, ds.$$

[By parallel to $\mathbf{c}'(t)$ we mean that $\mathbf{F}(\mathbf{c}(t)) = \lambda(t)\mathbf{c}'(t)$, where $\lambda(t) > 0$.]

Let $\vec{c}(t) : [a,b] \to \mathbb{R}^n$

$\vec{F} : \mathbb{R}^n \to \mathbb{R}^n$

$$\vec{F}(\vec{c}(t)) \cdot \vec{c}'(t) = \lambda(t)\, \vec{c}'(t) \cdot \vec{c}'(t) = \lambda(t) \|\vec{c}'(t)\|^2$$

$$= \lambda(t) \|\vec{c}'(t)\| \, \|\vec{c}'(t)\|$$

$$= \|\lambda(t)\, \vec{c}'(t)\| \, \|\vec{c}'(t)\|$$

$$= \|\vec{F}(\vec{c}(t))\| \, \|\vec{c}'(t)\|$$

$$\therefore \int_{\vec{c}} \vec{F} \cdot d\vec{s} = \int_a^b \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t)\, dt$$

$$= \int_a^b \|\vec{F}(\vec{c}(t))\| \, \|\vec{c}'(t)\|\, dt$$

$$= \int_{\vec{c}} \|\vec{F}(\vec{c}(t))\| \, ds \quad \text{by definition}$$

7. **7.** Suppose the path **c** has length l, and $\|\mathbf{F}\| \leq M$. Prove that

$$\left| \int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} \right| \leq Ml.$$

Since $\vec{F}(\vec{c}(t)) \cdot \vec{c}'(t) = \|\vec{F}(\vec{c}(t))\| \cdot \|\vec{c}'(t)\| \cos\theta$, for $\mathbb{R}^3$

$\therefore \quad |\vec{F}(\vec{c}(t)) \cdot \vec{c}'(t)| \leq \|\vec{F}(\vec{c}(t))\| \, \|\vec{c}'(t)\|$, by Cauchy-Schwartz $|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\| \, \|\vec{b}\|$

$$\leq M \|\vec{c}'(t)\|$$

$\therefore$ From $|a+b| \leq |a| + |b|$,

$$\left| \int_a^b \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t) \, dt \right| \leq \int_a^b |\vec{F}(\vec{c}(t)) \cdot \vec{c}'(t)| \, dt$$

$$\leq \int_a^b \|\vec{F}(\vec{c}(t))\| \, \|\vec{c}'(t)\| \, dt$$

$$\leq \int_c^b M \|\vec{c}'(t)\| \, dt = M \int_a^b \|\vec{c}'(t)\| \, dt = ML$$

$\therefore \quad \left| \int_{\vec{c}} \vec{F} \cdot d\vec{s} \right| = \left| \int_a^b \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t) \, dt \right| \leq ML$

8.

8. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{F}(x, y, z) = y\mathbf{i} + 2x\mathbf{j} + y\mathbf{k}$ and the path $\mathbf{c}$ is defined by $\mathbf{c}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$, $0 \le t \le 1$.

$\vec{c}'(t) = (1, 2t, 3t^2)$. $F(\vec{c}(t)) = (t^2, 2t, t^2)$

$\therefore \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t) = t^2 + 4t^2 + 3t^4 = 5t^2 + 3t^4$

$\therefore \int_0^1 5t^2 + 3t^4 \, dt = \frac{5}{3}t^3 + \frac{3}{5}t^5 \Big|_0^1 = \frac{5}{3} + \frac{3}{5} = \underline{\frac{34}{15}}$

9.

9. Evaluate

$$\int_C y\,dx + (3y^3 - x)\,dy + z\,dz$$

for each of the paths $\mathbf{c}(t) = (t, t^n, 0)$, $0 \le t \le 1$, where $n = 1, 2, 3, \ldots$.

$\vec{c}'(t) = (1, nt^{n-1}, 0)$

$\therefore (y, 3y^3 - x, z) \cdot (1, nt^{n-1}, 0) = (t^n, 3t^{3n} - t, 0) \cdot (1, nt^{n-1}, 0)$

$= t^n + 3nt^{3n+n-1} - nt^n$

$= (1-n)t^n + 3nt^{4n-1}$

$\therefore \int_0^1 (1-n)t^n + 3nt^{4n-1} \, dt = \frac{(1-n)t^{n+1}}{n+1} + \frac{3n}{4n}t^{4n} \Big|_{t=0}^{t=1}$

$= \frac{1-n}{n+1} + \frac{3}{4} - 0 = \underline{\frac{3}{4} - \frac{n-1}{n+1}}$

10.

10. This exercise refers to Example 12. Let L be a very long wire, a planar section of which (with the plane perpendicular to the wire) is shown in Figure 7.2.14.

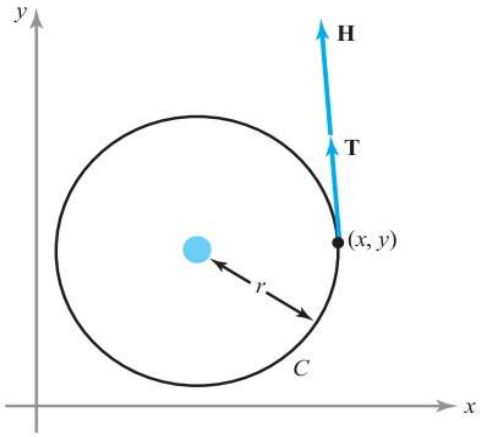


figure **7.2.14** A planar section of a long wire and a curve C about the wire.

Suppose this plane is the xy plane. Experiments show that $\mathbf{H}$ is tangent to every circle in the xy plane whose center is the axis of L, and that the magnitude of $\mathbf{H}$ is constant on every such circle C. Thus, $\mathbf{H} = H\mathbf{T}$, where $\mathbf{T}$ is a unit tangent vector to C and H is some scalar. Using this information, show that $H = I/2\pi r$, where r is the radius of circle C and I is the current flowing in the wire.

Note $\|\vec{H}\| = H\|\vec{T}\| = H$

$$\int_C \vec{H}\cdot d\vec{s} = I \qquad \text{Let the wire be at } (x,y)=(0,0).$$

$\therefore$ C can be described as $(r\cos\theta, r\sin\theta)$, $0 \le \theta \le 2\pi$

$$\therefore \quad \vec{c}\,'(t) = r(-\sin\theta, \cos\theta) = r\vec{T} = r\frac{\vec{H}}{H}$$

$$\therefore \int_C \vec{H}\cdot d\vec{s} = \int_0^{2\pi} \vec{H}\cdot\vec{c}\,'(t)\,dt = \int_0^{2\pi} \vec{H}\cdot\left(\frac{r}{H}\vec{H}\right)dt$$

$$= \int_0^{2\pi} r\frac{\|\vec{H}\|^2}{H}\,dt = \int_0^{2\pi} rH\,dt = rH(2\pi)$$

$$\therefore 2\pi r H = \int_C \vec{H}\cdot d\vec{s} = I \Rightarrow H = \frac{I}{2\pi r}$$

11.

11. The image of the path $t \mapsto (\cos^3 t, \sin^3 t)$, $0 \le t \le 2\pi$ in the plane is shown in Figure 7.2.15. Evaluate the integral of the vector field $\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$ around this curve.

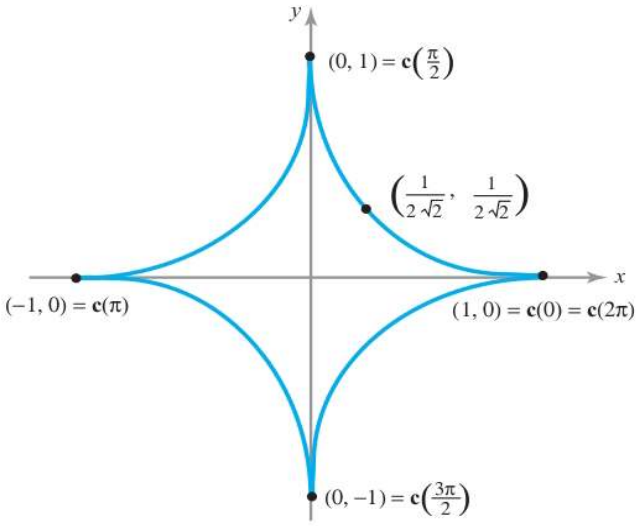


figure **7.2.15** The hypocycloid $\mathbf{c}(t) = (\cos^3 t, \sin^3 t)$ (Exercise 11).

$$\vec{F}(\vec{c}(t)) = (\cos^3 t, \sin^3 t)$$

$$\vec{c}\,'(t) = (-3\cos^2 t \sin t, 3\sin^2 t \cos t)$$

$$\vec{F}(\vec{c}) \cdot \vec{c}\,' = -3\cos^5 t \sin t + 3\sin^5 t \cos t$$

$$= 3\sin t \cos t (\sin^4 t - \cos^4 t)$$

$$= 3\sin t \cos t (\sin^2 t + \cos^2 t)(\sin^2 t - \cos^2 t)$$

$$= 3\sin t \cos t (\sin^2 t - \cos^2 t)$$

$$= 3\sin^3 t \cos t - 3\sin t \cos^3 t$$

$$\therefore \int_c \vec{F} \cdot d\vec{s} = \int_0^{2\pi} 3\sin^3 t \cos t - 3\sin t \cos^3 t \, dt$$

$$= \frac{3}{4}\sin^4 t + \frac{3}{4}\cos^4 t \Big|_0^{2\pi} = 0 + \frac{3}{4} - \left(0 + \frac{3}{4}\right) = \underline{0}$$

12. **12.** Suppose $\mathbf{c}_1$ and $\mathbf{c}_2$ are two paths with the same endpoints and $\mathbf{F}$ is a vector field. Show that $\int_{\mathbf{c}_1} \mathbf{F} \cdot d\mathbf{s} = \int_{\mathbf{c}_2} \mathbf{F} \cdot d\mathbf{s}$ is equivalent to $\int_C \mathbf{F} \cdot d\mathbf{s} = 0$, where C is the closed curve obtained by first moving along $\mathbf{c}_1$ and then moving along $\mathbf{c}_2$ in the opposite direction.

$$\int_{\vec{c}_2} \vec{F} \cdot d\vec{s} = -\int_{\vec{c}_2 \text{ opp}} \vec{F} \cdot d\vec{s} \quad \text{by Theorem 1, p. 364 of text.}$$

$$\therefore \int_{\vec{c}_1} \vec{F} \cdot d\vec{s} = -\int_{\vec{c}_2 \text{ opp}} \vec{F} \cdot d\vec{s} \iff$$

$$\int_{\vec{c}_1} \vec{F} \cdot d\vec{s} + \int_{\vec{c}_2 \text{ opp}} \vec{F} \cdot d\vec{s} = 0 \iff$$

$$\int_C \vec{F} \cdot d\vec{s} = 0$$

13. **13.** Let $\mathbf{c}(t)$ be a path and $\mathbf{T}$ the unit tangent vector. What is $\int_{\mathbf{c}} \mathbf{T} \cdot d\mathbf{s}$?

Let $\vec{c}(t): [a,b] \to \mathbb{R}^n$

$$\vec{T}(t) = \frac{\vec{c}\,'(t)}{\|\vec{c}\,'(t)\|} \quad \therefore \int_{\vec{c}} \vec{T} \cdot d\vec{s} =$$

$$\int_{\vec{c}} \frac{\vec{c}\,'(t)}{\|\vec{c}\,'(t)\|} \cdot d\vec{s} = \int_a^b \frac{\vec{c}\,'(t) \cdot \vec{c}\,'(t)}{\|\vec{c}\,'(t)\|}\, dt$$

$$= \int_a^b \frac{\|\vec{c}\,'(t)\|^2}{\|\vec{c}\,'(t)\|}\, dt = \int_a^b \|\vec{c}\,'(t)\|\, dt$$

$$= \text{length of path of } \vec{c}(t)$$

14. **14.** Let $\mathbf{F} = (z^3 + 2xy)\mathbf{i} + x^2\mathbf{j} + 3xz^2\mathbf{k}$. Show that the integral of $\mathbf{F}$ around the circumference of the unit square with vertices $(\pm 1, \pm 1)$ is zero.

Let $f(x,y,z) = xz^3 + x^2y$

$\therefore \vec{F}(x,y,z) = \nabla f(x,y,z) = (z^3 + 2xy,\ x^2,\ 3xz^2)$

$\therefore \int_c \vec{F} \cdot d\vec{s} = \int_c \nabla f \cdot d\vec{s} = f(\vec{c}(b)) - f(\vec{c}(a))$

where $\vec{c}(a)$ = start point, $\vec{c}(b)$ = end point

(i.e., $\vec{c} : [a,b] \to \mathbb{R}^3$)

But $\vec{c}(a) = \vec{c}(b)$. $\therefore f(\vec{c}(b)) - f(\vec{c}(a)) = 0$

$\therefore \int_c \vec{F} \cdot d\vec{s} = 0$.

15. **15.** Using the path in Exercise 11, observe that a C^1 map $\mathbf{c}\colon [a, b] \to \mathbb{R}^3$ can have an image that does not "look smooth." Do you think this could happen if $\mathbf{c}'(t)$ were always nonzero?

If $\vec{c}'(t) \neq 0$, then $\vec{c}'(t)$ has a defined direction.

The zero vector has no defined direction.

∴ The "points" of the path in Exercise 11, which have no direction, will be avoided.

The pointed parts of the path are where $\vec{c}'(t)$ must be zero, as the zero vector is the only vector without direction.

16. **16.** What is the value of the integral of a gradient field around a closed curve C?

It is 0, since if $\vec{F} = \nabla f$, $\int_C \nabla f \cdot d\vec{s} =$ $F(\text{end point}) - F(\text{start point}) = 0$, since end point = start point in a closed curve.

17. **17.** Evaluate the line integral

$$\int_C 2xyz\,dx + x^2 z\,dy + x^2 y\,dz,$$

where C is an oriented simple curve connecting (1, 1, 1) to (1, 2, 4).

Assume start point = (1,1,1), endpoint = (1,2,4)

Note that if $f(x,y,z) = x^2yz$, then

$\nabla f = (2xyz, x^2z, x^2y)$, which is $\vec{F}$ in the line integral $\int_C \vec{F}\cdot d\vec{s}$, so the exact path of C doesn't matter, just the endpoints.

$$\therefore \int_C \vec{F}\cdot d\vec{s} = \int_C \nabla f \cdot d\vec{s} = f(1,2,4) - f(1,1,1)$$

$$= 8 - 1 = \underline{7}$$

18. **18.** Suppose $\nabla f(x, y, z) = 2xyze^{x^2}\mathbf{i} + ze^{x^2}\mathbf{j} + ye^{x^2}\mathbf{k}$. If $f(0, 0, 0) = 5$, find $f(1, 1, 2)$.

$$\frac{\partial f}{\partial x} = 2xyze^{x^2} \Rightarrow f(x,y,z) = yze^{x^2} + g(y,z)$$

$$\frac{\partial f}{\partial y} = ze^{x^2} + \frac{\partial g}{\partial y} = ze^{x^2} \Rightarrow f(x,y,z) = yze^{x^2} + h(z)$$

$$\frac{\partial f}{\partial z} = ye^{x^2} + \frac{\partial h}{\partial z} = ye^{x^2} \Rightarrow f(x,y,z) = yze^{x^2} + K,$$ K a constant

$$f(0,0,0) = 5 \Rightarrow f(x,y,z) = yze^{x^2} + 5$$

$$\therefore f(1,1,2) = \underline{2e + 5}$$

19.

19. Consider the gravitational force field (with $G = m = M = 1$) defined [for $(x, y, z) \neq (0, 0, 0)$] by

$$\mathbf{F}(x, y, z) = -\frac{1}{(x^2 + y^2 + z^2)^{3/2}}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k}).$$

Show that the work done by the gravitational force as a particle moves from (x_1, y_1, z_1) to (x_2, y_2, z_2) along any path depends only on the radii $R_1 = \sqrt{x_1^2 + y_1^2 + z_1^2}$ and $R_2 = \sqrt{x_2^2 + y_2^2 + z_2^2}$.

Need to show $\vec{F} = \nabla f$ for some $f(x,y,z)$.

Then $\int_C \vec{F} \cdot d\vec{s}$ = work done along a path C =

$f(\text{endpoint}) - f(\text{startpoint})$

Look at $f(x,y,z) = (x^2+y^2+z^2)^{-\frac{1}{2}}$

Then $\nabla f = -(x^2+y^2+z^2)^{-\frac{3}{2}}(x, y, z) = \vec{F}$

$$\therefore \int_C \vec{F} \cdot d\vec{s} = \int_C \nabla f \cdot d\vec{s} = f(\text{endpoint}) - f(\text{startpoint})$$

Since start point $= (x_1, y_1, z_1)$, then

$$f(x_1, y_1, z_1) = \frac{1}{\sqrt{x_1^2+y_1^2+z_1^2}} = \frac{1}{R_1}$$

and end point $= (x_2, y_2, z_2)$, so

$$f(x_2, y_2, z_2) = \frac{1}{\sqrt{x_2^2+y_2^2+z_2^2}} = \frac{1}{R_2}$$

$$\therefore \int_C \vec{F} \cdot d\vec{s} = \frac{1}{R_2} - \frac{1}{R_1}$$

20.

20. A cyclist rides up a mountain along the path shown in Figure 7.2.16. She makes one complete revolution around the mountain in reaching the top, while her vertical rate of climb is constant. Throughout the trip she exerts a force described by the vector field

$$\mathbf{F}(x, y, z) = y\mathbf{i} + x\mathbf{j} + \mathbf{k}.$$

What is the work done by the cyclist in traveling from A to B? What is unrealistic about this model of a cyclist?

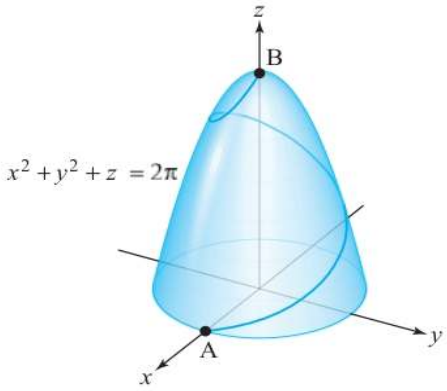


figure **7.2.16** How much work is done in cycling up this mountain?

(a) $\vec{F} = \nabla f(x,y,z)$, where $f(x,y,z) = xy + z$

Work done $= \int_C \vec{F} \cdot d\vec{s} = \int_C \nabla f \cdot d\vec{s}$

$= f(\text{end point}) - f(\text{start point})$

Start point $= (\sqrt{2\pi}, 0, 0)$

end point $= (0, 0, 2\pi)$.

$\therefore W = f(0,0,2\pi) - f(\sqrt{2\pi}, 0, 0)$

$= 2\pi - 0 = \underline{2\pi}$ units

(b) Near the top, $x \approx 0$, $y \approx 0$, so that $\vec{F}$ is virtually straight upward ("all $\hat{k}$"), an orientation highly unlikely for a cyclist.

21.

21. Let $\mathbf{c}\colon [a, b] \to \mathbb{R}^3$ be a path such that $\mathbf{c}'(t) \neq \mathbf{0}$. Recall from Section 4.1 that when this condition holds, $\mathbf{c}$ is said to be ***regular***. Let the function f be defined by the formula $f(x) = \int_a^x \|\mathbf{c}'(t)\|\, dt$.

(a) What is df/dx?

(a) By The Fundamental Theorem of Calculus,

$$\frac{d}{dx} f(x) = \frac{d}{dx} \int_a^x \|\vec{c}'(t)\|\, dt = \underline{\|\vec{c}'(x)\|}$$

(b)

(b) Using the answer to part (a), prove that $f\colon [a, b] \to [0, L]$, where L is the length of $\mathbf{c}$, has a differentiable inverse $g\colon [0, L] \to [a, b]$ satisfying $f \circ g(s) = s$, $g \circ f(x) = x$. (You may use the one-variable inverse function theorem stated at the beginning of Section 3.5.)

Since $f'(x) = \|\vec{c}'(x)\| > 0$, Then f is increasing on $[a, b]$, and so is one-to-one on $[a, b]$.

Since $f(b) = \int_a^b \|\vec{c}'(t)\|\, dt = L$, and $f(a) = 0$,

Then f is also onto

$\therefore$ An inverse function $g: [0, L] \to [a, b]$ exists

That is continuous, one-to-one and onto,

and $g'(s) = \dfrac{1}{f'(x)}$, $(f \circ g)(s) = f(g(s)) = s$,

$(g \circ f)(x) = g(f(x)) = x$.

(c)

(c) Compute dg/ds.

$$\frac{dg}{ds} = \frac{1}{f'(x)} = \underline{\frac{1}{\|c'(x)\|}}$$

(d) (d) Recall that a path $s \mapsto \mathbf{b}(s)$ is said to be of unit speed, or parametrized by arc length, if $\|\mathbf{b}'(s)\| = 1$. Show that the reparametrization of $\mathbf{c}$ given by $\mathbf{b}(s) = \mathbf{c} \circ g(s)$ is of unit speed. Conclude that any regular path can be reparametrized by arc length. (Thus, for example, the Frenet formulas in Exercise 23 of Section 4.2 can be applied to the reparametrization $\mathbf{b}$.)

$$\vec{b}'(s) = \vec{c}'(g(s)) \cdot g'(s)$$

$$= \vec{c}'(x) \cdot \frac{1}{\|\vec{c}'(x)\|} \qquad \text{since } g(s) = x$$

$$\therefore \|\vec{b}'(s)\| = \frac{\|\vec{c}'(x)\|}{\|\vec{c}'(x)\|} = 1.$$

(b) above showed a reparametrization function exists for $\vec{c}(t)$, if $\vec{c}'(t) \neq 0$, so that $\vec{b}(s)$ can always be created.

22. **22.** Along a "thermodynamic path" C in (V, T, P) space,

(i) The heat gained is $\int_C \Lambda_V \, dV + K_V \, dT$, where Λ_V, K_V are functions of (V, T, P), depending on the particular physical system.

(ii) The work done is $\int_C P \, dV$.

For a van der Waals gas, we have

$$P(V, T) = \frac{RT}{V - b} - \frac{a}{V^2}, \qquad J\Lambda_V = \frac{RT}{V - b},$$

and $\qquad K_V = \text{constant},$

where R, b, a, and J are known constants. Initially, the gas is at a temperature T_0 and volume V_0.

(a) An ***adiabatic*** process is a thermodynamic motion $(V(t), T(t), P(t))$ for which

$$\frac{dT}{dV} = \frac{dT/dt}{dV/dt} = -\frac{\Lambda_V}{K_V}.$$

If the van der Waals gas undergoes an adiabatic process in which the volume doubles to $2V_0$, compute

(1) the heat gained;
(2) the work done; and
(3) the final volume, temperature, and pressure.

(a) $\frac{dT}{dV} = -\frac{\Lambda_V}{K_V}$, $\therefore \Lambda_V \, dV + K_V \, dT = 0$

(1) Heat gained $= \int_C \Lambda_V \, dV + K_V \, dT = \int_C 0 = \underline{0}$

(2) Work done $= \int_C P\,dV = \int_C P \frac{dV}{dt}\,dt$

$P(V,T) = \frac{RT}{V-b} - \frac{a}{V^2}$, a function of V and T.

From $\frac{dT}{dt} = -\frac{\Lambda_V}{K_V}\frac{dV}{dt}$, and $\Lambda_V = \frac{RT}{J(V-b)}$,

$$\frac{dT}{dt} = -\frac{RT}{JK_V}\cdot\frac{1}{V-b}\frac{dV}{dt}, \text{ or}$$

$$\frac{1}{T}\frac{dT}{dt} = -\frac{R}{JK_V}\cdot\frac{1}{V-b}\frac{dV}{dt}$$

$$\therefore \int_{t_0}^{t} \frac{1}{T}\frac{dT}{dt} = -\frac{R}{JK_V}\int_{t_0}^{t} \frac{1}{V-b}\frac{dV}{dt}$$

$$\ln T\Big|_{t_0}^{t} = -\frac{R}{JK_V}\ln(V-b)\Big|_{t_0}^{t}$$

$$\therefore \ln\frac{T(t)}{T_0} = -\frac{R}{JK_V}\ln\left[\frac{V(t)-b}{V_0-b}\right]$$

where $T(t_0) = T_0$ = initial temperature

$V(t_0) = V_0$ = initial volume

Taking exponents, and using $T(t) = T$, $V(t) = V$

$$\frac{T}{T_0} = e^{-R/JK_v} \frac{V-b}{V_0-b}$$

$$\therefore T = \frac{T_0 e^{-R/JK_v}}{V_0-b}(V-b) \qquad [1]$$

Substituting this into $P(V,T)$:

$$P(V,T) = \frac{R}{V-b} \cdot \frac{T_0 e^{-R/JK_v}}{V_0-b}(V-b) - \frac{a}{V^2}$$

$$\frac{RT_0 e^{-R/JK_v}}{V_0-b} - \frac{a}{V^2} \qquad [2]$$

Thus P is just a function of V, given T_0 and V_0.

$$\therefore \int_C P\,dV = \int_{t_0}^{t_f} \left[\frac{RT_0 e^{-R/JK_v}}{V_0-b} - \frac{a}{V^2}\right]\frac{dV}{dt}\,dt$$

$$= \frac{RT_0 e^{-R/JK_v}}{V_0-b} V(t) + \frac{a}{V(t)}\Bigg|_{t_0}^{t_f}$$

and $V(t_0) = V_0$, $V(t_f) = 2V_0$

$$\therefore \int_C P\,dV = \frac{RT_0 e^{-R/JK_V}}{V_0 - b}\left(2V_0 - V_0\right) + \left(\frac{a}{2V_0} - \frac{a}{V_0}\right)$$

$$\therefore \text{Work done} = \frac{RT_0 V_0 e^{-R/JK_V}}{V_0 - b} - \frac{a}{2V_0}$$

(3) $V_{final} = 2V_0$

$$T_{final} = \frac{T_0 e^{-R/JK_V}}{V_0 - b}\left(2V_0 - b\right), \text{ from [1]}$$

$$P_{final} = \frac{RT_0 e^{-R/JK_V}}{V_0 - b} - \frac{a}{4V_0^2}, \text{ from [2]}$$

(6) (b) After the process indicated in part (a), the gas is cooled (or heated) at constant volume until the original temperature T_0 is achieved. Compute
(1) the heat gained;
(2) the work done; and
(3) the final volume, temperature, and pressure.

(1) Heat gained $= \int_C \Lambda_V\,dV + K_V\,dT$

Here $\frac{dV}{dt} = 0$ since V is constant.

$\therefore$ Heat gained $= \int_C K_V\,dT = K_V\left(T_f - T_i\right)$

Here, $T_f = T_0$, and T_i is, from (4)

$$T_i = \frac{T_0 e^{-R/5K_v}}{V_0 - b}(2V_0 - b)$$

$$\therefore \text{Heat gained} = K_v \left[T_0 - \frac{T_0 (2V_0 - b)}{V_0 - b} e^{-R/5K_v} \right]$$

$$= K_v T_0 \left[1 - \left(\frac{2V_0 - b}{V_0 - b} \right) e^{-R/5K_v} \right]$$

(2) Since $dV = 0$,

$$\text{Work done} = \int_c P\,dV = \underline{0}$$

(3) Final volume = $2V_0$

Final temperature = T_0

$$\text{Final pressure} = \frac{R(T_0)}{(2V_0) - b} - \frac{a}{(2V_0)^2}$$

$$= \frac{RT_0}{2V_0 - b} - \frac{a}{4V_0^2}$$

(C) (c) After the process indicated in part (b), the gas is compressed until the gas returns to its original volume V_0. The temperature is held constant throughout the process. Compute
(1) the heat gained;
(2) the work done; and
(3) the final volume, temperature, and pressure.

(1) Heat gained $= \int_C \Lambda_v \, dv + K_v \, dT = \int_C \Lambda_v \, dv$

since here $\frac{dT}{dt} = 0$ since temperature constant at T_0

$$\therefore \int_C \Lambda_v \, dv = \int_{2V_0}^{V_0} \frac{R}{J} \frac{T_0}{V-b} \, dv$$

$$\frac{RT_0}{J} \ln(V-b) \Big|_{V=2V_0}^{V=V_0} = \frac{RT_0}{J} \ln\left(\frac{V_0 - b}{2V_0 - b}\right)$$

(2) Work done $= \int_C p \, dv$

$$= \int_{2V_0}^{V_0} \frac{RT_0}{V-b} - \frac{a}{V^2} \, dv$$

$$= RT_0 \ln(V-b) + \frac{a}{V} \Big|_{V=2V_0}^{V=V_0}$$

$$= RT_0 \ln\left(\frac{V_0 - b}{2V_0 - b}\right) + \frac{a}{2V_0}$$

(3) Final volume = V_0

Final temperature = T_0

Final pressure = $\frac{RT_0}{V_0 - b} - \frac{a}{V_0^2}$

∴ All 3 parameters back to initial state before cycle began

(d)

(d) For the cyclic process described in parts (a), (b), (c), compute
(1) the total heat gained; and
(2) the total work done.

(1) Total heat gained = 0 from (a)

$+ K_V T_0 \left[1 - \left(\frac{2V_0 - b}{V_0 - b}\right) e^{-R/JK_V}\right]$ from (b)

$+ \frac{RT_0}{J} \ln\left(\frac{V_0 - b}{2V_0 - b}\right)$ from (c)

(2) Total work done = $\frac{RT_0 V_0 e^{-R/JK_V}}{V_0 - b} - \frac{a}{2V_0}$ (a)

$+ 0$ (b)

$+ RT_0 \ln\left(\frac{V_0 - b}{2V_0 - b}\right) + \frac{a}{2V_0}$ (c)

$$= RT_0 \left[\frac{V_0 e^{-R/JK_V}}{V_0 - b} + \ln\left(\frac{V_0 - b}{2V_0 - b}\right) \right]$$

7.3 Parametrized Surfaces

Note Title 2/20/2017

In Exercises 1 to 3, find an equation for the plane tangent to the given surface at the specified point.

1. **1.** $x = 2u, \quad y = u^2 + v, \quad z = v^2, \quad at\ (0, 1, 1)$

Let $\phi(u,v) = (2u, u^2+v, v^2) = (0,1,1) \Rightarrow u=0, v=1$

$x_u = 2 \quad y_u = 2u \quad z_u = 0 \quad \therefore \phi_u(0,1) = (2,0,0)$

$x_v = 0 \quad y_v = 1 \quad z_v = 2v \quad \therefore \phi_v(0,1) = (0,1,2)$

$$\therefore \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 0 \\ 0 & 1 & 2 \end{vmatrix} = (0, -4, 2)$$

$\therefore (0,-4,2) \cdot (x-0, y-1, z-1) = 0$, or

$-4(y-1) + 2(z-1) = 0$, or $z-1 = 2(y-1)$

or $\underline{z = 2(y-1)+1}$ or $2y - z - 1 = 0$

2. **2.** $x = u^2 - v^2, \quad y = u + v, \quad z = u^2 + 4v,$
$at\ (-\frac{1}{4}, \frac{1}{2}, 2)$

Let $\phi(u,v) = (u^2-v^2, u+v, u^2+4v) = (-\frac{1}{4}, \frac{1}{2}, 2)$

$\Rightarrow u+v = \frac{1}{2}. \quad \therefore u^2 - v^2 = (u+v)(u-v) = \frac{1}{2}(u-v) = -\frac{1}{4}$

$\Rightarrow u - v = -\frac{1}{2}$

$\therefore 2u = 0 \Rightarrow u = 0, \quad v = \frac{1}{2}$

check: $u^2 + 4v = 0 + 4(\frac{1}{2}) = 2 = z.$

$\phi_u = (2u, 1, 2u)$ $\quad \phi_u(0, \frac{1}{2}) = (0, 1, 0)$

$\phi_v = (-2v, 1, 4)$ $\quad \phi_v(0, \frac{1}{2}) = (-1, 1, 4)$

$$\therefore \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 0 \\ -1 & 1 & 4 \end{vmatrix} = (4, 0, 1)$$

$\therefore (4, 0, 1) \cdot (x + \frac{1}{4}, y - \frac{1}{2}, z - 2) = 0$

or $4x + 1 + z - 2 = 0$, or $4x + z - 1 = 0$

or $\underline{z = -4x + 1}$

3.

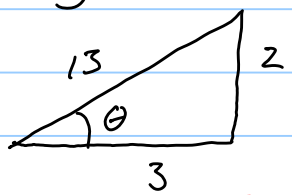
3. $x = u^2, \quad y = u \sin e^v, \quad z = \frac{1}{3} u \cos e^v,$ *at* $(13, -2, 1)$

Let $\phi(u, v) = (u^2, u \sin e^v, \frac{1}{3} u \cos e^v) = (13, -2, 1)$

$\Rightarrow u^2 = 13, \; u = \pm\sqrt{13}, \quad \mp\sqrt{13} \sin(e^v) = -2$

$\pm\sqrt{13} \cos(e^v) = 3$

$\therefore e^v = \pi - \text{Arcsin} \frac{2}{\sqrt{13}}$



$\theta = \text{Arcsin} \frac{2}{\sqrt{13}}$

$v = \ln(\pi - \text{Arcsin} \frac{2}{\sqrt{13}}) \quad u = -\sqrt{13}$

$\phi_u = (2u, \sin(e^v), \frac{1}{3}\cos(e^v))$

$\phi_v = (0, u e^v \cos(e^v), -\frac{1}{3} u e^v \sin(e^v))$

Calculating ϕ_u, ϕ_v at $(u,v) = (-\sqrt{13}, \ln(\pi - \text{Arcsin}\frac{2}{\sqrt{13}}))$ could get messy.

$\therefore$ Try $\phi_u \times \phi_v$ before evaluating at point.

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2u & \sin(e^v) & \frac{1}{3}\cos(e^v) \\ 0 & ue^v\cos(e^v) & -\frac{1}{3}ue^v\sin(e^v) \end{vmatrix}$$

$$= \left[-\frac{1}{3}ue^v\sin^2(e^v) - \frac{1}{3}ue^v\cos^2(e^v), \right.$$

$$\left. \frac{2}{3}u^2e^v\sin(e^v), 2u^2e^v\cos(e^v) \right]$$

$$= \left[-\frac{1}{3}ue^v, \frac{2}{3}u^2e^v\sin(e^v), 2u^2e^v\cos(e^v) \right]$$

$$= \frac{ue^v}{3}\left[-1, 2u\sin(e^v), 6u\cos(e^v) \right]$$

$$= \frac{ue^v}{3}\left[-1, -4, 18 \right]$$

using $u\sin(e^v) = -2$, $\frac{1}{3}u\cos(e^v) = 1$

Ignore $\frac{ue^v}{3}$ since that is a constant times the normal vector of $(-1, -4, 18)$.

$\therefore$ Plane $= (-1)(x-13) - 4(y+2) + 18(z-1) = 0$

or $-x - 4y + 18z + (13 - 8 - 18) = 0$

$$x + 4y - 18z + 13 = 0$$

4. **4.** At what points are the surfaces in Exercises 1 and 2 regular?

When $\phi_u \times \phi_v \neq 0$.

For #1: $\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2u & 0 \\ 0 & 1 & 2v \end{vmatrix} = (4uv, -4v, 2)$

$\therefore \phi_u \times \phi_v \neq 0$ for any (u, v).

$\therefore$ Regular for all points

For #2: $\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2u & 1 & 2u \\ -2v & 1 & 4 \end{vmatrix} = (4 - 2u, -8u - 4uv, 2u + 2v)$

$4 - 2u = 0$ if $u = 2$. $\therefore$ $-8(2) - 4(2)v = 0$ if $v = -2$

and $2(2) + 2(-2) = 0$.

$\therefore$ Regular for all points except $(u, v) = (2, -2)$,

or $(x, y, z) = (u^2 - v^2, u + v, u^2 + 4v)$

$= (4 - 4, 2 - 2, 4 - 8) = (0, 0, -4)$

$\therefore$ Regular everywhere except $(0, 0, -4)$

*In Exercises 5 and 6, find all points (u_0, v_0), where $S = \Phi(u_0, v_0)$ is **not** smooth (regular).*

5.

5. $\Phi(u, v) = (u^2 - v^2, u^2 + v^2, v)$

$\phi_u = (2u, 2u, 0) \quad \phi_v = (-2v, 2v, 1)$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2u & 2u & 0 \\ -2v & 2v & 1 \end{vmatrix} = (2u, -2u, 8uv)$$

$\phi_u \times \phi_v = \vec{0}$ only when $u = 0$

6.

6. $\Phi(u, v) = (u - v, u + v, 2uv)$

$\phi_u = (1, 1, 2v) \quad \phi_v = (-1, 1, 2u)$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \vec{k} \\ 1 & 1 & 2v \\ -1 & 1 & 2u \end{vmatrix} = (2u - 2v, -2u - 2v, 2)$$

$\therefore \phi_u \times \phi_v$ is never $\vec{0}$ due to $\hat{k}$ component

$\therefore$ S is smooth for all points (u, v)

7.

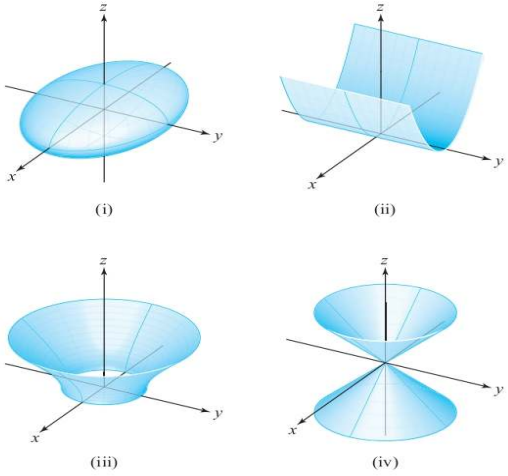
7. Match the following parameterizations to the surfaces shown in the figures.

(a) $\Phi(u, v) = ((2\sqrt{1 + u^2}) \cos v, (2\sqrt{1 + u^2}) \sin v, u)$

(b) $\Phi(u, v) = (3 \cos u \sin v, 2 \sin u \sin v, \cos v)$

(c) $\Phi(u, v) = (u, v, u^2)$

(d) $\Phi(u, v) = (u \cos v, u \sin v, u)$



(a) Note $x^2+y^2 = 4(1+u^2)(\sin^2 v + \cos^2 v) = 4 + 4u^2$

$\therefore$ when $u=0$, $z=0$, and just a circle of radius 2 in xy-plane.

As u increases/decreases, circles are bigger by a factor of $4u^2$.

$\therefore$ (iii) fits this description

(b) Note $\phi(0,0) = (0,0,1)$. Only (i) satisfies this condition.

Note also $4x^2 + 9y^2 = 36\cos^2 u \sin^2 v + 36 \sin^2 u \sin^2 v$
$= 36 \sin^2 v$

And $\therefore 4x^2 + 9y^2 + 36z^2 = 36\sin^2 v + 36\cos^2 v = 36$

$\therefore$ an ellipsoid, $\therefore$ (i)

(c) Note for $y=0$ (from $v=0$), $(u, 0, u^2)$ is a parabola, and also for any fixed value (or section from $y=$ constant) of v.

$\therefore$ paraboloid of one sheet: (ii)

(d) Note $x^2+y^2 = u^2$, $\therefore$ a circle of radius $|u|$

As z changes linearly, a cone. $\therefore$ (iv)

8.

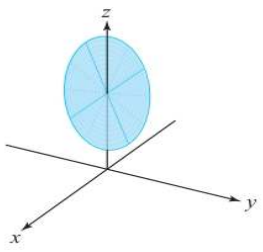
8. Match the following parametrizations to the surfaces shown in the figures.

(a) $\Phi(u, v) = (u\cos v,\ u\sin v,\ 4 - u\cos v - u\sin v)$; $u \in [0, 1],\ v \in [0, 2\pi]$

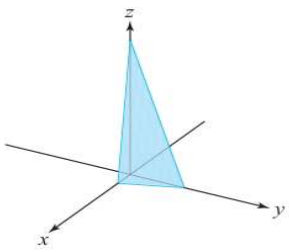
(b) $\Phi(u, v) = (u\cos v,\ u\sin v,\ 4 - u^2)$

(c) $\Phi(u, v) = (u,\ v,\ \frac{1}{3}(12 - 8u - 3v))$

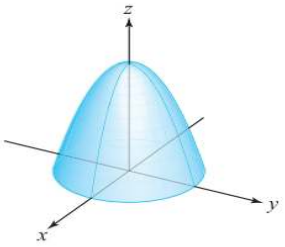
(d) $\Phi(u, v) = ((u^2 + 6u + 11)\cos v,\ u,\ (u^2 + 6u + 11)\sin v)$



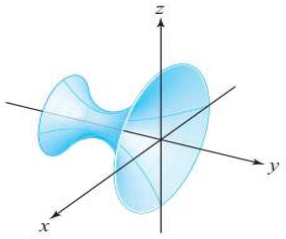
(i)



(ii)



(iii)



(iv)

(a) $x^2 + y^2 = u^2$, a circle of radius u, $0 \le u \le 1$

$z = 4 - x - y$, a plane, intersecting z-axis at $(0,0,4)$ with intercepts at $(4,0,0)$ and $(0,4,0)$

∴ Imagine the union of multiple cylinders (radii ≤ 1) perpendicular to xy-plane intersecting the angled plane of $z = 4 - x - y$.

∴ (i)

(b) $x = u\cos v$, $y = u\sin v$ is a circle of radius $|u|$
For a fixed u, the $z = 4 - u^2$ is a constant.
∴ level curves perpendicular to z-axis are circles.

∴ (iii).

(c) $z = \frac{1}{3}(12 - 8x - 3y)$ is a plane with intercepts $(0,0,4)$, $(3/2, 0, 0)$, $(0,4,0)$
x and y are independent of each other.

∴ (ii)

(d) Note y is independent. For a fixed y (fixed u)

$x^2 + z^2 = (K+11)^2$, where $K = u^2 + u$

∴ level curves perpendicular to y-axis are circles, whose radii act like parabolas as K (or u) changes.

∴ (iv)

9. **9.** Find an expression for a unit vector normal to the surface

$$x = \cos v \sin u, \qquad y = \sin v \sin u, \qquad z = \cos u$$

at the image of a point (u, v) for u in $[0, \pi]$ and v in $[0, 2\pi]$. Identify this surface.

(a) $\phi(u,v) = (\cos v \sin u, \sin v \sin u, \cos u)$

$\therefore \phi_u = (\cos v \cos u, \sin v \cos u, -\sin u)$

$\phi_v = (-\sin v \sin u, \cos v \sin u, 0)$

$\therefore$ Normal vector $= \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos v \cos u & \sin v \cos u & -\sin u \\ -\sin v \sin u & \cos v \sin u & 0 \end{vmatrix}$

$= (\sin^2 u \cos v, \sin^2 u \sin v, \cos^2 v \sin u \cos u + \sin^2 v \sin u \cos u)$

$= (\sin^2 u \cos v, \sin^2 u \sin v, \sin u \cos u)$

$\|\phi_u \times \phi_v\|^2 = \sin^4 u (\cos^2 v + \sin^2 v) + \sin^2 u \cos^2 u$

$= \sin^4 u + \sin^2 u \cos^2 u = \sin^2 u (\sin^2 u + \cos^2 u)$

$= \sin^2 u$

$\therefore \| \phi_u \times \phi_v \| = \sin u$, and for $u \in (0, \pi)$, $\sin u > 0$.

$\therefore$ Vector in direction of $\phi_u \times \phi_v$ has length factor of $\sin u$. If factor out the $(\sin u)$, then vector has length = 1 for all $u \in [0, \pi]$, and so is never 0, making S regular for all points.

This parametrization does not assure regularity at poles: $u = 0, \pi$. A different parametrization could make the surface regular at poles.

A unit normal vector: $(\sin u \cos v, \sin u \sin v, \cos u)$

$= (x, y, z)$, where $x^2 + y^2 + z^2 = 1$.

(5) For a fixed $u \in [0, \pi]$, $0 \le \sin u \le 1$,

x, y are on a circle of radius $\sin u$.

These are level curves for $z = \cos u$, and for $u \in [0, \pi]$, $-1 \le \cos u \le 1$.

For $\cos u = 1$ or -1, level curves are a point ($\sin u = 0$).

∴ This parametrization is the spherical coordinate system, $\rho = 1$, $u \equiv \phi$, $v \equiv \theta$

∴ Surface is a unit sphere, center at $(0,0,0)$.

10. **10.** Repeat Exercise 9 for the surface

$$x = 3\cos\theta\sin\phi, \qquad y = 2\sin\theta\sin\phi, \qquad z = \cos\phi$$

for θ in $[0, 2\pi]$ and ϕ in $[0, \pi]$.

(a) Let $c(\theta,\phi) = (3\cos\theta\sin\phi,\ 2\sin\theta\sin\phi,\ \cos\phi)$

$$\therefore c_\theta = (-3\sin\theta\sin\phi,\ 2\cos\theta\sin\phi,\ \cos\phi)$$

$$c_\phi = (3\cos\theta\cos\phi,\ 2\sin\theta\cos\phi,\ -\sin\phi)$$

$$\therefore c_\theta \times c_\phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3\sin\theta\sin\phi & 2\cos\theta\sin\phi & 0 \\ 3\cos\theta\cos\phi & 2\sin\theta\cos\phi & -\sin\phi \end{vmatrix}$$

$$= (-2\cos\theta\sin^2\phi,\ -3\sin\theta\sin^2\phi,\ -6\sin^2\theta\sin\phi\cos\phi - 6\cos^2\theta\sin\phi\cos\phi)$$

$$= (-2\cos\theta\sin^2\phi,\ -3\sin\theta\sin^2\phi,\ -6\sin\phi\cos\phi)$$

$$\|c_\theta \times c_\phi\|^2 = 4\cos^2\theta\sin^4\phi + 9\sin^2\theta\sin^4\phi + 36\sin^2\phi\cos^2\phi$$

$$= 4\sin^4\phi + 5\sin^2\theta\sin^4\phi + 36\sin^2\phi\cos^2\phi$$

$$= \sin^2\phi \left[4\sin^2\phi + 5\sin^2\theta\sin^2\phi + 36\cos^2\phi\right]$$

$$= \sin^2\phi \left[4 + 5\sin^2\theta\sin^2\phi + 32\cos^2\phi\right]$$

$$\therefore \|c_\theta \times c_\phi\| = \sin\phi\sqrt{4 + 5\sin^2\theta\sin^2\phi + 32\cos^2\phi}$$

$$\therefore \frac{c_\theta \times c_\phi}{\|c_\theta \times c_\phi\|} = \frac{(-2\cos\theta\sin\phi, -3\sin\theta\sin\phi, -6\cos\phi)}{\sqrt{4 + 5\sin^2\theta\sin^2\phi + 32\cos^2\phi}}$$

for $\phi \neq 0, \pi$

$$= \frac{\left(-\frac{2}{3}x, -\frac{3}{2}y, -6z\right)}{\sqrt{4 + \frac{5}{4}y^2 + 32z^2}}$$

or,

$$\frac{(4x, 9y, 36z)}{6\sqrt{4 + \frac{5}{4}y^2 + 32z^2}}$$

(5) $4x^2 + 9y^2 = 36\cos^2\theta\sin^2\phi + 36\sin^2\theta\sin^2\phi = 36\sin^2\phi$

$$\therefore 4x^2 + 9y^2 + 36z^2 = 36\sin^2\phi + 36\cos^2\phi = 36$$

$$\therefore \frac{x^2}{9} + \frac{y^2}{4} + z^2 = 1, \text{ an ellipsoid.}$$

11. **11.** Repeat Exercise 9 for the surface

$$x = \sin v, \qquad y = u, \qquad z = \cos v$$

for $0 \le v \le 2\pi$ and $-1 \le u \le 3$.

(a) Let $\phi(u, v) = (\sin v, u, \cos v)$

$\phi_u = (0, 1, 0) \quad \phi_v = (\cos v, 0, -\sin v)$

$$\therefore \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 0 \\ \cos v & 0 & -\sin v \end{vmatrix} = (-\sin v, 0, -\cos v)$$

$\therefore$ unit normal is $\underline{(\sin v, 0, \cos v)} = \underline{\dfrac{(x, 0, z)}{\sqrt{x^2 + z^2}}}$

(b) x and z describe a circle, parallel to y-axis, of radius 1.

$-1 \le y \le 3$. $\therefore$ A cylinder parallel to y-axis, from $y = -1$ to $y = 3$, with y-axis the axis.

12. **12.** Repeat Exercise 9 for the surface

$$x = (2 - \cos v)\cos u, \quad y = (2 - \cos v)\sin u, \quad z = \sin v$$

for $-\pi \le u \le \pi$, $-\pi \le v \le \pi$. Is this surface regular?

(a) Let $\phi(u, v) = [(2 - \cos v)\cos u, (2 - \cos v)\sin u, \sin v]$

$\therefore \phi_u = [-(2 - \cos v)\sin u, (2 - \cos v)\cos u, 0]$

$\phi_v = [\sin v \cos u, \sin v \sin u, \cos v]$

$$\therefore \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -(2 - \cos v)\sin u & (2 - \cos v)\cos u & 0 \\ \sin v \cos u & \sin v \sin u & \cos v \end{vmatrix}$$

$$= \Big[(2\cos v - \cos^2 v)\cos u,\ (2\cos v - \cos^2 v)\sin u,$$
$$-(2-\cos v)\sin v\,(\sin^2 u + \cos^2 u) \Big]$$

$$= (2-\cos v)\big[\cos v\cos u,\ \cos v \sin u,\ -\sin v\big]$$

$2-\cos v > 0$, and so is just a magnitude factor.

$\therefore$ Let $\vec{N} = (\cos v \cos u, \cos v \sin u, -\sin v)$

and $\|\vec{N}\| = 1$

(b) $\sin(v) = 0$ for $v = -\pi, 0, \pi$

For those values, $\cos v = \mp 1$.

$\therefore \vec{N} = \pm(\cos u, \sin u, 0)$ for $v = -\pi, 0, \pi$.

But $\cos u$ and $\sin u$ are never simultaneously zero.

$\therefore \vec{N} \neq \vec{0}$. $\therefore$ Surface is regular

(c) For a fixed v, $x = (2-\cos v)\cos u$, $y = (2-\cos v)\sin u$ describes a circle of positive radius $(2-\cos v)$.

$\therefore$ Level curves perpendicular to z axis are circles.

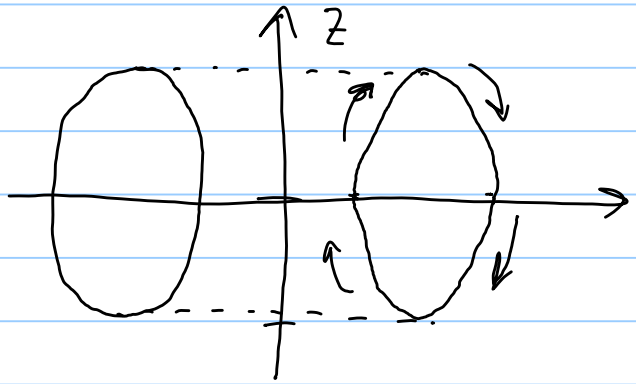
When $z = 0$ $(v = -\pi, 0, \pi)$, $(2-\cos v) = 3, 1, 3$

$z = 1 \left(v = \frac{\pi}{2}\right), \ (2 - \cos v) = 2$

$z = -1 \left(v = -\frac{\pi}{2}\right), \ (2 - \cos v) = 2$

$\therefore$ v: $-\pi, -\frac{\pi}{2}, 0, \frac{\pi}{2}, \pi$ circle radius: 3, 2, 1, 2, 3

z: 0, −1, 0, 1, 0 circle radius: 3, 2, 1, 2, 3



$\therefore$ a <u>torus</u> or doughnut perpendicular to z-axis.

13.

13. (a) Develop a formula for the plane tangent to the surface $x = h(y, z)$.

(b) Obtain a similar formula for $y = k(x, z)$.

(a) Let $\phi(u,v) = (h(u,v), u, v)$

$\therefore$ $\phi_u = (h_u, 1, 0) \quad \phi_v = (h_v, 0, 1)$

$\therefore$ $\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ h_u & 1 & 0 \\ h_v & 0 & 1 \end{vmatrix} = (1, -h_u, -h_v)$

This is the same as $(1, -h_y, -h_z)$

Let $x_0 = h(y_0, z_0)$.

$\therefore (1, -h_y(y_0,z_0), -h_z(y_0,z_0)) \cdot (x-x_0, y-y_0, z-z_0) = 0$

or $(x-x_0) - h_y(y_0,z_0)(y-y_0) - h_z(y_0,z_0)(z-z_0) = 0$

or $x = x_0 + h_y(y_0,z_0)(y-y_0) + h_z(y_0,z_0)(z-z_0)$

(5) $y = y_0 + K_x(x_0,z_0)(x-x_0) + K_z(x_0,z_0)(z-z_0)$

where $y_0 = k(x_0, z_0)$

14. **14.** Find the equation of the plane tangent to the surface $x = u^2$, $y = v^2$, $z = u^2 + v^2$ at the point $u = 1$, $v = 1$.

Let $\phi(u,v) = (u^2, v^2, u^2+v^2)$

$\therefore \phi_u = (2u, 0, 2u) \quad \phi_v = (0, 2v, 2v)$

$\therefore \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2u & 0 & 2u \\ 0 & 2v & 2v \end{vmatrix} = (-4uv, -4uv, 4uv)$

$\therefore \phi_u \times \phi_v (1,1) = (-4,-4,4)$. Let $\vec{N} = (-1,-1,1)$

$\phi(1,1) = (1,1,2)$.

$\therefore (-1,-1,1) \cdot (x-1, y-1, z-2) = 0$, or

$(1-x)+(1-y)+z-2=0$, or $\underline{z=x+y}$

15. **15.** Find a parametrization of the surface $z = 3x^2 + 8xy$ and use it to find the tangent plane at $x = 1$, $y = 0$, $z = 3$. Compare your answer with that using graphs.

Let $\phi(u,v) = (u, v, 3u^2+8uv)$

$\phi_u = (1, 0, 6u+8v)$ $\phi_v = (0, 1, 8u)$

$\therefore \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 6u+8v \\ 0 & 1 & 8u \end{vmatrix} = (-6u-8v, -8u, 1)$

$\therefore \phi_u \times \phi_v(1,0) = (-6, -8, 1)$ $\phi(1,0) = (1,0,3)$

$\therefore (-6,-8,1)\cdot(x-1, y, z-3) = 0$, or

$-6(x-1)-8y+z-3=0$, or $\underline{z-6x-8y+3=0}$

16. **16.** Find a parametrization of the surface $x^3 + 3xy + z^2 = 2$, $z > 0$, and use it to find the tangent plane at the point $x = 1$, $y = 1/3$, $z = 0$. Compare your answer with that using level sets.

Why stipulate $z>0$ when asking at point $z=0$?

(a) $z = \sqrt{2-x^3-3xy}$. If use $\phi(u,v) = (u, v, \sqrt{2-u^3-3uv})$,

Then ϕ_u and ϕ_v will have $(2-u^3-3uv)^{-\frac{1}{2}}$ in 3rd component, and can't divide by 0.

$\therefore$ Try $y = \frac{2 - z^2 - x^3}{3x}$.

Let $\phi(u,v) = \left(u, \frac{2 - v^2 - u^3}{3u}, v\right)$

$\therefore \phi(1,0) = \left(1, \frac{1}{3}, 0\right)$

$$\phi_u = \left[1, \frac{3u(-3u^2) - (2 - v^2 - u^3)(3)}{9u^2}, 0\right]$$

$$\phi_v = \left[0, -\frac{2v}{3u}, 1\right]$$

$$\therefore \phi_u(1,0) = \left[1, \frac{3(-3) - (1)(3)}{9}, 0\right] = \left[1, -\frac{4}{3}, 0\right]$$

$$\phi_v(1,0) = [0, 0, 1]$$

$$\therefore \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -4/3 & 0 \\ 0 & 0 & 1 \end{vmatrix} = \left(-\frac{4}{3}, -1, 0\right)$$

$\therefore \left(-\frac{4}{3}, -1, 0\right) \cdot \left(x - 1, y - \frac{1}{3}, z - 0\right) = 0$, or

$-\frac{4}{3}(x-1) - \left(y - \frac{1}{3}\right) = 0$, or

$-4(x-1) - 3\left(y - \frac{1}{3}\right) = 0$, or $\underline{4x + 3y = 5}$

(6) Using level sets, $f(x,y,z) = x^3 + 3xy + z^2 = 2$

Gradient is perpendicular to level set.

$\nabla f = (3x^2 + 3y,\ 3x,\ 2z)$. $\therefore \nabla(1, 1/3, 0) = (4, 3, 0)$

$\therefore (4,3,0) \cdot (x-1,\ y-1/3,\ z-0) = 0$, or

$4(x-1) + 3(y-\frac{1}{3}) + 0 = 0$, or

$4x - 4 + 3y - 1 = 0$, or $\underline{4x + 3y = 5}$

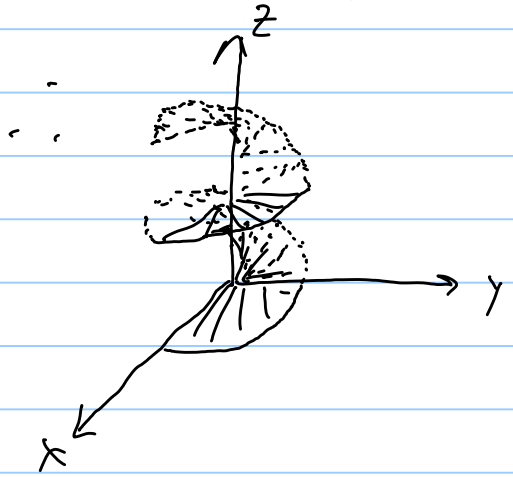
17. **17.** Consider the surface in $\mathbb{R}^3$ parametrized by

$$\Phi(r, \theta) = (r\cos\theta, r\sin\theta, \theta), \quad 0 \le r \le 1 \text{ and } 0 \le \theta \le 4\pi.$$

(a) (a) Sketch and describe the surface.

$(r\cos\theta, r\sin\theta)$ is a circle of radius r.

$\therefore$ as r varies from $0 \le r \le 1$, a line sweeps out a circle, as θ increases, and the "ray" elevates according to θ, and so sweeps out a sheet, like a screw, going around the start point $(1,0,0)$ twice, once at $\theta = 2\pi$, and again at $\theta = 4\pi$.



$\phi(r, \theta) = (r\cos\theta, r\sin\theta, \theta)$

$0 \le r \le 1$

$0 \le \theta \le 4\pi$

(5) (b) Find an expression for a unit normal to the surface.

$$\phi_r = (\cos\theta, \sin\theta, 0) \quad \phi_\theta = (-r\sin\theta, r\cos\theta, 1)$$

$$\therefore \phi_r \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & 0 \\ -r\sin\theta & r\cos\theta & 1 \end{vmatrix}$$

$$= (\sin\theta, -\cos\theta, r\cos^2\theta + r\sin^2\theta)$$

$$= (\sin\theta, -\cos\theta, r)$$

$$\|(\sin\theta, \cos\theta, r)\| = \sqrt{1+r^2}$$

$$\therefore \vec{N} = \frac{1}{\sqrt{1+r^2}} (\sin\theta, -\cos\theta, r)$$

(c) (c) Find an equation for the plane tangent to the surface at the point (x_0, y_0, z_0).

$$x_0 = r\cos\theta, \quad y_0 = r\sin\theta, \quad z_0 = \theta$$

$$\therefore x_0^2 + y_0^2 = r^2, \quad r = \sqrt{x_0^2 + y_0^2}$$

$$\therefore \vec{N} = \frac{1}{\sqrt{1+x_0^2+y_0^2}} \left(\sin(z_0), -\cos(z_0), \sqrt{x_0^2+y_0^2}\right)$$

$$\therefore \left(\sin(z_0), -\cos(z_0), \sqrt{x_0^2+y_0^2}\right) \cdot (x - x_0, y - y_0, z - z_0) = 0$$

or, $\sin(z_0)(x-x_0) - \cos(z_0)(y-y_0) + \sqrt{x_0^2+y_0^2}\,(z-z_0) = 0$

or, multiplying by $\sqrt{x_0^2+y_0^2}$ and noting

$\sin(z_0)\sqrt{x_0^2+y_0^2} = y_0 \quad (y = r\sin\theta = r\sin z)$

and $\cos(z_0)\sqrt{x_0^2+y_0^2} = x_0 \quad (x = r\cos\theta = r\cos z)$

$\therefore \; y_0(x-x_0) - x_0(y-y_0) + (x_0^2+y_0^2)(z-z_0) = 0$

(d)

(d) If (x_0, y_0, z_0) is a point on the surface, show that the horizontal line segment of unit length from the z axis through (x_0, y_0, z_0) is contained in the surface and in the plane tangent to the surface at (x_0, y_0, z_0).

(1) Let $\theta_0 = z_0$, $r_0 = \sqrt{x_0^2+y_0^2}$ $\therefore \phi(r_0, \theta_0) = (x_0, y_0, z_0)$

$= (r_0\cos\theta_0, r_0\sin\theta_0, \theta_0)$ is on the surface,

and by definition, all points $0 \le r_0$ at $z = \theta_0$

are on the surface. $\therefore$ The segment from

$(0, 0, \theta_0)$ to $(r_0\cos\theta_0, r_0\sin\theta_0, \theta_0) = (x_0, y_0, z_0)$

is on the surface.

(2) A vector parallel to segment is

$(r_0\cos\theta_0, r_0\sin\theta_0, \theta_0) - (0, 0, \theta_0) =$

$(r_0\cos\theta_0, r_0\sin\theta_0, 0)$, or $(\cos\theta_0, \sin\theta_0, 0)$

A vector parallel to $\vec{N}$ from (c) is

$(\sin\theta_0, -\cos\theta_0, r_0)$.

$\therefore (\cos\theta_0, \sin\theta_0, 0) \cdot (\sin\theta_0, -\cos\theta_0, 0)$

$= \cos\theta_0\sin\theta_0 - \sin\theta_0\cos\theta_0 = 0$

$\therefore$ All points on segment from z-axis to (x_0, y_0, z_0) are perpendicular to $\vec{N}$, and $\therefore$ are in the tangent plane.

18. **18.** Given a sphere of radius 2 centered at the origin, find the equation for the plane that is tangent to it at the point $(1, 1, \sqrt{2})$ by considering the sphere as:

(a) a surface parametrized by $\Phi(\theta, \phi) = (2\cos\theta\sin\phi, 2\sin\theta\sin\phi, 2\cos\phi)$;

(b) a level surface of $f(x, y, z) = x^2 + y^2 + z^2$; and

(c) the graph of $g(x, y) = \sqrt{4 - x^2 - y^2}$.

(a) $\phi_\theta = (-2\sin\theta\sin\phi, 2\cos\theta\sin\phi, 0)$

$\phi_\phi = (2\cos\theta\cos\phi, 2\sin\theta\cos\phi, -2\sin\phi)$

For the point $(1, 1, \sqrt{2})$, $\theta = \text{Arctan}\frac{y}{x} = \text{Arctan}\,1 = \frac{\pi}{4}$

$\phi = \text{Arccos}\frac{z}{\rho} = \text{Arccos}\frac{\sqrt{2}}{2} = \frac{\pi}{4}$

$\therefore \phi_\theta\left(\frac{\pi}{4}, \frac{\pi}{4}\right) = \left(-2\frac{\sqrt{2}}{2}\frac{\sqrt{2}}{2}, 2\frac{\sqrt{2}}{2}\frac{\sqrt{2}}{2}, 0\right) = (-1, 1, 0)$

$\phi_\phi\left(\frac{\pi}{4}, \frac{\pi}{4}\right) = \left(2\frac{\sqrt{2}}{2}\frac{\sqrt{2}}{2}, 2\frac{\sqrt{2}}{2}\frac{\sqrt{2}}{2}, -2\frac{\sqrt{2}}{2}\right) = (1, 1, -\sqrt{2})$

$\therefore \phi_\theta \times \phi_\phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 0 \\ 1 & 1 & -\sqrt{2} \end{vmatrix} = (-\sqrt{2}, -\sqrt{2}, -2)$

$\therefore (-\sqrt{2}, -\sqrt{2}, -2) \cdot (x-1, y-1, z-\sqrt{2}) = 0$, or

$-\sqrt{2}(x-1) - \sqrt{2}(y-1) - 2(z-\sqrt{2}) = 0$, or

$\sqrt{2}x + \sqrt{2}y + 2z = 4\sqrt{2}$, or

$x + y + \sqrt{2}z = 4$

(5) (b) a level surface of $f(x, y, z) = x^2 + y^2 + z^2$;

Gradients are perpendicular to level sets.

The point $(1, 1, \sqrt{2})$ means $f(1, 1, \sqrt{2}) = 4$

$\nabla f = (2x, 2y, 2z)$, so that $\nabla f(1, 1, \sqrt{2}) = (2, 2, 2\sqrt{2})$

$\therefore (2, 2, \sqrt{2}) \cdot (x-1, y-1, z-\sqrt{2}) = 0$, or

$2(x-1) + 2(y-1) + 2\sqrt{2}(z-\sqrt{2}) = 0$, or

$2x + 2y + 2\sqrt{2}z = 8$, or $x + y + \sqrt{2}z = 4$

(c) (c) the graph of $g(x, y) = \sqrt{4 - x^2 - y^2}$.

$g(1,1) = \sqrt{2} = z_0$

Tangent plane: $z = z_0 + g_x(x - x_0) + g_y(y - y_0)$

$g_x = \frac{1}{2}(4 - x^2 - y^2)^{-\frac{1}{2}}(-2x)$

$\therefore g_x(1,1) = \frac{1}{2}(4 - 1 - 1)^{-\frac{1}{2}}(-2) = -\frac{1}{\sqrt{2}}$

$g_y = \frac{1}{2}(4 - x^2 - y^2)^{-\frac{1}{2}}(-2y)$

$\therefore g_y(1,1) = \frac{1}{2}(4 - 1 - 1)^{-\frac{1}{2}}(-2) = -\frac{1}{\sqrt{2}}$

$\therefore z = \sqrt{2} - \frac{1}{\sqrt{2}}(x - 1) - \frac{1}{\sqrt{2}}(y - 1)$, or

$\sqrt{2}z = 2 - x + 1 - y + 1$, or

$x + y + \sqrt{2}z = 4$

19.

19. (a) Find a parametrization for the hyperboloid $x^2 + y^2 - z^2 = 25$.

(b) Find an expression for a unit normal to this surface.

(c) Find an equation for the plane tangent to the surface at $(x_0, y_0, 0)$, where $x_0^2 + y_0^2 = 25$.

(d) Show that the lines $(x_0, y_0, 0) + t(-y_0, x_0, 5)$ and $(x_0, y_0, 0) + t(y_0, -x_0, 5)$ lie in the surface *and* in the tangent plane found in part (c).

(a) Let $x = r\cos\theta$, $y = r\sin\theta$. $\therefore x^2 + y^2 = r^2$

Looking at $r^2 - z^2 = 25$ and noting

$\cosh^2(u) - \sinh^2(u) = 1$

Let $r = 5\cosh(u)$ $\quad z = 5\sinh(u)$

$\therefore x = 5\cosh(u)\cos\theta,\ y = 5\cosh(u)\sin\theta,$

$\therefore$ Let $\phi(u,\theta) = (5\cosh(u)\cos\theta, 5\cosh(u)\sin\theta, 5\sinh(u))$

for $-\infty < u < \infty,\ 0 \le \theta < 2\pi$

Another, noting $z^2 + 25 = x^2 + y^2$

$\therefore x = r\cos\theta,\ y = r\sin\theta,\ x^2 + y^2 = r^2$

$\therefore r = \sqrt{z^2+25}$

$\therefore \phi(\theta, z) = \left[\sqrt{z^2+25}\cos\theta, \sqrt{z^2+25}\sin\theta, z\right]$

(b) Use level sets. Let $f(x,y,z) = x^2 + y^2 - z^2 = 25$

$\nabla f = (2x, 2y, -2z)$ which is parallel to $(x, y, -z)$

$\therefore$ unit normal $= \dfrac{(x,y,-z)}{\sqrt{x^2+y^2+z^2}} = \dfrac{(x,y,-z)}{\sqrt{2z^2+25}}$

(c) (c) Find an equation for the plane tangent to the surface at $(x_0, y_0, 0)$, where $x_0^2 + y_0^2 = 25$.

From (b), normal to plane is $(x_0, y_0, 0)$.

$\therefore (x_0, y_0, 0) \cdot (x - x_0, y - y_0, z) = 0,$ or

$$x_0(x - x_0) + y_0(y - y_0) = 0, \text{ or}$$

$$x_0 x + y_0 y = x_0^2 + y_0^2 = 25$$

$$\therefore \quad x_0 x + y_0 y = 25$$

(d) (d) Show that the lines $(x_0, y_0, 0) + t(-y_0, x_0, 5)$ and $(x_0, y_0, 0) + t(y_0, -x_0, 5)$ lie in the surface *and* in the tangent plane found in part (c).

(1) From (c), normal to tangent plane at $(x_0, y_0, 0)$ is $(x_0, y_0, 0) = \vec{N}$.

Note $(-y_0, x_0, 5) \cdot \vec{N} = -x_0 y_0 + x_0 y_0 + 0 = 0$

and $(y_0, -x_0, 5) \cdot \vec{N} = x_0 y_0 - x_0 y_0 + 0 = 0$.

$\therefore$ Both vectors $(-y_0, x_0, 5)$ and $(y_0, -x_0, 5)$ are $\perp \vec{N} \Rightarrow$ parallel to tangent plane.

Since $(x_0, y_0, 0)$ is in the plane, then the line $(x_0, y_0, 0) + t(-y_0, x_0, 5)$, $t \in R$, goes through $(x_0, y_0, 0)$ $(t = 0)$ and is parallel to the tangent plane,

and so is in the tangent plane.

Similarly for the other line,

$$(x_0, y_0, 0) + t(y_0, -x_0, 5)$$

(2) Since $(x_0, y_0, 0)$ is on the surface, $x_0^2 + y_0^2 - 0 = 25$,

or $x_0^2 + y_0^2 = 25$ [1]

$$(x_0, y_0, 0) + t(-y_0, x_0, 5) = (x_0 - ty_0, y_0 + tx_0, 5t)$$

∴ To show every point on the line is on the surface, must show each point on line satisfies $x^2 + y^2 - z^2 = 25$

Since $(x_0 - ty_0)^2 + (y_0 + tx_0)^2 - (5t)^2 =$

$$x_0^2 - 2x_0y_0t + t^2y_0^2 + y_0^2 + 2x_0y_0t + t^2x_0^2 + 25t^2 =$$

$$(x_0^2 + y_0^2) + t^2(x_0^2 + y_0^2 - 25) =$$

$$25 + t^2(0) \quad \text{from [1]}$$

$$= 25.$$

∴ Each point on the line $(x_0, y_0, 0) + t(-y_0, x_0, 5)$ is on the surface.

Similarly for the line $(x_0, y_0, 0) + t(y_0, -x_0, 5)$

$= (x_0 + t y_0, y_0 - t x_0, 5t)$,

$(x_0 + t y_0)^2 + (y_0 - t x_0)^2 - (5t)^2 =$

$x_0^2 + y_0^2 + t^2(x_0^2 + y_0^2 - 25) =$

$25 + t^2(0)$ from [1]

$= 25$

$\therefore$ The line $(x_0, y_0, 0) + t(y_0, -x_0, 5)$ is also on the surface.

20. **20.** A parametrized surface is described by a differentiable function $\Phi: \mathbb{R}^2 \to \mathbb{R}^3$. According to Chapter 2, the derivative should give a linear approximation that yields a representation of the tangent plane. This exercise demonstrates that this is indeed the case.

(a) (a) Assuming $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$, show that the range of the linear transformation $\mathbf{D}\Phi(u_0, v_0)$ is the plane spanned by $\mathbf{T}_u$ and $\mathbf{T}_v$. [Here $\mathbf{T}_u$ and $\mathbf{T}_v$ are evaluated at (u_0, v_0).]

$D\phi(u_0, v_0)$ is a 3×2 matrix of the partial derivatives of ϕ.

Let $\phi(u, v) = (f(u,v), g(u,v), h(u,v))$

$\therefore \vec{T}_u = (f_u, g_u, h_u)$ $\vec{T}_v = (f_v, g_v, h_v)$

$$\therefore D\phi = \begin{bmatrix} f_u & f_v \\ g_u & g_v \\ h_u & h_v \end{bmatrix} = \begin{bmatrix} f_u(u_0,v_0) & f_v(u_0,v_0) \\ g_u(u_0,v_0) & g_v(u_0,v_0) \\ h_u(u_0,v_0) & h_v(u_0,v_0) \end{bmatrix}$$

$$\therefore \text{Range of } D\phi = \begin{bmatrix} f_u & f_v \\ g_u & g_v \\ h_u & h_v \end{bmatrix} \begin{bmatrix} u-u_0 \\ v-v_0 \end{bmatrix}, \text{ for all } (u,v)$$

$$= \begin{bmatrix} f_u(u_0,v_0)(u-u_0) + f_v(u_0,v_0)(v-v_0) \\ g_u(u_0,v_0)(u-u_0) + g_v(u_0,v_0)(v-v_0) \\ h_u(u_0,v_0)(u-u_0) + h_v(u_0,v_0)(v-v_0) \end{bmatrix}$$

$$= (u-u_0)\begin{bmatrix} f_u \\ g_u \\ h_u \end{bmatrix} + (v-v_0)\begin{bmatrix} f_v \\ g_v \\ h_v \end{bmatrix}$$

$$= (u-u_0)\vec{T}_u + (v-v_0)\vec{T}_v$$

($\vec{T}_u, \vec{T}_v$ viewed as column vectors).

= plane spanned by $\vec{T}_u$ and $\vec{T}_v$

(6) (b) Show that $\mathbf{w} \perp (\mathbf{T}_u \times \mathbf{T}_v)$ if and only if $\mathbf{w}$ is in the range of $\mathbf{D\Phi}(u_0, v_0)$.

(1) Suppose $\vec{w}$ is in the range of $D\phi(u_0,v_0)$

From (a), this means that $\exists (u,v)$

s.t. $\vec{w} = (u-u_0)\vec{T}_u + (v-v_0)\vec{T}_v$

$$\therefore \vec{w} \cdot (\vec{T}_u \times \vec{T}_v) = \left[(u-u_0)\vec{T}_u + (v-v_0)\vec{T}_v\right] \cdot (\vec{T}_u \times \vec{T}_v)$$

$$= (u-u_0)\vec{T}_u \cdot (\vec{T}_u \times \vec{T}_v) + (v-v_0)\vec{T}_v \cdot (\vec{T}_u \times \vec{T}_v)$$

$$= 0 + 0 = 0,$$

since $\vec{T}_u \cdot (\vec{T}_u \times \vec{T}_v) = 0$ and $\vec{T}_v \cdot (\vec{T}_u \times \vec{T}_v) = 0$

$$\left[\vec{a} \cdot (\vec{a} \times \vec{b}) = 0\right]$$

(2) Suppose $\vec{w} \perp (\vec{T}_u \times \vec{T}_v)$

$\therefore \vec{w} \cdot (\vec{T}_u \times \vec{T}_v) = 0$, and so $\vec{w}$ is in the plane spanned by $\vec{T}_u$ and $\vec{T}_v$ through (u_0, v_0). $\therefore \exists (u,v)$ s.t.

$$\vec{w} = (u-u_0)\vec{T}_u + (v-v_0)\vec{T}_v$$

From (a), this means $\vec{w}$ is in the range of $D\phi(u_0, v_0)$

(c) (c) Show that the tangent plane as defined in this section is the same as the "parametrized plane"

$$(u, v) \mapsto \mathbf{\Phi}(u_0, v_0) + \mathbf{D\Phi}(u_0, v_0)\begin{bmatrix} u - u_0 \\ v - v_0 \end{bmatrix}.$$

As shown in (a), $D\phi(u_0, v_0)\begin{bmatrix} u-u_0 \\ v-v_0 \end{bmatrix}$ represents

The range spanned by $\vec{T}_u(u_0, v_0)$ and $\vec{T}_v(u_0, v_0)$.

$\therefore$ This is the same plane referenced above, as both planes contain the point $\phi(u_0, v_0)$.

21.

21. Consider the surfaces $\Phi_1(u, v) = (u, v, 0)$ and $\Phi_2(u, v) = (u^3, v^3, 0)$.

(a)

(a) Show that the image of Φ_1 and of Φ_2 is the *xy* plane.

The xy-plane is characterized by all points with z-component 0: $(x, y, 0)$, $x, y \in \mathbb{R}$.

(1) $\therefore$ For ϕ_1, Given any x, y, let $u = x$, $v = y$.
$\therefore$ Any point in xy-plane can be represented by ϕ_1, so xy-plane $\subseteq$ Image ϕ_1

And every $\phi(u, v)$ is in the xy-plane, as $\phi(u, v) = (u, v, 0)$. $\therefore$ Image $\phi_2 \subseteq$ xy-plane

$\therefore$ Image of ϕ_1 = xy-plane.

(2) For ϕ_2, given any x, y, let $u = \sqrt[3]{x}$, $v = \sqrt[3]{y}$

$\therefore \phi_2(u, v) = (x, y, 0)$. $\therefore$ xy-plane $\subseteq$ Image ϕ_2

And every $\phi_2(u, v)$ is in xy-plane since z-component of $\phi_2(u, v)$ is zero.

$\therefore$ Image $\phi_2 \subseteq$ xy-plane.

$\therefore$ Image ϕ_2 = xy-plane.

(6) (b) Show that Φ_1 describes a regular surface, yet Φ_2 does not. Conclude that the notion of regularity of a surface S depends on the existence of at least one regular parametrization for S.

(1) $\phi_{1u} = (1,0,0)$ $\phi_{1v} = (0,1,0)$.

$\therefore \phi_{1u} \times \phi_{1v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = (0,0,1) \neq \vec{0}$

$\therefore$ By def., ϕ_1 is regular.

(2) $\phi_{2u} = (3u^2, 0, 0)$ $\phi_{2v} = (0, 3v^2, 0)$.

For $u=0, v=0$, $\phi_{2u} \times \phi_{2v}(0,0) = (0,0,0)$.

$\therefore$ ϕ_2 not regular at $(u,v) = (0,0)$.

(C) (c) Prove that the tangent plane of S is well defined independently of the regular (one-to-one) parametrization (you will need to use the inverse function theorem from Section 3.5).

Problem is not sufficiently stated to understand what is being asked. Inverse function Theorem uses $R^n \to R^n$. ϕ_1 and ϕ_2 above are $R^2 \to R^3$. I don't see when enough information has been given to state a Jacobian determinant is nonzero.

(d) (d) After these remarks, do you think you can find a regular parametrization of the cone of Figure 7.3.7? – you mean 7.3.8

No - A regular parametrization can't be found making cone smooth at origin.

22. **22.** The image of the parametrization

$$\Phi(u, v) = (x(u, v), y(u, v), z(u, v))$$
$$= (a \sin u \cos v,\ b \sin u \sin v,\ c \cos u)$$

with $b < a$, $0 \le u \le \pi$, $0 \le v \le 2\pi$ parametrizes an ellipsoid.

(a) Show that all points in the image of Φ satisfy:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

(the Cartesian equation of an ellipsoid).

(a) $x^2 = a^2 \sin^2 u \cos^2 v$

$y^2 = b^2 \sin^2 u \sin^2 v$

$z^2 = c^2 \cos^2 u$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{a^2 \sin^2 u \cos^2 v}{a^2} + \frac{b^2 \sin^2 u \sin^2 v}{b^2}$$

$$= \sin^2 u (\cos^2 v + \sin^2 v) = \sin^2 u$$

$$\therefore \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = \sin^2 u + \frac{c^2 \cos^2 u}{c^2} = \sin^2 u + \cos^2 u = 1$$

(b) (b) Show that the image surface is regular at all points. Assume $a, b, c \neq 0$.

$\phi_u = (a \cos u \cos v,\ b \cos u \sin v,\ -c \sin u)$

$\phi_v = (-a \sin u \sin v,\ b \sin u \cos v,\ 0)$

$$\therefore \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a\cos u\cos v & b\cos u\sin v & -c\sin u \\ -a\sin u\sin v & b\sin u\cos v & 0 \end{vmatrix}$$

$$= (cb\sin^2 u\cos v,\ ac\sin^2 u\sin v,$$

$$ab\sin u\cos u\cos^2 v + ab\sin u\cos u\sin^2 v)$$

$$= (cb\sin^2 u\cos v,\ ac\sin^2 u\sin v,\ ab\sin u\cos u)$$

Exclude $u = 0, \pi$, so $\sin u \neq 0$.

$\therefore cb\sin^2 u \neq 0$, $ac\sin^2 u \neq 0$ (x, y components).

$\cos v$ and $\sin v$ can't be 0, so either

the x-component or y-component $\neq 0$.

$\therefore \phi_u \times \phi_v \neq 0$ for $u \neq 0, \pi$.

$\therefore$ Ellipsoid regular at all points not

equal to $(0,0,c)$ and $(0,0,-c)$.

For the two z-polar points, use the parametrization

$$\phi(u,v) = (a\sin u\cos v,\ b\cos u,\ c\sin u\sin v),$$

$$0 \leq u \leq \pi,\quad 0 \leq v \leq 2\pi$$

$\therefore$ each point satisfies $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

This time, the parametrization shows all points regular (including $(0,0,\pm c)$) except possibly $(0, \pm b, 0)$. But these two polar points were shown to be regular with the first parametrization.

$\therefore$ All points are regular.

23.

23. The image of the parametrization

$$\Phi(u, v) = (x(u, v), y(u, v), z(u, v))$$
$$= ((R + r\cos u)\cos v, (R + r\cos u)\sin v, r\sin u)$$

with $0 \le u, v \le 2\pi$, $0 < r < 1$ parametrizes a torus (or doughnut) S.

(a) Show that all points in the image (x, y, z) satisfy:

$$(\sqrt{x^2 + y^2} - R)^2 + z^2 = r^2.$$

(b) Show that the image surface is regular at all points.

$R \ge r$, where
R = distance from center of torus to center of ring,
r = radius of ring.

Problem fails to specify R.

(a) $x^2 + y^2 = (R + r\cos u)^2 \cos^2 v + (R + r\cos u)^2 \sin^2 v$

$= (R + r\cos u)^2$

$\therefore \sqrt{x^2 + y^2} = |R + r\cos u| = R + r\cos u$ as $R \ge r\cos u$

$\therefore \sqrt{x^2 + y^2} - R = r\cos u$

$$\therefore \left(\sqrt{x^2+y^2} - R\right)^2 + z^2 = (r\cos u)^2 + (r\sin u)^2$$

$$= r^2$$

(6) $\phi_u = [-r\sin u\cos v,\ -r\sin u\sin v,\ r\cos u]$

$\phi_v = [-(R+r\cos u)\sin v,\ (R+r\cos u)\cos v,\ 0]$

$$\therefore \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -r\sin u\cos v & -r\sin u\sin v & r\cos u \\ -(R+r\cos u)\sin v & (R+r\cos u)\cos v & 0 \end{vmatrix}$$

$$= [-r(R+r\cos u)\cos u\cos v,\ -r(R+r\cos u)\cos u\sin v,$$

$$-r(R+r\cos u)\sin u\cos^2 v - r(R+r\cos u)\sin u\sin^2 v]$$

$$= -r(R+r\cos u)[\cos u\cos v,\ \cos u\sin v,\ \sin u]$$

If $\sin u = 0$, then $\cos u \neq 0$, and $\cos v$ and $\sin v$ can't both be 0.

$\therefore [\cos u\cos v,\ \cos u\sin v,\ \sin u] \neq [0,0,0]$.

Also $r > 0$.

If $R = r$, then $R + r\cos u$ could be 0. (i.e., torus has no hole).

$\therefore$ Specify $R > r$, and $\therefore \phi_u \times \phi_v \neq 0$.

$\therefore$ Torus regular everywhere.

Really, should state: $0 \le u \le 2\pi$
$0 \le v \le 2\pi$
$0 < r < R$

24. **24.** Let Φ be a regular surface at (u_0, v_0); that is, Φ is of class C^1 and $\mathbf{T}_u \times \mathbf{T}_v \neq \mathbf{0}$ at (u_0, v_0).

(a) Use the implicit function theorem (Section 3.5) to show that the image of Φ *near* (u_0, v_0) is the graph of a C^1 function of two variables. If the z component of $\mathbf{T}_u \times \mathbf{T}_v$ is nonzero, we can write it as $z = f(x, y)$.

(c) Show that the tangent plane at $\Phi(u_0, v_0)$ defined by the plane spanned by $\mathbf{T}_u$ and $\mathbf{T}_v$ coincides with the tangent plane of the graph of $z = f(x, y)$ at this point.

should → 5e (6)

$\phi(u,v) = [x(u,v), y(u,v), z(u,v)]$

(a) $\vec{T_u} = (x_u, y_u, z_u)$ $\vec{T_v} = (x_v, y_v, z_v)$.

$$\vec{T_u} \times \vec{T_v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_u & y_u & z_u \\ x_v & y_v & z_v \end{vmatrix} = \begin{bmatrix} y_u z_v - y_v z_u \\ x_v z_u - x_u z_v \\ x_u y_v - x_v y_u \end{bmatrix}$$

$\vec{T_u} \times \vec{T_v} \neq \vec{0} \Rightarrow$ at least one of the components is non-zero. Suppose, for simplicity, it is the z-component: $x_u y_v - x_v y_u \neq 0$.

$\therefore$ The determinant $\begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} \neq 0$ at (u_0, v_0)

$\therefore$ Consider $F(u,v) = (x(u,v), y(u,v)) = (x, y)$

where $F_1(u,v) = x(u,v)$, $F_2(u,v) = y(u,v)$

Near $F(u_0, v_0) = (x_0, y_0)$, the Jacobian determinant of F at (u_0, v_0) is $\begin{vmatrix} x_u(u_0,v_0) & x_v(u_0,v_0) \\ y_u(u_0,v_0) & y_v(u_0,v_0) \end{vmatrix} \neq 0$

∴ By the Inverse Function Theorem (a special case of the Implicit Function Theorem) as stated on p. 209 of the text, there is a C^1 function $g: \mathbb{R}^2 \to \mathbb{R}^2$ s.t.

$g(x,y) = (u,v)$ for x near x_0, y near y_0.

∴ The graph is $(x, y, g(x,y))$, where $z = g(x,y)$

If the y-component of $\vec{T}_u \times \vec{T}_v \neq 0$, then the graph becomes $(x, z, g(x,z))$, $y = g(x,y)$

If x-component of $\vec{T}_u \times \vec{T}_2 \neq 0$, then $(y, z, g(y,z))$, $x = g(y,z)$

(6) (c) Show that the tangent plane at $\Phi(u_0, v_0)$ defined by the plane spanned by $\mathbf{T}_u$ and $\mathbf{T}_v$ coincides with the tangent plane of the graph of $z = f(x, y)$ at this point.

The tangent plane at $\phi(u_0, v_0)$ is

$$\phi(u_0, v_0) + \begin{bmatrix} x_u & x_v \\ y_u & y_v \\ z_u & z_v \end{bmatrix} \begin{bmatrix} u - u_0 \\ v - v_0 \end{bmatrix}$$

$$= \begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + \begin{bmatrix} x_u \\ y_u \\ z_u \end{bmatrix}(u-u_0) + \begin{bmatrix} x_v \\ y_v \\ z_v \end{bmatrix}(v-v_0)$$

$$= \phi(u_0,v_0) + \vec{T}_u(u-u_0) + \vec{T}_v(v-v_0)$$

For the graph $(x, y, g(x,y))$, the tangent plane at $\phi(u_0,v_0) = (x_0, y_0, z_0)$ is

$$\begin{bmatrix} x_0 \\ y_0 \\ z_0 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ g_x(x_0,y_0) \end{bmatrix}(x-x_0) + \begin{bmatrix} 0 \\ 1 \\ g_y(x_0,y_0) \end{bmatrix} y - y_0$$

The normal to this plane is

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & g_x(x_0,y_0) \\ 0 & 1 & g_y(x_0,y_0) \end{vmatrix} = \{-g_x(x_0,y_0), -g_y(x_0,y_0), 1\}$$

From (a), $F(g(x,y)) = (x,y)$ and $g(F(u,v)) = (u,v)$

$$g(F(u,v)) = g(F_1(u,v), F_2(u,v)) = g(x(u,v), y(u,v))$$

$$\therefore g_u = \frac{\partial g}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial g}{\partial y}\cdot\frac{\partial y}{\partial u} = g_x x_u + g_y y_u \quad [1]$$

$$g_v = \frac{\partial g}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial g}{\partial y}\cdot\frac{\partial y}{\partial v} = g_x x_v + g_y y_v \quad [2]$$

But $z = g(x,y) = g(x(u,v), y(u,v))$

$\therefore z_u = g_u \;,\; z_v = g_v$

$\therefore$ [1], [2] become $\begin{bmatrix} x_u & y_u \\ x_v & y_v \end{bmatrix} \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} z_u \\ z_v \end{bmatrix}$

Since $x_u y_v - x_v y_u \neq 0$ (From (4)), Then

$$\begin{bmatrix} g_x \\ g_y \end{bmatrix} = \frac{1}{x_u y_v - x_v y_u} \begin{bmatrix} y_v & -y_u \\ -x_v & x_u \end{bmatrix} \begin{bmatrix} z_u \\ z_v \end{bmatrix}$$

$$= \begin{bmatrix} z_u y_v - z_v y_u \\ z_v x_u - z_u x_v \end{bmatrix} \cdot \frac{1}{x_u y_v - x_v y_u}$$

$$\therefore \begin{bmatrix} -g_x \\ -g_y \\ 1 \end{bmatrix} = \frac{1}{x_u y_v - x_v y_u} \begin{bmatrix} z_v y_u - z_u y_v \\ z_v x_u - z_u x_v \\ x_u y_v - x_v y_u \end{bmatrix}$$

i.e., by ignoring the scaling factor, the normal to the plane is parallel to:

$$\begin{bmatrix} y_u z_v - y_v z_u \\ x_u z_v - x_v z_u \\ x_u y_v - x_v y_u \end{bmatrix}$$

which is the same vector as $\vec{T}_u \times \vec{T}_v$

$\therefore$ Near $\phi(u_0, v_0) = (x_0, y_0, z_0)$, the plane spanned by $\vec{T}_u$ and $\vec{T}_v$ that includes (x_0, y_0, z_0) is the same plane with normal $\begin{bmatrix} -g_x \\ -g_y \\ 1 \end{bmatrix}$ that also passes through (x_0, y_0, z_0),

where $z = g(x, y)$ was defined by the inverse function theorem near (x_0, y_0, z_0).

7.4 Area of a Surface

Note Title 3/1/2017

1.

1. Find the surface area of the unit sphere S represented parametrically by Φ: $D \to S \subset \mathbb{R}^3$, where D is the rectangle $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi$ and Φ is given by the equations

$$x = \cos\theta \sin\phi, \qquad y = \sin\theta \sin\phi, \qquad z = \cos\phi.$$

Note that we can represent the entire sphere parametrically, but we cannot represent it in the form $z = f(x, y)$.

(1) D is an elementary region

(2) $\phi(\theta, \phi)$ is C^1 and one-to-one except at $\theta = 2\pi$ (the boundary)

(3) The image is regular, under this parametrization, except at $\phi = 0, \pi$ (finite # of points).

$$\vec{T}_\theta = (-\sin\theta\sin\phi, \cos\theta\sin\phi, 0) = (x_\theta, y_\theta, z_\theta)$$

$$\vec{T}_\phi = (\cos\theta\cos\phi, \sin\theta\cos\phi, -\sin\phi) = (x_\phi, y_\phi, z_\phi)$$

$$\vec{T}_\theta \times \vec{T}_\phi = (-\cos\theta\sin^2\phi, -\sin\theta\sin^2\phi, -\sin^2\theta\sin\phi\cos\phi - \cos^2\theta\sin\phi\cos\phi)$$

$$= (-\cos\theta\sin^2\phi, -\sin\theta\sin^2\phi, -\sin\phi\cos\phi)$$

(Note for $\phi = 0, \pi$, $\vec{T}_\theta \times \vec{T}_\phi = \vec{0}$)

$$\therefore \left|\frac{\partial(x,y)}{\partial(\theta,\phi)}\right|^2 = \sin^2\phi\cos^2\phi \quad \text{(from the } z\text{-coordinate)}$$

$$\left|\frac{\partial(x,z)}{\partial(\theta,\phi)}\right|^2 = \sin^2\theta\sin^4\phi \quad \text{(from the y-coordinate)}$$

$$\left|\frac{\partial(y,z)}{\partial(\theta,\phi)}\right|^2 = \cos^2\theta\sin^4\phi \quad \text{(from the x-coordinate)}$$

$$\therefore \|\vec{T}_\theta \times \vec{T}_\phi\|^2 = \sin^2\phi\cos^2\phi + (\sin^2\theta+\cos^2\theta)\sin^4\phi$$

$$= \sin^2\phi\,(\cos^2\phi+\sin^2\phi)$$

$$= \sin^2\phi$$

$$\therefore \|\vec{T}_\theta \times \vec{T}_\phi\| = \sin\phi, \text{ and } \sin\phi \geq 0 \text{ for } 0\leq\phi\leq\pi$$

$$\therefore A(S) = \iint_D \sin\phi\, d\theta d\phi = \int_0^{\pi}\int_0^{2\pi} \sin\phi\, d\theta d\phi$$

$$= 2\pi\int_0^{\pi}\sin\phi\, d\phi = 2\pi\left[-\cos\phi\right]\Big|_{\phi=0}^{\phi=\pi}$$

$$= 2\pi\left[1-(-1)\right] = \underline{4\pi}$$

2. **2.** In Exercise 1, what happens if we allow ϕ to vary from $-\pi/2$ to $\pi/2$? From 0 to 2π? Why do we obtain different answers?

(a) (1) D is still an elementary region
(2) ϕ is still C^1 but is not one-to-one for points in D and not on the boundary

[e.g., $\phi = \pm \frac{\pi}{6}$, $\cos\left(\frac{\pi}{6}\right) = \cos\left(-\frac{\pi}{6}\right)$ and

$\cos\theta \sin\left(\frac{\pi}{6}\right) = \cos(\pi+\theta)\sin\left(-\frac{\pi}{6}\right)$

$\sin\theta \sin\left(\frac{\pi}{6}\right) = \sin(\pi+\theta)\sin\left(-\frac{\pi}{6}\right)$

$\therefore$ an infinite # of non-boundary points for which $\phi(\theta,\phi)$ is not one-to-one.]

However, this can be addressed by breaking up $\phi(\theta,\phi)$ to be ϕ_1 for $-\pi/2 \le \phi \le 0$, and ϕ_2 for $0 \le \phi \le \pi/2$.

(3) The image is regular except at $\phi = 0$ (finite # points).

for $-\pi/2 \le \phi \le 0$, $\| \vec{T}_\theta \times \vec{T}_\phi \| = -\sin\phi$
since $\sin\phi \le 0$.

for $0 \le \phi \le \pi/2$, $\| \vec{T}_\theta \times \vec{T}_\phi \| = \sin\phi$

$$\therefore A(S) = \iint_D \| \vec{T}_\theta \times \vec{T}_\phi \| \, d\theta d\phi$$

$$= \int_0^{2\pi} \int_{-\pi/2}^{0} -\sin\phi \, d\theta d\phi + \int_0^{2\pi} \int_0^{\pi/2} \sin\phi \, d\theta d\phi$$

$$= -2\pi \int_{-\pi/2}^{0} \sin\phi \, d\phi + 2\pi \int_{0}^{\pi/2} \sin\phi \, d\phi$$

$$= -2\pi \left[-\cos\phi \right]_{-\pi/2}^{0} + 2\pi \left[-\cos\phi \right]_{0}^{\pi/2}$$

$$= -2\pi \left[-1 - (0) \right] + 2\pi \left[0 - (-1) \right]$$

$$= \underline{4\pi}$$

(5) $\phi(\theta, \phi)$ is not one-to-one for an infinite # of non-boundary points, as $0 \le \phi \le \pi$ covers the sphere once, and $\pi \le \phi \le 2\pi$ cover it again.

This can be addressed by letting

$$\phi(\theta,\phi) = \begin{cases} \phi_1(\theta,\phi) : 0 \le \phi \le \pi, \ 0 \le \theta \le 2\pi \\ \phi_2(\theta,\phi) : \pi \le \phi \le 2\pi, \ 0 \le \theta \le 2\pi \end{cases}$$

Note $\| \vec{T}_\theta \times \vec{T}_\phi \| = \sin\phi, \quad 0 \le \phi \le \pi$

$$= -\sin\phi, \quad \pi \le \phi \le 2\pi$$

$$\therefore A(\sigma) = \int_0^{\pi} \int_0^{2\pi} \sin\phi \, d\theta \, d\phi + \int_{\pi}^{2\pi} -\sin\phi \, d\theta \, d\phi$$

$$= 2\pi \int_0^{\pi} \sin\phi \, d\phi + 2\pi \int_{\pi}^{2\pi} \sin\phi \, d\phi$$

$$= 2\pi \left[-\cos\phi \right]_0^{\pi} - 2\pi \left[-\cos\phi \right]_{\pi}^{2\pi}$$

$$= 2\pi \left[1 - (-1) \right] - 2\pi \left[-1 - (1) \right]$$

$$= 4\pi + 4\pi = \underline{8\pi}$$

Different answers because (a) covers the sphere once, and (b) covers it twice.

3. **3.** Find the area of the helicoid in Example 2 if the domain D is $0 \le r \le 1$ and $0 \le \theta \le 3\pi$.

As in Example 2, $A(S) = \iint_D \| \vec{T}_r \times \vec{T}_\theta \| \, dr \, d\theta$

$$= \int_0^{3\pi} \int_0^1 \sqrt{r^2+1} \, dr \, d\theta = 3\pi \int_0^1 \sqrt{r^2+1} \, dr$$

From Table of Integrals #43 from text,

$$= 3\pi \left[\frac{r}{2}\sqrt{r^2+1} + \frac{1}{2} \log \left| r + \sqrt{r^2+1} \right| \right]_{r=0}^{r=1}$$

$$= 3\pi \left[\frac{1}{2}\sqrt{2} + \frac{1}{2} \log(1+\sqrt{2}) \right]$$

4.

4. The torus T can be represented parametrically by the function Φ: $D \to \mathbb{R}^3$, where Φ is given by the coordinate functions $x = (R + \cos\phi)\cos\theta$, $y = (R + \cos\phi)\sin\theta$, $z = \sin\phi$; D is the rectangle $[0, 2\pi] \times [0, 2\pi]$, that is, $0 \le \theta \le 2\pi$, $0 \le \phi \le 2\pi$; and $R > 1$ is fixed (see Figure 7.4.6). Show that $A(T) = (2\pi)^2 R$, first by using formula (3) and then by using formula (6).

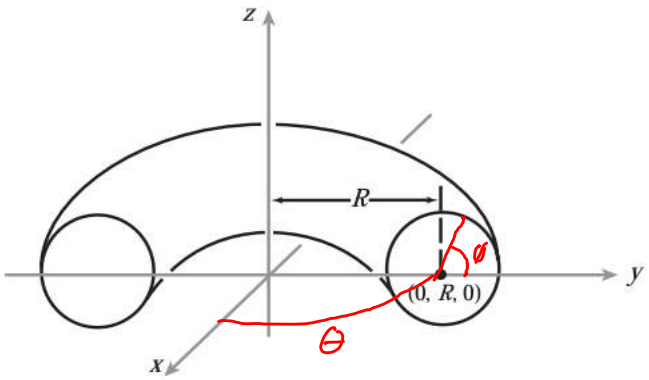


figure **7.4.6** A cross section of a torus.

Radius of cross sectional circle is 1, $\therefore R > 1$.

(a) Using $\iint_D \sqrt{\left(\frac{\partial(x,y)}{\partial(\theta,\phi)}\right)^2 + \left(\frac{\partial(y,z)}{\partial(\theta,\phi)}\right)^2 + \left(\frac{\partial(x,z)}{\partial(\theta,\phi)}\right)^2}$

$$\left(\frac{\partial(x,y)}{\partial(\theta,\phi)}\right)^2 = \begin{vmatrix} -(R+\cos\phi)\sin\theta & -\sin\phi\cos\theta \\ (R+\cos\phi)\cos\theta & -\sin\phi\sin\theta \end{vmatrix}^2$$

$$= \left[(R+\cos\phi)\sin\phi\sin^2\theta + (R+\cos\phi)\sin\phi\cos^2\theta\right]^2$$

$$= (R+\cos\phi)^2\sin^2\phi$$

$$\left(\frac{\partial(y,z)}{\partial(\theta,\phi)}\right)^2 = \begin{vmatrix} (R+\cos\phi)\cos\theta & -\sin\phi\sin\theta \\ 0 & \cos\phi \end{vmatrix}^2$$

$$= (R+\cos\phi)^2\cos^2\theta\cos^2\phi$$

$$\left(\frac{\partial(x,z)}{\partial(\theta,\phi)}\right)^2 = \begin{vmatrix} -(R+\cos\phi)\sin\theta & -\sin\phi\sin\theta \\ 0 & \cos\phi \end{vmatrix}^2$$

$$= (R+\cos\phi)^2\sin^2\theta\cos^2\phi$$

$$\therefore \frac{\partial(x,y)}{\partial(\theta,\phi)}^2 + \frac{\partial(y,z)}{\partial(\theta,\phi)}^2 + \frac{\partial(x,z)}{\partial(\theta,\phi)}^2 =$$

$$(R+\cos\phi)^2\sin^2\phi + (R+\cos\phi)^2\cos^2\theta\cos^2\phi + (R+\cos\phi)^2\sin^2\theta\cos^2\phi$$

$$= (R+\cos\phi)^2\sin^2\phi + (R+\cos\phi)^2\cos^2\phi$$

$$= (R+\cos\phi)^2$$

Since $R>1$, $R+\cos\phi > 0$, $\therefore \sqrt{(R+\cos\phi)^2} = R+\cos\phi$

$$\therefore A(T) = \int_0^{2\pi}\int_0^{2\pi} (R+\cos\phi)\, d\theta\, d\phi$$

$$= 2\pi \int_0^{2\pi} (R+\cos\phi)\, d\phi = 2\pi \left[R\phi + \sin\phi \right]_{\phi=0}^{\phi=2\pi}$$

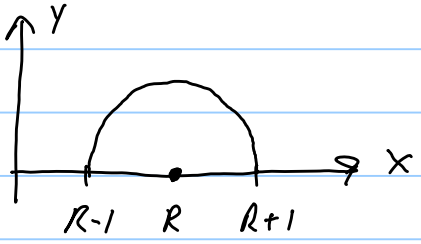
$$= \underline{4\pi^2 R}$$

(5) Using $A = 2\pi \int_a^b |x| \sqrt{1+f'(x)^2}\, dx$

Here, $y = f(x) = \sqrt{1-(x-R)^2}$, $R>1$

$a = R-1$, $b = R+1$

$\therefore x > 0$ for $x \in [R-1, R+1]$



This will give area of top half of torus.

$$f'(x) = \frac{1}{2}\left(1-(x-R)^2\right)^{-\frac{1}{2}}\left(-2(x-R)\right) = -\frac{x-R}{\sqrt{1-(x-R)^2}}$$

$$\therefore 1+f'(x)^2 = 1+\frac{(x-R)^2}{1-(x-R)^2} = \frac{1}{1-(x-R)^2}$$

$$\therefore A(T) = 4\pi\int_{R-1}^{R+1}\frac{x}{\sqrt{1-(x-R)^2}}\,dx$$

Let $u = x-R$, $\therefore du = dx$, $x = u+R$

$x = R-1 \Rightarrow u = -1$, $x = R+1 \Rightarrow u = 1$

$$= 4\pi\int_{-1}^{1}\frac{u+R}{\sqrt{1-u^2}}\,du = 4\pi\int_{-1}^{1}\frac{u\,du}{\sqrt{1-u^2}} + 4\pi R\int_{-1}^{1}\frac{du}{\sqrt{1-u^2}}$$

$$= 4\pi\left[-\sqrt{1-u^2}\right]_{-1}^{1} + 4\pi R\left[\operatorname{Arcsin} u\right]_{-1}^{1}$$

$$= 0 + 4\pi R\left[\frac{\pi}{2}-\left(-\frac{\pi}{2}\right)\right]$$

$$= \underline{4\pi^2 R}$$

5.

5. Let $\Phi(u, v) = (e^u \cos v,\ e^u \sin v,\ v)$ be a mapping from $D = [0, 1] \times [0, \pi]$ in the uv plane onto a surface S in xyz space.

(a) Find $T_u \times T_v$.

(b) Find the equation for the tangent plane to S when $(u, v) = (0, \frac{\pi}{2})$.

(c) Find the area of $\Phi(D)$.

(a) $\vec{T}_u = (e^u \cos v,\ e^u \sin v,\ 0)$

$$\vec{T}_v = (-e^u \sin v,\ e^u \cos v,\ 1)$$

$$\vec{T}_u \times \vec{T}_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ e^u \cos v & e^u \sin v & 0 \\ -e^u \sin v & e^u \cos v & 1 \end{vmatrix}$$

$$= (e^u \sin v,\ -e^u \cos v,\ e^{2u})$$

(6) (b) Find the equation for the tangent plane to S when $(u, v) = (0, \frac{\pi}{2})$.

$$\phi(0, \pi/2) = (0, 1, \pi/2),\quad \vec{T}_u(0, \pi/2) \times \vec{T}_v(0, \pi/2) = (1, 0, 1)$$

$$\therefore \vec{T}_u \times \vec{T}_v \cdot (x - x_0, y - y_0, z - z_0) = 0 \Rightarrow$$

$$(1, 0, 1) \cdot (x - 0, y - 1, z - \pi/2) = 0 \Rightarrow$$

$$x + z - \frac{\pi}{2} = 0$$

(C) (c) Find the area of $\Phi(D)$.

Note $\phi(u, v)$ is one-to-one and regular on $[0,1] \times [0, \pi]$

$$\| \vec{T}_u \times \vec{T}_v \| = \sqrt{e^{2u} \sin^2 v + e^{2u} \cos^2 v + e^{4u}} = \sqrt{e^{2u} + e^{4u}}$$

$$\therefore A(\phi(D)) = \int_0^1 \int_0^{\pi} \sqrt{e^{2u} + e^{4u}}\, dv\, du$$

$$= \pi \int_0^1 \sqrt{e^{2u} + e^{4u}}\, du$$

From Wolfram: $\int_0^1 \sqrt{\exp(2x)+\exp(4x)}\,dx = \frac{1}{2}\left(-\sqrt{2}+e\sqrt{1+e^2}-\sinh^{-1}(1)+\sinh^{-1}(e)\right) \approx 3.6515$

$\therefore A(\phi(D)) \approx \pi(3.6515)$

Note: Answer in back of text says

$\|T_u \times T_v\| = (e^u \sin v, -e^u \cos v, e^u)$

This makes $A(\phi(D)) = \int_0^1 \int_0^{\pi} \sqrt{e^{2u}+e^{2u}} = \sqrt{2}\pi(e-1)$

But the z-component is e^{2u}, not e^u

6. **6.** Find the area of the surface defined by $z = xy$ and $x^2 + y^2 \le 2$.

Use the parametrization $x = r\cos\theta$, $y = r\sin\theta$, $z = r^2\cos\theta\sin\theta = \frac{1}{2}r^2\sin 2\theta$

$0 \le r \le \sqrt{2}$, $0 \le \theta \le 2\pi$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r$$

$$\frac{\partial(x,z)}{\partial(r,\theta)} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ r\sin 2\theta & r^2\cos 2\theta \end{vmatrix} = r^2(\cos\theta\cos 2\theta + \sin 2\theta\sin\theta)$$
$$= r^2(\cos^3\theta - \cos\theta\sin^2\theta + 2\sin^2\theta\cos\theta)$$
$$= r^2(\cos^3\theta + \sin^2\theta\cos\theta)$$
$$= r^2\cos\theta$$

$$\frac{\partial(y,z)}{\partial(r,\theta)} = \begin{vmatrix} \sin\theta & r\cos\theta \\ r\sin 2\theta & r^2\cos 2\theta \end{vmatrix} = r^2(\sin\theta\cos 2\theta - \cos\theta\sin 2\theta)$$

$$= r^2(\sin\theta\cos^2\theta - \sin^3\theta - 2\sin\theta\cos^2\theta)$$

$$= -r^2(\sin\theta\cos^2\theta + \sin^3\theta)$$

$$= -r^2\sin\theta$$

$$\therefore A(S) = \int_0^{2\pi}\int_0^{\sqrt{2}} \sqrt{r^2 + r^4\cos^2\theta + r^4\sin^2\theta}\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\sqrt{2}} \sqrt{r^2 + r^4}\, dr\, d\theta = 2\pi\int_0^{\sqrt{2}} r\sqrt{1+r^2}\, dr$$

$$= 2\pi\left[\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)(1+r^2)^{3/2}\right]_{r=0}^{r=\sqrt{2}}$$

$$= \frac{2\pi}{3}\left[3^{3/2} - 1\right] = \frac{2\pi}{3}\left[3\sqrt{3} - 1\right]$$

7.

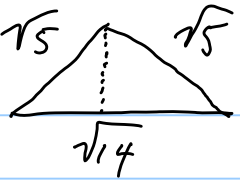
7. Use a surface integral to find the area of the triangle T in $\mathbb{R}^3$ with vertices at $(1, 1, 0)$, $(2, 1, 2)$, and $(2, 3, 3)$. Verify your answer by finding the lengths of the sides and using classical geometry.
[HINT: Write the triangle as the graph $z = g(x, y)$ over a triangle T^* in the xy plane.]

(a) Classical geometry

Sides: $\sqrt{(2-1)^2 + (1-1)^2 + (2-0)^2} = \sqrt{5}$

$\sqrt{(2-2)^2 + (3-1)^2 + (3-2)^2} = \sqrt{5}$

$\sqrt{(2-1)^2+(3-1)^2+(3-0)^2} = \sqrt{14}$ $\therefore$



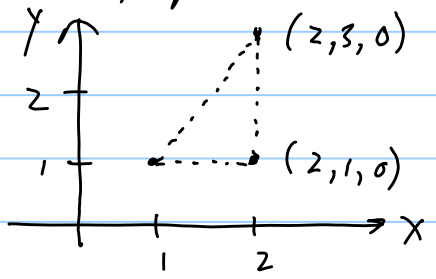
Isosceles, $\therefore$ height $= \sqrt{(\sqrt{5})^2 - \left(\frac{\sqrt{14}}{2}\right)^2} = \sqrt{5 - 7/2} = \sqrt{3/2}$

$\therefore \frac{1}{2}\sqrt{14}\left(\sqrt{3/2}\right) = \frac{1}{2}\sqrt{42/2} = \underline{\frac{1}{2}\sqrt{21}}$

(6) The triangle projects onto the xy-plane as

$(1,1,0)$, $(2,1,0)$, $(2,3,0)$

a right triangle



$\therefore$ Can use orthogonal (independent) parameters

(u,v), $1 \le u \le 2$, $1 \le v \le 2u-1$

as $y = 2x-1$ is the line connecting $(1,1)$ to $(2,3)$.

The surface of the triangle in R^3 is a plane connecting the 3 points. $\therefore z = g(x,y)$ will be equation of this plane.

Normal to plane: $[(2,1,2)-(1,1,0)] \times [(2,3,3)-(1,1,0)]$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 2 \\ 1 & 2 & 3 \end{vmatrix} = (-4,-1,2)$$

$\therefore$ plane is $(4,1,2) \cdot (x-1, y-1, z-0) = 0$, or

$$4x + y + 2z - 5 = 0, \text{ or } z = -2x - \tfrac{1}{2}y + \tfrac{5}{2}$$

$$\therefore g(x,y) = -2x - \tfrac{1}{2}y + \tfrac{5}{2}$$

$\therefore$ Parametrization of plane is:

$$\phi(u,v) = \left(u, v, -2u - \tfrac{v}{2} + \tfrac{5}{2}\right).$$

$$\phi_u = (1, 0, -2) \qquad \phi_v = (0, 1, -\tfrac{1}{2})$$

$$\therefore \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -2 \\ 0 & 1 & -\frac{1}{2} \end{vmatrix} = \left(2, \tfrac{1}{2}, 1\right)$$

$$\therefore \|\phi_u \times \phi_v\| = \sqrt{2^2 + \left(\tfrac{1}{2}\right)^2 + 1^2} = \sqrt{21/4} = \frac{\sqrt{21}}{2}$$

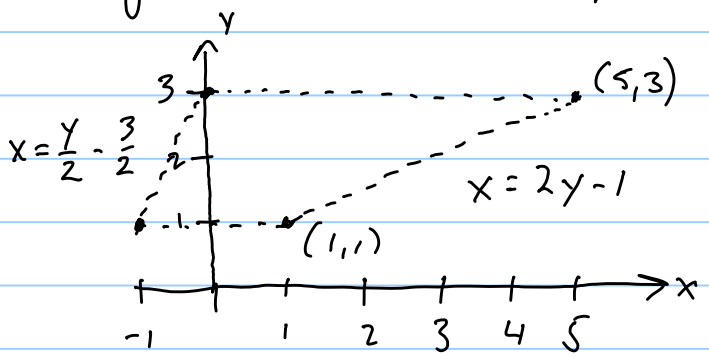
$$\therefore A(S) = \iint_D \frac{\sqrt{21}}{2}\, du\, dv = \int_1^2 \int_1^{2u-1} \frac{\sqrt{21}}{2}\, dv\, du$$

$$= \frac{\sqrt{21}}{2} \int_1^2 v \Big|_1^{2u-1} du = \frac{\sqrt{21}}{2} \int_1^2 2u - 2\, du$$

$$= \sqrt{21} \left[\tfrac{1}{2}(u-1)^2\right]\Big|_{u=1}^{u=2} = \frac{\sqrt{21}}{2}$$

8. **8.** Use a surface integral to find the area of the quadrilateral D in $\mathbb{R}^3$ with vertices at $(-1, 1, 2)$, $(1, 1, 2)$, $(0, 3, 5)$, and $(5, 3, 5)$. Verify your answer by finding the lengths of the sides and using classical geometry. [HINT: See the hint in the previous problem.]

(a) Projection onto xy-plane is: $(0,3,0)$ $(5,3,0)$ $(-1,1,0)$ $(1,1,0)$



Let $y = v$, $1 \le v \le 3$

$x = u$, $\frac{v}{2} - \frac{3}{2} \le u \le 2v - 1$

Making x a function of y due to shape

Need expression of plane containing the quadrilateral.

Normal to plane: $[(1,1,2)-(-1,1,2)] \times [(0,3,5)-(-1,1,2)]$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 0 \\ 1 & 2 & 3 \end{vmatrix} = (0,-6,4), \text{ or } (0,-3,2)$$

$\therefore (0,-3,2) \cdot (x-(-1), y-1, z-2) = 0$, or

$-3y + 2z - 1 = 0$, or $z = \frac{3}{2}y + \frac{1}{2}$ (parallel to x-axis)

Parametrization of surface plane $z = g(x,y) = \frac{3}{2}y + \frac{1}{2}$

is $\phi(u,v) = (u, v, \frac{3}{2}v + \frac{1}{2})$ for

$$\frac{v}{2} - \frac{3}{2} \le u \le 2v-1, \quad 1 \le v \le 3$$

$$\phi_u = (1,0,0) \qquad \phi_v = (0,1,\tfrac{3}{2})$$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 0 & 1 & 3/2 \end{vmatrix} = (0, -\tfrac{3}{2}, 1)$$

$$\therefore \|\phi_u \times \phi_v\| = \sqrt{\tfrac{9}{4}+1} = \frac{\sqrt{13}}{2}$$

$$\therefore A(S) = \iint_D \|\phi_u \times \phi_v\| \, du\, dv = \int_1^3 \int_{\frac{v}{2}-\frac{3}{2}}^{2v-1} \frac{\sqrt{13}}{2} \, du\, dv$$

$$= \frac{\sqrt{13}}{2} \int_1^3 (2v-1) - \left(\tfrac{v}{2} - \tfrac{3}{2}\right) du = \frac{\sqrt{13}}{2} \int_1^3 \tfrac{3}{2}v + \tfrac{1}{2} \, dv$$

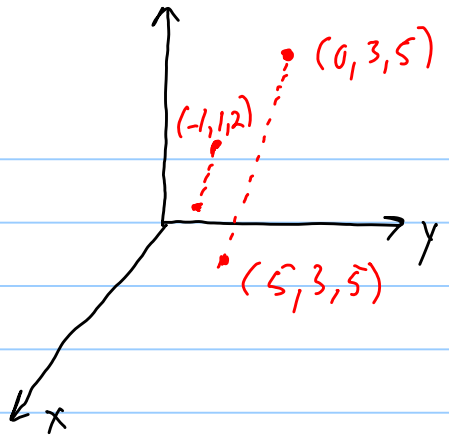
$$= \frac{\sqrt{13}}{2} \left[\tfrac{3}{4}v^2 + \tfrac{v}{2} \right]_{v=1}^{v=3}$$

$$= \frac{\sqrt{13}}{2} \left[\tfrac{27}{4} + \tfrac{3}{2} - \left(\tfrac{3}{4} + \tfrac{1}{2}\right) \right] = \frac{\sqrt{13}}{2} \left[6 + 1 \right]$$

$$= \underline{\underline{\tfrac{7}{2}\sqrt{13}}}$$

(5) Classical geometry

Side lengths: $(-1,1,2)$ to $(1,1,2)$: $\sqrt{2^2+0^2+0^2} = 2$



$(1,1,2)$ to $(0,3,5)$: $\sqrt{1^2+2^2+3^2} = \sqrt{14}$

$(0,3,5)$ to $(5,3,5)$: $\sqrt{5^2+0^2+0^2} = 5$

$(5,3,5)$ to $(-1,1,2)$: $\sqrt{6^2+2^2+3^2} = 7$

Sides of length 2 & 5 are parallel:

vector $(-1,1,2)$ to $(1,1,2)$ = $(2,0,0)$

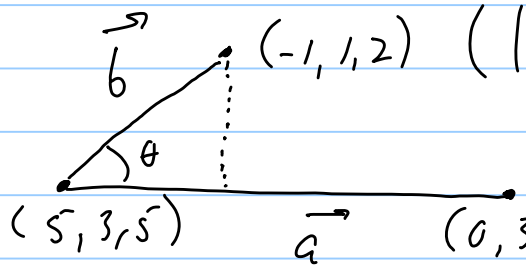
vector $(0,3,5)$ to $(5,3,5)$ = $(5,0,0)$

∴ One is a multiple of the other.

Average of lengths = $\frac{1}{2}(2+5) = \underline{\frac{7}{2}}$

Height = distance from one side to other

Find projection of one point to other line:



$(|\vec{b}|\cos\theta)\frac{\vec{a}}{|\vec{a}|} = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}|}\left(\frac{\vec{a}}{|\vec{a}|}\right)$

$= \frac{\vec{a}\cdot\vec{b}}{\vec{a}\cdot\vec{a}}\vec{a} = \vec{p}$

$\vec{b} = (-1,1,2) - (5,3,5) = (-6,-2,-3)$ $\quad \vec{a} = (-5,0,0)$

$\therefore \frac{(-5,0,0)\cdot(-6,-2,-3)}{(-5,0,0)\cdot(-5,0,0)}(-5,0,0) = \frac{30}{25}(-5,0,0) =$

$(-6, 0, 0) = \vec{p}$.

$\therefore \vec{b} - \vec{p}$ = perpendicular vector from point to line

$= (-6, -2, -3) - (-6, 0, 0) = (0, -2, -3)$

$\therefore$ height $= \|\vec{b} - \vec{p}\| = \sqrt{2^2 + 3^2} = \sqrt{13}$

$\therefore$ Area = (average of parallel bases)(height)

$= \frac{7}{2}\sqrt{13}$

7.

9. Let $\Phi(u, v) = (u - v, u + v, uv)$ and let D be the unit disc in the uv plane. Find the area of $\Phi(D)$.

$D = \{(u,v): -1 \le u \le 1, -\sqrt{1-u^2} \le v \le \sqrt{1-u^2}\}$

$\phi_u = (1, 1, v) \quad \phi_v = (-1, 1, u)$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & v \\ -1 & 1 & u \end{vmatrix} = (u - v, -u - v, 2)$$

$\|\phi_u \times \phi_v\| = \sqrt{(u-v)^2 + (-u-v)^2 + 2^2} = \sqrt{2u^2 + 2v^2 + 4}$

$\therefore$

$$A(S) = \iint_D \sqrt{2u^2 + 2v^2 + 4}\, du\, dv$$

$$= \int_{-1}^{1} \int_{-\sqrt{1-u^2}}^{\sqrt{1-u^2}} \sqrt{2u^2+2v^2+4}\, dv\, du$$

As this is messy, let $u = r\cos\theta$, $v = r\sin\theta$,
$0 \le \theta \le 2\pi$, $0 \le r \le 1$.

Jacobian of $T(r,\theta) = (r\cos\theta, r\sin\theta)$ is $(r\,dr\,d\theta)$,

and $f(u,v) = \sqrt{2u^2+2v^2+4}$ becomes

$$f(r\cos\theta, r\sin\theta) = \sqrt{2r^2\cos^2\theta + 2r^2\sin^2\theta + 4}$$

$$= \sqrt{2r^2+4}$$

$$\therefore A(S) = \int_0^{2\pi} \int_0^1 \sqrt{2r^2+4}\; r\, dr\, d\theta$$

$$= \sqrt{2}(2\pi) \int_0^1 r\sqrt{r^2+2}\, dr$$

$$= 2\sqrt{2}\,\pi \left[\left(\tfrac{2}{3}\right)\left(\tfrac{1}{2}\right)\left(r^2+2\right)^{3/2} \right]_{r=0}^{r=1}$$

$$= \tfrac{2}{3}\sqrt{2}\,\pi \left[3\sqrt{3} - 2\sqrt{2} \right]$$

$$= 2\sqrt{6}\,\pi - \tfrac{8}{3}\pi$$

10. **10.** Find the area of the portion of the unit sphere that is cut out by the cone $z \geq \sqrt{x^2 + y^2}$ (see Exercise 1).

Using spherical coordinates $\vec{r}(\theta,\phi) = \begin{bmatrix} \cos\theta \sin\phi \\ \sin\theta \sin\phi \\ \cos\phi \end{bmatrix}$

$0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq \frac{\pi}{4}$

as $\phi = \frac{\pi}{4}$ represents $z = \sqrt{x^2+y^2}$

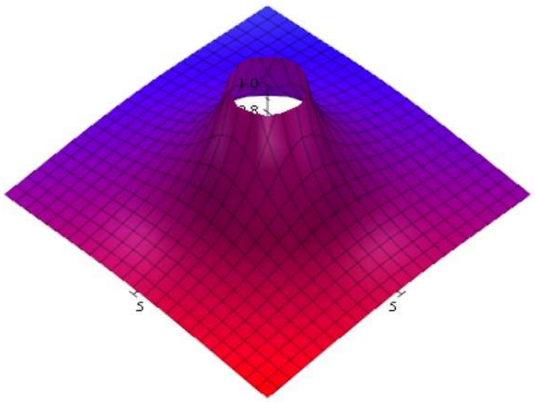
From problem #1, $\| \vec{r}_\theta \times \vec{r}_\phi \| = \sin\phi$

$$\therefore A(S) = \int_0^{\frac{\pi}{4}} \int_0^{2\pi} \sin\phi \, d\theta \, d\phi$$

$$= 2\pi \int_0^{\pi/4} \sin\phi \, d\phi = 2\pi \left[-\cos\phi \right]_0^{\pi/4}$$

$$= 2\pi \left[-\frac{\sqrt{2}}{2} - (-1) \right] = \pi (2 - \sqrt{2})$$

11. **11.** Show that the surface $x = 1/\sqrt{y^2 + z^2}$, where $1 \leq x < \infty$, can be filled but not painted!



Basically, a horn going to ∞.
(x-axis is pointing up)

Need to show volume (surface of revolution) is finite, but surface area is infinite.

The surface of revolution is obtained by letting $y = f(x) = \frac{1}{x}$. $y^2 + z^2$ is a unit circle, and the rotation of $f(x)$ about the x-axis yields the surface $x = 1/\sqrt{y^2 + z^2}$

(a) $$A(S) = 2\pi \int_1^\infty |f(x)| \sqrt{1 + f'(x)^2}\, dx$$

$$= 2\pi \int_1^\infty \frac{1}{x} \sqrt{1 + \left(-\frac{1}{x^2}\right)^2}\, dx = 2\pi \int_1^\infty \sqrt{\frac{x^4+1}{x^6}}\, dx$$

$$= 2\pi \lim_{b\to\infty} \int_1^b \sqrt{\frac{x^4+1}{x^6}}\, dx$$

Note since $x > 0$, $\sqrt{\frac{x^4+1}{x^6}} > \sqrt{\frac{x^4}{x^6}} = \frac{1}{x}$

$$\therefore 2\pi \lim_{b\to\infty} \int_1^b \sqrt{\frac{x^4+1}{x^6}}\, dx \geq 2\pi \lim_{b\to\infty} \int_1^b \frac{1}{x}\, dx$$

$$= 2\pi \lim_{b\to\infty} \ln(b) = \underline{\infty}$$

$\therefore A(6) \geq 2\pi \lim_{6\to\infty} \ln(6) = \infty$, so

$A(s)$ doesn't exist

(b) Volume of revolution of $y = \frac{1}{x}$ is:

$$V = \pi \int_1^\infty f(x)^2\,dx = \pi \int_1^\infty \frac{dx}{x^2}$$

$$= \pi \lim_{b\to\infty} \int_1^b \frac{dx}{x^2} = \pi \lim_{b\to\infty} \left\{-\frac{1}{b} + 1\right\} = \pi$$

$\therefore$ Volume exists

12. **12.** Find a parametrization of the surface $x^2 - y^2 = 1$, where $x > 0$, $-1 \leq y \leq 1$ and $0 \leq z \leq 1$. Use your answer to express the area of the surface as an integral.

(a) From $\cosh^2 u - \sinh^2 u = 1$, let $x = \cosh(u)$, $y = \sinh(u)$

$\therefore \phi(u, v) = (\cosh(u), \sinh(u), v)$, $0 \leq v \leq 1$.

For $-1 \leq y \leq 1$, $-1 \leq \sinh(u) \leq 1$, $-1 \leq \frac{e^u - e^{-u}}{2} \leq 1$,

$-2 \leq e^u - e^{-u} \leq 2$. Let $x = e^u \therefore e^{-u} = \frac{1}{x}$

For $e^u - e^{-u} = 2$, $x = 2 + \frac{1}{x}$, $x^2 - 2x + 1$,

$x = 1 \pm \sqrt{2}$, so $x = 1 + \sqrt{2}$ as $e^u > 0$.

$\therefore e^u = 1+\sqrt{2}, \quad u = \log(1+\sqrt{2})$

For $-2 = e^u - e^{-u}$, $\frac{1}{x} - 2 = x$, $x^2 + 2x - 1$, $x = -1 \pm \sqrt{2}$,

$\therefore e^u = -1+\sqrt{2}, \quad u = \log(-1+\sqrt{2})$.

$\therefore \log(-1+\sqrt{2}) \le u \le \log(1+\sqrt{2})$

$\therefore \phi(u,v) = (\cosh(u), \sinh(u), v)$, for

$\log(-1+\sqrt{2}) \le u \le \log(1+\sqrt{2}), \quad 0 \le v < 1$

(6) $\phi_u = (\sinh(u), \cosh(u), 0) \quad \phi_v = (0,0,1)$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \sinh(u) & \cosh(u) & 0 \\ 0 & 0 & 1 \end{vmatrix} = (\cosh(u), -\sinh(u), 0)$$

$\therefore \|\phi_u \times \phi_v\| = \sqrt{\cosh^2 u + \sinh^2 u}$

$\therefore$

$$A(S) = \int_0^1 \int_{\log(-1+\sqrt{2})}^{\log(1+\sqrt{2})} \sqrt{\cosh^2 u + \sinh^2 u}\, du\, dv$$

$$= \int_{\log(-1+\sqrt{2})}^{\log(1+\sqrt{2})} \sqrt{\cosh^2 u + \sinh^2 u}\, du$$

13. **13.** Represent the ellipsoid E:

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$

parametrically and write out the integral for its surface area $A(E)$. (Do not evaluate the integral.)

Use spherical coordinates

$T(\theta,\phi) = (a\cos\theta\sin\phi,\ b\sin\theta\sin\phi,\ c\cos\phi)$,

for $0 \le \phi \le \pi$, $0 \le \theta \le 2\pi$

Check: $\frac{(a\cos\theta\sin\phi)^2}{a^2} + \frac{(b\sin\theta\sin\phi)^2}{b^2} + \frac{(c\cos\phi)^2}{c^2}$

$= (\cos^2\theta + \sin^2\theta)\sin^2\phi + \cos^2\phi = 1$

$$\frac{\partial(x,y)}{\partial(\theta,\phi)} = \begin{vmatrix} -a\sin\theta\sin\phi & a\cos\theta\cos\phi \\ b\cos\theta\sin\phi & b\sin\theta\cos\phi \end{vmatrix}$$

$= -ab\sin^2\theta\sin\phi\cos\phi - ab\cos^2\theta\sin\phi\cos\phi$

$= -ab\sin\phi\cos\phi$

$$\frac{\partial(x,z)}{\partial(\theta,\phi)} = \begin{vmatrix} -a\sin\theta\sin\phi & a\cos\theta\cos\phi \\ 0 & -c\sin\phi \end{vmatrix}$$

$= ac\sin\theta\sin^2\phi$

$$\frac{\partial(y,z)}{\partial(\theta,\phi)} = \begin{vmatrix} b\cos\theta\sin\phi & b\sin\theta\cos\phi \\ 0 & -c\sin\phi \end{vmatrix}$$

$$= -bc\cos\theta\sin^2\phi$$

$$\therefore A(S) = \iint_D \sqrt{a^2b^2\sin^2\phi\cos^2\phi + a^2c^2\sin^2\theta\sin^4\phi + b^2c^2\cos^2\theta\sin^4\phi}$$

$$= \int_0^{\pi}\int_0^{2\pi} \sqrt{a^2b^2\sin^2\phi\cos^2\phi + a^2c^2\sin^2\theta\sin^4\phi + b^2c^2\cos^2\theta\sin^4\phi}\, d\theta\, d\phi$$

14.

14. Let the curve $y = f(x)$, $a \le x \le b$, be rotated about the y axis. Show that the area of the surface swept out is given by equation (6); that is,

$$A = 2\pi \int_a^b |x|\sqrt{1 + [f'(x)]^2}\, dx.$$

Interpret the formula geometrically using arc length and slant height.

Assuming $f(x)$ is C^1, for a given $\Delta x = x_{i+1} - x_i$,

by the mean value theorem, $f(x_{i+1}) - f(x_i) = f'(x^*)\cdot\Delta x$,

$x_i \le x_i^* \le x_{i+1}$, or $\Delta y = f'(x_i^*)\Delta x$

$\therefore$ The length of the curve from $f(x_i)$ to $f(x_{i+1})$

is about $\sqrt{\Delta y^2 + \Delta x^2}$ by Pythagorean theorem.

$$\sqrt{\Delta y^2 + \Delta x^2} = \sqrt{f'(x_i^*)^2\Delta x^2 + \Delta x^2} = \left(\sqrt{f'(x_i^*)^2 + 1}\right)\Delta x$$

If revolved around the y-axis, the area of

this rim is $2\pi r\ell = 2\pi |x_i^*| \sqrt{1+f'(x_i^*)^2}\,\Delta x$

Taking a Riemann sum of all of these rims,

$$\sum_{i=0}^{n-1} 2\pi |x_i^*| \sqrt{1+f'(x_i^*)^2}\,\Delta x =$$

$$2\pi \int_a^b |x| \sqrt{1+f'(x)^2}\,dx$$

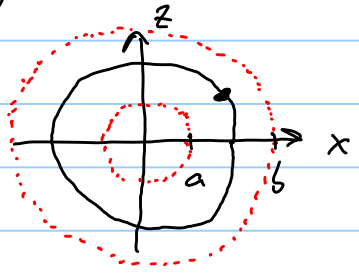
Interpreting using arc length, $ds = \sqrt{\Delta x^2 + \Delta y^2}$ above, so $ds = \sqrt{1+f'(x)^2}\,dx$. Each rim has surface area of $2\pi|x|\,ds$, where $|x|$ = radius of rim, and $2\pi|x|$ is the circumference.

$$\therefore \quad 2\pi \int_a^b |x| \sqrt{1+f'(x)^2}\,dx = 2\pi \int_c |x|\,ds$$

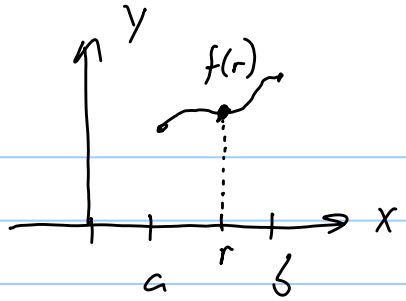
For a parametrization of the surface, rotating about y-axis creates circles parallel to xz-plane.

$\therefore$ Let $x = r\cos\theta \quad z = r\sin\theta$

$a \le r \le b, \quad 0 \le \theta \le 2\pi$



Here, $y = f(r)$



$$\therefore \phi(r,\theta) = (r\cos\theta,\ f(r),\ r\sin\theta)$$

$$\phi_r = (\cos\theta,\ f'(r),\ \sin\theta)$$

$$\phi_\theta = (-r\sin\theta,\ 0,\ r\cos\theta)$$

$$\phi_r \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & f'(r) & \sin\theta \\ -r\sin\theta & 0 & r\cos\theta \end{vmatrix}$$

$$= (r\cos\theta\, f'(r),\ -r\sin^2\theta - r\cos^2\theta,\ r\sin\theta\, f'(r))$$

$$= (r\cos\theta\, f'(r),\ -r,\ r\sin\theta\, f'(r))$$

$$\therefore \|\phi_r \times \phi_\theta\| = \sqrt{r^2\cos^2\theta\, f'(r)^2 + r^2 + r^2\sin^2\theta\, f'(r)^2}$$

$$= \sqrt{r^2 + r^2 f'(r)^2} = |r|\sqrt{1 + f'(r)^2}$$

$$\therefore A(s) = \iint_D \|\phi_r \times \phi_\theta\|\, dr\, d\theta = \int_0^{2\pi}\int_a^b |r|\sqrt{1 + f'(r)^2}\, dr\, d\theta$$

$$= 2\pi \int_a^b |r|\sqrt{1 + f'(r)^2}\, dr$$

15. **15.** Find the area of the surface obtained by rotating the curve $y = x^2$, $0 \le x \le 1$, about the y axis.

$$A(S) = 2\pi \int_0^1 x\sqrt{1+(2x)^2}\,dx \text{, as } f'(x) = 2x,\ |x| = x$$

$$= \frac{2\pi}{8}\int_0^1 8x\sqrt{1+4x^2}\,dx = \frac{\pi}{4}\left[\frac{2}{3}(1+4x^2)^{\frac{3}{2}}\right]_{x=0}^{x=1}$$

$$= \frac{\pi}{6}\left\{5^{3/2} - 1\right\} = \frac{\pi}{6}(5\sqrt{5} - 1)$$

16. **16.** Use formula (4) to compute the surface area of the cone in Example 1.

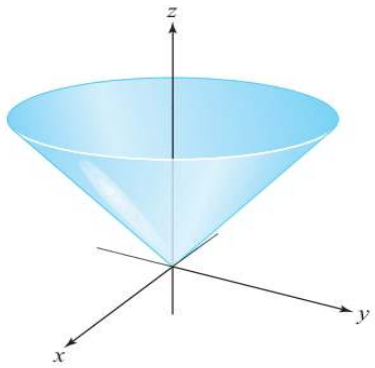


figure **7.3.8** The surface $z = \sqrt{x^2 + y^2}$ is a cone. It is not regular at its tip.

$$\phi(x,y) = (x, y, \sqrt{x^2+y^2})$$

$$\frac{\partial z}{\partial x} = x(x^2+y^2)^{-\frac{1}{2}}$$

$$\frac{\partial z}{\partial y} = y(x^2+y^2)^{-\frac{1}{2}}$$

Note $-1 \le x \le 1$, $-\sqrt{1-x^2} \le y \le \sqrt{1-x^2}$ from Example 1, p. 385 of text

$$A(S) = \iint_D \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1}\,dA$$

$$= \int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \sqrt{\frac{x^2}{x^2+y^2} + \frac{y^2}{x^2+y^2} + 1}\,dy\,dx$$

$$= \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \sqrt{2}\, dy\, dx = \int_{-1}^{1} 2\sqrt{2}\sqrt{1-x^2}\, dx$$

$$= 2\sqrt{2}\left[\frac{x}{2}\sqrt{1-x^2} + \frac{1}{2}\arcsin x \right]_{x=-1}^{x=1}$$

$$= 2\sqrt{2}\left[\frac{1}{2}\arcsin(1) - \frac{1}{2}\arcsin(-1) \right]$$

$$= 2\sqrt{2}\left[\frac{\pi}{4} + \frac{\pi}{4} \right] = \underline{\sqrt{2}\pi}$$

17. **17.** Find the area of the surface defined by $x + y + z = 1$, $x^2 + 2y^2 \le 1$.

An ellipse in xy-plane : $x^2 + \frac{y^2}{1/2} \le 1$

$\therefore$ Let $x = r\cos\theta$, $y = \frac{r\sin\theta}{\sqrt{2}}$, $0 \le r \le 1$, $0 \le \theta \le 2\pi$

$\therefore r^2\cos^2\theta + 2\left(\frac{r^2\sin^2\theta}{2}\right) = r^2$

$z = 1 - x - y$.

$\therefore \phi(r, \theta) = \left(r\cos\theta, \frac{r\sin\theta}{\sqrt{2}}, 1 - r\cos\theta - \frac{r\sin\theta}{\sqrt{2}} \right)$

$\phi_r = \left(\cos\theta, \frac{\sin\theta}{\sqrt{2}}, -\cos\theta - \frac{\sin\theta}{\sqrt{2}} \right)$

$\phi_\theta = \left(-r\sin\theta, \frac{r\cos\theta}{\sqrt{2}}, r\sin\theta - \frac{r\cos\theta}{\sqrt{2}} \right)$

$$\phi_r \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \frac{\sin\theta}{\sqrt{2}} & -r\cos\theta - \frac{\sin\theta}{\sqrt{2}} \\ -r\sin\theta & \frac{r\cos\theta}{\sqrt{2}} & r\sin\theta - \frac{r\cos\theta}{\sqrt{2}} \end{vmatrix}$$

$$= \left(\frac{r\sin^2\theta}{\sqrt{2}} - \frac{r\sin\theta\cos\theta}{2} + \frac{r\cos^2\theta}{\sqrt{2}} + \frac{r\cos\theta\sin\theta}{2}, \right.$$

$$r\cos\theta\sin\theta + \frac{r\sin^2\theta}{\sqrt{2}} - r\cos\theta\sin\theta + \frac{r\cos^2\theta}{\sqrt{2}},$$

$$\left. \frac{r\cos^2\theta}{\sqrt{2}} + \frac{r\sin^2\theta}{\sqrt{2}} \right)$$

$$= \left(\frac{r}{\sqrt{2}}, \frac{r}{\sqrt{2}}, \frac{r}{\sqrt{2}} \right)$$

$$\therefore \| \phi_r \times \phi_\theta \| = \sqrt{\frac{r^2}{2} + \frac{r^2}{2} + \frac{r^2}{2}} = r\sqrt{\frac{3}{2}}$$

$$\therefore A(S) = \int_0^{2\pi} \int_0^1 \sqrt{\frac{3}{2}}\, r \, dr\, d\theta = \sqrt{\frac{3}{2}}\, 2\pi \int_0^1 r\, dr$$

$$= \sqrt{\frac{3}{2}}\,(2\pi)\left[\frac{r^2}{2}\right]_{r=0}^{r=1} = \pi\sqrt{\frac{3}{2}} = \frac{\pi}{2}\sqrt{6}$$

18. **18.** Show that for the vectors $\mathbf{T}_u$ and $\mathbf{T}_v$, we have the formula

$$\|\mathbf{T}_u \times \mathbf{T}_v\| = \sqrt{\left[\frac{\partial(x, y)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(y, z)}{\partial(u, v)}\right]^2 + \left[\frac{\partial(x, z)}{\partial(u, v)}\right]^2}.$$

$$T(u,v) = (x(u,v), y(u,v), z(u,v))$$

$$\vec{T}_u = (x_u, y_u, z_u), \quad \vec{T}_v = (x_v, y_v, z_v).$$

$$\vec{T}_u \times \vec{T}_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_u & y_u & z_u \\ x_v & y_v & z_v \end{vmatrix}$$

$$= (y_u z_v - y_v z_u, \; x_v z_u - x_u z_v, \; x_u y_v - x_v y_u)$$

$$= \left(\begin{vmatrix} y_u & y_v \\ z_u & z_v \end{vmatrix}, - \begin{vmatrix} x_u & x_v \\ z_u & z_v \end{vmatrix}, \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} \right)$$

$$= \left(\frac{\partial(y,z)}{\partial(u,v)}, - \frac{\partial(x,z)}{\partial(u,v)}, \frac{\partial(x,y)}{\partial(u,v)} \right)$$

$$\therefore \left\| \vec{T}_u \times \vec{T}_v \right\| = \sqrt{\frac{\partial(x,y)^2}{\partial(u,v)^2} + \frac{\partial(x,z)^2}{\partial(u,v)} + \frac{\partial(y,z)^2}{\partial(u,v)}}$$

19.

19. Compute the area of the surface given by

$$x = r\cos\theta, \quad y = 2r\cos\theta, \quad z = \theta,$$
$$0 \le r \le 1, \quad 0 \le \theta \le 2\pi.$$

Sketch.

For x and y, this is $y = 2x$. Projection onto xy-plane is the segment $-1 \le x \le 1$, $-2 \le y \le 2$ with slope = 2. Initial point ($\theta = 0$, $r = 1$) is $(1, 2, 0)$. Point at $\theta = \frac{\pi}{2}$ is $(0, 0, \frac{\pi}{2})$. A straight line

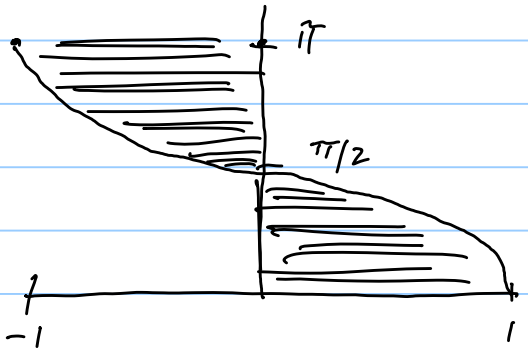
equation would be $(1,2,0) + t\left(-1,-2,\frac{\pi}{2}\right)$, $0 \le t \le 1$

For $\theta = \frac{\pi}{3}$ $(r=1)$, the point is $\left(\frac{1}{2}, 1, \frac{\pi}{3}\right)$

But for $t = \frac{1}{2}$, the point on the line is $\left(\frac{1}{2}, 1, \frac{\pi}{4}\right)$.

∴ The actual image is a curve.

An en-face view of the surface for $0 \le \theta \le \pi$ is:

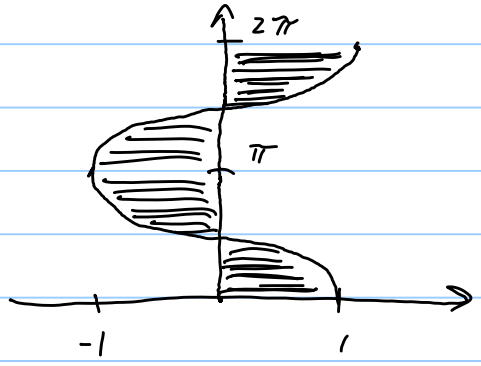


En-face view, looking at plane of $y = 2x$.

Note that $0 \le r \le 1$ means for a given $z = \theta$, x and y vary between the z-axis and the respective values for x and y.

i.e., the "rays" fan out from the z-axis, not the point (0,0,0).

∴ For $0 \le \theta \le 2\pi$:



$$\vec{T}(r,\theta) = (r\cos\theta, 2r\cos\theta, \theta)$$

$$\therefore \vec{T}_r = (\cos\theta, 2\cos\theta, 0) \quad \vec{T}_\theta = (-r\sin\theta, -2r\sin\theta, 1)$$

$$\therefore \vec{T}_r \times \vec{T}_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & 2\cos\theta & 0 \\ -r\sin\theta & -2r\sin\theta & 1 \end{vmatrix}$$

$$= (2\cos\theta, -\cos\theta, 0)$$

$$\therefore \| \vec{T}_r \times \vec{T}_\theta \| = \sqrt{5\cos^2\theta} = \sqrt{5}\,|\cos\theta|$$

$$\therefore A(S) = \int_0^1 \int_0^{2\pi} \sqrt{5}\,|\cos\theta|\, d\theta\, dr$$

$\cos\theta \geq 0$: $0 \leq \theta \leq \frac{\pi}{2}$, $\frac{3}{2}\pi \leq \theta \leq 2\pi$

$\cos\theta \leq 0$: $\frac{\pi}{2} \leq \theta \leq \frac{3}{2}\pi$

$$= \sqrt{5}\int_0^{\pi/2} \cos\theta\, d\theta - \sqrt{5}\int_{\frac{\pi}{2}}^{\frac{3}{2}\pi} \cos\theta\, d\theta + \sqrt{5}\int_{\frac{3}{2}\pi}^{2\pi} \cos\theta\, d\theta$$

$$= \sqrt{5}\left\{ \sin\theta\Big|_0^{\frac{\pi}{2}} - \sin\theta\Big|_{\frac{\pi}{2}}^{\frac{3}{2}\pi} + \sin\theta\Big|_{\frac{3}{2}\pi}^{2\pi} \right\}$$

$$= \sqrt{5}\left[1 - (-1-1) + (0-(-1)) \right]$$

$$= \sqrt{5}\left[1 + 2 + 1 \right] = \underline{4\sqrt{5}}$$

Note: $\Omega = \{r, \theta : 0 \leq r \leq 1, 0 \leq \theta \leq 2\pi\}$ is elementary

$T(r,\theta)$ is one-to-one and C^1

$T(\Omega)$ is regular except at $\theta = \frac{\pi}{2}, \frac{3}{2}\pi$

20.

20. Prove *Pappus' theorem*: Let $\mathbf{c}$: $[a, b] \to \mathbb{R}^2$ be a C^1 path whose image lies in the right half plane and is a simple closed curve. The area of the lateral surface generated by rotating the image of $\mathbf{c}$ about the y axis is equal to $2\pi \bar{x} l(\mathbf{c})$, where $\bar{x}$ is the average value of the x coordinates of points on $\mathbf{c}$ and $l(\mathbf{c})$ is the length of $\mathbf{c}$. (See Exercises 16 to 19, Section 7.1, for a discussion of average values.)

Let $f(x,y,z) = x$ (i.e., the x-coordinate of a point).

From section 7.1, Exercises 16-19, the average value of $f(x,y,z)$ along $\vec{c}$ is $\dfrac{\int_{\vec{c}} f(x,y,z)\,ds}{\ell(\vec{c})}$

and $\ell(\vec{c}) = \int_{\vec{c}} \|\vec{c}\,'(t)\|\,dt$

Let $\vec{c}(t) = [x(t), y(t)]$, and making $f(x,y,z) = x$ be the x-coordinates of $\vec{c}(t)$, then

$$\int_{\vec{c}} f(x,y,z)\,ds = \int_a^b x(t)\,\|\vec{c}\,'(t)\|\,dt \quad \text{from section 7.1.}$$

Note $x(t) \geq 0$ for $\vec{c}(t)$ since $\vec{c}$ is in the right half of the xy-plane.

$$\therefore \bar{x} = \frac{\int_{\vec{c}} x\,ds}{\int_{\vec{c}} \|\vec{c}\,'(t)\|\,dt} = \frac{\int_a^b x(t)\,\|\vec{c}\,'(t)\|\,dt}{\int_a^b \|\vec{c}\,'(t)\|\,dt}$$

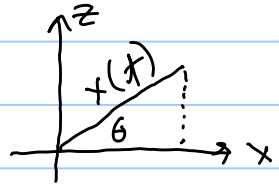
$$\therefore \bar{x} \int_a^b \|\vec{c}\,'(t)\|\,dt = \int_a^b x(t)\,\|\vec{c}\,'(t)\|\,dt$$

$$\therefore 2\pi\bar{x} \int_a^b \|\vec{c}\,'(t)\|\,dt = 2\pi \int_a^b x(t)\,\|\vec{c}\,'(t)\|\,dt, \text{ or}$$

$$2\pi\bar{x}\,\ell(\vec{c}) = 2\pi \int_a^b x(t)\,\|\vec{c}\,'(t)\|\,dt \qquad [1]$$

A parametrization of $\vec{c}(t)$ rotating around the y-axis is $\phi(t,\theta) = (x(t)\cos\theta,\ y(t),\ x(t)\sin\theta)$

$a \le t \le b,\ 0 \le \theta \le 2\pi,$



θ rotates in xz-plane.

$$\therefore \phi_t = (x'\cos\theta,\ y',\ x'\sin\theta) \quad \phi_\theta = (-x\sin\theta,\ 0,\ x\cos\theta)$$

$$\therefore \phi_t \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x'\cos\theta & y' & x'\sin\theta \\ -x\sin\theta & 0 & x\cos\theta \end{vmatrix}$$

$$= (xy'\cos\theta,\ -xx'\sin^2\theta - xx'\cos^2\theta,\ xy'\sin\theta)$$

$$= (xy'\cos\theta,\ -xx',\ xy'\sin\theta)$$

$$\therefore \|\phi_t \times \phi_\theta\| = \sqrt{x^2(y')^2\cos^2\theta + x^2(x')^2 + x^2(y')^2\sin^2\theta}$$

$$= \sqrt{x(t)^2 y'(t)^2 + x(t)^2 x'(t)^2}$$

$$= x(t)\sqrt{x'(t)^2 + y'(t)^2} = x(t)\,\|\vec{c}\,'(t)\|$$

Note $\sqrt{x(t)^2} = x(t)$ since in right half of xy-plane

$$\therefore A(S) = \int\int_D \|\phi_t \times \phi_\theta\|\, dA = \int_a^b \int_0^{2\pi} x(t)\,\|\vec{c}\,'(t)\|\, dt$$

$$= 2\pi \int_a^b x(t)\,\|\vec{c}\,'(t)\|\, dt$$

$\therefore$ [1] becomes

$$2\pi\,\bar{x}\,\ell(\vec{c}) = 2\pi \int_a^b x(t)\,\|\vec{c}\,'(t)\|\, dt = A(S)$$

21.

21. The cylinder $x^2 + y^2 = x$ divides the unit sphere S into two regions S_1 and S_2, where S_1 is inside the cylinder and S_2 outside. Find the ratio of areas $A(S_2)/A(S_1)$.

$x^2 + y^2 = x \Rightarrow x^2 - x + y^2 = 0 \Rightarrow \left(x - \frac{1}{2}\right)^2 + y^2 = \frac{1}{4}$, a circle centered at $(\frac{1}{2}, 0)$, radius $\frac{1}{2}$

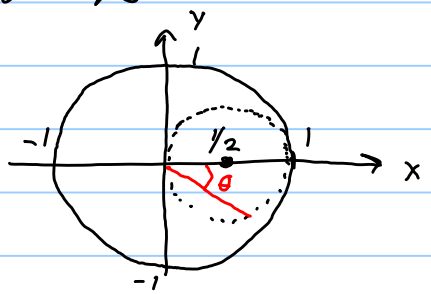
Unit sphere has surface area 4π.

$\therefore A(S_1) + A(S_2) = 4\pi$, $\therefore \dfrac{A(S_2)}{A(S_1)} = \dfrac{4\pi - A(S_1)}{A(S_1)}$

$\therefore$ Just find $A(S_1)$. Using symmetry, just find top half of $A(S_1)$.

Top half of sphere is $z = \sqrt{1 - x^2 - y^2}$

Need to find a parametrization of D.



Let $x = r\cos\theta$, $y = r\sin\theta$

$$-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$$

From $x^2 + y^2 = x$, $r^2\cos^2\theta + r^2\sin^2\theta = r\cos\theta$,

or $r = \cos\theta$ at the rim of the inscribed circle.

$\therefore$ $0 \le r \le \cos\theta$.

$$z = \sqrt{1 - (r\cos\theta)^2 - (r\sin\theta)^2} = \sqrt{1 - r^2}$$

$$\therefore \phi(r, \theta) = \left(r\cos\theta, r\sin\theta, \sqrt{1 - r^2}\right)$$

$$-\pi/2 \le \theta \le \pi/2, \quad 0 \le r \le \cos\theta$$

$$\therefore \phi_r = \left[\cos\theta, \sin\theta, -r(1 - r^2)^{-\frac{1}{2}}\right]$$

$$\phi_\theta = \left[-r\sin\theta, r\cos\theta, 0\right]$$

$$\therefore \phi_r \times \phi_\theta = \left[r^2(1 - r^2)^{-\frac{1}{2}}\cos\theta, r^2(1 - r^2)^{-\frac{1}{2}}\sin\theta, r\right]$$

$$\therefore \|\phi_r \times \phi_\theta\| = \left[\frac{r^4\cos^2\theta}{(1-r^2)} + \frac{r^4\sin^2\theta}{(1-r^2)} + r^2 \right]^{1/2}$$

$$= \left[\frac{r^4 + r^2 - r^4}{(1-r^2)} \right]^{1/2} = \frac{r}{\sqrt{1-r^2}}$$

$$\therefore A(S) = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \int_0^{\cos\theta} \frac{r}{\sqrt{1-r^2}}\, dr\, d\theta$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} -(1-r^2)^{\frac{1}{2}} \Big|_0^{\cos\theta} d\theta \qquad \sqrt{1-\cos^2\theta} = \sin\theta,\ 0 \le \theta \le \pi/2$$
$$\sqrt{1-\cos^2\theta} = -\sin\theta \quad -\pi/2 \le \theta \le 0$$

$$= \int_0^{\pi/2} -\sin\theta - (-1)\, d\theta + \int_{-\pi/2}^{0} \sin\theta - (-1)\, d\theta$$

$$= \cos\theta + \theta \Big|_0^{\frac{\pi}{2}} + \left(-\cos\theta + \theta \Big|_{-\pi/2}^{0} \right)$$

$$= \left(0 + \frac{\pi}{2}\right) - (1+0) + \left[(-1+0) - \left(0 - \frac{\pi}{2}\right) \right]$$

$$= \frac{\pi}{2} - 1 + \left(-1 + \frac{\pi}{2}\right) = \pi - 2$$

$$\therefore A(S_1) = 2(\pi - 2) = 2\pi - 4 \qquad 4\pi - A(S_1) = 2\pi + 4$$

$$\therefore \frac{A(S_2)}{A(S_1)} = \frac{4\pi - A(S_1)}{A(S_1)} = \frac{2\pi + 4}{2\pi - 4} = \frac{\pi + 2}{\pi - 2}$$

22.

22. Suppose a surface S that is the graph of a function $z = f(x, y)$, where $(x, y) \in D \subset \mathbb{R}^2$ can also be described as the set of $(x, y, z) \in \mathbb{R}^3$ with $F(x, y, z) = 0$ (a level surface). Derive a formula for $A(S)$ that involves only F.

$$F(x,y,z) = z - f(x,y) = 0.$$

$$\therefore F_x = -f_x \quad F_y = -f_y \quad F_z = 1$$

$$\therefore (F_x)^2 = (-f_x)^2, \ (F_y)^2 = (-f_y)^2, \ (F_z)^2 = 1$$

$$\therefore \therefore \nabla F \cdot \nabla F = f_x^2 + f_y^2 + 1$$

$$\therefore A(S) = \iint_D \sqrt{f_x^2 + f_y^2 + 1}\, dA$$

$$= \iint_D \sqrt{\nabla F \cdot \nabla F}\, dA$$

$$= \iint_D \|\nabla F\|\, dA$$

23.

23. Calculate the area of the frustum shown in Figure 7.4.7 using (a) geometry alone and, second, (b) a surface-area formula.

(a) Surface area of the slanted side of a cone (i.e., ignoring base area)

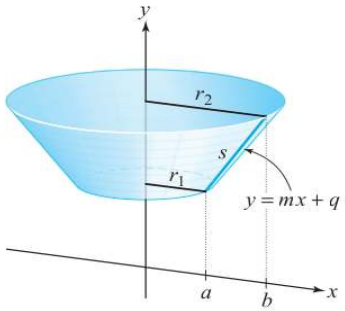
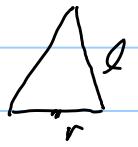


figure **7.4.7** A line segment revolved around the y axis becomes a frustum of a cone.

is $\pi r l$, where r = radius of circular base,



l = length of slant height

$\therefore$ For this frustrum, $\pi r_2(l+s) - \pi r_1 l =$

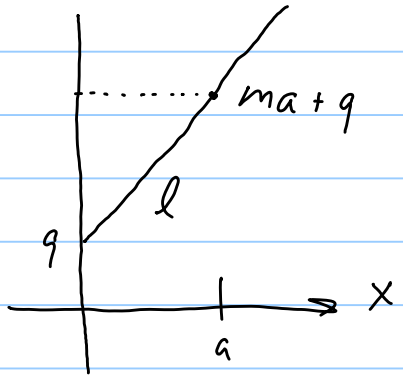
area cone with base r_2, slant height $l+s$
− area cone with base r_1, slant height l

$= \pi l(r_2 - r_1) + \pi r_2 s = \pi l(b-a) + \pi b s$

Here, $s^2 = (b-a)^2 + [(mb+q)-(ma+q)]^2$

$= (b-a)^2 + m^2(b-a)^2 = (b-a)^2(1+m^2)$

$\therefore s = (b-a)\sqrt{m^2+1}$



$l^2 = [(ma+q)-q]^2 + a^2$

$= m^2a^2 + a^2 = a^2(m^2+1)$

$\therefore l = a\sqrt{m^2+1}$

$\therefore$ Area of frustrum $= \pi(a\sqrt{m^2+1})(b-a) + \pi b(b-a)\sqrt{m^2+1}$

$= \pi(b-a)\sqrt{m^2+1}(a+b) = \pi(b^2-a^2)\sqrt{m^2+1}$

(b) This is a surface of revolution about y-axis.

$$\therefore A = 2\pi \int_a^b |x| \sqrt{1 + f'(x)^2}\, dx$$

$|x| = x$ since $0 \le a < b$, $f(x) = mx + q \Rightarrow f'(x) = m$

$$\therefore A = 2\pi \int_a^b x\sqrt{1+m^2}\, dx = 2\pi\sqrt{1+m^2}\left[\frac{x^2}{2}\right]_{x=a}^{x=b}$$

$$= \underline{\pi\sqrt{1+m^2}\,(b^2-a^2)} = \underline{\pi\sqrt{1+m^2}\,(b+a)(b-a)}$$

24.

24. A cylindrical hole of radius 1 is bored through a solid ball of radius 2 to form a ring coupler, as shown in Figure 7.4.8. Find the volume and outer surface area of this coupler.

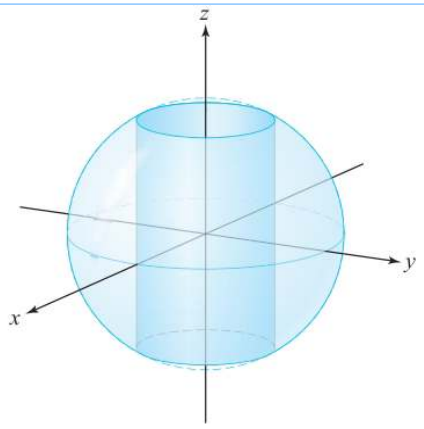


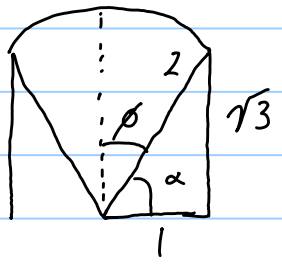
figure **7.4.8** Find the outer surface area and volume of the shaded region.

(a) Volume

Volume of sphere $= \frac{4}{3}\pi(2)^3 = \frac{32}{3}\pi$

Volume of bored cylinder:

$\cos\alpha = \frac{1}{2} \Rightarrow \alpha = \frac{\pi}{3}$, $\phi = \pi/6$



Use spherical coordinates, find volume of upper half of bored cylinder.

Volume of "ice cream cone" part is:

$$\int_0^{2\pi}\int_0^{\pi/6}\int_0^2 \rho^2 \sin\phi\, d\rho\, d\phi\, d\theta =$$

$$\int_0^{2\pi} d\theta \int_0^{\frac{\pi}{6}} \sin\phi \, d\phi \int_0^2 \rho^2 \, d\rho = 2\pi \left[-\cos\phi\right]_0^{\frac{\pi}{6}} \left[\frac{\rho^3}{3}\right]_0^2$$

$$= 2\pi \left[-\frac{\sqrt{3}}{2} - (-1)\right] \left[\frac{8}{3}\right] = \underline{\frac{8}{3}\pi(2-\sqrt{3})}$$

Volume of side part:

Note $\rho \sin\phi = 1$ for $\frac{\pi}{6} \le \phi \le \frac{\pi}{2}$

$$\therefore \int_0^{2\pi} \int_{\pi/6}^{\pi/2} \int_0^{\frac{1}{\sin\phi}} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$$

$$= 2\pi \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \sin\phi \, \frac{\rho^3}{3}\Big|_{\rho=0}^{\rho=\frac{1}{\sin\phi}} d\phi = \frac{2\pi}{3} \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{d\phi}{\sin^2\phi}$$

$$= \frac{2\pi}{3} \left[-\cot\phi\right]_{\pi/6}^{\pi/2} = \frac{2\pi}{3}\left[0 - (-\sqrt{3})\right] = \underline{\frac{2\sqrt{3}}{3}\pi}$$

$\therefore$ Volume of top half of bore =

$$\frac{8}{3}\pi(2-\sqrt{3}) + \frac{2\sqrt{3}}{3}\pi = \frac{\pi}{3}(16 - 8\sqrt{3} + 2\sqrt{3})$$

$$= \frac{\pi}{3}(16 - 6\sqrt{3}) = \frac{2\pi}{3}(8 - 3\sqrt{3})$$

$\therefore$ Volume of bored cylinder $= \frac{4}{3}\pi(8-3\sqrt{3})$

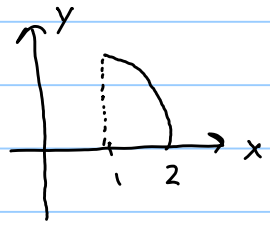
$\therefore$ Volume of coupler $= \frac{32}{3}\pi - \frac{4}{3}\pi(8-3\sqrt{3})$

$$= \frac{32}{3}\pi - \frac{32}{3}\pi + 4\sqrt{3}\,\pi = \underline{4\sqrt{3}\,\pi}$$

Cylindrical coordinates may have been easier

(5) Surface area of outer coupler

View surface as a circular arc rotated about y-axis. $x^2+y^2=4 \Rightarrow y=\sqrt{4-x^2}$, $1 \le x \le 2$



Just compute top half of coupler.

$$y' = \tfrac{1}{2}(4-x^2)^{-\frac{1}{2}}(-2x) = \frac{x}{\sqrt{4-x^2}}$$

$$\therefore A(S) = 2\pi \int_1^2 x\sqrt{1+\left(\frac{x}{\sqrt{4-x^2}}\right)^2}\,dx$$

$$= 2\pi \int_1^2 x\sqrt{\frac{4-x^2+x^2}{4-x^2}}\,dx = 4\pi \int_1^2 \frac{x}{\sqrt{4-x^2}}\,dx$$

$$= 4\pi\left[-(4-x^2)^{\frac{1}{2}}\right]_{x=1}^{x=2} = 4\pi\left[0-(-\sqrt{3})\right]$$

$$= 4\sqrt{3}\,\pi$$

$\therefore$ Coupler surface area $= \underline{8\sqrt{3}\,\pi}$

25. **25.** Find the area of the graph of the function $f(x, y) = \frac{2}{3}(x^{3/2} + y^{3/2})$ that lies over the domain $[0, 1] \times [0, 1]$.

$f_x = x^{\frac{1}{2}} \qquad f_y = y^{\frac{1}{2}}$

$$A(S) = \iint_D \sqrt{f_x^2 + f_y^2 + 1}\, dA = \int_0^1 \int_0^1 \sqrt{x + y + 1}\, dx\, dy$$

$$= \int_0^1 \frac{2}{3}(x + y + 1)^{3/2} \Big|_{x=0}^{x=1} dy$$

$$= \int_0^1 \frac{2}{3}(y + 2)^{3/2} - \frac{2}{3}(y + 1)^{\frac{3}{2}}\, dy$$

$$= \left(\frac{2}{3}\right)\left(\frac{2}{5}\right)(y + 2)^{5/2} \Big|_{y=0}^{y=1} - \left(\frac{2}{3}\right)\left(\frac{2}{5}\right)(y + 1)^{5/2} \Big|_{y=0}^{y=1}$$

$$= \frac{4}{15}\left[3^{5/2} - 2^{5/2}\right] - \frac{4}{15}\left[2^{5/2} - 1\right]$$

$$= \frac{4}{15}\left[9\sqrt{3} - 4\sqrt{2} - 4\sqrt{2} + 1\right]$$

$$= \frac{4}{15}\left(9\sqrt{3} - 8\sqrt{2} + 1\right)$$

26.

26. Express the surface area of the following graphs over the indicated region D as a double integral. Do not evaluate.

(a) $(x+2y)^2$; $D=[-1,2]\times[0,2]$

(b) $xy+x/(y+1)$; $D=[1,4]\times[1,2]$

(c) $xy^3e^{x^2y^2}$; $D=$ unit circle centered at the origin

(d) $y^3\cos^2 x$; $D=$ triangle with vertices $(-1,1)$, $(0,2)$, and $(1,1)$

$$A(S)=\iint_D \sqrt{f_x^2+f_y^2+1}\,dA$$

(a) $f_x = 2(x+2y)$ $\quad f_y = 4(x+2y)$

$f_x^2 = 4(x+2y)^2$ $\quad f_y^2 = 16(x+2y)^2$

$$\therefore A(S)=\int_{-1}^{2}\int_{0}^{2}\sqrt{20(x+2y)^2}\,dy\,dx$$

$$=2\sqrt{5}\int_{-1}^{2}\int_{0}^{2}\sqrt{(x+2y)^2}\,dy\,dx$$

(b) $f_x = y+\frac{1}{y+1}=\frac{y^2+y+1}{y+1}$ $\quad f_y = x-\frac{x}{(y+1)^2}=\frac{x(y^2+2y)}{(y+1)^2}$

$f_x^2 = \frac{(y^2+y+1)^2}{(y+1)^2}$ $\quad f_y^2 = \frac{x^2(y^2+2y)^2}{(y+1)^4}$

$$f_x^2+f_y^2+1=\frac{(y^2+y+1)^2(y+1)^2+x^2(y^2+2y)^2+(y+1)^4}{(y+1)^4}$$

$$\therefore A(S)=\int_{1}^{4}\int_{1}^{2}\frac{\sqrt{(y^2+y+1)^2(y+1)^2+x^2(y^2+2y)^2+(y+1)^4}}{(y+1)^2}\,dy\,dx$$

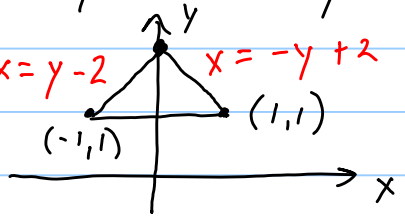
(c) $f_x = y^3 e^{x^2y^2} + 2x^2y^5 e^{x^2y^2} = (y^3 + 2x^2y^5)e^{x^2y^2}$

$f_y = 3xy^2 e^{x^2y^2} + 2x^3y^4 e^{x^2y^2} = (3xy^2 + 2x^3y^4)e^{x^2y^2}$

$$\therefore A(S) = \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \sqrt{1 + \left[(y^3 + 2x^2y^5)^2 + (3xy^2 + 2x^3y^4)\right] e^{2x^2y^2}}\, dy\, dx$$

(d) $f_x = -2y^3 \cos x \sin x \qquad f_y = 3y^2 \cos^2 x$

$1 + f_x^2 + f_y^2 = 1 + 4y^6 \cos^2 x \sin^2 x + 9y^4 \cos^4 x$

$D =$ 

an x-simple region

$$\therefore A(S) = \int_{1}^{2} \int_{y-2}^{-y+2} \sqrt{1 + 4y^6 \cos^2 x \sin^2 x + 9y^4 \cos^4 x}\, dx\, dy$$

27.

27. Show that the surface area of the upper hemisphere of radius R, $z = \sqrt{R^2 - x^2 - y^2}$, can be computed by formula (4), evaluated as an improper integral.

$z_x = -x(R^2 - x^2 - y^2)^{-\frac{1}{2}} \qquad z_y = -y(R^2 - x^2 - y^2)^{-\frac{1}{2}}$

$$1 + z_x^2 + z_y^2 = 1 + \frac{x^2}{R^2 - x^2 - y^2} + \frac{y^2}{R^2 - x^2 - y^2} = \frac{R^2}{R^2 - x^2 - y^2}$$

D = circle in xy-plane of radius R, center $(0,0)$.

$$\therefore A(S) = \int_{-1}^{1} \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} \frac{R}{\sqrt{R^2-x^2-y^2}} \, dy\, dx$$

Switch to polar coordinates: $x = r\cos\theta$, $y = r\sin\theta$

Jacobian $= r\,dr\,d\theta$, $0 \le r \le R$, $0 \le \theta \le 2\pi$

$$R^2 - x^2 - y^2 = R^2 - r^2\cos^2\theta - r^2\sin^2\theta = R^2 - r^2$$

$$\therefore A(S) = R \int_0^{2\pi} \int_0^{R} \frac{r}{\sqrt{R^2-r^2}} \, dr\, d\theta$$

$$= 2\pi R \lim_{x \to R^-} \int_0^{x} \frac{r}{\sqrt{R^2-r^2}} \, dr$$

$$= 2\pi R \lim_{x \to R^-} \left[-(R^2-r^2)^{\frac{1}{2}} \right]_{r=0}^{r=x}$$

$$= 2\pi R \lim_{x \to R^-} \left[-\sqrt{R^2-x^2} - (-R) \right]$$

$$= 2\pi R \left[0 + R \right] = \underline{2\pi R^2}$$

7.5 Integrals of Scalar Functions Over Surfaces

Note Title 3/15/2017

1.

1. Evaluate the integral of the function $f(x, y, z) = x + y$ over the surface S given by:

$$\Phi(u, v) = (2u\cos v, 2u\sin v, u), \quad u \in [0, 4], v \in [0, \pi]$$

$$\phi_u = (2\cos v, 2\sin v, 1) \qquad \phi_v = (-2u\sin v, 2u\cos v, 0)$$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2\cos v & 2\sin v & 1 \\ -2u\sin v & 2u\cos v & 0 \end{vmatrix}$$

$$= (-2u\cos v, -2u\sin v, 4u\cos^2 v + 4u\sin^2 v)$$

$$= (-2u\cos v, -2u\sin v, 4u)$$

$$\|\phi_u \times \phi_v\| = \sqrt{4u^2 + 16u^2} = 2\sqrt{5}\,u$$

$$f(x(u,v), y(u,v), z(u,v)) = 2u\cos v + 2u\sin v$$

$$\therefore \iint_S f\,ds = \int_0^4 \int_0^{\pi} (2u\cos v + 2u\sin v)\, 2\sqrt{5}\,u\, dv\, du$$

$$= \int_0^4 4\sqrt{5}\,u^2\, du \int_0^{\pi} (\cos v + \sin v)\, dv$$

$$= \left(4\sqrt{5}\,\frac{u^3}{3}\Big|_0^4\right)\left(\sin v - \cos v\Big|_0^{\pi}\right)$$

$$= \left(\frac{256\sqrt{5}}{3}\right)(2) = \frac{512\sqrt{5}}{3}$$

2.

2. Evaluate the integral of the function $f(x, y, z) = z + 6$ over the surface S given by:

$$\Phi(u, v) = (u, \frac{v}{3}, v), \quad u \in [0, 2], v \in [0, 3].$$

$$f(x(u,v), y(u,v), z(u,v)) = v + 6$$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 0 & 1/3 & 1 \end{vmatrix} = (0, -1, 1/3)$$

$$\therefore \|\phi_u \times \phi_v\| = \sqrt{1 + 1/9} = \sqrt{10}/3$$

$$\therefore \iint_S f\,ds = \int_0^2 \int_0^3 \frac{\sqrt{10}}{3}(v + 6)\,dv\,du$$

$$= \frac{2\sqrt{10}}{3}\left(\frac{v^2}{2} + 6v\right)\Big|_{v=0}^{v=3} = \frac{2\sqrt{10}}{3}\left(\frac{9}{2} + 18\right)$$

$$= 15\sqrt{10}$$

3.

3. Evaluate the integral

$$\iint_S (3x - 2y + z)\,dS,$$

where S is the portion of the plane $2x + 3y + z = 6$ that lies in the first octant.

Normal to the plane = $(2, 3, 1)$. Unit $\vec{n} = \frac{1}{\sqrt{14}}(2, 3, 1)$.

$\therefore \vec{n} \cdot \vec{k} = \cos\theta = \frac{1}{\sqrt{14}}$. The plane is $z = g(x,y) = 6 - 2x - 3y$

$$\therefore \iint_S f\,dS = \iint_D \frac{f(x,y,g(x,y))}{\cos\theta}\,dx\,dy$$

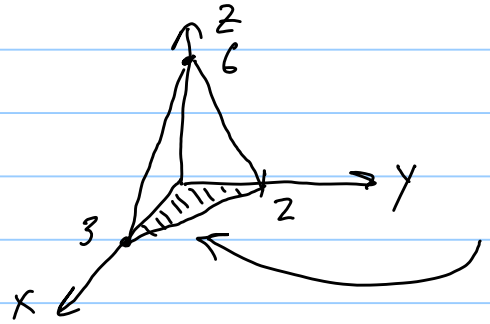
$$= \iint_D \sqrt{14}\,[3x - 2y + (6 - 2x - 3y)]\,dx\,dy$$

$$= \iint_D \sqrt{14}\,(x - 5y + 6)\,dx\,dy$$

For first octant: $2x + 3(0) + (0) = 6 \Rightarrow x = 3$

$2(0) + 3y + (0) = 6 \Rightarrow y = 2$

$2(0) + 3(0) + z = 6 \Rightarrow z = 6$



$y = -\frac{2}{3}x + 2$

$\therefore D = \{(x,y) : 0 \le x \le 3,\ 0 \le y \le -\frac{2}{3}x + 2\}$

$$\therefore \sqrt{14}\int_0^3 \int_0^{-\frac{2}{3}x+2} (x - 5y + 6)\,dy\,dx = \sqrt{14}\int_0^3 xy - \frac{5}{2}y^2 + 6y \Big|_0^{-\frac{2}{3}x+2}\,dx$$

$$= \sqrt{14}\int_0^3 -\frac{2}{3}x^2 + 2x - \frac{5}{2}\left(-\frac{2}{3}x + 2\right)^2 + 6\left(-\frac{2}{3}x + 2\right)\,dx$$

$-4x + 12$

$$= \sqrt{14} \int_0^3 -\frac{2}{3}x^2 - 2x + 12 - \frac{5}{2}\left(-\frac{2}{3}x + 2\right)^2 dx$$

$$= \sqrt{14} \left[-\frac{2}{9}x^3 - x^2 + 12x - \frac{5}{2}\left(\frac{1}{3}\right)\left(-\frac{3}{2}\right)\left(-\frac{2}{3}x + 2\right)^3 \right]_{x=0}^{x=3}$$

$$= \sqrt{14} \left[-6 - 9 + 36 - 0 - \left(0 - 0 + 0 + \frac{5}{4}(8)\right) \right]$$

$$= \sqrt{14} \left[21 - 10 \right] = \underline{11\sqrt{14}}$$

4.

4. Evaluate the integral

$$\iint_S (x + z)\, dS,$$

where S is the part of the cylinder $y^2 + z^2 = 4$ with $x \in [0, 5]$.

Let $\phi(x, \theta) = (x, 2\cos\theta, 2\sin\theta)$ be a parametrization of the cylinder, $0 \le x \le 5$, $0 \le \theta \le 2\pi$

$$\phi_x \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 0 & -2\sin\theta & 2\cos\theta \end{vmatrix} = (0, -2\cos\theta, -2\sin\theta)$$

$$\therefore \|\phi_x \times \phi_\theta\| = 2$$

$$\therefore \iint_S (x + z)\, ds = \int_0^5 \int_0^{2\pi} (x + 2\sin\theta)(2)\, d\theta\, dx$$

$$= \int_0^5 2x\theta - 4\cos\theta \Big|_0^{2\pi} \, dx$$

$$= \int_0^5 (4\pi x - 4) - (0 - 4)\, dx = \int_0^5 4\pi x \, dx = 2\pi x^2 \Big|_0^5$$

$$= \underline{50\pi}$$

5.

5. Let S be the surface defined by

$$\Phi(u, v) = (u + v,\ u - v,\ uv).$$

(a) Show that the image of S is in the graph of the surface $4z = x^2 - y^2$.

(b) Evaluate $\iint_S x\, dS$ for all points on the graph S, over $x^2 + y^2 \le 1$.

(a) $x = u+v$, $y = u-v$, $x^2 = u^2 + 2uv + v^2$, $y^2 = u^2 - 2uv + v^2$

$\therefore x^2 - y^2 = 4uv = 4z$

$\therefore$ every point of $\phi(u,v)$ is a point on $4z = x^2 - y^2$

Also, given (x,y,z), let $u = \frac{x+y}{2}$, $v = \frac{x-y}{2}$

$\therefore u+v = x$, $u-v = y$, and $uv = z$. $\therefore$ Every point of the graph is in the image of ϕ.

(b) $\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & v \\ 1 & -1 & u \end{vmatrix} = (u+v,\ v-u,\ -2)$

$$\therefore \|\phi_u \times \phi_v\| = \sqrt{2u^2 + 2v^2 + 4}$$

For $x^2 + y^2 \le 1$, $(u+v)^2 + (u-v)^2 \le 1$, or $u^2 + v^2 \le \frac{1}{2}$

$\therefore D$ is a circular disc of radius $\frac{1}{\sqrt{2}}$

$$\iint_S x \, dS = \iint_D (u+v)\sqrt{2u^2+2v^2+4}\, dA$$

Make a polar substition : $0 \le r \le \frac{1}{\sqrt{2}}$, $0 \le \theta \le 2\pi$

$$u = r\cos\theta, \quad v = r\sin\theta$$

$$\therefore \sqrt{2u^2+2v^2+4} = \sqrt{2r^2+4}$$

$$\therefore \iint_S x \, dS = \int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}} (r\cos\theta + r\sin\theta)\sqrt{2r^2+4}\; r\, dr\, d\theta$$

$$= \sqrt{2}\int_0^{2\pi}\int_0^{\frac{1}{\sqrt{2}}} (\cos\theta + \sin\theta)\, r^2\sqrt{r^2+2}\, dr\, d\theta$$

$$= \sqrt{2}\int_0^{2\pi} (\cos\theta + \sin\theta)\, d\theta \int_0^{\frac{1}{\sqrt{2}}} r^2\sqrt{r^2+2}\, dr$$

$$= \sqrt{2}\,(\sin\theta - \cos\theta)\Big|_0^{2\pi} \int_0^{\frac{1}{\sqrt{2}}} r^2\sqrt{r^2+2}\, dr$$

$$= \sqrt{2}\left[0 - 1 - (0 - 1)\right] \int_0^{\frac{1}{\sqrt{2}}} r^2\sqrt{r^2+2}\, dr$$

$$= 0 \int_0^{\frac{1}{\sqrt{2}}} r^2\sqrt{r^2+2}\, dr = \underline{0}$$

6. **6.** Evaluate the integral

$$\iint_S (x^2 z + y^2 z)\, dS,$$

where S is the part of the plane $z = 4 + x + y$ that lies inside the cylinder $x^2 + y^2 = 4$.

Normal to plane is $(1, 1, -1)$, unit $\vec{n} = \frac{1}{\sqrt{3}}(-1, -1, 1)$

$\therefore \cos\theta = \vec{n} \cdot \hat{K} = \frac{1}{\sqrt{3}}$

$$\therefore \iint_S z(x^2+y^2)\, dS = \sqrt{3} \iint_D z(x^2+y^2)\, dA$$

Now make a polar conversion: $x = r\cos\theta$, $y = r\sin\theta$,

$0 \le r \le 2$, $0 \le \theta \le 2\pi$, $z = 4 + r\cos\theta + r\sin\theta$

$$\therefore \sqrt{3} \int_0^{2\pi} \int_0^2 (4 + r\cos\theta + r\sin\theta)(r^2)\, r\, dr\, d\theta$$

$$= \sqrt{3} \int_0^{2\pi} 4r^3 + (\cos\theta + \sin\theta) r^4 \Big|_0^2 \, d\theta$$

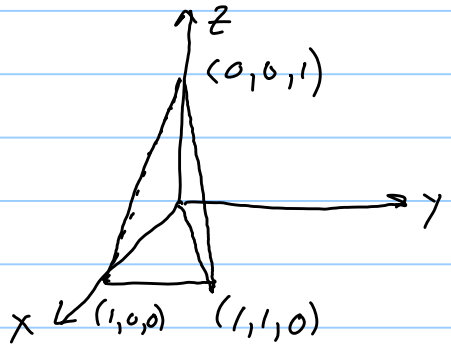
$$= \sqrt{3} \int_0^{2\pi} 32 + 16(\cos\theta + \sin\theta)\, d\theta$$

$$= \sqrt{3} \left[32\theta + 16(\sin\theta - \cos\theta) \right]_0^{2\pi}$$

$$= 64\sqrt{3}\pi + 16\sqrt{3}\left[0 - 1 - (0 - 1)\right]$$

$$= \underline{64\sqrt{3}\pi}$$

7. **7.** Compute $\iint_S xy\, dS$, where S is the surface of the tetrahedron with sides $z = 0$, $y = 0$, $x + z = 1$, and $x = y$.



Since 4 sides, this is the sum of 4 surface integrals:

(1) xy-plane (3) $x + z = 1$ plane

(2) xz-plane (4) $x = y$ plane

Note $f(x,y,z) = xy$

(1) xy-plane Here, $\vec{n} = \hat{k}$. $\therefore \cos\theta = \vec{n} \cdot \hat{k} = 1$

$$\therefore \iint_S f(x,y,z)\, dS = \iint_D \frac{f(x,y,g(x,y))}{\cos\theta}\, dx\, dy$$

$$= \iint_D f(x,y,0)\, dx\, dy = \iint_D xy\, dx\, dy$$

$$D = \{(x,y) : 0 \le x \le 1,\ 0 \le y \le x\}$$

$$\therefore \int_0^1 \int_0^x xy \, dy\, dx = \int_0^1 x \frac{y^2}{2} \Big|_{y=0}^{y=x} dx = \int_0^1 \frac{x^3}{2} dx$$

$$= \frac{x^4}{8} \Big|_0^1 = \underline{\frac{1}{8}}$$

(2) xz-plane Here $f(x,y,z) = 0$ as $y = 0$, so $xy = 0$

$$\therefore \iint_S f(x,y,z)\, dS = \underline{0}$$

(3) $x + z = 1$ plane Here, $\vec{n} = \frac{1}{\sqrt{2}}(1,0,1)$.

$$\therefore \vec{n} \cdot \hat{k} = \frac{1}{\sqrt{2}} = \cos\theta$$

$$\therefore \iint_S f(x,y,z)\, dS = \iint_D \frac{f(x,y,g(x,y))}{\cos\theta} \, dx\, dy$$

$$= \sqrt{2} \iint_D xy \, dx\, dy$$

D is the projection of this plane onto the xy-plane, and so is $D = \{(x,y) : 0 \le x \le 1,\ 0 \le y \le x\}$

$$\therefore \sqrt{2}\int_0^1\int_0^x xy\,dy\,dx = \sqrt{2}\left(\frac{1}{8}\right) \quad \text{from (1)}$$

(4) $x = y$ plane. A parametrization of this side is $\phi(u,v) = (u, u, v)$, $0 \le u \le 1$, $0 \le v \le 1-u$

as the upper boundary of the z component comes from the $x+z=1$ plane, or $z = 1-x$.

$$\therefore \quad \phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = (1, -1, 0)$$

$$\therefore \|\phi_u \times \phi_v\| = \sqrt{2} \qquad f(x(u,v), y(u,v), z(u,v)) = u^2$$

$$\therefore \iint_S f\,dS = \int_0^1\int_0^{1-u} u^2(\sqrt{2})\,dv\,du$$

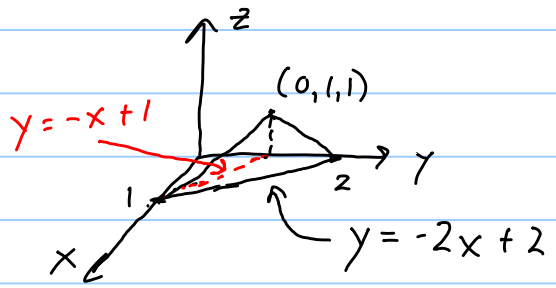
$$= \int_0^1 \sqrt{2}\,u^2 v\Big|_{v=0}^{v=1-u} du = \int_0^1 \sqrt{2}\,u^2(1-u)\,du$$

$$= \sqrt{2}\left[\frac{u^3}{3} - \frac{u^4}{4}\right]_0^1 = \sqrt{2}\left(\frac{1}{3} - \frac{1}{4}\right) = \frac{\sqrt{2}}{12}$$

$$\therefore (1)+(2)+(3)+(4) = \frac{1}{8} + 0 + \frac{\sqrt{2}}{8} + \frac{\sqrt{2}}{12} = \frac{3}{24} + \frac{3\sqrt{2}}{24} + \frac{2\sqrt{2}}{24}$$

$$= \frac{5\sqrt{2}+3}{24}$$

8. **8.** Evaluate $\iint_S xyz\,dS$, where S is the triangle with vertices $(1, 0, 0)$, $(0, 2, 0)$, and $(0, 1, 1)$.



Projection of triangle onto the xy-plane is bounded below by $y = -x+1$ and above by $y = -2x+2$, $0 \le x \le 1$.

$\therefore D = \{(x,y) : 0 \le x \le 1, -x+1 \le y \le -2x+2\}$

A normal to the triangular surface is:

$[(0,1,1) - (1,0,0)] \times [(0,1,1) - (0,2,0)]$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ 0 & 1 & -1 \end{vmatrix} = (2,1,1)$$

$\therefore$ plane of triangle is $2x+y+z = K$, so choose a point of triangle (in plane),

$\therefore 2(1)+(0)+(0) = K$, $K = 2$.

$\therefore 2x+y+z = 2$, so $z = 2-2x-y$

$\therefore$ unit normal $= \frac{1}{\sqrt{6}}(2,1,1) = \vec{n}$

$\therefore \vec{n} \cdot \hat{k} = \frac{1}{\sqrt{6}} = \cos\theta$

$$\therefore \iint_S xyz\,dS = \iint_D \frac{xyz}{\cos\theta}\,dx\,dy = \sqrt{6}\iint_D xyz\,dx\,dy$$

$$= \sqrt{6}\int_0^1 \int_{-x+1}^{-2x+2} xy(2-2x-y)\,dy\,dx$$

$$= \sqrt{6}\int_0^1 \int_{-x+1}^{-2x+2} 2xy - 2x^2y - xy^2\,dy\,dx$$

$$= \sqrt{6}\int_0^1 (x-x^2)y^2 - \frac{xy^3}{3}\Big|_{y=-x+1}^{y=-2x+2}\,dx$$

$$= \sqrt{6}\int_0^1 (x-x^2)\left[4x^2-8x+4-(x^2-2x+1)\right] - x\left[-8x^3+24x^2-24x+8-(-x^3+3x^2-3x+1)\right]dx$$

($3x^2-6x+3$; $-7x^3+21x^2-21x+7$)

$$= \sqrt{6}\int_0^1 \left[3x^3-6x^2+3x-3x^4+6x^3-3x^2+(7x^4-21x^3+21x^2-7x)\right]dx$$

$$= \sqrt{6}\int_0^1 4x^4-12x^3+12x^2-4x\,dx$$

$$= \sqrt{6}\left[\frac{4}{5}x^5-3x^4+4x^3-2x^2\right]_0^1$$

$$= 76\left(\frac{4}{5} - 3 + 4 - 2\right) = -\frac{76}{5}$$

9.

9. Evaluate $\iint_S z\,dS$, where S is the upper hemisphere of radius a, that is, the set of (x, y, z) with $z = \sqrt{a^2 - x^2 - y^2}$.

Use spherical coordinates: $x = a\cos\theta\sin\phi$, $y = a\sin\theta\sin\phi$, $z = a\cos\phi$

$0 \le \theta \le 2\pi$, $0 \le \phi \le \pi/2$

$\therefore \vec{T}(\theta,\phi) = (a\cos\theta\sin\phi, a\sin\theta\sin\phi, a\cos\phi)$

describes the hemisphere

$$\vec{T}_\theta \times \vec{T}_\phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -a\sin\theta\sin\phi & a\cos\theta\sin\phi & 0 \\ a\cos\theta\cos\phi & a\sin\theta\cos\phi & -a\sin\phi \end{vmatrix}$$

$$= (-a^2\cos\theta\sin^2\phi, -a^2\sin\theta\sin^2\phi, -a^2\sin^2\theta\sin\phi\cos\phi - a^2\cos^2\theta\sin\phi\cos\phi)$$

$$= (-a^2\cos\theta\sin^2\phi, -a^2\sin\theta\sin^2\phi, -a^2\sin\phi\cos\phi)$$

$$\therefore \|\vec{T}_\theta \times \vec{T}_\phi\| = \sqrt{a^4\cos^2\theta\sin^4\phi + a^4\sin^2\theta\sin^4\phi + a^4\sin^2\phi\cos^2\phi}$$

$$= a^2\sqrt{\sin^4\phi + \sin^2\phi\cos^2\phi} = a^2\sqrt{\sin^2\phi}$$

$$= a^2\sin\phi$$

$$\therefore \iint_S z\,dS = \int_0^{2\pi}\int_0^{\pi/2} (a\cos\phi)\,a^2\sin\phi\,d\phi\,d\theta$$

$$= 2\pi a^3 \int_0^{\frac{\pi}{2}} \cos\phi \sin\phi \, d\phi = 2\pi a^3 \left[\frac{\sin^2\phi}{2} \right]_{\phi=0}^{\phi=\pi/2}$$

$$= \underline{\pi a^3}$$

10. **10.** Evaluate $\iint_S (x + y + z)\, dS$, where S is the boundary of the unit ball B; that is, S is the set of (x, y, z) with $x^2 + y^2 + z^2 = 1$. (HINT: Use the symmetry of the problem.)

Should be 0: Octant I $(x,y,z>0)$ cancels octant where $x,y,z<0$. Similarly for other octants.

To check: use spherical coordinates.

$$T(\theta,\phi) = (\cos\theta\sin\phi, \sin\theta\sin\phi, \cos\phi), \quad 0 \le \theta \le 2\pi, \quad 0 \le \phi \le \pi$$

From #9 above, $\| \vec{T}_\theta \times \vec{T}_\phi \| = \sin\phi$

$$\therefore \iint_S (x+y+z)\, dS = \int_0^{2\pi}\int_0^{\pi} (\cos\theta\sin\phi + \sin\theta\sin\phi + \cos\phi)\sin\phi \, d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} (\cos\theta + \sin\theta)\sin^2\phi + \cos\phi\sin\phi \, d\phi\, d\theta$$

$$= \int_0^{2\pi} (\cos\theta + \sin\theta)\, d\theta \int_0^{\pi} \sin^2\phi\, d\phi + \int_0^{2\pi}\int_0^{\pi} \cos\phi\sin\phi \, d\phi\, d\theta$$

$$= (\sin\theta - \cos\theta)\Big|_0^{2\pi} \int_0^{\pi} \sin^2\phi \, d\phi + 2\pi \left[\frac{\sin^2\phi}{2} \right]_{\phi=0}^{\phi=\pi}$$

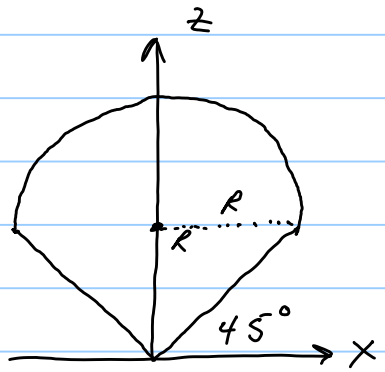
$$= [0 - 1 - (0 - 1)] \int_0^{\pi} \sin^2\phi \, d\phi + 2\pi(0)$$

$$= 0 + 0 = \underline{0}$$

11.

11. (a) Compute the area of the portion of the cone $x^2 + y^2 = z^2$ with $z \geq 0$ that is inside the sphere $x^2 + y^2 + z^2 = 2Rz$, where R is a positive constant.

(b) What is the area of that portion of the sphere that is inside the cone?



(a) $x^2 + y^2 + z^2 - 2Rz = x^2 + y^2 + (z - R)^2 = R^2$

∴ The sphere has center $(0, 0, R)$ and radius R.

Choose polar coordinates. The cone can be described as $\phi(r, \theta) = (r\cos\theta, r\sin\theta, r)$

since the cone makes a 45° angle with z-axis

$0 \leq r \leq R$, $0 \leq \theta \leq 2\pi$. Cone + sphere meet at (x, y, R)

$$\phi_r \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & 1 \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (-r\cos\theta, -r\sin\theta, r(\cos^2\theta + \sin\theta))$$

$$= (-r\cos\theta, -r\sin\theta, r)$$

$$\therefore \|\phi_r \times \phi_\theta\| = \sqrt{r^2 + r^2} = \sqrt{2}\, r$$

$$\therefore A(s) = \int_0^{2\pi}\int_0^{R} \sqrt{2}\, r \, dr\, d\theta = 2\sqrt{2}\,\pi \left[\frac{r^2}{2}\right]_0^{R}$$

$$= \underline{\sqrt{2}\,\pi R^2}$$

(5) It's just the top half of a sphere of radius R

$$= \frac{4\pi R^2}{2} = \underline{2\pi R^2}$$

12. **12.** Verify that in spherical coordinates, on a sphere of radius R,

$$\|\mathbf{T}_\phi \times \mathbf{T}_\theta\|\, d\phi\, d\theta = R^2 \sin\phi\, d\phi\, d\theta.$$

As in problem #9, sphere of radius R can be described as $T(\theta, \phi) = (R\cos\theta\sin\phi,\ R\sin\theta\sin\phi,\ R\cos\phi)$ using spherical coordinates.

$$\therefore \vec{T_\theta} \times \vec{T_\phi} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -R\sin\theta\sin\phi & R\cos\theta\sin\phi & 0 \\ R\cos\theta\cos\phi & R\sin\theta\cos\phi & -R\sin\phi \end{vmatrix}$$

$$= \left(-R^2\cos\theta\sin^2\phi,\ -R^2\sin\theta\sin^2\phi,\ -R^2\sin^2\theta\sin\phi\cos\phi - R^2\cos^2\theta\sin\phi\cos\phi\right)$$

$= (-R^2\cos\theta\sin^2\phi, -R^2\sin\theta\sin^2\phi, -R^2\sin\phi\cos\phi)$

$\therefore \|\vec{T}_\theta \times \vec{T}_\phi\| = \sqrt{R^4\cos^2\theta\sin^4\phi + R^4\sin^2\theta\sin^4\phi + R^4\sin^2\phi\cos^2\phi}$

$= R^2\sqrt{\sin^4\phi + \sin^2\phi\cos^2\phi} = R^2\sqrt{\sin^2\phi}$

$= R^2\sin\phi$

$\therefore \|\vec{T}_\theta \times \vec{T}_\phi\| \, d\phi\, d\theta = R^2\sin\phi \, d\phi d\theta$

13.

13. Evaluate $\iint_S z\,dS$, where S is the surface $z = x^2 + y^2$, $x^2 + y^2 \le 1$.

$z = x^2+y^2$ is a paraboloid, $x^2+y^2 \le 1$ means within unit disc in xy-plane.

Use polar coordinates. A parametrization of the paraboloid is $\phi(r,\theta) = (r\cos\theta, r\sin\theta, r^2)$

since $x^2+y^2 = (r\cos\theta)^2 + (r\sin\theta)^2 = r^2 = z$

$0 \le \theta \le 2\pi$, $0 \le r \le 1$ as $r^2 = x^2+y^2 \le 1 \Rightarrow r \le 1$.

$\therefore \phi_r \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & 2r \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (-2r^2\cos\theta, -2r^2\sin\theta, r)$

as $r\cos^2\theta + r\sin^2\theta = r$

$$\therefore \|\phi_r \times \phi_\theta\| = \sqrt{4r^4\cos^2\theta + 4r^4\sin^2\theta + r^2}$$

$$= \sqrt{4r^4 + r^2} = r\sqrt{4r^2+1}$$

$$\therefore \iint_S z\,dS = \int_0^{2\pi}\int_0^1 (r^2)\, r\sqrt{4r^2+1}\,dr\,d\theta$$

$$= 2\pi \int_0^1 r^3\sqrt{4r^2+1}\,dr$$

Let $s = 2r$, $s^2 = 4r^2$, $r^3 = \frac{s^3}{8}$

$r = 0 \Rightarrow s = 0$, $r = 1 \Rightarrow s = 2$

$ds = 2dr$

$$= 2\pi\left(\frac{1}{16}\right)\int_0^2 s^3\sqrt{s^2+1}\,ds$$

Now use table of integrals #69 from text

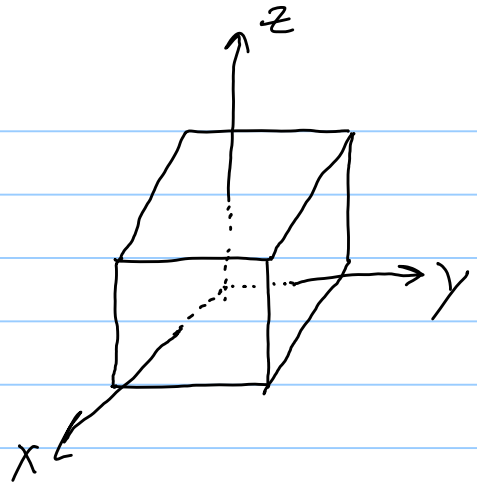
$$= \frac{\pi}{8}\left[\left(\frac{s^2}{5} - \frac{2}{15}\right)\sqrt{(1+s^2)^3}\,\Big|_{s=0}^{s=2}\right.$$

$$= \frac{\pi}{8}\left[\left(\frac{4}{5} - \frac{2}{15}\right)\sqrt{125} - \left(-\frac{2}{15}\right)\sqrt{1}\right]$$

$$= \frac{\pi}{8}\left[\frac{4}{5}\,5\sqrt{5} - \frac{2}{15}\,5\sqrt{5} + \frac{2}{15}\right] = \frac{\pi}{8}\left[4\sqrt{5} - \frac{2}{3}\sqrt{5} + \frac{2}{15}\right]$$

$$= \frac{\pi}{4}\left[2\sqrt{5} - \frac{\sqrt{5}}{3} + \frac{1}{15}\right] = \frac{\pi}{4}\left[\frac{25\sqrt{5}+1}{15}\right]$$

14. **14.** Evaluate the surface integral $\iint_S z^2\,dS$, where S is the boundary of the cube $C = [-1, 1] \times [-1, 1] \times [-1, 1]$. (HINT: Do each face separately and add the results.)



By symmetry $\iint_S z^2\,dS$ is the same over the top surface as for the bottom surface.

Each side is also the same and z^2 is the same over each side.

$\therefore$ Only need to compute $\iint_S z^2\,dS$ for the top surface and one side surface.

Answer = 4 (side surface) + 2 (top surface).

For top surface, $\phi(x,y) = (x, y, 1)$, $-1 \le x \le 1$, $-1 \le y \le 1$.

$$\therefore \phi_x \times \phi_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = (0, 0, 1).$$

$\therefore \|\phi_x \times \phi_y\| = 1$. Note $z^2 = 1$.

$$\therefore \text{Top Surface} = \iint_S z^2 \|\phi_x \times \phi_y\|\,dx\,dy = \int_{-1}^{1}\int_{-1}^{1} dx\,dy = \underline{\underline{4}}$$

For a side surface, use $\phi(y,z) = (1, y, z)$,

$-1 \le y \le 1, \quad -1 \le z \le 1.$

$$\therefore \phi_y \times \phi_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = (1, 0, 0).$$

$$\therefore \| \phi_y \times \phi_z \| = 1.$$

$$\therefore \text{Side surface} = \iint_S z^2 \, \| \phi_y \times \phi_z \| \, dy \, dz$$

$$= \int_{-1}^{1} \int_{-1}^{1} z^2 \, dy \, dz = 2 \int_{-1}^{1} z^2 \, dz = 2 \left[\frac{z^3}{3} \right]_{z=-1}^{z=1}$$

$$= 2 \left[\frac{1}{3} - \left(-\frac{1}{3} \right) \right] = \underline{\frac{4}{3}}$$

$$\therefore 2(4) + 4\left(\frac{4}{3}\right) = 8 + \frac{16}{3} = \underline{13\frac{1}{3}} = \frac{40}{3}$$

15.

15. Find the mass of a spherical surface S of radius R such that at each point $(x, y, z) \in S$ the mass density is equal to the distance of (x, y, z) to some fixed point $(x_0, y_0, z_0) \in S$.

Using symmetry, assume $(x_0, y_0, z_0) = (0, 0, R)$,

$\therefore m(x,y,z) = \sqrt{x^2 + y^2 + (z-R)^2}$, since $(x_0, y_0, z_0) \in S$.

Use spherical coordinates for S: $\vec{T}(\theta, \phi) =$

$(R\cos\theta\sin\phi,\ R\sin\theta\sin\phi,\ R\cos\phi)$, $0\le\theta\le 2\pi$, $0\le\phi\le\pi$

$\therefore$ From problem #12, $\|\vec{T}_\theta \times \vec{T}_\phi\| = R^2\sin\phi$

$m(R\cos\theta\sin\phi,\ R\sin\theta\sin\phi,\ R\cos\phi) =$

$$\sqrt{R^2(\cos^2\theta+\sin^2\theta)\sin^2\phi + (R\cos\phi - R)^2}$$

$$= \sqrt{R^2\sin^2\phi + R^2\cos^2\phi - 2R^2\cos\phi + R^2}$$

$$= \sqrt{2R^2 - 2R^2\cos\phi} = \sqrt{2}R\sqrt{1-\cos\phi}$$

$\therefore$

$$M = \iint_S m(x,y,z)\,dS = \iint_D m(\theta,\phi)\,\|\vec{T}_\theta\times\vec{T}_\phi\|\,d\phi\,d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \sqrt{2}R\sqrt{1-\cos\phi}\,(R^2\sin\phi)\,d\phi\,d\theta$$

$$= \sqrt{2}R^3(2\pi)\left(\tfrac{2}{3}\right)(1-\cos\phi)^{3/2}\Big|_{\phi=0}^{\phi=\pi}$$

$$= \sqrt{2}\,\tfrac{4}{3}\pi R^3\left[(1+1)^{3/2} - (1-1)^{3/2}\right]$$

$$= \sqrt{2}\,\tfrac{4}{3}\pi R^3(2\sqrt{2}) = \underline{\underline{\tfrac{16}{3}\pi R^3}}$$

16.

16. A metallic surface S is in the shape of a hemisphere $z = \sqrt{R^2 - x^2 - y^2}$, where (x, y) satisfies $0 \le x^2 + y^2 \le R^2$. The mass density at $(x, y, z) \in S$ is given by $m(x, y, z) = x^2 + y^2$. Find the total mass of S.

Using spherical coordinates, $0 \le \theta \le 2\pi$, $0 \le \phi \le \frac{\pi}{2}$,

$$\vec{T}(\theta, \phi) = \{R\cos\theta\sin\phi, R\sin\theta\sin\phi, R\cos\phi\}.$$

$$\| \vec{T}_\theta \times \vec{T}_\phi \| = R^2 \sin\phi.$$

$$m(x,y,z) \Rightarrow m(\theta,\phi) = (R\cos\theta\sin\phi)^2 + (R\sin\theta\sin\phi)^2$$

$$= R^2\sin^2\phi$$

$$\therefore M = \iint_S m(x,y,z)\,dS = \iint_D m(\theta,\phi)\,\| \vec{T}_\theta \times \vec{T}_\phi \|\,d\theta\,d\phi$$

$$= \int_0^{2\pi}\int_0^{\pi/2} (R^2\sin^2\phi)(R^2\sin\phi)\,d\phi\,d\theta$$

$$= 2\pi R^4 \int_0^{\pi/2} \sin^3\phi\,d\phi = 2\pi R^4 \int_0^{\frac{\pi}{2}} (1-\cos^2\phi)\sin\phi\,d\phi$$

$$= 2\pi R^4 \int_0^{\pi/2} \sin\phi\,d\phi + 2\pi R^4 \int_0^{\pi/2} \cos^2\phi\,(-\sin\phi)\,d\phi$$

$$= 2\pi R^4 \left[-\cos\phi\right]_0^{\pi/2} + 2\pi R^4 \left[\frac{\cos^3\phi}{3}\right]_0^{\pi/2}$$

$$= 2\pi R^4 \left[0-(-1)\right] + 2\pi R^4 \left[0 - \frac{1}{3}\right]$$

$$= 2\pi R^4 \left[1 - \frac{1}{3}\right] = \frac{4}{3}\pi R^4$$

17.

17. Let S be the sphere of radius R.

(a) Argue by symmetry that

$$\iint_S x^2\, dS = \iint_S y^2\, dS = \iint_S z^2\, dS.$$

(a) The summation of a function solely along an axis is the same irrespective of the chosen axis because of the symmetry of the sphere relative to any axis.

(b)

(b) Use this fact and some clever thinking to evaluate, with very little computation, the integral

$$\iint_S x^2\, dS.$$

$$x^2 + y^2 + z^2 = R^2$$

$$\therefore \iint_S (x^2+y^2+z^2)\, dS = \iint_S R^2\, ds = R^2 \iint_S ds$$

$$= R^2\left(4\pi R^2\right) = 4\pi R^4$$

$$\iint_S (x^2+y^2+z^2)\, ds = \iint_S x^2\, ds + \iint_S y^2\, ds + \iint_S z^2\, ds$$

$$= 3\iint_S x^2\, ds$$

$$\therefore \quad 4\pi R^4 = \iint_S (x^2+y^2+z^2)\,dS = 3\iint_S x^2\,dS$$

$$\therefore \quad \iint_S x^2\,dS = \underline{\underline{\frac{4}{3}\pi R^4}}$$

(c) (c) Does this help in Exercise 16?

In #16, integrating only over half a sphere.

If integrating over entire sphere,

$$\text{Mass} = \iint_S (x^2+y^2)\,dS = \iint_S x^2\,dS + \iint_S y^2\,dS$$

$$= 2\iint_S x^2\,dS = 2\left(\frac{4}{3}\pi R^4\right)$$

From symmetry, mass of whole sphere = twice mass of hemisphere.

$$\therefore \text{Hemisphere mass} = \frac{1}{2}(2)\left(\frac{4}{3}\pi R^4\right) = \underline{\underline{\frac{4}{3}\pi R^4}}$$

So, yes, this does help in #16.

18.

18. (a) Use Riemann sums to justify the formula

$$\frac{1}{A(S)} \iint_S f(x, y, z)\, dS$$

for the *average value* of f over the surface S.

If $\phi(u,v)$ is a paramitrization of S, $\phi : D \to S$, and R_{ij} is a small rectangle in D, $\phi(R_{ij}) = S_{ij}$ a portion of S corrrsponding to R_{ij} through ϕ, Then the total sum of $f(x,y,z)$ over S is $S_n = \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} f(\phi(u_i, v_i))\, A(S_{ij})$, where

$A(S_{ij})$ = area of S_{ij}. The approximation of the total sum, S_n, becomes better as $n \to \infty$.

If $A(S)$ = total surface area, an approximation of the average of $f()$ over S is: $\frac{1}{A(S)}[S_n]$.

$\therefore$ As $n \to \infty$, the average of $f()$ over S is $\frac{1}{A(S)} \iint_S f(x,y,z)\, dS$ since $\lim_{n\to\infty} S_n = \iint_S f(x,y,z)\, dS$.

(6) (b) In Example 3 of this section, show that the average of $f(x, y, z) = z^2$ over the sphere is $1/3$.

From Example 3, p. 395 of text, $\iint_S z^2\,dS = \frac{4\pi}{3}$

where S is a unit sphere.

But $A(S) = 4\pi(1)^2 = 4\pi$.

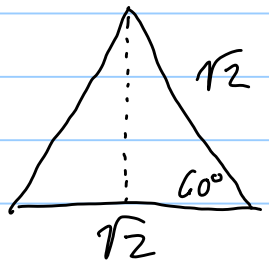
$\therefore$ Ave value $= \frac{4\pi}{3} \Big/ A(S) = \frac{4\pi}{3} \Big/ 4\pi = \frac{1}{3}$

(c) (c) Define the ***center of gravity*** $(\bar{x}, \bar{y}, \bar{z})$ of a surface S to be such that $\bar{x}$, $\bar{y}$, and $\bar{z}$ are the average values of the x, y, and z coordinates on S. Show that the center of gravity of the triangle in Example 4 of this section is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

i.e., $\bar{x} = \dfrac{\iint_S x\,dS}{A(S)}$

From Example 4, p. 397 of text, $\iint_S x\,dS = \frac{\sqrt{3}}{6}$

The triangle is an equilateral triangle with side length $= \sqrt{(1-0)^2 + (0-1)^2 + (0-0)^2} = \sqrt{2}$

$\therefore$ 

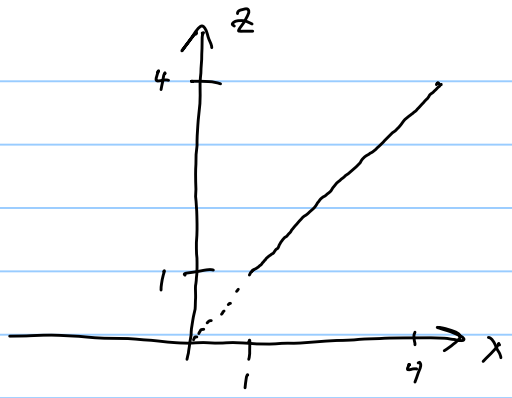
Height $= \sqrt{2}\sin 60° = \sqrt{2}\left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{6}}{2}$

$\therefore$ Area $= \frac{1}{2}(\sqrt{2})\left(\frac{\sqrt{6}}{2}\right) = \frac{\sqrt{12}}{4}$

$\therefore \bar{x} = \frac{\sqrt{3}}{6} \Big/ \frac{\sqrt{12}}{4} = \frac{4}{6}\frac{\sqrt{3}}{\sqrt{12}} = \frac{2}{3}\sqrt{\frac{1}{4}} = \frac{2}{3}\cdot\frac{1}{2} = \frac{1}{3}$

By symmetry, $\bar{y} = \frac{1}{3}$, $\bar{z} = \frac{1}{3}$ $\therefore (\bar{x}, \bar{y}, \bar{z}) = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$

19. **19.** Find the average value of $f(x, y, z) = x + z^2$ on the truncated cone $z^2 = x^2 + y^2$, with $1 \le z \le 4$.



$$\text{Ave} = \frac{\iint_S (x + z^2)\, dS}{A(S)}$$

(a) From geometry, or using problem #23 of section 7.4, p. 392 of text, $A(S) = \pi(b^2 - a^2)\sqrt{m^2 + 1}$

$= \pi(16 - 1)\sqrt{1 + 1} = 15\sqrt{2}\,\pi$, as $b = 4$, $a = 1$, $m = 1$.

(b) Find parametrization of S using cylindrical coordinates.

$x = r\cos\theta$, $y = r\sin\theta$, $z = z$, $0 \le \theta \le 2\pi$, $1 \le z \le 4$,

$x^2 + y^2 = r^2 = z^2$, $\therefore r = z$.

$\therefore \phi(\theta, z) = (z\cos\theta, z\sin\theta, z)$

$$\therefore \phi_\theta \times \phi_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -z\sin\theta & z\cos\theta & 0 \\ \cos\theta & \sin\theta & 1 \end{vmatrix} = (z\cos\theta,\ z\sin\theta,\ -z\sin^2\theta - z\cos^2\theta)$$

$$= (z\cos\theta, z\sin\theta, -z)$$

$$\therefore \|\phi_\theta \times \phi_z\| = \sqrt{z^2 + z^2} = \sqrt{2}\, z$$

$$\therefore \iint_S (x+z^2)\,dS = \int_0^{2\pi}\int_1^4 (z\cos\theta + z^2)(\sqrt{2}\,z)\,dz\,d\theta$$

$$= \sqrt{2}\int_0^{2\pi}\int_1^4 (z^2\cos\theta + z^3)\,dz\,d\theta = \sqrt{2}\int_0^{2\pi}\left[\cos\theta\,\frac{z^3}{3} + \frac{z^4}{4}\right]_{z=1}^{4} d\theta$$

$$= \sqrt{2}\int_0^{2\pi}\left[\frac{64}{3}\cos\theta + \frac{256}{4} - \left(\frac{\cos\theta}{3} + \frac{1}{4}\right)\right] d\theta$$

$$= \sqrt{2}\int_0^{2\pi}\left(21\cos\theta + \frac{255}{4}\right) d\theta = \sqrt{2}\left[21\sin\theta + \frac{255}{4}\theta\right]_0^{2\pi}$$

$$= \sqrt{2}\left[0 + \frac{255}{2}\pi - (0+0)\right] = \frac{255\sqrt{2}\,\pi}{2}$$

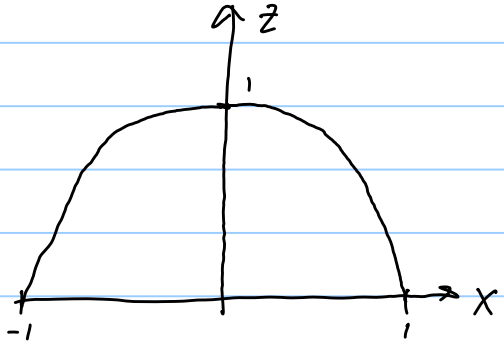
(c) $$\text{Ave} = \frac{\frac{255\sqrt{2}\,\pi}{2}}{15\sqrt{2}\,\pi} = \frac{17}{2}$$

20.

20. Evaluate the integral

$$\iint_S (1-z)\,dS,$$

where S is the graph of $z = 1 - x^2 - y^2$, with $x^2 + y^2 \le 1$.



A paraboloid. Use cylindrical coordinates

Let $x = r\cos\theta$, $y = r\sin\theta$, $z = 1-(x^2+y^2) = 1-r^2$

$0 \le \theta \le 2\pi, \quad 0 \le r \le 1.$

Let $\phi(r, \theta) = (r\cos\theta, r\sin\theta, 1-r^2)$

$$\therefore \phi_r \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & -2r \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (2r^2\cos\theta, 2r^2\sin\theta, r\cos^2\theta + r\sin^2\theta)$$

$$= (2r^2\cos\theta, 2r^2\sin\theta, r)$$

$$\|\phi_r \times \phi_\theta\| = \sqrt{4r^4 + r^2} = r\sqrt{4r^2+1}$$

$$\therefore \iint_S (1-z)\,dS = \int_0^{2\pi}\int_0^1 (1-r^2)(r\sqrt{4r^2+1})\,dr\,d\theta$$

$$= 2\pi \int_0^1 r\sqrt{4r^2+1}\,dr - 2\pi \int_0^1 r^3\sqrt{4r^2+1}\,dr$$

$$= \frac{2\pi}{8}\int_0^1 8r\sqrt{4r^2+1}\,dr - \frac{\pi}{4}\left[\frac{25\sqrt{5}+1}{15}\right]$$

See answer to #13 above

$$= \frac{\pi}{4}\left(\frac{2}{3}\right)(4r^2+1)^{3/2}\Big|_{r=0}^{r=1} - \frac{\pi}{4}\left[\frac{25\sqrt{5}+1}{15}\right]$$

$$= \frac{\pi}{4}\left(\frac{2}{3}\right)\left[5\sqrt{5}-1\right] - \frac{\pi}{4}\left[\frac{5\sqrt{5}}{3} + \frac{1}{15}\right]$$

$$= \frac{\pi}{4}\left[\frac{10\sqrt{5}}{3} - \frac{2}{3} - \frac{5\sqrt{5}}{3} - \frac{1}{15}\right]$$

$$= \frac{\pi}{4}\left[\frac{5\sqrt{5}}{3} - \frac{11}{15}\right]$$

21. **21.** Find the x, y, and z coordinates of the center of gravity of the octant of the solid sphere of radius R and centered at the origin determined by $x \geq 0$, $y \geq 0$, $z \geq 0$. (HINT: Write this octant as a parametrized surface—see Example 3 of this section and Exercise 18.)

Interpret "solid sphere" to be the shell, not the ball.

$$\therefore \bar{x} = \frac{\iint_S x\,dS}{A(S)}, \quad \bar{y} = \frac{\iint_S y\,dS}{A(S)}, \quad \bar{z} = \frac{\iint_S z\,dS}{A(S)},$$

using #18(c) above. By symmetry, $\bar{x} = \bar{y} = \bar{z}$.

$A(S) = 4\pi R^2/8 = \frac{\pi R^2}{2}$, as there are 8 octants to a sphere.

For the octant $x \geq 0, y \geq 0, z \geq 0$, let

$$T(\theta, \phi) = (R\cos\theta\sin\phi, R\sin\theta\sin\phi, R\cos\phi),$$

$$0 \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \phi \leq \frac{\pi}{2}.$$

$\|\vec{T}_\theta \times \vec{T}_\phi\| = R^2\sin\phi$ as seen from Example 3 and pages 395-396 of text.

$$\therefore \iint_S x\,dS = \int_0^{\pi/2}\int_0^{\pi/2} (R\cos\theta\sin\phi)(R^2\sin\phi)\,d\phi\,d\theta$$

$$= R^3 \int_0^{\frac{\pi}{2}} \cos\theta\,d\theta \int_0^{\frac{\pi}{2}} \sin^2\phi\,d\phi \qquad \text{use } \sin^2\theta = \frac{1-\cos 2\theta}{2}$$

$$= R^3 \left[\sin\theta\right]_0^{\pi/2} \left[\frac{1}{2}\theta - \frac{\sin 2\theta}{4}\right]_0^{\pi/2}$$

$$= R^3[1-0]\left[\frac{\pi}{4} - 0 - (0-0)\right] = \frac{\pi R^3}{4}$$

$$\therefore \bar{x} = \frac{\frac{\pi R^3}{4}}{\frac{\pi R^2}{2}} = \frac{R}{2}$$

$$\therefore (\bar{x}, \bar{y}, \bar{z}) = \left(\frac{R}{2}, \frac{R}{2}, \frac{R}{2}\right)$$

22. **22.** Find the z coordinate of the center of gravity (the average z coordinate) of the surface of a hemisphere ($z \le 0$) with radius r (see Exercise 18). Argue by symmetry that the average x and y coordinates are both zero.

From #21 above, $\bar{z}$ for each octant with $z \le 0$ is $-\frac{r}{2}$. $\therefore$ The entire hemisphere with $z \le 0$ will have $\bar{z} = -\frac{r}{2}$.

From symmetry, $|\bar{x}| = |\bar{y}| = |\bar{z}|$ for the 4

octants below the xy-plane. If you add each $(\bar{x}, \bar{y}, \bar{z})$ and divide by 4, you will get

$$\left[\left(\frac{r}{2}, \frac{r}{2}, -\frac{r}{2}\right) + \left(-\frac{r}{2}, \frac{r}{2}, -\frac{r}{2}\right) + \left(\frac{r}{2}, -\frac{r}{2}, -\frac{r}{2}\right) + \left(-\frac{r}{2}, -\frac{r}{2}, -\frac{r}{2}\right)\right]/4$$

$$= \left(0, 0, -\frac{r}{2}\right).$$

Alternatively, $\bar{z} = \iint_S z\,dS \,/\, A(S)$ from #18 above, and $\iint_S z\,dS = -\pi r^3$ from #9.

$$A(S) = \frac{4\pi r^2}{2} = 2\pi r^2.$$

$$\therefore \bar{z} = -\pi r^3 / 2\pi r^2 = -\frac{r}{2}$$

23.

23. Let $\Phi: D \subset \mathbb{R}^2 \to \mathbb{R}^3$ be a parametrization of a surface S defined by

$$x = x(u, v), \qquad y = y(u, v), \qquad z = z(u, v).$$

(a)

(a) Let

$$\frac{\partial \Phi}{\partial u} = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right) \quad \text{and} \quad \frac{\partial \Phi}{\partial v} = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right),$$

that is, $\partial \Phi / \partial u = \mathbf{T}_u$ and $\partial \Phi / \partial v = \mathbf{T}_v$, and set

$$E = \left\| \frac{\partial \Phi}{\partial u} \right\|^2, \qquad F = \frac{\partial \Phi}{\partial u} \cdot \frac{\partial \Phi}{\partial v}, \qquad G = \left\| \frac{\partial \Phi}{\partial v} \right\|^2.$$

Show that

$$\sqrt{EG - F^2} = \|\mathbf{T}_u \times \mathbf{T}_v\|,$$

and that the surface area of S is

$$A(S) = \iint_D \sqrt{EG - F^2}\, du\, dv.$$

In this notation, how can we express $\iint_S f\,dS$ for a general function of f?

$$\vec{T_u} \times \vec{T_v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_u & y_u & z_u \\ x_v & y_v & z_v \end{vmatrix} = (y_u z_v - y_v z_u,\ x_v z_u - x_u z_v,\ x_u y_v - x_v y_u)$$

$$\therefore \left\| \vec{T}_u \times \vec{T}_v \right\|^2 = (x_u y_v - x_v y_u)^2 + (x_v z_u - x_u z_v)^2 + (y_u z_v - y_v z_u)^2$$

$$E = x_u^2 + y_u^2 + z_u^2 \qquad G = x_v^2 + y_v^2 + z_v^2$$

$$F = x_u x_v + y_u y_v + z_u z_v$$

One could go through the algebraic manipulations, but note from Exercise # 24, Section 1.3, p. 50 of the text that $(\vec{u} \times \vec{v}) \cdot (\vec{u}' \times \vec{v}') = \begin{vmatrix} \vec{u} \cdot \vec{u}' & \vec{u} \cdot \vec{v}' \\ \vec{u}' \cdot \vec{v} & \vec{v} \cdot \vec{v}' \end{vmatrix}$

so that $\left\| \vec{T}_u \times \vec{T}_v \right\|^2 = (\vec{T}_u \times \vec{T}_v) \cdot (\vec{T}_u \times \vec{T}_v)$

$$= \begin{vmatrix} \vec{T}_u \cdot \vec{T}_u & \vec{T}_u \cdot \vec{T}_v \\ \vec{T}_u \cdot \vec{T}_v & \vec{T}_v \cdot \vec{T}_v \end{vmatrix}$$

$$= \left\| \vec{T}_u \right\|^2 \left\| \vec{T}_v \right\|^2 - (\vec{T}_u \cdot \vec{T}_v)(\vec{T}_u \cdot \vec{T}_v)$$

$$= (E)(G) - (F)(F)$$

$$\therefore \left\| \vec{T}_u \times \vec{T}_v \right\|^2 = EG - F^2 \Rightarrow \underline{\left\| \vec{T}_u \times \vec{T}_v \right\| = \sqrt{EG - F^2}}$$

$$\therefore \underline{A(S) = \iint_D \left\| \vec{T}_u \times \vec{T}_v \right\| du\,dv = \iint_D \sqrt{EG - F^2}\, du\,dv}$$

$$\therefore \iint_S f\,dS = \iint_D f(x(u,v), y(u,v), z(u,v)) \sqrt{EG - F^2}\,du\,dv$$

$$= \iint_D f(\phi(u,v)) \sqrt{EG - F^2}\,du\,dv$$

(6) (b) What does the formula for $A(S)$ become if the vectors $\partial\Phi/\partial u$ and $\partial\Phi/\partial v$ are orthogonal?

$$\vec{T_u} \perp \vec{T_v} \Rightarrow \vec{T_u} \cdot \vec{T_v} = 0 \Rightarrow F = 0$$

$$\therefore A(S) = \iint_D \sqrt{EG}\,du\,dv$$

(c) (c) Use parts (a) and (b) to compute the surface area of a sphere of radius a.

Let $T(\theta, \phi) = (a\cos\theta\sin\phi, a\sin\theta\sin\phi, a\cos\phi)$.

$$\vec{T_\theta} = (-a\sin\theta\sin\phi, a\cos\theta\sin\phi, 0)$$

$$\vec{T_\phi} = (a\cos\theta\cos\phi, a\sin\theta\cos\phi, -a\sin\phi)$$

Note $\vec{T_\theta} \cdot \vec{T_\phi} = -a^2\sin\theta\cos\theta\sin\phi\cos\phi + a^2\sin\theta\cos\theta\sin\phi\cos\phi + 0 = 0$

$$E = a^2\sin^2\theta\sin^2\phi + a^2\cos^2\theta\sin^2\phi + 0$$

$$= a^2 \sin^2\phi \left(\sin^2\theta + \cos^2\theta\right) = a^2 \sin^2\phi$$

$$G = a^2 \cos^2\theta \cos^2\phi + a^2 \sin^2\theta \cos^2\phi + a^2 \sin^2\phi$$

$$= a^2 \cos^2\phi \left(\cos^2\theta + \sin^2\theta\right) + a^2 \sin^2\phi$$

$$= a^2 \cos^2\phi + a^2 \sin^2\phi = a^2$$

$$\therefore EG = \left(a^2 \sin^2\phi\right)\left(a^2\right) = a^4 \sin^2\phi$$

$$\therefore A(S) = \iint_D \sqrt{EG}\, d\theta\, d\phi = \iint_D a^2 \sin\phi\, d\theta\, d\phi$$

$$= a^2 \int_0^{2\pi} \int_0^{\pi} \sin\phi\, d\phi\, d\theta = 2\pi a^2 \int_0^{\pi} \sin\phi\, d\phi$$

$$= 2\pi a^2 \left[-\cos\phi\right]_0^{\pi} = 2\pi a^2 \left[1 - (-1)\right]$$

$$= 4\pi a^2$$

24.

24. ***Dirichlet's functional*** for a parametrized surface $\mathbf{\Phi}\colon D \to \mathbb{R}^3$ is defined by[11]

$$J(\mathbf{\Phi}) = \frac{1}{2} \iint_D \left(\left\| \frac{\partial \mathbf{\Phi}}{\partial u} \right\|^2 + \left\| \frac{\partial \mathbf{\Phi}}{\partial v} \right\|^2 \right) du\, dv.$$

Use Exercise 23 to argue that the area $A(\mathbf{\Phi}) \le J(\mathbf{\Phi})$ and equality holds if

(a) $\left\| \dfrac{\partial \mathbf{\Phi}}{\partial u} \right\|^2 = \left\| \dfrac{\partial \mathbf{\Phi}}{\partial v} \right\|^2$ and (b) $\dfrac{\partial \mathbf{\Phi}}{\partial u} \cdot \dfrac{\partial \mathbf{\Phi}}{\partial v} = 0.$

Compare these equations with Exercise 23 and the remarks at the end of Section 7.4. A parametrization $\mathbf{\Phi}$ that satisfies conditions (a) and (b) is said to be ***conformal***.

$$A(\phi) = \iint_D \sqrt{EG - F^2}\, du\, dv \quad \text{and}$$

$$J(\phi) = \frac{1}{2}\iint_D (E + G)\, du\, dv$$

where $E = \|\phi_u\|^2$, $G = \|\phi_v\|^2$, $F = \phi_u \cdot \phi_v$ (dot product)

(1) If $\phi_u \cdot \phi_v = 0$, then $F = 0$ and if $\|\phi_u\|^2 = \|\phi_v\|^2$, then $E = G$, so

$$\sqrt{EG - F^2} = \sqrt{E^2} = \sqrt{G^2} = E = G$$

$$\therefore A(\phi) = \iint_D E\, du\, dv = \iint_D G\, du\, dv \quad \text{and}$$

$$J(\phi) = \frac{1}{2}\iint_D (E + E)\, du\, dv = \iint_D E\, du\, dv$$

$$\therefore A(\phi) = J(\phi)$$

(2) Since $E \geq 0$, $G \geq 0$, $F^2 \geq 0$, then

$$EG - F^2 \leq EG \Rightarrow \sqrt{EG - F^2} \leq \sqrt{EG}$$

$$\sqrt{EG} \leq \frac{1}{2}(E + G) \Leftrightarrow 2\sqrt{EG} \leq (E + G)$$

$$\Leftrightarrow 4EG \leq (E + G)^2 = E^2 + 2EG + G^2$$

$$\iff 0 \le E^2 - 2EG + G^2$$

$$\iff 0 \le (E-G)^2$$

Since $(E-G)^2 \ge 0$, then $\sqrt{EG} \le \frac{1}{2}(E+G)$

$\therefore \sqrt{EG-F^2} \le \frac{1}{2}(E+G)$

$\therefore$ Given any partition of D, for each R_{ij} of D, $\sqrt{EG-F^2} \le \frac{1}{2}(E+G)$.

$$\therefore A_n = \sum_{i=1}^{n-1}\sum_{j=1}^{n-1} \sqrt{EG-F^2}\,\Delta u \Delta v \le \sum_{i=1}^{n-1}\sum_{j=1}^{n-1} \frac{1}{2}(E+G)\Delta u \Delta v = J_n$$

where E, G, and F are evaluated over each R_{ij}.

$$\therefore A(\phi) = \lim_{n\to\infty} A_n \le \lim_{n\to\infty} J_n = J(\phi)$$

25.

25. Let $D \subset \mathbb{R}^2$ and $\Phi\colon D \to \mathbb{R}^2$ be a smooth function $\Phi(u, v) = (x(u, v), y(u, v))$ satisfying conditions (a) and (b) of Exercise 16 and assume that

$$\det \begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} > 0.$$

Show that x and y satisfy the ***Cauchy–Riemann equations*** $\partial x/\partial u = \partial y/\partial v$, $\partial x/\partial v = -\partial y/\partial u$. Conclude that $\nabla^2 \Phi = 0$ (i.e., each component of Φ is harmonic).

Error: (a) and (b) of Exercise 24

Exercise 24 was Exercise 16 in the 5th edition.

For ease of visualization, let $\phi_u = (x_u, y_u) = (a, b)$

$\phi_v = (x_v, y_v) = (c, d)$.

Condition (a) $\Rightarrow \|\phi_u\|^2 = \|\phi_v\|^2 \Rightarrow a^2 + b^2 = c^2 + d^2$ [1]

Condition (b) $\Rightarrow \phi_u \cdot \phi_v = 0 \Rightarrow (a, b) \cdot (c, d) = 0 \Rightarrow$

$$ac + bd = 0 \quad [2]$$

$$\det \begin{bmatrix} x_u & x_v \\ y_u & y_v \end{bmatrix} > 0 \Rightarrow ad - bc > 0 \quad [3]$$

Need to prove: $a = d$ and $b = -c$

(1) If $a = 0$, then [2] $\Rightarrow bd = 0 \Rightarrow b = 0$ or $d = 0$

If $b = 0$, then $ad - bc = 0$, contradicting [3]

$\therefore d = 0$, and [1] $\Rightarrow b^2 = c^2 \Rightarrow b = c$ or $b = -c$

If $b = c$, then [3] becomes $-b^2 > 0$, a contradiction

$\therefore b = -c$, no contradiction in [1], [2], [3]

$\therefore a = 0 \Rightarrow d = 0 \Rightarrow b = -c$, so

$\underline{a = d}$ and $\underline{b = -c}$ and Theorem true.

(2) Assume $a \neq 0$ $\therefore$ [2] $\Rightarrow c = -\frac{bd}{a}$

$\therefore$ [1] $\Rightarrow a^2 + b^2 = \left(-\frac{bd}{a}\right)^2 + d^2$, or

$a^4 + a^2b^2 = b^2d^2 + a^2d^2$, or

$a^2(a^2 - d^2) = b^2(d^2 - a^2)$ [4]

If $a \neq d$, then $a^2 - d^2 \neq 0$, so [4] $\Rightarrow$

$a^2 = -b^2$. But $a^2 > 0$ and it can't be true that $-b^2 > 0$.

$\therefore a = d$ and $\therefore d \neq 0$

$\therefore$ [2] $\Rightarrow ac + ba = 0 \Rightarrow c + b = 0 \Rightarrow b = -c$

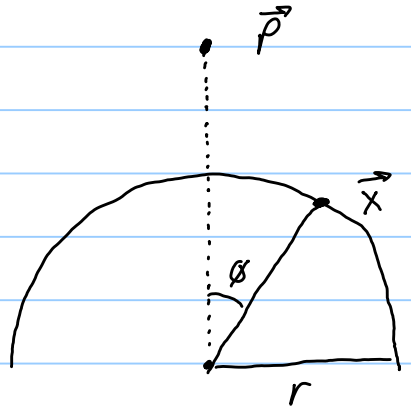
$\therefore a \neq 0 \Rightarrow$ $a = d$ and $b = -c$ and the Theorem is true.

26.

26. Let S be a sphere of radius r and $\mathbf{p}$ be a point inside or outside the sphere (but not on it). Show that

$$\iint_S \frac{1}{\|\mathbf{x} - \mathbf{p}\|}\, dS = \begin{cases} 4\pi r & \text{if } \mathbf{p} \text{ is inside } S \\ 4\pi r^2/d & \text{if } \mathbf{p} \text{ is outside } S, \end{cases}$$

where d is the distance from $\mathbf{p}$ to the center of the sphere and the integration is over the sphere. [HINT: Assume $\mathbf{p}$ is on the z-axis. Then change variables and evaluate. Why is this assumption on $\mathbf{p}$ justified?]



Using symmetry, can assume $\vec{p}$ is on any axis.

So, assume it is on the z-axis for ease of calculations.

$\vec{x}$ is on S.

Using spherical coordinates, $\vec{x}$ on S can be described as $(r\cos\theta\sin\phi, r\sin\theta\sin\phi, r\cos\phi)$.

$\vec{p}$ can be described as $(0,0,d)$

$$\therefore \|\vec{x}-\vec{p}\| = \sqrt{(r^2\cos^2\theta\sin^2\phi)+(r^2\sin^2\theta\sin^2\phi)+(r\cos\phi-d)^2}$$

$$= \sqrt{r^2\sin^2\phi + r^2\cos^2\phi - 2rd\cos\phi + d^2}$$

$$= \sqrt{r^2 - 2rd\cos\phi + d^2}$$

This expression holds for $\vec{p}$ inside or outside the sphere. If $\vec{p}$ were on the sphere, $r=d$. If $\vec{p}$ is at the center, $d=0$, so $\|\vec{x}-\vec{p}\| = r$.

For a sphere using spherical coordinates as a parametrization, $\|\vec{T}_\theta \times \vec{T}_\phi\| = r^2\sin\phi$

$$\therefore \iint_S \frac{1}{\|\vec{x}-\vec{p}\|}\,dS = \int_0^{2\pi}\int_0^{\pi} \frac{r^2\sin\phi}{\sqrt{r^2+d^2-2rd\cos\phi}}\,d\phi\,d\theta$$

$$= 2\pi r^2 \int_0^{\pi} \frac{\sin\phi}{\sqrt{r^2+d^2-2rd\cos\phi}}\,d\phi \qquad [1]$$

If $d=0$, this becomes $2\pi r^2 \int_0^{\pi} \frac{\sin\phi}{\sqrt{r^2}}\, d\phi$

$= 2\pi r \int_0^{\pi} \sin\phi\, d\phi = 2\pi r \left[-\cos\phi\right]_0^{\pi} = 2\pi r\left[1-(-1)\right]$

$= \underline{4\pi r}$

For $d \neq 0$, [1] becomes $\frac{2\pi r^2}{2rd} \int_0^{\pi} \frac{2rd\sin\phi}{\sqrt{r^2+d^2-2rd\cos\phi}}\, d\phi$

$= \frac{\pi r}{d}\left[(2)(r^2+d^2-2rd\cos\phi)^{\frac{1}{2}}\right]_{\phi=0}^{\phi=\pi}$

$= \frac{2\pi r}{d}\left[(r^2+d^2+2rd)^{\frac{1}{2}} - (r^2+d^2-2rd)^{\frac{1}{2}}\right]$

$= \frac{2\pi r}{d}\left[\sqrt{(r+d)^2} - \sqrt{(r-d)^2}\right]$ [2]

Inside S, $d<r$ and $\sqrt{(r-d)^2} = r-d$

$\therefore$ [2] becomes $\frac{2\pi r}{d}\left[(r+d)-(r-d)\right] = \underline{4\pi r}$

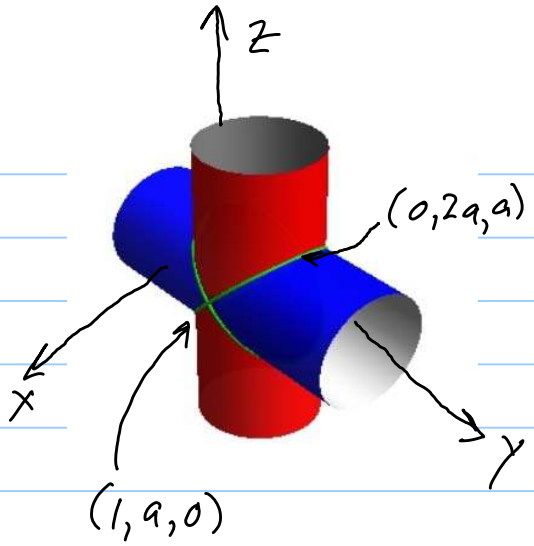
$4\pi r$ is also the result for $d=0$.

Outside S, $d>r$, so $\sqrt{(r-d)^2} = \sqrt{(d-r)^2} = d-r$

$\therefore$ [2] becomes $\frac{2\pi r}{d}\left[(r+d)-(d-r)\right] = \underline{\frac{4\pi r^2}{d}}$

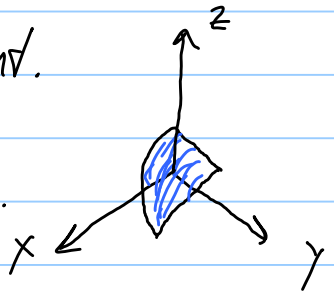
27.

27. Find the surface area of that part of the cylinder $x^2 + z^2 = a^2$ that is inside the cylinder $x^2 + y^2 = 2ay$ and also in the positive octant ($x \geq 0$, $y \geq 0$, $z \geq 0$). Assume $a > 0$.



$x^2 + y^2 = 2ay \Leftrightarrow x^2 + (y-a)^2 = a^2$,
a cylinder perpendicular to the xy-plane, centred at $(0, a, z)$

From the image, it would be the top portion of the blue pipe in the positive octant.

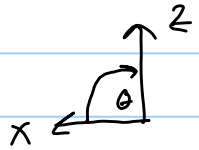


Cylinders suggest cylindrical coordinates.

$T(\theta, y) = (a\cos\theta, y, a\sin\theta) = (x, y, z)$.

where T represents the "blue" cylinder.

Here, $0 \leq \theta \leq \frac{\pi}{2}$:



From $x^2 + (y-a)^2 = a^2$, $(y-a)^2 = a^2 - x^2$,

$y - a = \sqrt{a^2 - x^2}$ since in positive octant.

$\therefore y = a + \sqrt{a^2 - x^2} = a + \sqrt{a^2 - (a\cos\theta)^2} = a + a\sin\theta$

Note when $\theta = 0$, $y = a$, so $(x, y, z) = (1, a, 0)$

when $\theta = \frac{\pi}{2}$, $y = 2a$, so $(x, y, z) = (0, 2a, a)$

$\therefore$ For $0 \le \theta \le \frac{\pi}{2}$, $y(\theta) = a + a\sin\theta$, so $a \le y \le 2y$

$\therefore$ Will be getting surface area of "outer" half of blue cylinder. $\therefore$ Multiply by 2.

$\therefore$

$$A(S) = 2\int_0^{\frac{\pi}{2}} \int_a^{a+a\sin\theta} \| \vec{T}_\theta \times \vec{T}_y \| \, dy \, d\theta$$

$$\vec{T}_\theta \times \vec{T}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -a\sin\theta & 0 & a\cos\theta \\ 0 & 1 & 0 \end{vmatrix} = (-a\cos\theta, \; 0, \; -a\sin\theta)$$

$$\therefore \| \vec{T}_\theta \times \vec{T}_y \| = \sqrt{a^2\cos^2\theta + 0 + a^2\sin^2\theta} = a$$

$$\therefore A(S) = 2\int_0^{\frac{\pi}{2}} \int_a^{a+a\sin\theta} a \, dy \, d\theta$$

$$= 2a \int_0^{\frac{\pi}{2}} \left[y \right]_a^{a+a\sin\theta} d\theta$$

$$= 2a \int_0^{\frac{\pi}{2}} a\sin\theta = 2a^2 \left[-\cos\theta \right]_{\theta=0}^{\frac{\pi}{2}}$$

$$= 2a^2 \left[0 - (-1) \right] = \underline{2a^2}$$

28.

28. Let a surface S be defined implicitly by $F(x, y, z) = 0$ for (x, y) in a domain D of $\mathbb{R}^2$. Show that

$$\iint_S \left|\frac{\partial F}{\partial z}\right| dS = \iint_D \sqrt{\left(\frac{\partial F}{\partial x}\right)^2 + \left(\frac{\partial F}{\partial y}\right)^2 + \left(\frac{\partial F}{\partial z}\right)^2}\, dx\, dy.$$

Compare with Exercise 22 of Section 7.4.

For $A(S) = \iint_S f(x,y,z)\,dS$, here $f(x,y,z) = \left|\frac{\partial F}{\partial z}\right|$

Since S is over D in xy-plane, it is assumed there is a function $z = g(x,y)$.

$\therefore$ as in p. 395 of text,

$$\iint_S f(x,y,z)\,dS = \iint_D \left|\frac{\partial F}{\partial z}\right| \sqrt{1 + g_x^2 + g_y^2}\, dx\, dy \quad [1]$$

Since $F(x,y,z) = 0$, by the chain rule,

$$\frac{\partial F}{\partial x} = 0 = \frac{\partial F}{\partial x}\cdot\frac{\partial x}{\partial x} + \frac{\partial F}{\partial y}\cdot\frac{\partial y}{\partial x} + \frac{\partial F}{\partial z}\cdot\frac{\partial z}{\partial x}$$

$$\therefore \quad 0 = \frac{\partial F}{\partial x}\cdot 1 + 0 + \frac{\partial F}{\partial z}\cdot\frac{\partial z}{\partial x}$$

$$\text{or,} \quad \frac{\partial z}{\partial x} = \frac{-\partial F}{\partial x} \Big/ \frac{\partial F}{\partial z}$$

Similarly, $\frac{\partial z}{\partial y} = -\frac{\partial F}{\partial y} \Big/ \frac{\partial F}{\partial z}$

But $\frac{\partial z}{\partial x} = g_x$, $\frac{\partial z}{\partial y} = g_y$

$$\therefore \sqrt{1 + g_x^2 + g_y^2} = \sqrt{1 + \left(\partial z/\partial x\right)^2 + \left(\partial z/\partial y\right)^2}$$

$$= \sqrt{1 + \left(-\frac{\partial F}{\partial x} \Big/ \frac{\partial F}{\partial z}\right)^2 + \left(-\frac{\partial F}{\partial y} \Big/ \frac{\partial F}{\partial z}\right)^2}$$

$$= \sqrt{\frac{\left(\frac{\partial F}{\partial z}\right)^2 + \left(\frac{\partial F}{\partial x}\right)^2 + \left(\frac{\partial F}{\partial y}\right)^2}{\left(\frac{\partial F}{\partial z}\right)^2}}$$

$$= \frac{\sqrt{\left(\partial F/\partial x\right)^2 + \left(\partial F/\partial y\right)^2 + \left(\partial F/\partial z\right)^2}}{\left|\partial F/\partial z\right|}$$

$\therefore$ [1] becomes

$$\iint_D \left|\frac{\partial F}{\partial z}\right| \sqrt{F_x^2 + F_y^2 + F_z^2} \Big/ |F_z| \, dx\,dy$$

$$= \iint_D \sqrt{\left(\partial F/\partial x\right)^2 + \left(\partial F/\partial y\right)^2 + \left(\partial F/\partial z\right)^2} \, dx\,dy$$

7.6 Surface Integrals of Vector Fields

Note Title 4/12/2017

1.

1. Consider the closed surface S consisting of the graph $z = 1 - x^2 - y^2$ with $z \geq 0$, and also the unit disc in the xy plane. Give this surface an outer normal. Compute:

$$\iint_S \mathbf{F} \cdot d\mathbf{S}$$

where $\mathbf{F}(x, y, z) = (2x, 2y, z)$.

Use cylindrical coordinates: $0 \leq \theta \leq 2\pi$, $0 \leq r \leq 1$

$\therefore x = r\cos\theta$, $y = r\sin\theta$, $z = 1 - x^2 - y^2 = 1 - r^2$

(a) $\vec{T}(r,\theta) = (x, y, 1-x^2-y^2) = (r\cos\theta, r\sin\theta, 1-r^2)$

$$\vec{T}_r \times \vec{T}_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & -2r \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (2r^2\cos\theta,\ 2r^2\sin\theta,\ r\cos^2\theta + r\sin\theta)$$

$\therefore \vec{N} = (2r^2\cos\theta, 2r^2\sin\theta, r)$ for paraboloid

Note: $\hat{k}$ component positive, so points up/out.

(b)

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F} \cdot \vec{N}\, dr\,d\theta$$

$$= \int_0^{2\pi} \int_0^1 (2r\cos\theta, 2r\sin\theta, 1-r^2) \cdot (2r^2\cos\theta, 2r^2\sin\theta, r)\, dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^1 4r^3\cos^2\theta + 4r^3\sin^2\theta + r - r^3 \, dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^1 3r^3 + r \, dr\,d\theta = 2\pi\int_0^1 3r^3 + r \, dr$$

$$= 2\pi\left[\frac{3}{4}r^4 + \frac{r^2}{2}\right]_0^1 = 2\pi\left[\frac{3}{4} + \frac{1}{2}\right] = \underline{\frac{5}{2}\pi}$$

(c) Since over "closed" surface, must compute

$$\iint_S \vec{F}\cdot d\vec{S} \quad \text{where } S \text{ is unit disk.}$$

Again use cylindrical coordinates, $0 \le \theta \le 2\pi$, $0 \le r \le 1$.

$$T(r,\theta) = (r\cos\theta, r\sin\theta, 0)$$

$$\vec{T}_r \times \vec{T}_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & 0 \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (0,0,r)$$

To make $\vec{N}$ point "out", choose $\vec{N} = (0,0,-r)$

$$\therefore \iint_S \vec{F}\cdot d\vec{S} = \int_0^{2\pi}\int_0^1 \vec{F}\cdot\vec{N}\, dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^1 (2r\cos\theta, 2r\sin\theta, 0)\cdot(0,0,-r)\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^1 0\,dr\,d\theta = \underline{0}$$

(d) $\therefore \frac{5}{2}\pi + 0 = \underline{\frac{5}{2}\pi}$

2.

2. Evaluate the surface integral

$$\iint_S \mathbf{F}\cdot d\mathbf{S}$$

where $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z^2\mathbf{k}$ and S is the surface parameterized by $\Phi(u, v) = (2\sin u, 3\cos u, v)$, with $0 \le u \le 2\pi$ and $0 \le v \le 1$.

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2\cos u & -3\sin u & 0 \\ 0 & 0 & 1 \end{vmatrix} = (-3\sin u, -2\cos u, 0)$$

$\vec{F}$, at the surface, is $(2\sin u, 3\cos u, v^2)$

$$\therefore \vec{F}\cdot(\phi_u \times \phi_v) = -6\sin^2 u - 6\cos^2 u + 0 = -6$$

$$\therefore \iint_S \vec{F}\cdot d\vec{S} = \iint_D (-6)\,du\,dv = -6\int_0^{2\pi}\int_0^1 dv\,du$$

$$= \underline{-12\pi}$$

3.

3. Let $\mathbf{F}(x, y, z) = (x, y, z)$. Evaluate

$$\iint_S \mathbf{F} \cdot d\mathbf{S},$$

where S is:

(a) the upper hemisphere of radius 3, centered at the origin.

(b) the entire sphere of radius 3, centered at the origin.

For the sphere, let $T(\phi, \theta) = (R\cos\theta\sin\phi, R\sin\theta\sin\phi, R\cos\phi)$, here, $R = 3$

$$\therefore \vec{T_\phi} \times \vec{T_\theta} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ R\cos\theta\cos\phi & R\sin\theta\cos\phi & -R\sin\phi \\ -R\sin\theta\sin\phi & R\cos\theta\sin\phi & 0 \end{vmatrix}$$

$$= (R^2\cos\theta\sin^2\phi, R^2\sin\theta\sin^2\phi, R^2\cos\phi\sin\phi)$$

$$= R\sin\phi\,(R\cos\theta\sin\phi, R\sin\theta\sin\phi, R\cos\phi)$$

$$= R\sin\phi\,(x, y, z) = R\sin\phi\,\vec{r} = R^2\sin\phi\,\vec{n},$$

where $\vec{n} = (\cos\theta\sin\phi, \sin\theta\sin\phi, \cos\phi)$

$$\therefore \|\vec{T_\phi} \times \vec{T_\theta}\| = R^2\sin\phi$$

Also, $\vec{F}(x, y, z) = (R\cos\theta\sin\phi, R\sin\theta\sin\phi, R\cos\phi) = R\vec{n}$

(a) $$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F} \cdot (\vec{T_\phi} \times \vec{T_\theta})\, d\phi\, d\theta$$

$$= \iint_D (R\vec{n}) \cdot (R^2 \sin\phi\, \vec{n})\, d\phi\, d\theta \quad , \text{ note } \vec{n}\cdot\vec{n} = 1$$

$$= R^3 \iint_D (\vec{n}\cdot\vec{n}) \sin\phi\, d\phi\, d\theta = R^3 \int_0^{2\pi}\int_0^{\pi/2} \sin\phi\, d\phi\, d\theta$$

$$= 2\pi R^3 \left[-\cos\phi\right]_0^{\pi/2} = 2\pi R^3 \left[0 - (-1)\right] = 2\pi R^3$$

As $R = 3$, $2\pi R^3 = \underline{54\pi}$

(5) $R^3 \int_0^{2\pi}\int_0^{\pi} \sin\phi\, d\phi\, d\theta = 4\pi R^3 = \underline{108\pi}$

4. **4.** Let $\mathbf{F}(x, y, z) = 2x\mathbf{i} - 2y\mathbf{j} + z^2\mathbf{k}$. Evaluate

$$\iint_S \mathbf{F} \cdot d\mathbf{S},$$

where S is the cylinder $x^2 + y^2 = 4$ with $z \in [0, 1]$.

Using cylindrical coordinates, the surface S is

$$T(\theta, z) = (2\cos\theta, 2\sin\theta, z), \quad 0 \le \theta \le 2\pi, \ 0 \le z \le 1.$$

$$\therefore \vec{F}(T(\theta, z)) = (4\cos\theta, -4\sin\theta, z^2)$$

$$\vec{T}_\theta \times \vec{T}_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2\sin\theta & 2\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

$$= (2\cos\theta, 2\sin\theta, 0)$$

$$\therefore \iint_S \vec{F}\cdot d\vec{S} = \iint_D \vec{F}\cdot(\vec{T}_\theta \times \vec{T}_z)\, d\theta\, dz$$

$$= \int_0^{2\pi}\int_0^1 (4\cos\theta, -4\sin\theta, z^2)\cdot(2\cos\theta, 2\sin\theta, 0)\, dz\, d\theta$$

$$= \int_0^{2\pi}\int_0^1 8\cos^2\theta - 8\sin^2\theta \; dz\, d\theta \qquad \cos^2\theta - \sin^2\theta = \cos 2\theta$$

$$= 8\int_0^{2\pi} \cos 2\theta\, d\theta = 4\sin 2\theta \Big|_0^{2\pi} = \underline{0}$$

5. **5.** Let the temperature of a point in $\mathbb{R}^3$ be given by $T(x, y, z) = 3x^2 + 3z^2$. Compute the heat flux across the surface $x^2 + z^2 = 2$, $0 \le y \le 2$, if $k = 1$.

From p. 407 of text, $\vec{F} = -K\nabla T = -(6x, 0, 6z)$

Use cylindrical coordinates for S:

$$T(\theta, y) = (\sqrt{2}\cos\theta, y, \sqrt{2}\sin\theta), \quad 0 \le \theta \le 2\pi, \quad 0 \le y \le 2$$

$$\therefore \vec{F}(T(\theta, y)) = (-6\sqrt{2}\cos\theta, y, -6\sqrt{2}\sin\theta)$$

$$\vec{T}_\theta \times \vec{T}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\sqrt{2}\sin\theta & 0 & \sqrt{2}\cos\theta \\ 0 & 1 & 0 \end{vmatrix}$$

$$= (-\sqrt{2}\cos\theta,\ 0,\ -\sqrt{2}\sin\theta)$$

$$\therefore \vec{F}(T(\theta,y)) \cdot (\vec{T}_\theta \times \vec{T}_y) = 12\cos^2\theta + 0 + 12\sin^2\theta = 12$$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \iint_D 12\, d\theta dy = \int_0^{2\pi}\int_0^2 12\, dy d\theta$$

$$= \underline{48\pi}$$

6. **6.** Compute the heat flux across the unit sphere S if $T(x, y, z) = x$. Can you interpret your answer physically?

$$\vec{F} = -K\nabla T = -K(1,0,0)$$

For a unit sphere, $\vec{T}(\phi,\theta) = (\cos\theta\sin\phi, \sin\theta\sin\phi, \cos\phi)$

$$\vec{T}_\phi \times \vec{T}_\theta = (\sin^2\phi\cos\theta, \sin^2\phi\sin\theta, \sin\phi\cos\phi)$$

$$\therefore \vec{F}(T(\phi,\theta)) \cdot (\vec{T}_\phi \times \vec{T}_\theta) = -k\sin^2\phi\cos\theta$$

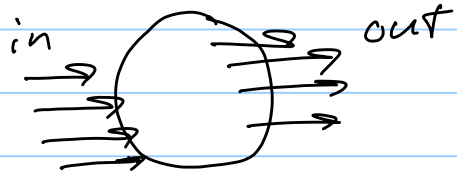
$$\therefore \iint_S \vec{F} \cdot d\vec{S} = -k\int_0^{2\pi}\int_0^{\pi} \sin^2\phi\cos\theta\, d\phi d\theta$$

$$= -k\int_0^{2\pi} \cos\theta\, d\theta \int_0^{\pi} \sin^2\phi\, d\phi$$

$$= -k[\sin\theta]_0^{2\pi} \int_0^{\pi} \sin^2\phi \, d\phi = -k[0-0]\int_0^{\pi} \sin^2\phi \, d\phi$$

$$= \underline{\underline{0}}$$

As $T(x,y,z) = x$ only depends on x-coordinate, temp. gradient is one-way. As sphere is symmetrical, flow in on one side equals flow out on other side. Net is 0.



7. **7.** Let S be the closed surface that consists of the hemisphere $x^2 + y^2 + z^2 = 1$, $z \geq 0$, and its base $x^2 + y^2 \leq 1$, $z = 0$. Let $\mathbf{E}$ be the electric field defined by $\mathbf{E}(x, y, z) = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$. Find the electric flux across S. (HINT: Break S into two pieces S_1 and S_2 and evaluate $\iint_{S_1} \mathbf{E} \cdot d\mathbf{S}$ and $\iint_{S_2} \mathbf{E} \cdot d\mathbf{S}$ separately.)

(a) Upper hemisphere: $T(\phi,\theta) = (\cos\theta\sin\phi, \sin\theta\sin\theta, \cos\phi)$

$0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq \pi/2$

$$\vec{E}(T(\phi,\theta)) = (2\cos\theta\sin\phi, 2\sin\theta\sin\phi, 2\cos\phi)$$

$$\vec{T}_\phi \times \vec{T}_\theta = (\sin^2\phi\cos\theta, \sin^2\phi\sin\theta, \sin\phi\cos\phi)$$

$$\therefore \vec{E} \cdot d\vec{S} = 2\sin^3\phi\cos^2\theta + 2\sin^3\phi\sin^2\theta + 2\sin\phi\cos^2\phi$$

$$= 2\sin^3\phi + 2\sin\phi\cos^2\phi$$

$$= 2\sin\phi$$

$$\therefore \iint_S \vec{E} \cdot d\vec{s} = \iint_D 2\sin\phi \, d\phi \, d\theta = \int_0^{2\pi} \int_0^{\frac{\pi}{2}} 2\sin\phi \, d\phi \, d\theta$$

$$= 4\pi \left[-\cos\phi \right]_0^{\pi/2} = 4\pi \left[0 - (-1) \right] = \underline{4\pi}$$

(b) Bottom base: $T(\theta, r) = (r\cos\theta, r\sin\theta, 0)$

$0 \le \theta \le 2\pi, \quad 0 \le r \le 1$

$$\vec{E}(T(\theta, r)) = (2r\cos\theta, 2r\sin\theta, 0)$$

$$\vec{T}_\theta \times \vec{T}_r = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -r\sin\theta & r\cos\theta & 0 \\ \cos\theta & \sin\theta & 0 \end{vmatrix} = (0, 0, -r)$$

Note $\vec{T}_\theta \times \vec{T}_r$ points down or "out".

$$\therefore \vec{E} \cdot (\vec{T}_\theta \times \vec{T}_r) = 0$$

$$\therefore \iint_S \vec{E} \cdot d\vec{S} = 0, \text{ which makes sense since}$$

$\vec{E}(x, y, 0)$ is parallel to xy-plane and has no component ($\hat{z}$-coord) crossing bottom base.

(c) (a) + (b) $= 4\pi + 0 = \underline{4\pi}$

8. **8.** Let the velocity field of a fluid be described by $\mathbf{F} = \sqrt{y}\mathbf{i}$ (measured in meters per second). Compute how many cubic meters of fluid per second are crossing the surface $x^2 + z^2 = 1, 0 \le y \le 1, 0 \le x \le 1$.

Cylinder can be parametrized as:

$$\phi(y,\theta) = (\cos\theta, y, \sin\theta), \quad -\frac{\pi}{2} \le \theta \le \frac{\pi}{2}, \quad 0 \le y \le 1$$

$$\therefore \vec{F}(\phi(y,\theta)) = (\sqrt{y}, 0, 0)$$

$$\phi_y \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{vmatrix} = (\cos\theta, 0, \sin\theta)$$

$\therefore \phi_y \times \phi_\theta$ points "out"

$$\therefore \vec{F} \cdot (\phi_y \times \phi_\theta) = \sqrt{y}\cos\theta + 0 + 0$$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \int_{-\pi/2}^{\pi/2} \int_0^1 \sqrt{y}\cos\theta \, dy \, d\theta$$

$$= \int_{-\pi/2}^{\pi/2} \cos\theta \, d\theta \int_0^1 \sqrt{y} \, dy = \sin\theta \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2}{3} y^{3/2} \Big|_0^1$$

$$= [1 - (-1)]\left[\frac{2}{3} - 0\right] = \frac{4}{3}$$

$\therefore \frac{4}{3}$ meters^3/sec flows out of the half cylinder

9.

9. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where S is the surface $x^2 + y^2 + 3z^2 = 1$, $z \le 0$ and $\mathbf{F}$ is the vector field $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + zx^3y^2\mathbf{k}$. (Let $\mathbf{n}$, the unit normal, be upward pointing.)

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y & -x & zx^3y^2 \end{vmatrix} = (2x^3yz - 0,\ 0 - 3x^2y^2z,\ -1 - 1)$$

$$= (2x^3yz, -3x^2y^2z, -2)$$

For S: $z^2 = \frac{1 - x^2 - y^2}{3}$, $z \le 0$, so $z = -\sqrt{\frac{1-x^2-y^2}{3}}$

Let $T(x,y) = \left(x, y, -\left(\frac{1-x^2-y^2}{3}\right)^{\frac{1}{2}}\right)$, $-1 \le x \le 1$, $-\sqrt{1-x^2} \le y \le \sqrt{1-x^2}$

$$\vec{T}_x \times \vec{T}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & \frac{x}{3}\left(\frac{1-x^2-y^2}{3}\right)^{-\frac{1}{2}} \\ 0 & 1 & \frac{y}{3}\left(\frac{1-x^2-y^2}{3}\right)^{-\frac{1}{2}} \end{vmatrix}$$

$$= \left[-\frac{x}{3}\left(\frac{1-x^2-y^2}{3}\right)^{-\frac{1}{2}}, -\frac{y}{3}\left(\frac{1-x^2-y^2}{3}\right)^{-\frac{1}{2}}, 1 \right]$$

Note $\vec{T}_x \times \vec{T}_y$ points "up" as $\hat{k}$ component $= 1$.

$$\therefore (\nabla \times \vec{F}) \cdot (\vec{T}_x \times \vec{T}_y) =$$

$$-\frac{2x^4yz}{3}\left(\frac{1-x^2-y^2}{3}\right)^{-\frac{1}{2}} + x^2y^3z\left(\frac{1-x^2-y^2}{3}\right)^{-\frac{1}{2}} - 2$$

$$= z\left(x^2y^3 - \frac{2x^4y}{3}\right)\left(\frac{1-x^2-y^2}{3}\right)^{-\frac{1}{2}} - 2$$

Note: $z = -\left(\frac{1-x^2-y^2}{3}\right)^{\frac{1}{2}}$

$$= \frac{2x^4y}{3} - x^2y^3 - 2$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot dS = \int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \left(\frac{2x^4y}{3} - x^2y^3 - 2 \right) dy\, dx$$

$$= \int_{-1}^{1} \left. \frac{x^4y^2}{3} - \frac{x^2y^4}{4} - 2y \right|_{y=-\sqrt{1-x^2}}^{y=\sqrt{1-x^2}} dx$$

$$= \int_{-1}^{1} \frac{x^4(1-x^2)}{3} - \frac{x^2(1-x^2)^2}{4} - 2\sqrt{1-x^2} - \left[\frac{x^4(1-x^2)}{3} - \frac{x^2(1-x^2)^2}{4} + 2\sqrt{1-x^2} \right] dx$$

Text Table of Integrals, # 38

$$= -4 \int_{-1}^{1} \sqrt{1-x^2}\, dx = -4 \left[\frac{x}{2}\sqrt{1-x^2} + \frac{x^2}{2} \operatorname{Arcsin} x \right]_{-1}^{1}$$

$$= -4 \left[\tfrac{1}{2} \operatorname{Arcsin}(1) - \tfrac{1}{2} \operatorname{Arcsin}(-1) \right]$$

$$= -4 \left[\frac{\pi}{4} + \frac{\pi}{4} \right] = -2\pi$$

10.

10. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where $\mathbf{F} = (x^2 + y - 4)\mathbf{i} + 3xy\mathbf{j} + (2xz + z^2)\mathbf{k}$ and S is the surface $x^2 + y^2 + z^2 = 16$, $z \geq 0$. (Let $\mathbf{n}$, the unit normal, be upward pointing.)

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2+y-4 & 3xy & 2xz+z^2 \end{vmatrix} = (0-0,\ 0-2z,\ 3y-1)$$

$$= (0, -2z, 3y-1)$$

Let $T(x,y) = \left(x, y, \sqrt{16-x^2-y^2}\right)$, $-4 \le x \le 4$, $-\sqrt{16-x^2} \le y \le \sqrt{16-x^2}$

$$\vec{T}_x \times \vec{T}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -x\sqrt{16-x^2-y^2} \\ 0 & 1 & -y\sqrt{16-x^2-y^2} \end{vmatrix}$$

$$= \left(x\sqrt{16-x^2-y^2},\ y\sqrt{16-x^2-y^2},\ 1\right)$$

Note $\vec{T}_x \times \vec{T}_y$ points "up" as $\hat{k}$ component is 1.

$$\therefore (\nabla \times \vec{F}) \cdot (\vec{T}_x \times \vec{T}_y) = 0 + \left(-2zy\sqrt{16-x^2-y^2}\right) + 3y - 1$$

$$= 3y - 2y\left(\sqrt{16-x^2-y^2}\right)\left(\sqrt{16-x^2-y^2}\right) - 1$$

$$= 3y - 2y(16-x^2-y^2) - 1 = -29y + 2x^2y + 2y^3 - 1$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{-4}^{4} \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} (-29y + 2x^2y + 2y^3 - 1)\, dy\, dx$$

$$= \int_{-4}^{4} -\frac{29}{2}y^2 + x^2y^2 + \frac{y^4}{2} - y \Big|_{y=-\sqrt{16-x^2}}^{y=\sqrt{16-x^2}} dx$$

Using #38 Text table of integrals

$$= \int_{-4}^{4} -2\sqrt{16-x^2}\, dx = -2\left[\frac{x}{2}\sqrt{16-x^2} + 8\,\text{Arcsin}\frac{x}{4}\right]_{x=-4}^{x=4}$$

$= -2\left[8\,\text{Arcsin}(1) - 8\,\text{Arcsin}(-1)\right]$

$= -2\left[8\left(\frac{\pi}{2}\right) - 8\left(-\frac{\pi}{2}\right)\right] = -16\pi$

11.

11. Calculate the integral $\iint_S \mathbf{F}\cdot d\mathbf{S}$, where S is the entire surface of the solid half ball $x^2 + y^2 + z^2 \le 1$, $z \ge 0$, and $\mathbf{F} = (x + 3y^5)\mathbf{i} + (y + 10xz)\mathbf{j} + (z - xy)\mathbf{k}$. (Let S be oriented by the outward-pointing normal.)

(a) Top surface: Use spherical coordinates

$T(\phi,\theta) = (\cos\theta\sin\phi,\ \sin\theta\sin\phi,\ \cos\phi)$, $0 \le \theta \le 2\pi$, $0 \le \phi \le \frac{\pi}{2}$

$T_\phi \times T_\theta = (\sin^2\phi\cos\theta,\ \sin^2\phi\sin\theta,\ \sin\phi\cos\phi)$ $\sin\phi\cos\phi\,\hat{k}$ points "out"

$\vec{F}(\phi,\theta) = \left[\cos\theta\sin\phi + 3(\sin\theta\sin\phi)^5,\ \sin\theta\sin\phi + 10\cos\theta\sin\phi\cos\phi,\ \cos\phi - \cos\theta\sin\theta\sin^2\phi\right]$

$\therefore \vec{F}(\phi,\theta)\cdot(T_\phi \times T_\theta) =$

$\sin^3\phi\cos^2\theta + 3\sin^5\theta\cos\theta\sin^7\phi$

$+ \sin^3\phi\sin^2\theta + 10\sin^3\phi\cos\phi\sin\theta\cos\theta$

$+ \sin\phi\cos^2\phi - \sin\theta\cos\theta\sin^3\phi\cos\phi$

$= \sin^3\phi + 3\sin^5\theta\cos\theta\sin^7\phi + 10\sin^3\phi\cos\phi\sin\theta\cos\theta$

$+ \sin\phi\cos^2\phi - \sin\theta\cos\theta\sin^3\phi\cos\phi$

Integrating each term of $\vec{F}\cdot(T_\phi \times T_\theta)$ yields many

cancellations.

$$\int_0^{2\pi}\int_0^{\pi/2} 3\sin^5\theta\cos\theta\sin^7\phi\,d\phi\,d\theta = \frac{3\sin^6\theta}{6}\Big|_{\theta=0}^{\theta=2\pi}\int_0^{\frac{\pi}{2}}\sin^7\phi\,d\phi$$

$$= (0)\int_0^{\frac{\pi}{2}}\sin^7\phi\,d\phi = 0 \qquad \sin(2\pi)=\sin(0)=0$$

$$\int_0^{2\pi}\int_0^{\frac{\pi}{2}} 10\sin^3\phi\cos\phi\sin\theta\cos\theta\,d\phi\,d\theta = \frac{\sin^2\theta}{2}\Big|_0^{2\pi}\int_0^{\pi/2}10\sin^3\phi\cos\phi\,d\phi$$

$$= (0)\int_0^{\pi/2}10\sin^3\phi\cos\phi\,d\phi = 0 \qquad \sin(2\pi)=\sin(0)=0$$

Similarly, $\int_0^{2\pi}\int_0^{\frac{\pi}{2}}\sin\theta\cos\theta\sin^3\phi\cos\phi\,d\phi\,d\theta = 0$

$$\therefore \int_0^{2\pi}\int_0^{\frac{\pi}{2}}\vec{F}\cdot(T_\phi\times T_\theta) = \int_0^{2\pi}\int_0^{\pi/2}(\sin^3\phi+\sin\phi\cos^2\phi)\,d\phi\,d\theta$$

$$= 2\pi\int_0^{\pi/2}(\sin^3\phi+\sin\phi\cos^2\phi)\,d\phi$$

$$= 2\pi\int_0^{\pi/2}(1-\cos^2\phi)\sin\phi+\sin\phi\cos^2\phi\,d\phi$$

$$= 2\pi \int_0^{\pi/2} \sin\phi \, d\phi = 2\pi \left[-\cos\phi \right]_0^{\pi/2} = \underline{2\pi}$$

$$\therefore \text{ Top Surface } \iint_S \vec{F} \cdot d\vec{S} = \underline{2\pi}$$

(6) Bottom Surface

$$T(\theta, r) = (r\cos\theta, r\sin\theta, 0), \quad 0 \le r \le 1 \quad 0 \le \theta \le 2\pi$$

$$T_\theta = (-r\sin\theta, r\cos\theta, 0) \qquad T_r = (\cos\theta, \sin\theta, 0)$$

$$T_\theta \times T_r = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -r\sin\theta & r\cos\theta & 0 \\ \cos\theta & \sin\theta & 0 \end{vmatrix} = (0, 0, -r)$$

$-r\hat{k}$ points "out"

$$\vec{F}(\theta, r) = (r\cos\theta + 3r^5\sin^5\theta, r\sin\theta, -r^2\sin\theta\cos\theta)$$

$$\therefore \vec{F} \cdot (T_\theta \times T_r) = r^3 \sin\theta\cos\theta$$

$$\therefore \int_0^{2\pi} \int_0^1 r^3 \sin\theta\cos\theta \, dr\, d\theta = \frac{\sin^2\theta}{2}\Big|_0^{2\pi} \frac{r^4}{4}\Big|_0^1 = \underline{0}$$

as $\sin(2\pi) = \sin(0) = 0$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = 2\pi + 0 = \underline{2\pi}$$

12.

12.* A restaurant is being built on the side of a mountain. The architect's plans are shown in Figure 7.6.11.

(a) The vertical curved wall of the restaurant is to be built of glass. What will be the surface area of this wall?

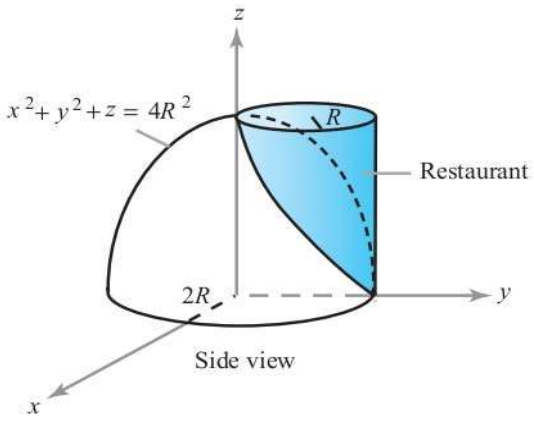
(b) To be large enough to be profitable, the consulting engineer informs the developer that the volume of the interior must exceed $\pi R^4/2$. For what R does the proposed structure satisfy this requirement?

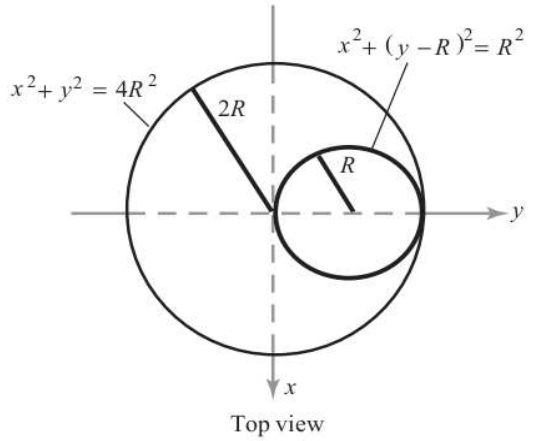
(c) During a typical summer day, the environs of the restaurant are subject to a temperature field given by

$$T(x, y, z) = 3x^2 + (y - R)^2 + 16z^2.$$

A heat flux density $\mathbf{V} = -k\nabla T$ (k is a constant depending on the grade of insulation to be used) through all sides of the restaurant (including the top and the contact with the hill) produces a heat flux.

What is this total heat flux? (Your answer will depend on R and k.)





The region of interest is completely below the top disk, so use the disk for parametrizations - for surface area & volume.

(a) The disk admits a polar parametrization:

$x = R\cos\theta$, $y = R + R\sin\theta$, $0 \le \theta \le 2\pi$, where θ is measured from line parallel to x-axis.

$\therefore$ A parametrization of the portion of cylinder

of interest is: $T(\theta, z) = (R\cos\theta, R + R\sin\theta, z)$.

Need to find a range for z, the vertical height.

The cylinder "height" depends on θ: i.e., where x and y are located. A line of "height" height goes from the paraboloid to the xy disk.

For the paraboloid, $z = 4R^2 - (x^2+y^2)$, and from the cylinder, $x^2 + y^2 - 2yR + R^2 = R^2$, so $x^2+y^2 = 2yR$

$\therefore$ At intersection of cylinder and paraboloid,

$$z = 4R^2 - (2yR) = 4R^2 - 2(R + R\sin\theta)R$$

$$= 2R^2 - 2R^2\sin\theta$$

$$\therefore\ 2R^2 - 2R^2\sin\theta \le z \le 4R^2$$

So, for

$\theta =$	0	$\pi/2$	π	$\frac{3}{2}\pi$
$z =$	$2R^2$	0	$2R^2$	$4R^2$

The above table gives the lower boundary of the cylinder portion in question.

$$A(S) = \iint_D \| \vec{T}_\theta \times \vec{T}_z \| \, d\theta\, dz$$

$$\vec{T}_\theta \times \vec{T}_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -R\sin\theta & R\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = (R\cos\theta,\ R\sin\theta,\ 0)$$

$$\therefore \left\| \vec{T}_\theta \times \vec{T}_z \right\| = R$$

$$\therefore A(S) = \int_0^{2\pi} \int_{2R^2 - 2R^2\sin\theta}^{4R^2} R\, dz\, d\theta$$

$$= R \int_0^{2\pi} 2R^2 + 2R^2\sin\theta\, d\theta = 4\pi R^3 + 2R^3 \int_0^{2\pi} \sin\theta\, d\theta$$

$$= 4\pi R^3 + \left[-\cos\theta\right]_0^{2\pi} = 4\pi R^2 + 0$$

$\therefore$ Surface area of glass wall $= \underline{4\pi R^3}$

(6) (b) To be large enough to be profitable, the consulting engineer informs the developer that the volume of the interior must exceed $\pi R^4/2$. For what R does the proposed structure satisfy this requirement?

Modifying parametrization in (a), the volume is described as $T(r,\theta,z) = (r\cos\theta,\ R + r\sin\theta,\ z)$, $0 \le r \le R$, $0 \le \theta \le 2\pi$

For z: $z = 4R^2 - (x^2+y^2) = 4R^2 - (r\cos\theta)^2 - (R + r\sin\theta)^2$

$= 4R^2 - r^2\cos^2\theta - R^2 - 2Rr\sin\theta - r^2\sin^2\theta$

$= 3R^2 - r^2 - 2Rr\sin\theta$

$$\therefore\ 3R^2 - r^2 - 2Rr\sin\theta \le z \le 4R^2$$

$$\text{Jacobian } \frac{\partial(x,y,z)}{\partial(r,\theta,z)} \text{ from } T() = \begin{vmatrix} \cos\theta & -r\sin\theta & 0 \\ \sin\theta & r\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = r$$

$$\therefore V = \iiint_W dx\,dy\,dz = \iiint_{W^*} r\,dr\,d\theta\,dz$$

$$= \int_0^{2\pi}\int_0^{R}\int_{3R^2 - r^2 - 2Rr\sin\theta}^{4R^2} r\,dz\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^{R} r\left(R^2 + r^2 + 2Rr\sin\theta\right) dr\,d\theta$$

$$= \int_0^{2\pi} R^2\frac{r^2}{2} + \frac{r^4}{4} + Rr^2\sin\theta \Big|_{r=0}^{r=R} d\theta$$

$$= \int_0^{2\pi} \left(\frac{3}{4}R^4 + R^4\sin\theta\right) d\theta = \frac{3}{4}R^4(2\pi) + 0$$

$$= \frac{3}{2}\pi R^4$$

$\therefore$ Volume $= \frac{3}{2}\pi R^4 > \frac{\pi}{2}R^4$, so any $R > 0$ works.

(c) (c) During a typical summer day, the environs of the restaurant are subject to a temperature field given by

$$T(x, y, z) = 3x^2 + (y - R)^2 + 16z^2.$$

A heat flux density $\mathbf{V} = -k\nabla T$ (k is a constant depending on the grade of insulation to be used) through all sides of the restaurant (including the top and the contact with the hill) produces a heat flux.

What is this total heat flux? (Your answer will depend on R and k.)

$$\nabla T = (6x, 2y - 2R, 32z)$$

$$\therefore \vec{V} = -k(6x, 2y - 2R, 32z)$$

(1) Top surface: parametrization is,

$$T(r, \theta) = (r\cos\theta, R + r\sin\theta, 4R^2)$$

$$0 \le \theta \le 2\pi, \quad 0 \le r \le R$$

$$\therefore \vec{T_r} \times \vec{T_\theta} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & 0 \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (0, 0, r)$$

which points "out"

$$\therefore \vec{V} \cdot (\vec{T_r} \times \vec{T_\theta}) = -32kzr = -32k(4R^2)r$$

$$= -128kR^2 r$$

$$\therefore \iint_T \vec{V} \cdot (\vec{T_r} \times \vec{T_\theta})\, dr\, d\theta = \int_0^{2\pi} \int_0^R -128kR^2 r\, dr\, d\theta$$

$$= -256\pi kR^2 \left[\frac{r^2}{2}\right]_0^R = \underline{-128\pi kR^4}$$

(2) Side surface: parametrization is, as in (a),

$$T(\theta, z) = (R\cos\theta,\ R + R\sin\theta,\ z),\quad 0 \le \theta \le 2\pi,$$

$$2R^2 - 2R^2\sin\theta \le z \le 4R^2$$

$$\therefore \vec{T}_\theta \times \vec{T}_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -R\sin\theta & R\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = (R\cos\theta,\ R\sin\theta,\ 0)$$

Note this points "out".

$$\therefore \vec{V} \cdot (\vec{T}_\theta \times \vec{T}_z) = -k(6x,\ 2y - 2R,\ 3z^2) \cdot (R\cos\theta, R\sin\theta, 0)$$

$$= -k\left(6xR\cos\theta + 2yR\sin\theta - 2R^2\sin\theta\right)$$

$$= -k\left[6(R\cos\theta)R\cos\theta + 2(R + R\sin\theta)R\sin\theta - 2R^2\sin\theta\right]$$

$$= -k\left[6R^2\cos^2\theta + 2R^2\sin^2\theta\right]$$

$$= -k\left[4R^2\cos^2\theta + 2R^2\right]$$

$$\therefore \iint_T \vec{V} \cdot (\vec{T}_\theta \times \vec{T}_z)\, d\theta\, dz = \int_0^{2\pi} \int_{2R^2 - 2R^2\sin\theta}^{4R^2} -k\left[4R^2\cos^2\theta + 2R^2\right] dz\, d\theta$$

$$= -k\int_0^{2\pi} \left(4R^2\cos^2\theta + 2R^2\right)\left(2R^2 + 2R^2\sin\theta\right) d\theta$$

$$(2)(2)\left(2R^2\cos^2\theta + R^2\right)\left(R^2\sin\theta + R^2\right)$$

$$= -4K \int_0^{2\pi} 2R^4\cos^2\theta\sin\theta + R^4\sin\theta + 2R^4\cos^2\theta + R^4 \, d\theta$$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$$= -4KR^4 \left[-\frac{2\cos^3\theta}{3} - \cos\theta + \theta + \frac{\sin 2\theta}{2} + \theta \right]_0^{2\pi}$$

$$= -4KR^4 \left[-\frac{2}{3} - 1 + 0 + 4\pi - \left(-\frac{2}{3} - 1 + 0 \right) \right]$$

$$= \underline{\underline{-16\pi KR^4}}$$

(3) Paraboloid surface: a preliminary parametrization is:

$$T(r,\theta) = (r\cos\theta, R + r\sin\theta, z), \quad 0 \le \theta \le 2\pi, \quad 0 \le r \le R$$

This time, $z = 0$ is the bottom, and the paraboloid is the top.

Here, $z = 4R^2 - (x^2+y^2) = 4R^2 - (r\cos\theta)^2 - (R + r\sin\theta)^2$

$$= 4R^2 - r^2\cos^2\theta - R^2 - 2Rr\sin\theta - r^2\sin^2\theta$$

$$= 3R^2 - r^2 - 2Rr\sin\theta$$

$$\therefore T(r,\theta) = (r\cos\theta, R + r\sin\theta, 3R^2 - r^2 - 2Rr\sin\theta)$$

$$\vec{T}_\theta \times \vec{T}_r = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -r\sin\theta & r\cos\theta & -2Rr\cos\theta \\ \cos\theta & \sin\theta & -2r - 2R\sin\theta \end{vmatrix}$$

$$= (-2r^2\cos\theta - 2Rr\sin\theta\cos\theta + 2Rr\sin\theta\cos\theta,$$
$$-2Rr\cos^2\theta - 2r^2\sin\theta - 2Rr\sin^2\theta,$$
$$-r\sin^2\theta - r\cos^2\theta)$$

$$= (-2r^2\cos\theta, -2Rr - 2r^2\sin\theta, -r)$$

Notice this points down ($-r$) and in, and so points "out", consistent with the normals to the other two sides.

$$\therefore \vec{V} \cdot (\vec{T}_\theta \times \vec{T}_r) =$$

$$-K(6x, 2y - 2R, 32z) \cdot (-2r^2\cos\theta, -2Rr - 2r^2\sin\theta, -r)$$

$$= -K(6r\cos\theta, 2r\sin\theta, 96R^2 - 32r^2 - 64Rr\sin\theta) \cdot$$
$$(-2r^2\cos\theta, -2Rr - 2r^2\sin\theta, -r)$$

$$= K[12r^3\cos^2\theta + 4Rr^2\sin\theta + 4r^3\sin^2\theta$$
$$+ 96R^2r - 32r^3 - 64Rr^2\sin\theta]$$

$$= K[8r^3\cos^2\theta - 28r^3 - 60Rr^2\sin\theta + 96R^2r]$$

$$\therefore \iint_T \vec{V} \cdot (\vec{T}_r \times \vec{T}_\theta)\, dr\, d\theta =$$

$$4k\int_0^{2\pi}\int_0^{R}(2r^3\cos^2\theta - 7r^3 - 15Rr^2\sin\theta + 24R^2r)\,dr\,d\theta$$

$$= 4k\int_0^{2\pi} \frac{r^4}{2}\cos^2\theta - \frac{7}{4}r^4 - 5Rr^3\sin\theta + 12R^2r^2\Big|_0^R\,d\theta$$

$$= 4k\int_0^{2\pi} \frac{R^4}{2}\cos^2\theta - \frac{7R^4}{4} - 5R^4\sin\theta + 12R^4\,d\theta$$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$$= 4k\left[\frac{R^4}{4}\theta + \frac{R^4}{2}\frac{\sin 2\theta}{4} - \frac{7R^4}{4}\theta + 5R^4\cos\theta + 12R^4\theta\right]_0^{2\pi}$$

$$= 4k\left[\frac{R^4\pi}{2} + 0 - \frac{7R^4\pi}{2} + 5R^4 + 24R^4\pi - (0+0-0+5R^4+0)\right]$$

$$= 4k\left[\frac{R^4\pi}{2} - \frac{7R^4\pi}{2} + 24R^4\pi\right]$$

$$= 2kR^4\pi - 14R^4\pi + 96kR^4\pi = \underline{84\pi kR^4}$$

$\therefore (1)+(2)+(3) = -128\pi kR^4 - 16\pi kR^4 + 84\pi kR^4$

$$= \underline{60\pi kR^4}$$

13. **13.** Find the flux of the vector field $\mathbf{V}(x, y, z) = 3xy^2\mathbf{i} + 3x^2y\mathbf{j} + z^3\mathbf{k}$ out of the unit sphere.

Using spherical coordinates, unit sphere is described by

$$T(\theta, \phi) = (\cos\theta\sin\phi, \sin\theta\sin\phi, \cos\phi) = (x, y, z)$$

As seen in Example 1, p. 401 of text,

$$\vec{T}_\phi \times \vec{T}_\theta = -\vec{T}_\theta \times \vec{T}_\phi = (\sin^2\phi\cos\theta, \sin^2\phi\sin\theta, \sin\phi\cos\phi).$$

$$= \sin\phi\,(x, y, z)$$

$$\therefore \vec{V} \cdot (\vec{T}_\phi \times \vec{T}_\theta) = \sin\phi\,(3xy^2, 3x^2y, z^3) \cdot (x, y, z)$$

$$= \sin\phi\left[3x^2y^2 + 3x^2y^2 + z^4\right]$$

$$= \sin\phi\,(6x^2y^2 + z^4)$$

$$= \sin\phi\left[6(\cos^2\theta\sin^2\phi)(\sin^2\theta\sin^2\phi) + \cos^4\phi\right]$$

$$= \sin\phi\left[6\cos^2\theta\sin^2\theta\sin^4\phi + \cos^4\phi\right]$$

$$= 6\cos^2\theta\sin^2\theta\sin^5\phi + \cos^4\phi\sin\phi$$

$$\therefore \iint_T \vec{V} \cdot (\vec{T}_\phi \times \vec{T}_\theta)\, d\theta\, d\phi =$$

$$\int_0^{2\pi} \int_0^{\pi} \left[6\cos^2\theta\sin^2\theta\sin^5\phi + \cos^4\phi\sin\phi\right] d\phi\, d\theta$$

$$= \int_0^{2\pi} 6\cos^2\theta \sin^2\theta (1-\cos^2\phi)^2 \sin\phi \, d\phi \, d\theta + 2\pi \left[-\frac{\cos^5\phi}{5} \right]_{\phi=0}^{\phi=\pi}$$

$$= \int_0^{2\pi} \int_0^{\pi} 6\cos^2\theta \sin^2\theta \left(1 - 2\cos^2\phi + \cos^4\phi\right) \sin\phi \, d\phi \, d\theta + 2\pi \left[\frac{1}{5} - \left(-\frac{1}{5}\right) \right]$$

$$= \int_0^{2\pi} 6\cos^2\theta \sin^2\theta \left[-\cos\phi + \frac{2}{3}\cos^3\phi - \frac{\cos^5\phi}{5} \right]_{\phi=0}^{\phi=\pi} d\theta + \frac{4}{5}\pi$$

$$= \int_0^{2\pi} 6\cos^2\theta \sin^2\theta \left[1 - \frac{2}{3} + \frac{1}{5} - \left(-1 + \frac{2}{3} - \frac{1}{5}\right) \right] d\theta + \frac{4}{5}\pi$$

$$= \int_0^{2\pi} 6\left(\frac{16}{15}\right) \cos^2\theta \sin^2\theta \, d\theta + \frac{4}{5}\pi$$

$$= \int_0^{2\pi} \frac{8}{5} 4\cos^2\theta \sin^2\theta \, d\theta + \frac{4}{5}\pi = \frac{8}{5}\int_0^{2\pi} \sin^2 2\theta \, d\theta + \frac{4}{5}\pi$$

$\frac{1-\cos 2\theta}{2} = \sin^2\theta$

$$= \frac{8}{5}\int_0^{2\pi} \frac{1-\cos 4\theta}{2} d\theta + \frac{4}{5}\pi = \frac{4}{5}\pi + \frac{8}{5}\left[\frac{\theta}{2} - \frac{\sin 4\theta}{8} \right]_0^{2\pi}$$

$$= \frac{4}{5}\pi + \frac{8}{5}\left[\pi - 0 - (0-0) \right] = \frac{4}{5}\pi + \frac{8}{5}\pi$$

$$= \frac{12}{5}\pi$$

14.

14. Evaluate the surface integral $\iint_S \mathbf{F} \cdot \mathbf{n}\, dA$, where $\mathbf{F}(x, y, z) = \mathbf{i} + \mathbf{j} + z(x^2 + y^2)^2\mathbf{k}$ and S is the surface of the cylinder $x^2 + y^2 \le 1$, $0 \le z \le 1$.

The cylinder has a circular side, a top and bottom.

(a) Side: $T(\theta, z) = (\cos\theta, \sin\theta, z)$, $0 \le \theta \le 2\pi$, $0 \le z \le 1$.

$$\therefore \vec{T_\theta} \times \vec{T_z} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = (\cos\theta, \sin\theta, 0) = \vec{n},$$

and $\|\vec{n}\| = 1$

$$\therefore \vec{F} \cdot \vec{n} = (1, 1, z(x^2+y^2)^2) \cdot (x, y, 0) = x + y + 0 = x + y$$

$$\therefore \vec{F} \cdot \vec{n} = \cos\theta + \sin\theta$$

$$\therefore \iint_S \vec{F} \cdot \vec{n}\, dA = \int_0^{2\pi} \int_0^1 (\cos\theta + \sin\theta)\, dz\, d\theta$$

$$= \int_0^{2\pi} (\cos\theta + \sin\theta)\, d\theta = \Big[\sin\theta - \cos\theta \Big]_0^{2\pi}$$

$$= [0 - 1 - (0 - 1)] = \underline{0}$$

(b) Top: $T(r, \theta) = (r\cos\theta, r\sin\theta, 1)$, $0 \le r \le 1$, $0 \le \theta \le 2\pi$

$$\vec{T_r} \times \vec{T_\theta} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & 0 \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (0, 0, r) = r\hat{k}$$

Note $x^2 + y^2 = r^2$

$$\therefore \vec{n} = \hat{k}$$

$$\therefore \vec{F} \cdot \vec{n} = (1, 1, z(x^2+y^2)^2) \cdot (0,0,1) = z(x^2+y^2)^2$$

$$= r^4$$

$$\therefore \iint_S \vec{F} \cdot \vec{n}\, dA = \int_0^{2\pi} \int_0^1 r^4 \, dr\, d\theta = 2\pi \left[\frac{r^5}{5} \right]_0^1$$

$$= \frac{2}{5}\pi$$

(c) Bottom: $T(r,\theta) = (r\cos\theta, r\sin\theta, 0)$, $0 \le r \le 1, 0 \le \theta \le 2\pi$

Keeping with orientation of "out", $\vec{n} = -\hat{k}$

$$\therefore \vec{F} \cdot \vec{n} = (1, 1, z(x^2+y^2)^2) \cdot (0,0,-1) = -z(x^2+y^2)^2$$

$$= 0 \quad \text{since} \quad z = 0$$

$$\therefore \iint_S \vec{F} \cdot \vec{n}\, dA = 0$$

$$\therefore (a) + (b) + (c) = \frac{2}{5}\pi$$

15. **15.** Let S be the surface of the unit sphere. Let $\mathbf{F}$ be a vector field and F_r its radial component. Prove that

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \int_{\theta=0}^{2\pi} \int_{\phi=0}^{\pi} F_r \sin\phi \, d\phi \, d\theta.$$

What is the corresponding formula for real-valued functions f?

Using spherical coordinates, a parametrization for S

is: $T(\phi,\theta) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi) = (x,y,z) = \vec{r}$

$0 \le \phi \le \pi, \quad 0 \le \theta \le 2\pi$

As in Example 1, p. 401 of text,

$$\vec{T}_\phi \times \vec{T}_\theta = \sin\phi(\cos\theta, \sin\theta, \cos\phi) = \sin\phi\,\vec{r},$$

where $\vec{r}$ (i.e., the normal) points "out".

$$\therefore \vec{F}\cdot(\vec{T}_\phi \times \vec{T}_\theta) = \vec{F}\cdot(\sin\phi\,\vec{r}) = \sin\phi(\vec{F}\cdot\vec{r})$$

$$= \sin\phi\, F_r,$$

where F_r = component of $\vec{F}$ along $\vec{r}$, by geometric interpretation of dot product.

$$\therefore \iint_S \vec{F}\cdot d\vec{S} = \iint_D \vec{F}\cdot(\vec{T}_\phi \times \vec{T}_\theta)\, d\phi\, d\theta$$

$$= \iint_D F_r \sin\phi\, d\phi\, d\theta = \int_0^{2\pi}\int_0^{\pi} F_r \sin\phi\, d\phi\, d\theta$$

For real valued functions, $\|\vec{T}_\phi \times \vec{T}_\theta\| = \|\sin\phi\,\vec{r}\| = \sin\phi$ since $\|\vec{r}\| = 1$

$$\therefore \iint_S F\, dS = \iint_D F(T(\phi,\theta))\,\|\vec{T}_\phi \times \vec{T}_\theta\|\, d\phi\, d\theta$$

$$= \iint_D F(T(\phi,\theta)) \sin\phi \, d\phi d\theta$$

16.

16. Prove the following mean-value theorem for surface integrals: If $\mathbf{F}$ is a continuous vector field, then

$$\iint_S \mathbf{F}\cdot\mathbf{n}\, dS = [\mathbf{F}(Q)\cdot\mathbf{n}(Q)]\, A(S)$$

for some point $Q \in S$, where $A(S)$ is the area of S. [HINT: Prove it for real functions first, by reducing the problem to one of a double integral: Show that if $g \geq 0$, then

$$\iint_D fg\, dA = f(Q) \iint_D g\, dA$$

for some $Q \in D$ (do it by considering $(\iint_D fg\, dA)/(\iint_D g\, dA)$ and using the intermediate-value theorem).]

Assume S is a C^1 surface, so that the parametrization $\phi(u,v): \mathbb{R}^2 \to \mathbb{R}^3$ is class C^1.

$\therefore \phi_u \times \phi_v$ is continuos, and so is $\|\phi_u \times \phi_v\|$, and $\therefore$

$\vec{F} \cdot \left(\frac{\phi_u \times \phi_v}{\|\phi_u \times \phi_v\|}\right) = \vec{F}\cdot\vec{n}$ is continuous on D as $\vec{F}$ is continuous. Let $G(u,v) = \vec{F}(\phi(u,v))\cdot\vec{n}(u,v)$ and assume D is closed. $\therefore$ G contains a maximum and minimum on D.

Let $Q_{min} \in D$ s.t. $G(Q_{min})$ = the minimum.

$Q_{max} \in D$ s.t. $G(Q_{max})$ = the maximum.

$\therefore G(Q_{min}) \leq G(u,v) \leq G(Q_{max})$, $(u,v) \in D$.

$\therefore$ Given a partition of D,

$$\sum_u \sum_v G(Q_{min}) \le \sum_u \sum_v G(u,v) \le \sum_u \sum_v G(Q_{max}),$$

or $$\iint_D G(Q_{min})\,dS \le \iint_S G\,dS \le \iint_S G(Q_{max})\,dS$$

or $$G(Q_{min}) \iint_S dS \le \iint_S G\,dS \le G(Q_{max}) \iint dS$$

or $$G(Q_{min})\,A(S) \le \iint_S G\,dS \le G(Q_{max})\,A(S)$$

$\therefore$ $$G(Q_{min}) \le \frac{\iint_S G\,dS}{A(S)} \le G(Q_{max})$$

By Intermediate Value Theorem for continuous functions on a closed domain, there exists an element $Q \in D$, s.t.

$$G(Q) = \frac{\iint_D G\,dS}{A(S)}$$

But $G(Q) = \vec{F}(\phi(Q)) \cdot \vec{n}(Q)$

$\therefore$ There exists a $Q \in D$ s.t.

$$\left[\vec{F}(\phi(Q)) \cdot \vec{n}(Q)\right] A(S) = \iint_S \vec{F} \cdot \vec{n}\, dS$$

17. **17.** Work out a formula like that in Exercise 15 for integration over the surface of a cylinder.

A parametrization for a unit cylinder perpendicular to the xy-plane is: $T(\theta, z) = (\cos\theta, \sin\theta, z)$, $0 \le \theta \le 2\pi$, $a \le z \le b$ where the cylinder has a height from a to b.

∴ Normal to cylinder is:

$$\vec{T}_\theta \times \vec{T}_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = (\cos\theta, \sin\theta, 0)$$

and this normal points "out" from origin.

Let $\vec{n} = (\cos\theta, \sin\theta, 0)$.

Let $\vec{F} = (F_x, F_y, F_z)$. ∴ $\vec{F} \cdot \vec{n} = F_x \cos\theta + F_y \sin\theta$

But $\vec{F} \cdot \vec{n}$ is the radial component of $\vec{F}$: i.e., the component in the xy-plane parallel to $\vec{n}$.

∴ Let $F_r = F_x \cos\theta + F_y \sin\theta$.

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F} \cdot \vec{n}\, dA = \int_0^{2\pi} \int_a^b F_r\, dz\, d\theta$$

18. **18.** Let S be a surface in $\mathbb{R}^3$ that is actually a subset D of the xy plane. Show that the integral of a scalar function $f(x, y, z)$ over S reduces to the double integral of $f(x, y, z)$ over D. What does the surface integral of a vector field over S become? (Make sure your answer is compatible with Example 6.)

(a) A parametrization for S is $T(x,y) = (x, y, 0)$.

$$\vec{T}_x \times \vec{T}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = (0,0,1) = \hat{k}$$

$$\therefore \| \vec{T}_x \times \vec{T}_y \| = 1.$$

By definition,

$$\iint_S f(x,y,z)\, dS = \iint_D f(T(x,y)) \| \vec{T}_x \times \vec{T}_y \| \, dx\,dy$$

$$= \iint_D f(x,y,0)\, dx\,dy$$

(b) The surface integral of $\vec{F} = (F_x, F_y, F_z)$ is:

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F} \cdot (\vec{T}_x \times \vec{T}_y)\, dx\,dy = \iint_D F_z \, dx\,dy$$

19.

19. Let the velocity field of a fluid be described by $\mathbf{F} = \mathbf{i} + x\mathbf{j} + z\mathbf{k}$ (measured in meters per second). Compute how many cubic meters of fluid per second are crossing the surface described by $x^2 + y^2 + z^2 = 1, z \geq 0$.

Using spherical coordinates, a parametrization of the surface is: $T(\phi, \theta) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi) = (x, y, z)$

$0 \leq \phi \leq \frac{\pi}{2}, \quad 0 \leq \theta \leq 2\pi.$

As in Example 1, p. 401 of the text,

$$\vec{T}_\phi \times \vec{T}_\theta = \sin\phi(\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi)$$

$$= \sin\phi(x, y, z).$$

$$\therefore \vec{F} \cdot (\vec{T}_\phi \times \vec{T}_\theta) = (1, x, z) \cdot [\sin\phi(x, y, z)]$$

$$= \sin\phi(x + xy + z^2)$$

$$= \sin\phi[\sin\phi\cos\theta + (\sin\phi\cos\theta)(\sin\phi\sin\theta) + \cos^2\phi]$$

$$= \sin^2\phi\cos\theta + \sin^3\phi\cos\theta\sin\theta + \sin\phi\cos^2\phi$$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F} \cdot (\vec{T}_\phi \times \vec{T}_\theta)\, d\phi\, d\theta = \int_0^{2\pi}\int_0^{\frac{\pi}{2}} \sin^2\phi\cos\theta\, d\phi\, d\theta$$

$$+ \int_0^{2\pi}\int_0^{\frac{\pi}{2}} \sin^3\phi\cos\theta\sin\theta\, d\phi\, d\theta + \int_0^{2\pi}\int_0^{\frac{\pi}{2}} \sin\phi\cos^2\phi\, d\phi\, d\theta$$

$$= \int_0^{2\pi} \cos\theta\, d\theta \int_0^{\frac{\pi}{2}} \sin^2\phi\, d\phi + \int_0^{2\pi} \cos\theta \sin\theta\, d\theta \int_0^{\frac{\pi}{2}} \sin^3\phi\, d\phi + 2\pi \int_0^{\frac{\pi}{2}} \sin\phi \cos^2\phi\, d\phi$$

$$= \sin\theta \Big|_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin^2\phi\, d\phi + \frac{\sin^2\theta}{2}\Big|_0^{2\pi} \int_0^{\frac{\pi}{2}} \sin^3\phi\, d\phi + 2\pi \left[-\frac{\cos^3\phi}{3} \right]_0^{\frac{\pi}{2}}$$

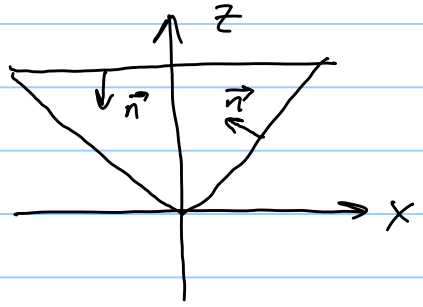
$$= 0 \cdot \int_0^{\pi/2} \sin^2\phi\, d\phi + 0 \cdot \int_0^{\pi/2} \sin^3\phi\, d\phi + 2\pi \left[0 - \left(-\frac{1}{3} \right) \right]$$

$$= 0 + 0 + \frac{2}{3}\pi = \frac{2\pi}{3}$$

20.

20. (a) A uniform fluid that flows vertically downward (heavy rain) is described by the vector field $\mathbf{F}(x, y, z) = (0, 0, -1)$. Find the total flux through the cone $z = (x^2 + y^2)^{1/2}$, $x^2 + y^2 \le 1$.

(b) The rain is driven sideways by a strong wind so that it falls at a 45° angle, and it is described by $\mathbf{F}(x, y, z) = -(\sqrt{2}/2, 0, \sqrt{2}/2)$. Now what is the flux through the cone?



(a) Choose normals (orientation) of surfaces that point inward to the cone, and ∴ computing flux "into" the cone.

(1) Top surface, a disk, with parametrization

$$\vec{T}(r, \theta) = (r\cos\theta, r\sin\theta, 1), \quad 0 \le r \le 1, \ 0 \le \theta \le 2\pi$$

$$\therefore \vec{T}_r \times \vec{T}_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & 0 \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (0, 0, r)$$

We want $\vec{n}$ to point down, $\therefore$ take

$$\vec{T}_\theta \times \vec{T}_r = -\vec{T}_r \times \vec{T}_\theta = (0,0,-r)$$

$$\therefore \vec{F} \cdot (\vec{T}_\theta \times \vec{T}_r) = (0,0,-1)\cdot(0,0,-r) = r$$

$$\therefore \iint_S \vec{F}\cdot d\vec{S} = \iint_D \vec{F}\cdot(\vec{T}_\theta \times \vec{T}_r)\, dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^1 r\, dr\, d\theta = 2\pi\left[\frac{r^2}{2}\right]_0^1 = \underline{\pi}$$

(2) Side of cone: parametrization is:

$$T(r,\theta) = (r\cos\theta, r\sin\theta, z), \text{ where}$$

$$z = (x^2+y^2)^{\frac{1}{2}} = (r^2\cos^2\theta + r^2\sin^2\theta)^{\frac{1}{2}} = r$$

$$\therefore T(r,\theta) = (r\cos\theta, r\sin\theta, r),\ 0\le r\le 1,\ 0\le\theta\le 2\pi$$

$$\vec{T}_r \times \vec{T}_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & 1 \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (-r\cos\theta,\ -r\sin\theta,\ r)$$

This normal points up ($r\hat{k}$ component) and in $(-r\cos\theta\,\hat{i} - r\sin\theta\,\hat{j})$, which is toward inside of cone.

$$\therefore \vec{F} \cdot (\vec{T}_r \times \vec{T}_\theta) = (0,0,-1) \cdot (-r\cos\theta, -r\sin\theta, r)$$

$$= -r.$$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \iint_D (-r)\, dr d\theta = \int_0^{2\pi} \int_0^1 (-r)\, dr d\theta$$

$$= -\pi$$

$$(1) + (2) = \pi - \pi = 0$$

(5) (b) The rain is driven sideways by a strong wind so that it falls at a 45° angle, and it is described by $\mathbf{F}(x, y, z) = -(\sqrt{2}/2, 0, \sqrt{2}/2)$. Now what is the flux through the cone?

(1) Top surface

From (a) $\vec{T}_\theta \times \vec{T}_r = (0,0,-r)$

$$\therefore \vec{F} \cdot (\vec{T}_\theta \times \vec{T}_r) = \left(-\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}\right) \cdot (0,0,-r) = \frac{\sqrt{2}}{2} r$$

$$\therefore \int_0^{2\pi} \int_0^1 \frac{\sqrt{2}}{2} r\, dr d\theta = \frac{\sqrt{2}}{2} \pi$$

(2) Side of cone

From (a), $\vec{T}_r \times \vec{T}_\theta = (-r\cos\theta, -r\sin\theta, r)$

$$\therefore \vec{F} \cdot (\vec{T}_r \times \vec{T}_\theta) = \left(-\frac{\sqrt{2}}{2}, 0, -\frac{\sqrt{2}}{2}\right) \cdot (-r\cos\theta, -r\sin\theta, r)$$

$$= \frac{\sqrt{2}}{2} r\cos\theta - \frac{\sqrt{2}}{2} r$$

$$\therefore \int_0^{2\pi} \int_0^1 \left(\frac{\sqrt{2}}{2} r\cos\theta - \frac{\sqrt{2}}{2} r\right) dr\,d\theta$$

$$= \int_0^{2\pi} \cos\theta\, d\theta \int_0^1 \frac{\sqrt{2}}{2} r\, dr - \int_0^{2\pi} \int_0^1 \frac{\sqrt{2}}{2} r\, dr\, d\theta$$

$$= 0 \cdot \frac{\sqrt{2}}{4} - 2\pi\left(\frac{\sqrt{2}}{4}\right) = \underline{-\frac{\sqrt{2}}{2}\pi}$$

$$(1) + (2) = \frac{\sqrt{2}}{2}\pi - \frac{\sqrt{2}}{2}\pi = \underline{0}$$

21.

21. For $a > 0$, $b > 0$, $c > 0$, let S be the upper half ellipsoid

$$S = \left\{(x, y, z) \,\middle|\, \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1, z \geq 0\right\},$$

with orientation determined by the upward normal. Compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F}(x, y, z) = (x^3, 0, 0)$.

Let $T(r, \theta) = (ar\cos\theta, br\sin\theta, z)$, $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$

where $z = c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}} = c\sqrt{1 - \frac{a^2r^2\cos^2\theta}{a^2} - \frac{b^2r^2\sin^2\theta}{b^2}}$

$$= c\sqrt{1 - r^2}$$

$$\therefore \vec{T}(r,\theta) = \left(ar\cos\theta,\ br\sin\theta,\ c\sqrt{1-r^2}\right)$$

$$\therefore \vec{T}_r \times \vec{T}_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a\cos\theta & b\sin\theta & -cr(1-r^2)^{-\frac{1}{2}} \\ -ar\sin\theta & br\cos\theta & 0 \end{vmatrix}$$

$$= \left(bcr^2(1-r^2)^{-\frac{1}{2}}\cos\theta,\ acr^2(1-r^2)^{-\frac{1}{2}},\ abr\right)$$

Since $abr \geq 0$, $abr\hat{k}$ points up.

$$\therefore \vec{F}\cdot(\vec{T}_r \times \vec{T}_\theta) = x^3\left[bcr^2(1-r^2)^{-\frac{1}{2}}\cos\theta\right]$$

$$= \left(a^3r^3\cos^3\theta\right)\left(bcr^2(1-r^2)^{-\frac{1}{2}}\cos\theta\right)$$

$$= \frac{a^3bcr^5\cos^4\theta}{\sqrt{1-r^2}}$$

$$\therefore \iint_S \vec{F}\cdot d\vec{S} = a^3bc\int_0^{2\pi}\int_0^1 \frac{\cos^4\theta\, r^5}{\sqrt{1-r^2}}\,dr\,d\theta$$

$$= a^3bc\int_0^{2\pi}\cos^4\theta\,d\theta \int_0^1 \frac{r^5}{\sqrt{1-r^2}}\,dr$$

Let $r = \sin\phi$
$\therefore dr = \cos\phi\,d\phi$
$\sqrt{1-r^2} = \cos\phi$

$$= a^3bc\int_0^{2\pi}\cos^2\theta(1-\sin^2\theta)\,d\theta \int_0^{\frac{\pi}{2}}\sin^5\phi\,d\phi$$

$\sin^5\phi = \sin^4\phi\sin\phi$
$= (1-\cos^2\phi)^2\sin\phi$

$$= a^3bc \int_0^{2\pi} \cos^2\theta - (\cos\theta\sin\theta)^2 \, d\theta \int_0^{\frac{\pi}{2}} (1 - 2\cos^2\phi + \cos^4\phi)\sin\phi \, d\phi$$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$, $\sin 2\theta = 2\sin\theta\cos\theta$

$$= a^3bc \int_0^{2\pi} \left[\frac{1+\cos 2\theta}{2} - \frac{\sin^2 2\theta}{4}\right] d\theta \left[-\cos\phi + \frac{2}{3}\cos^3\phi - \frac{\cos^5\phi}{5}\right]_0^{\frac{\pi}{2}}$$

$\sin^2 x = \frac{1-\cos 2x}{2}$

$$= a^3bc \int_0^{2\pi} \frac{1}{2} + \frac{\cos 2\theta}{2} - \left(\frac{1-\cos 4\theta}{8}\right) d\theta \left[0 - \left(-1 + \frac{2}{3} - \frac{1}{5}\right)\right]$$

$$= a^3bc \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} - \frac{\theta}{8} + \frac{\sin 4\theta}{32}\right]_0^{2\pi} \left[\frac{8}{15}\right]$$

$$= a^3bc \left[\pi + 0 - \frac{\pi}{4} + 0\right]\left[\frac{8}{15}\right] = a^3bc\left(\frac{3}{4}\pi\right)\left(\frac{8}{15}\right)$$

$$= \frac{2}{5} a^3bc\pi$$

22. **22.** If S is the upper hemisphere $\{(x, y, z) \mid x^2 + y^2 + z^2 = 1, z \geq 0\}$ oriented by the normal pointing out of the sphere, compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$ for parts (a) and (b).

Let $\vec{r}(\phi, \theta) = (\sin\phi\cos\theta, \sin\phi\sin\theta, \cos\phi)$, $0 \leq \phi \leq \frac{\pi}{2}$, $0 \leq \theta \leq 2\pi$

$\therefore \vec{r}_\phi \times \vec{r}_\theta = (\sin^2\phi\cos\theta, \sin^2\phi\sin\theta, \sin\phi\cos\phi)$, as in Example 1, p. 401 of the text. This normal points "out."

(a) $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j}$

$$\vec{F} = (\sin\phi\cos\theta, \sin\phi\sin\theta, 0).$$

$$\therefore \vec{F} \cdot (\vec{T}_\phi \times \vec{T}_\theta) = \sin^3\phi\cos^2\theta + \sin^3\phi\sin^2\theta = \sin^3\phi$$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \int_0^{2\pi}\int_0^{\pi} \sin^3\phi \, d\phi \, d\theta = 2\pi \int_0^{\pi} \sin^3\phi \, d\phi$$

$$= 2\pi \int_0^{\pi} (1-\cos^2\phi)\sin\phi \, d\phi = 2\pi \left[-\cos\phi + \frac{\cos^3\phi}{3} \right]_0^{\pi}$$

$$= 2\pi \left[1 - \frac{1}{3} - \left(-1 + \frac{1}{3}\right) \right] = 2\pi\left(\frac{4}{3}\right) = \underline{\underline{\frac{8}{3}\pi}}$$

(b) $\mathbf{F}(x, y, z) = y\mathbf{i} + x\mathbf{j}$

$$\vec{F} = (\sin\phi\sin\theta, \sin\phi\cos\theta, 0)$$

$$\therefore \vec{F} \cdot (\vec{T}_\phi \times \vec{T}_\theta) = \sin^3\phi\sin\theta\cos\theta + \sin^3\phi\sin\theta\cos\theta$$

$$= \sin^3\phi\,(2\sin\theta\cos\theta) = \sin^3\phi\sin 2\theta$$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \int_0^{2\pi}\int_0^{\pi} \sin^3\phi\sin 2\theta \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \sin 2\theta \, d\theta \int_0^{\pi} \sin^3\phi \, d\phi$$

$$= \left[-\frac{\cos 2\theta}{2} \Big|_0^{2\pi} \right] \begin{bmatrix} 4 \\ 3 \end{bmatrix} \qquad \text{from (a), } \int_0^{\pi} \sin^3\phi \, d\phi = \frac{4}{3}$$

$$= \left[-\frac{1}{2} - \left(-\frac{1}{2}\right) \right] \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 0 \cdot \frac{4}{3} = \underline{0}$$

(c) For each of these vector fields, compute $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ and $\int_C \mathbf{F} \cdot d\mathbf{s}$, where C is the unit circle in the xy plane traversed in the counterclockwise direction (as viewed from the positive z axis). (Notice that C is the boundary of S. The phenomenon illustrated here will be studied more thoroughly in the next chapter, using Stokes' theorem.)

(1) $\vec{F} = (x, y, 0)$

$$\therefore \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & 0 \end{vmatrix} = (0, 0, 0)$$

$$\therefore (\nabla \times \vec{F}) \cdot d\vec{S} = 0$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \underline{0}$$

(2) $\vec{F} = (y, x, 0)$ $\quad \therefore \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & x & 0 \end{vmatrix} = (0, 0, 1-1) = (0, 0, 0)$

$$\therefore (\nabla \times \vec{F}) \cdot d\vec{S} = 0 \quad \therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \underline{0}$$

(3) Let $\vec{c}(t) = (\cos t, \sin t, 0)$, $0 \le t \le 2\pi$

$\therefore \vec{c}\,'(t) = (-\sin t, \cos t, 0)$.

$\vec{F}(\vec{c}(t)) \cdot \vec{c}\,'(t) = (\cos t, \sin t, 0) \cdot (-\sin t, \cos t, 0)$

$= -\sin t \cos t + \sin t \cos t = 0$

$\therefore \iint_C \vec{F} \cdot d\vec{s} = \int_0^{2\pi} 0 \, dt = \underline{0}$

(4) Use $\vec{c}(t)$ as in (3)

$\vec{F}(\vec{c}(t)) = (\sin t, \cos t, 0)$

$\therefore \vec{F}(\vec{c}(t)) \cdot \vec{c}\,'(t) = (\sin t, \cos t, 0) \cdot (-\sin t, \cos t, 0)$

$= \cos^2 t - \sin^2 t = \cos 2t$

$\therefore \iint_C \vec{F} \cdot d\vec{s} = \int_0^{2\pi} \cos 2t \, dt = \frac{\sin 2t}{2} \Big|_0^{2\pi} = \underline{0}$

7.7 Applications to Differential Geometry, Physics, and Forms of Life

Note Title 4/24/2017

1.

1. The helicoid can be described by

$$\Phi(u, v) = (u\cos v,\ u\sin v,\ bv), \text{ where } b \neq 0.$$

Show that $H = 0$ and that $K = -b^2/(b^2 + u^2)^2$. In Figures 7.7.1 and 7.7.5, we see that the helicoid is actually a soap film surface. Surfaces in which $H = 0$ are called ***minimal surfaces.***

$$\phi_u = (\cos v, \sin v, 0) \qquad \phi_v = (-u\sin v, u\cos v, b)$$

$$\phi_{uu} = (0,0,0) \qquad \phi_{vv} = (-u\cos v, -u\sin v, 0)$$

$$\phi_{uv} = (-\sin v, \cos v, 0)$$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos v & \sin v & 0 \\ -u\sin v & u\cos v & b \end{vmatrix} = (b\sin v, -b\cos v, u)$$

$$\| \phi_u \times \phi_v \| = \sqrt{b^2 + u^2} \qquad \therefore W = b^2 + u^2$$

$$\therefore \ell(u,v) = \vec{N}(u,v) \cdot \phi_{uu} = 0$$

$$m(u,v) = \vec{N}(u,v) \cdot \phi_{uv} = \frac{(b\sin v, -b\cos v, u)}{\sqrt{b^2+u^2}} \cdot (-\sin v, \cos v, 0)$$

$$= \frac{-b}{\sqrt{b^2+u^2}}$$

$$n(u,v) = \vec{N}(u,v) \cdot \phi_{vv} = \frac{(b\sin v, -b\cos v, u)}{\sqrt{b^2+u^2}} \cdot (-u\cos v, -u\sin v, 0)$$

$$= \frac{-ub\sin v\cos v + ub\cos v\sin v + 0}{\sqrt{b^2+u^2}} = 0$$

$$\therefore K = \frac{ln - m^2}{w} = \frac{(0)\left(\frac{-2bu\cos v\sin v}{\sqrt{b^2+u^2}}\right) - \left(\frac{-b}{\sqrt{b^2+u^2}}\right)^2}{b^2+u^2}$$

$$= -\frac{b^2}{(b^2+u^2)^2}$$

$$H = \frac{Gl + En - 2Fm}{2w}$$

$$= \frac{\|\phi_v\|^2 \overset{l=0}{(0)} + \|\phi_u\|^2 \overset{n=0}{(0)} - 2\phi_u\cdot\phi_v\left(\frac{-b}{\sqrt{b^2+u^2}}\right)}{2(b^2+u^2)}$$

$$= \frac{-2(\cos v, \sin v, 0)\cdot(-u\sin v, u\cos v, b)\left(\frac{-b}{\sqrt{b^2+u^2}}\right)}{2(b^2+u^2)}$$

$$= \frac{-ub\cos v\sin v + ub\sin v\cos v + 0}{(b^2+u^2)(\sqrt{b^2+u^2})} = 0$$

$$\therefore \underline{H = 0}$$

2.

2. Consider the saddle surface $z = xy$. Show that

$$K = \frac{-1}{(1 + x^2 + y^2)^2},$$

and that

$$H = \frac{-xy}{(1 + x^2 + y^2)^{3/2}}.$$

Let $\phi(x,y) = (x, y, xy)$

$\therefore \phi_x = (1, 0, y) \quad \phi_y = (0, 1, x)$

$\phi_{xx} = (0,0,0) \quad \phi_{yy} = (0,0,0) \quad \phi_{xy} = (0,0,1)$

$E = \|\phi_x\|^2 = 1 + y^2 \quad G = \|\phi_y\|^2 = 1 + x^2$

$F = \phi_x \cdot \phi_y = xy$

$$\phi_x \times \phi_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & y \\ 0 & 1 & x \end{vmatrix} = (-y, -x, 1)$$

$W = \|\phi_x \times \phi_y\|^2 = 1 + x^2 + y^2$

$$\vec{N} = \frac{\phi_x \times \phi_y}{\|\phi_x \times \phi_y\|} = \frac{1}{\sqrt{1 + x^2 + y^2}}(-y, -x, 1)$$

$\therefore l = \vec{N} \cdot \phi_{xx} = \vec{N} \cdot (0,0,0) = 0$

$m = \vec{N} \cdot \phi_{xy} = \frac{(-y, -x, 1)}{\sqrt{1 + x^2 + y^2}} \cdot (0,0,1) = \frac{1}{\sqrt{1 + x^2 + y^2}}$

$n = \vec{N} \cdot \phi_{yy} = \vec{N} \cdot (0,0,0) = 0$

$$\therefore K = \frac{ln - m^2}{w} = \frac{(0)(0) - \frac{1}{1+x^2+y^2}}{1+x^2+y^2} = -\frac{1}{(1+x^2+y^2)^2}$$

$$H = \frac{Gl + En - 2Fm}{2w} = \frac{G(0) + E(0) - 2(xy)\left(\frac{1}{\sqrt{1+x^2+y^2}}\right)}{2(1+x^2+y^2)}$$

$$= \frac{-xy}{(1+x^2+y^2)^{3/2}}$$

3. **3.** Show that $\Phi(u, v) = (u, v, \log\cos v - \log\cos u)$ has mean curvature zero (and is thus a minimal surface; see Exercise 1).

$$\phi_u = (1, 0, \tan u) \qquad \phi_v = (0, 1, -\tan v)$$

$$\phi_{uu} = (0, 0, \sec^2 u) \qquad \phi_{vv} = (0, 0, -\sec^2 v)$$

$$\phi_{uv} = (0, 0, 0)$$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & \tan u \\ 0 & 1 & \tan v \end{vmatrix} = (-\tan u, -\tan v, 1)$$

$$\|\phi_u \times \phi_v\| = \sqrt{1 + \tan^2 u + \tan^2 v}, \quad W = 1 + \tan^2 u + \tan^2 v$$

$$E = \|\phi_u\|^2 = 1 + \tan^2 u \qquad G = \|\phi_v\|^2 = 1 + \tan^2 v$$

$$F = \phi_u \cdot \phi_v = -\tan u \tan v$$

$$\vec{N} = \frac{(-\tan u, -\tan v, 1)}{\sqrt{1+\tan^2 u + \tan^2 v}}$$

$$l = \vec{N} \cdot \phi_{uu} = \frac{(-\tan u, -\tan v, 1)}{\sqrt{1+\tan^2 u + \tan^2 v}} \cdot (0, 0, \sec^2 u)$$

$$= \frac{\sec^2 u}{\sqrt{1+\tan^2 u + \tan^2 v}} = \frac{\sec^2 u}{\sqrt{W}}$$

$$m = \vec{N} \cdot \phi_{uv} = \vec{N} \cdot (0,0,0) = 0$$

$$n = \vec{N} \cdot \phi_{vv} = \frac{(-\tan u, -\tan v, 1)}{\sqrt{1+\tan^2 u + \tan^2 v}} \cdot (0, 0, -\sec^2 v)$$

$$= \frac{-\sec^2 v}{\sqrt{1+\tan^2 u + \tan^2 v}} = \frac{-\sec^2 v}{\sqrt{W}}$$

$$\therefore H = \frac{Gl + En - 2Fm}{2W} \qquad m = 0$$

$$= \frac{\frac{(1+\tan^2 v)(\sec^2 u)}{\sqrt{W}} + \frac{(1+\tan^2 u)(-\sec^2 v)}{\sqrt{W}}}{2W}$$

$$\sec^2 x = 1 + \tan^2 x$$

$$= \frac{(1+\tan^2 v)(1+\tan^2 u) - (1+\tan^2 u)(1+\tan^2 v)}{2W^2}$$

$$= 0$$

4. **4.** Find the Gauss curvature of the elliptic paraboloid

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}.$$

Let $\phi(x,y) = \left(x, y, \frac{x^2}{a^2} + \frac{y^2}{b^2}\right)$

$\therefore \phi_x = \left(1, 0, \frac{2x}{a^2}\right) \quad \phi_y = \left(0, 1, \frac{2y}{b^2}\right)$

$\phi_{xx} = \left(0, 0, \frac{2}{a^2}\right) \quad \phi_{yy} = \left(0, 0, \frac{2}{b^2}\right)$

$\phi_{xy} = (0,0,0)$

$$\phi_x \times \phi_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 2x/a^2 \\ 0 & 1 & 2y/b^2 \end{vmatrix} = \left(-2x/a^2, -2y/b^2, 1\right)$$

$$\therefore \vec{N} = \frac{\left(-\frac{2x}{a^2}, -\frac{2y}{b^2}, 1\right)}{\sqrt{1 + \frac{4x^2}{a^4} + \frac{4y^2}{b^2}}} = \frac{\left(-\frac{2x}{a^2}, -\frac{2y}{b^2}, 1\right)}{\sqrt{W}}$$

$$l = \vec{N} \cdot \phi_{xx} = \frac{\left(-\frac{2x}{a^2}, -\frac{2y}{b^2}, 1\right)}{\sqrt{W}} \cdot \left(0, 0, \frac{2}{a^2}\right) = \frac{2}{a^2\sqrt{W}}$$

$$m = \vec{N} \cdot \phi_{xy} = 0$$

$$n = \vec{N} \cdot \phi_{yy} = \frac{\left(-\frac{2x}{a^2}, -\frac{2y}{b^2}, 1\right)}{\sqrt{W}} \cdot \left(0, 0, \frac{2}{b^2}\right) = \frac{2}{b^2\sqrt{W}}$$

$$\therefore K = \frac{ln - m^2}{w} = \frac{\left(\frac{2}{a^2\sqrt{w}}\right)\left(\frac{2}{b^2\sqrt{w}}\right) - 0}{w}$$

$$= \frac{4}{a^2b^2w^2} = \frac{4}{a^2b^2\left(1 + \frac{4x^2}{a^4} + \frac{4y^2}{b^4}\right)^2}$$

$$= \frac{4}{a^2b^2\left(\frac{a^4b^4 + 4b^4x^2 + 4a^4y^2}{a^4b^4}\right)^2}$$

$$= \frac{4a^6b^6}{(a^4b^4 + 4b^4x^2 + 4a^4y^2)^2}$$

5. **5.** Find the Gauss curvature of the hyperbolic paraboloid

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}.$$

Let $\phi(x,y) = \left(x, y, \frac{x^2}{a^2} - \frac{y^2}{b^2}\right)$

$\phi_x = (1, 0, 2x/a^2)$ $\phi_y = (0, 1, -2y/b^2)$

$\phi_{xx} = (0, 0, \frac{2}{a^2})$ $\phi_{yy} = (0, 0, -\frac{2}{b^2})$

$\phi_{xy} = (0, 0, 0)$

$$\phi_x \times \phi_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 2x/a^2 \\ 0 & 1 & -2y/b^2 \end{vmatrix} = (-2x/a^2, 2y/b^2, 1)$$

$$W = 1 + \frac{4x^2}{a^4} + \frac{4y^2}{b^4} = \frac{a^4b^4 + 4b^4x^2 + 4a^4y^2}{a^4b^4}$$

$$\vec{N} = \frac{\left(-\frac{2x}{a^2}, \frac{2y}{b^2}, 1\right)}{\sqrt{W}}$$

$$l = \vec{N} \cdot \phi_{xx} = \frac{\left(-\frac{2x}{a^2}, \frac{2y}{b^2}, 1\right)}{\sqrt{W}} \cdot \left(0, 0, \frac{2}{a^2}\right) = \frac{2}{a^2\sqrt{W}}$$

$$m = \vec{N} \cdot \phi_{xy} = 0$$

$$n = \vec{N} \cdot \phi_{yy} = \frac{\left(-\frac{2x}{a^2}, \frac{2y}{b^2}, 1\right)}{\sqrt{W}} \cdot \left(0, 0, -\frac{2}{b^2}\right) = -\frac{2}{b^2\sqrt{W}}$$

$$\therefore K = \frac{ln - m^2}{W} = \frac{\left(\frac{2}{a^2\sqrt{W}}\right)\left(-\frac{2}{b^2\sqrt{W}}\right) - 0}{W}$$

$$= -\frac{4}{a^2b^2W^2} = \frac{-4}{a^2b^2\left(\frac{a^4b^4 + 4b^4x^2 + 4a^4y^2}{a^4b^4}\right)^2}$$

$$= \frac{-4a^6b^6}{\left(a^4b^4 + 4b^4x^2 + 4a^4y^2\right)^2}$$

C.

6. Compute the Gauss curvature of the ellipsoid

$$\frac{x^2}{a^2}+\frac{y^2}{a^2}+\frac{z^2}{c^2}=1.$$

Let $x = a\cos\theta\sin\phi$, $y = a\sin\theta\sin\phi$, $z = c\cos\phi$

$$\therefore \frac{x^2}{a^2}+\frac{y^2}{a^2} = \frac{a^2\sin^2\phi(\cos^2\theta+\sin^2\theta)}{a^2} = \sin^2\phi$$

$$\therefore \frac{x^2}{a^2}+\frac{y^2}{a^2}+\frac{z^2}{c^2} = \sin^2\phi+\frac{c^2\cos^2\phi}{c^2} = 1.$$

$$\therefore T(\theta,\phi) = (a\cos\theta\sin\phi,\ a\sin\theta\sin\phi,\ c\cos\phi),$$
$$0\le\theta\le 2\pi,\ 0\le\phi\le\pi.$$

$T_\theta = (-a\sin\theta\sin\phi,\ a\cos\theta\sin\phi,\ 0)$ $\quad a^2\sin^2\phi$

$T_\phi = (a\cos\theta\cos\phi,\ a\sin\theta\cos\phi,\ -c\sin\phi)$ $\quad a^2\cos^2\phi + c^2\sin^2\phi$

$T_{\theta\theta} = (-a\cos\theta\sin\phi,\ -a\sin\theta\sin\phi,\ 0) = (-x,-y,0)$

$T_{\phi\phi} = (-a\cos\theta\sin\phi,\ -a\sin\theta\sin\phi,\ -c\cos\phi) = (-x,-y,-z)$

$T_{\theta\phi} = (-a\sin\theta\cos\phi,\ a\cos\theta\cos\phi,\ 0)$

$$T_\theta \times T_\phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -a\sin\theta\sin\phi & a\cos\theta\sin\phi & 0 \\ a\cos\theta\cos\phi & a\sin\theta\cos\phi & -c\sin\phi \end{vmatrix}$$

$$= (-ac\cos\theta\sin^2\phi,\ -ac\sin\theta\sin^2\phi,\ -a^2\sin\phi\cos\phi)$$

$$\|T_\theta \times T_\phi\|^2 = a^2c^2\sin^4\phi + a^4\sin^2\phi\cos^2\phi = W$$

$$\therefore \vec{N} = \frac{(-ac\cos\theta\sin^2\phi,\ -ac\sin\theta\sin^2\phi,\ -a^2\sin\phi\cos\phi)}{\sqrt{W}}$$

$$= -\frac{c\sin\phi}{\sqrt{W}}\left(x, y, \frac{a^2}{c^2}z\right)$$

$$\therefore l = \vec{N}\cdot T_{\theta\theta} = -\frac{c\sin\phi}{\sqrt{W}}\left(x, y, \frac{a^2}{c^2}z\right)\cdot(-x, -y, 0)$$

$$= \frac{c\sin\phi}{\sqrt{W}}\left(x^2+y^2\right)$$

$$= \frac{c\sin\phi}{\sqrt{W}}\left(a^2\sin^2\phi\right) = \frac{a^2c\sin^3\phi}{\sqrt{W}}$$

$$m = \vec{N}\cdot T_{\theta\phi} = \frac{(-ac\cos\theta\sin^2\phi,\ -ac\sin\theta\sin^2\phi,\ -a^2\sin\phi\cos\phi)}{\sqrt{W}}\cdot$$
$$(-a\sin\theta\cos\phi,\ a\cos\theta\cos\phi,\ 0)$$

$$= \frac{a^2c\sin\theta\cos\theta\sin^2\phi\cos\phi - a^2c\sin\theta\cos\theta\sin^2\phi\cos\phi + 0}{\sqrt{W}} = 0$$

$$n = \vec{N}\cdot T_{\phi\phi} = -\frac{c\sin\phi}{\sqrt{W}}\left(x, y, \frac{a^2}{c^2}z\right)\cdot(-x, -y, -z)$$

$$= \frac{c\sin\phi}{\sqrt{W}}\left(x^2+y^2+\frac{a^2}{c^2}z^2\right)$$

$$= \frac{c\sin\phi}{\sqrt{W}}\left[x^2+y^2+a^2\left(1-\frac{x^2}{a^2}-\frac{y^2}{a^2}\right)\right] = \frac{a^2c\sin\phi}{\sqrt{W}}$$

$$\therefore K = \frac{ln - m^2}{w} = \frac{\left(\frac{a^2 c \sin^3\phi}{\sqrt{w}}\right)\left(\frac{a^2 c \sin\phi}{\sqrt{w}}\right) - 0}{w}$$

$$= \frac{a^4 c^2 \sin^4\phi}{w^2} = \frac{a^4 c^2 \sin^4\phi}{(a^2 c^2 \sin^4\phi + a^4 \sin^2\phi \cos^2\phi)^2}$$

$$= \frac{a^4 c^2 \sin^4\phi}{[(a^2\sin^2\phi)(c^2\sin^2\phi + a^2\cos^2\phi)]^2}$$

$$= \frac{c^2}{(c^2\sin^2\phi + a^2\cos^2\phi)^2}$$

7. **7.** After finding K in Exercise 6, integrate K to show that:

$$\frac{1}{2\pi}\iint_S K\, dA = 2.$$

From #6 above, $K = \dfrac{c^2}{(c^2\sin^2\phi + a^2\cos^2\phi)^2}$

$$\frac{1}{2\pi}\iint_S 1\, dS = \frac{1}{2\pi}\iint_S K\, dA = \frac{1}{2\pi}\iint_S K \|T_\theta \times T_\phi\|\, d\theta d\phi$$

From #6 above, $\|T_\theta \times T_\phi\|^2 = a^2c^2\sin^4\phi + a^4\sin^2\phi\cos^2\phi$,

so $\|T_\theta \times T_\phi\| = a\sin\phi\,(c^2\sin^2\phi + a^2\cos^2\phi)^{\frac{1}{2}}$

$$\therefore K\|T_\theta \times T_\phi\| = \frac{ac^2\sin\phi\,(c^2\sin^2\phi + a^2\cos^2\phi)^{\frac{1}{2}}}{(c^2\sin^2\phi + a^2\cos^2\phi)^2}$$

$$= \frac{ac^2 \sin\phi}{(c^2\sin^2\phi + a^2\cos^2\phi)^{3/2}}$$

$$\therefore \frac{1}{2\pi}\iint_S K\,dA = \frac{1}{2\pi}\int_0^{2\pi}\int_0^{\pi} \frac{ac^2\sin\phi}{(c^2\sin^2\phi + a^2\cos^2\phi)^{3/2}}\,d\phi\,d\theta$$

$$= ac^2 \int_0^{\pi} \frac{\sin\phi}{(c^2\sin^2\phi + a^2\cos^2\phi)^{3/2}}\,d\phi \qquad [1]$$

Note $\sin\phi = -d(\cos\phi)$. From text table of integrals,

#42: $\int \frac{dx}{(a^2-x^2)^{3/2}} = \frac{x}{a^2\sqrt{a^2-x^2}}$

and from other tables, $\int \frac{dx}{(a^2+x^2)^{3/2}} = \frac{x}{a^2\sqrt{a^2+x^2}}$

$$\therefore \frac{\sin\phi}{(c^2\sin^2\phi + a^2\cos^2\phi)^{3/2}} = \frac{\sin\phi}{[c^2(1-\cos^2\phi) + a^2\cos^2\phi]^{3/2}}$$

$$= \frac{\sin\phi}{[c^2 - (c^2-a^2)\cos^2\phi]^{3/2}} = \frac{\sin\phi}{(c^2-a^2)^{3/2}\left(\frac{c^2}{c^2-a^2} - \cos^2\phi\right)^{3/2}} \quad c>a$$

or $$= \frac{\sin\phi}{[c^2 + (a^2-c^2)\cos^2\phi]^{3/2}} = \frac{\sin\phi}{(a^2-c^2)^{3/2}\left(\frac{c^2}{a^2-c^2} + \cos^2\phi\right)^{3/2}} \quad a>c$$

$\therefore$ Let $x = -\cos\phi \;\therefore\; dx = \sin\phi\,d\phi$

$\phi = 0 \Rightarrow x = -1,\ \phi = \pi \Rightarrow x = 1$

(a) Assume $c > a$

$\therefore$ [1] becomes $\dfrac{ac^2}{(c^2-a^2)^{3/2}} \displaystyle\int_{-1}^{1} \frac{dx}{\left(\frac{c^2}{c^2-a^2} - x^2\right)^{3/2}} =$

$$\frac{ac^2}{(c^2-a^2)^{3/2}} \cdot \frac{x}{\left(\frac{c^2}{c^2-a^2}\right)\sqrt{\frac{c^2}{c^2-a^2} - x^2}} \Bigg|_{x=-1}^{x=1}$$

using $\displaystyle\int \frac{du}{(a^2-u^2)^{3/2}}$

$$= \frac{2ac^2}{(c^2-a^2)^{3/2}\left(\frac{c^2}{c^2-a^2}\right)\sqrt{\frac{c^2}{c^2-a^2} - 1}}$$

$$= \frac{2a}{(c^2-a^2)(c^2-a^2)^{1/2}\left(\frac{1}{c^2-a^2}\right)\sqrt{\frac{a^2}{c^2-a^2}}}$$

$$= \frac{2a}{(c^2-a^2)^{1/2}\frac{1}{\sqrt{c^2-a^2}} \cdot a} = \underline{2}$$

$\therefore \dfrac{1}{2\pi} \displaystyle\iint_S K\,dA = 2$, assuming $c > a$.

(b) Now assume $a < c$

$\therefore$ [1] becomes $\dfrac{ac^2}{(a^2-c^2)^{3/2}} \displaystyle\int_{-1}^{1} \frac{dx}{\left(\frac{c^2}{a^2-c^2} + x^2\right)^{3/2}}$

$$= \frac{ac^2}{(a^2-c^2)^{3/2}} \cdot \frac{x}{\left(\frac{c^2}{a^2-c^2}\right)\sqrt{\frac{c^2}{a^2-c^2}+x^2}} \Bigg|_{x=-1}^{x=1}$$

using $\int \frac{du}{(a^2+u^2)^{3/2}}$

$$= \frac{2ac^2}{(a^2-c^2)^{3/2}\left(\frac{c^2}{a^2-c^2}\right)\sqrt{\frac{c^2}{a^2-c^2}+1}}$$

$$= \frac{2a}{(a^2-c^2)(a^2-c^2)^{1/2}\left(\frac{1}{a^2-c^2}\right)\sqrt{\frac{a^2}{a^2-c^2}}}$$

$$= \frac{2a}{(a^2-c^2)^{1/2}\frac{1}{\sqrt{a^2-c^2}}\cdot a} = \underline{2}$$

$$\therefore \frac{1}{2\pi}\iint_S K\,dA = 2, \quad \text{assuming } a<c$$

(c) For $a=c$, the ellipsoid is a sphere of radius a.

Example 2 on p. 415 of text showed $K = \frac{1}{a^2}$.

$$\therefore \frac{1}{2\pi}\iint_S K\,dA = \frac{1}{2\pi a^2}\iint_S dA = \frac{1}{2\pi a^2}(4\pi a^2) = 2$$

$$\therefore \frac{1}{2\pi}\iint_S K\,dA = 2 \quad \text{for } a<c,\ a>c,\ a=c.$$

8.

8. Find the curvature K of:

(a) the cylinder $\Phi(u, v) = (2\cos v, 2\sin v, u)$

(b) the surface $\Phi(u, v) = (u, v, u^2)$

(a) $\phi_u = (0,0,1)$ $\phi_v = (-2\sin v, 2\cos v, 0)$

$\phi_{uu} = (0,0,0)$ $\phi_{vv} = (-2\cos v, -2\sin v, 0)$ $\phi_{uv} = (0,0,0)$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ -2\sin v & 2\cos v & 0 \end{vmatrix} = (-2\cos v, -2\sin v, 0)$$

$$\|\phi_u \times \phi_v\|^2 = 4\cos^2 v + 4\sin^2 v = 4 = W$$

$$\vec{N} = \frac{\phi_u \times \phi_v}{\sqrt{W}} = (-\cos v, -\sin v, 0)$$

$$l = \vec{N} \cdot \phi_{uu} = \vec{N} \cdot (0,0,0) = 0$$

$$m = \vec{N} \cdot \phi_{uv} = \vec{N} \cdot (0,0,0) = 0$$

$$n = \vec{N} \cdot \phi_{vv} = (-\cos v, -\sin v, 0) \cdot (-2\cos v, -2\sin v, 0) = 2$$

$$\therefore K = \frac{ln - m^2}{W} = \frac{0 \cdot 2 - 0}{4} = \underline{0}$$

(b) $\phi_u = (1,0,2u)$ $\phi_v = (0,1,0)$

$\phi_{uu} = (0,0,2)$ $\phi_{vv} = (0,0,0)$ $\phi_{uv} = (0,0,0)$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 2u \\ 0 & 1 & 0 \end{vmatrix} = (-2u, 0, 1)$$

$$\| \phi_u \times \phi_v \|^2 = 4u^2 + 1 = w$$

$$\vec{N} = \frac{\phi_u \times \phi_v}{\sqrt{w}} = \frac{1}{\sqrt{4u^2+1}}(-2u, 0, 1)$$

$$l = \vec{N} \cdot \phi_{uu} = \frac{1}{\sqrt{4u^2+1}}(-2u, 0, 1) \cdot (0, 0, 2) = \frac{2}{4u^2+1}$$

$$n = \vec{N} \cdot \phi_{uv} = \vec{N} \cdot (0, 0, 0) = 0$$

$$m = \vec{N} \cdot \phi_{vv} = \vec{N} \cdot (0, 0, 0) = 0$$

$$K = \frac{ln - m^2}{w} = \frac{l(0) - (0)^2}{w} = \underline{0}$$

9.

9. Show that Enneper's surface

$$\Phi(u, v) = \left(u - \frac{u^3}{3} + uv^2,\ v - \frac{v^3}{3} + u^2 v,\ u^2 - v^2 \right)$$

is a minimal surface ($H = 0$).

$$\phi_u = (1 - u^2 + v^2,\ 2uv,\ 2u) \qquad \phi_v = (2uv,\ 1 - v^2 + u^2,\ -2v)$$

$$\phi_{uu} = (-2u, 2v, 2) \qquad \phi_{vv} = (2u, -2v, -2) \qquad \phi_{uv} = (2v, 2u, 0)$$

$$\begin{aligned} E &= (1 - u^2 + v^2)^2 + 4u^2v^2 + 4u^2 \\ &= 1 - 2u^2 + 2v^2 - 2u^2v^2 + u^4 + v^4 + 4u^2v^2 + 4u^2 \\ &= 1 + 2u^2 + 2v^2 + 2u^2v^2 + u^4 + v^4 \end{aligned}$$

$$F = (2uv - 2u^3v + 2uv^3) + (2uv - 2uv^3 + 2u^3v) - 4uv$$
$$= 0$$

$$G = 4u^2v^2 + (1 - v^2 + u^2)^2 + 4v^2$$
$$= 4u^2v^2 + (1 + 2u^2 - 2v^2 - 2u^2v^2 + u^4 + v^4) + 4v^2$$
$$= 1 + 2u^2 + 2v^2 + 2u^2v^2 + u^4 + v^4$$

$\therefore E = G$ Also note $\phi_{uu} = -\phi_{vv}$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ (-u^2 + v^2) & 2uv & 2u \\ 2uv & 1 - v^2 + u^2 & -2v \end{vmatrix}$$

$$= (-2u - 2uv^2 - 2u^3,\ 2v + 2u^2v + 2v^3,\ 1 - 2u^2v^2 - u^4 - v^4)$$

$$\| \phi_u \times \phi_v \|^2 = EG - F^2 = E^2 - 0 = E^2 = W$$

$$\vec{N} = \frac{\phi_u \times \phi_v}{\sqrt{W}}$$

$$l = \vec{N} \cdot \phi_{uu} = \vec{N} \cdot (-\phi_{vv}) = -\vec{N} \cdot \phi_{vv} = -n$$

$$m = \vec{N} \cdot \phi_{uv}$$

$$H = \frac{Gl + En - 2Fm}{2W} = \frac{Gl + (G)(-l) - 2(0)m}{2W}$$

$$= \frac{0 - 0}{2W} = \frac{0}{\text{—}}$$

10. **10.** Consider the torus T given in Exercise 4, Section 7.4. Compute its Gauss curvature and verify the theorem of Gauss–Bonnet. [HINT: Show that $\|T_\theta \times T_\phi\|^2 = (R+\cos\phi)^2$ and $K = \cos\phi/(R+\cos\phi)$.]

(a) From #4 of Section 7.4,

$$\bar{T}(\theta,\phi) = [(R+\cos\phi)\cos\theta, (R+\cos\phi)\sin\theta, \sin\phi]$$

$0 \le \theta \le 2\pi,\ 0 \le \phi \le 2\pi,\ R > 1$

And, from that problem,

$$\|T_\theta \times T_\phi\|^2 = \left[\frac{\partial(x,y)}{\partial(\theta,\phi)}\right]^2 + \left[\frac{\partial(y,z)}{\partial(\theta,\phi)}\right]^2 + \left[\frac{\partial(x,z)}{\partial(\theta,\phi)}\right]^2$$

$$= (R+\cos\phi)^2 = W$$

$$T_\theta = [-(R+\cos\phi)\sin\theta, (R+\cos\phi)\cos\theta, 0]$$

$$T_\phi = [-\cos\theta\sin\phi, -\sin\theta\sin\phi, \cos\phi]$$

$$T_{\theta\theta} = [-(R+\cos\phi)\cos\theta, -(R+\cos\phi)\sin\theta, 0]$$

$$T_{\phi\phi} = [-\cos\theta\cos\phi, -\sin\theta\cos\phi, -\sin\phi]$$

$$T_{\theta\phi} = [\sin\theta\sin\phi, -\cos\theta\sin\phi, 0]$$

$$T_\theta \times T_\phi = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -(R+\cos\phi)\sin\theta & (R+\cos\phi)\cos\theta & 0 \\ -\cos\theta\sin\phi & -\sin\theta\sin\phi & \cos\phi \end{vmatrix}$$

$$[(R+\cos\phi)\cos\theta\cos\phi, (R+\cos\phi)\sin\theta\cos\phi, -(R+\cos\phi)\sin\phi]$$

$$\vec{N} = \frac{T_\theta \times T_\phi}{\sqrt{W}} = \frac{T_\theta \times T_\phi}{R + \cos\phi} \qquad (\text{Note } R > 1 \Rightarrow R + \cos\phi > 0)$$

$$= (\cos\theta\cos\phi, \sin\theta\cos\phi, -\sin\phi)$$

$$l = \vec{N} \cdot T_{\theta\theta} = (\cos\theta\cos\phi, \sin\theta\cos\phi, -\sin\phi) \cdot$$
$$-(R+\cos\phi)[\cos\theta, \sin\theta, 0]$$
$$= -(R+\cos\phi)\cos\phi$$

$$n = \vec{N} \cdot T_{\phi\phi} = (\cos\theta\cos\phi, \sin\theta\cos\phi, -\sin\phi) \cdot$$
$$[-\cos\theta\cos\phi, -\sin\theta\cos\phi, -\sin\phi]$$

$$= -\cos^2\theta\cos^2\phi - \sin^2\theta\cos^2\phi - \sin^2\phi$$
$$= -\cos^2\phi - \sin^2\phi$$
$$= -1$$

$$m = \vec{N} \cdot T_{\theta\phi} = (\cos\theta\cos\phi, \sin\theta\cos\phi, -\sin\phi) \cdot$$
$$(\sin\theta\sin\phi, -\cos\theta\sin\phi, 0) = 0$$

$$\therefore K = \frac{ln - m^2}{W} = \frac{-(R+\cos\phi)\cos\phi\,(-1) - 0}{(R+\cos\phi)^2}$$

$$= \frac{\cos\phi}{(R+\cos\phi)}$$

(6) $$\frac{1}{2\pi}\iint_S K\,dA = \frac{1}{2\pi}\int_0^{2\pi}\int_0^{2\pi} K\,\|T_\theta \times T_\phi\|\,d\theta\,d\phi$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^{2\pi} \frac{\cos\phi}{(R+\cos\phi)} (R+\cos\phi)\, d\theta\, d\phi$$

$$= \frac{2\pi}{2\pi} \int_0^{2\pi} \cos\phi\, d\phi = \sin\phi \Big|_0^{2\pi} = \underline{\underline{0}}$$

11. **11.** Let $\Phi(u, v) = (u, h(u)\cos v, h(u)\sin v)$, $h > 0$, be a surface of revolution. Show that $K = -h''/h\{1 + (h')^2\}^2$.

Given x-coordinate u, get $h(u)$, and $h(u)$ revolves around x-axis.

$$\phi_u = (1, h'\cos v, h'\sin v) \qquad \phi_v = (0, -h\sin v, h\cos v)$$

$$\phi_{uu} = (0, h''\cos v, h''\sin v)$$

$$\phi_{vv} = (0, -h\cos v, -h\sin v)$$

$$\phi_{uv} = (0, -h'\sin v, h'\cos v)$$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & h'\cos v & h'\sin v \\ 0 & -h\sin v & h\cos v \end{vmatrix}$$

$$= (hh', -h\cos v, -h\sin v)$$

$$\|\phi_u \times \phi_v\|^2 = (hh')^2 + h^2 = h^2[1 + (h')^2] = W$$

$$\vec{N} = \frac{\phi_u \times \phi_v}{\sqrt{w}} = \frac{(h', -\cos v, -\sin v)}{\sqrt{1+(h')^2}}, \text{ as } h > 0.$$

$$\ell = \vec{N} \cdot \phi_{uu} = \frac{1}{\sqrt{1+(h')^2}} (h', -\cos v, -\sin v) \cdot (0, h''\cos v, h''\sin v)$$

$$= \frac{-h''}{\sqrt{1+(h')^2}}$$

$$n = \vec{N} \cdot \phi_{vv} = \frac{1}{\sqrt{1+(h')^2}} (h', -\cos v, -\sin v) \cdot (0, -h\cos v, -h\sin v)$$

$$= \frac{h}{\sqrt{1+(h')^2}}$$

$$m = \vec{N} \cdot \phi_{uv} = \frac{1}{\sqrt{1+(h')^2}} (h', -\cos v, -\sin v) \cdot (0, -h'\sin v, h'\cos v)$$

$$= \frac{0 + h'\cos v \sin v - h'\sin v \cos v}{\sqrt{1+(h')^2}} = 0$$

$$\therefore K = \frac{\ell n - m^2}{w} = \frac{\left(\frac{-h''}{\sqrt{1+(h')^2}}\right)\left(\frac{h}{\sqrt{1+(h')^2}}\right) - 0}{h^2\left[1+(h')^2\right]}$$

$$= \frac{-h''}{h\left[1+(h')^2\right]^2}$$

12. **12.** A parametrization Φ of a surface S is said to be ***conformal*** (see Section 7.4), provided that $E = G$, $F = 0$. Assume that Φ conformally parametrizes S.[19] Show that if H and K vanish identically, then S must be part of a plane in $\mathbb{R}^3$.

$$K=0 \Rightarrow \frac{\ell n - m^2}{w} = 0 \Rightarrow \ell n = m^2$$

$$H=0,\ E=G,\ F=0 \Rightarrow \frac{G\ell + En - 2(0)m}{2w} = 0$$

$$\Rightarrow G(\ell + n) = 0$$

(1) If $E = G = 0$, then $\phi_u = 0$ and $\phi_v = 0$

$\phi_u = 0 \Rightarrow \phi(u,v) = (f(v), g(v), h(v))$

$\phi_v = 0 \Rightarrow \phi(u,v) = (a,b,c)$, a,b,c constants.

So S is a point

(2) If $E \neq 0$, $G \neq 0$, then $G(\ell + n) = 0 \Rightarrow \ell = -n$.

$\therefore \ell n = m^2 \Rightarrow -n^2 = m^2$, so $\ell = n = m = 0$.

$\therefore \vec{N} \cdot \phi_{uu} = 0$, $\vec{N} \cdot \phi_{vv} = 0$, $\vec{N} \cdot \phi_{uv} = 0$

$\therefore \vec{N} \perp \phi_{uu}$, $\vec{N} \perp \phi_{vv}$, $\vec{N} \perp \phi_{uv}$ [0]

Since $F = 0$, $\phi_u \cdot \phi_v = 0$, and assuming

$\phi_u \neq 0$ and $\phi_v \neq 0$ (S is regular or smooth), then ϕ_u and ϕ_v form a basis for the plane determined by ϕ_u and ϕ_v ($\phi_u \times \phi_v \neq 0$ for a regular surface).

Since $\phi_u \perp \vec{N}$, $\phi_v \perp \vec{N}$, then $\phi_{uu}, \phi_{vv}, \phi_{uv}$ are in the same plane of ϕ_u and ϕ_v from [0]

$\therefore \phi_{uu} = \alpha_1 \phi_u + \beta_1 \phi_v$ [1]

$\phi_{vv} = \alpha_2 \phi_u + \beta_2 \phi_v$ [2]

$\phi_{uv} = \alpha_3 \phi_u + \beta_3 \phi_v$ [3]

where α_i, β_i are constants.

To show S is in a plane (or is planar), it suffices to show $\vec{N}$ is constant (i.e., does not depend on u or v from $S = \phi(u,v)$).

$\therefore$ Need to show $\frac{\partial}{\partial u}\vec{N} = \vec{0}$ and $\frac{\partial}{\partial v}\vec{N} = \vec{0}$

$\vec{N} = \frac{\phi_u \times \phi_v}{\|\phi_u \times \phi_v\|}$, by definition

and $\|\phi_u \times \phi_v\| = \sqrt{EG - F^2}$ (p. 414 of text)

$= \sqrt{E^2} = E = G$

and $E = \|\phi_u\|^2 = \phi_u \cdot \phi_u = G = \|\phi_v\|^2 = \phi_v \cdot \phi_v$

$\therefore \vec{N} = \dfrac{\phi_u \times \phi_v}{\phi_u \cdot \phi_u} = \dfrac{\phi_u \times \phi_v}{\phi_v \cdot \phi_v}$

$\therefore$ Look at $\dfrac{\partial \vec{N}}{\partial u} = \dfrac{\partial}{\partial u} \dfrac{\phi_u \times \phi_v}{\phi_u \cdot \phi_u}$

$$= \frac{(\phi_u \cdot \phi_u)[\phi_{uu} \times \phi_v + \phi_u \times \phi_{uv}] - (\phi_u \times \phi_v)(2\phi_{uu} \cdot \phi_u)}{[\phi_u \cdot \phi_u]^2} \quad [4]$$

From [1], $\phi_{uu} \times \phi_v = (\alpha_1 \phi_u + \beta_1 \phi_v) \times \phi_v$ $\quad \phi_v \times \phi_v = 0$

$= \alpha_1 \phi_u \times \phi_v$ $\quad [4.1]$

From [3], $\phi_u \times \phi_{uv} = \phi_u \times (\alpha_3 \phi_u + \beta_3 \phi_v)$ $\quad \phi_u \times \phi_u = 0$

$= \beta_3 \phi_u \times \phi_v$ $\quad [4.2]$

and $\phi_{uu} \cdot \phi_u = (\alpha_1 \phi_u + \beta_1 \phi_v) \cdot \phi_u$ $\quad F = 0 = \phi_u \cdot \phi_v$

$= \alpha_1 \phi_u \cdot \phi_u$ $\quad [4.3]$

$\therefore$ [4] becomes $\dfrac{(\phi_u \cdot \phi_u)(\alpha_1 + \beta_3)\phi_u \times \phi_v - 2\alpha_1(\phi_u \cdot \phi_u)\phi_u \times \phi_v}{[\phi_u \cdot \phi_u]^2}$

$$\therefore \frac{\partial \vec{N}}{\partial u} = (\beta_3 - \alpha_1)\frac{\phi_u \times \phi_v}{\phi_u \cdot \phi_u} \qquad [5]$$

Now using $E = G \Rightarrow \phi_u \cdot \phi_u = \phi_v \cdot \phi_v$,

$$\frac{\partial \vec{N}}{\partial u} = \frac{\partial}{\partial u}\frac{\phi_u \times \phi_v}{\phi_v \cdot \phi_v}$$

$$= \frac{(\phi_v \cdot \phi_v)[\phi_{uu} \times \phi_v + \phi_u \times \phi_{uv}] - (\phi_u \times \phi_v)(2\phi_{uv} \cdot \phi_v)}{[\phi_v \cdot \phi_v]^2}$$

using [4.1] and [4.2]

$$= \frac{(\phi_v \cdot \phi_v)(\alpha_1 + \beta_3)\phi_u \times \phi_v - (2\phi_{uv} \cdot \phi_v)\phi_u \times \phi_v}{[\phi_v \cdot \phi_v]^2} \qquad [5.1]$$

From [3], $\phi_{uv} = \alpha_3 \phi_u + \beta_3 \phi_v$, so

$$\phi_{uv} \cdot \phi_v = (\alpha_3 \phi_u + \beta_3 \phi_v) \cdot \phi_v \qquad F = 0 = \phi_u \cdot \phi_v$$

$$= \beta_3(\phi_v \cdot \phi_v)$$

$\therefore$ [5.1] becomes $\dfrac{(\phi_v \cdot \phi_v)(\alpha_1 + \beta_3)\phi_u \times \phi_v - 2\beta_3(\phi_v \cdot \phi_v)\phi_u \times \phi_v}{[\phi_v \cdot \phi_v]^2}$

$$\therefore \frac{\partial \vec{N}}{\partial u} = (\alpha_1 - \beta_3)\frac{\phi_u \times \phi_v}{\phi_v \cdot \phi_v} = (\alpha_1 - \beta_3)\frac{\phi_u \times \phi_v}{\phi_u \cdot \phi_u} \qquad [6]$$

$E = G \Rightarrow \phi_u \cdot \phi_u = \phi_v \cdot \phi_v$

$\therefore$ [5], [6] $\Rightarrow \dfrac{\partial \vec{N}}{\partial u} = (\beta_3 - \alpha_1)\dfrac{\phi_u \times \phi_v}{\phi_u \cdot \phi_u} = (\alpha_1 - \beta_3)\dfrac{\phi_u \times \phi_v}{\phi_u \cdot \phi_u}$

so $\beta_3 - \alpha_1 = \alpha_1 - \beta_3 \Rightarrow \alpha_1 = \beta_3$

so $\dfrac{\partial \vec{N}}{\partial u} = \vec{0}$ [7]

Now look at $\dfrac{\partial \vec{N}}{\partial v} = \dfrac{\partial}{\partial v}\dfrac{\phi_u \times \phi_v}{\phi_u \cdot \phi_u}$

$$= \frac{(\phi_u \cdot \phi_u)\left[\phi_{uv} \times \phi_v + \phi_u \times \phi_{vv}\right] - \phi_u \times \phi_v (2\phi_{uv} \cdot \phi_u)}{\left[\phi_u \cdot \phi_u\right]^2} \quad [8]$$

From [3], $\phi_{uv} \times \phi_v = (\alpha_3 \phi_u + \beta_3 \phi_v) \times \phi_v$ $\quad \phi_v \times \phi_v = 0$

$= \alpha_3(\phi_u \times \phi_v)$ [8.1]

and $\phi_{uv} \cdot \phi_u = (\alpha_3 \phi_u + \beta_3 \phi_v) \cdot \phi_u$ $\quad F = 0 = \phi_u \cdot \phi_v$

$= \alpha_3(\phi_u \cdot \phi_u)$ [8.2]

From [2], $\phi_u \times \phi_{vv} = \phi_u \times (\alpha_2 \phi_u + \beta_2 \phi_v)$ $\quad \phi_u \times \phi_u = 0$

$= \beta_2(\phi_u \times \phi_v)$ [8.3]

From [8.1], [8.2], [8.3], [8] becomes

$$\frac{(\phi_u \cdot \phi_u)(\alpha_3 + \beta_2)\,\phi_u \times \phi_v - (2\alpha_3)(\phi_u \cdot \phi_u)\,\phi_u \times \phi_v}{[\phi_u \cdot \phi_u]^2}$$

$$= (\beta_2 - \alpha_3)\,\frac{\phi_u \times \phi_v}{\phi_u \cdot \phi_u}$$

$$\therefore \frac{\partial \vec{N}}{\partial v} = (\beta_2 - \alpha_3)\,\frac{\phi_u \times \phi_v}{\phi_u \cdot \phi_u} \qquad [9]$$

Now using $E = G = \phi_u \cdot \phi_u = \phi_v \cdot \phi_v$,

$$\frac{\partial \vec{N}}{\partial v} = \frac{\partial}{\partial v}\,\frac{\phi_u \times \phi_v}{\phi_v \cdot \phi_v}$$

$$= \frac{(\phi_v \cdot \phi_v)[\phi_{uv} \times \phi_v + \phi_u \times \phi_{vv}] - (\phi_u \times \phi_v)(2\phi_{vv} \cdot \phi_v)}{[\phi_v \cdot \phi_v]^2} \qquad [10]$$

From [8.1], [8.3],

$$\phi_{uv} \times \phi_v + \phi_u \times \phi_{vv} = (\alpha_3 + \beta_2)\,\phi_u \times \phi_v \qquad [10.1]$$

From [2] $\phi_{vv} \cdot \phi_v = (\alpha_2 \phi_u + \beta_2 \phi_v) \cdot \phi_v$ $\phi_u \cdot \phi_v = 0 = F$

$$= \beta_2 (\phi_v \cdot \phi_v) \qquad [10.2]$$

From [10.1], [10.2], [10] becomes

$$\frac{(\phi_v \cdot \phi_v)(\alpha_3 + \beta_2)\,\phi_u \times \phi_v - (2\beta_2)(\phi_v \cdot \phi_v)\,\phi_u \times \phi_v}{[\phi_v \cdot \phi_v]^2}$$

$$= (\alpha_3 - \beta_2)\,\frac{\phi_u \times \phi_v}{\phi_v \cdot \phi_v}$$

$$\therefore \frac{\partial \vec{N}}{\partial v} = (\alpha_3 - \beta_2)\,\frac{\phi_u \times \phi_v}{\phi_v \cdot \phi_v} \qquad [11]$$

$\therefore$ [9], [11] mean $\frac{\partial \vec{N}}{\partial v} = (\beta_2 - \alpha_3)\frac{\phi_u \times \phi_v}{\phi_u \cdot \phi_u} = (\alpha_3 - \beta_2)\frac{\phi_u \times \phi_v}{\phi_v \cdot \phi_v}$

and since $\phi_u \cdot \phi_u = \phi_v \cdot \phi_v$ as $E = G$,

$$\beta_2 - \alpha_3 = \alpha_3 - \beta_2 \Rightarrow \alpha_3 = \beta_2 \Rightarrow \frac{\partial \vec{N}}{\partial v} = \vec{0} \qquad [12]$$

$\therefore$ [7], [12] $\frac{\partial \vec{N}}{\partial u} = \vec{0}$, $\frac{\partial \vec{N}}{\partial v} = \vec{0}$

$\therefore \vec{N}$, the normal vector to S, is a constant vector for all u, v.

$\therefore$ S is part of a plane (S is planar).

Review Exercises for Chapter 7

Note Title 5/1/2017

1.

1. Integrate $f(x, y, z) = xyz$ along the following paths:

(a) $\mathbf{c}(t) = (e^t \cos t,\ e^t \sin t,\ 3),\ 0 \le t \le 2\pi$

(b) $\mathbf{c}(t) = (\cos t,\ \sin t,\ t),\ 0 \le t \le 2\pi$

(c) $\mathbf{c}(t) = \frac{3}{2}t^2\mathbf{i} + 2t^2\mathbf{j} + t\mathbf{k},\ 0 \le t \le 1$

(d) $\mathbf{c}(t) = t\mathbf{i} + (1/\sqrt{2})\,t^2\mathbf{j} + \frac{1}{3}t^3\mathbf{k},\ 0 \le t \le 1$

(a) $\vec{c}\,'(t) = (e^t\cos t - e^t\sin t,\ e^t\sin t + e^t\cos t,\ 0)$

$$\|\vec{c}\,'(t)\| = \left[e^{2t}\cos^2 t - 2e^{2t}\cos t\sin t + e^{2t}\sin^2 t + e^{2t}\cos^2 t + 2e^{2t}\cos t\sin t + e^{2t}\sin^2 t + 0\right]^{\frac{1}{2}}$$

$$= \left[e^{2t} + e^{2t}\right]^{\frac{1}{2}} = \sqrt{2}\,e^t$$

$$f(x, y, z) = (e^t\cos t)(e^t\sin t)(3) = 3e^{2t}\cos t\sin t$$

$$\therefore f(x(t), y(t), z(t))\,\|\vec{c}\,'(t)\| = 3\sqrt{2}\,e^{3t}\cos t\sin t = \frac{\sqrt{2}}{2}\sin 2t\,(3e^{3t})$$

$$\therefore \int_{\vec{c}} f\,ds = \frac{\sqrt{2}}{2}\int_0^{2\pi} \sin 2t\,(3e^{3t})\,dt \qquad [1]$$

$u = \sin 2t \quad v = e^{3t}$

$dv = 3e^{3t}dt$

$$= \frac{\sqrt{2}}{2}e^{3t}\sin 2t\Big|_0^{2\pi} - \frac{\sqrt{2}}{2}\int_0^{2\pi} e^{3t}\,2\cos 2t\,dt$$

$\int u\,dv = uv - \int v\,du$

$$= 0 - \frac{\sqrt{2}}{3}\int_0^{2\pi} 3e^{3t}\cos 2t\,dt$$

$$= -\frac{\sqrt{2}}{3} e^{3t}\cos 2t \Big|_0^{2\pi} + \frac{\sqrt{2}}{3}\int_0^{2\pi} e^{3t}(-2\sin 2t)\,dt$$

$$= \left[-\frac{\sqrt{2}}{3}e^{6\pi} - \left(-\frac{\sqrt{2}}{3}\right)\right] - \frac{2\sqrt{2}}{3}\int_0^{2\pi} \left(\frac{1}{3}\right) 3e^{3t}\sin 2t\,dt$$

$$= \frac{\sqrt{2}}{3}\left(1 - e^{6\pi}\right) - \frac{2\sqrt{2}}{9}\int_0^{2\pi} 3e^{3t}\sin 2t\,dt \qquad [2]$$

Combining [1] and [2],

$$\therefore \left(\frac{\sqrt{2}}{2} + \frac{2\sqrt{2}}{9}\right)\int_0^{2\pi} 3e^{3t}\sin 2t\,dt = \frac{\sqrt{2}}{3}\left(1 - e^{6\pi}\right)$$

$$\therefore \left(1 + \frac{4}{9}\right)\frac{\sqrt{2}}{2}\int_0^{2\pi} 3e^{3t}\sin 2t\,dt = \frac{\sqrt{2}}{3}\left(1 - e^{6\pi}\right)$$

$$\therefore \int_{\vec{c}} f\,ds = \frac{\sqrt{2}}{2}\int_0^{2\pi} 3e^{3t}\sin 2t\,dt = \frac{9}{13}\left(\frac{\sqrt{2}}{3}\right)\left(1 - e^{6\pi}\right)$$

$$= \frac{3\sqrt{2}}{13}\left(1 - e^{6\pi}\right)$$

(6) (b) $\mathbf{c}(t) = (\cos t, \sin t, t),\ 0 \le t \le 2\pi$

$\vec{c}\,'(t) = (-\sin t, \cos t, 1)$ $\therefore \|\vec{c}\,'(t)\| = \sqrt{2}$

$$f(x,y,z) = (\cos t)(\sin t)(t)$$

$$\therefore \int_{\vec{C}} f\,ds = \int_0^{2\pi} \sqrt{2}\, t \cos t \sin t\, dt$$

$\sin 2\theta = 2\sin\theta\cos\theta$

$$\frac{\sqrt{2}}{2}\int_0^{2\pi} t\sin(2t)\,dt = -\frac{\sqrt{2}}{4}\int_0^{2\pi} t\, d(\cos(2t))$$

$$= -\frac{\sqrt{2}}{4}\left[t\cos(2t)\right]_0^{2\pi} + \frac{\sqrt{2}}{4}\int_0^{2\pi}\cos(2t)\,dt$$

$\int u\,dv = uv - \int v\,du$

$$= -\frac{\sqrt{2}}{4}\left\{2\pi - 0\right\} + \frac{\sqrt{2}}{4}\left.\frac{\sin 2t}{2}\right|_0^{2\pi}$$

$$= -\frac{\sqrt{2}}{2}\pi$$

(c) (c) $\mathbf{c}(t) = \frac{3}{2}t^2\mathbf{i} + 2t^2\mathbf{j} + t\mathbf{k},\ 0 \le t \le 1$

$$\vec{C}'(t) = (3t, 4t, 1). \quad \|\vec{C}'(t)\| = \sqrt{9t^2 + 16t^2 + 1} = \sqrt{1+25t^2}$$

$$f(x,y,z) = \left(\tfrac{3}{2}t^2\right)(2t^2)(t) = 3t^5$$

$$\therefore \int_{\vec{C}} f\,ds = \int_0^1 3t^5\sqrt{1+25t^2}\,dt$$

From Wolfram: $\int x^5\sqrt{1+25x^2}\,dx = \frac{(25x^2+1)^{3/2}(9375x^4-300x^2+8)}{1640625} + \text{constant}$

$$\therefore \int_{\vec{c}} f\,ds = \frac{3(25t^2+1)^{3/2}(9375t^4-300t^2+8)}{1,640,625}\Bigg|_0^1$$

$$= \frac{3(26)^{3/2}(9375-300+8) - 3(1)(8)}{1,640,625} \qquad 1,640,625 = 3(35)(25^3)$$

$$= \frac{3(26)\sqrt{26}(9083) - 24}{3(35)(25)^3} \qquad (26)(9083) = 236,158$$

$$= \frac{(236,158)\sqrt{26} - 8}{35(25)^3}$$

(d) (d) $\mathbf{c}(t) = t\mathbf{i} + (1/\sqrt{2})\,t^2\mathbf{j} + \frac{1}{3}t^3\mathbf{k},\ 0 \le t \le 1$

$$\vec{c}\,'(t) = (1, \sqrt{2}t, t^2)$$

$$\therefore \|\vec{c}(t)\| = \sqrt{1+2t^2+t^4} = \sqrt{(1+t^2)^2} = 1+t^2$$

$$f(x,y,z) = f(\vec{c}(t)) = (t)\left(\frac{t^2}{\sqrt{2}}\right)\left(\frac{1}{3}t^3\right) = \frac{t^6}{3\sqrt{2}}$$

$$\therefore \int_{\vec{c}} f\,ds = \int_0^1 \left(\frac{t^6}{3\sqrt{2}}\right)(1+t^2)\,dt$$

$$= \frac{1}{3\sqrt{2}} \int_0^1 t^6 + t^8 \, dt = \frac{1}{3\sqrt{2}} \left[\frac{t^7}{7} + \frac{t^9}{9} \right]_0^1$$

$$= \frac{1}{3\sqrt{2}} \left(\frac{1}{7} + \frac{1}{9} \right) = \frac{1}{3\sqrt{2}} \left(\frac{16}{63} \right) = \frac{8\sqrt{2}}{189}$$

2.

2. Compute the integral of f along the path $\mathbf{c}$ in each of the following cases:

(a) $f(x, y, z) = x + y + yz$; $\mathbf{c}(t) = (\sin t, \cos t, t)$, $0 \le t \le 2\pi$

(a) $\vec{c}'(t) = (\cos t, -\sin t, 1)$ $\quad \|\vec{c}'(t)\| = \sqrt{2}$

$f(\vec{c}(t)) = \sin t + \cos t + t\cos t$

$$\therefore \int_{\vec{c}} f \, ds = \sqrt{2} \int_0^{2\pi} \sin t + \cos t + t\cos t \, dt$$

$$= 0 + 0 + \sqrt{2} \int_0^{2\pi} t\cos t \, dt = \sqrt{2} \int_0^{2\pi} t \, d(\sin t)$$

$$= \sqrt{2}\, t\sin t \Big|_0^{2\pi} - \sqrt{2} \int_0^{2\pi} \sin t \, dt$$

$$= 0 - 0 = \underline{0}$$

(6) (b) $f(x, y, z) = x + \cos^2 z$; $\mathbf{c}(t) = (\sin t, \cos t, t)$, $0 \le t \le 2\pi$

$\vec{c}'(t) = (\cos t, -\sin t, 1) \quad \|\vec{c}'(t)\| = \sqrt{2}$

$f(\vec{c}(t)) = \sin t + \cos^2 t$

$$\therefore \int_{\vec{c}} f\,ds = \sqrt{2}\int_0^{2\pi} \sin t + \cos^2 t\,dt = \sqrt{2}\int_0^{2\pi} \cos^2 t\,dt$$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1$

$$= \frac{\sqrt{2}}{2}\int_0^{2\pi} (1 + \cos 2t)\,dt = \frac{\sqrt{2}}{2}\left[t + \frac{\sin 2t}{2}\right]_0^{2\pi}$$

$$= \frac{\sqrt{2}}{2}(2\pi) = \underline{\pi\sqrt{2}}$$

(c) (c) $f(x, y, z) = x + y + z$; $\mathbf{c}(t) = (t, t^2, \frac{2}{3}t^3)$, $0 \le t \le 1$

$\vec{c}'(t) = (1, 2t, 2t^2)$

$\|\vec{c}'(t)\| = \sqrt{1 + 4t^2 + 4t^4} = \sqrt{(1 + 2t^2)^2} = 1 + 2t^2$

$f(\vec{c}(t)) = t + (t^2) + (\frac{2}{3}t^3)$

$$\therefore \int_{\vec{c}} f\,ds = \int_0^1 f(\vec{c}(t))\,\|\vec{c}'(t)\|\,dt$$

$$= \int_0^1 \left[t + t^2 + \frac{2}{3}t^3\right](1 + 2t^2)\,dt$$

$$= \int_0^1 t + t^2 + \frac{8}{3}t^3 + 2t^4 + \frac{4}{3}t^5\,dt$$

$$= \frac{t^2}{2} + \frac{t^3}{3} + \frac{2}{3}t^4 + \frac{2}{5}t^5 + \frac{2}{9}t^6 \Big|_0^1$$

$$= \frac{1}{2} + \frac{1}{3} + \frac{2}{3} + \frac{2}{5} + \frac{2}{9} = \frac{3}{2} + \frac{2}{5} + \frac{2}{9} = \frac{135 + 36 + 20}{90}$$

$$= \frac{191}{90}$$

3.

3. Compute each of the following line integrals:

(a) $\int_C (\sin \pi x)\,dy - (\cos \pi y)\,dz$, where C is the triangle whose vertices are (1, 0, 0), (0, 1, 0), and (0, 0, 1), in that order

(b) $\int_C (\sin z)\,dx + (\cos z)\,dy - (xy)^{1/3}\,dz$, where C is the path $\mathbf{c}(\theta) = (\cos^3 \theta, \sin^3 \theta, \theta)$, $0 \le \theta \le 7\pi/2$

(a) $C_1: (1,0,0)$ to $(0,1,0)$: $(1,0,0) + t[(0,1,0) - (1,0,0)]$, $0 \le t \le 1$

$\therefore \vec{c}_1(t) = (1,0,0) + t(-1,1,0) = (1-t, t, 0) = [x(t), y(t), z(t)]$

$\vec{c}_1'(t) = (-1, 1, 0) = (x', y', z')$

$\therefore (\sin \pi x)\,dy - (\cos \pi y)\,dz =$

$$[\sin \pi(1-t)](1) - [\cos \pi t](0) = \sin \pi(1-t)$$

$$\therefore \int_{C_1} \sin \pi x \, dy - \cos \pi y \, dz = \int_0^1 \sin \pi(1-t) \, dt$$

$$= \frac{\cos \pi(1-t)}{\pi} \Big|_0^1 = \frac{1}{\pi} \left[\cos(0) - \cos(\pi) \right]$$

$$= \underline{\frac{2}{\pi}}$$

$C_2 : (0,1,0) \text{ to } (0,0,1) : (0,1,0) + t[(0,0,1) - (0,1,0)], \; 0 \le t \le 1$

$$\therefore \vec{C}_2(t) = (0,1,0) + t(0,-1,1) = (0, 1-t, t) = (x, y, z)$$

$$\vec{C}'_2(t) = (0,-1,1) = (x', y', z')$$

$$\therefore (\sin \pi x) \, dy - (\cos \pi y) \, dz =$$

$$(\sin \pi(0))(-1) - [\cos \pi(1-t)](1) = -\cos \pi(1-t)$$

$$\therefore \int_{C_2} \sin \pi x \, dy - \cos \pi y \, dz = \int_0^1 -\cos(\pi - \pi t) \, dt$$

$$= \frac{\sin(\pi - \pi t)}{\pi} \Big|_0^1 = \frac{1}{\pi} \left[\sin(0) - \sin(\pi) \right] = \underline{0}$$

C_3 : $(0,0,1)$ to $(1,0,0)$: $(0,0,1) + t[(1,0,0) - (0,0,1)]$, $0 \le t \le 1$

$\therefore \vec{C}_3(t) = (0,0,1) + t(1,0,-1) = (t, 0, 1-t) = (x, y, z)$

$\vec{C}_3'(t) = (1,0,-1) = (x', y', z')$

$\therefore (\sin \pi x)\,dy - (\cos \pi y)\,dz =$

$(\sin \pi t)(0) - (\cos \pi(0))(-1) = 1$

$$\therefore \int_{C_3} \sin \pi x \, dy - \cos \pi y \, dz = \int_0^1 1 \, dt = \underline{1}$$

$$\therefore C_1 + C_2 + C_3 = \frac{2}{\pi} + 0 + 1 = \underline{\underline{1 + \frac{2}{\pi}}}$$

(6) (b) $\int_C (\sin z)\,dx + (\cos z)\,dy - (xy)^{1/3}\,dz$, where C is the path $\mathbf{c}(\theta) = (\cos^3\theta, \sin^3\theta, \theta)$, $0 \le \theta \le 7\pi/2$

$\vec{C}(\theta) = (\cos^3\theta, \sin^3\theta, \theta) = (x, y, z)$

$\vec{C}'(\theta) = (-3\cos^2\theta \sin\theta, 3\sin^2\theta \cos\theta, 1) = (x', y', z')$

$\therefore \sin z\,dx = (\sin\theta)(-3\cos^2\theta\sin\theta) = -3\cos^2\theta\sin^2\theta$

$\cos z\,dy = (\cos\theta)(3\sin^2\theta\cos\theta) = 3\cos^2\theta\sin^2\theta$

$(xy)^{1/3} = (\cos^3\theta \sin^3\theta)^{1/3} = \cos\theta\sin\theta$

$$\therefore \int_C \sin z\,dx + \cos z\,dy - (xy)^{\frac{1}{3}} = \int_0^{\frac{7\pi}{2}} -\cos\theta\sin\theta\,d\theta$$

$$= -\frac{1}{2}\int_0^{7\pi/2} \sin 2\theta\,d\theta = \frac{1}{4}\cos 2\theta \Big|_0^{\frac{7\pi}{2}}$$

$$= \frac{1}{4}\left[\cos(7\pi) - \cos(0)\right] = \frac{1}{4}\left[-1-1\right] = -\frac{1}{2}$$

4.

4. If $\mathbf{F}(\mathbf{x})$ is orthogonal to $\mathbf{c}'(t)$ at each point on the curve $\mathbf{x} = \mathbf{c}(t)$, what can you say about $\int_C \mathbf{F}\cdot d\mathbf{s}$?

$$\int_{\vec{C}} \vec{F}\cdot d\vec{s} = \int_{\vec{C}} \vec{F}(\vec{c}(t))\cdot \vec{c}'(t) = \int_{\vec{C}} (0)\,dt = 0$$

5.

5. Find the work done by the force $\mathbf{F}(x, y) = (x^2 - y^2)\mathbf{i} + 2xy\mathbf{j}$ in moving a particle counterclockwise around the square with corners $(0, 0)$, $(a, 0)$, (a, a), $(0, a)$, $a > 0$.

Let $C = C_1 + C_2 + C_3 + C_4$

$C_1 : (0,0)$ to $(a,0) : (0,0) + t\left[(a,0) - (0,0)\right]$, $0 \le t \le 1$

$\therefore \vec{C}_1(t) = (at, 0) = (x, y)$

$\vec{C}_1'(t) = (a, 0) = (x', y')$

$\vec{F}(\vec{C}_1(t)) = (a^2t^2, 0)$

$$\therefore \vec{F}(\vec{c}_1(t)) \cdot \vec{c}_1'(t) = a^3 t^2$$

$$\therefore \int_{C_1} \vec{F} \cdot d\vec{s} = \int_0^1 a^3 t^2 \, dt = \frac{a^3}{3} t^3 \Big|_0^1 = \underline{\frac{a^3}{3}}$$

$C_2 : (a,0) \text{ to } (a,a) : (a,0) + t[(a,a) - (a,0)] =$
$(a,0) + t(0,a), \quad 0 \le t \le 1$

$$\therefore \vec{c}_2(t) = (a, at) = (x, y)$$

$$\vec{c}_2'(t) = (0, a) = (x', y')$$

$$\therefore \vec{F}(\vec{c}_2(t)) = (a^2 - a^2 t^2, 2a^2 t)$$

$$\therefore \vec{F}(\vec{c}_2(t)) \cdot \vec{c}_2'(t) = 2a^3 t$$

$$\therefore \int_{C_2} \vec{F} \cdot d\vec{s} = \int_0^1 2a^3 t = a^3 t^2 \Big|_0^1 = a^3$$

$C_3 : (a,a) \text{ to } (0,a) : (a,a) + t[(0,a) - (a,a)] =$
$(a,a) + t(-a, 0), \quad 0 \le t \le 1$

$$\therefore \vec{c}_3(t) = (a - at, a) = (x, y)$$

$$\vec{c}_3'(t) = (-a, 0) = (x', y')$$

$$\vec{F}(\vec{c}_3(t)) = [(a - at)^2 - a^2, 2a(a - at)]$$

$$= (a^2t^2 - 2a^2t,\ 2a^2 - 2a^2t)$$

$$\therefore \vec{F}(\vec{c}_3(t)) \cdot \vec{c}_3'(t) = 2a^3t - a^3t^2$$

$$\int_{C_3} \vec{F} \cdot d\vec{s} = \int_0^1 2a^3t - a^3t^2 = a^3t^2 - \frac{a^3}{3}t^3 \Big|_0^1$$

$$= a^3 - \frac{a^3}{3}$$

$$C_4 : (0,a) \text{ to } (0,0) : (0,a) + t[(0,0) - (0,a)] =$$
$$(0,a) + t(0,-a),\ 0 \le t \le 1$$

$$\therefore \vec{c}_4(t) = (0,\ a - at) = (x,\ y)$$

$$\vec{c}_4'(t) = (0,\ -a) = (x',\ y')$$

$$\vec{F}(\vec{c}_4(t)) = [0 - (a-at)^2,\ 2(0)(a-at)]$$

$$= [-a^2 + 2a^2t - a^2t^2,\ 0]$$

$$\therefore \vec{F}(\vec{c}_4(t)) \cdot \vec{c}_4'(t) = 0$$

$$\int_{C_4} \vec{F} \cdot d\vec{s} = \int_0^1 0\, dt = \underline{0}$$

$$\therefore C_1 + C_2 + C_3 + C_4 = \frac{a^3}{3} + a^3 + \left(a^3 - \frac{a^3}{3}\right) + 0 = \underline{2a^3}$$

6. **6.** A ring in the shape of the curve $x^2 + y^2 = a^2$ is formed of thin wire weighing $|x| + |y|$ grams per unit length at (x, y). Find the mass of the ring.

Using symmetry, only need to calculate for Quadrant I, then multiply by 4.

$\therefore F(x,y) = x + y. \quad \vec{C}(\theta) = (a\cos\theta, a\sin\theta) = (x,y)$

$0 \le \theta \le \pi/2$

$\therefore \vec{C}'(\theta) = (-a\sin\theta, a\cos\theta) \quad \|\vec{C}'(\theta)\| = a$

$$\therefore \int_C F\,ds = \int_0^{\pi/2} F(\vec{C}(\theta))\,\|\vec{C}'(\theta)\|\,d\theta$$

$$= \int_0^{\pi/2} (a\cos\theta + a\sin\theta)\,a\,d\theta$$

$$= a^2 \int_0^{\pi/2} (\cos\theta + \sin\theta)\,d\theta = a^2 \left[\sin\theta - \cos\theta\right]_0^{\pi/2}$$

$$= a^2 \left[1 - 0 - (0 - 1)\right] = 2a^2$$

$\therefore$ Mass $= \underline{8a^2}$

7.

7. Find a parametrization for each of the following surfaces:

(a) $x^2 + y^2 + z^2 - 4x - 6y = 12$

(b) $2x^2 + y^2 + z^2 - 8x = 1$

(c) $4x^2 + 9y^2 - 2z^2 = 8$

(a) $(x^2 - 4x + 4) + (y^2 - 6y + 9) + z^2 = 12 + 4 + 9$

$\therefore (x-2)^2 + (y-3)^2 + z^2 = 5^2$, a sphere, or

$\| (x,y,z) - (2,3,0) \| = 5$

Use spherical coordinates

$x' = \rho \sin\phi \cos\theta$, $y' = \rho \sin\phi \sin\theta$, $z' = \rho \cos\phi$

Here, $\rho = 5$, and (x', y', z') centered at $(0,0,0)$.

$\therefore (x,y,z) = (x',y',z') + (2,3,0)$

$\therefore x = 2 + 5\sin\phi\cos\theta \quad y = 3 + 5\sin\phi\sin\theta \quad z = 5\cos\phi$

Or, $T(\theta,\phi) = (2 + 5\sin\phi\cos\theta, 3 + 5\sin\phi\sin\theta, 5\cos\phi)$,

$0 \le \phi \le \pi$, $0 \le \theta \le 2\pi$

(b) $2x^2 - 8x = 2(x^2 - 4x) = 2(x^2 - 4x + 4) - 8 = 2(x-2)^2 - 8$

$= \underbrace{(x-2)^2}_{\frac{1}{2}} - 8$

$\therefore \underbrace{(x-2)^2}_{\frac{1}{2}} + y^2 + z^2 = 9$, or

$$\frac{(x-2)^2}{\left(\frac{3}{\sqrt{2}}\right)^2} + \frac{y^2}{3^2} + \frac{z^2}{3^2} = 1 \qquad [1]$$

$\therefore$ an ellipsoid centred at $(2,0,0)$, with minor axes of $\frac{3}{\sqrt{2}}$ for x-axis, 3 for y,z axes.

$\therefore$ Use spherical coordinates as a framework.

From $\frac{(x-2)^2}{\left(\frac{\sqrt{3}}{2}\right)^2}$, let $x = 2 + \frac{\sqrt{3}}{2} f(\theta,\phi)$

$\therefore$ $\frac{(x-2)^2}{\left(\frac{\sqrt{3}}{2}\right)^2}$ becomes $f^2(\theta,\phi)$

If $y = 3g(\theta,\phi)$, $z = 3h(\theta,\phi)$, then [1] becomes $f^2 + g^2 + h^2 = 1$.

Using the sphere as an analogy,

$f(\theta,\phi) = \cos\theta\sin\phi$, $g(\theta,\phi) = \sin\theta\sin\phi$,

$h(\theta,\phi) = \cos\phi$

$\therefore$ $x = 2 + \frac{\sqrt{3}}{2}\cos\theta\sin\phi$, $y = 3\sin\theta\sin\phi$, $z = 3\cos\phi$

$\therefore$ The "ρ" coefficient is the length of the

relevant minor axis.

$\therefore T(\theta,\phi) = \left[2 + \frac{\sqrt{3}}{2}\cos\theta\sin\phi,\ 3\sin\theta\sin\phi,\ 3\cos\phi\right]$,

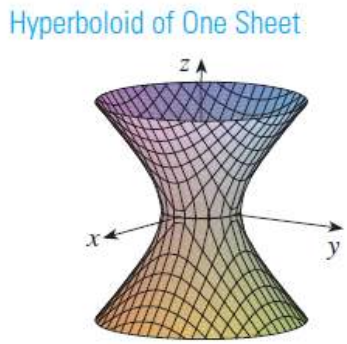
$0 \le \theta \le 2\pi,\ 0 \le \phi \le \pi$

(C) (c) $4x^2 + 9y^2 - 2z^2 = 8$

$\frac{4}{8}x^2 + \frac{9}{8}y^2 - \frac{2}{8}z^2 = 1,\quad \frac{X^2}{(\sqrt{2})^2} + \frac{Y^2}{\left(\frac{2\sqrt{2}}{3}\right)^2} - \frac{Z^2}{2^2} = 1$

Analogous to an hyperboloid of one sheet, each horizontal section an ellipse

Use cylindrical coordinates



$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

Horizontal traces are ellipses.

Vertical traces are hyperbolas.

The axis of symmetry corresponds to the variable whose coefficient is negative.

from Stewart - Calculus, 7th ed.

$4x^2 + 9y^2 = 8 + 2z^2,\quad \frac{x^2}{\frac{8+2z^2}{4}} + \frac{y^2}{\frac{8+2z^2}{9}} = 1$

Using $\cos^2\theta + \sin^2\theta = 1,\ -\infty < z < \infty$

$\therefore \frac{X^2}{\left(\frac{\sqrt{8+2z^2}}{2}\right)^2} + \frac{Y^2}{\left(\frac{\sqrt{8+2z^2}}{3}\right)^2} = 1$

$\therefore$ Let $x = \left(\frac{\sqrt{8+2z^2}}{2}\right)\cos\theta,\quad y = \left(\frac{\sqrt{8+2z^2}}{3}\right)\sin\theta$

$$\therefore T(\theta, z) = \left[\frac{\sqrt{8+2z^2}}{2}\cos\theta, \frac{\sqrt{8+2z^2}}{3}\sin\theta, z \right],$$

$$0 \le \theta \le 2\pi, \quad -\infty < z < \infty$$

8.

8. Find the area of the surface defined by Φ: $(u, v) \mapsto (x, y, z)$, where

$$x = h(u, v) = u + v, \qquad y = g(u, v) = u,$$
$$z = f(u, v) = v;$$

$0 \le u \le 1, 0 \le v \le 1$. Sketch.

$$\phi(u,v) = (u+v, u, v)$$

$$\phi_u = (1,1,0) \quad \phi_v = (1,0,1)$$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = (1,-1,-1)$$

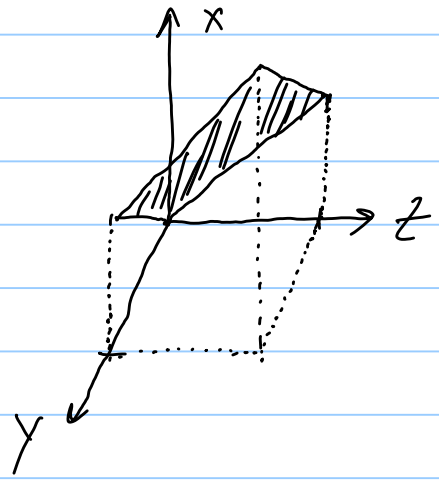
$$\therefore \|\phi_u \times \phi_v\| = \sqrt{3}$$

$$\therefore A(S) = \iint_D \|\phi_u \times \phi_v\| \, du\,dv = \int_0^1 \int_0^1 \sqrt{3} \, du\,dv$$

$$= \sqrt{3}$$

Much like the graph of $z = g(x,y)$, this is the graph of $x = g(y,z)$, so using yz plane as

domain D, and surface above/below that.



$x = y + z$
a plane

9.

9. Write a formula for the surface area of $\Phi\colon (r, \theta) \mapsto (x, y, z)$, where

$$x = r\cos\theta, \qquad y = 2r\sin\theta, \qquad z = r;$$

$0 \le r \le 1, 0 \le \theta \le 2\pi$. Describe the surface.

$$\phi(r,\theta) = (r\cos\theta, 2r\sin\theta, r)$$

$$\phi_r = (\cos\theta, 2\sin\theta, 1) \qquad \phi_\theta = (-r\sin\theta, 2r\cos\theta, 0)$$

$$\phi_r \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & 2\sin\theta & 1 \\ -r\sin\theta & 2r\cos\theta & 0 \end{vmatrix} = (-2r\cos\theta, -r\sin\theta, 2r)$$

$$\therefore \|\phi_r \times \phi_\theta\| = \sqrt{4r^2\cos^2\theta + r^2\sin^2\theta + 4r^2}$$

$$= \sqrt{3r^2\cos^2\theta + 5r^2} = r\sqrt{3\cos^2\theta + 5}$$

$$\therefore A(S) = \int_0^{2\pi}\int_0^1 r\sqrt{3\cos^2\theta + 5}\, dr\, d\theta$$

$$= \int_0^{2\pi} \sqrt{3\cos^2\theta + 5}\, d\theta \left[\frac{r^2}{2} \right]_{r=0}^{r=1}$$

$$= \frac{1}{2} \int_0^{2\pi} \sqrt{3\cos^2\theta + 5}\, d\theta$$

Note: $2x = 2r\cos\theta \quad \therefore (2x)^2 + y^2 = 4r^2 = 4z^2$

$\therefore 4x^2 + y^2 = 4z^2, \quad z^2 = x^2 + \frac{1}{4}y^2$

$f(x,y) = x^2 + \frac{1}{4}y^2$ is a series of ellipsis.

Note $z \geq 0$ since $0 \leq r \leq 1$.

$\therefore z^2 = x^2 + \frac{1}{4}y^2$ is the upper portion of a cone in which each horizontal section is an ellipse of form $x^2 + \frac{1}{4}y^2 = K$, a constant.

10.

10. Suppose $z = f(x, y)$ and $(\partial f/\partial x)^2 + (\partial f/\partial y)^2 = c$, $c > 0$. Show that the area of the graph of f lying over a region D in the xy plane is $\sqrt{1 + c}$ times the area of D.

Given the parametrization $\phi(x,y) = (x, y, f(x,y))$,

$\| \phi_x \times \phi_y \| = \sqrt{1 + f_x^2 + f_y^2} = \sqrt{1 + c}$

$$\therefore A(S) = \iint_D \sqrt{1+c}\, dx\,dy = \sqrt{1+c} \iint_D dx\,dy$$

$$= \sqrt{1+c}\, A(D)$$

11. **11.** Compute the integral of $f(x, y, z) = x^2 + y^2 + z^2$ over the surface in Review Exercise 8.

From #8, $\|\phi_u \times \phi_v\| = \sqrt{3}$, $\phi(u,v) = (u+v, u, v)$

$$\therefore f(\phi(u,v)) = (u+v)^2 + u^2 + v^2 = 2u^2 + 2v^2 + 2uv$$

$$\therefore \iint_S f\,ds = \int_0^1 \int_0^1 \sqrt{3}(2u^2 + 2v^2 + 2uv)\,du\,dv$$

$$= 2\sqrt{3} \int_0^1 \int_0^1 u^2 + v^2 + uv\,du\,dv$$

$$= 2\sqrt{3} \int_0^1 \frac{u^3}{3} + uv^2 + \frac{u^2 v}{2} \Big|_{u=0}^{u=1} dv$$

$$= 2\sqrt{3} \int_0^1 \left(\frac{1}{3} + v^2 + \frac{v}{2}\right) dv = 2\sqrt{3} \left[\frac{v}{3} + \frac{v^3}{3} + \frac{v^2}{4}\right]_0^1$$

$$= 2\sqrt{3}\left(\frac{1}{3} + \frac{1}{3} + \frac{1}{4}\right) = \frac{22}{12}\sqrt{3} = \frac{11\sqrt{3}}{6}$$

12.

12. Find $\iint_S f\,dS$ in each of the following cases:

(a) $f(x, y, z) = x$; S is the part of the plane $x + y + z = 1$ in the positive octant defined by $x \geq 0, y \geq 0, z \geq 0$

(a) $z = 1 - x - y \quad \therefore \phi(x,y) = (x, y, 1-x-y)$

The plane has axis intercepts: $(1,0,0), (0,1,0), (0,0,1)$.

$\therefore 0 \leq x \leq 1, \quad 0 \leq y \leq 1.$

$\phi_x = (1, 0, -1) \quad \phi_y = (0, 1, -1)$

$$\phi_x \times \phi_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ 0 & 1 & -1 \end{vmatrix} = (1, 1, 1)$$

$$\|\phi_x \times \phi_y\| = \sqrt{3}$$

$$\therefore \iint_S f\,ds = \int_0^1 \int_0^1 (x)(\sqrt{3})\,dx\,dy$$

$$= \int_0^1 \frac{\sqrt{3}}{2}\,dy = \underline{\underline{\frac{\sqrt{3}}{2}}}$$

(b) $f(x, y, z) = x^2$; S is the part of the plane $x = z$ inside the cylinder $x^2 + y^2 = 1$

For S, $\phi(r, \theta) = (r\cos\theta, r\sin\theta, r\cos\theta)$

$0 \leq r \leq 1, \quad 0 \leq \theta \leq 2\pi$

$\therefore f(\phi(r, \theta)) = r^2\cos^2\theta$

$$\phi_r = (\cos\theta, \sin\theta, \cos\theta) \qquad \phi_\theta = (-r\sin\theta, r\cos\theta, -r\sin\theta)$$

$$\phi_r \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & \cos\theta \\ -r\sin\theta & r\cos\theta & -r\sin\theta \end{vmatrix} = (-r, 0, r)$$

$$\|\phi_r \times \phi_\theta\| = \sqrt{2}\, r$$

$$\therefore \iint_S f\, dS = \int_0^{2\pi}\int_0^1 (r^2\cos^2\theta)(\sqrt{2}\, r)\, dr\, d\theta$$

$$= \sqrt{2}\int_0^{2\pi} \cos^2\theta\, d\theta \int_0^1 r^3\, dr = \sqrt{2}\int_0^{2\pi} \cos^2\theta\, d\theta \left[\frac{r^4}{4}\right]_0^1$$

$\cos^2\theta - \sin^2\theta = \cos 2\theta = 2\cos^2\theta - 1$

$$= \frac{\sqrt{2}}{4}\int_0^{2\pi} \cos^2\theta\, d\theta = \frac{\sqrt{2}}{4}\int_0^{2\pi} \frac{1+\cos 2\theta}{2}\, d\theta$$

$$= \frac{\sqrt{2}}{4}\left[\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_0^{2\pi} = \frac{\sqrt{2}}{4}\left[\pi + 0 - (0)\right]$$

$$= \frac{\sqrt{2}}{4}\pi$$

(c)

(c) $f(x, y, z) = x$; S is the part of the cylinder $x^2 + y^2 = 2x$ with $0 \le z \le \sqrt{x^2 + y^2}$

$x^2 + y^2 = 2x \iff (x-1)^2 + y^2 = 1$, cylinder with

center at $(1,0,0)$, radius 1.

$z = \sqrt{x^2+y^2}$ is upper half of cone.

Let $x = 1+\cos\theta$, $y = \sin\theta$, $0 \le \theta \le 2\pi$

$\therefore (x-1)^2 + y^2 = 1.$

For z, $z = \sqrt{(1+\cos\theta)^2 + \sin^2\theta} = \sqrt{1+2\cos\theta+1}$

$= \sqrt{2(1+\cos\theta)} = \sqrt{4\left(\frac{1+\cos\theta}{2}\right)} = \sqrt{4\cos^2\frac{\theta}{2}}$

$= 2\left|\cos\frac{\theta}{2}\right|$ for $0 \le \theta \le 2\pi$

$\therefore 0 \le z \le 2\cos\frac{\theta}{2}$ for $-\pi \le \theta \le \pi$

$\therefore T(\theta, z) = (1+\cos\theta, \sin\theta, z)$, $-\pi \le \theta \le \pi$, $0 \le z \le 2\cos\frac{\theta}{2}$

$f(T(\theta,z)) = 1+\cos\theta$

$T_\theta = (-\sin\theta, \cos\theta, 0)$ $T_z = (0,0,1)$

$$T_\theta \times T_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = (\cos\theta, \sin\theta, 0)$$

$\|T_\theta \times T_z\| = 1$

$$\therefore \iint_S f\,dS = \int_{-\pi}^{\pi}\int_0^{2\cos\frac{\theta}{2}} (1+\cos\theta)(1)\,dz\,d\theta$$

$$= \int_{-\pi}^{\pi} (1+\cos\theta)\left(2\cos\frac{\theta}{2}\right) d\theta$$

$$\cos\theta = \cos\left(\frac{\theta}{2}+\frac{\theta}{2}\right) = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}$$

$$= \int_{-\pi}^{\pi} 2\cos\frac{\theta}{2} + 2\cos^3\frac{\theta}{2} - 2\sin^2\frac{\theta}{2}\cos\frac{\theta}{2}\, d\theta$$

$$= \int_{-\pi}^{\pi} 2\cos\frac{\theta}{2} + 2\left(1-\sin^2\frac{\theta}{2}\right)\cos\frac{\theta}{2} - 2\sin^2\frac{\theta}{2}\cos\frac{\theta}{2}\, d\theta$$

$$= \int_{-\pi}^{\pi} 4\cos\frac{\theta}{2} - 4\sin^2\frac{\theta}{2}\cos\frac{\theta}{2}\, d\theta$$

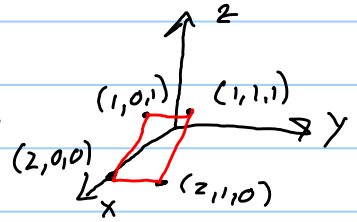
$$= 8\sin\frac{\theta}{2}\Big|_{-\pi}^{\pi} - \frac{8}{3}\sin^3\frac{\theta}{2}\Big|_{-\pi}^{\pi}$$

$$= 8[1-(-1)] - \frac{8}{3}[1-(-1)] = 16 - \frac{16}{3} = \underline{\frac{32}{3}}$$

13.

13. Compute the integral of $f(x, y, z) = xyz$ over the rectangle with vertices (1, 0, 1), (2, 0, 0), (1, 1, 1), and (2, 1, 0).

Note the order →



Assume this is a surface integral over entire rectangular area, not the path integral.

Find normal to plane: $(2,1,0)-(2,0,0)=(0,1,0)$

$(1,0,1)-(2,0,0)=(-1,0,1)$

$$\therefore \vec{N} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = (1,0,1) \qquad \|\vec{N}\| = \sqrt{2}$$

$$\therefore \text{Plane}: \quad (1,0,1)\cdot[(x,y,z)-(2,0,0)]$$

$$= (1,0,1)\cdot(x-2,y,z) = 0$$

Or, $x-2+z=0$, $z=2-x$

$$\therefore \phi(x,y) = (x,y,2-x), \quad 1\le x\le 2, \quad 0\le y\le 1$$

$$\|\phi_x \times \phi_y\| = \|\vec{N}\| = \sqrt{2}$$

$$f(\phi(x,y)) = (x)(y)(2-x) = 2xy - x^2y$$

$$\therefore \iint_S f\,ds = \int_1^2 \int_0^1 (2xy - x^2y)(\sqrt{2})\,dy\,dx$$

$$= \sqrt{2}\int_1^2 xy^2 - \frac{x^2y^2}{2}\Big|_{y=0}^{y=1} dx = \sqrt{2}\int_1^2 x - \frac{x^2}{2}\,dx$$

$$= \sqrt{2}\left[\frac{x^2}{2} - \frac{x^3}{6}\right]\Big|_1^2 = \sqrt{2}\left[2 - \frac{4}{3} - \left(\frac{1}{2} - \frac{1}{6}\right)\right]$$

$$= \sqrt{2}\left(\frac{2}{3} - \frac{1}{3}\right) = \underline{\sqrt{2}/3}$$

14. **14.** Compute the integral of $x + y$ over the surface of the unit sphere.

$T(\theta, \phi) = (\cos\theta \sin\phi, \sin\theta \sin\phi, \cos\phi), \quad 0 \le \theta \le 2\pi, \quad 0 \le \phi \le \pi$

$f(T(\theta, \phi)) = \cos\theta \sin\phi + \sin\theta \sin\phi$

$T_\theta = (-\sin\theta \sin\phi, \cos\theta \sin\phi, 0)$

$T_\phi = (\cos\theta \cos\phi, \sin\theta \cos\phi, -\sin\phi)$

As shown in Example 1, p. 401 of text,

$T_\theta \times T_\phi = (-\sin^2\phi \cos\theta, -\sin^2\phi \sin\theta, -\sin\phi \cos\phi)$

$\therefore \| T_\theta \times T_\phi \| = \sqrt{\sin^4\phi (\cos^2\theta + \sin^2\theta) + \sin^2\phi \cos^2\phi}$

$= \sin\phi, \text{ and } \sin\phi \ge 0 \text{ for } 0 \le \phi \le \pi$

$\therefore \iint_S f\,ds = \int_0^{2\pi} \int_0^{\pi} (\cos\theta \sin\phi + \sin\theta \sin\phi)(\sin\phi)\, d\phi\, d\theta$

$= \int_0^{2\pi} \int_0^{\pi} \sin^2\phi (\cos\theta + \sin\theta)\, d\phi\, d\theta$

$= \int_0^{2\pi} (\cos\theta + \sin\theta)\, d\theta \int_0^{\pi} \sin^2\phi\, d\phi$

$$= (0) \int_0^{\pi} \sin^2\phi \, d\phi = \underline{0}$$

15.

15. Compute the surface integral of x over the triangle with vertices (1, 1, 1), (2, 1, 1), and (2, 0, 3).

Assume surface integral over area of triangle.

Normal to plane of triangle: $(2,1,1)-(1,1,1) = (1,0,0)$

$(2,0,3)-(1,1,1) = (1,-1,2)$

$\therefore \vec{N} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 0 \\ 1 & -1 & 2 \end{vmatrix} = (0,-2,-1)$ Choose $\vec{N}$ s.t. coefficient to $\hat{k}$ is positive

$\therefore \vec{N} = (0,2,1)$

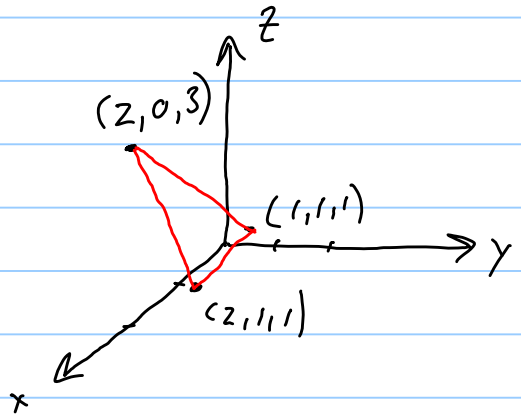
$\therefore \vec{n}$ = unit normal $= \frac{1}{\sqrt{5}}(0,2,1)$

$\cos\theta$ = angle of plane of triangle with $\hat{k}$

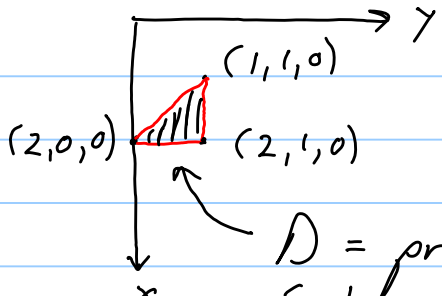
$= \vec{n} \cdot \hat{k} = \frac{1}{\sqrt{5}}$

$$\therefore \iint_S f \, dS = \iint_D \frac{f(x,y,g(x,y))}{\cos\theta} \, dx\,dy$$

$$= \sqrt{5} \iint_D x \, dx\,dy$$



D in the xy-plane is described as: $1 \le x \le 2$



D = projection of triangle onto xy plane

For y: slanted line is $y = -x + 2$

$\therefore -x + 2 \le y \le 1$

$$\therefore \iint_S f\,dS = \sqrt{5} \int_1^2 \int_{-x+2}^{1} x\,dy\,dx$$

$$= \sqrt{5} \int_1^2 xy \Big|_{y=-x+2}^{y=1} dx = \sqrt{5} \int_1^2 x - (-x^2 + 2x)\,dx$$

$$= \sqrt{5} \int_1^2 x^2 - x\,dx = \sqrt{5} \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_1^2$$

$$= \sqrt{5} \left[\frac{8}{3} - 2 - \left(\frac{1}{3} - \frac{1}{2} \right) \right] = \sqrt{5} \left[\frac{7}{3} - \frac{3}{2} \right]$$

$$= \frac{5}{6}\sqrt{5}$$

16C.

16. A paraboloid of revolution S is parametrized by $\Phi(u, v) = (u\cos v, u\sin v, u^2)$, $0 \le u \le 2$, $0 \le v \le 2\pi$.

(a) Find an equation in x, y, and z describing the surface.

(a) $z = u^2 = (u\cos v)^2 + (u\sin v)^2 = x^2 + y^2$

$\therefore z = x^2 + y^2, \quad -2 \le x \le 2, \quad -\sqrt{4-x^2} \le y \le \sqrt{4-x^2}$

(b) What are the geometric meanings of the parameters u and v?

(6) v represents the radian angle around a circle of radius u

(c) Find a unit vector orthogonal to the surface at $\Phi(u, v)$.

(C) $\phi_u = (\cos v, \sin v, 2u) \quad \phi_v = (-u\sin v, u\cos v, 0)$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos v & \sin v & 2u \\ -u\sin v & u\cos v & 0 \end{vmatrix} = (-2u^2\cos v, -2u^2\sin v, u)$$

$$\|\phi_u \times \phi_v\| = \sqrt{4u^4\cos^2 v + 4u^4\sin^2 v + u^2}$$

$$= u\sqrt{4u^2+1}$$

$$\therefore \vec{n}(u,v) = \frac{1}{\sqrt{4u^2+1}}(-2u\cos v, -2u\sin v, 1)$$

(d) Find the equation for the tangent plane at $\Phi(u_0, v_0) = (1, 1, 2)$ and express your answer in the following two ways:

(i) parametrized by u and v; and

(ii) in terms of x, y, and z.

(d) $\phi(u_0, v_0) = (1,1,2) = (u_0\cos v_0, u_0\sin v_0, u_0^2)$

$\therefore u_0^2 = 2, \quad u_0 = \sqrt{2} \quad$ as $0 \le u \le 2$

$\therefore \sqrt{2}\cos v_0 = 1, \sqrt{2}\sin v_0 = 1, \therefore v_0 = \frac{\pi}{4}$

$\therefore (u_0, v_0) = (\sqrt{2}, \frac{\pi}{4})$.

$\phi_u(\sqrt{2}, \frac{\pi}{4}) = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 2\sqrt{2})$

$\phi_v(\sqrt{2}, \frac{\pi}{4}) = (-\sqrt{2}\frac{\sqrt{2}}{2}, \sqrt{2}\frac{\sqrt{2}}{2}, 0) = (-1, 1, 0)$

(i) $Plane = (1,1,2) + u(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 2\sqrt{2}) + v(-1,1,0)$

(ii) $\vec{n}(\sqrt{2}, \frac{\pi}{4}) = \frac{1}{3}(-2\sqrt{2}\cdot\frac{\sqrt{2}}{2}, -2\sqrt{2}\cdot\frac{\sqrt{2}}{2}, 1)$

$= \frac{1}{3}(-2, -2, 1)$,

so just use $\vec{N} = (-2,-2,1)$

$\therefore (-2,-2,1)\cdot(x-1, y-1, z-2) = 0$

or $-2(x-1) - 2(y-1) + z - 2 = 0$,

or $z = 2 + 2x - 2 + 2y - 2$,

or $z = 2x + 2y - 2$

(e) (e) Find the area of S.

$$A(S) = \iint_D \|\phi_u \times \phi_v\| \, du\,dv = \int_c^{2\pi}\int_0^1 u\sqrt{4u^2+1}\, du\,dv$$

$$= 2\pi \int_0^1 u\sqrt{4u^2+1}\,du = \frac{\pi}{4}\int_0^1 8u(4u^2+1)^{1/2}\,du$$

$$= \frac{\pi}{4}\left[\frac{2}{3}(4u^2+1)^{3/2}\right]_0^1 = \frac{\pi}{6}\left[5^{3/2}-1\right]$$

$$= \frac{\pi}{6}(5\sqrt{5}-1)$$

17.

17. Let $f(x, y, z) = xe^y \cos \pi z$.

(a) Compute $\mathbf{F} = \nabla f$.

(b) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{c}(t) = (3\cos^4 t, 5\sin^7 t, 0)$, $0 \le t \le \pi$.

(a) $\nabla f = (f_x, f_y, f_z) = (e^y \cos \pi z,\ xe^y \cos \pi z,\ -\pi x e^y \sin \pi z)$

(b) $$\int_C \vec{F} \cdot d\vec{s} = \int_C \nabla f \cdot d\vec{s} = \int_0^{\pi} \nabla f(\vec{c}(t)) \cdot \vec{c}\,'(t)\,dt$$

$$= \int_0^{\pi} (f \circ \vec{c})'(t)\,dt = (f \circ \vec{c})(t)\Big|_{t=0}^{t=\pi}$$

$$= f(\vec{c}(\pi)) - f(\vec{c}(0)) = f(3,0,0) - f(3,0,0) = 0$$

18.

18. Let $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where S is the upper hemisphere of the unit sphere $x^2 + y^2 + z^2 = 1$.

Let $T(\phi, \theta) = (\cos\theta \sin\phi, \sin\theta \sin\phi, \cos\phi)$,

$0 \le \theta \le 2\pi$, $0 \le \phi \le \frac{\pi}{2}$

$\therefore$ as in Example 1, p. 401 of the text,

$T_\phi \times T_\theta = (\sin^2\phi\cos\theta, \sin^2\phi\sin\theta, \sin\phi\cos\phi)$

and $\vec{F} \cdot (T_\phi \times T_\theta) = \sin\phi$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \int_0^{2\pi}\int_0^{\pi/2} \sin\phi \, d\phi \, d\theta$$

$$= \int_0^{2\pi} \left[-\cos\phi\right]_{\phi=0}^{\phi=\frac{\pi}{2}} d\theta = \int_0^{2\pi} \left[1 - (0)\right] d\theta = \underline{2\pi}$$

19. **19.** Let $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Evaluate $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{c}(t) = (e^t, t, t^2)$, $0 \le t \le 1$.

$\vec{c}\,'(t) = (e^t, 1, 2t)$

$\vec{F} \cdot \vec{c}\,'(t) = (e^t, t, t^2) \cdot (e^t, 1, 2t) = e^{2t} + t + 2t^3$

$$\therefore \int_{\vec{c}} \vec{F} \cdot d\vec{s} = \int_0^1 (e^{2t} + t + 2t^3)\, dt = \frac{e^{2t}}{2} + \frac{t^2}{2} + \frac{t^4}{2}\Big|_0^1$$

$$= \frac{e^2}{2} + \frac{1}{2} + \frac{1}{2} - \left(\frac{1}{2} + 0 + 0\right) = \frac{e^2}{2} + \frac{1}{2} = \underline{\frac{1}{2}(e^2 + 1)}$$

20. **20.** Let $\mathbf{F} = \nabla f$ for a given scalar function. Let $\mathbf{c}(t)$ be a closed curve, that is, $\mathbf{c}(b) = \mathbf{c}(a)$. Show that $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s} = 0$.

$$\int_{\vec{c}} \vec{F} \cdot d\vec{s} = \int_a^b \nabla f \cdot \vec{c}\,'(t)\,dt = \int_a^b (f \circ \vec{c})'(t)\,dt$$

$$= (f \circ \vec{c})(t)\Big|_{t=a}^{t=b} = f(\vec{c}(b)) - f(\vec{c}(a)) = \underline{0}$$

21. **21.** Consider the surface $\Phi(u, v) = (u^2 \cos v,\ u^2 \sin v,\ u)$. Compute the unit normal at $u = 1$, $v = 0$. Compute the equation of the tangent plane at this point.

(a) $\phi_u = (2u\cos v, 2u\sin v, 1)$ $\phi_v = (-u^2\sin v, u^2\cos v, 0)$

$\phi_u(1,0) = (2,0,1)$ $\phi_v(1,0) = (0,1,0)$

$$\phi_u \times \phi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} = (-1,0,2) = \vec{N}$$

$\therefore \|\vec{N}\| = \sqrt{5}$

$\therefore$ unit normal $\vec{n}(1,0) = \underline{\frac{1}{\sqrt{5}}(-1,0,2)}$

(b) $\phi(1,0) = (1,0,1)$

$\therefore \vec{N} \cdot [(x,y,z) - (1,0,1)] = 0$

$(-1,0,2) \cdot (x-1, y, z-1) = 0$

$$\text{or, } 1 - x + 2z - 2 = 0, \quad \underline{2z - x = 1}$$

22. **22.** Let S be the part of the cone $z^2 = x^2 + y^2$ with z between 1 and 2 oriented by the normal pointing out of the cone. Compute $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = (x^2, y^2, z^2)$.

Let $x = r\cos\theta$, $y = r\sin\theta$, $\therefore x^2 + y^2 = r^2 = z^2 \Rightarrow z = r$

$\therefore T(r, \theta) = (r\cos\theta, r\sin\theta, r)$, $0 \le \theta \le 2\pi$, $1 \le r \le 2$

$T_r = (\cos\theta, \sin\theta, 1) \quad T_\theta = (-r\sin\theta, r\cos\theta, 0)$

$$\therefore T_r \times T_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & 1 \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (-r\cos\theta, -r\sin\theta, r)$$

$T_r \times T_\theta$ points "inward" and "upward" as $r > 0$.

$\therefore$ Choose $\vec{N} = T_\theta \times T_r = (r\cos\theta, r\sin\theta, -r)$

$\vec{F} = (r^2\cos^2\theta, r^2\sin^2\theta, r^2)$

$\therefore \vec{F} \cdot \vec{N} = r^3\cos^3\theta + r^3\sin^3\theta - r^3 = r^3(\cos^3\theta + \sin^3\theta - 1)$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \int_0^{2\pi} \int_1^2 r^3(\cos^3\theta + \sin^3\theta - 1)\, dr\, d\theta$$

$$= \int_0^{2\pi} (\cos^3\theta + \sin^3\theta - 1) \left. \frac{r^4}{4} \right|_{r=1}^{r=2} d\theta$$

$$= \frac{15}{4}\int_0^{2\pi} (1-\sin^2\theta)\cos\theta + (1-\cos^2\theta)\sin\theta - 1 \, d\theta$$

$$= \frac{15}{4}\int_0^{2\pi} \cos\theta + \sin\theta - 1 - \sin^2\theta\cos\theta - \cos^2\theta\sin\theta \, d\theta$$

$$= \frac{15}{4}\left[0 + 0 - 2\pi\right] - \frac{15}{4}\int_0^{2\pi} \left[\sin^2\theta\cos\theta + \cos^2\theta\sin\theta\right] d\theta$$

$$= -\frac{30}{4}\pi - \frac{15}{4}\left[\frac{\sin^3\theta}{3} - \frac{\cos^3\theta}{3}\right]_0^{2\pi}$$

$$= -\frac{15}{2}\pi - \frac{15}{4}\left[0 - \frac{1}{3} - \left(0 - \frac{1}{3}\right)\right] = -\frac{15}{2}\pi$$

23. **23.** Let $\mathbf{F} = x\mathbf{i} + x^2\mathbf{j} + yz\mathbf{k}$ represent the velocity field of a fluid (velocity measured in meters per second). Compute how many cubic meters of fluid per second are crossing the xy plane through the square $0 \le x \le 1$, $0 \le y \le 1$.

The square in the xy-plane can be described as

$$\phi(x,y) = (x,y,0), \quad 0 \le x \le 1, \quad 0 \le y \le 1$$

$$\phi_x = (1,0,0) = \hat{i} \qquad \phi_y = (0,1,0) = \hat{j}$$

$$\phi_x \times \phi_y = \hat{i} \times \hat{j} = \hat{k} = (0,0,1)$$

$$\vec{F}(\phi(x,y)) = (x, x^2, 0) \quad \therefore \vec{F} \cdot (\phi_x \times \phi_y) = 0$$

$$\therefore \int_{\phi} \vec{F} \cdot d\vec{S} = \underline{0}$$

which makes sense as $\vec{F}$ flows parallel to xy-plane at $z = 0$.

24.

24. Show that the surface area of the part of the sphere $x^2 + y^2 + z^2 = 1$ lying above the rectangle $[-a, a] \times [-a, a]$, where $2a^2 < 1$, in the xy plane is

$$A = 2\int_{-a}^{a} \sin^{-1}\left(\frac{a}{\sqrt{1-x^2}}\right) dx.$$

A parametrization of the upper hemisphere is

$$\phi(x,y) = (x, y, \sqrt{1-x^2-y^2}), \quad -a \le x \le a, \quad -a \le y \le a, \quad 2a^2 < 1.$$

$$\phi_x = \left[1, 0, -x(1-x^2-y^2)^{-\frac{1}{2}}\right]$$

$$\phi_y = \left[0, 1, -y(1-x^2-y^2)^{-\frac{1}{2}}\right]$$

Let $w = \sqrt{1-x^2-y^2}$

$$\therefore \phi_x \times \phi_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -x/w \\ 0 & 1 & -y/w \end{vmatrix} = \left(\frac{x}{w}, \frac{y}{w}, 1\right)$$

$$\therefore \|\phi_x \times \phi_y\| = \frac{\sqrt{x^2+y^2+w^2}}{w} = \frac{1}{w} = \frac{1}{\sqrt{1-x^2-y^2}}$$

$$\therefore A(S) = \iint_D \|\phi_x \times \phi_y\| \, dx\,dy$$

$$= \int_{-a}^{a}\int_{-a}^{a} \frac{1}{\sqrt{1-x^2-y^2}}\,dy\,dx = \int_{-a}^{a}\int_{-a}^{a} \frac{1}{\sqrt{c^2-y^2}}\,dy\,dx$$

letting $c^2 = 1-x^2$

$$= \int_{-a}^{a} \operatorname{Arcsin}\frac{y}{c}\Big|_{y=-a}^{y=a}\,dx$$

$$= \int_{-a}^{a} \operatorname{Arcsin}\left(\frac{a}{c}\right) - \operatorname{Arcsin}\left(\frac{-a}{c}\right)\,dx$$

$\operatorname{Arcsin}(-x) = -\operatorname{Arcsin}(x)$

$$= 2\int_{-a}^{a} \operatorname{Arcsin}\left(\frac{a}{c}\right)\,dx = 2\int_{-a}^{a} \operatorname{Arcsin}\left(\frac{a}{\sqrt{1-x^2}}\right)\,dx$$

25.

25. Let S be a surface and C a closed curve bounding S. Verify the equality

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_C \mathbf{F} \cdot d\mathbf{s}$$

if $\mathbf{F}$ is a gradient field (use Review Exercise 20).

If $\vec{F}$ is a gradient of some function f, $\nabla f = \vec{F}$,

then $\nabla \times (\nabla f) = \vec{0}$ (see Theorem 1, p. 252 of text).

$\therefore$ $\nabla \times \vec{F} = \vec{0} \Rightarrow \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = 0$

But by #20 above, $\int_C \vec{F} \cdot d\vec{s} = 0$.

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_C \vec{F} \cdot d\vec{s}$$

26.

26. Calculate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = (x, y, -y)$ and S is the cylindrical surface defined by $x^2 + y^2 = 1$, $0 \le z \le 1$, with normal pointing out of the cylinder.

Let $\phi(r, \theta) = (\cos\theta, \sin\theta, r)$, $0 \le \theta \le 2\pi$, $0 \le r \le 1$

$\therefore \phi_r = (0, 0, 1)$ $\phi_\theta = (-\sin\theta, \cos\theta, 0)$

$$\phi_r \times \phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ -\sin\theta & \cos\theta & 0 \end{vmatrix} = (-\cos\theta, -\sin\theta, 0)$$

$\phi_r \times \phi_\theta$ points "in", so choose $\phi_\theta \times \phi_r = (\cos\theta, \sin\theta, 0)$

$\vec{F}(\phi(r, \theta)) = (\cos\theta, \sin\theta, -\sin\theta)$

$\therefore \vec{F} \cdot (\phi_\theta \times \phi_r) = \cos^2\theta + \sin^2\theta + 0 = 1$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \int_0^{2\pi} \int_0^1 (1)\, dr\, d\theta = \underline{2\pi}$$

27.

27. Let S be the portion of the cylinder $x^2 + y^2 = 4$ between the planes $z = 0$ and $z = x + 3$. Compute the following:

(a) $\iint_S x^2\, dS$

(b) $\iint_S y^2\, dS$

(c) $\iint_S z^2\, dS$

Define S by $\phi(\theta, z) = (2\cos\theta, 2\sin\theta, z)$

$0 \le \theta \le 2\pi$, $0 \le z \le 2\cos\theta + 3$

$$\therefore \phi_\theta = (-2\sin\theta, 2\cos\theta, 0) \quad \phi_z = (0,0,1)$$

$$\therefore \phi_\theta \times \phi_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2\sin\theta & 2\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = (2\cos\theta, 2\sin\theta, 0)$$

$$\| \phi_\theta \times \phi_z \| = 2$$

(a) $$\iint_S x^2 \, dS = \int_0^{2\pi} \int_0^{2\cos\theta + 3} 4\cos^2\theta \, (2) \, dz \, d\theta$$

$$= 8 \int_0^{2\pi} \cos^2\theta \, (2\cos\theta + 3) \, d\theta = 8 \int_0^{2\pi} 2\cos^3\theta + 3\cos^2\theta \, d\theta$$

$\cos 2\theta = 2\cos^2\theta - 1$

$$= 16 \int_0^{2\pi} (1 - \sin^2\theta)\cos\theta \, d\theta + 24 \int_0^{2\pi} \frac{1 + \cos 2\theta}{2} \, d\theta$$

$$= 16 \left[\sin\theta - \frac{\sin^3\theta}{3} \right]_0^{2\pi} + 24 \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^{2\pi}$$

$$= 16(0) + 24 \left[\pi + 0 - (0+0) \right] = \underline{24\pi}$$

(5) $$\iint_S y^2 \, dS = \int_C \int_0^{2\cos\theta + 3} (4\sin^2\theta)(2) \, dz \, d\theta$$

$$= 8\int_0^{2\pi} \sin^2\theta\,(2\cos\theta + 3)\,d\theta = 8\int_0^{2\pi} (2\sin^2\theta\cos\theta + 3\sin^2\theta)\,d\theta$$

$\cos 2\theta = 1 - 2\sin^2\theta$

$$= 16\int_0^{2\pi} \sin^2\theta\cos\theta\,d\theta + 24\int_0^{2\pi} \frac{1-\cos 2\theta}{2}\,d\theta$$

$$= 16\left[\frac{\sin^3\theta}{3}\right]_0^{2\pi} + 24\left[\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right]_0^{2\pi}$$

$$= 16(0) + 24\left[\pi - 0 - (0-0)\right] = \underline{24\pi}$$

(c) $$\iint_S z^2\,dS = \int_0^{2\pi}\int_0^{2\cos\theta+3} (z^2)(2)\,dz\,d\theta$$

$$= 2\int_0^{2\pi} \frac{z^3}{3}\Big|_{z=0}^{2\cos\theta+3}\,d\theta$$

$$= \frac{2}{3}\int_0^{2\pi} 8\cos^3\theta + 36\cos^2\theta + 54\cos\theta + 27\,d\theta \qquad [1]$$

From (a), $\int_0^{2\pi} \cos^3\theta\,d\theta = 0$, $\int_0^{2\pi} \cos^2\theta\,d\theta = \pi$

and $\int_0^{2\pi} \cos\theta\,d\theta = 0$

$\therefore$ [1] becomes $\frac{2}{3}\left[8(0) + 36(\pi) + 54(0) + 27(2\pi)\right]$

$= \frac{2}{3}\left[36\pi + 54\pi\right] = \underline{60\pi}$

28.

28. Let Γ be the curve of intersection of the plane $z = ax + by$, with the cylinder $x^2 + y^2 = 1$. Find all values of the real numbers a and b such that $a^2 + b^2 = 1$ and

$$\int_\Gamma y\,dx + (z - x)\,dy - y\,dz = 0.$$

Let $\Gamma(\theta) = (\cos\theta, \sin\theta, a\cos\theta + b\sin\theta)$, $0 \le \theta \le 2\pi$

$dx = -\sin\theta\,d\theta \quad dy = \cos\theta\,d\theta \quad dz = (-a\sin\theta + b\cos\theta)\,d\theta$

$\therefore y\,dx = -\sin^2\theta\,d\theta$

$(z - x)\,dy = (a\cos\theta + b\sin\theta - \cos\theta)\cos\theta\,d\theta$

$= (a\cos^2\theta + b\sin\theta\cos\theta - \cos^2\theta)\,d\theta$

$-y\,dz = -\sin\theta(-a\sin\theta + b\cos\theta)\,d\theta$

$= (a\sin^2\theta - b\sin\theta\cos\theta)\,d\theta$

$\therefore y\,dx + (z - x)\,dy - y\,dz = a\cos^2\theta + a\sin^2\theta - \sin^2\theta - \cos^2\theta$

$= a - 1$

$$\int_\Gamma y\,dx + (z - x)\,dy - y\,dz = \int_\Gamma a - 1 = \int_0^{2\pi} (a - 1)\,d\theta$$

$= 2\pi(a-1) = 0 \Rightarrow a = 1$

$\therefore a^2 + b^2 = 1 \Rightarrow b = 0$

$\therefore a = 1, \; b = 0$

29.

29. A circular helix that lies on the cylinder $x^2 + y^2 = R^2$ with pitch p may be described parametrically by

$$x = R\cos\theta, \qquad y = R\sin\theta, \qquad z = p\theta, \qquad \theta \geq 0.$$

A particle slides under the action of gravity (which acts parallel to the z axis) without friction along the helix. If the particle starts out at the height $z_0 > 0$, then when it reaches the height z, $0 \leq z < z_0$, along the helix, its speed is given by

$$\frac{ds}{dt} = \sqrt{(z_0 - z)2g},$$

where s is arc length along the helix, g is the constant of gravity, and t is time.

(a) Find the length of the part of the helix between the planes $z = z_0$ and $z = z_1$, $0 \leq z_1 < z_0$.

(b) Compute the time T_0 it takes the particle to reach the plane $z = 0$.

(a) Let $\phi(\theta) = (R\cos\theta, R\sin\theta, p\theta)$, $\theta \geq 0$

$\phi'(\theta) = (-R\sin\theta, R\cos\theta, p)$

$\|\phi'(\theta)\| = \sqrt{R^2\sin^2\theta + R^2\cos^2\theta + p^2} = \sqrt{R^2 + p^2}$

$\therefore$

$$\text{Arc length} = \int_{z_1/p}^{z_0/p} \sqrt{R^2 + p^2}\, d\theta = \frac{\sqrt{R^2 + p^2}}{p}(z_0 - z_1)$$

Note $z_0 = p\theta_0$, $z_1 = p\theta_1$, $\therefore \theta_0 = z_0/p$, $\theta_1 = z_1/p$

(b) $\dfrac{\Delta \text{length (at point } z)}{\Delta \text{ time (at point } z)} \approx \text{speed, instantaneous (at } z)$

$\therefore \dfrac{\Delta \text{ Arc length}}{\text{speed (instantaneous)}} \approx \Delta \text{ time at point } z$

$\therefore \dfrac{\Delta \text{ Arc length} / \Delta z}{\text{speed (instantaneous)}} \approx \Delta \text{time} / \Delta z$

$\therefore \displaystyle\int_0^{z_0} \left(\frac{\Delta \text{Time}}{\Delta z}\right) dz = T_0$, as quantities are functions of z.

From (a), $\text{Arc length}(z) = \dfrac{\sqrt{R^2+\rho^2}}{\rho}(z_0 - z)$

$\therefore \dfrac{\Delta \text{length}}{\Delta z} = -\dfrac{d}{dz}\left[\dfrac{\sqrt{R^2+\rho^2}}{\rho}(z_0 - z)\right]$,

the minus sign appearing as the length gets shorter with increasing z toward z_0, the maximum value of z.

$\therefore -\dfrac{d}{dz}\,\text{Arc length} = -\dfrac{d}{dz}\left[\dfrac{\sqrt{R^2+\rho^2}}{\rho}(z_0 - z)\right] = \dfrac{\sqrt{R^2+\rho^2}}{\rho}$

$\therefore \dfrac{\Delta \text{ Arc length} / \Delta z}{\text{speed (instantaneous)}} \approx \dfrac{\dfrac{\sqrt{R^2+\rho^2}}{\rho}}{\sqrt{2g(z_0 - z)}}$

$$= \frac{1}{\rho}\sqrt{\frac{R^2+\rho^2}{2g}} \cdot \frac{1}{\sqrt{z_0 - z}}$$

$$\therefore T_0 = \frac{1}{\rho}\sqrt{\frac{R^2+\rho^2}{2g}} \int_0^{z_0} \frac{1}{\sqrt{z_0 - z}}\, dz$$

$$= -\frac{1}{\rho}\sqrt{\frac{R^2+\rho^2}{2g}} \left[2\sqrt{z_0 - z} \right] \Bigg|_{z=0}^{z=z_0}$$

$$= \frac{1}{\rho}\sqrt{\frac{R^2+\rho^2}{2g}} \cdot 2\sqrt{z_0} = \sqrt{\frac{4z_0(R^2+\rho^2)}{2g\rho^2}}$$

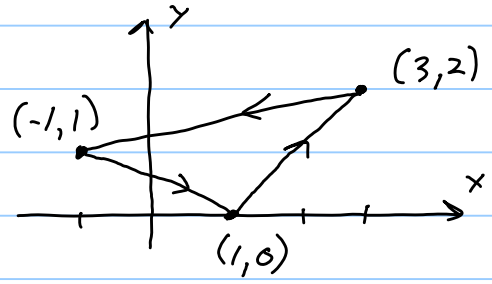
$$= \sqrt{\frac{2z_0(R^2+\rho^2)}{g\rho^2}}$$

8.1 Green's Theorem

Note Title 5/18/2017

1.

1. Let D be the triangle in the xy plane with vertices at $(-1, 1)$, $(1, 0)$, and $(3, 2)$. Describe the boundary ∂D as a piecewise smooth curve, oriented counterclockwise.



$(-1,1)$ to $(1,0)$: $(-1,1) + t[(1,0) - (-1,1)]$

$= (-1,1) + t(2,-1) = (-1+2t, 1-t)$, $0 \le t \le 1$

$(1,0)$ to $(3,2)$: $(1,0) + t[(3,2) - (1,0)]$, $0 \le t \le 1$

$= (1,0) + t(2,2)$, $0 \le t \le 1$

$= (1,0) + (t-1)(2,2)$, $1 \le t \le 2$

$= (1,0) + (2t-2, 2t-2)$, $1 \le t \le 2$

$= (2t-1, 2t-2)$, $1 \le t \le 2$

$(3,2)$ to $(-1,1)$: $(3,2) + t[(-1,1) - (3,2)]$, $0 \le t \le 1$

$= (3,2) + t(-4,-1)$, $0 \le t \le 1$

$= (3,2) + (t-2)(-4,-1)$, $2 \le t \le 3$

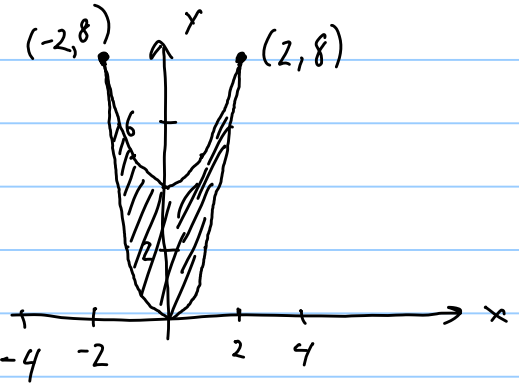
$= (3,2) + (-4t+8, -t+2)$, $2 \le t \le 3$

$= (-4t+11, -t+4)$, $2 \le t \le 3$

$\therefore \vec{c}(t) = \begin{cases} (2t-1, -t+1), & 0 \le t \le 1 \\ (2t-1, 2t-2), & 1 \le t \le 2 \\ (-4t+11, -t+4), & 2 \le t \le 3 \end{cases}$

2.

2. Let D be the region in the xy plane lying between the curves $y = x^2 + 4$ and $y = 2x^2$. Describe the boundary ∂D as a piecewise smooth curve, oriented counterclockwise.



Let $\phi_1(t) = (t, 2t^2)$, $-2 \le t \le 2$

$\phi_2(t) = (t, t^2+4)$, $-2 \le t \le 2$

Shift $\phi_1(t) = [t-2, 2(t-2)^2]$, $0 \le t \le 4$

Shift $\phi_2(t) = [t-6, (t-6)^2+4]$, $4 \le t \le 8$

Now make $\phi_2(t)$ "counterclockwise":

$$\phi_2(t) = [6-t, (6-t)^2+4], \quad 4 \le t \le 8$$

$$\therefore \vec{C}(t) = \begin{cases} [t-2, 2(t-2)^2] & 0 \le t \le 4 \\ [6-t, (6-t)^2+4] & 4 \le t \le 8 \end{cases}$$

In Exercises 3 to 6, verify Green's theorem for the indicated region D and boundary ∂D, and functions P and Q.

3. **3.** $D = [-1, 1] \times [-1, 1]$, $P(x, y) = -y$, $Q(x, y) = x$

$$\iint_D (Q_x - P_y)\,dx\,dy = \int_{-1}^{1}\int_{-1}^{1} [1-(-1)]\,dx\,dy = 2\int_{-1}^{1}\int_{-1}^{1} dx\,dy = 8$$

$\partial D =$

C_1: $(-1,-1) + s[(1,-1)-(-1,-1)] = (-1,-1) + s(2,0)$
$= (-1+2s, -1)$ $\quad 0 \le s \le 1$

C_2: $(1,-1) + (s-1)[(1,1)-(1,-1)] = (1,-1) + (s-1)(0,2)$
$= (1, 2s-3)$ $\quad 1 \le s \le 2$

C_3: $(1,1) + (s-2)[(-1,1)-(1,1)] = (1,1) + (s-2)(-2,0)$
$= (5-2s, 1)$ $\quad 2 \le s \le 3$

$$C_4: (-1,1) + (s-3)\left[(-1,-1)-(-1,1)\right] = (-1,1) + (s-3)(0,-2)$$
$$= (-1, 7-2s) \quad 3 \le s \le 4$$

$$\therefore C_1' = (2,0) \quad C_2' = (0,2) \quad C_3' = (-2,0) \quad C_4' = (0,-2)$$

$$\therefore \int_C P\,dx + Q\,dy = \int_{C_1} 2P\,ds + \int_{C_2} 2Q\,ds + \int_{C_3} -2P\,ds + \int_{C_4} -2Q\,dy$$

$$= \int_0^1 2\left[-(-1)\right] ds + \int_1^2 2[1]\,ds + \int_2^3 -2\left[-(1)\right] ds + \int_3^4 -2[-1]\,ds$$

$$= 2 + 2 + 2 + 2 = \underline{8}$$

4.

4. $D = [-1, 1] \times [-1, 1], \quad P(x, y) = x, \quad Q(x, y) = y$

From #3 above, $\partial D = \begin{cases} C_1: (-1+2s, -1) & 0 \le s \le 1 \\ C_2: (1, 2s-3) & 1 \le s \le 2 \\ C_3: (5-2s, 1) & 2 \le s \le 3 \\ C_4: (-1, 7-2s) & 3 \le s \le 4 \end{cases}$

$$\therefore C_1' = (2,0) \quad C_2' = (0,2) \quad C_3' = (-2,0) \quad C_4' = (0,-2)$$

$$\therefore \int_C P\,dx + Q\,dy = \int_{C_1} 2P\,ds + \int_{C_2} 2Q\,ds + \int_{C_3} -2P\,ds + \int_{C_4} -2Q\,dy$$

$$= \int_0^1 2\left[-1+2s\right] ds + \int_1^2 2\left[2s-3\right] ds + \int_2^3 -2\left[5-2s\right] ds + \int_3^4 -2\left[7-2s\right] ds$$

$$= \int_0^1 -2+4s\,ds + \int_1^2 4s-6\,ds + \int_2^3 -10+4s + \int_3^4 -14+4s\,ds$$

$$= -2s+2s^2\Big|_0^1 + 2s^2-6s\Big|_1^2 + -10s+2s^2\Big|_2^3 + -14s+2s^2\Big|_3^4$$

$$= (0) + [8-12-(2-6)] + [-30+18-(-20+8)] + [-56+32-(-42+18)]$$

$$= 0 + 0 + (-12+12) + (-24-(-24)) = \underline{0}$$

$$\iint_D Q_x - P_y\,dx\,dy = \int_{-1}^1\int_{-1}^1 (0-0)\,dx\,dy = \underline{0}$$

5. **5.** $D = [-1, 1] \times [-1, 1]$, $P(x, y) = x - y$, $Q(x, y) = x + y$
[HINT: Use 3 and 4.]

(a) From #3 above, $\partial D = \begin{cases} C_1: (-1+2s, -1) & 0 \le s \le 1 \\ C_2: (1, 2s-3) & 1 \le s \le 2 \\ C_3: (5-2s, 1) & 2 \le s \le 3 \\ C_4: (-1, 7-2s) & 3 \le s \le 4 \end{cases}$

$\therefore C_1' = (2,0)\quad C_2' = (0,2)\quad C_3' = (-2,0)\quad C_4' = (0,-2)$

$$\therefore \int_C P\,dx + Q\,dy = \int_{C_1} 2P\,ds + \int_{C_2} 2Q\,ds + \int_{C_3} -2P\,ds + \int_{C_4} -2Q\,dy$$

$$= \int_0^1 2[2s]\,ds + \int_1^2 2[-2+2s]\,ds + \int_2^3 -2[4-2s]\,ds + \int_3^4 -2[6-2s]\,ds$$

$$= 2s^2\Big|_0^1 + -4s+2s^2\Big|_1^2 + -8s+2s^2\Big|_2^3 + -12s+2s^2\Big|_3^4$$

$$= (2-0) + [0-(-2)] + [-6-(-8)] + [-16-(-18)]$$

$$= 2+2+2+2 = \underline{8}$$

(b) $$\iint_D (Q_x - P_y)\,dx\,dy = \int_{-1}^{1}\int_{-1}^{1} (1+1)\,dx\,dy = \underline{8}$$

6. **6.** $D = \left[0, \frac{\pi}{2}\right] \times \left[0, \frac{\pi}{2}\right]$, $P(x, y) = \sin x$, $Q(x, y) = \cos y$

(a) $$\iint_D Q_x - P_y\,dx\,dy = \int_0^{\frac{\pi}{2}}\int_0^{\frac{\pi}{2}} (0-0)\,dx\,dy = \underline{0}$$

(b) ∂D $\quad C_1(s) = \left(\frac{\pi}{2}s, 0\right) \quad 0 \le s \le 1$

$C_2(s) = \left(\frac{\pi}{2}, \frac{\pi}{2}(s-1)\right) = \left(\frac{\pi}{2}, \frac{\pi}{2}s - \frac{\pi}{2}\right) \quad 1 \le s \le 2$

$C_3(s) = \left(\frac{\pi}{2}(2-s) + \frac{\pi}{2}, \frac{\pi}{2}\right) = \left(\frac{3}{2}\pi - \frac{\pi}{2}s, \frac{\pi}{2}\right) \quad 2 \le s \le 3$

$C_4(s) = \left(0, \frac{\pi}{2}(3-s) + \frac{\pi}{2}\right) = \left(0, 2\pi - \frac{\pi}{2}s\right) \quad 3 \le s \le 4$

$\therefore C_1' = \left(\frac{\pi}{2}, 0\right) \quad C_2' = \left(0, \frac{\pi}{2}\right) \quad C_3' = \left(-\frac{\pi}{2}, 0\right) \quad C_4' = \left(0, -\frac{\pi}{2}\right)$

$$\therefore \int_C P\,dx + Q\,dy = \int_{C_1} \frac{\pi}{2} P\,ds + \int_{C_2} \frac{\pi}{2} Q\,ds + \int_{C_3} -\frac{\pi}{2} P\,ds + \int_{C_4} -\frac{\pi}{2} Q\,ds$$

$$= \frac{\pi}{2}\int_0^1 \sin\left(\frac{\pi}{2}s\right)ds + \frac{\pi}{2}\int_1^2 \cos\left(\frac{\pi}{2}s - \frac{\pi}{2}\right)ds$$

$$+ \frac{\pi}{2}\int_2^3 -\sin\left(\frac{3}{2}\pi - \frac{\pi}{2}s\right)ds + \frac{\pi}{2}\int_3^4 -\cos\left(2\pi - \frac{\pi}{2}s\right)ds$$

$$= \frac{\pi}{2}\left[-\frac{2}{\pi}\cos\left(\frac{\pi}{2}s\right)\Big|_0^1 + \frac{2}{\pi}\sin\left(\frac{\pi}{2}s - \frac{\pi}{2}\right)\Big|_1^2 \right.$$

$$\left. -\frac{2}{\pi}\cos\left(\frac{3}{2}\pi - \frac{\pi}{2}s\right)\Big|_2^3 + \frac{2}{\pi}\sin\left(2\pi - \frac{\pi}{2}s\right)\Big|_3^4 \right]$$

$$= -\cos\left(\frac{\pi}{2}s\right)\Big|_0^1 + \sin\left(\frac{\pi}{2}s - \frac{\pi}{2}\right)\Big|_1^2 - \cos\left(\frac{3}{2}\pi - \frac{\pi}{2}s\right)\Big|_2^3 + \sin\left(2\pi - \frac{\pi}{2}s\right)\Big|_3^4$$

$$= [0-(-1)] + [1-0] + [-1-(0)] + [0-(1)]$$

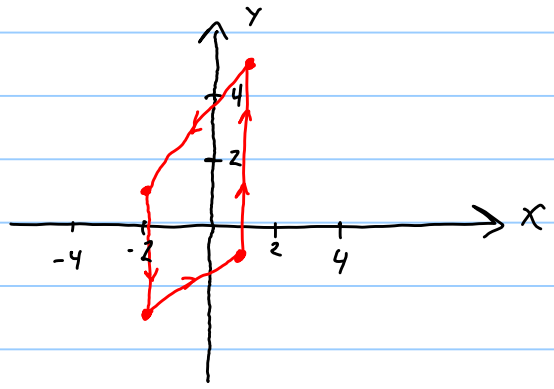
$$= 1 + 1 - 1 - 1 = \underline{0}$$

$$\therefore \int_{\vec{C}} P\,dx + Q\,dy = \iint_D Q_x - P_y\,dx\,dy$$

7.

7. Let C be the closed, piecewise smooth curve formed by traveling in straight lines between the points $(-2, 1)$, $(-2, -3)$, $(1, -1)$, $(1, 5)$, and back to $(-2, 1)$, in that order. Use Green's theorem to evaluate the integral:

$$\int_C (2xy)\,dx + (xy^2)\,dy$$



Probably easier to evaluate $\iint_D Q_x - P_y \, dx\,dy$

D is described as $-2 \le x \le 1$.

$y_1 = \phi_1(x)$ is line from $(-2,-3)$ to $(1,-1)$

slope: $\frac{-1-(-3)}{1-(-2)} = \frac{2}{3}$ $\therefore y = \frac{2}{3}x + C$, $(-3) = \frac{2}{3}(-2) + C$,

$-9 = -4 + 3C$, $C = -\frac{5}{3}$

$\therefore \phi_1(x) = \frac{2}{3}x - \frac{5}{3}$

$y_2 = \phi_2(x)$ is line from $(-2,1)$ to $(1,5)$

slope: $\frac{5-1}{1-(-2)} = \frac{4}{3}$ $\therefore y = \frac{4}{3}x + C$, $5 = \frac{4}{3}(1) + C$

$15 = 4 + 3C$, $C = \frac{11}{3}$

$\therefore \phi_2 = \frac{4}{3}x + \frac{11}{3}$

$\therefore D = \frac{2}{3}x - \frac{5}{3} \le y \le \frac{4}{3}x + \frac{11}{3}$, $-2 \le x \le 1$

$Q_x = y^2$, $P_y = 2x$

$$\therefore \int_{-2}^{1} \int_{\frac{2}{3}x - \frac{5}{3}}^{\frac{4}{3}x + \frac{11}{3}} (y^2 - 2x)\, dy\, dx$$

$$= \int_{-2}^{1} \left. \frac{y^3}{3} - 2xy \right|_{y = \frac{2}{3}x - \frac{5}{3}}^{y = \frac{4}{3}x + \frac{11}{3}} dx$$

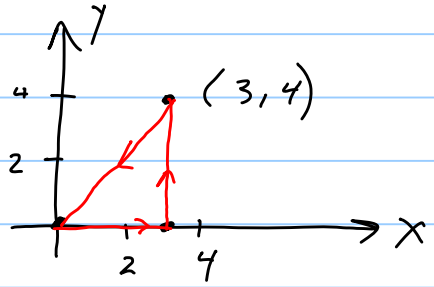
$$= \int_{-2}^{1} \frac{\left(\frac{4}{3}x + \frac{11}{3}\right)^3}{3} - 2x\left(\frac{4}{3}x + \frac{11}{3}\right) - \frac{\left(\frac{2}{3}x - \frac{5}{3}\right)^3}{3} + 2x\left(\frac{2}{3}x - \frac{5}{3}\right) dx$$

$$= \int_{-2}^{1} \frac{56}{81}x^3 + \frac{160}{27}x^2 + \frac{146}{27}x + \frac{1456}{81}\, dx = 61$$

using Wolfram for the computations

8.

8. A particle travels across a flat surface, moving due east for 3 m, then due north for 4 m, and then returns to its origin. A force field acts on the particle, given by $\mathbf{F}(x, y) = (3x + 4y^2)\mathbf{i} + (10xy)\mathbf{j}$. (Here we assume that $\mathbf{j}$ points north.) Use Green's theorem to find the work done on the particle by $\mathbf{F}$.



Let $D = \{(x,y) : 0 \le x \le 3,\ 0 \le y \le \frac{4}{3}x\}$

$$\therefore \int_C \vec{F} \cdot d\vec{s} = \int_{\partial D} F_1\, dx + F_2\, dy = \iint_D Q_x - P_y\, dy\, dx$$

$$Q_x = \frac{\partial}{\partial x}(10xy) = 10y \qquad P_y = \frac{\partial}{\partial y}(3x + 4y^2) = 8y$$

$$\therefore \int_0^3 \int_0^{\frac{4}{3}x} (10y - 8y)\, dy\, dx = \int_0^3 \int_0^{\frac{4}{3}x} 2y\, dy\, dx$$

$$= \int_0^3 y^2 \Big|_0^{\frac{4}{3}x} dx = \int_0^3 \frac{16}{9}x^2\, dx = \frac{16}{27}x^3 \Big|_0^3$$

$= 16$

9. **9.** Evaluate $\int_C y\,dx - x\,dy$, where C is the boundary of the square $[-1, 1] \times [-1, 1]$ oriented in the counterclockwise direction, using Green's theorem.

$D = -1 \le x \le 1,\ -1 \le y \le 1 \qquad P(x,y) = y,\ Q(x,y) = -x$

$P_y = 1 \qquad Q_x = -1$

$$\therefore \int_C y\,dx - x\,dy = \iint_D (Q_x - P_y)\,dx\,dy = \int_{-1}^{1}\int_{-1}^{1} (-1-1)\,dy\,dx$$

$$= -2(2)(2) = -8$$

10. **10.** Find the area of the disc D of radius R using Green's theorem.

$$A = \frac{1}{2}\int_{\partial D} x\,dy - y\,dx, \quad P(x,y) = -y,\ Q(x,y) = x$$

Let $\phi(\theta) = (R\cos\theta, R\sin\theta), \quad 0 \le \theta \le 2\pi$

where $\partial D = \phi(\theta) \qquad \therefore \phi'(\theta) = (-R\sin\theta, R\cos\theta)$

$$\therefore x\,dy - y\,dx = (R\cos\theta)(R\cos\theta\,d\theta) - (R\sin\theta)(-R\sin\theta\,d\theta)$$

$$\therefore A = \frac{1}{2}\int_0^{2\pi} R^2\cos^2\theta + R^2\sin^2\theta\,d\theta = \frac{1}{2}R^2\int_0^{2\pi} d\theta = \pi R^2$$

11.

11. Verify Green's theorem for the disc D with center $(0, 0)$ and radius R and the functions:

(a) $P(x, y) = xy^2,\ Q(x, y) = -yx^2$

(b) $P(x, y) = x + y,\ Q(x, y) = y$

(c) $P(x, y) = xy = Q(x, y)$

(d) $P(x, y) = 2y,\ Q(x, y) = x$

$$\partial D = \phi(\theta) = (R\cos\theta, R\sin\theta),\ 0 \le \theta \le 2\pi$$

(a) (1) $$\int_{\partial D} P\,dx + Q\,dy =$$

$$\int_0^{2\pi} (R\cos\theta)(R^2\sin^2\theta)(-R\sin\theta) - (R\sin\theta)(R^2\cos^2\theta)(R\cos\theta)\,d\theta$$

$$= R^4 \int_0^{2\pi} -\sin^3\theta\cos\theta - \cos^3\theta\sin\theta$$

$$= R^4 \left[-\frac{\sin^4\theta}{4} + \frac{\cos^4\theta}{4} \right]_0^{2\pi} = R^4\left[0+1-(0+1)\right] = \underline{0}$$

(2) Let $D = \{(x,y) : -R \le x \le R,\ -\sqrt{R^2-x^2} \le y \le \sqrt{R^2-x^2}\}$

$$\iint_D Q_x - P_y\,dx\,dy = \iint_D (-2xy) - (2xy)\,dx\,dy$$

$$= \iint_D -4xy\,dx\,dy = \int_{-R}^{R} \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} -4xy\,dy\,dx$$

$$= \int_{-R}^{R} \left(-xy^2 \Big|_{y=-\sqrt{R^2-x^2}}^{y=\sqrt{R^2-x^2}} \right) dx = \int_{-R}^{R} 0 \, dx = \underline{0}$$

(5) (b) $P(x, y) = x + y,\ Q(x, y) = y$

(1) $$\int_{\partial D} P\,dx + Q\,dy = \int_0^{2\pi} (R\cos\theta + R\sin\theta)(-R\sin\theta) + (R\sin\theta)(R\cos\theta)\,d\theta$$

$$= \int_0^{2\pi} -R^2\cos\theta\sin\theta - R^2\sin^2\theta + R^2\sin\theta\cos\theta \, d\theta$$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta$

$$= \int_0^{2\pi} -R^2\sin^2\theta \, d\theta = -R^2 \int_0^{2\pi} \frac{1-\cos 2\theta}{2} \, d\theta$$

$$= -R^2 \left[\frac{\theta}{2} - \frac{\sin 2\theta}{2} \right]_0^{2\pi} = \underline{-\pi R^2}$$

(2) Let $D = \{(x,y) : -R \le x \le R,\ -\sqrt{R^2-x^2} \le y \le \sqrt{R^2-x^2}\}$

$Q_x = 0 \quad P_y = 1$

$$\therefore \iint_D Q_x - P_y \, dx\,dy = \int_{-R}^{R} \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} (-1)\,dy\,dx$$

$$= -2 \int_{-R}^{R} \sqrt{R^2-x^2}\,dx = -2 \left[\frac{x}{2}\sqrt{R^2-x^2} + \frac{R^2}{2}\operatorname{Arcsin}\frac{x}{R} \right]_{-R}^{R}$$

$$= -2\left[0 + \frac{R^2}{2}\text{Arcsin}(1) - \left(0 + \frac{R^2}{2}\text{Arcsin}(-1)\right)\right]$$

$$= -2\left[R^2\text{Arcsin}(1)\right] = \underline{-R^2\pi}$$

(c) (c) $P(x, y) = xy = Q(x, y)$

(1) $$\int_{\partial D} P\,dx + Q\,dy =$$

$$\int_0^{2\pi} (R\cos\theta)(R\sin\theta)(-R\sin\theta) + (R\cos\theta)(R\sin\theta)(R\cos\theta)\,d\theta$$

$$= R^3\int_0^{2\pi} -\sin^2\theta\cos\theta + \cos^2\theta\sin\theta\,d\theta$$

$$= R^3\left[-\frac{\sin^3\theta}{3} - \frac{\cos^3\theta}{3}\right]_0^{2\pi} = R^3\left[0 - \frac{1}{3} - \left(0 - \frac{1}{3}\right)\right]$$

$$= \underline{0}$$

(2) $Q_x = y \quad P_y = x$ Use polar coordinate substitution $x = r\cos\theta,\ y = r\sin\theta,\ 0 \le r \le 1,\ 0 \le \theta \le 2\pi$

$$\iint_D (y - x)\,dx\,dy = \int_0^{2\pi}\int_0^1 (r\sin\theta - r\cos\theta)(r)\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^1 (\sin\theta - \cos\theta)\, r^2\, dr\, d\theta$$

$$= \int_0^{2\pi} (\sin\theta - \cos\theta)\, d\theta \int_0^1 r^2\, dr$$

$$= -\cos\theta - \sin\theta \Big|_0^{2\pi} \frac{r^3}{3}\Big|_0^1 = (0)\left(\frac{1}{3}\right) = \underline{0}$$

(d) (d) $P(x, y) = 2y,\ Q(x, y) = x$

(1) $$\int_{\partial D} P\,dx + Q\,dy = \int_0^{2\pi} (2R\sin\theta)(-R\sin\theta) + (R\cos\theta)(R\cos\theta)\, d\theta$$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta$

$$= R^2\int_0^{2\pi} (\cos^2\theta - \sin^2\theta) - \sin^2\theta\, d\theta = R^2\int_0^{2\pi} \cos 2\theta + \left(\frac{\cos 2\theta - 1}{2}\right) d\theta$$

$$= R^2\left[\frac{\sin 2\theta}{2} + \frac{\sin 2\theta}{4} - \frac{\theta}{2}\right]_0^{2\pi} = \underline{-\pi R^2}$$

(2) $Q_x = 1 \quad P_y = 2$

$$\iint_D (1-2)\, dx\, dy = -\iint_D dx\, dy = \underline{-\pi R^2}$$

(area of disk is πR^2).

12. **12.** Using the divergence theorem, show that $\int_{\partial D} \mathbf{F} \cdot \mathbf{n}\, ds = 0$, where $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$ and D is the unit disc. Verify this directly.

(1) $$\int_{\partial D} \vec{F} \cdot \vec{n}\, ds = \iint_D \operatorname{div} \vec{F}\, dA = \iint_D \frac{\partial}{\partial x}(y) + \frac{\partial}{\partial y}(-x)\, dA$$

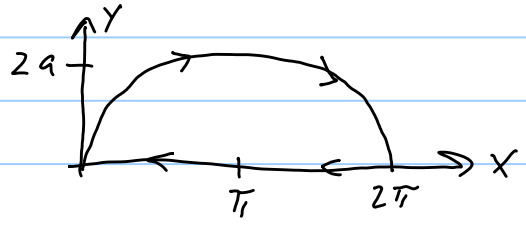
$$= \iint_D 0 + 0\, dA = \underline{0}$$

(2) Let $\vec{c}(\theta) = (\cos\theta, \sin\theta)$. $\therefore \vec{n} = (\cos\theta, \sin\theta)$

$\therefore \vec{F} \cdot \vec{n} = (\sin\theta, -\cos\theta) \cdot (\cos\theta, \sin\theta) = 0$

$\therefore \int_{\partial D} \vec{F} \cdot \vec{n}\, ds = \int_{\partial D} 0\, ds = \underline{0}$

13. **13.** Find the area bounded by one arc of the cycloid $x = a(\theta - \sin\theta)$, $y = a(1 - \cos\theta)$, where $a > 0$, and $0 \le \theta \le 2\pi$, and the x axis (use Green's theorem).



From Green's Theorem, $A = \frac{1}{2} \int_{\partial D} x\,dy - y\,dx$,

where ∂D is a simple closed curve of positive orientation.

For negative orientation, $A = -\frac{1}{2} \int_{\partial D} x\,dy - y\,dx$

The top curve is negatively oriented for $0 \le \theta \le 2\pi$,

$$\vec{c}(\theta) = [a(\theta - \sin\theta),\ a(1-\cos\theta)], \quad 0 \le \theta \le 2\pi$$

For the bottom of the closed curve (x-axis),

$$\vec{c}(\theta) = [4\pi - \theta, 0], \quad 2\pi \le \theta \le 4\pi$$

(1) Top curve: $dx = (a - a\cos\theta)\,d\theta \quad dy = a\sin\theta\,d\theta$

$$\therefore A = -\frac{1}{2}\int_0^{2\pi} [(a\theta - a\sin\theta)(a\sin\theta) - (a - a\cos\theta)(a - a\cos\theta)]\,d\theta$$

$a^2\theta\sin\theta - a^2\sin^2\theta - a^2 + a^2\cos\theta + a^2\cos\theta - a^2\cos^2\theta$

$$-\frac{a^2}{2}\int_0^{2\pi} \theta\sin\theta - 2 + 2\cos\theta\,d\theta$$

$u = \theta \quad dv = \sin\theta\,d\theta \quad v = -\cos\theta \quad \int u\,dv = uv - \int v\,du$

$$= -\frac{a^2}{2}\left[-\theta\cos\theta\Big|_0^{2\pi} + \int_0^{2\pi}\cos\theta\,d\theta - 4\pi + 0\right]$$

$$= -\frac{a^2}{2}\left[-2\pi + 0 - 4\pi + 0\right] = \underline{3\pi a^2}$$

(2) Bottom curve: $dx = -d\theta \quad dy = 0\,d\theta$

$$\therefore A = -\frac{1}{2}\int_{2\pi}^{4\pi} (a\theta - a\sin\theta)(0) - (0)(-1)\,d\theta = \underline{0}$$

$$\therefore A = 3\pi a^2 + 0 = \underline{3\pi a^2}$$

14.

14. Under the conditions of Green's theorem, prove that

(a) $\displaystyle\int_{\partial D} PQ\,dx + PQ\,dy = \iint_D \left[Q\left(\frac{\partial P}{\partial x} - \frac{\partial P}{\partial y}\right) + P\left(\frac{\partial Q}{\partial x} - \frac{\partial Q}{\partial y}\right)\right] dx\,dy$

(a) $$\int_{\partial D} PQ\,dx + PQ\,dy = \iint_D \frac{\partial}{\partial x}(PQ) - \frac{\partial}{\partial y}(PQ)\,dx\,dy$$

$$= \iint_D P_x Q + P Q_x - (P_y Q + P Q_y)\,dx\,dy$$

$$= \iint_D (P_x - P_y) Q + P(Q_x - Q_y)\,dx\,dy$$

(b) $\displaystyle\int_{\partial D} \left(Q\frac{\partial P}{\partial x} - P\frac{\partial Q}{\partial x}\right) dx + \left(P\frac{\partial Q}{\partial y} - Q\frac{\partial P}{\partial y}\right) dy$
$\displaystyle = 2\iint_D \left(P\frac{\partial^2 Q}{\partial x\,\partial y} - Q\frac{\partial^2 P}{\partial x\,\partial y}\right) dx\,dy$

(b) $$\int_{\partial D} \left(Q\frac{\partial P}{\partial x} - P\frac{\partial Q}{\partial x}\right) dx + \left(P\frac{\partial Q}{\partial y} - Q\frac{\partial P}{\partial y}\right) dy$$

$= P'$ $\qquad$ $= Q'$

$$= \iint_D \frac{\partial}{\partial x}\left(P\frac{\partial Q}{\partial y} - Q\frac{\partial P}{\partial y}\right) - \frac{\partial}{\partial y}\left(Q\frac{\partial P}{\partial x} - P\frac{\partial Q}{\partial x}\right) dx\,dy$$

Q' $\qquad$ P'

using Green's Th.

$$= \iint_D \frac{\partial P}{\partial x}\frac{\partial Q}{\partial y} + P\frac{\partial^2 Q}{\partial x \partial y} - \frac{\partial Q}{\partial x}\frac{\partial P}{\partial y} - Q\frac{\partial^2 P}{\partial x \partial y} - \frac{\partial Q}{\partial y}\frac{\partial P}{\partial x} - Q\frac{\partial^2 P}{\partial y \partial x} + \frac{\partial P}{\partial y}\frac{\partial Q}{\partial x} + P\frac{\partial^2 Q}{\partial y \partial x}\, dx\,dy$$

$$= \iint_D P\frac{\partial^2 Q}{\partial x y} - Q\frac{\partial^2 P}{\partial x \partial y} - Q\frac{\partial^2 P}{\partial y \partial x} + P\frac{\partial^2 Q}{\partial y \partial x}\, dx\,dy$$

$$= 2\iint P\frac{\partial^2 Q}{\partial x \partial y} - Q\frac{\partial^2 P}{\partial x \partial y}\, dx\,dy$$

$P, Q\ C^2 \Rightarrow P_{xy} = P_{yx},\ Q_{xy} = Q_{yx}$

15.

15. Evaluate the line integral

$$\int_C (2x^3 - y^3)\, dx + (x^3 + y^3)\, dy,$$

where C is the unit circle, and verify Green's theorem for this case.

Let $\vec{c}(\theta) = (\cos\theta, \sin\theta)$, $0 \le \theta \le 2\pi$, $\therefore dx = -\sin\theta\, d\theta$, $dy = \cos\theta\, d\theta$

(a) $$\int_C (2\cos^3\theta - \sin^3\theta)(-\sin\theta) + (\cos^3\theta + \sin^3\theta)(\cos\theta)\, d\theta$$

$$= \int_0^{2\pi} -2\cos^3\theta\sin\theta + \sin^3\theta\cos\theta + \sin^4\theta + \cos^4\theta\, d\theta$$

$$= \frac{1}{2}\cos^4\theta\Big|_0^{2\pi} + \frac{\sin^4\theta}{4}\Big|_0^{2\pi} + \int_0^{2\pi} \sin^2\theta(1-\cos^2\theta) + \cos^2\theta(1-\sin^2\theta)\,d\theta$$

$$= 0 + 0 + \int_0^{2\pi} (\sin^2\theta + \cos^2\theta) - 2\sin^2\theta\cos^2\theta\,d\theta$$

$$= 2\pi - \frac{1}{2}\int_0^{2\pi} (2\sin\theta\cos\theta)^2\,d\theta = 2\pi - \int_0^{2\pi} \sin^2 2\theta\,d\theta$$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta \Rightarrow \sin^2\theta = \frac{1-\cos 2\theta}{2}$

$$= 2\pi - \frac{1}{2}\int_0^{2\pi} \frac{1-\cos 4\theta}{2}\,d\theta = 2\pi - \frac{\theta}{4}\Big|_0^{2\pi} + \frac{\sin 4\theta}{16}\Big|_0^{2\pi}$$

$$= 2\pi - \frac{\pi}{2} + 0 = \underline{\frac{3}{2}\pi}$$

(5) $P(x,y) = 2x^3 - y^3 \quad Q(x,y) = x^3 + y^3 \quad P_y = -3y^2 \quad Q_x = 3x^2$

$$\therefore \iint_D Q_x - P_y\,dx\,dy = \iint_D 3x^2 + 3y^2\,dx\,dy$$

using $x = r\cos\theta$, $y = r\sin\theta$, $0 \le r \le 1$, $0 \le \theta \le 2\pi$

$$= \int_0^{2\pi}\int_0^1 (3r^2\cos^2\theta + 3r^2\sin^2\theta)\,r\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^1 3r^3\,dr\,d\theta = (2\pi)\frac{3r^4}{4}\Big|_0^1 = \underline{\frac{3}{2}\pi}$$

16.

16. Prove the following generalization of Green's theorem: Let D be a region in the xy plane with boundary a finite number of oriented simple closed curves. Suppose that by means of a finite number of line segments parallel to the coordinate axes, D can be decomposed into a finite number of simple regions D_i with the boundary of each D_i oriented counterclockwise (see Figure 8.1.5). Then if P and Q are of class C^1 on D,

$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \int_{\partial D} P\,dx + Q\,dy,$$

where ∂D is the oriented boundary of D. (HINT: Apply Green's theorem to each D_i.)

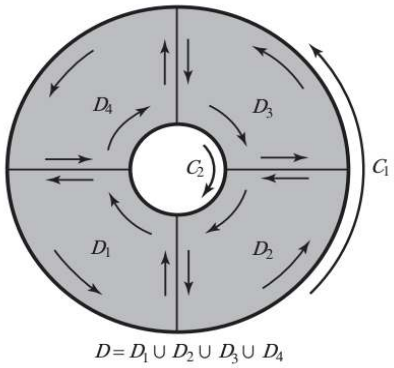


figure **8.1.5** Green's theorem applies to $D = D_1 \cup D_2 \cup D_3 \cup D_4$.

This seems to be an attempt to verbalize the concept shown in Figure 8.1.5.

Assume $D = D_1 \cup D_2 \ldots \cup D_n$ so that adjacent sides of D_i, D_j cancel out.

$$\therefore \int_{\partial D} P\,dx + Q\,dy = \int_{\partial D_1} P\,dx + Q\,dy + \ldots + \int_{\partial D_n} P\,dx + Q\,dy$$

$$= \iint_{D_1} (Q_x - P_y)\,dx\,dy + \ldots + \iint_{D_n} (Q_x - P_y)\,dx\,dy$$

$$= \iint_{D} (Q_x - P_y)\,dx\,dy$$

Using linearity of single and double integrals.

17. **17.** Verify Green's theorem for the integrand of Exercise 15 (that is, with $P = 2x^3 - y^3$ and $Q = x^3 + y^3$) and the annular region D described by $a \le x^2 + y^2 \le b$, with boundaries oriented as in Figure 8.1.5.

(a) Let $\vec{c}_1(\theta) = (\sqrt{b}\cos\theta, \sqrt{b}\sin\theta)$ $dx = -\sqrt{b}\sin\theta\, d\theta$ $dy = \sqrt{b}\cos\theta\, d\theta$

$\vec{c}_2(\theta) = (\sqrt{a}\cos\theta, \sqrt{a}\sin\theta)$ $dx = -\sqrt{a}\sin\theta\, d\theta$ $dy = \sqrt{a}\cos\theta\, d\theta$

$$\therefore \int_C P\,dx + Q\,dy = \int_{C_1^+} P\,dx + Q\,dy + \int_{C_2^-} P\,dx + Q\,dy$$

$$= \int_{C_1^+} P\,dx + Q\,dy - \int_{C_2^+} P\,dx + Q\,dy$$

$$= \int_0^{2\pi} \left(2b^{\frac{3}{2}}\cos^3\theta - b^{\frac{3}{2}}\sin^3\theta\right)\left(-b^{\frac{1}{2}}\sin\theta\right) + \left(b^{\frac{3}{2}}\cos^3\theta + b^{\frac{3}{2}}\sin^3\theta\right)\left(b^{\frac{1}{2}}\cos\theta\right) d\theta$$

$$- \int_0^{2\pi} \left(2a^{\frac{3}{2}}\cos^3\theta - a^{\frac{3}{2}}\sin^3\theta\right)\left(-a^{\frac{1}{2}}\sin\theta\right) + \left(a^{\frac{3}{2}}\cos^3\theta + a^{\frac{3}{2}}\sin^3\theta\right)\left(a^{\frac{1}{2}}\cos\theta\right) d\theta$$

$$= (b^2 - a^2)\int_0^{2\pi} -2\cos^3\theta\sin\theta + \sin^3\theta\cos\theta + \sin^4\theta + \cos^4\theta\, d\theta$$

$$= (b^2 - a^2)\left(\frac{3}{2}\pi\right)$$ using calculations in #15(a)

(6) As in #15(6),

$$\iint_D (Q_x - P_y)\,dx\,dy = \iint_D 3x^2 + 3y^2\,dx\,dy$$

using $x = r\cos\theta$, $y = r\sin\theta$, $a^{\frac{1}{2}} \le r \le b^{\frac{1}{2}}$, $0 \le \theta \le 2\pi$

$$= \int_0^{2\pi} \int_{a^{1/2}}^{b^{1/2}} (3r^2\cos^2\theta + 3r^2\sin^2\theta)\, r\,dr\,d\theta$$

$$= (2\pi)\ \frac{3}{4} r^4 \Big|_{a^{\frac{1}{2}}}^{b^{\frac{1}{2}}} = \frac{3}{2}\pi(b^2 - a^2)$$

18.

18. Let D be a region for which Green's theorem holds. Suppose f is harmonic; that is,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

on D. Prove that

$$\int_{\partial D} \frac{\partial f}{\partial y}\,dx - \frac{\partial f}{\partial x}\,dy = 0.$$

Let $f: \mathbb{R}^2 \to \mathbb{R}$, $P(x,y) = \dfrac{\partial f}{\partial y}$, $Q(x,y) = -\dfrac{\partial f}{\partial x}$

$\therefore$ $Q_x = -\dfrac{\partial^2 f}{\partial x^2}$, $P_y = \dfrac{\partial^2 f}{\partial y^2}$

By Green's Theorem,

$$\int_{\partial D} \frac{\partial f}{\partial y}\,dx - \frac{\partial f}{\partial x}\,dy = \int_{\partial D} P\,dx + Q\,dy$$

$$= \iint_D (Q_x - P_y)\,dx\,dy = \iint_D -\frac{\partial^2 f}{\partial x^2} - \frac{\partial^2 f}{\partial y^2}\,dx\,dy$$

$$= \iint_D -\left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}\right) dx\,dy = \iint_D (0)\,dx\,dy = 0$$

19.

19. (a) Verify the divergence theorem for $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$ and D the unit disc $x^2 + y^2 \le 1$.

(b) Evaluate the integral of the normal component of $2xy\mathbf{i} - y^2\mathbf{j}$ around the ellipse defined by $x^2/a^2 + y^2/b^2 = 1$.

(a) (1) Let $\vec{c}(\theta) = (\cos\theta, \sin\theta)$ $\quad \vec{c}\,'(\theta) = (-\sin\theta, \cos\theta)$

$\therefore \vec{n}(\theta) = (\cos\theta, \sin\theta)$ $\quad \vec{F}(\theta) = (\cos\theta, \sin\theta)$

$$\therefore \int_{\partial D} \vec{F}\cdot\vec{n}\,ds = \int_0^{2\pi} \cos^2\theta + \sin^2\theta\,d\theta = \underline{2\pi}$$

(2) $$\iint_D \operatorname{div} F\,dA = \iint_D (1+1)\,dA = 2\iint_D dA$$

$= 2(\pi(1)^2) = \underline{2\pi}$ since area unit disc $= \pi(1)^2$

(5) Let $\vec{c}(\theta) = (a\cos\theta, b\sin\theta)$, $0 \le \theta \le 2\pi$

$\therefore \vec{c}\,'(\theta) = (-a\sin\theta, b\cos\theta)$, $\|\vec{c}\,'(\theta)\| = \sqrt{a^2\cos^2\theta + b^2\sin^2\theta}$

$$\therefore \vec{n}(\theta) = \frac{(b\cos\theta, a\sin\theta)}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}$$

$$\vec{F}(\theta) = \left[2(a\cos\theta)(b\sin\theta), -b^2\sin^2\theta\right]$$

$$= \left[2ab\cos\theta\sin\theta, -b^2\sin\theta\right]$$

$$\therefore \vec{F}(\theta)\cdot\vec{n}(\theta) = \frac{2ab^2\cos^2\theta\sin\theta - ab^2\sin^2\theta}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}$$

$$\therefore \int_{\partial D} \vec{F}\cdot\vec{n}\,ds = \int_0^{2\pi} \left(\frac{2ab^2\cos^2\theta\sin\theta - ab^2\sin^2\theta}{\sqrt{a^2\sin^2\theta + b^2\cos^2\theta}}\right)\left(\sqrt{a^2\cos^2\theta + b^2\sin^2\theta}\right)d\theta$$

By the Divergence Theorem,

$$\int_{\partial D} \vec{F}\cdot\vec{n}\,ds = \iint_D \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y}\right)dx\,dy$$

$$= \iint_D (2y - 2y)\,dx\,dy \quad , \text{ since } \frac{\partial}{\partial x}(2xy) = 2y, \frac{\partial}{\partial y}(-y^2) = -2y$$

$$= \iint_D (0)\,dx\,dy = \underline{0}$$

$\therefore$ avoided an ugly trigonometric integration

20. **20.** Let $P(x, y) = -y/(x^2 + y^2)$ and $Q(x, y) = x/(x^2 + y^2)$. Assuming D is the unit disc, investigate why Green's theorem fails for this P and Q.

P and Q are not C^1 at origin $(x,y) = (0,0)$.

For example, $P_y = \dfrac{(x^2+y^2)(-1) - (-y)(2y)}{(x^2+y^2)^2} = \dfrac{y^2 - x^2}{(x^2+y^2)^2}$

P_y is not continuous at $(0,0)$

Along $y = x$, $\lim\limits_{(x,y)\to(0,0)} P_y = 0$

Along $x = 0$, $\lim\limits_{(x,y)\to(0,0)} P_y = \lim\limits_{(x,y)\to(0,0)} \dfrac{1}{y^2} = +\infty$ (i.e., does not exist)

∴ Can't use Lemma 1 on p. 429 of text, as

$\displaystyle\int_{\phi_1(x)}^{\phi_2(x)} P_y(x,y)\,dy$ does not exist on the unit disc,

where $\phi_1(x) = -\sqrt{1-x^2}$, $\phi_2(x) = \sqrt{1-x^2}$

21. **21.** Use Green's theorem to evaluate $\int_{C^+} (y^2 + x^3)\,dx + x^4\,dy$, where C^+ is the perimeter of the square $[0, 1] \times [0, 1]$ in the counterclockwise direction.

$$\int_{C^+} (y^2 + x^3)\,dx + x^4\,dy = \iint_D \left[\frac{\partial}{\partial x} x^4 - \frac{\partial}{\partial y}(y^2 + x^3)\right] dx\,dy$$

$$= \iint_D (4x^3 - 2y)\,dx\,dy = \int_0^1 \int_0^1 (4x^3 - 2y)\,dx\,dy$$

$$= \int_0^1 x^4 - 2xy \Big|_0^1 \,dy = \int_0^1 (1 - 2y)\,dy = y - y^2 \Big|_0^1 = \underline{0}$$

22.

22. Verify Theorem 3 by showing that $(\nabla \times \mathbf{F}) \cdot \mathbf{k} = \partial Q/\partial x - \partial P/\partial y$.

Let $\vec{F}(x, y, z) = [P(x,y), Q(x,y), 0]$

$$\therefore \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & 0 \end{vmatrix}$$

$$= (0 - Q_z,\ P_z - 0,\ Q_x - P_y)$$

$$= (0, 0, Q_x - P_y) \quad \text{as} \quad Q_z = 0,\ P_z = 0.$$

$$\therefore (\nabla \times \vec{F}) \cdot \hat{k} = (0, 0, Q_x - P_y) \cdot (0, 0, 1)$$

$$= Q_x - P_y$$

23.

23. Use Theorem 2 to compute the area inside the ellipse $x^2/a^2 + y^2/b^2 = 1$.

Let $\vec{c}(\theta) = (a\cos\theta, b\sin\theta)$, $0 \le \theta \le 2\pi$

$$\therefore dx = -a\sin\theta\,d\theta, \quad dy = b\cos\theta\,d\theta$$

$$\therefore A = \frac{1}{2}\int_{\partial D} x\,dy - y\,dx =$$

$$= \frac{1}{2}\int_0^{2\pi} (a\cos\theta)(b\cos\theta) - (b\sin\theta)(-a\sin\theta)\,d\theta$$

$$= \frac{1}{2}\int_0^{2\pi} ab\cos^2\theta + ab\sin^2\theta\,d\theta = \frac{ab}{2}\int_0^{2\pi} d\theta$$

$$= \frac{ab}{2}(2\pi) = \underline{\pi ab}$$

24. **24.** Use Theorem 2 to recover the formula $A = \frac{1}{2}\int_a^b r^2\,d\theta$ for a region in polar coordinates.

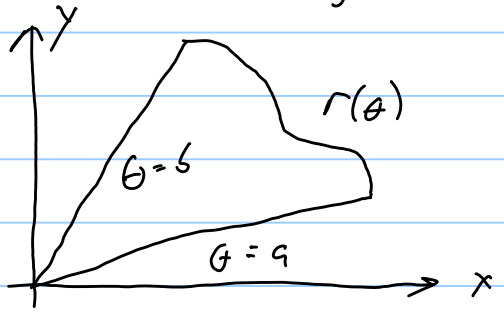
Let $r(\theta)$ be a continuous non-negative function for $a \le \theta \le b$, let D be the region bounded by $r(\theta)$ and the lines $\theta = a$ and $\theta = b$.

Let C be the counterclockwise curve bounding D.

$C_1 : t\,[r(a)\cos(a), r(a)\sin(a)]$, $0 \le t \le 1$

$C_2 : [r(\theta)\cos\theta, r(\theta)\cos\theta]$, $a \le \theta \le b$

$C_3 : (1-t)\,[r(b)\cos(b), r(b)\sin(b)]$, $0 \le t \le 1$



$\therefore C = C_1 + C_2 + C_3$

Using Theorem 2, $A = \frac{1}{2}\int_{\partial A} x\,dy - y\,dx$

$$= \frac{1}{2}\int_{C_1} x\,dy - y\,dx + \frac{1}{2}\int_{C_2} x\,dy - y\,dx + \frac{1}{2}\int_{C_3} x\,dy - y\,dx$$

For C_1: $x(t) = t\,r(a)\cos(a)$ $\quad dx = r(a)\cos(a)\,dt$

$y(t) = t\,r(a)\sin(a)$ $\quad dy = r(a)\sin(a)\,dt$

$$\therefore \frac{1}{2}\int_0^1 [t\,r(a)\cos(a)]\,r(a)\sin(a) - [t\,r(a)\sin(a)]\,r(a)\cos(a)\,dt$$

$$= \frac{1}{2}\int_0^1 0\,dt = 0$$

For C_3: $x(t) = (1-t)\,r(b)\cos(b)$ $\quad dx = -r(b)\cos(b)\,dt$

$y(t) = (1-t)\,r(b)\sin(b)$ $\quad dy = -r(b)\sin(b)\,dt$

$$\therefore \frac{1}{2}\int_0^1 [(1-t)\,r(b)\cos(b)][-r(b)\sin(b)] - [(1-t)\,r(b)\sin(b)][-r(b)\cos(b)]\,dt$$

$$= \frac{1}{2}\int_0^1 (0)\,dt = 0$$

For C_2: $x(\theta) = r(\theta)\cos\theta$ $\quad dx = r'(\theta)\cos\theta - r(\theta)\sin\theta\, d\theta$

$y(\theta) = r(\theta)\sin\theta$ $\quad dy = r'(\theta)\sin\theta + r(\theta)\cos\theta\, d\theta$

$$\therefore \frac{1}{2}\int_a^b r(\theta)\cos\theta\left[r'(\theta)\sin\theta + r\theta\cos\theta\right] - r(\theta)\sin\theta\left[r'(\theta)\cos\theta - r(\theta)\sin\theta\right] d\theta$$

$$= \frac{1}{2}\int_a^b r(\theta)r'(\theta)\cos\theta\sin\theta - r(\theta)r'(\theta)\cos\theta\sin\theta + r^2(\theta)\cos^2\theta + r^2(\theta)\sin^2\theta \, d\theta$$

$$= \frac{1}{2}\int_a^b r^2(\theta)\, d\theta$$

$$\therefore A = 0 + \frac{1}{2}\int_a^b r^2(\theta)\, d\theta + 0$$

$$= \frac{1}{2}\int_a^b r^2(\theta)\, d\theta$$

25.

25. Sketch the proof of Green's theorem for the region shown in Figure 8.1.10.

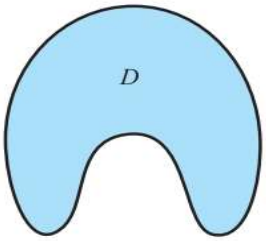
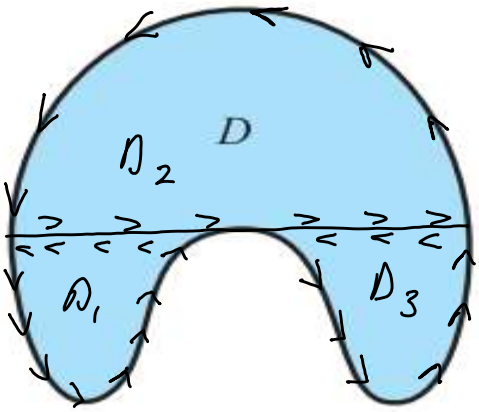


figure **8.1.10** Prove Green's theorem for this region.



Break D into 3 subregions: D_1, D_2, D_3

D_2 is bounded by two y-simple curves.

D_1 and D_3 are each bounded by two x-simple curves.

$\therefore D = D_1 \cup D_2 \cup D_3$, and by linearity of integrals,

$$\int_{\partial D} F\,dA = \int_{\partial D_1} F\,dA + \int_{\partial D_2} F\,dA + \int_{\partial D_3} F\,dA$$

where $F = F(x,y) = [P(x,y), Q(x,y)]$.

By Green's Theorem for simple closed curves,

$$\int_{D_1} F\,dA + \int_{D_2} F\,dA + \int_{D_3} F\,dA =$$

$$\iint_{D_1} Q_x - P_y\,dA + \iint_{D_2} Q_x - P_y\,dA + \iint_{D_3} Q_x - P_y\,dA$$

$$= \iint_D Q_x - P_y \, dA \text{ , by linearity of double integrals.}$$

$$\therefore \int_{\partial D} P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dx\,dy$$

26. **26.** Prove the identity

$$\int_{\partial D} \phi \nabla\phi \cdot \mathbf{n}\, ds = \iint_D (\phi \nabla^2 \phi + \nabla\phi \cdot \nabla\phi)\, dA.$$

From The Divergence Theorem,

$$\int_{\partial D} \vec{F} \cdot \vec{n}\, ds = \iint_D \operatorname{div} \vec{F}\, dA$$

Here, $\vec{F} = \phi \nabla\phi = \phi(x,y)\left[\phi_x(x,y), \phi_y(x,y)\right]$

$$= \left[\phi\phi_x, \phi\phi_y\right]$$

And: $\operatorname{div}(\phi\nabla\phi) = \operatorname{div}\left[\phi\phi_x, \phi\phi_y\right]$

$$= \frac{\partial}{\partial x}\phi\phi_x + \frac{\partial}{\partial y}\phi\phi_y$$

$$= \phi_x\phi_x + \phi\phi_{xx} + \phi_y\phi_y + \phi\phi_{yy}$$

$$\therefore \int_{\partial D} \phi\nabla\phi \cdot \vec{n}\, ds = \iint_D \phi_x\phi_x + \phi\phi_{xx} + \phi_y\phi_y + \phi\phi_{yy}\, dA \quad [1]$$

But $\phi\nabla^2\phi + \nabla\phi\cdot\nabla\phi = \phi(\phi_{xx}+\phi_{yy}) + (\phi_x,\phi_y)\cdot(\phi_x,\phi_y)$

$= \phi\phi_{xx} + \phi\phi_{yy} + \phi_x\phi_x + \phi_y\phi_y$

$\therefore \iint_D \phi\nabla^2\phi + \nabla\phi\cdot\nabla\phi \, dA$

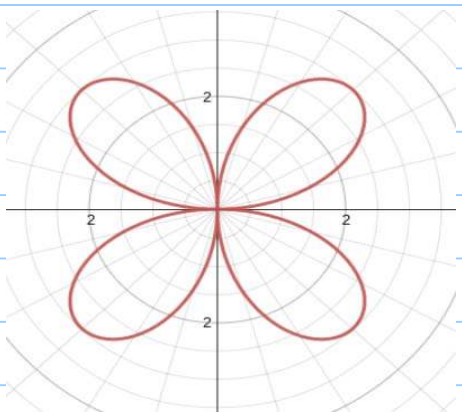
$= \iint_D \phi\phi_{xx} + \phi\phi_{yy} + \phi_x\phi_x + \phi_y\phi_y \, dA$ [2]

[1] and [2] are equal.

$\therefore \int_{\partial D} \phi\nabla\phi\cdot\vec{n}\, ds = \iint_D (\phi\nabla^2\phi + \nabla\phi\cdot\nabla\phi)\, dA$

27.

27. Use Green's theorem to find the area of one loop of the four-leafed rose $r = 3\sin 2\theta$. (HINT: $x\,dy - y\,dx = r^2\,d\theta$.)



$x = r\cos\theta = 3\sin 2\theta\cos\theta$

$y = r\sin\theta = 3\sin 2\theta\sin\theta$

$\therefore dx = (6\cos 2\theta\cos\theta - 3\sin 2\theta\sin\theta)\, d\theta$

$dy = (6\cos 2\theta\sin\theta + 3\sin 2\theta\cos\theta)\, d\theta$

$\therefore x\,dy = 3\sin 2\theta\cos\theta(6\cos 2\theta\sin\theta + 3\sin 2\theta\cos\theta)\, d\theta$

$$= (18\sin 2\theta \cos 2\theta \sin\theta\cos\theta + 9\sin^2 2\theta \cos^2\theta)\, d\theta$$

$$y\,dx = 3\sin 2\theta \sin\theta (6\cos 2\theta\cos\theta - 3\sin 2\theta \sin\theta)\, d\theta$$

$$= (18\sin 2\theta \cos 2\theta \sin\theta\cos\theta - 9\sin^2 2\theta \sin^2\theta)\, d\theta$$

$$\therefore x\,dy - y\,dx = (9\sin^2 2\theta \cos^2\theta + 9\sin^2 2\theta \sin^2\theta)\, d\theta$$

$$= (9\sin^2 2\theta)\, d\theta$$

$$= (r^2)\, d\theta$$

One loop, $0 \le \theta \le \frac{\pi}{2}$, is a simple closed curve.

$$\therefore A = \frac{1}{2}\int_{\partial D} x\,dy - y\,dx = \frac{1}{2}\int_0^{\frac{\pi}{2}} r^2\, d\theta$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} 9\sin^2 2\theta\, d\theta$$

$1 - 2\sin^2\theta = \cos 2\theta$,

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

$$= \frac{9}{4}\int_0^{\frac{\pi}{2}} 1 - \cos 4\theta\, d\theta = \frac{9}{4}\left[\theta - \frac{\sin 4\theta}{4}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{9}{4}\left(\frac{\pi}{2}\right) = \frac{9\pi}{8}$$

28.

28. Show that if C is a simple closed curve that bounds a region to which Green's theorem applies, then the area of the region D bounded by C is

$$A = \int_{\partial D} x\,dy = -\int_{\partial D} y\,dx.$$

Show how this implies Theorem 2.

Let $F(x,y) = [P(x,y), Q(x,y)] = [-y, 0]$ (i.e., $P = -y, Q = 0$)

Then by Green's Theorem,

(1) $$\int_{\partial D} P\,dx + Q\,dy = \int_{\partial D} P\,dx = \iint_D (Q_x - P_y)\,dx\,dy$$

or, $$\int_{\partial D} (-y)\,dx = -\int_{\partial D} y\,dx = \iint_D 0 - (-1)\,dx\,dy$$

$$\therefore -\int_{\partial D} y\,dx = \iint_D dx\,dy = A$$

$$\therefore A = -\int_{\partial D} y\,dx$$

(2) Let $F(x,y) = [0, x]$ (i.e., $P = 0, Q = x$)

$\therefore$ By Green's Theorem, $\int_{\partial D} P\,dx + Q\,dy =$

$$\int_{\partial D} x\,dy = \iint_D (Q_x - P_y)\,dx\,dy = \iint_D dx\,dy = A$$

$$\therefore A = \int_{\partial D} x\,dy$$

(3) From (1), (2), $$2A = -\int_{\partial D} y\,dx + \int_{\partial D} x\,dy,$$

$$\therefore A = \frac{1}{2}\int_{\partial D} x\,dy - y\,dx$$

Exercises 29 to 37 illustrate the application of Green's theorem to partial differential equations. (Further applications are given in the Internet supplement.) They are particularly concerned with solutions to Laplace's equation, that is, with harmonic functions. For these exercises, let D be an open region in $\mathbb{R}^2$ with boundary ∂D. Let u: $D \cup \partial D \to \mathbb{R}$ be a continuous function that is of class C^2 on D. Suppose $\mathbf{p} \in D$ and the closed discs $B_\rho = B_\rho(\mathbf{p})$ of radius ρ centered at $\mathbf{p}$ are contained in D for $0 < \rho \leq R$. Define $I(\rho)$ by

$$I(\rho) = \frac{1}{\rho}\int_{\partial B_\rho} u\ ds.$$

What is R? Since D is open, there is an $r > 0$ s.t. the open ball $B_r(\vec{p}) \subset D$. Now let $0 < R < r$, and $\therefore$ the closed disc $B_R(\vec{p}) \subset D$. $\therefore$ For all ρ, $0 < \rho \leq R$, consider the set of all closed discs, $B_\rho(\vec{p})$.

29. **29.** Show that $\lim_{\rho \to 0} I(\rho) = 2\pi u(\mathbf{p})$.

Let $\vec{c}(\theta) = \vec{p} + (\rho\cos\theta, \rho\sin\theta)$, $0 \le \theta \le 2\pi$,

a parametrization of the rim of $B_\rho(\vec{p}) = \partial B_\rho$

$\therefore \vec{c}\,'(\theta) = (-\rho\sin\theta, \rho\cos\theta) \quad \|\vec{c}\,'(\theta)\| = \rho$

$$\therefore I(\rho) = \frac{1}{\rho}\int_{\partial B_\rho} u\,ds = \frac{1}{\rho}\int_{\vec{c}} u(\vec{c}(\theta))\,\|\vec{c}\,'(\theta)\|\,d\theta$$

$$= \frac{1}{\rho}\int_0^{2\pi} u\left[\vec{p} + (\rho\cos\theta, \rho\sin\theta)\right]\rho\,d\theta$$

$$= \int_0^{2\pi} u\left[\vec{p} + (\rho\cos\theta, \rho\sin\theta)\right]d\theta$$

Thus, for a fixed $\rho > 0$, $0 < \rho \le R$

$$\therefore I(\rho) = \int_0^{2\pi} u\left[\vec{r}(\rho,\theta)\right]d\theta, \quad [1]$$

where $\vec{r}(\rho,\theta) = \vec{p} + (\rho\cos\theta, \rho\sin\theta) = \partial B_\rho$, $0 \le \theta \le 2\pi$

Since u is continuous on D, $\lim_{\rho \to 0} u\left[\vec{p} + (\rho\cos\theta, \rho\sin\theta)\right] = u(\vec{p})$

$\therefore$ Seems natural that

$$\lim_{\rho \to 0} \int_0^{2\pi} u[\,]\, d\theta = 2\pi\, u(\vec{p})$$

Proof: Let $\epsilon > 0$. Since u is continuous on D,

$\exists\, \delta > 0$ s.t. if $\vec{r} \in D$ and $0 < \|\vec{r} - \vec{p}\| < \delta$

Then $|u(\vec{r}) - u(\vec{p})| < \frac{\epsilon}{2\pi}$

$\therefore$ if $\rho < \delta$, then for $\vec{r} \in \partial B_\rho$

$\vec{r}(\rho, \theta) = \vec{p} + (\rho\cos\theta, \rho\sin\theta)$ for $0 < \rho < \delta$, $0 \le \theta \le 2\pi$

$0 < \|\vec{r}(\rho,\theta) - \vec{p}\| < \rho < \delta$, so $|u(\vec{r}(\rho,\theta)) - u(\vec{p})| < \frac{\epsilon}{2\pi}$

Note: $\vec{r}$, a boundary point, is a distance $\rho < \delta$ from $\vec{p}$.

$\therefore$ From [13],

$$|I(\rho) - 2\pi u(\vec{p})| = \left| \int_0^{2\pi} u[\vec{r}(\rho,\theta)]\, d\theta - 2\pi u(\vec{p}) \right|$$

$$= \left| \int_0^{2\pi} u[\vec{r}(\rho,\theta)]\, d\theta - \int_0^{2\pi} u(\vec{p})\, d\theta \right| = \left| \int_0^{2\pi} \left(u[\vec{r}(\rho,\theta)] - u(\vec{p}) \right) d\theta \right|$$

$$\le \int_0^{2\pi} |u[\vec{r}(\rho,\theta)] - u(\vec{p})|\, d\theta \qquad \left| \int_a^b f \right| \le \int_a^b |f|$$

$$< \int_0^{2\pi} \frac{\epsilon}{2\pi}\, d\theta = \frac{\epsilon}{2\pi}(2\pi) = \epsilon$$

$\therefore$ Given any $\epsilon > 0$, $\exists\, \delta > 0$ s.t. if $0 < \rho < \delta$, Then

$|I(\rho) - 2\pi u(\vec{p})| < \epsilon$, where $I(\rho) = \int_0^{2\pi} u[\vec{r}(\rho,\theta)]\, d\theta$, $\vec{r} \in \partial B_\rho$

$\therefore \lim_{\rho \to 0} I(\rho) = 2\pi u(\vec{p})$

30. **30.** Let $\mathbf{n}$ denote the outward unit normal to ∂B_ρ and $\partial u/\partial n = \nabla u \cdot \mathbf{n}$. Show that

$$\int_{\partial B_\rho} \frac{\partial u}{\partial n}\, ds = \iint_{B_\rho} \nabla^2 u\, dA.$$

From The Divergence Theorem (p. 436 of text),

$$\int_{\partial D} \vec{F} \cdot \vec{n}\, ds = \iint_D \operatorname{div} \vec{F}\, dA$$

Here, let $\vec{F} = \nabla u$. $\therefore \int_{\partial D} \nabla u \cdot \vec{n}\, ds = \iint_D \operatorname{div}(\nabla u)\, dA$

But $\operatorname{div}(\nabla u) = \nabla \cdot (\nabla u) = \nabla^2 u$

$\therefore \int_{\partial B_\rho} \frac{\partial u}{\partial n}\, ds = \int_{\partial B_\rho} (\nabla u \cdot \vec{n})\, ds = \iint_{B_\rho} \nabla^2 u\, dA$

31. **31.** Using Exercise 30, show that $I'(\rho) = (1/\rho) \iint_{B_\rho} \nabla^2 u \, dA$.

Using the parametrization $\vec{c}(\theta) = \vec{\rho} + (\rho\cos\theta, \rho\sin\theta)$ for B_ρ, $0 < \rho$, $0 \le \theta \le 2\pi$, let $\vec{\rho} = (\rho_1, \rho_2)$,

Note $\|\vec{c}\,'(\theta)\| = \sqrt{\rho^2(-\sin\theta)^2 + \rho^2\cos^2\theta} = \rho$

Then from #29, $\displaystyle\int_0^{2\pi} u(\rho_1 + \rho\cos\theta, \rho_2 + \rho\sin\theta)\, d\theta$

Now use differentiation under integral, i.e.,

if $F(x) = \displaystyle\int_a^b f(x,\theta)\, d\theta$, $F'(x) = \int_a^b \frac{\partial}{\partial x} f(x,\theta)\, d\theta$

$\therefore \dfrac{d}{d\rho} I(\rho) = \displaystyle\int_0^{2\pi} u_x(\rho_1 + \rho\cos\theta, \rho_2 + \rho\sin\theta)\cos\theta +$ using chain rule

$u_y(\rho_1 + \rho\cos\theta, \rho\sin\theta)\sin\theta \, d\theta$

$= \displaystyle\int_0^{2\pi} \nabla u(\vec{\rho}, \rho, \theta) \cdot (\cos\theta, \sin\theta)\, d\theta$

$= \displaystyle\int_0^{2\pi} \nabla u \cdot \vec{n}\, d\theta$

$\vec{n} = (\cos\theta, \sin\theta)$ - outward normal

$$= \frac{1}{\rho}\int_0^{2\pi} (\nabla u \cdot \vec{n})\,\rho\, d\theta = \frac{1}{\rho}\int_0^{2\pi} (\nabla u \cdot \vec{n})\,\|\vec{c}\,'(\theta)\|\, d\theta$$

$$= \frac{1}{\rho}\int_{\partial B_\rho} (\nabla u \cdot \vec{n})\, ds$$

$$= \frac{1}{\rho}\iint_{B_\rho} \operatorname{div}(\nabla u)\, dA \qquad \text{Divergence Theorem}$$

$$= \frac{1}{\rho}\iint_{B_\rho} \nabla^2 u\, dA$$

$$\therefore I'(\rho) = \frac{1}{\rho}\iint_{B_\rho} \nabla^2 u\, dA$$

32.

32. Suppose u satisfies Laplace's equation: $\nabla^2 u = 0$ on D. Use the preceding exercises to show that

$$u(\mathbf{p}) = \frac{1}{2\pi R}\int_{\partial B_R} u\, ds.$$

(This expresses the fact that the value of a harmonic function at a point is the average of its values on the circumference of any disc centered about it.)

Since $\nabla^2 u = 0$ on D, From #31 above,

$$I'(\rho) = \frac{1}{\rho}\iint_D \nabla^2 u\, dA = 0$$

$\therefore I(\rho)$ is a constant for all $\vec{\rho} \in D$, $\rho > 0$,

and $B_\rho(\vec{p}) \subset D$.

By #29, $\lim_{\rho \to 0} I(\rho) = 2\pi u(\vec{p})$

Since $I(\rho)$ is continuous for $\rho > 0$, $I(\rho)$ must equal $2\pi u(\vec{p})$ for all $\rho > 0$, $B_\rho(\vec{p}) \subset D$. If not, choose $\epsilon < |I(\rho) - 2\pi u(\vec{p})|$. This will contradict $\lim_{\rho \to 0} I(\rho) = 2\pi u(\vec{p})$

$\therefore I(\rho) = 2\pi u(\vec{p})$

$\therefore I(\rho) = \frac{1}{\rho} \int_{\partial B_\rho} u \, ds = 2\pi u(\vec{p})$

$\therefore u(\vec{p}) = \frac{1}{2\pi\rho} \int_{\partial B_\rho} u \, ds$ for $0 < \rho \leq R$

or $u(\vec{p}) = \frac{1}{2\pi R} \int_{\partial B_R} u \, ds$

33.

33. Use Exercise 32 to show that if u is harmonic (i.e., if $\nabla^2 u = 0$), then $u(\mathbf{p})$ can be expressed as an area integral

$$u(\mathbf{p}) = \frac{1}{\pi R^2} \iint_{B_R} u \, dA.$$

Assume A is a disc of radius R centered at $\vec{p}$

This disc, B_R, can be parametrized as

$$\vec{c}(\rho,\theta) = \vec{p} + (\rho\cos\theta, \rho\sin\theta), \quad 0 \le \rho \le R, \; 0 \le \theta \le 2\pi$$

and the Jacobian is $\left|\frac{\partial(x,y)}{\partial(\rho,\theta)}\right| = \begin{vmatrix} \cos\theta & -\rho\sin\theta \\ \sin\theta & \rho\cos\theta \end{vmatrix} = \rho$

$$\therefore \iint_{B_R} u \, dA = \int_0^R \int_0^{2\pi} u(\vec{c}(\rho,\theta)) \, \rho \, d\theta \, d\rho$$

But $\rho = \|\vec{c}\,'(\rho,\theta)\|$, so the path integral

$$\int_{\partial B_\rho} u \, ds = \int_0^{2\pi} u(\vec{c}) \|\vec{c}\,'\| \, d\theta = \int_0^{2\pi} u(\vec{c}) \, \rho \, d\theta$$

$$\therefore \iint_{B_R} u \, dA = \int_0^R \left(\int_0^{2\pi} u(\vec{c}) \, \rho \, d\theta \right) d\rho$$

$$= \int_0^R \left[\int_{\partial B_\rho} u \, ds \right] d\rho$$

$$= \int_0^R \left[2\pi\rho\, u(\vec{\rho}) \right] d\rho$$ from #32 above

$$= 2\pi u(\vec{\rho}) \int_0^R \rho\, d\rho = 2\pi u(\vec{\rho}) \left(\frac{R^2}{2} \right)$$

$$= \pi R^2 u(\vec{\rho})$$

$$\therefore \quad u(\vec{\rho}) = \frac{1}{\pi R^2} \iint_{B_R} u\, dA$$

Essentially, the average of the values of u over A.

34. **34.** Suppose u is a harmonic function defined on D (i.e., $\nabla^2 u = 0$ on D) and that u has a local maximum (or minimum) at a point $\mathbf{p}$ in D.

(a) Show that u must be constant on some disc centered at $\mathbf{p}$. (HINT: Use the results of Exercise 25.)

→ 5th edition text. For the 6th edition, use #33.

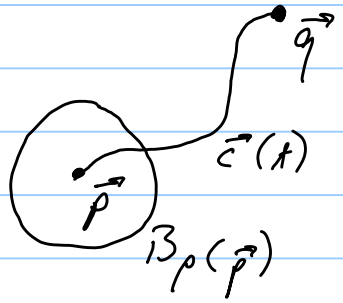
(a) If $u(\vec{\rho})$ is a max for some $\vec{\rho} \in D$, from #33 above, $u(\vec{\rho})$ is an average of its values on some disc around $\vec{\rho}$. The fact that $u(\vec{\rho})$ is a max means then that all values of u on the disc must be the same value, $u(\vec{\rho})$, to make the

average equal to the maximum.

(6)

(b) Suppose that D is path-connected [i.e., for any points $\mathbf{p}$ and $\mathbf{q}$ in D, there is a continuous path $\mathbf{c}$: $[0, 1] \to D$ such that $\mathbf{c}(0) = \mathbf{p}$ and $\mathbf{c}(1) = \mathbf{q}$] and that for some $\mathbf{p}$ the maximum or minimum at $\mathbf{p}$ is absolute; thus, $u(\mathbf{q}) \le u(\mathbf{p})$ or $u(\mathbf{q}) \ge u(\mathbf{p})$ for every $\mathbf{q}$ in D. Show that u must be constant on D.

(The result in this Exercise is called a ***strong maximum*** or ***minimum principle*** for harmonic functions. Compare this with Exercises 46 to 50 in Section 3.3.)



The idea is to establish constancy of u on $B_\rho(\vec{p})$, and then to use continuity to extend the zone of constancy along $\vec{c}(t)$. The B_ρ are closed, and choosing new balls at points on rims, which overlap old ball, and using continuity, extends zone of constancy. Hard too see if this sequence converges on $\vec{q}$. A rigorous proof, using least upper bound concept on intervals of $\vec{c}(t)$, is given in #35(b). Proof doesn't depend on $\nabla^2 u = 0$; i.e., $\nabla^2 u \ge 0$ or $\nabla^2 u \le 0$ is acceptable.

35.

35. A function is said to be ***subharmonic*** on D if $\nabla^2 u \geq 0$ everywhere in D. It is said to be ***superharmonic*** if $\nabla^2 u \leq 0$.

(a) Derive a strong maximum principle for subharmonic functions.

(b) Derive a strong minimum principle for superharmonic functions.

(a) If D is path connected, and for some $\vec{\rho} \in D$ $u(\vec{\rho})$ is an absolute maximum, then u is constant on D if $\nabla^2 u \geq 0$ everywhere in D.

(1) If $\nabla^2 u \geq 0$ on D and for $\vec{\rho} \in D$, $u(\vec{\rho})$ is a local maximum, there exists an $R > 0$ s.t. u is constant on $B_R(\vec{\rho}) \subset D$.

Pf: Since D is open, and $\vec{\rho} \in D$, there exists an $r > 0$ s.t. $B_r(\vec{\rho}) \subset D$. Since $u(\vec{\rho})$ is a local maximum, there exists an $R > 0$, $0 < R < r$, s.t. $B_R(\vec{\rho}) \subset D$ and for all $\vec{q} \in B_R(\vec{\rho})$, $\vec{q} \neq \vec{\rho}$, $u(\vec{\rho}) \geq u(\vec{q})$. Now consider all $B_\rho(\vec{\rho})$, $0 < \rho \leq R$. Note the B_ρ are closed.

By #31, $I'(\rho) = \frac{1}{\rho} \iint_{B_\rho} \nabla^2 u \, dA$

Since $\nabla^2 u \geq 0$ everywhere in D, then $\nabla^2 u \geq 0$

everywhere in B_ρ.

$\therefore \iint_{B_\rho} \nabla^2 u \, dA \geq 0$ since this is a sum of $\nabla^2 u$ on B_ρ. $\therefore I'(\rho) \geq 0$, so $I(\rho)$ is a non-decreasing function for $0 < \rho \leq R$. $\therefore I(\rho) \leq I(R)$

#29 showed $\lim_{\rho \to 0} I(\rho) = 2\pi u(\vec{p})$

Since $I(\rho)$ is continuous (it's differentiable),

Then $2\pi u(\vec{p}) \leq I(\rho)$, as $\rho > 0$.

$\therefore 2\pi u(\vec{p}) \leq I(\rho) \leq I(R)$, $0 < \rho \leq R$

Since the rim of any $B_\rho(\vec{p})$ can be parameterized as $\vec{p} + \rho(\cos\theta, \sin\theta)$, Then for every $\vec{q} \in B_R(\vec{p})$, there exists ρ, θ_0 s.t. $\vec{q} = \vec{p} + \rho(\cos\theta_0, \sin\theta_0)$.

Suppose $u(\vec{q}) < u(\vec{p})$ for $\vec{q} \in B_R(\vec{p})$ [*]

Since u is continuous at $u(\vec{q})$, given any $\epsilon > 0$ there exists a $\delta > 0$ s.t. if $\theta \in [\theta_0 - \delta, \theta_0 + \delta]$,

and $\vec{x} = \vec{p} + \rho(\cos\theta, \sin\theta)$, where ρ is from $\vec{q}$,

then $|u(\vec{x}) - u(\vec{q})| < \epsilon$. $\therefore$ Let $\epsilon = \dfrac{u(\vec{p}) - u(\vec{q})}{2}$.

$\therefore u(\vec{x}) - u(\vec{q}) < \epsilon \Rightarrow u(\vec{x}) < u(\vec{q}) + \epsilon$

$\therefore u(\vec{x}) + \epsilon < u(\vec{q}) + 2\epsilon = u(\vec{q}) + u(\vec{p}) - u(\vec{q}) = u(\vec{p})$

$\therefore$ for $\theta \in [\theta_0 - \delta, \theta_0 + \delta]$, $u(\vec{x}) < u(\vec{p}) - \epsilon$

That is, for the fixed ρ from $\vec{q}$, along the arc $[\theta_0 - \delta, \theta_0 + \delta]$ of the rim of $B_\rho(\vec{p})$, $u < u(\vec{p}) - \epsilon$

$$\therefore 2\pi u(\vec{p}) \le I(\rho) = \int_0^{2\pi} u[\vec{p} + \rho(\cos\theta, \sin\theta)]\, d\theta$$

see solution to #29

$$= \int_0^{\theta_0 - \delta} u\, d\theta + \int_{\theta_0 + \delta}^{2\pi} u\, d\theta + \int_{\theta_0 - \delta}^{\theta_0 + \delta} u\, d\theta$$

Keeping ρ from $\vec{q}$ fixed

$$< \int_0^{\theta_0 - \delta} u(\vec{p})\, d\theta + \int_{\theta_0 + \delta}^{2\pi} u(\vec{p})\, d\theta + \int_{\theta_0 - \delta}^{\theta_0 + \delta} [u(\vec{p}) - \epsilon]\, d\theta$$

$u \le u(\vec{p})$, a local maximum

$u < u(\vec{p}) - \epsilon$

$$= (2\pi - 2\delta)\, u(\vec{p}) + 2\delta\, [u(\vec{p}) - \epsilon]$$

$u(\vec{p})$, $u(\vec{p}) - \epsilon$ are constants

$$= 2\pi u(\vec{p}) - 2\delta\epsilon$$

Or, $2\pi u(\vec{\rho}) < 2\pi u(\vec{\rho}) - 2\delta\epsilon$

a clear contradiction to [*], since $\delta, \epsilon > 0$.

$\therefore$ For $u(\vec{q}) \leq u(\vec{\rho})$, can't have $u(\vec{q}) < u(\vec{\rho})$,

$\therefore u(\vec{q}) = u(\vec{\rho})$

Since $\vec{q}$ was arbitrary in $B_R(\vec{\rho})$, $\vec{q} \neq \vec{\rho}$,

u is constant on $B_R(\vec{\rho})$.

(2) Suppose D is path connected, and for some $\vec{\rho} \in D$, $u(\vec{\rho})$ is an absolute maximum.
Then u is constant everywhere on D.

Pf: Let $\vec{q}$ be any point in D, $\vec{q} \neq \vec{\rho}$, and let

$\vec{c}(t): [0,1] \to D$ be a continuous path from

$\vec{\rho}$ to $\vec{q}$, s.t. $\vec{c}(0) = \vec{\rho}$, $\vec{c}(1) = \vec{q}$.

Let $\mathcal{B} = \{ b : 0 < b \leq 1 \text{ s.t. for } t \in [0,b), u[\vec{c}(t)] = u(\vec{\rho}) \}$

$\cup \{0\}$. Thus, $\mathcal{B}$ is the set of all intervals

$t \in [0,b) \cup [0]$ for which $u[\vec{c}(t)] = u(\vec{\rho})$

$\mathcal{B}$ is not empty, as $b = 0 \in \mathcal{B}$, and $\mathcal{B}$ is

bounded above by 1. ∴ B has a least upper bound, call it b_0.

Since $\vec{c}$ is continuous on $[0,1]$, $u[\vec{c}(t)]$ is continuous, and so $u[\vec{c}(b_0)] = u(\vec{p})$,

i.e., if $b_0 = 0$, then $u[\vec{c}(b_0)] = u[\vec{c}(0)] = u[\vec{p}]$

and if $b_0 > 0$, $\lim_{t \to b_0^-} u[\vec{c}(t)] = u(\vec{p})$, for by the definition of b_0, if $0 < |t - b_0| < \delta$,

$$|u[\vec{c}(t)] - u(\vec{p})| = |u(\vec{p}) - u(\vec{p})| = 0 < \text{any } \epsilon.$$

Since $u[\vec{c}(b_0)] = u(\vec{p})$, $u[\vec{c}(b_0)]$ is a local maximum as $u(\vec{p})$ is an absolute maximum.

By (1) above, there is a disc around $\vec{c}(b_0)$, say of radius $\rho > 0$, s.t. for $\vec{x} \in B_\rho(\vec{c}(b_0))$, $u(\vec{x}) = u(\vec{p})$

Suppose $b_0 \neq 1$.

Since $\vec{c}$ is continuous, given $\rho > 0$, $\exists\, \delta > 0$ s.t. for $t \in [b_0, b_0 + \delta)$, $\vec{c}(t) \in B_\rho(\vec{c}(b_0))$

and $[b_0, b_0+\delta) \subset [0,1)$

But this means there are elements $b \in B$ s.t. $b > b_0$ for which $t \in [0,b]$ and $u[\vec{c}(t)] = u(\vec{p})$.

This contradicts b_0 as the least upper bound.

$\therefore b_0 = 1 \Rightarrow u[\vec{c}(1)] = u(\vec{q}) = u(\vec{p})$

Since $\vec{q} \neq \vec{p}$ was any point in D, u is constant on D.

(b) If D is path connected, and for some $\vec{p} \in D$ $u(\vec{p})$ is an absolute minimum, then u is constant on D if $\nabla^2 u \leq 0$ everywhere in D.

Pf: The proof is the same as for (a), except

(1) reverse inequalities

(2) replace word "maximum" with "minimum"

(3) replace word "non-decreasing" with "non-increasing".

36.

36. Suppose D is the disc $\{(x, y) \mid x^2 + y^2 < 1\}$ and C is the circle $\{(x, y) \mid x^2 + y^2 = 1\}$. In the Internet supplement, we shall show that if f is a continuous real-valued function on C, then there is a continuous function u on $D \cup C$ that agrees with f on C and is harmonic on D. That is, f has a harmonic extension to the disc. Assuming this, show the following:

(a) If q is a nonconstant continuous function on $D \cup C$ that is subharmonic (but not harmonic) on D, then there is a continuous function u on $D \cup C$ that is harmonic on D such that u agrees with q on C and $q < u$ everywhere on D.

(a) Note q is continuous on C. From the statement of the problem, there is a harmonic extension, u, to $D \cup C$ s.t. $u = q$ on C, and $\nabla^2 u = 0$ on D and u is continuous on $D \cup C$.

Since u is continuous on a closed set ($D \cup C$), it has an absolute maximum on $D \cup C$.

From #47, section 3.3, the absolute maximum of u is on C.

Since $D \cup C$ is path-connected, #34(b) above says u is constant on $D \cup C$. $\therefore$ Call this constant $M = u(D \cup C)$.

From #46, section 3.3, q cannot have a maximum on D, so it must be on C. Since $u = q$ on C, $q_{max} = M$. $\therefore$ $q < M = u$ on D.

(6) (b) The same assertion holds if "subharmonic" is replaced by "superharmonic" and "$q < u$" by "$q > u$."

As in (a), there is a harmonic extension of f on $D \cup C$, call it u, s.t. $u = g$ on C, and $\nabla^2 u = 0$ on D, and u is continuous on $D \cup C$.

(1) u continuous on closed $D \cup C \Rightarrow u$ has an absolute minimum on $D \cup C$.

(2) #49, section 3.3, $\Rightarrow$ absolute minimum of u is on C.

(3) $D \cup C$ is path-connected, so #34(b) above $\Rightarrow u$ is constant on $D \cup C$. Call this constant $m = u(D \cup C)$.

(4) #48, section 3.3 $\Rightarrow g$ cannot have a minimum on D; i.e., it must be on C. Since $u = g$ on C, $g_{min} = m \therefore g > m$ on D.

37.

37. Let D be as in Exercise 36. Let $f\colon D \to \mathbb{R}$ be continuous. Show that a solution to the equation $\nabla^2 u = 0$ satisfying $u(\mathbf{x}) = f(\mathbf{x})$ for all $\mathbf{x} \in \partial D$ is unique.

Assume u is not unique. Let u_1, u_2 be s.t.

$$\nabla^2 u_1 = 0, \quad \nabla^2 u_2 = 0, \quad u_1 = f \text{ on } \partial D, \quad u_2 = f \text{ on } \partial D.$$

Consider $\phi = u_1 - u_2$. Since $u_1 = f$ and $u_2 = f$ on ∂D, then $\phi = 0$ on ∂D, and

$$\nabla^2 \phi = \nabla^2 u_1 - \nabla^2 u_2 = 0 \text{ on } D$$

Letting $\vec{F} = \phi \nabla \phi = (\phi \phi_x, \phi \phi_y)$, from #26 above,

$$\int_{\partial D} \phi \nabla \phi \cdot \vec{n}\, ds = \iint_D (\phi \nabla^2 \phi + \nabla \phi \cdot \nabla \phi)\, dA$$

But on ∂D, $\phi = 0$. $\therefore \int_{\partial D} \phi \nabla \phi \cdot \vec{n}\, ds = 0$.

Also, $\nabla^2 \phi = 0$ on D means $\iint_D \phi \nabla^2 \phi\, dA = 0$.

$$\therefore \iint_D \nabla \phi \cdot \nabla \phi\, dA = 0. \qquad [*]$$

But $\nabla \phi \cdot \nabla \phi = \|\nabla \phi\|^2 \geq 0$. Since [*] is

a summation of non-negative terms, to get 0, $\nabla\phi$ must be 0 on D. $\therefore \phi$ is constant on D, and 0 on ∂D.

Since ϕ is continuous on $D \cup \partial D$, and $\phi = 0$ on ∂D, $\phi = 0$ on $D \cup \partial D$. $\therefore u_1 = u_2$

38.

38. Use Green's theorem to prove the change of variables formula in the following special case:

$$\iint_D dx\,dy = \iint_{D^*} \left|\frac{\partial(x, y)}{\partial(u, v)}\right| du\,dv$$

for a transformation $(u, v) \mapsto (x(u, v), y(u, v))$.

From $\int_{\partial D} P\,dx + Q\,dy = \iint_D (Q_x - P_y)\,dx\,dy = \iint_D dx\,dy$

we want $Q_x - P_y = 1$. $\therefore$ Create $F = (P, Q)$ so that $Q_x = 1$, $P_y = 0$ or $Q_x = 0$, $P_y = -1$.

Choose $Q(x,y) = x$, $P(x,y) = 0$, $\therefore Q_x = 1$, $P_y = 0$

$\therefore \int_{\partial D} P\,dx + Q\,dy = \int_{\partial D} x\,dy = \iint_D dx\,dy = A(D)$

where $A(D)$ = area of D.

∴ We've used Green's Theorem to convert an area integral, $\iint_D dx\,dy$, to a line integral, $\int_{\partial D} x\,dy$.

The idea is to convert the line integral to one from ∂D to one evaluating over the new boundary, ∂D^*, and then using Green's Theorem, to convert that to an area integral, which will be over the new region, D^*.

When evaluating $\int_{\partial D} x\,dy$, we choose a

parametrization for ∂D, such as $\vec{c}(t) : I \to \mathbb{R}^2$, where I is some interval, such as $I = [a, b]$, and $\vec{c}(t) = [x(t), y(t)]$. ∴ $\vec{c}\,'(t) = [x'(t), y'(t)]$.
∴ Converting differential form to standard form,

$$\int_{\partial D} x\,dy = \int_{\vec{c}} x(t)\frac{dy(t)}{dt}\,dt = \int_a^b x(t)\,y'(t)\,dt$$

Now consider the transformation

$$T(u,v) = [x(u,v), y(u,v)],$$

assumed to be one-to-one and onto, $T: D^* \to D$, and T is class C^2, so that $x(u,v)$, $y(u,v)$ are C^2.

The new region, D^*, will have an orientation preserving parametrization for ∂D^*, call it

$s(\theta): I^* \to R^2$. Thus, u and v become functions of θ, and so do x and y as a result:

$$s(\theta) = [u(\theta), v(\theta)].$$

Using the chain rule, $\frac{dy}{d\theta} = \frac{\partial y}{\partial u}\cdot\frac{du}{d\theta} + \frac{\partial y}{\partial v}\cdot\frac{dv}{d\theta}$,

or, using differential form, $dy = \frac{\partial y}{\partial u}du + \frac{\partial y}{\partial v}dv$

$$\therefore \int_{\partial D} x\,dy = \int_{\partial D^*} x(u,v)\left[\frac{\partial y}{\partial u}du + \frac{\partial y}{\partial v}dv\right] \quad [*]$$

where ∂D^* is parametrized by $s(\theta): I^* \to R^2$

$$= \int_{\partial D^*} x(u,v)\frac{\partial y}{\partial u}\,du + x(u,v)\frac{\partial y}{\partial v}\,dv$$

analagous to $P\,dx + Q\,dy$

$$= \iint_{D^*} \frac{\partial}{\partial u}\left(x(u,v)\frac{\partial y}{\partial v}\right) - \frac{\partial}{\partial v}\left(x(u,v)\frac{\partial y}{\partial u}\right) du\,dv$$

using Green's Theorem

$$= \iint_{D^*} \frac{\partial x(u,v)}{\partial u}\cdot\frac{\partial y}{\partial v} + x(u,v)\cdot\frac{\partial^2 y}{\partial u\partial v} - \left(\frac{\partial x(u,v)}{\partial v}\cdot\frac{\partial y}{\partial u} + x(u,v)\cdot\frac{\partial^2 y}{\partial v\partial u}\right) du\,dv$$

$$= \iint_{D^*} \left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \right) du\,dv$$

$\frac{\partial^2 y}{\partial v \partial u} = \frac{\partial^2 y}{\partial u \partial v}$ since T is C^2

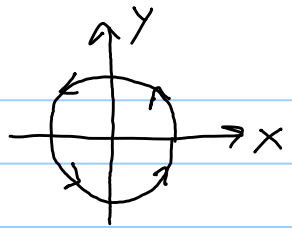
$$= \iint_{D^*} \left| \begin{matrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{matrix} \right| du\,dv = \iint_{D^*} \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du\,dv$$

$$\therefore \iint_D dx\,dy = \iint_{D^*} \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du\,dv$$

[*] $$\int_{\partial D} x\,dy = \int_{\partial D^*} x(u,v) \left[\frac{\partial y}{\partial u}\,du + \frac{\partial y}{\partial v}\,dv \right]$$

requires a bit of explanation. The evaluation of $x\,dy$ along the curve ∂D in the xy-plane requires one parametrization, call it $\vec{c}(t)$, and the result should be the same as the right integral, along the curve ∂D^* in the uv-plane, which requires a different parametrization, call it $\vec{s}(t)$.

A concrete example helps to explain.



Consider the unit disc in the

xy-plane: $\iint_D dx\,dy = \pi = \int_{\partial D} x\,dy$

Using counterclockwise rotation, let

$\vec{c}(t) = (-t, \sqrt{1-t^2})$, $t \in [-1, 1]$ for top curve

$\vec{c}(t) = (t, -\sqrt{1-t^2})$, $t \in [-1, 1]$ for bottom curve

For top curve: $\vec{c}\,'(t) = \left[-1, -t(1-t^2)^{-\frac{1}{2}}\right]$

$$\therefore \int_{\partial D_{top}} x\,dy = \int_{\vec{c}} (-t)\left(-t(1-t^2)^{-\frac{1}{2}}\right) dt = \int_{-1}^{1} \frac{t^2}{\sqrt{1-t^2}}\,dt$$

$$= -\frac{t}{2}\sqrt{1-t^2} + \frac{1}{2}\arcsin t \Big|_{-1}^{1} = \arcsin(1) = \frac{\pi}{2}$$

using table of integrals #55 of text

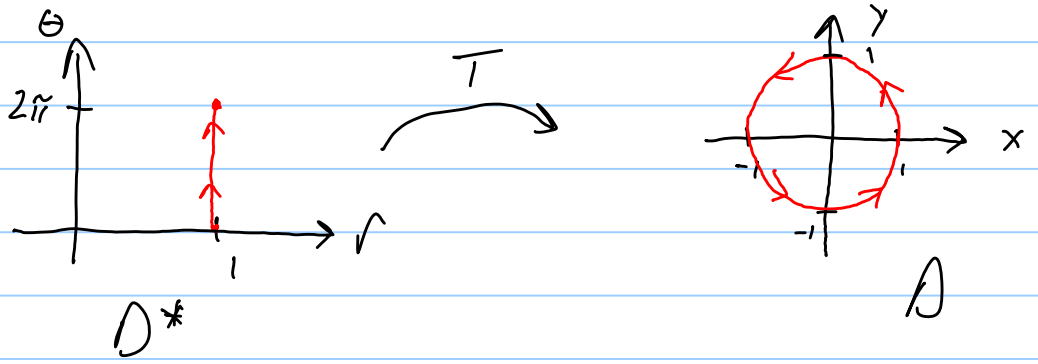
For bottom curve: $\vec{c}\,'(t) = \left[1, t(1-t^2)^{-\frac{1}{2}}\right]$

$$\therefore \int_{\partial D_{Bottom}} x\,dy = \int_{-1}^{1} \frac{t^2}{\sqrt{1-t^2}}\,dt = \frac{\pi}{2}$$

$$\therefore \int_{\partial D} x\,dy = \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

Now consider the curve under the transformation

$$T(r,\theta) = \begin{cases} (r\cos\theta, r\sin\theta) = (x(r,\theta), y(r,\theta)), & r > 0 \\ (0,0), & r = 0 \end{cases}$$



Let $\vec{s}(t) = (1, t) = (r(t), \theta(t))$, $0 \le t \le 2\pi$

$\vec{s}(t)$ is a different parametrization, located in the $r\theta$-plane, and is orientation preserving, counterclockwise.

$$\therefore (T \circ s)(t) = (\cos t, \sin t) = [x(r(t),\theta(t)), y(r(t),\theta(t))]$$

Note $\vec{s}\,'(t) = (0, 1) = (r'(t), \theta'(t))$

Also, $\dfrac{dy}{dt} = \dfrac{\partial y}{\partial r}\cdot\dfrac{dr}{dt} + \dfrac{\partial y}{\partial \theta}\cdot\dfrac{d\theta}{dt}$, or

using differential notation,

$$dy = \frac{\partial y}{\partial r}\cdot dr + \frac{\partial y}{\partial \theta}\cdot d\theta$$

$\therefore$ [*] becomes: $\displaystyle\int_{\partial D} x\,dy = \int_{\partial D^*} x(r,\theta)\left(\frac{\partial y}{\partial r}dr + \frac{\partial y}{\partial \theta}d\theta\right)$

$$= \int_{\partial D^*} x(r,\theta)\frac{\partial y}{\partial r}dr + x(r,\theta)\frac{\partial y}{\partial \theta}d\theta$$

$$= \int_{\vec{S}} \left[x\left(r(t),\theta(t)\right)\frac{\partial y}{\partial r}\frac{dr}{dt} + x\left(r(t),\theta(t)\right)\frac{\partial y}{\partial \theta}\frac{d\theta}{dt}\right]dt$$

$$= \int_0^{2\pi} \left[(\cos t)(\sin\theta(t))(0) + (\cos t)(r\cos t)(1)\right]dt$$

$\frac{dr}{dt} = 0$; $r = 1$

$$= \int_0^{2\pi} \cos^2 t\,dt = \int_0^{2\pi} \frac{1+\cos 2t}{2}\,dt = \frac{t}{2} + \frac{\sin 2t}{4}\Big|_0^{2\pi} = \underline{\pi}$$

Note: did not rely upon Jacobian, just the chain rule for dy.

This is not a reparametrization of the

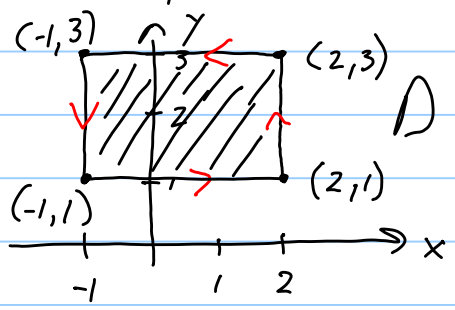
original $\vec{c}(t)$ in the xy-plane, but a transformation $D^* \to D$, and using a completely different parametrization $\vec{s}(t)$ in the $r\theta$-plane. This is analogous to variable substitution in integration in single variable calculus, with change of limits of integration.

8.2 Stokes' Theorem

Note Title 6/22/2017

1. **1.** Let S be the portion of the plane $2x + 3y + z = 5$ lying between the points $(-1, 1, 4)$, $(2, 1, -2)$, $(2, 3, -8)$, and $(-1, 3, 2)$. Find parameterizations for both the surface S and its boundary ∂S. Be sure that their respective orientations are compatible with Stokes' theorem.

Note that the 4 points project onto the xy-plane as $(-1,1,0)$, $(2,1,0)$, $(2,3,0)$, $(-1,3,0)$, a rectangle. Note the last point should be $(-1,3,-2)$ not $(-1,3,2)$, so that all points satisfy $2x+3y+z=5$ and lie in the plane.



For the surface S, note $z = 5 - 2x - 3y$.

$\therefore \phi(x,y) = (x, y, 5-2x-3y)$ will describe S for $-1 \le x \le 2$, $1 \le y \le 3$.

A normal to S is $(2,3,1)$ from the coefficients of $2x+3y+z=5$. As the coefficient of $\hat{k}$ is 1, this points "up". $\therefore$ ∂D should be

counterclockwise. Starting from $(-1,1,0)$, create parametrizations of ∂D, one side at a time, and use $z = 5-2x-3y$ to generate ∂S.

For $(-1,1)$ to $(2,1)$: $(-1,1) + t[(2,1)-(-1,1)]$

$= (-1,1) + t(3,0) = (-1+3t, 1)$, $0 \le t \le 1$.

Now $z = 5-2(-1+3t)-3(1) = 4-6t$

$\therefore \gamma(t) = (-1+3t, 1, 4-6t)$, $0 \le t \le 1$

For $(2,1)$ to $(2,3)$: $(2,1) + (t-1)[(2,3)-(2,1)]$, $1 \le t \le 2$

Could use t instead of $(t-1)$, but then would need $0 \le t \le 1$, and we want t to pick up after $0 \le t \le 1$ from first segment above.

$\therefore (2,1) + (t-1)(0,2) = (2, 2t-1)$

$\therefore z = 5-2(2)-3(2t-1) = 4-6t$

$\therefore \gamma(t) = (2, -1+2t, 4-6t)$, $1 \le t \le 2$

For $(2,3)$ to $(-1,3)$: $(2,3) + (t-2)[(-1,3)-(2,3)]$, $2 \le t \le 3$

$= (2,3) + (t-2)(-3,0) = (-3t+8, 3)$

$\therefore z = 5 - 2(-3t+8) - 3(3) = 6t - 20$

$\therefore \gamma(t) = (8-3t, 3, -20+6t)$, $2 \le t \le 3$

For $(-1,3)$ to $(-1,1)$: $(-1,3) + (t-3)[(-1,1) - (-1,3)]$, $3 \le t \le 4$

$= (-1,3) + (t-3)(0,-2) = (-1, 9-2t)$

$\therefore z = 5 - 2(-1) - 3(9-2t) = -20 + 6t$

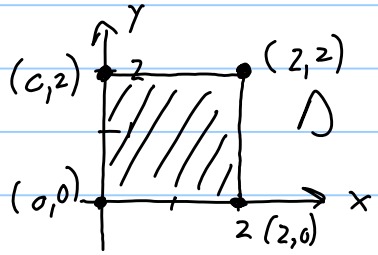
$\therefore \gamma(t) = (-1, 9-2t, -20+6t)$, $3 \le t \le 4$

$\therefore$ ∂S Summary:

$$\gamma(t) = \begin{cases} (-1+3t, 1, 4-6t), & 0 \le t \le 1 \\ (2, -1+2t, 4-6t), & 1 \le t \le 2 \\ (8-3t, 3, -20+6t), & 2 \le t \le 3 \\ (-1, 9-2t, -20+6t), & 3 \le t \le 4 \end{cases}$$

2.

2. Let S be the portion of the surface $z = x^2 + y^2$ lying between the points $(0, 0, 0)$, $(2, 0, 4)$, $(0, 2, 4)$, and $(2, 2, 8)$. Find parameterizations for both the surface S and its boundary ∂S. Be sure that their respective orientations are compatible with Stokes' theorem.



Note all 4 points lie on the surface $z = x^2 + y^2$, and the points project onto the xy-plane, giving D and ∂D,

as a square.

S can be parametrized as $\phi(u,v) = (u, v, u^2+v^2)$, $0 \le u \le 2$, $0 \le v \le 2$

∂S can be parametrized by "lifting" ∂D up to S, as was done in #1.

$\therefore$ Specify ∂D and use $z = x^2+y^2$ for the 3rd component.

For S, find the normal $T_u \times T_v$.

$$T_u = \left[\frac{\partial u}{\partial u}, \frac{\partial v}{\partial u}, \frac{\partial (u^2+v^2)}{\partial u}\right] = (1, 0, 2u)$$

$$T_v = \left[\frac{\partial u}{\partial v}, \frac{\partial v}{\partial v}, \frac{\partial (u^2+v^2)}{\partial v}\right] = (0, 1, 2v)$$

$$\therefore T_u \times T_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 2u \\ 0 & 1 & 2v \end{vmatrix} = (-2u, -2v, 1)$$

$\hat{k}$ component is 1, so $T_u \times T_v$ points "up".

$\therefore$ move around ∂D in counterclockwise fashion.

$\therefore$ $(0,0) \to (2,0) \to (2,2) \to (0,2) \to (0,0)$

For $(0,0)$ to $(2,0)$: $(0,0) + t[(2,0)-(0,0)]$, $0 \le t \le 1$

$$= (0,0) + t(2,0) = (2t, 0)$$

$$\therefore z = (2t)^2 + (0)^2 = 4t^2$$

$$\therefore \gamma(t) = (2t, 0, 4t^2), \quad 0 \le t \le 1$$

For $(2,0)$ to $(2,2)$: $(2,0) + (t-1)[(2,2)-(2,0)]$, $1 \le t \le 2$

$= (2,0) + (t-1)(0,2) = (2, 2t-2)$

$\therefore z = (2)^2 + (2t-2)^2 = 4t^2 - 8t + 8$

$\therefore \gamma(t) = (2, 2t-2, 4t^2-8t+8)$, $1 \le t \le 2$

For $(2,2)$ to $(0,2)$: $(2,2) + (t-2)[(0,2)-(2,2)]$, $2 \le t \le 3$

$= (2,2) + (t-2)(-2,0) = (-2t+6, 2)$

$\therefore z = (-2t+6)^2 + (2)^2 = 4t^2 - 24t + 40$

$\therefore \gamma(t) = (-2t+6, 2, 4t^2-24t+40)$, $2 \le t \le 3$

For $(0,2)$ to $(0,0)$: $(0,2) + (t-3)[(0,0)-(0,2)]$, $3 \le t \le 4$

$= (0,2) + (t-3)(0,-2) = (0, -2t+8)$

$\therefore z = (0)^2 + (-2t+8)^2 = 4t^2 - 32t + 64$

$\therefore \gamma(t) = (0, -2t+8, 4t^2-32t+64)$, $3 \le t \le 4$

Summary for ∂S:

$$\gamma(t) = \begin{cases} (2t, 0, 4t^2), & 0 \le t \le 1 \\ (2, 2t-2, 4t^2-8t+8), & 1 \le t \le 2 \\ (-2t+6, 2, 4t^2-24t+40), & 2 \le t \le 3 \\ (0, -2t+8, 4t^2-32t+64), & 3 \le t \le 4 \end{cases}$$

In Exercises 3 to 6, verify Stokes' theorem for the given surface S and boundary ∂S, and vector fields $\mathbf{F}$.

3.

3. $S = \{(x, y, z) : x^2 + y^2 + z^2 = 1, \quad z \geq 0\}$
(oriented as a graph)
$\partial S = \{(x, y) : x^2 + y^2 = 1\}$
$\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix} = (0, 0, 0).$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = 0$$ i.e., don't need to find a parametrization for S.

For ∂S: let $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$, $0 \leq \theta \leq 2\pi$
$\therefore \vec{c}\,'(\theta) = (-\sin\theta, \cos\theta, 0)$

$$\therefore \vec{F}(\vec{c}(\theta)) \cdot \vec{c}\,'(\theta) = (\cos\theta, \sin\theta, 0) \cdot (-\sin\theta, \cos\theta, 0) = 0.$$

$$\therefore \int_{\partial S} \vec{F} \cdot d\vec{s} = \int_0^{2\pi} 0 \, d\theta = 0.$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s} = \underline{0}$$

4.

4. S as in (1), and $\mathbf{F} = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & z & x \end{vmatrix} = (-1, -1, -1)$$

For S: $T(\phi,\theta) = (\cos\theta\sin\phi, \sin\theta\sin\phi, \cos\phi)$, $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi/2$

$T_\phi = (\cos\theta\cos\phi, \sin\theta\cos\phi, -\sin\phi)$

$T_\theta = (-\sin\theta\sin\phi, \cos\theta\sin\phi, 0)$

$$T_\phi \times T_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta\cos\phi & \sin\theta\cos\phi & -\sin\phi \\ -\sin\theta\sin\phi & \cos\theta\sin\phi & 0 \end{vmatrix} = (\cos\theta\sin^2\phi, \sin\theta\sin^2\phi, \cos\phi\sin\phi)$$

Since $\cos\phi\sin\phi \ge 0$ for $0 \le \phi \le \frac{\pi}{2}$, $T_\phi \times T_\theta$ points "up and out"

$$\therefore \left[(\nabla \times \vec{F})(T(\phi,\theta))\right] \cdot (T_\phi \times T_\theta) =$$

$$(-1,-1,-1) \cdot (\cos\theta\sin^2\phi, \sin\theta\sin^2\phi, \cos\phi\sin\phi) =$$

$$-(\cos\theta + \sin\theta)\sin^2\phi - \cos\phi\sin\phi$$

$$\therefore \iint_S \nabla \times \vec{F} \cdot d\vec{S} = \int_0^{2\pi}\int_0^{\frac{\pi}{2}} \left[-(\cos\theta+\sin\theta)\sin^2\phi - \cos\phi\sin\phi\right] d\phi\, d\theta$$

$$= \int_0^{2\pi} -(\cos\theta+\sin\theta)\,d\theta \int_0^{\frac{\pi}{2}} \sin^2\phi\, d\phi \; - \; 2\pi \int_0^{\pi/2} \cos\phi\sin\phi\, d\phi$$

$$= 0 \cdot \int_0^{\frac{\pi}{2}} \sin^2\phi\, d\phi \; - \; 2\pi \left.\frac{\sin^2\phi}{2}\right|_0^{\frac{\pi}{2}} = \underline{-\pi}$$

For ∂S: using $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$, $0 \le \theta \le 2\pi$
$\vec{c}'(\theta) = (-\sin\theta, \cos\theta, 0)$ ← counterclockwise

$$\vec{F}(\vec{c}(\theta)) \cdot \vec{c}'(\theta) = (\sin\theta, 0, \cos\theta) \cdot (-\sin\theta, \cos\theta, 0)$$

$$= -\sin^2\theta$$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta$
$\therefore -\sin^2\theta = -\frac{1}{2} - \frac{\cos 2\theta}{2}$

$$\therefore \int_{\partial S} \vec{F} \cdot d\vec{s} = \int_0^{2\pi} (-\sin^2\theta)\, d\theta$$

$$= \int_0^{2\pi} -\frac{1}{2} - \frac{\cos 2\theta}{2}\, d\theta = -\pi - \frac{\sin 2\theta}{4}\Big|_0^{2\pi} = -\pi - 0 = \underline{-\pi}$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s} = \underline{-\pi}$$

5. **5.** $S = \{(x, y, z) : z = 1 - x^2 - y^2, \quad z \ge 0\}$
(oriented as a graph)
$\partial S = \{(x, y) : x^2 + y^2 = 1\}$
$\mathbf{F} = z\mathbf{i} + x\mathbf{j} + (2zx + 2xy)\mathbf{k}$

For S: Let $T(r, \theta) = (r\cos\theta, r\sin\theta, 1 - r^2)$, $0 \le r \le 1$, $0 \le \theta \le 2\pi$

$$\therefore T_r = (\cos\theta, \sin\theta, -2r) \quad T_\theta = (-r\sin\theta, r\cos\theta, 0)$$

$$\therefore T_r \times T_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & -2r \\ -r\sin\theta & r\cos\theta & 0 \end{vmatrix} = (2r^2\cos\theta, 2r^2\sin\theta, r)$$

As $r \ge 0$, $T_r \times T_\theta$ "points" up as $\hat{k}$ component is r.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ z & x & 2zx + 2xy \end{vmatrix} = (2x, 1 - 2z - 2y, 1)$$

$$\therefore (\nabla \times \vec{F})(T(r,\theta)) = [2(r\cos\theta), 1 - 2(1-r^2) - 2(r\sin\theta), 1]$$

$$= [2r\cos\theta, -1 + 2r^2 - 2r\sin\theta, 1]$$

$$\therefore (\nabla \times \vec{F}) \cdot (T_r \times T_\theta) =$$

$$[2r\cos\theta, -1 + 2r^2 - 2r\sin\theta, 1] \cdot (2r^2\cos\theta, 2r^2\sin\theta, r)$$

$$= 4r^3\cos^2\theta - 2r^2\sin\theta + 4r^4\sin\theta - 4r^3\sin^2\theta + r$$

$$= 4r^3(\cos^2\theta - \sin^2\theta) - 2r^2\sin\theta + 4r^4\sin\theta + r$$

$\cos^2\theta - \sin^2\theta = \cos 2\theta$

$$\therefore \iint_S \nabla \times F \cdot d\vec{S} =$$

$$\int_0^{2\pi} \int_0^1 4r^3\cos 2\theta - 2r^2\sin\theta + 4r^4\sin\theta + r \, dr\, d\theta$$

$$= \int_0^{2\pi} \cos 2\theta \, r^4 - \frac{2}{3} r^3 \sin\theta + \frac{4}{5} r^5 \sin\theta + \frac{r^2}{2} \Big|_{r=0}^{r=1} d\theta$$

$$= \int_0^{2\pi} \cos 2\theta - \frac{2}{3}\sin\theta + \frac{4}{5}\sin\theta + \frac{1}{2} \, d\theta$$

$$= 0 - \frac{2}{3}(0) + \frac{4}{5}(0) + \frac{1}{2}(2\pi) = \underline{\pi}$$

For ∂S: Let $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$, $0 \le \theta \le 2\pi$

$\vec{c}\,'(\theta) = (-\sin\theta, \cos\theta, 0)$.

Note $\vec{c}(\theta)$ is counterclockwise

$$\therefore \int_{\partial S} \vec{F} \cdot d\vec{s} = \int_{\vec{c}} \vec{F}(\vec{c}(\theta)) \cdot \vec{c}\,'(\theta)\, d\theta$$

$$= \int_0^{2\pi} [0, \cos\theta, 2(0)(\cos\theta) + 2(\cos\theta)(\sin\theta)] \cdot (-\sin\theta, \cos\theta, 0)\, d\theta$$

$\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1$

$$= \int_0^{2\pi} \cos^2\theta\, d\theta = \int_0^{2\pi} \frac{1 + \cos 2\theta}{2}\, d\theta = \frac{1}{2}\theta + \frac{\sin 2\theta}{4}\Big|_0^{2\pi} = \underline{\pi}$$

$$\therefore \iint_S \nabla \times \vec{F} \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s} = \underline{\pi}$$

C.

6. S as in (3), and $\mathbf{F} = z^2\mathbf{i} + x\mathbf{j} + y^2\mathbf{k}$

For S: Let $T(\phi, \theta) = (\cos\theta\sin\phi, \sin\theta\sin\phi, \cos\phi)$

$0 \le \phi \le \pi/2$, $0 \le \theta \le 2\pi$

$\therefore T_\phi = [\cos\theta\cos\phi, \sin\theta\cos\phi, -\sin\phi)$

$T_\theta = [-\sin\theta\sin\phi, \cos\theta\sin\phi, 0)$

$$\therefore T_\phi \times T_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta\cos\phi & \sin\theta\cos\phi & -\sin\phi \\ -\sin\theta\sin\phi & \cos\theta\sin\phi & 0 \end{vmatrix}$$

$$= \left[\cos\theta\sin^2\phi,\ \sin\theta\sin^2\phi,\ \cos\phi\sin\phi\right]$$

for $0 \le \phi \le \frac{\pi}{2}$, $\cos\phi\sin\phi \ge 0$, which is $\hat{k}$ component, so "points" up.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ z^2 & x & y^2 \end{vmatrix} = (2y, 2z, 1)$$

$$\therefore \nabla \times \vec{F}\,(T(\phi,\theta)) = \left[2(\sin\theta\sin\phi),\ 2(\cos\phi),\ 1\right]$$

$$\therefore \nabla \times \vec{F} \cdot (T_\phi \times T_\theta) =$$

$$\left[2\sin\theta\sin\phi,\ 2\cos\phi,\ 1\right] \cdot \left[\cos\theta\sin^2\phi,\ \sin\theta\sin^2\phi,\ \cos\phi\sin\phi\right]$$

$$= 2\cos\theta\sin\theta\sin^3\phi + 2\sin\theta\cos\phi\sin^2\phi + \cos\phi\sin\phi$$

$2\cos\theta\sin\theta = \sin 2\theta$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} =$$

$$\int_0^{2\pi}\int_0^{\pi/2} \sin 2\theta\sin^3\phi + 2\sin\theta\cos\phi\sin^2\phi + \cos\phi\sin\phi \; d\phi\, d\theta$$

$$= \int_0^{2\pi} \sin 2\theta\, d\theta \int_0^{\frac{\pi}{2}} \sin^3\phi\, d\phi + \int_0^{2\pi} 2\sin\theta\, d\theta \int_0^{\frac{\pi}{2}} \cos\phi\sin^2\phi\, d\phi + \int_0^{2\pi}\int_0^{\frac{\pi}{2}} \cos\phi\sin\phi \; d\phi\, d\theta$$

$$= 0 \int_0^{\frac{\pi}{2}} \sin^3\phi \, d\phi + 0 \int_0^{\frac{\pi}{2}} \cos\phi \sin^2\phi \, d\phi + \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \cos\phi \sin\phi \, d\phi \, d\theta$$

$$= 0 + 0 + 2\pi \left[\frac{\sin^2\phi}{2} \right]_{\phi=0}^{\phi=\frac{\pi}{2}} = \underline{\pi}$$

For ∂S: From #3, $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$, $0 \le \theta \le 2\pi$

$\vec{c}'(\theta) = (-\sin\theta, \cos\theta, 0)$

$$\therefore \vec{F}(\vec{c}(\theta)) \cdot \vec{c}'(\theta) = [(\theta)^2, \cos\theta, (\sin\theta)^2] \cdot (-\sin\theta, \cos\theta, 0)$$

$$= \cos^2\theta$$

$$\therefore \int_{\partial S} \vec{F} \cdot d\vec{s} = \int_0^{2\pi} \cos^2\theta \, d\theta = \int_0^{2\pi} \frac{1+\cos 2\theta}{2} \, d\theta = \underline{\pi}$$

(same computation as in #5).

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s} = \pi$$

7.

7. Let C be the closed, piecewise smooth curve formed by traveling in straight lines between the points (0, 0, 0), (2, 0, 4), (3, 2, 6), (1, 2, 2), and back to the origin, in that order. (Thus the surface S lying interior to C is contained in the plane $z = 2x$.) Use Stokes' theorem to evaluate the integral:

$$\int_C (z \cos x)\, dx + (x^2 yz)\, dy + (yz)\, dz$$

The points delineate a parallelogram in the $z = 2x$ plane.

A parametrization for S bounded by the parallelogram is:

$$T(u,v) = \left(u + \frac{v}{2}, v, 2u + v\right), \quad 0 \le u \le 2, \; 0 \le v \le 2$$

The problem suggests evaluating $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$

$$T_u = (1, 0, 2) \qquad T_v = \left(\tfrac{1}{2}, 1, 1\right)$$

$$T_u \times T_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 2 \\ 1/2 & 1 & 1 \end{vmatrix} = (-2, 0, 1)$$, which points "up" as $\hat{k}$ component is $1 > 0$.

"Up" is also the direction, using the right-hand rule, of following the points from $(0,0,0) \to (2,0,4) \to (3,2,6) \to$ etc.

$$\vec{F} = (z\cos x, x^2 yz, yz)$$

$$\therefore \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ z\cos x & x^2 yz & yz \end{vmatrix} = (z - x^2 y, \cos x, 2xyz)$$

$$\therefore \nabla \times \vec{F} \cdot (T_u \times T_v) = -2z + 2x^2 y + 2xyz$$

$$\therefore \left[\nabla \times \vec{F} \cdot (T_u \times T_v)\right] \circ T(u,v) =$$

$$-2(2u+v) + 2\left(u + \frac{v}{2}\right)^2 (v) + 2\left(u + \frac{v}{2}\right)(v)(2u+v)$$

$$= -4u - 2v + 2\left(u^2 v + uv^2 + \frac{v^3}{4}\right) + 4u^2 v + 4uv^2 + v^3$$

$$= -4u - 2v + 6u^2 v + 6uv^2 + \frac{3}{2} v^3$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_0^2 \int_0^2 -4u - 2v + 6u^2v + 6uv^2 + \frac{3}{2}v^3 \, du\,dv$$

$$= \int_0^2 -2u^2 - 2uv + 2u^3v + 3u^2v^2 + \frac{3}{2}uv^3 \Big|_{u=0}^{u=2} \, dv$$

$$= \int_0^2 -8 - 4v + 16v + 12v^2 + 3v^3 \, dv$$

$$= -8v - 2v^2 + 8v^2 + 4v^3 + \frac{3}{4}v^4 \Big|_0^2$$

$$= -16 - 8 + 32 + 32 + 12 = \underline{52}$$

8.

8. Let C be the closed, piecewise smooth curve formed by traveling in straight lines between the points (0, 0, 0), (2, 1, 5), (1, 1, 3), and back to the origin, in that order. Use Stokes' theorem to evaluate the integral:

$$\int_C (xyz)\,dx + (xy)\,dy + (x)\,dz$$

$\vec{F} = (xyz, xy, x)$ $\quad \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xyz & xy & x \end{vmatrix} = (0, xy-1, y-xz)$

Let $\vec{a} = (2,1,5) - (0,0,0)$, $\vec{b} = (1,1,3) - (0,0,0)$

$\therefore \vec{a} \times \vec{b}$ is a normal to the plane of the 3 points, and points in the direction used in following the path of $(0,0,0) \to (2,1,5) \to (1,1,3) \to (0,0,0)$.

$$\therefore \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 5 \\ 1 & 1 & 3 \end{vmatrix} = (-2, -1, 1)$$

$\therefore$ equation of plane: $-2x - y + z = D$,

$\therefore -2(0) - (0) + (0) = 0 = D.$

$\therefore$ Plane : $z = 2x + y.$

$\therefore$ Parametrization of S: $T(u,v) = (u, v, 2u + v)$

and the limits, from the extremes of the x, y coordinates of the points: $0 \le u \le 2$, $0 \le v \le 1$.

$\therefore T_u = (1, 0, 2) \quad T_v = (0, 1, 1)$

$$T_u \times T_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 2 \\ 0 & 1 & 1 \end{vmatrix} = (-2, -1, 1)$$

Note $T_u \times T_v$ has positive $\hat{k}$ component, so points in direction ("up") as suggested by direction of path.

$$\therefore (\nabla \times \vec{F}) \circ T(u,v) = [0, (u)(v) - 1, (v) - (u)(2u+v)]$$

$$= [0, uv - 1, v - 2u^2 - 2uv]$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \iint_S (\nabla \times \vec{F}) \circ T(u,v) \cdot (T_u \times T_v)$$

$$= \iint_D [0,\ uv-1,\ v-2u^2-2uv] \cdot (-2,-1,1)\ du\,dv$$

$$= \int_0^2 \int_0^1 1 - uv + v - 2u^2 - 2uv\ dv\,du$$

$$= 1 - 3uv + v - 2u^2$$

$$= \int_0^2 v - \frac{3}{2}uv^2 + \frac{v^2}{2} - 2u^2 v \Big|_0^1\ du$$

$$= \int_0^2 1 - \frac{3}{2}u + \frac{1}{2} - 2u^2\ du = \int_0^2 \frac{3}{2} - \frac{3}{2}u - 2u^2\ du$$

$$= \frac{3}{2}u - \frac{3}{4}u^2 - \frac{2}{3}u^3 \Big|_0^2 = 3 - 3 - \frac{16}{3} = -\frac{16}{3}$$

9.

9. Redo Exercise 9 of Section 7.6 using Stokes' theorem.

9. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where S is the surface $x^2 + y^2 + 3z^2 = 1$, $z \le 0$ and $\mathbf{F}$ is the vector field $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + zx^3y^2\mathbf{k}$. (Let $\mathbf{n}$, the unit normal, be upward pointing.)

From #9, Section 7.6, $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = -2\pi$

From $S = x^2 + y^2 + 3z^2 = 1$, letting $z = 0$, $\partial S: x^2 + y^2 = 1$

$\therefore$ Let $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$ $0 \le \theta \le 2\pi$

$\therefore$ $\vec{c}(\theta)$ for ∂S is counterclockwise, consistent with $\vec{n}$ pointing "up".

$\therefore$ $\vec{c}\,'(\theta) = (-\sin\theta, \cos\theta, 0)$

$\therefore$ $\vec{F}(\vec{c}(\theta)) = [\sin\theta, -\cos\theta, 0]$

$\therefore$ $\vec{F}(\vec{c}(\theta)) \cdot \vec{c}\,'(\theta) = -\sin^2\theta - \cos^2\theta = -1$

$\therefore$ $$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s} = \int_0^{2\pi} (-1)\, d\theta = \underline{-2\pi}$$

10. **10.** Redo Exercise 10 of Section 7.6 using Stokes' theorem.

10. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where $\mathbf{F} = (x^2 + y - 4)\mathbf{i} + 3xy\mathbf{j} + (2xz + z^2)\mathbf{k}$ and S is the surface $x^2 + y^2 + z^2 = 16$, $z \ge 0$. (Let $\mathbf{n}$, the unit normal, be upward pointing.)

From #10, Section 7.6, $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = -16\pi$

Letting $z = 0$, $\partial S: x^2 + y^2 = 16$

$\therefore$ Let $\vec{c}(\theta) = (4\cos\theta, 4\sin\theta, 0)$, $0 \le \theta \le 2\pi$

$\therefore$ $\vec{c}(\theta)$ for ∂S is counterclockwise, consistent with $\vec{n}$ for S pointing "up".

$$\therefore \vec{c}\,'(\theta) = (-4\sin\theta, 4\cos\theta, 0)$$

$$\vec{F}(\vec{c}(\theta)) = \langle 16\cos^2\theta + 4\sin\theta - 4,\ 48\cos\theta\sin\theta,\ 0 \rangle$$

$$\therefore \vec{F}(\vec{c}(\theta)) \cdot \vec{c}\,'(\theta) = -64\cos^2\theta\sin\theta - 16\sin^2\theta + 16\sin\theta + 192\cos^2\theta\sin\theta$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s} = \int_0^{2\pi} \vec{F}(\vec{c}(\theta)) \cdot \vec{c}\,'(\theta)\, d\theta$$

$$= \int_0^{2\pi} -64\cos^2\theta\sin\theta - 16\sin^2\theta + 16\sin\theta + 192\cos^2\theta\sin\theta\, d\theta$$

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$

$$= \frac{64}{3}\cos^3\theta\Big|_0^{2\pi} - 8\Big|_0^{2\pi} - \sin 2\theta\Big|_0^{2\pi} - 16\cos\theta\Big|_0^{2\pi} - \frac{192}{3}\cos^3\theta\Big|_0^{2\pi}$$

$$= 0 \quad - 16\pi \quad - 0 \quad - 0 \quad - 0$$

$$= \underline{-16\pi}$$

11. **11.** Verify Stokes' theorem for the upper hemisphere $z = \sqrt{1 - x^2 - y^2}$, $z \geq 0$, and the radial vector field $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

(a) $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix} = (0,0,0)$$

$\therefore$ if $T(u,v)$ is a parametrization for S,

$$(\nabla \times \vec{F}) \cdot (T_u \times T_v) = 0 \Rightarrow \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \underline{0}$$

(b) $\int_{\partial S} \vec{F} \cdot d\vec{s}$ Let $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$, $0 \le \theta \le 2\pi$

$\therefore \vec{c}'(\theta) = (-\sin\theta, \cos\theta, 0)$

$\vec{F}(\vec{c}(\theta)) = (\cos\theta, \sin\theta, 0)$.

$\therefore \vec{F} \cdot \vec{c}'(\theta) = (-\cos\theta \sin\theta) + (\cos\theta \sin\theta) + (0) = 0$

$\therefore \int_{\partial S} \vec{F} \cdot d\vec{s} = \underline{0}$

(a) & (b) $\Rightarrow \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s} = 0$

12. **12.** Let S be a surface with boundary ∂S, and suppose $\mathbf{E}$ is an electric field that is perpendicular to ∂S. Show that the induced magnetic flux across S is constant in time. (HINT: Use Faraday's law.)

The magnetic flux across S is defined as

$\iint_S \vec{H} \cdot d\vec{S}$. To show this is constant in time, need to show $\frac{\partial}{\partial t} \iint_S \vec{H} \cdot d\vec{S} = 0$

Use differentiation under integral,

$$\frac{\partial}{\partial t} \iint_S \vec{H} \cdot d\vec{S} = \frac{\partial}{\partial t} \iint_D \vec{H}(T(u,v)) \cdot (T_u \times T_v)\, du\, dv$$

where $T(u,v)$ is a parametrization of S, $T: D \to S$

$$= \iint_D \frac{\partial}{\partial t} \left[\vec{H} \cdot (T_u \times T_v) \right] du\, dv$$

and $\frac{\partial}{\partial t}(T_u \times T_v) = 0$ since $T_u \times T_v$ is time independent

$$= \iint_D \frac{\partial \vec{H}}{\partial t} \cdot (T_u \times T_v)\, du\, dv$$

$$= \iint_D (-\nabla \times \vec{E}) \cdot (T_u \times T_v)\, du\, dv \qquad \text{Faraday's Law}$$

$$= \int_{\partial S} (-\vec{E}) \cdot d\vec{S} \qquad \text{Stokes' Theorem}$$

$= -\int_{\vec{C}} \vec{E}(\vec{c}(\theta)) \cdot \vec{c}'(\theta)\, d\theta$ $\quad \vec{c}(\theta)$ a parametrization of ∂S

$= 0$ since $\vec{E} \perp \vec{c}'(\theta)$ as was given.

$\therefore \frac{\partial}{\partial t} \iint_S \vec{H} \cdot d\vec{S} = 0 \Rightarrow$ magnetic flux is constant.

13.

13. Let S be the capped cylindrical surface shown in Figure 8.2.13. S is the union of two surfaces, S_1 and S_2, where S_1 is the set of (x, y, z) with $x^2 + y^2 = 1$, $0 \le z \le 1$, and S_2 is the set of (x, y, z) with $x^2 + y^2 + (z-1)^2 = 1$, $z \ge 1$. Set $\mathbf{F}(x, y, z) = (zx + z^2 y + x)\mathbf{i} + (z^3 yx + y)\mathbf{j} + z^4 x^2 \mathbf{k}$. Compute $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$. (HINT: Stokes' theorem holds for this surface.)

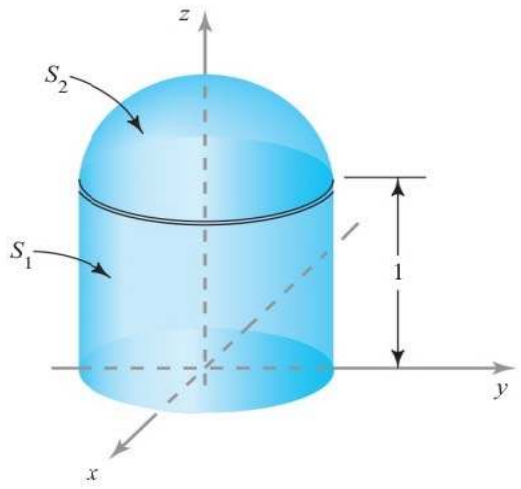


figure **8.2.13** The capped cylinder is the union of S_1 and S_2.

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s}$$

where ∂S is the circle $x^2+y^2=1$ in the $z=0$ plane.

Parametrize ∂S as $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$, $0 \le \theta \le 2\pi$

$\therefore \vec{c}'(\theta) = (-\sin\theta, \cos\theta, 0)$

$\vec{F}(\vec{c}(\theta)) = [\cos\theta, \sin\theta, 0]$, as z-component of $\vec{c}(\theta) = 0$.

$\therefore \vec{F}(\vec{c}(\theta)) \cdot \vec{c}'(\theta) = 0 \quad \therefore \int_{\partial S} \vec{F} \cdot d\vec{s} = 0 = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$

14. **14.** Let $\mathbf{c}$ consist of straight lines joining (1, 0, 0), (0, 1, 0), and (0, 0, 1) and let S be the triangle with these vertices. Verify Stokes' theorem directly with $\mathbf{F} = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ yz & xz & xy \end{vmatrix} = (x-x,\ y-y,\ z-z) = (0,0,0).$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = 0$$

Now parametrize the triangle

$(1,0,0)$ to $(0,1,0)$: $(1,0,0) + t[(0,1,0)-(1,0,0)]$

$= (1,0,0) + t(-1,1,0) = (1-t,\ 1,\ 0)$, $0 \le t \le 1$

$(0,1,0)$ to $(0,0,1)$: $(0,1,0) + (t-1)[(0,0,1)-(0,1,0)]$

$= (0,1,0) + (t-1)(0,-1,1) = (0,\ 2-t,\ t-1)$, $1 \le t \le 2$

$(0,0,1)$ to $(1,0,0)$: $(0,0,1) + (t-2)[(1,0,0)-(0,0,1)]$

$= (0,0,1) + (t-2)(1,0,-1) = (t-2,\ 0,\ 3-t)$, $2 \le t \le 3$

$$\therefore \vec{F}(\vec{c}(t)) = \begin{cases} (0,0,1-t) & 0 \le t \le 1 \\ (-t^2+3t-2,\ 0,\ 0) & 1 \le t \le 2 \\ (0,\ -t^2+5t-6,\ 0) & 2 \le t \le 3 \end{cases}$$

$$\vec{c}'(t) = \begin{cases} (-1,0,0) & 0 \le t \le 1 \\ (0,-1,1) & 1 \le t \le 2 \\ (1,0,-1) & 2 \le t \le 3 \end{cases}$$

$$\therefore \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t) = \begin{cases} 0+0+0=0 & 0 \le t \le 1 \\ 0+0+0=0 & 1 \le t \le 2 \\ 0+0+0=0 & 2 \le t \le 3 \end{cases}$$

$$\therefore \int_0^1 \vec{F} \cdot \vec{c}'\, dt + \int_1^2 \vec{F} \cdot \vec{c}'\, dt + \int_2^3 \vec{F} \cdot \vec{c}'\, dt = 0+0+0=0$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = 0 = \int_{\partial S} \vec{F} \cdot d\vec{s}$$

15.

15. Evaluate the integral $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where S is the portion of the surface of a sphere defined by

$x^2 + y^2 + z^2 = 1$ and $x + y + z \ge 1$, and where $\mathbf{F} = \mathbf{r} \times (\mathbf{i} + \mathbf{j} + \mathbf{k})$, $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

Use symmetry to make problem analogous to an orientation in which it is easier to parametrize S and ∂S. $\vec{F}$ is radially symmetric, so choose an orientation making $\hat{i}+\hat{j}+\hat{k}$ the axis.

$\therefore$ Make the z-axis analogous to $\hat{i}+\hat{j}+\hat{k}$.

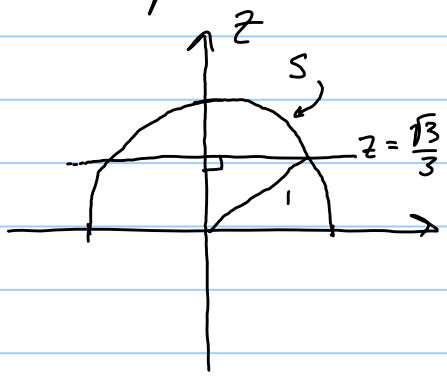
$\|\hat{i}+\hat{j}+\hat{k}\| = \sqrt{3}$ $\therefore \sqrt{3}\hat{k} \equiv \hat{i}+\hat{j}+\hat{k}$.

$\therefore \vec{F} = \vec{r} \times (\sqrt{3}\hat{k}) = (x\hat{i}+y\hat{j}+z\hat{k}) \times \sqrt{3}\hat{k}$

$= -\sqrt{3}x\hat{j} + \sqrt{3}y\hat{i} = (\sqrt{3}y, -\sqrt{3}x, 0)$

Since the plane $x+y+z=1$ has $(1,1,1)$ as a normal, choose a plane $z=c$, which is normal to $\sqrt{3}\hat{k}$, just as $x+y+z=1$ is normal to $\hat{i}+\hat{j}+\hat{k}$, and c = distance to origin. The distance of $(0,0,0)$ to the plane $x+y+z-1=0$ is

$$\frac{|-1|}{\sqrt{1+1+1}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \quad \text{(see text, p. 43)}.$$



$\therefore$ Choose the plane $z = \frac{\sqrt{3}}{3}$.

$\therefore$ radius of ∂S is $\sqrt{1-\left(\frac{\sqrt{3}}{3}\right)^2}$

$= \sqrt{\frac{2}{3}} = \frac{\sqrt{6}}{3}$.

$\therefore$ Parametrization of ∂S: $\vec{c}(\theta) = \left[\frac{\sqrt{6}}{3}\cos\theta, \frac{\sqrt{6}}{3}\sin\theta, 0\right]$, $0 \le \theta \le 2\pi$

$\therefore \vec{c}'(\theta) = \left[-\frac{\sqrt{6}}{3}\sin\theta, \frac{\sqrt{6}}{3}\cos\theta, 0\right]$.

$\vec{F}(\vec{c}(\theta)) = \left[\sqrt{3}\left(\frac{\sqrt{6}}{3}\sin\theta\right), -\sqrt{3}\left(\frac{\sqrt{6}}{3}\cos\theta\right), 0\right]$

$$= \left[\sqrt{2}\sin\theta, -\sqrt{2}\cos\theta, 0\right]$$

$$\therefore \vec{F}\cdot\vec{c}\,'(\theta) = -\frac{\sqrt{12}}{3}\sin^2\theta - \frac{\sqrt{12}}{3}\cos^2\theta = -\frac{2\sqrt{3}}{3}$$

$$\therefore \iint_S (\nabla\times\vec{F})\cdot d\vec{S} = \int_{\vec{C}} \vec{F}\cdot d\vec{s} = \int_0^{2\pi}\left(-\frac{2\sqrt{3}}{3}\right)d\theta = \underline{\underline{-\frac{4\sqrt{3}}{3}\pi}}$$

Alternate solution

$$\int_{\vec{C}} \vec{F}\cdot d\vec{s} = \int_{\vec{C}} \vec{F}\cdot\frac{\vec{c}\,'}{\|\vec{c}\,'\|}\|\vec{c}\,'\| = \int_{\vec{C}} \vec{F}\cdot\vec{n}'\,ds = \vec{F}\cdot\vec{n}'\int_{\vec{C}} ds$$

where $\vec{n}'$ is the unit tangent vector to the circle ∂S in the counterclockwise direction (i.e., converting line integral → path integral).

But $\int_{\vec{C}} ds$ = circumference of $\partial S = 2\pi r$, $r = \partial S$ radius

From above, radius of $\partial S = \sqrt{\frac{2}{3}}$.

$$\therefore \iint_S (\nabla\times\vec{F})\cdot d\vec{S} = \vec{F}\cdot\vec{n}'\left(2\pi\sqrt{\frac{2}{3}}\right) = \|\vec{F}\|\,\|\vec{n}'\|\cos\theta\left(2\pi\sqrt{\frac{2}{3}}\right)$$

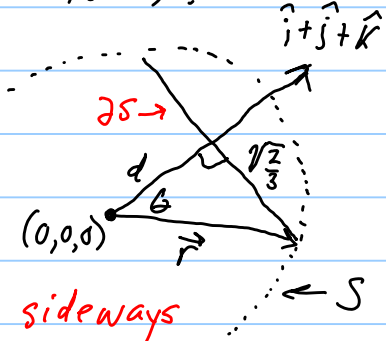
$$= \|\vec{F}\|\cos\theta\left(2\pi\sqrt{\frac{2}{3}}\right) \text{ as } \|\vec{n}'\| = 1$$

But $\vec{F} = \vec{r}\times(\hat{i}+\hat{j}+\hat{k})$, and from the geometry of the problem, $\vec{r}\times(\hat{i}+\hat{j}+\hat{k})$ points in opposite direction of $\vec{n}'$.

$\therefore \cos\theta =$ angle between $\vec{F}$ and $\vec{n}' = -1 \quad (\theta = 180°)$.

$\|\vec{F}\| = \|\vec{r}\| \, \|\hat{i}+\hat{j}+\hat{k}\| \sin\theta = (1)(\sqrt{3})\left(\frac{\sqrt{\frac{2}{3}}}{1}\right) = \sqrt{2}$

as $\|\vec{r}\| = 1$ on sphere S (d = distance from $(0,0,0)$ to plane containing ∂S).



sideways view of ∂S

$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = (\sqrt{2})(-1)\left(2\pi\sqrt{\frac{2}{3}}\right) = -\frac{4\pi}{\sqrt{3}} = -\frac{4\sqrt{3}}{3}\pi$

16.

16. Show that the calculation in Exercise 15 can be simplified by observing that $\int_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \int_{\partial \Sigma} \mathbf{F} \cdot d\mathbf{r}$ for any other surface Σ. By picking Σ appropriately, $\iint_\Sigma (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$ may be easy to compute. Show that this is the case if Σ is taken to be the portion of the plane $x + y + z = 1$ inside the circle ∂S.

In Stokes' Theorem, the shape of S does not matter, as long as the parametrization $\phi : D \subset \mathbb{R}^2 \to S$ is one-to-one. $\therefore$ Different surfaces with the same boundary will have the same calculation since $\partial S = \partial \Sigma$

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{r} = \int_{\partial \Sigma} \vec{F} \cdot d\vec{r} = \iint_\Sigma (\nabla \times \vec{F}) \cdot d\vec{\Sigma}$$

where $d\vec{r}$ is used here instead of $d\vec{s}$ ($\vec{r}$ and $\vec{s}$ point

to the path of the boundary of a surface, so $d\vec{r} \approx \Delta\vec{r}$, $d\vec{S} \approx \Delta\vec{S}$).

$\therefore$ in #15, the surface integral $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$ could be computed using the plane $x+y+z=1$ instead of the surface of the sphere above the plane.

Doing this for #15 above, let S be the disc in the plane $x+y+z=1$.

Note that $\vec{F} = \vec{r} \times (\hat{i}+\hat{j}+\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ 1 & 1 & 1 \end{vmatrix} = (y-z, z-x, x-y)$

$\therefore \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y-z & z-x & x-y \end{vmatrix} = (-2, -2, -2)$

$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n}\, dS$

where $\vec{n}$ = unit normal to S, the plane $= \frac{1}{\sqrt{3}}(1,1,1)$ for the plane $x+y+z=1$.

$\therefore (\nabla \times \vec{F}) \cdot \vec{n} = (-2,-2,-2) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) = -\frac{6}{\sqrt{3}}$

$\therefore \iint_S (\nabla \times \vec{F}) \cdot \vec{n}\, dS = \iint_S \left(-\frac{6}{\sqrt{3}}\right) dS = -\frac{6}{\sqrt{3}} \iint_S dS$

But $\iint_S dS$ = area of the circle that is the boundary for S. $\therefore \iint_S dS = \pi r^2$, where r = radius of the circle ∂S.

Computing the distance of $(0,0,0)$ to $x+y+z=1$, the radius was computed in #15 to be $\sqrt{\frac{2}{3}}$.

$\therefore$ Area bounded by $\partial S = \frac{2}{3}\pi$

$\therefore \iint_S (\nabla \times \vec{F}) \cdot \vec{n}\, dS = -\frac{6}{\sqrt{3}} \left(\frac{2}{3}\pi\right) = -\frac{4\pi}{\sqrt{3}}$

as in #15.

17. **17.** Calculate the surface integral $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where S is the hemisphere $x^2+y^2+z^2 = 1$, $x \geq 0$, and $\mathbf{F} = x^3\mathbf{i} - y^3\mathbf{j}$.

The boundary ∂S can be parametrized as:

$\vec{C}(\theta) = (0, \cos\theta, \sin\theta)$, $0 \leq \theta \leq 2\pi$ and is

counterclockwise viewed from yz-plane, using "up" = $\hat{i}$.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^3 & -y^3 & 0 \end{vmatrix} = (0,0,0)$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \underline{0}$$ (don't even need $\vec{c}(\theta)$ to compute line integral).

18. **18.** Find $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where S is the ellipsoid $x^2 + y^2 + 2z^2 = 10$ and $\mathbf{F}$ is the vector field $\mathbf{F} = (\sin xy)\mathbf{i} + e^x\mathbf{j} - yz\mathbf{k}$.

Break the ellipsoid into top & bottom. $\therefore$ ∂S will be the circle $x^2+y^2=10$ for top & bottom, but the orientation will be opposite for each.

$\therefore$ Top ($z \geq 0$): $$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s}$$

Bottom ($z \leq 0$): $$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = -\int_{\partial S} \vec{F} \cdot d\vec{s}$$

$\therefore$ Total: $$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot ds - \int_{\partial S} \vec{F} \cdot d\vec{s} = \underline{0}$$

19. **19.** Let $\mathbf{F} = y\mathbf{i} - x\mathbf{j} + zx^3y^2\mathbf{k}$. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n}\, dA$, where S is the surface defined by $x^2 + y^2 + z^2 = 1$, $z \le 0$.

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n}\, dA = \int_{\partial S} \vec{F} \cdot d\vec{s}$$

where, if $T(u,v)$ is a parametrization of S, then $\vec{n} = \frac{T_u \times T_v}{\|T_u \times T_v\|}$, so $dA = du\,dv$, and $\vec{n}$ could be up or down.

Rather than going through the tedious computations of using spherical coordinates and computing over the hemisphere, just compute the line integral or do the surface integral over the disc at $z = 0$.

$\therefore$ Let $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$, $\vec{c}\,'(\theta) = (-\sin\theta, \cos\theta, 0)$.

$0 \le \theta \le 2\pi$ for $\vec{n}$ "up", $0 \le \theta \le -2\pi$ for $\vec{n}$ "down".

$\therefore \vec{F}(\vec{c}(\theta)) = (\sin\theta, -\cos\theta, 0)$, so $\vec{F} \cdot \vec{c}\,'(\theta) = -\sin^2\theta - \cos^2\theta + 0 = -1$

$$\therefore \int_{\partial S} \vec{F} \cdot d\vec{s} = \int_0^{2\pi} (-1)\, d\theta = -2\pi, \text{ or } \int_0^{-2\pi} (-1)\, d\theta = 2\pi$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot \vec{n}\, dA = \underline{\pm 2\pi} \quad \text{depending on choice of orientation.}$$

20.

20. A hot-air balloon has the truncated spherical shape shown in Figure 8.2.14. The hot gases escape through the porous envelope with a velocity vector field

$$\mathbf{V}(x, y, z) = \nabla \times \mathbf{\Phi}(x, y, z)$$
$$\text{where} \qquad \mathbf{\Phi}(x, y, z) = -y\mathbf{i} + x\mathbf{j}.$$

If $R = 5$, compute the volume flow rate of the gases through the surface.

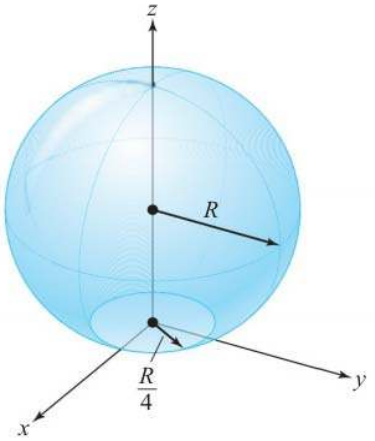


figure **8.2.14** A hot-air balloon.

The flow rate through a patch $d\vec{S}$ would be $\vec{V} \cdot d\vec{S}$

$$\therefore \iint_S \vec{V} \cdot d\vec{S} = \int_S (\nabla \times \vec{\phi}) \cdot d\vec{S} = \int_{\partial S} \vec{\phi} \cdot d\vec{s}$$

A parametrization for ∂S: $\vec{c}(\theta) = \left(\frac{R}{4}\cos\theta, \frac{R}{4}\sin\theta, 0\right)$, $0 \le \theta \le 2\pi$

$$\therefore \vec{c}\,'(\theta) = \left(-\frac{R}{4}\sin\theta, \frac{R}{4}\cos\theta, 0\right).$$

$$\vec{\phi}(\vec{c}(\theta)) = \left(-\frac{R}{4}\sin\theta, \frac{R}{4}\cos\theta, 0\right).$$

$$\therefore \vec{\phi}(\vec{c}(\theta)) \cdot \vec{c}\,'(\theta) = \frac{R^2}{16}\sin^2\theta + \frac{R^2}{16}\cos^2\theta = \frac{R^2}{16}$$

$$\therefore \int_{\partial S} \vec{\phi} \cdot d\vec{s} = \int_0^{2\pi} \vec{\phi} \cdot \vec{c}\,' \, d\theta = \int_0^{2\pi} \frac{R^2}{16}\, d\theta = \underline{\frac{\pi}{8} R^2}$$

21. **21.** Prove that Faraday's law implies $\nabla \times \mathbf{E} = -\partial \mathbf{H}/\partial t$.

Let S be a surface with boundary C.

Faraday's Law: Voltage around C = negative rate of change of magnetic flux through S.

where $\int_C \vec{E} \cdot d\vec{s}$ = voltage around C

$\iint_S \vec{H} \cdot d\vec{S}$ = magnetic flux across S

$$\therefore \int_C \vec{E} \cdot d\vec{s} = -\frac{\partial}{\partial t} \iint_S \vec{H} \cdot d\vec{S}$$

$\vec{E}, \vec{H}$ assumed to be C^1 functions

From Stokes' Theorem $\int_C \vec{E} \cdot d\vec{s} = \iint_S (\nabla \times \vec{E}) \cdot d\vec{S}$

$$\therefore \iint_S (\nabla \times \vec{E}) \cdot d\vec{S} = -\frac{\partial}{\partial t} \iint_S \vec{H} \cdot d\vec{S} \qquad [1]$$

If $T(u,v)$ is a parametrization of S, $T: D \to S$, then

$$-\frac{\partial}{\partial t} \iint_S \vec{H} \cdot d\vec{S} = -\frac{\partial}{\partial t} \left[\lim_{n \to \infty} \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} \vec{H}(T(u_i^*, v_j^*)) \cdot (T_{u_i} \times T_{v_j}) \Delta u \Delta v \right]$$

where D is partitioned into n^2 rectangles R_{ij}, $(u_i^*, v_j^*) \in R_{ij}$, and $T_{u_i} \cdot \Delta u$ and $T_{v_j} \cdot \Delta v$ are tangent vectors to S at $T(u_i, v_j)$.

$$= \lim_{n\to\infty} \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} -\frac{\partial}{\partial t}\left[\vec{H}(T(u_i^*, v_j^*)) \cdot (T_{u_i} \times T_{v_j})\, \Delta u \Delta v\right] \quad [2]$$

But $$\frac{\partial}{\partial t}\left[\vec{H} \cdot (T_u \times T_v)\right] = \frac{\partial \vec{H}}{\partial t} \cdot (T_u \times T_v) + \vec{H} \cdot \frac{\partial}{\partial t}(T_u \times T_v)$$

and $\frac{\partial}{\partial t}(T_u \times T_v) = 0$ as $T(u,v)$ is time independent.

$\therefore$ [2] becomes

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} \sum_{j=0}^{n-1} -\frac{\partial}{\partial t} \vec{H}(T(u_i^*, v_j^*)) \cdot (T_{u_i} \times T_{v_j})\, \Delta u \Delta v$$

$$= \iint_S \left(-\frac{\partial}{\partial t} \vec{H}\right) \cdot d\vec{S}$$

$\therefore$ [1] becomes

$$\iint_S (\nabla \times \vec{E}) \cdot d\vec{S} = \iint_S \left(-\frac{\partial}{\partial t} \vec{H}\right) \cdot d\vec{S},$$

or $$\iint_S \left(\nabla \times \vec{E} + \frac{\partial}{\partial t} \vec{H}\right) \cdot d\vec{S} = 0 \qquad [3]$$

This implies $\nabla \times \vec{E} + \frac{\partial}{\partial t}\vec{H} = \vec{0}$, for if it were nonzero at some point, by continuity, there would be some small patch of S for which $\nabla \times \vec{E} + \frac{\partial}{\partial t}\vec{H}$ would be nonzero for all points in the patch, so that [3], which is a summation, would be nonzero over the patch. This would contradict Faradays law, which leads to [3].

22.

22. Let S be a surface and let $\mathbf{F}$ be perpendicular to the tangent to the boundary of S. Show that

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0.$$

What does this mean physically if $\mathbf{F}$ is an electric field?

By Stokes' Theorem $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s}$

$= \int_{\partial S} \vec{F}(\vec{c}(\theta)) \cdot \vec{c}'(\theta)\, d\theta$, $\vec{c}(\theta)$ a parametrization of ∂S.

But $\vec{F} \cdot \vec{c}' = 0$ since $\vec{F} \perp$ tangent to ∂S, and $\vec{c}'(\theta)$ is tangent to ∂S.

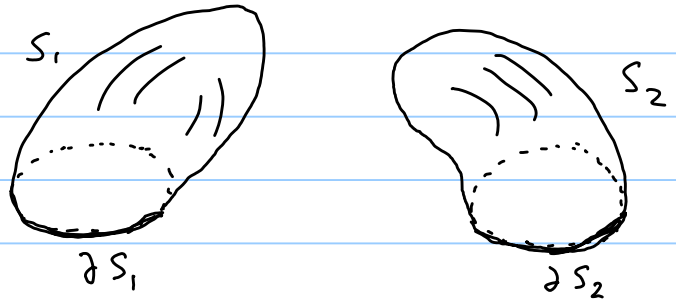
$\therefore \int_{\partial S} \vec{F} \cdot d\vec{s} = 0 = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$

If $\vec{F}$ is an electric field, then by Faraday's Law, the voltage around ∂S is $0 \Rightarrow$ rate of change of magnetic flux through S is 0, so magnetic flux through S is constant over time.

23. **23.** Consider two surfaces S_1, S_2 with the same boundary ∂S. Describe with sketches how S_1 and S_2 must be oriented to ensure that

$$\iint_{S_1} (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \iint_{S_2} (\nabla \times \mathbf{F}) \cdot d\mathbf{S}.$$

S_1 and S_2 must be s.t. orientation of ∂S_1 = orientation of ∂S_2



24. **24.** For a surface S and a fixed vector $\mathbf{v}$, prove that

$$2 \iint_S \mathbf{v} \cdot \mathbf{n}\, dS = \int_{\partial S} (\mathbf{v} \times \mathbf{r}) \cdot d\mathbf{s},$$

where $\mathbf{r}(x, y, z) = (x, y, z)$.

By Stokes' Theorem, $\int_{\partial S} (\vec{v} \times \vec{r}) \cdot ds = \iint_S [\nabla \times (\vec{v} \times \vec{r})] \cdot d\vec{S}$

$= \iint_S [\nabla \times (\vec{v} \times \vec{r})] \cdot (T_u \times T_v)\, du\, dv = \iint_S [\nabla \times (\vec{v} \times \vec{r}) \cdot \frac{(T_u \times T_v)}{\|T_u \times T_v\|} \cdot \|T_u \times T_v\|\, du\, dv$

$= \iint_S [\nabla \times (\vec{v} \times \vec{r})] \cdot \vec{n}\, dS$, a surface integral of a scalar valued function,

where $\vec{n} = \frac{T_u \times T_v}{\|T_u \times T_v\|}$, a unit normal to S,

and $T(u,v)$ is a parametrization of S;

Let $\vec{v} = (a, b, c)$, a fixed vector

$$\therefore \vec{v} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a & b & c \\ x & y & z \end{vmatrix} = (bz - cy,\ cx - az,\ ay - bx)$$

$$\therefore \nabla \times (\vec{v} \times \vec{r}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ bz - cy & cx - az & ay - bx \end{vmatrix} = (a - (-a),\ b - (-b),\ c - (-c))$$

$$= (2a, 2b, 2c) = 2\vec{v}$$

$$\therefore \int_{\partial S} (\vec{v} \times \vec{r}) \cdot d\vec{s} = \iint_S [\nabla \times (\vec{v} \times \vec{r})] \cdot \vec{n}\, dS$$

$$= \iint_S (2\vec{v}) \cdot \vec{n}\, dS = 2 \iint_S \vec{v} \cdot \vec{n}\, dS$$

25. **25.** Argue informally that if S is a closed surface, then

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0$$

(see Exercise 23). (A *closed surface* is one that forms the boundary of a region in space; thus, for example, a sphere is a closed surface.)

Divide S into 2 surfaces with the same boundary, ∂S, such as "top" and "bottom", connected by ∂S, perhaps created by a plane intersecting the "middle" of S. If S_1 and S_2 are the surfaces s.t. $S_1 \cup S_2 = S$, then $\partial S_1 = \partial S_2$, but the orientation is opposite.

$$\therefore \int_{\partial S_1} \vec{F} \cdot d\vec{S}_1 = -\int_{-\partial S_1} \vec{F} \cdot d\vec{S}_1 = -\int_{\partial S_2} \vec{F} \cdot d\vec{S}_1$$

$$\therefore \text{By Stokes' Theorem, } \iint_{S_1} (\nabla \times \vec{F}) \cdot d\vec{S}_1 = -\iint_{S_2} (\nabla \times \vec{F}) \cdot d\vec{S}_2$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \iint_{S_1} (\nabla \times \vec{F}) \cdot d\vec{S}_1 + \iint_{S_2} (\nabla \times \vec{F}) \cdot d\vec{S}_2 = 0$$

26. **26.** If C is a closed curve that is the boundary of a surface S, and f and g are C^2 functions, show that

(a) $\int_C f\nabla g \cdot d\mathbf{s} = \iint_S (\nabla f \times \nabla g) \cdot d\mathbf{S}$

(b) $\int_C (f\nabla g + g\nabla f) \cdot d\mathbf{s} = 0$

Note for a function f, $\text{curl}(\text{grad } f) = \vec{0}$,

or $\nabla \times (\nabla f) = \vec{0}$ (text p. 252)

Also, $\text{curl}(f\vec{F}) = f[\text{curl}(\vec{F})] + \nabla f \times \vec{F}$ (text p. 255, #10)

$\therefore$ Let $\vec{G} = \nabla g$ (gradient of g). $\therefore \text{curl}(\vec{G}) = \vec{0}$

(a) $\therefore \int_C f\nabla g \cdot d\vec{s} = \int_C (f\vec{G}) \cdot d\vec{s} = \iint_S [\nabla \times (f\vec{G})] \cdot d\vec{S}$

$= \iint_S \left(f[\text{curl}(\vec{G})] + \nabla f \times \vec{G} \right) \cdot d\vec{S}$

($\text{curl}(\vec{G}) = \vec{0}$; $\vec{G} = \nabla g$)

$= \iint_S (\nabla f \times \nabla g) \cdot d\vec{S}$

(b) Note $\nabla(fg) = f\nabla g + g\nabla f$ (text p. 255, #3)

$\therefore \int_C (f\nabla g + g\nabla f) \cdot d\vec{s} = \int_C \nabla(fg) \cdot d\vec{s}$

$$= \iint_S \nabla \times (\nabla(fg)) \cdot d\vec{S} = \iint_S \vec{0} \cdot d\vec{S} = 0$$

curl(grad f) = $\vec{0}$

27.

27. (a) If C is a closed curve that is the boundary of a surface S, and $\mathbf{v}$ is a constant vector, show that

$$\int_C \mathbf{v} \cdot d\mathbf{s} = 0.$$

(b) Show that this is true even if C is not the boundary of a surface S.

(a) By Stokes' Theorem, $\int_C \vec{v} \cdot d\vec{s} = \iint_S (\nabla \times \vec{v}) \cdot d\vec{S}$

But since $\vec{v}$ is a constant vector, $\nabla \times \vec{v} = \vec{0}$

$$\therefore \iint_S (\nabla \times \vec{v}) \cdot d\vec{S} = \iint_S \vec{0} \cdot d\vec{S} = 0 = \int_C \vec{v} \cdot d\vec{s}$$

(b) Let $\vec{c}(\theta)$ be a parametrization of C, $a \le \theta \le b$ and $\vec{c}(a) = \vec{c}(b)$, and $\vec{c} : [a,b] \to \mathbb{R}^3$

Let $\vec{v} = (x,y,z)$, where x, y, and z are constants.

And let $\vec{c}(\theta) = [f(\theta), g(\theta), h(\theta)]$, $f, g, h : [a,b] \to \mathbb{R}$.

$\therefore$ By definition, $\int_C \vec{v} \cdot d\vec{s} = \int_a^b \vec{v} \cdot \vec{c}\,'(\theta)\, d\theta$

$$= \int_a^b (x,y,z) \cdot [f'(\theta), g'(\theta), h'(\theta)]\, d\theta$$

$$= \int_a^b x f'(\theta) + y g'(\theta) + z h'(\theta)\, d\theta$$

$$= x\int_a^b f'(\theta)\,d\theta + y\int_a^b g'(\theta)\,d\theta + z\int_a^b h'(\theta)\,d\theta$$

using Fundamental Theorem of Calculus

$$= x[f(b)-f(a)] + y[g(b)-g(a)] + z[h(b)-h(a)]$$

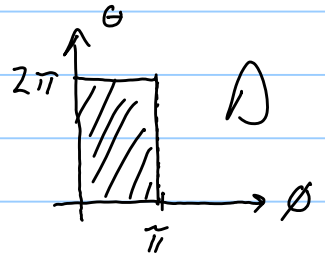
$$= (x,y,z) \cdot [(f(b), g(b), h(b)) - (f(a), g(a), h(a))]$$

$$= \vec{v} \cdot [\vec{c}(b) - \vec{c}(a)] = \vec{v} \cdot \vec{0} = 0$$

$$\therefore \int_C \vec{v} \cdot d\vec{s} = 0$$

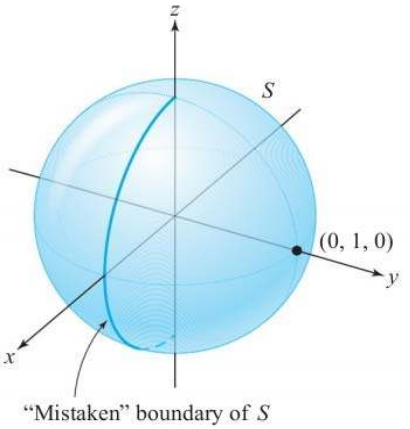
28. **28.** Show that $\Phi: D \to \mathbb{R}^3$, $D = [0, \pi] \times [0, 2\pi]$, $\Phi(\phi, \theta) = (\cos\theta \sin\phi, \sin\theta \sin\phi, \cos\phi)$, which parametrizes the unit sphere, takes the boundary of D to half of a great circle on S.

Consider each edge of D:



(a) $\theta = 0$, $0 \le \phi \le \pi$ $\therefore \Phi(\phi, 0) = (\sin\phi, 0, \cos\phi)$

This is the half circle from $(0,0,1)$ to $(0,0,-1)$ in the xz-plane, including the poles.



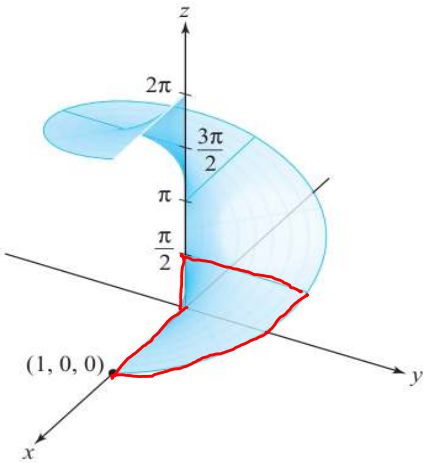
(b) $\theta = 2\pi$, $0 \le \phi \le \pi$ This is the same as (a) since $\cos(0) = \cos(2\pi) = 1$ and $\sin(0) = \sin(2\pi) = 0$

(c) $\phi = 0$, $0 \le \theta \le 2\pi$ $\therefore \Phi(0,\theta) = (0,0,1)$ or just the top pole of the sphere, or top of the half great circle in the non-negative xz-plane.

(d) $\phi = \pi$, $0 \le \theta \le 2\pi$ $\therefore \Phi(\pi,\theta) = (0,0,-1)$ or just the bottom pole of the sphere, or bottom of the half great circle in the non-negative xz-plane.

29.

29. Verify Theorem 6 for the helicoid $\Phi(r,\theta) = (r\cos\theta, r\sin\theta, \theta)$, $(r,\theta) \in [0,1] \times [0,\pi/2]$, and the vector field $\mathbf{F}(x,y,z) = (z,x,y)$.



just the surface bounded by the red 2S

(a) $\Phi_r = (\cos\theta, \sin\theta, 0)$

$\Phi_\theta = (-r\sin\theta, r\cos\theta, 1)$

$$\Phi_r \times \Phi_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta & \sin\theta & 0 \\ -r\sin\theta & r\cos\theta & 1 \end{vmatrix}$$

$= (\sin\theta, -\cos\theta, r)$ $\hat{k}$ component $= r$.

Since $r \geq 0$, $\vec{\Phi}_r \times \vec{\Phi}_\theta$ points "up" toward the positive z-axis.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ z & x & y \end{vmatrix} = (1,1,1)$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_0^{\frac{\pi}{2}} \int_0^1 (1,1,1) \cdot (\sin\theta, -\cos\theta, r)\, dr\, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^1 (\sin\theta - \cos\theta + r)\, dr\, d\theta = \int_0^{\frac{\pi}{2}} \left(\sin\theta - \cos\theta + \frac{1}{2}\right) d\theta$$

$$= -\cos\theta - \sin\theta + \frac{\theta}{2} \Big|_0^{\frac{\pi}{2}} = \left[0 - 1 + \frac{\pi}{4}\right] - \left[-1 - 0 + 0\right]$$

$$= \frac{\pi}{4}$$

(b) To keep the orientation consistent with (a), move along ∂S by starting at $(0,0,0) \to (1,0,0) \to (0,1,\frac{\pi}{2}) \to (0,0,\frac{\pi}{2})$ back to $(0,0,0)$: essentially counterclockwise

(1) $(0,0,0) \to (1,0,0)$ Let $\vec{c}_1(t) = (t,0,0)$ $0 \leq t \leq 1$

$\therefore \vec{c}_1'(t) = (1,0,0)$ $\therefore \vec{F}(\vec{c}_1(t)) = (0,t,0)$

$$\therefore \vec{F} \cdot \vec{c}_1' = (0,t,0) \cdot (1,0,0) = 0$$

$$\therefore \int_{\vec{c}_1} \vec{F} \cdot d\vec{s} = \underline{0}$$

(2) $(1,0,0) \to (0,1,\frac{\pi}{2})$ Let $\vec{c}_2(t) = (\cos t, \sin t, t)$, $0 \le t \le \frac{\pi}{2}$

$$\therefore \vec{c}_2'(t) = (-\sin t, \cos t, 1) \quad \vec{F}(\vec{c}_2(t)) = (t, \cos t, \sin t)$$

$$\therefore \vec{F} \cdot \vec{c}_2' = -t\sin t + \cos^2 t + \sin t$$

$$\therefore \int_{\vec{c}_2} \vec{F} \cdot d\vec{s} = \int_0^{\frac{\pi}{2}} (-t\sin t + \cos^2 t + \sin t)\, dt$$

$\cos^2 t = \frac{1+\cos 2t}{2}$

$$= t\cos t - \sin t + \frac{t}{2} + \frac{\sin 2t}{4} - \cos t \Big|_0^{\frac{\pi}{2}}$$

$$= \left(0 - 1 + \frac{\pi}{4} + 0 - 0\right) - (0 - 0 + 0 + 0 - 1) = \underline{\pi/4}$$

(3) $(0,1,\frac{\pi}{2}) \to (0,0,\frac{\pi}{2})$ Let $\vec{c}_3(t) = (0, 1-t, \frac{\pi}{2})$, $0 \le t \le 1$

$$\therefore \vec{c}_3'(t) = (0,-1,0) \qquad \vec{F}(\vec{c}_3(t)) = (\tfrac{\pi}{2}, 0, 1-t)$$

$$\therefore \vec{F} \cdot \vec{c}_3' = (\tfrac{\pi}{2}, 0, 1-t) \cdot (0,-1,0) = 0$$

$$\therefore \int_{\vec{c}_3} \vec{F} \cdot d\vec{s} = \underline{0}$$

(4) $(0,0,\frac{\pi}{2}) \to (0,0,0)$ Let $\vec{c}_4(t) = (0,0,\frac{\pi}{2}-t)$, $0 \le t \le \frac{\pi}{2}$

$\therefore \vec{c}_4'(t) = (0,0,-1)$ $\quad \vec{F}(\vec{c}_4(t)) = (\frac{\pi}{2}-t, 0, 0)$

$\therefore \vec{F} \cdot \vec{c}_4' = (\frac{\pi}{2}-t, 0, 0) \cdot (0,0,-1) = 0$

$\therefore \int_{\vec{c}_4} \vec{F} \cdot d\vec{s} = 0$

$\therefore \int_{\vec{c}_1} F \cdot d\vec{s} + \int_{\vec{c}_2} \vec{F} \cdot d\vec{s} + \int_{\vec{c}_3} \vec{F} \cdot d\vec{s} + \int_{\vec{c}_4} \vec{F} \cdot d\vec{s} = \int_{\partial S} \vec{F} \cdot d\vec{s}$

$= 0 + \frac{\pi}{4} + 0 + 0 = \frac{\pi}{4}$

$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s} = \frac{\pi}{4}$

30. **30.** Prove Theorem 6.

Theorem 6 Stokes' Theorem: Parametrized Surfaces Let S be an oriented surface defined by a one-to-one parametrization Φ: $D \subset \mathbb{R}^2 \to S$, where D is a region to which Green's theorem applies. Let ∂S denote the oriented boundary of S and let $\mathbf{F}$ be a C^1 vector field on S. Then

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \int_{\partial S} \mathbf{F} \cdot d\mathbf{s}.$$

If S has no boundary, and this includes surfaces such as the sphere, then the integral on the left is zero (see Exercise 25).

Instead of $\Phi: D \to S$, use $\vec{T}: D \subset \mathbb{R}^2 \to S$ as the 1-to-1 parametrization. $\therefore$ if $(u,v) \in D$, $\vec{T}(u,v) \in S$.

Let $\vec{T}(u,v) = [f(u,v), g(u,v), h(u,v)]$, and $\vec{T}$ is C^2.

Assume S has an oriented boundary, defined by $\vec{T}[\vec{c}(t)] = \vec{T}[u(t), v(t)]$, where $\vec{c}: [a,b] \to \mathbb{R}^2$ and $\vec{c}(t) = [u(t), v(t)]$ is a parametrization of ∂D.

Let $\vec{\rho}(t) = \partial S$, so that $\vec{\rho}(t) = \vec{T}(\vec{c}(t)) = \vec{T}[u(t), v(t)]$

$$\therefore \vec{\rho}(t) = \left[f(u(t), v(t)),\ g(u(t), v(t)),\ h(u(t), v(t)) \right]$$

$$\therefore \vec{\rho}\,'(t) = \left[f_u \frac{du}{dt} + f_v \frac{dv}{dt},\ g_u \frac{du}{dt} + g_v \frac{dv}{dt},\ h_u \frac{du}{dt} + h_v \frac{dv}{dt} \right]$$

Let $\vec{F} = (F, G, H) = [F(x,y,z), G(x,y,z), H(x,y,z)]$

On surface S, $\vec{F} = [F(x(u,v), y(u,v), z(u,v)), G(x(u,v), y(u,v), z(u,v)), H(x(u,v), y(u,v), z(u,v))]$

$$\therefore \vec{T}_u \times \vec{T}_v = (g_u h_v - g_v h_u,\ f_v h_u - f_u h_v,\ f_u g_v - f_v g_u)$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F & G & H \end{vmatrix} = (G_z - H_y,\ F_z - H_x,\ G_x - F_y)$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \iint_D (G_z - H_y,\ F_z - H_x,\ G_x - F_y) \cdot (\vec{T}_u \times \vec{T}_v)\, du\, dv$$

$$= \iint_D (G_z - H_y)(g_u h_v - g_v h_u) + (F_z - H_x)(f_v h_u - f_u h_v) + (G_x - F_y)(f_u g_v - f_v g_u)\, du\, dv \qquad [1]$$

Also, $$\int_{\partial S} \vec{F} \cdot d\vec{s} = \int_{\vec{\rho}} \vec{F} \cdot d\vec{s} = \int_a^b \vec{F}(\vec{\rho}(t)) \cdot \vec{\rho}\,'(t)\, dt$$

$$= \int_a^b \left[F(f(u,v), g(u,v), h(u,v)),\ G(f(u,v), g(u,v), h(u,v)),\ H(f(u,v), g(u,v), h(u,v)) \right] \cdot \vec{\rho}\,'(t)\, dt$$

$$= \int_a^b F\left(f_u \frac{du}{dt} + f_v \frac{dv}{dt}\right) + G\left(g_u \frac{du}{dt} + g_v \frac{dv}{dt}\right) + H\left(h_u \frac{du}{dt} + h_v \frac{dv}{dt}\right) dt$$

$$= \int_a^b \left[F f_u + G g_u + H h_u \right] \frac{du}{dt} + \left[F f_v + G g_v + H h_v \right] \frac{dv}{dt}\, dt$$

$$= \int_{\partial D} \left[F f_u + G g_u + H h_u \right] du + \left[F f_v + G g_v + H h_v \right] dv$$

using Green's Theorem on D

$$= \iint_D \frac{\partial}{\partial u}\left[F f_v + G g_v + H h_v \right] - \frac{\partial}{\partial v}\left[F f_u + G g_u + H h_u \right] du\, dv$$

and remembering F, G, H are functions of f, g, h

$$= \iint_D \Big[F_x f_u + F_y g_u + F_z h_u \Big] f_v + F f_{vu} \quad \text{(cancel)}$$
$$- \Big[F_x f_v + F_y g_v + F_z h_v \Big] f_u - F f_{uv} \quad \Big\} \; \frac{\partial}{\partial u} F f_v - \frac{\partial}{\partial v} F f_u$$

$$+ \Big[G_x f_u + G_y g_u + G_z h_u \Big] g_v + G g_{vu} \quad \text{(cancel)}$$
$$- \Big[G_x f_v + G_y g_v + G_z h_v \Big] g_u - G g_{uv} \quad \Big\} \; \frac{\partial}{\partial u} G g_v - \frac{\partial}{\partial v} G g_u$$

$$+ \Big[H_x f_u + H_y g_u + H_z h_u \Big] h_v + H h_{vu} \quad \text{(cancel)}$$
$$- \Big[H_x f_v + H_y g_v + H_z h_v \Big] h_u - H h_{uv} \quad \Big\} \; \frac{\partial}{\partial u} H h_v - \frac{\partial}{\partial v} H h_u$$

$$du\,dv$$

$$= \iint_D F_y \Big[f_v g_u - f_u g_v \Big] + F_z \Big[f_v h_u - f_u h_v \Big] \quad \text{(combine)} \quad \text{(combine)}$$
$$+ G_x \Big[f_u g_v - f_v g_u \Big] + G_z \Big[g_v h_u - g_u h_v \Big] \quad \text{(combine)}$$
$$+ H_x \Big[f_u h_v - f_v h_u \Big] + H_y \Big[g_u h_v - g_v h_u \Big] \; du\,dv$$

$$= \iint_D (G_z - H_y) \Big[g_v h_u - g_u h_v \Big] + (F_z - H_x) \Big[f_v h_u - f_u h_v \Big]$$
$$+ (G_x - F_y) \Big[f_u g_v - f_v g_u \Big] \; du\,dv$$

which is [1] from above

$$= \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} \qquad \therefore \int_{\partial S} \vec{F} \cdot d\vec{s} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

If S does not have an oriented boundary, then from #25, $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = 0$

31.

31. Let $\mathbf{F} = x^2\mathbf{i} + (2xy + x)\mathbf{j} + z\mathbf{k}$. Let C be the circle $x^2 + y^2 = 1$ and S the disc $x^2 + y^2 \leq 1$ within the plane $z = 0$.

(a) Determine the flux of $\mathbf{F}$ out of S.

(b) Determine the circulation of $\mathbf{F}$ around C.

(c) Find the flux of $\nabla \times \mathbf{F}$. Verify Stokes' theorem directly in this case.

(a) Better stated as "Flux of $\vec{F}$ through S".

Flux through S is the component of $\vec{F}$ along a perpendicular to S. Choose either "up" or "down" as the orientation.

$\therefore$ If $\vec{T}(r,\theta): D \to S$ is a parametrization of S,

then $\vec{T}_r \times \vec{T}_\theta$ is perpendicular to S.

$\therefore$ $\vec{F} \cdot (\vec{T}_r \times \vec{T}_\theta)$ would be the flux, summed over D.

$\therefore$ Let $\vec{T}(r,\theta) = (r\cos\theta, r\sin\theta, 0)$, $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$

$\vec{T}_r = (\cos\theta, \sin\theta, 0)$

$\vec{T}_\theta = (-r\sin\theta, r\cos\theta, 0)$ $\therefore$ $\vec{T}_r \times \vec{T}_\theta = (0, 0, r)$

$\therefore$ Flux $= \iint_S \vec{F} \cdot d\vec{S} = \iint_D \vec{F} \cdot (\vec{T}_r \times \vec{T}_\theta)\, dA$

$$= \int_0^{2\pi} \int_0^1 (r^2\cos^2\theta,\ 2r^2\cos\theta\sin\theta + r\cos\theta,\ 0)\cdot(0,0,r)\,dr\,d\theta$$

$$= \int_0^{2\pi} \int_0^1 0\,dr\,d\theta = \underline{0}$$

(b) Here, we want component of $\vec{F}$ tangential to $\partial S = C$.

$\therefore$ If $\vec{c}(\theta) : [0, 2\pi] \to \mathbb{R}^2$ is a parametrization of C, then $\vec{F}\cdot\vec{c}'(\theta)$ summed over the length of C will give the circulation.

$\therefore$ Let $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$, $0 \le \theta \le 2\pi$

$\therefore$ Circulation $= \int_C \vec{F}\cdot d\vec{s} = \int_0^{2\pi} \vec{F}(c(\theta))\cdot\vec{c}'(\theta)\,d\theta$

$$= \int_0^{2\pi} (\cos^2\theta,\ 2\cos\theta\sin\theta + \cos\theta,\ 0)\cdot(-\sin\theta, \cos\theta, 0)\,d\theta$$

$$= \int_0^{2\pi} -\cos^2\theta\sin\theta - 2\sin^2\theta\cos\theta + \cos^2\theta + 0\,d\theta$$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$$= \frac{\cos^3\theta}{3} - \frac{2}{3}\sin^3\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}\Big|_0^{2\pi}$$

$$= (1-1) - (0-0) + (\pi - 0) + (0-0) = \underline{\pi}$$

(c) Flux $= \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$ Choose same parametrization as in (a)

$$= \iint_D (\nabla \times \vec{F}) \cdot (\vec{T}_r \times \vec{T}_\theta)\, dr\, d\theta$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 & 2xy + x & z \end{vmatrix}$$

$$= (0, 0, 2y + 1)$$

$$= \int_0^{2\pi} \int_0^1 (0, 0, 2r\sin\theta + 1) \cdot (0, 0, r)\, dr\, d\theta$$

$$= \int_0^{2\pi} \int_0^1 2r^2 \sin\theta + r \, dr\, d\theta = \int_0^{2\pi} \frac{2}{3} r^3 \sin\theta + \frac{r^2}{2} \Big|_0^1 d\theta$$

$$= \int_0^{2\pi} \left(\frac{2}{3}\sin\theta + \frac{1}{2}\right) d\theta = -\frac{2}{3}\cos\theta + \frac{\theta}{2} \Big|_0^{2\pi}$$

$$= \left(-\frac{2}{3} + \frac{2}{3}\right) + (\pi - 0) = \underline{\pi}$$

Since $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s}$ and from (b),

$\int_{\partial S} \vec{F} \cdot d\vec{s} = \pi$, Stokes' Theorem is verified.

32.

32. Let S be a surface with boundary ∂S, and suppose that $\mathbf{E}$ is an electric field that is perpendicular to ∂S. Use Faraday's law to show that the induced magnetic flux across S is constant in time.

Let $\vec{c}(t)$ be a parametrization for ∂S.

$\therefore \vec{c}\,'(t)$ is tangential to ∂S, and so $\vec{E} \perp \vec{c}\,'(t)$,

so that $\vec{E}(\vec{c}(t)) \cdot \vec{c}\,'(t) = 0$. Let $\vec{c}(t) : [a,b] \to \partial S$

$$\therefore \int_{\partial S} \vec{E} \cdot d\vec{s} = \int_a^b \vec{E}(\vec{c}(t)) \cdot \vec{c}\,'(t)\, dt = 0$$

As shown in Example 5, text p. 449,

$$\int_{\partial S} \vec{E} \cdot d\vec{s} = -\frac{\partial}{\partial t} \iint_S \vec{H} \cdot d\vec{S}$$

$$\therefore \frac{\partial}{\partial t} \iint_S \vec{H} \cdot d\vec{S} = 0 \Rightarrow \iint_S \vec{H} \cdot d\vec{S} \text{ is constant in time}$$

$\therefore$ Magnetic flux across $S = \iint_S \vec{H} \cdot d\vec{S}$ is constant in time

33.

33. Integrate $\nabla \times \mathbf{F}$, $\mathbf{F} = (3y, -xz, -yz^2)$ over the portion of the surface $2z = x^2 + y^2$ below the plane $z = 2$, both directly and by using Stokes' theorem.

(a) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3y & -xz & -yz^2 \end{vmatrix} = (x - z^2, 0, -z - 3)$

Let $\vec{T}(r,\theta) = \left(r\cos\theta, r\sin\theta, \frac{r^2}{2}\right)$, $0 \le r \le 2$, $0 \le \theta \le 2\pi$

$\therefore T : D \subset \mathbb{R}^2 \to S \subset \mathbb{R}^3$

$\vec{T}_r = (\cos\theta, \sin\theta, r)$ $\quad \therefore \vec{T}_r \times \vec{T}_\theta = (-r^2\cos\theta, -r^2\sin\theta, r)$

$\vec{T}_\theta = (-r\sin\theta, r\cos\theta, 0)$

$\therefore$ Choosing orientation "up" ($\hat{k}$ component is $r \ge 0$).

$(\nabla \times \vec{F})(\vec{T}(r,\theta)) = \left(r\cos\theta - \frac{r^4}{4}, 0, -\frac{r^2}{2} - 3\right)$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_0^{2\pi}\int_0^2 (\nabla \times \vec{F})(\vec{T}(r,\theta)) \cdot (\vec{T}_r \times \vec{T}_\theta)\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^2 \left(r\cos\theta - \frac{r^4}{4}, 0, -\frac{r^2}{2} - 3\right) \cdot (-r^2\cos\theta, -r^2\sin\theta, r)\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^2 \left(-r^3\cos^2\theta + \frac{r^6}{4}\cos\theta - \frac{r^3}{2} - 3r\right) dr\, d\theta$$

$$= \int_0^{2\pi} -\frac{r^4}{4}\cos^2\theta + \frac{r^7}{28}\cos\theta - \frac{r^4}{8} - \frac{3}{2}r^2 \Big|_{r=0}^{r=2} \, d\theta$$

$$= \int_0^{2\pi} -4\cos^2\theta + \frac{2^7}{28}\cos\theta - 8 \, d\theta$$

$\cos^2\theta = \frac{1+\cos 2\theta}{2}$

$$= -4\left[\frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_0^{2\pi} + \frac{2^7}{28}\sin\theta \Big|_0^{2\pi} - 8\theta \Big|_0^{2\pi}$$

$$= -4[\pi + 0] + 0 - 16\pi = -20\pi$$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \underline{-20\pi}$$ (or 20π if orientation reversed using $\vec{T}_\theta \times \vec{T}_r$).

(5) Using Stokes' Theorem, $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s}$

so compute the line integral.

Parametrize ∂S as $\vec{c}(\theta) = (2\cos\theta, 2\sin\theta, 2)$, $0 \le \theta \le 2\pi$

Note This is counterclockwise and normal to S pointing "up" is on the left using the right-hand rule.

$\therefore \vec{c}\,'(\theta) = (-2\sin\theta, 2\cos\theta, 0)$.

$$\vec{F}(\vec{c}(\theta)) = (6\sin\theta, -4\cos\theta, -8\sin\theta)$$

$$\therefore \vec{F}\cdot\vec{c}\,' = -12\sin^2\theta - 8\cos^2\theta = -4\sin^2\theta - 8$$

$$\therefore \int_{\partial S}\vec{F}\cdot d\vec{s} = \int_0^{2\pi} -4\sin^2\theta - 8\, d\theta \qquad \sin^2\theta = \frac{1-\cos 2\theta}{2}$$

$$= \int_0^{2\pi}(-10 + 2\cos 2\theta)\, d\theta = -10\theta + \sin 2\theta\Big|_0^{2\pi} = \underline{-20\pi}$$

$$\therefore \int_{\partial S}\vec{F}\cdot d\vec{s} = \underline{-20\pi}$$

The other orientation would be $2\pi \to 0$, and the computation would be $+20\pi$

34.

34. ***Ampère's law*** states that if the electric current density is described by a vector field **J** and the induced magnetic field is **H**, then the circulation of **H** around the boundary C of a surface S equals the integral of **J** over S (i.e., the total current crossing S). See Figure 8.2.15. Show that this is implied by the steady-state *Maxwell equation* $\nabla \times \mathbf{H} = \mathbf{J}$.

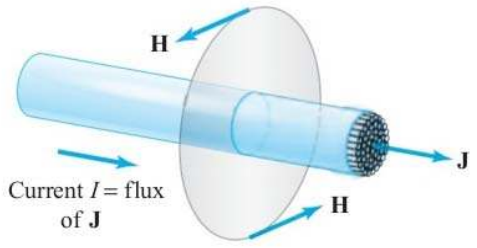


figure **8.2.15** Ampère's law.

The circulation of $\vec{H}$ around a boundary ∂S is defined as: $\int_{\partial S}\vec{H}\cdot d\vec{s}$, a line integral which sums $\vec{H}$ along a tangent to ∂S.

By Stokes' Theorem,

$$\int_{\partial S} \vec{H} \cdot d\vec{s} = \iint_S (\nabla \times \vec{H}) \cdot d\vec{S}$$, and by the Maxwell equation $\nabla \times \vec{H} = \vec{J}$, then

$$\int_{\partial S} \vec{H} \cdot d\vec{s} = \iint_S (\nabla \times \vec{H}) \cdot d\vec{S} = \iint_S \vec{J} \cdot d\vec{S},$$

the latter expression being the integral sum of $\vec{J}$ over S. $\therefore$ $$\int_{\partial S} \vec{H} \cdot d\vec{s} = \iint_S \vec{J} \cdot d\vec{S}$$

35.

35. Faraday's law relates the line integral of the electric field around a loop C to the surface integral of the rate of change of the magnetic field over a surface S with boundary C. Regarding the equation $\nabla \times \mathbf{E} = -\partial \mathbf{H}/\partial t$ as the basic equation, Faraday's law is a consequence of Stokes' theorem, as we have seen in Example 4.

Suppose we are given electric and magnetic fields in space that satisfy $\nabla \times \mathbf{E} = -\partial \mathbf{H}/\partial t$. Suppose C is the boundary of the Möbius band shown in Figures 7.6.3 and 7.6.4. Because the Möbius band cannot be oriented, Stokes' theorem does not apply. What becomes of Faraday's law? What do you guess $\int_C \mathbf{E} \cdot d\mathbf{s}$ equals?

Stokes' Theorem can't be applied to the Möbius surface, but the Möbius boundary curve C can be the boundary to another surface that is oriented: imagine a relatively flat sheet

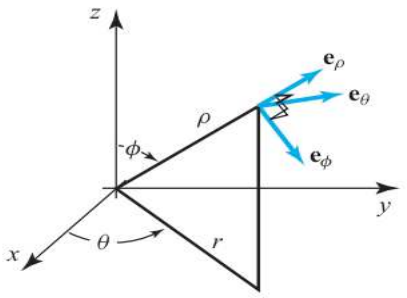
that is perpendicular to most of the Möbius band, that includes C. Then Stokes' Theorem applies, Faraday's Law holds, and $\int_C \vec{E} \cdot d\vec{s}$ equals the negative rate of flux of $\vec{H}$ through this new surface.

36.

36. (a) If in spherical coordinates, we write

$$\mathbf{e}_r = \alpha\mathbf{i} + \beta\mathbf{j} + \gamma\mathbf{k}, \text{ find } \alpha, \beta, \text{ and } \gamma.$$

(b) Find similar formulas for $\mathbf{e}_\phi$ and $\mathbf{e}_\theta$.



(a) Use $\hat{e}_\rho$ instead of $\hat{e}_r$

The x-coordinate of a vector $\vec{r}$ is $\|\vec{r}\| \cos\theta \sin\phi$

$\therefore x = \|\vec{r}\| \cos\theta \sin\phi$. Here, $\|\vec{r}\| = \|\hat{e}_\rho\| = 1$.

$\therefore x = \cos\theta \sin\phi = \alpha$

Similarly, $y = \sin\theta \sin\phi = \beta$, $z = \cos\phi = \gamma$

$\therefore \hat{e}_\rho = \cos\theta \sin\phi\, \hat{i} + \sin\theta \sin\phi\, \hat{j} + \cos\phi\, \hat{k}$

(b) For $\hat{e}_\theta$, note it is parallel to the xy-plane, so it's angle with the z-axis is $\frac{\pi}{2}$

$\therefore \cos\phi = 0$. Note $\hat{e}_\theta$ is rotated $\frac{\pi}{2}$ radians counterclockwise from r in the figure, so its angle with the x-axis is $\theta + \frac{\pi}{2}$, where θ is the angle r (projection of ρ onto xy-plane) makes with the x-axis.

$$\therefore \text{x-coordinate} = \|\hat{e}_\theta\| \cos\left(\theta + \tfrac{\pi}{2}\right) \sin\left(\tfrac{\pi}{2}\right)$$

$$= (1)(-\sin\theta)(1) = -\sin\theta$$

$$\text{y-coordinate} = \|\hat{e}_\theta\| \sin\left(\theta + \tfrac{\pi}{2}\right) \sin\left(\tfrac{\pi}{2}\right)$$

$$= (1)(\cos\theta)(1) = \cos\theta$$

$$\text{z-coordinate} = \|\hat{e}_\theta\| \cos\phi = (1)(0) = 0$$

$$\therefore \underline{\hat{e}_\theta = (-\sin\theta)\,\hat{i} + (\cos\theta)\,\hat{j}}$$

For $\hat{e}_\phi$, note its projection onto xy-plane is similar to projection of $\rho = r$.

$$\therefore \text{x-coordinate} = \|\hat{e}_\phi\| \cos\theta \sin\phi = \cos\theta\sin\phi$$

$$\text{y-coordinate} = \|\hat{e}_\phi\| \sin\theta \sin\phi = \sin\theta\sin\phi$$

For z-coordinate, it's angle with the z-axis is $\frac{\pi}{2}$ radians more than angle of ρ with z-axis.

$$\therefore z\text{-coordinate} = \|\hat{e}_\phi\| \cos\left(\phi + \frac{\pi}{2}\right) = -\sin\phi$$

$$\therefore \hat{e}_\phi = (\cos\theta\sin\phi)\,\hat{i} + (\sin\theta\sin\phi)\,\hat{j} - \sin\phi\,\hat{k}$$

Summary:

$$\hat{e}_\rho = \cos\theta\sin\phi\,\hat{i} + \sin\theta\sin\phi\,\hat{j} + \cos\phi\,\hat{k}$$

$$\hat{e}_\theta = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$$

$$\hat{e}_\phi = \cos\theta\sin\phi\,\hat{i} + \sin\theta\sin\phi\,\hat{j} - \sin\phi\,\hat{k}$$

8.3 Conservative Fields

Note Title 7/3/2017

1.

1. Determine which of the following vector fields **F** in the plane is the gradient of a scalar function f. If such an f exists, find it.

(a) $\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$

(b) $\mathbf{F}(x, y) = xy\mathbf{i} + xy\mathbf{j}$

(c) $\mathbf{F}(x, y) = (x^2 + y^2)\mathbf{i} + 2xy\mathbf{j}$

(a) $\frac{\partial(y)}{\partial x} = 0 = \frac{\partial(x)}{\partial y}$ $\therefore$ $\vec{F}$ is a gradient

Let $f(x,y)$ be the scalar function s.t. $\nabla f = \vec{F}$

$\therefore f_x = x$ $\therefore f(x,y) = \frac{x^2}{2} + g(y)$

$\therefore f_y = g'(y) = y$, so $g(y) = \frac{y^2}{2} + C$, C some constant

$\therefore f(x,y) = \frac{x^2}{2} + \frac{y^2}{2} + C$

(b) $\frac{\partial(xy)}{\partial x} = y$ $\frac{\partial(xy)}{\partial y} = x$ $x \neq y$

$\therefore \vec{F}$ is not a gradient

(c) $\frac{\partial(2xy)}{\partial x} = 2y = \frac{\partial}{\partial y}(x^2+y^2)$ $\therefore \vec{F}$ is a gradient

Let $f(x,y)$ be the scalar function s.t. $\nabla f = \vec{F}$

$\therefore f_x = x^2 + y^2 \Rightarrow f(x,y) = \frac{x^3}{3} + xy^2 + g(y)$

$\therefore f_y = 2xy + g'(y) \Rightarrow g'(y) = 0 \Rightarrow g(y) = C$, a constant

$\therefore f(x,y) = \frac{x^3}{3} + xy^2 + C$

2.

2. Repeat Exercise 1 for the following vector fields:

(a) $\mathbf{F}(x, y) = (\cos xy - xy \sin xy)\mathbf{i} - (x^2 \sin xy)\mathbf{j}$

(b) $\mathbf{F}(x, y) = (x\sqrt{x^2y^2 + 1})\mathbf{i} + (y\sqrt{x^2y^2 + 1})\mathbf{j}$

(c) $\mathbf{F}(x, y) = (2x \cos y + \cos y)\mathbf{i} - (x^2 \sin y + x \sin y)\mathbf{j}$

(a) $\frac{\partial}{\partial x}(-x^2 \sin xy) = -2x \sin xy - x^2 y \cos xy$

$\frac{\partial}{\partial y}(\cos xy - xy \sin xy) = -x \sin xy - x \sin xy - x^2 y \cos xy$

$= -2x \sin xy - x^2 y \cos xy$

As $\frac{\partial}{\partial x}(\,) = \frac{\partial}{\partial y}(\,)$, $\vec{F}$ is a gradient field.

Let $f(x,y)$ be the scalar function s.t. $\nabla f = \vec{F}$

$\therefore f_y = -x^2 \sin xy \Rightarrow f(x,y) = x \cos xy + g(x)$

$\therefore f_x = \cos xy - xy \sin xy + g'(x) \Rightarrow g(x) = C$

$\therefore f(x,y) = x \cos xy + C$, C a constant

(b) $\frac{\partial}{\partial x}(y\sqrt{x^2y^2 + 1}) = xy^3(x^2y^2 + 1)^{-\frac{1}{2}}$

$$\frac{\partial}{\partial y}\left(x\sqrt{x^2y^2+1}\right) = x^3y\left(x^2y^2+1\right)^{-\frac{1}{2}}$$

$\frac{\partial}{\partial x}(\,) \neq \frac{\partial}{\partial y}(\,) \Rightarrow \vec{F}$ is not a gradient field

(c) $\frac{\partial}{\partial x}\left(-x^2\sin y - x\sin y\right) = -2x\sin y - \sin y$

$\frac{\partial}{\partial y}\left(2x\cos y + \cos y\right) = -2x\sin y - \sin y$

$\frac{\partial}{\partial x}(\,) = \frac{\partial}{\partial y}(\,) \Rightarrow \vec{F}$ is a gradient field.

$\therefore$ Let $f(x,y)$ be the scalar function s.t. $\nabla f = \vec{F}$

$\therefore f_x = 2x\cos y + \cos y \Rightarrow f(x,y) = x^2\cos y + x\cos y + g(y)$

$\therefore f_y = -x^2\sin y - x\sin y + g'(y) \Rightarrow g'(y) = 0 \Rightarrow g(y) = C$

$\therefore f(x,y) = x^2\cos y + x\cos y + C$, C a constant

3.

3. For each of the following vector fields $\mathbf{F}$, determine (i) if there exists a function g such that $\nabla g = \mathbf{F}$, and (ii) if there exists a vector field $\mathbf{G}$ such that curl $\mathbf{G} = \mathbf{F}$. (It is not necessary to find g or $\mathbf{G}$.)

(a) $\mathbf{F}(x, y, z) = (4xz - x, -4yz, z - 2y)$

(b) $\mathbf{F}(x, y, z) = (e^x \sin y, e^x \cos y, z^2)$

(c) $\mathbf{F}(x, y, z) = (\log(z^2 + 1) + y^2, 2xy, \frac{2xz}{z^2+1})$

(d) $\mathbf{F}(x, y, z) = (x^2 + x \sin z, y \cos z - 2xy, \cos z + \sin z)$

(a) (i) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 4xz - x & -4yz & z - 2y \end{vmatrix} = (-2 + 4y, 4x, 0) \neq \vec{0}$

$\therefore$ No such function g s.t. $\nabla g = \vec{F}$

(ii) $\operatorname{div} \vec{F} = (4z - 1) + (-4z) + (1) = 0$

$\therefore$ yes, $\exists$ a $\vec{G}$ s.t. $\nabla \times \vec{G} = \vec{F}$

(b) (i) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^x \sin y & e^x \cos y & z^2 \end{vmatrix} = (0, 0, e^x \cos y - e^x \cos y) = \vec{0}$

yes, there is a function g s.t. $\nabla g = \vec{F}$

(ii) $\operatorname{div} \vec{F} = e^x \sin y + (-e^x \sin y) + 2z \neq 0$

$\therefore$ No, no vector field $\vec{G}$ s.t. $\vec{F} = \nabla \times \vec{G}$

(c) (i) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \log(z^2+1) + y^2 & 2xy & \frac{2xz}{z^2+1} \end{vmatrix} = \left(0, \frac{2z}{z^2+1} - \frac{2z}{z^2+1}, 0\right) = \vec{0}$

$\therefore$ Yes, $\exists$ a function g s.t. $\nabla g = \vec{F}$

(ii) $\operatorname{div} \vec{F} = 0 + 2x + \frac{(z^2+1)(2x) - (2xz)(2z)}{(z^2+1)^2} \neq 0$

$\therefore$ No, no $\vec{G}$ s.t. $\vec{F} = \nabla \times \vec{G}$

(d) (i) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 + x\sin z & y\cos z - 2xy & \cos z + \sin z \end{vmatrix}$

$= (0 + y\sin z, -, -) \neq \vec{0}$

No, no function g s.t. $\nabla g = \vec{F}$

(ii) $\text{div } \vec{F} = (2x + \sin z) + (\cos z - 2x) + (-\sin z + \cos z)$

$= 2\cos z \neq 0 \quad \therefore$ No $\vec{G}$ s.t. $\vec{F} = \nabla \times \vec{G}$

4.

4. For each of the following vector fields **F**, determine (i) if there exists a function g such that $\nabla g = \mathbf{F}$, and (ii) if there exists a vector field **G** such that curl **G** = **F**. (It is not necessary to find g or **G**.)

(a) $\mathbf{F}(x, y, z) = (e^x \cos y, -e^x \sin y, \pi)$

(b) $\mathbf{F}(x, y, z) = \left(\frac{y}{z^2+4}, \frac{x}{z^2+4}, \frac{-2xyz}{z^4+8z^2+16}\right)$

(c) $\mathbf{F}(x, y, z) = (x^2y^2z^2, ye^x, xy\cos z)$

(d) $\mathbf{F}(x, y, z) = (6z^5y^5, 9x^8z^2, 4x^3y^3)$

(a) (i) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^x\cos y & -e^x\sin y & \pi \end{vmatrix} = (0, 0, -e^x\sin y - e^x\sin y)$

$= \vec{0}$

$\therefore$ Yes, There is a function g s.t. $\nabla g = \vec{F}$

(ii) $\text{div } \vec{F} = (e^x\cos y) + (-e^x\cos y) + (0) = 0$

$\therefore$ Yes, There is a $\vec{G}$ s.t. $\nabla \times \vec{G} = \vec{F}$

(6)(i) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \frac{y}{z^2+4} & \frac{x}{z^2+4} & \frac{-2xyz}{(z^2+4)^2} \end{vmatrix}$

$= \left[\frac{-2xz}{(z^2+4)^2} - \frac{-2xz}{(z^2+4)^2}, \frac{-2yz}{(z^2+4)^2} - \frac{-2yz}{(z^2+4)^2}, \frac{1}{z^2+4} - \frac{1}{z^2+4} \right] = \vec{0}$

$\therefore$ Yes, There is a g s.t. $\nabla g = \vec{F}$

(ii) $div\, \vec{F} = (0) + (0) + \left[\frac{(z^2+4)^2(-2xy) - (-2xyz)(2(z^2+4)(2z))}{(z^2+4)^4} \right.$

$= \frac{(z^2+4)(-2xyz^2 - 8xy) + 8xyz^2(z^2+4)}{(z^2+4)^4} \neq 0$

$\therefore$ No, there is no $\vec{G}$ s.t. $\nabla \times \vec{G} = \vec{F}$

(c)(i) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2y^2z^2 & ye^x & xy\cos z \end{vmatrix} = (x\cos z, -, -) \neq \vec{0}$

$\therefore$ No, there is no g s.t. $\nabla g = \vec{F}$

(ii) $div\, \vec{F} = (2xy^2z^2) + (e^x) + (-xy\sin z) \neq 0$

$\therefore$ No, There is no $\vec{G}$ s.t. $\nabla \times \vec{G} = \vec{F}$

(d)(i) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 6z^5y^5 & 9x^8z^2 & 4x^3y^3 \end{vmatrix} = (12x^3y^2 - 18x^8z, -, -) \neq \vec{0}$

$\therefore$ No, there is no g s.t. $\nabla g = \vec{F}$

(ii) $\operatorname{div} \vec{F} = (0) + (0) + (0) = 0$

$\therefore$ Yes, there is a $\vec{G}$ s.t. $\nabla \times \vec{G} = \vec{F}$

5. **5.** Show that any two potential functions for a vector field on $\mathbb{R}^3$ differ at most by a constant.

Let $f(x,y,z)$ and $g(x,y,z)$ be two potential functions s.t. $\nabla f = \vec{F} = \nabla g \Rightarrow (f_x, f_y, f_z) = (g_x, g_y, g_z)$.

$\therefore f_x = g_x \Rightarrow f(x,y,z) = g(x,y,z) + h(y,z)$,

$h(y,z)$ some function of y,z only.

$\therefore f_y = g_y + h_y(y,z) \Rightarrow h_y(y,z) = 0 \Rightarrow h(y,z) = j(z)$,

where $j(z)$ is some function of z only.

$\therefore f(x,y,z) = g(x,y,z) + j(z)$.

But $f_z = g_z + j'(z) \Rightarrow j'(z) = 0 \Rightarrow j(z) = C$,

where C is some constant.

$\therefore f(x,y,z) = g(x,y,z) + C$, C a constant

6. **6.** (a) Let $\mathbf{F}(x, y) = (xy, y^2)$ and let $\mathbf{c}$ be the path $y = 2x^2$ joining $(0, 0)$ to $(1, 2)$ in $\mathbb{R}^2$. Evaluate $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$.

(b) Does the integral in part (a) depend on the path joining $(0, 0)$ to $(1, 2)$?

(a) Let $\vec{c}(t) = (t, 2t^2)$, $0 \le t \le 1$

$\therefore \vec{c}\,'(t) = (1, 4t)$ $\quad \vec{F}(\vec{c}(t)) = (2t^3, 4t^4)$

$$\therefore \int_C \vec{F} \cdot d\vec{s} = \int_0^1 (2t^3, 4t^4) \cdot (1, 4t)\, dt$$

$$= \int_0^1 2t^3 + 16t^5\, dt = \frac{t^4}{2} + \frac{8}{3}t^6 \Big|_0^1 = \frac{1}{2} + \frac{8}{3}$$

$$= \frac{3}{6} + \frac{16}{6} = \underline{\frac{19}{6}}$$

(b) Since $\nabla \times \vec{F} = \left[0, 0, \frac{\partial(y^2)}{\partial x} - \frac{\partial(xy)}{\partial y}\right] = (0, 0, -x) \neq \vec{0}$,

then $\vec{F}$ is <u>not</u> conservative.

$\therefore$ Integral in (a) <u>does</u> depend on path.

7. **7.** Let $\mathbf{F}(x, y, z) = (2xyz + \sin x)\mathbf{i} + x^2 z\mathbf{j} + x^2 y\mathbf{k}$. Find a function f such that $\mathbf{F} = \nabla f$.

$f_x = 2xyz + \sin x$, $\therefore f(x, y, z) = x^2yz - \cos x + g(y, z)$

$\therefore f_y = x^2 z + g_y(y, z) = x^2 z \Rightarrow g_y(y, z) = 0$

$\therefore g(y,z) = h(z) \Rightarrow f(x,y,z) = x^2yz - \cos x + h(z)$

$\therefore f_z = x^2y + h'(z) = x^2y \Rightarrow h'(z) = C$, a constant.

$\therefore f(x,y,z) = x^2yz - \cos x + C$, C some constant

8. **8.** Evaluate $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{c}(t) = (\cos^5 t, \sin^3 t, t^4)$, $0 \le t \le \pi$, and $\mathbf{F}$ is as in Exercise 7.

$$\int_C \vec{F} \cdot d\vec{s} = \int_0^{\pi} \nabla f \cdot \vec{c}\,'(t)\,dt = f(\vec{c}(\pi)) - f(\vec{c}(0))$$

$$f(-1, 0, \pi^4) - f(1,0,0) = [0 - \cos(-1)] - [0 - \cos(1)]$$

$= 0$ since $\cos(-a) = \cos(a)$, $a \in \mathbb{R}$.

9. **9.** If $f(x)$ is a smooth function of one variable, must $\mathbf{F}(x, y) = f(x)\mathbf{i} + f(y)\mathbf{j}$ be a gradient?

Yes, because $f_x(y) = f_y(x) = 0$, so $\nabla \times \vec{F} = 0$.

10. **10.** (a) Show that $\mathbf{F} = -\mathbf{r}/\|\mathbf{r}\|^3$ is the gradient of $f(x, y, z) = 1/r$.

(b) What is the work done by the force $\mathbf{F} = -\mathbf{r}/\|\mathbf{r}\|^3$ in moving a particle from a point $\mathbf{r}_0 \in \mathbb{R}^3$ "to ∞," where $\mathbf{r}(x, y, z) = (x, y, z)$?

(a) $\vec{r} = (x,y,z)$ $\therefore \|\vec{r}\| = \sqrt{x^2+y^2+z^2} = r$

$$\therefore f(x,y,z) = \frac{1}{r} = \frac{1}{\sqrt{x^2+y^2+z^2}} = (x^2+y^2+z^2)^{-\frac{1}{2}}$$

$$\therefore f_x = -\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}}(2x) = -x(x^2+y^2+z^2)^{-\frac{3}{2}} = -x\left(\frac{1}{r^3}\right)$$

Similarly, $f_y = -y(x^2+y^2+z^2)^{-\frac{3}{2}} = -y\left(\frac{1}{r^3}\right)$

$$f_z = -z(x^2+y^2+z^2)^{-\frac{3}{2}} = -z\left(\frac{1}{r^3}\right)$$

$$\therefore \nabla f = \left(-\frac{x}{r^3}, -\frac{y}{r^3}, -\frac{z}{r^3}\right) = -\frac{\vec{r}}{r^3} = \underline{\underline{-\frac{\vec{r}}{\|\vec{r}\|^3}}}$$

(b) $$\int_{\vec{C}} \vec{F}\cdot d\vec{s} = f(\text{endpoint}) - f(\text{startpoint})$$

$$= f(\infty) - f(\vec{r}_0) = \text{``}\frac{1}{\infty}\text{''} - \frac{1}{\|\vec{r}_0\|}$$

$$= \left(\lim_{r\to\infty}\frac{1}{r}\right) - \frac{1}{\|\vec{r}_0\|} = \underline{\underline{-\frac{1}{\|\vec{r}_0\|}}}$$

11. **11.** Let $\mathbf{F}(x, y, z) = xy\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Can there exist a function f such that $\mathbf{F} = \nabla f$?

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy & y & z \end{vmatrix} = (0, 0, -x) \neq \vec{0}$$

$\therefore$ No, no function f s.t. $\nabla f = \vec{F}$

12.

12. Let $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ and suppose each F_k satisfies the homogeneity condition

$$F_k(tx, ty, tz) = tF_k(x, y, z), \qquad k = 1, 2, 3.$$

Suppose also $\nabla \times \mathbf{F} = \mathbf{0}$. Prove that $\mathbf{F} = \nabla f$, where

$$2f(x, y, z) = xF_1(x, y, z) + yF_2(x, y, z) + zF_3(x, y, z).$$

[HINT: Use Review Exercise 31, Chapter 2.]

$$2f(\vec{x}) = x F_1(\vec{x}) + y F_2(\vec{x}) + z F_3(\vec{x})$$

$$\therefore \nabla(2f(\vec{x})) = 2\nabla f(\vec{x}) = \nabla\left[x F_1(\vec{x}) + y F_2(\vec{x}) + z F_3(\vec{x})\right]$$

$$= \left[\left(F_1 + x\frac{\partial F_1}{\partial x}\right) + y\frac{\partial F_2}{\partial x} + z\frac{\partial F_3}{\partial x},\right.$$

$$x\frac{\partial F_1}{\partial y} + \left(F_2 + y\frac{\partial F_2}{\partial y}\right) + z\frac{\partial F_3}{\partial y},$$

$$\left.x\frac{\partial F_1}{\partial z} + y\frac{\partial F_2}{\partial z} + \left(F_3 + z\frac{\partial F_3}{\partial z}\right)\right]$$

$$= \left\{F_1, F_2, F_3\right\} + \left[(x,y,z)\cdot\left(\frac{\partial F_1}{\partial x}, \frac{\partial F_2}{\partial x}, \frac{\partial F_3}{\partial x}\right),\right.$$

$$(x,y,z)\cdot\left(\frac{\partial F_1}{\partial y}, \frac{\partial F_2}{\partial y}, \frac{\partial F_3}{\partial y}\right),$$

$$\left.(x,y,z)\cdot\left(\frac{\partial F_1}{\partial z}, \frac{\partial F_2}{\partial z}, \frac{\partial F_3}{\partial z}\right)\right] \quad [1]$$

Note $(x,y,z) = \vec{x}$ and each $\frac{\partial F_i}{\partial x}, \frac{\partial F_i}{\partial y}, \frac{\partial F_i}{\partial z}$ is

evaluated at $\vec{x} = (x, y, z)$.

From $\nabla \times \vec{F} = \vec{0}$, $\left[\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right] = \vec{0}$

so $\frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z}$ $\frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x}$ $\frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y}$ [2]

Substituting the equations from [2] into [1],

$2 \nabla f(\vec{x}) = \vec{F} + \left[\vec{x} \cdot \left(\frac{\partial F_1}{\partial x}, \frac{\partial F_1}{\partial y}, \frac{\partial F_1}{\partial z} \right), \frac{\partial F_2}{\partial x} = \frac{\partial F_1}{\partial y} \quad \frac{\partial F_3}{\partial x} = \frac{\partial F_1}{\partial z} \right.$

$\vec{x} \cdot \left(\frac{\partial F_2}{\partial x}, \frac{\partial F_2}{\partial y}, \frac{\partial F_2}{\partial z} \right), \frac{\partial F_1}{\partial y} = \frac{\partial F_2}{\partial x} \quad \frac{\partial F_3}{\partial y} = \frac{\partial F_2}{\partial z}$

$\left. \vec{x} \cdot \left(\frac{\partial F_3}{\partial x}, \frac{\partial F_3}{\partial y}, \frac{\partial F_3}{\partial z} \right) \right] \frac{\partial F_1}{\partial z} = \frac{\partial F_3}{\partial x} \quad \frac{\partial F_2}{\partial z} = \frac{\partial F_3}{\partial y}$

$= \vec{F} + \left[\vec{x} \cdot \nabla F_1(\vec{x}), \vec{x} \cdot \nabla F_2(\vec{x}), \vec{x} \cdot \nabla F_3(\vec{x}) \right]$ [3]

and using $\vec{x} \cdot \nabla g(\vec{x}) = g(\vec{x})$ for homogeneous $g(\vec{x})$ from Exercise #31 of Review Exercises Chapter 2, (text p. 146), [3] becomes

$2 \nabla f(\vec{x}) = \vec{F} + \left[F_1(\vec{x}), F_2(\vec{x}), F_3(\vec{x}) \right] = \vec{F} + \vec{F}$

as $\vec{x} \cdot \nabla F_i(\vec{x}) = F_i(\vec{x})$ for homogeneous F_i

$\therefore 2\nabla f(\vec{x}) = 2\vec{F}(\vec{x})$, so $\nabla f = \vec{F}$

13.

13. Let $\mathbf{F}(x, y, z) = (e^x \sin y)\mathbf{i} + (e^x \cos y)\mathbf{j} + z^2\mathbf{k}$. Evaluate the integral $\int_{\mathbf{c}} \mathbf{F} \cdot d\mathbf{s}$, where $\mathbf{c}(t) = (\sqrt{t}, t^3, \exp\sqrt{t})$, $0 \le t \le 1$.

Since $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^x \sin y & e^x \cos y & z^2 \end{vmatrix} = (0-0,\ 0-0,\ e^x \cos y - e^x \cos y) = \vec{0}$,

Then $\exists f(x,y,z)$ s.t. $\nabla f = \vec{F}$

$f_x = e^x \sin y \Rightarrow f(x,y,z) = e^x \sin y + g(y,z)$,

g some function of y, z

$\therefore f_y = e^x \cos y + g_y(y,z) = e^x \cos y \Rightarrow g_y(y,z) = 0$

$\Rightarrow g(y,z) = h(z)$, h some function of z.

$\therefore f(x,y,z) = e^x \sin y + h(z)$

$f_z = h'(z) = z^2 \Rightarrow h(z) = \frac{z^3}{3} + C$, C a constant

$\therefore f(x,y,z) = e^x \sin y + \frac{z^3}{3} + C$

$\therefore \int_{\vec{c}} \vec{F} \cdot d\vec{s} = \int_0^1 \nabla f(\vec{c}(t)) \cdot \vec{c}'(t)\, dt =$

$f(\vec{c}(1)) - f(\vec{c}(0))$

$$= f(1,1,e) - f(0,0,1)$$

$$= e\sin(1) + \frac{e^3}{3} - \left(\frac{1}{3}\right) = e\sin(1) + \frac{e^3}{3} - \frac{1}{3}$$

14. **14.** Let a fluid have the velocity field $\mathbf{F}(x, y, z) = xy\mathbf{i} + yz\mathbf{j} + xz\mathbf{k}$. What is the circulation around the unit circle in the xy plane? Interpret your answer.

Parametrize circle as $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$, $0 \le \theta \le 2\pi$

$$\therefore \vec{c}\,'(\theta) = (-\sin\theta, \cos\theta, 0)$$

$$\therefore \vec{F}(\vec{c}(\theta)) \cdot \vec{c}\,'(\theta) = (\cos\theta\sin\theta, 0, 0) \cdot (-\sin\theta, \cos\theta, 0)$$

$$= -\sin^2\theta\cos\theta$$

$$\therefore \int_{\vec{c}} \vec{F} \cdot d\vec{s} = \int_0^{2\pi} (-\sin^2\theta\cos\theta)\, d\theta = -\frac{\sin^3\theta}{3}\Big|_0^{2\pi} = 0$$

$$\therefore \int_{\vec{c}} \vec{F} \cdot d\vec{s} = \underline{0}$$

This means the net circulation around the circle is zero. Note that $\vec{F}$ is not a conservative field, as $\nabla \times \vec{F} \neq \vec{0}$. Still, the line integral around a closed loop is zero.

15. **15.** The mass of the earth is approximately 6×10^{27} g and that of the sun is 330,000 times as much. The gravitational constant is 6.7×10^{-8} cm^3/s$^2 \cdot$g. The distance of the earth from the sun is about 1.5×10^{12} cm. Compute, approximately, the work necessary to increase the distance of the earth from the sun by 1 cm.

$$V(r_1) - V(r_1 + 1\text{cm}) = GM_{sun}m_{earth}\left(\frac{1}{r_1} - \frac{1}{r_1 + 1\text{cm}}\right)$$

Since $r_1 \approx r_1 + 1$ cm, the term in parenthesis becomes:

$$\left(\frac{1}{r_1} - \frac{1}{r_1+1}\right) = \left(\frac{r_1+1 - r_1}{r_1(r_1+1)}\right) \approx \frac{1}{r_1^2}$$

$$\therefore \frac{GM_{sun}m_{earth}}{(r_{sun\text{-}earth})^2} = \frac{(6.7\times10^{-8})(3.3\times10^{5})(6.0\times10^{27})^2}{(1.5\times10^{12})^2}$$

$$= \frac{354\times10^{51}}{10^{24}} = 354\times10^{27} = \underline{3.54\times10^{29}\ \text{ergs}}$$

16. **16.** (a) Show that $\int_C (x\,dy - y\,dx)/(x^2+y^2) = 2\pi$, where C is the unit circle.

(b) Conclude that the associated vector field $[-y/(x^2+y^2)]\mathbf{i} + [x/(x^2+y^2)]\mathbf{j}$ is not a conservative field.

(c) Show, however, that $\partial P/\partial y = \partial Q/\partial x$. Does this contradict the corollary to Theorem 7? If not, why not?

(a) Let $\vec{c}(\theta) = (\cos\theta, \sin\theta)$, $0 \le \theta \le 2\pi$

$\therefore x^2 + y^2 = 1$, $x\,dy = (\cos\theta)\cos\theta\,d\theta$

$-y\,dx = (-\sin\theta)(-\sin\theta)\,d\theta$

$$\therefore \frac{x\,dy - y\,dx}{x^2+y^2} = \frac{\cos^2\theta + \sin^2\theta}{1}\,d\theta = d\theta$$

$$\therefore \int_C \frac{x\,dy - y\,dx}{x^2+y^2} = \int_0^{2\pi} d\theta = \underline{2\pi}$$

(b) if $\vec{F} = \frac{-y}{x^2+y^2}\hat{i} + \frac{x}{x^2+y^2}\hat{j}$ were conservative, then

$\int_C \vec{F}\cdot d\vec{s} = 0$, where C is a closed circle.

But (a) showed this line integral is $2\pi \neq 0$

$\therefore \vec{F}$ is not conservative.

(c) $$\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}\left(\frac{-y}{x^2+y^2}\right) = \frac{(x^2+y^2)(-1) - (-y)(2y)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2}\right) = \frac{(x^2+y^2)(1) - (x)(2x)}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$\therefore P_y = Q_x$ (i.e., $\nabla \times \vec{F} = \vec{0}$).

Theorem 7 doesn't apply in this case b/c $\vec{F}$ has an exceptional point: $(0,0)$, which is not allow for R^2.

17.

17. Determine if the following vector fields $\mathbf{F}$ are gradient fields. If there exists a fuction f such that $\nabla f = \mathbf{F}$, find f.

(a) $\mathbf{F}(x, y, z) = (2xyz, x^2 z, x^2 y)$

(b) $\mathbf{F}(x, y) = (x \cos y, x \sin y)$

(c) $\mathbf{F}(x, y, z) = (x^2 e^y, xyz, e^z)$

(d) $\mathbf{F}(x, y) = (2x \cos y, -x^2 \sin y)$

(a) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xyz & x^2 z & x^2 y \end{vmatrix} = (x^2 - x^2,\ 2xy - 2xy,\ 2xz - 2xz)$

$= \vec{0}$ $\therefore$ yes

$f_x = 2xyz \Rightarrow f = x^2 yz + g(y, z)$

$f_y = x^2 z + g_y(y, z) \Rightarrow g_y(y, z) = 0 \Rightarrow g(y, z) = h(z)$

$f_z = x^2 y + h'(z) \Rightarrow h'(z) = 0 \Rightarrow h(z) = C$, a constant

$\therefore f(x, y, z) = x^2 yz + C$, C a constant

(b) $Q_x - P_y = \sin y - (-x \sin y) \neq 0$. $\therefore$ no

(c) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 e^y & xyz & e^z \end{vmatrix} = (0 - xy,\ 0 - 0,\ yz - x^2 e^y)$

$\neq \vec{0}$

$\therefore$ no

(d) $Q_x - P_y = (-2x \sin y) - (-2x \sin y) = 0$. $\therefore$ yes

$f_x = 2x \cos y \Rightarrow f(x, y) = x^2 \cos y + g(y)$

$f_y = -x^2 \sin y + g'(y) \Rightarrow g'(y) = 0 \Rightarrow g(y)$ constant

$\therefore f(x,y) = x^2 \cos y + C$, C a constant

18.

18. Determine if the following vector fields **F** are gradient fields. If there exists a function f such that $\nabla f = \mathbf{F}$, find f.

(a) $\mathbf{F}(x, y) = (2x + y^2 - y\sin x, 2xyz + \cos x)$

(b) $\mathbf{F}(x, y, z) = (6x^2z^2, 5x^2y^2, 4y^2z^2)$

(c) $\mathbf{F}(x, y) = (y^3 + 1, 3xy^2 + 1)$

(d) $\mathbf{F}(x, y) = (xe^{(x^2+y^2)} + 2xy,\ ye^{(x^2+y^2)} + 4y^3z,\ y^4)$ ← should be $\vec{F}(x,y,z)$

(a) $Q_x - P_y = (2yz - \sin x) - (2y - \sin x) \neq 0$ $\therefore$ no

(b) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 6x^2z & 5x^2y^2 & 4y^2z^2 \end{vmatrix} = (8yz^2 - 0, \text{—}, \text{—}) \neq \vec{0}$

$\therefore$ no

(c) $Q_x - P_y = (3y^2) - (3y^2) = 0$ $\therefore$ yes

$f_x = y^3 + 1 \Rightarrow f = xy^3 + x + g(y)$

$\therefore f_y = 3xy^2 + g'(y) \Rightarrow g'(y) = 1 \Rightarrow g(x) = y + C$

$\therefore f(x,y) = xy^3 + x + y + C$, C a constant

(d) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xe^{(x^2+y^2)} + 2xy & ye^{(x^2+y^2)} + 4y^3z & y^4 \end{vmatrix}$

$= (4y^3 - 4y^3, 0 - 0, 2xye^{(x^2+y^2)} - 2xye^{(x^2+y^2)} - 2x)$

$= (0, 0, -2x) \neq \vec{0}$. $\therefore$ no

19.

19. Show that the following vector fields are conservative. Calculate $\int_C \mathbf{F} \cdot d\mathbf{s}$ for the given curve.

(a) $\mathbf{F} = (xy^2 + 3x^2y)\mathbf{i} + (x + y)x^2\mathbf{j}$; C is the curve consisting of line segments from (1, 1) to (0, 2) to (3, 0).

(b) $\mathbf{F} = \dfrac{2x}{y^2+1}\mathbf{i} - \dfrac{2y(x^2+1)}{(y^2+1)^2}\mathbf{j}$; C is parametrized by $x = t^3 - 1$, $y = t^6 - t$, $0 \le t \le 1$.

(c) $\mathbf{F} = [\cos(xy^2) - xy^2 \sin(xy^2)]\mathbf{i} - 2x^2 y \sin(xy^2)\mathbf{j}$; C is the curve (e^t, e^{t+1}), $-1 \le t \le 0$.

(a) $Q_x - P_y = (3x^2 + 2xy) - (2xy + 3x^2) = 0$.

Rather than tediously computing the line integral by using parametrizations for C, find f s.t. $\nabla f = \vec{F}$.

$\therefore f_x = xy^2 + 3x^2y \Rightarrow f = \frac{x^2y^2}{2} + x^3y + g(y)$

$\therefore f_y = x^2y + x^3 + g'(y) \Rightarrow g'(y) = 0 \Rightarrow g(y)$ constant

$\therefore f(x,y) = \frac{x^2y^2}{2} + x^3y + C$, C a constant

$\therefore \int_C \vec{F} \cdot d\vec{s} = f(3,0) - f(1,1) = 0 - (\frac{1}{2} + 1) = -\frac{3}{2}$

(b) $Q_x - P_y = \left[-\frac{4xy}{(y^2+1)^2} \right] - \left[-\frac{4xy}{(y^2+1)^2} \right] = 0$

When $t=0$, $(x,y) = (-1,0)$

$t=1$, $(x,y) = (0,0)$

Find f s.t. $\nabla f = \vec{F}$

$\therefore f_x = \frac{2x}{y^2+1} \Rightarrow f = \frac{x^2}{y^2+1} + g(y)$

$\therefore f_y = -\frac{2x^2y}{(y^2+1)^2} + g'(y) \Rightarrow g'(y) = \frac{-2y}{(y^2+1)^2}$

$\Rightarrow g(y) = \frac{1}{y^2+1} + C$, C a constant

$\therefore f(x,y) = \frac{x^2}{y^2+1} + \frac{1}{y^2+1} + C$, C a constant

$\therefore \int_C \vec{F} \cdot d\vec{s} = f(0,0) - f(-1,0) = (0+1) - (1+1)$

$= -1$

(c)

(c) $\mathbf{F} = [\cos(xy^2) - xy^2 \sin(xy^2)]\mathbf{i} - 2x^2 y \sin(xy^2)\mathbf{j}$;
C is the curve (e^t, e^{t+1}), $-1 \le t \le 0$.

$$Q_x - P_y = [-4xy\sin(xy^2) - 2x^2y^3\cos(xy^2)]$$
$$- [-2xy\sin(xy^2) - 2xy\sin(xy^2) - 2x^2y^3\cos(xy^2)]$$
$$= \underline{0}$$

$\therefore$ Find f s.t. $\nabla f = \vec{F}$

$f_x = \cos(xy^2) - xy^2\sin(xy^2) \Rightarrow f = x\cos(xy^2) + g(y)$

$\therefore f_y = -2x^2y\sin(xy^2) + g'(y) \Rightarrow g'(y) = 0 \Rightarrow g(y) = K$

$\therefore f(x,y) = x\cos(xy^2) + K$, K a constant

At $t = -1$, (x,y) for C is $(e^{-1}, 1)$

$t = 0$, (x,y) for C is $(1, e)$

$\therefore \int_C \vec{F} \cdot d\vec{s} = f(1,e) - f(e^{-1}, 1)$
$$= \cos(e^2) - e^{-1}\cos(e^{-1})$$

20.

20. Prove Theorem 8. [HINT: Define $\mathbf{G} = G_1\mathbf{i} + G_2\mathbf{j} + G_3\mathbf{k}$ by

$$G_1(x, y, z) = \int_0^z F_2(x, y, t)\, dt - \int_0^y F_3(x, t, 0)\, dt$$
$$G_2(x, y, z) = -\int_0^z F_1(x, y, t)\, dt$$

and $G_3(x, y, z) = 0$.]

It is given that $\vec{F} = (F_1, F_2, F_3)$ and

$$\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 0 \quad (\operatorname{div} \vec{F} = 0) \qquad [0]$$

We need to show $\nabla \times \vec{G} = \vec{F}$, given $\vec{G}$ in the problem.

$$\nabla \times \vec{G} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ G_1 & G_2 & G_3 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ G_1 & G_2 & 0 \end{vmatrix}$$

$$= \left(-\frac{\partial G_2}{\partial z}, \frac{\partial G_1}{\partial z}, \frac{\partial G_2}{\partial x} - \frac{\partial G_1}{\partial y} \right)$$

$\therefore$ Need to find these 4 different derivatives.

Using the Fundamental Theorem of Calculus,

$$\frac{\partial G_2}{\partial z} = \frac{\partial}{\partial z} - \int_0^z F_1(x,y,t)\,dt = -F_1(x,y,z)$$

$$\therefore -\frac{\partial G_2}{\partial z} = F_1(x,y,z) \qquad [1]$$

$$\frac{\partial G_1}{\partial z} = \frac{\partial}{\partial z}\left[\int_0^z F_2(x,y,t)\,dt - \int_0^y F_3(x,t,0)\,dt \right]$$

$$= F_2(x,y,z) - 0 = F_2 \quad \therefore \frac{\partial G_1}{\partial z} = F_2(x,y,z) \qquad [2]$$

Now using differention under the integral,

$$\frac{\partial G_2}{\partial x} = \frac{\partial}{\partial x}\left(-\int_0^z F_1(x,y,t)\,dt\right) = \int_0^z -\frac{\partial F_1}{\partial x}(x,y,t)\,dt$$

$$\frac{\partial G_1}{\partial y} = \frac{\partial}{\partial y}\left(\int_0^z F_2(x,y,t)\,dt - \int_0^y F_3(x,t,0)\,dt\right)$$

$$= \int_0^z \frac{\partial F_2}{\partial y}(x,y,t)\,dt - F_3(x,y,0)$$

$$\therefore \frac{\partial G_2}{\partial x} - \frac{\partial G_1}{\partial y} = \int_0^z \left(-\frac{\partial F_1}{\partial x}(x,y,t) - \frac{\partial F_2}{\partial y}(x,y,t)\right)dt + F_3(x,y,0)$$

From [0], $\operatorname{div}\vec{F} = 0$, $-\frac{\partial F_1}{\partial x} - \frac{\partial F_2}{\partial y} = \frac{\partial F_3}{\partial z}$

$$\therefore \frac{\partial G_2}{\partial x} - \frac{\partial G_1}{\partial y} = \int_0^z \frac{\partial F_3}{\partial z}(x,y,t)\,dt + F_3(x,y,0)$$

$$= F_3(x,y,t)\Big|_{t=0}^{t=z} + F_3(x,y,0)$$

$$= F_3(x,y,z) - F(x,y,0) + F_3(x,y,0) = F_3(x,y,z)$$

$$\therefore \frac{\partial G_2}{\partial x} - \frac{\partial G_1}{\partial y} = F_3(x,y,z) \qquad [3]$$

$\therefore$ From [1], [2], [3], $\nabla \times \vec{G} = (F_1, F_2, F_3) = \vec{F}$

21. **21.** Is each of the following vector fields the curl of some other vector field? If so, find the vector field.

(a) $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

(b) $\mathbf{F} = (x^2 + 1)\mathbf{i} + (z - 2xy)\mathbf{j} + y\mathbf{k}$

(a) $\operatorname{div} \vec{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = \frac{\partial (x)}{\partial x} + \frac{\partial (y)}{\partial y} + \frac{\partial (z)}{\partial z}$

$= 1 + 1 + 1 = 3 \neq 0 \quad \therefore$ No

(b) $\operatorname{div} \vec{F} = (2x) + (-2x) + (0) = 0 \quad \therefore$ Yes

Use method of #20 above. Let $\vec{G}$ be s.t. $\nabla \times \vec{G} = \vec{F}$

$\therefore$ Let $G_3 = 0$

$$G_2 = -\int_0^z F_1(x, y, t)\, dt = -\int_0^z (x^2 + 1)\, dt = -x^2 z - z$$

$$G_1 = \int_0^z F_2(x, y, t)\, dt - \int_0^y F_3(x, t, 0)\, dt$$

$$= \int_0^z (t - 2xy)\, dt - \int_0^y t\, dt$$

$$= \frac{t^2}{2} - 2xyt \Big|_{t=0}^{t=z} - \frac{t^2}{2} \Big|_0^y$$

$$= \frac{z^2}{2} - 2xyz - \frac{y^2}{2}$$

$$\therefore \vec{G} = \left[\frac{z^2}{2} - 2xyz - \frac{y^2}{2},\ -x^2 z - z,\ 0 \right]$$

22. **22.** Let $\mathbf{F} = xz\mathbf{i} - yz\mathbf{j} + y\mathbf{k}$. Verify that $\nabla \cdot \mathbf{F} = 0$. Find a $\mathbf{G}$ such that $\mathbf{F} = \nabla \times \mathbf{G}$.

$$\nabla \cdot \vec{F} = \frac{\partial (xz)}{\partial x} + \frac{\partial (-yz)}{\partial y} + \frac{\partial (y)}{\partial z} = (z) + (-z) + (0) = 0$$

Use method of #20 to create $\vec{G}$ s.t. $\nabla \times \vec{G} = \vec{F}$

$\therefore G_3 = 0$

$$G_2 = -\int_0^z F_1(x, y, t)\, dt = -\int_0^z xt\, dt = -\frac{xt^2}{2}\Big|_{t=0}^{t=z}$$

$$= -\frac{xz^2}{2}$$

$$G_1 = \int_0^z F_2(x, y, t)\, dt - \int_0^y F_3(x, t, 0)\, dt$$

$$= \int_0^z -yt\, dt - \int_0^y t\, dt = -y\frac{t^2}{2}\Big|_{t=0}^{t=z} - \frac{t^2}{2}\Big|_0^y$$

$$= -y\frac{z^2}{2} - \frac{y^2}{2}$$

$$\therefore \vec{G} = \left[-\frac{yz^2}{2} - \frac{y}{2}, -\frac{xz^2}{2}, 0 \right]$$

23. **23.** Repeat Exercise 22 for $\mathbf{F} = y^2\mathbf{i} + z^2\mathbf{j} + x^2\mathbf{k}$.

$$\nabla \cdot \vec{F} = \frac{\partial (y^2)}{\partial x} + \frac{\partial (z^2)}{\partial y} + \frac{\partial (x^2)}{\partial z} = 0 + 0 + 0 = \underline{0}$$

Use method from #20 to get $\vec{G}$ s.t. $\nabla \times \vec{G} = \vec{F}$

$$\therefore G_3 = 0$$

$$G_2 = -\int_0^z F_1(x, y, t)\, dt = -\int_0^z y^2\, dt = -y^2 z$$

$$G_1 = \int_0^z F_2(x, y, t)\, dt - \int_0^y F_3(x, t, 0)\, dt$$

$$= \int_0^z t^2\, dt - \int_0^y x^2\, dt = \frac{z^3}{3} - x^2 y$$

$$\therefore \vec{G} = \left[\frac{z^3}{3} - x^2 y, -y^2 z, 0 \right]$$

Alternatively, try using simplicity:

$$\nabla \times \vec{G} = \left\{ \frac{\partial G_3}{\partial y} - \frac{\partial G_2}{\partial z}, \frac{\partial G_1}{\partial z} - \frac{\partial G_3}{\partial x}, \frac{\partial G_2}{\partial x} - \frac{\partial G_1}{\partial y} \right\}$$

Try $-\frac{\partial G_2}{\partial z} = \frac{\partial G_3}{\partial x} = \frac{\partial G_1}{\partial y} = 0$ so that

$$\nabla \times \vec{G} = \left\{ \frac{\partial G_3}{\partial y}, \frac{\partial G_1}{\partial z}, \frac{\partial G_2}{\partial x} \right\} = \{ y^2, z^2, x^2 \}$$

$$\therefore G_3 = \frac{y^3}{3}, \; G_1 = \frac{z^3}{3}, \; G_2 = \frac{x^3}{3}$$

$$\therefore \vec{G} = \left\{ \frac{z^3}{3}, \frac{x^3}{3}, \frac{y^3}{3} \right\}$$

24. **24.** Let $\mathbf{F} = xe^y\mathbf{i} - (x\cos z)\mathbf{j} - ze^y\mathbf{k}$. Find a $\mathbf{G}$ such that $\mathbf{F} = \nabla \times \mathbf{G}$.

$$\nabla \cdot \vec{F} = \frac{\partial (xe^y)}{\partial x} + \frac{\partial (-x\cos z)}{\partial y} + \frac{\partial (-ze^y)}{\partial z} = (e^y) + (0) + (-e^y) = 0$$

Use method from #20 to get $\vec{G}$ s.t. $\nabla \vec{G} = \vec{F}$

$$\therefore G_3 = 0$$

$$G_2 = -\int_0^z F_1(x, y, t)\, dt = -\int_0^z xe^y\, dt = -xze^y$$

$$G_1 = \int_0^z F_2(x,y,t)\,dt - \int_0^y F_3(x,t,0)\,dt$$

$$= \int_0^z -x\cos t\,dt - \int_0^y 0\,dt = -x\sin t\Big|_{t=0}^{t=z}$$

$$= -x\sin z$$

$$\therefore \vec{G} = (-x\sin z, -xze^y, 0)$$

Check: $\nabla \times \vec{G} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -x\sin z & -xze^y & 0 \end{vmatrix}$

$$= (xe^y, -x\cos z, -ze^y) = \vec{F}$$

25. **25.** Let $\mathbf{F} = (x\cos y)\mathbf{i} - (\sin y)\mathbf{j} + (\sin x)\mathbf{k}$. Find a $\mathbf{G}$ such that $\mathbf{F} = \nabla \times \mathbf{G}$.

$$\nabla \cdot \vec{F} = \frac{\partial(x\cos y)}{\partial x} + \frac{\partial(-\sin y)}{\partial y} + \frac{\partial(\sin x)}{\partial z} = (\cos y) + (-\cos y) + 0$$
$$= \underline{0}$$

Using method from #20 above to find $\vec{G}$ s.t. $\nabla \times \vec{G} = \vec{F}$,

$G_3 = 0$

$$G_2 = -\int_0^z F_1(x,y,t)\,dt = -\int_0^z x\cos y\,dt = -xz\cos y$$

$$G_1 = \int_0^z F_2(x,y,t)\,dt - \int_0^y F_3(x,t,0)\,dt$$

$$= \int_0^z -\sin y\,dt - \int_0^y \sin x\,dt$$

$$= -z\sin y - y\sin x$$

$$\therefore \vec{G}(x,y,z) = \underline{\underline{[-z\sin y - y\sin x,\ -xz\cos y,\ 0]}}$$

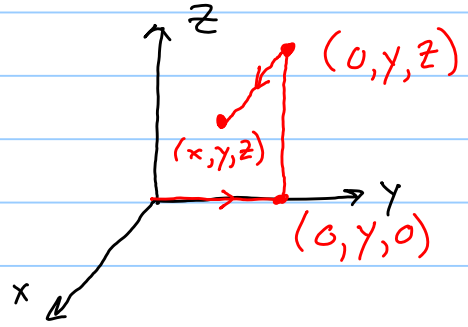
Check: $\frac{\partial G_3}{\partial y} - \frac{\partial G_2}{\partial z} = 0 - (-x\cos y) = x\cos y = F_1$

$$\frac{\partial G_1}{\partial z} - \frac{\partial G_3}{\partial x} = (-\sin y) - (0) = -\sin y = F_2$$

$$\frac{\partial G_2}{\partial x} - \frac{\partial G_1}{\partial y} = (-z\cos y) - (-z\cos y - \sin x) = \sin x = F_3$$

26.

26. By using different paths from (0, 0, 0) to (x, y, z), show that the function f defined in the proof of Theorem 7 for "condition (ii) implies condition (iii)" satisfies $\partial f/\partial x = F_1$ and $\partial f/\partial y = F_2$.



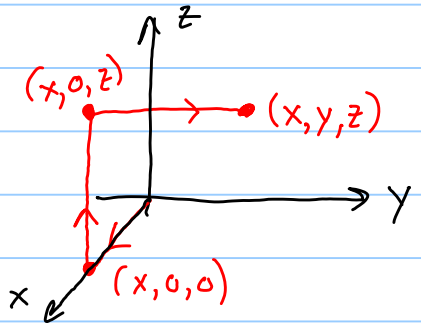
Let $f(x,y,z) =$

$$\int_0^x F_1(t,y,z)\,dt + \int_0^y F_2(0,t,0)\,dt + \int_0^z F_3(0,y,t)\,dt$$

$(0,y,z) \to (x,y,z)$ $\quad (0,0,0) \to (0,y,0)$ $\quad (0,y,0) \to (0,y,z)$

$$\therefore \frac{\partial f}{\partial x} = F_1(x,y,z) + 0 + 0 = F_1$$

Now let $f(x,y,z) =$



$$\int_0^x F_1(t,0,0)\,dt + \int_0^y F_2(x,t,z)\,dt + \int_0^z F_3(x,0,t)\,dt$$

$(0,0,0) \to (x,0,0)$ $\quad (x,0,z) \to (x,y,z)$ $\quad (x,0,0) \to (x,0,z)$

$$\therefore \frac{\partial f}{\partial y} = 0 + F_2(x,y,z) + 0 = F_2$$

27.

27. Let $\mathbf{F}$ be the vector field on $\mathbb{R}^3$ given by $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$.

(a) Show that $\mathbf{F}$ is rotational, that is, $\mathbf{F}$ is not irrotational.

(b) Suppose $\mathbf{F}$ represents the velocity vector field of a fluid. Show that if we place a cork in this fluid, it will revolve in a plane parallel to the xy plane, in a circular trajectory about the z axis.

(c) In what direction does the cork revolve?

(a) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -y & x & 0 \end{vmatrix} = (0, 0, 1-(-1)) = (0,0,2)$

$\neq \vec{0}$

(b) Let $\vec{c}(t) = [x(t), y(t), z(t)]$ be a parametrization for a point in the fluid at time t.

The point will follow along $\vec{F}$ (flow line)

$\therefore \vec{F}(\vec{c}(t)) = [-y(t), x(t), 0] = \vec{c}'(t) = (x', y', z')$

$\therefore z'(t) = 0 \Rightarrow z(t) = K$, a constant

$$x'(t) = -y(t) \qquad [1]$$

$$y'(t) = x(t) \qquad [2]$$

From [1], $x''(t) = -y'(t)$

and $-y'(t) = -x(t)$ from [2]

$\therefore x''(t) = -x(t)$, or $x''(t) + x(t) = 0$ [3]

Similarly, $y''(t) = x'(t)$ from [2]

$\therefore y''(t) = -y(t)$ from [1]

or $y''(t) + y(t) = 0$ [4]

$\therefore x(t) = A\sin t + B\cos t$, A, B constants

$y(t) = C\sin t + D\cos t$, C, D constants

But $\therefore x''(t) = -A\sin t - B\cos t$

and since $x''(t) = -y'(t) = -C\cos t + D\sin t$

then $-A\sin t - B\cos t = -C\cos t + D\sin t$

Letting $t = 0$, $-B = -C$, or $C = B$

$t = \frac{\pi}{2}$, $-A = D$

$\therefore x(t) = A\sin t + B\cos t$

$y(t) = B\sin t - A\cos t$

$\therefore \vec{c}(t) = [A\sin t + B\cos t,\ B\sin t - A\cos t,\ K]$

Note $[x(t)]^2 + [y(t)]^2 =$

$A^2\sin^2 t + 2AB\sin t\cos t + B^2\cos^2 t$
$+ B^2\sin^2 t - 2AB\sin t\cos t + A^2\cos^2 t$

$= (A^2+B^2)\sin^2 t + (A^2+B^2)\cos^2 t = A^2+B^2$

which is the equation of a circle of

radius $\sqrt{A^2+B^2}$

$\therefore \vec{c}(t)$ moves in a circle in a plane parallel to the xy-plane, through point $(0,0,k)$.

(c) As viewed from above (positive z, looking down at xy-plane), $\vec{F}$ at the point $(x,y) = (1,0)$ is

$\vec{F} = (-0)\hat{i} + (1)\hat{j} = \hat{j}$, and so is pointing counterclockwise.

28.

28. Let $\mathbf{G}$ be the vector field on $\mathbb{R}^3\backslash\{z \text{ axis}\}$ defined by

$$\mathbf{G} = \frac{-y}{x^2+y^2}\,\mathbf{i} + \frac{x}{x^2+y^2}\,\mathbf{j}.$$

(a) Show that $\mathbf{G}$ is irrotational.

(b) Show that the result of Exercise 27(b) holds for $\mathbf{G}$ also.

(c) How can we resolve the fact that the trajectories of $\mathbf{F}$ and $\mathbf{G}$ are both the same (circular about the z axis) yet $\mathbf{F}$ is rotational and $\mathbf{G}$ is not? [HINT: The property of being rotational is a local condition, that is, a property of the fluid in the neighborhood of a point.]

(a) $\nabla \times \vec{G} = \left[\dfrac{\partial\left(\frac{x}{x^2+y^2}\right)}{\partial x} - \dfrac{\partial\left(\frac{-y}{x^2+y^2}\right)}{\partial y}\right]\hat{k}$

$= \left[\dfrac{(x^2+y^2)(1)-(x)(2x)}{(x^2+y^2)^2} - \dfrac{(x^2+y^2)(-1)-(-y)(2y)}{(x^2+y^2)^2}\right]\hat{k}$

$= \left[\dfrac{2(x^2+y^2)-2x^2-2y^2}{(x^2+y^2)^2}\right]\hat{k} = 0\hat{k} = \vec{0}$

$\nabla \times \vec{G} = \vec{0} \Rightarrow \vec{G}$ is irrotational

(b) Let $\vec{c}(t) = [x(t), y(t), z(t)]$ be a parametrization for a point in the fluid at time t. The point will follow along $\vec{G}$ (flow line).

$$\therefore \vec{G}(\vec{c}(t)) = \vec{c}'(t) = [x', y', z']$$

$$\therefore \left[\frac{-y}{x^2+y^2}, \frac{x}{x^2+y^2}, 0 \right] = [x', y', z']$$

$$\therefore z' = 0 \Rightarrow z(t) = k, \text{ a constant.}$$

$$x' = \frac{-y}{x^2+y^2} \qquad y' = \frac{x}{x^2+y^2}$$

$$\therefore x'' = \frac{(x^2+y^2)(-y') - (-y)(2xx' + 2yy')}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2)\left(-\frac{x}{x^2+y^2}\right) + 2xy\left(\frac{-y}{x^2+y^2}\right) + 2y^2\left(\frac{x}{x^2+y^2}\right)}{(x^2+y^2)^2}$$

$$= \frac{-x}{(x^2+y^2)^2} \qquad \therefore x'' + \frac{x}{(x^2+y^2)^2} = 0 \quad [i]$$

$$y'' = \frac{(x^2+y^2)(x') - (x)(2xx' + 2yy')}{(x^2+y^2)^2}$$

$$= \frac{(x^2+y^2)\left(\frac{-y}{x^2+y^2}\right) - 2x^2\left(\frac{-y}{x^2+y^2}\right) - 2xy\left(\frac{x}{x^2+y^2}\right)}{(x^2+y^2)^2}$$

$$= \frac{-y}{(x^2+y^2)^2} \quad \therefore y'' + \frac{y}{(x^2+y^2)^2} = 0 \quad [2]$$

A solution to [1] and [2] is:

$$x(t) = \cos\left(t + n\tfrac{\pi}{2}\right) \quad y(t) = \sin\left(t + n\tfrac{\pi}{2}\right)$$

where n = any real number

so if $n = 0$, $x = \cos t$, $y = \sin t$
$n = 1$, $x = -\sin t$, $y = \cos t$
etc.

and $x^2 + y^2 = 1$, so $x'' + x = 0$, $y'' + y = 0$.

$$\therefore \vec{c}(t) = \left[\cos\left(t + n\tfrac{\pi}{2}\right), \sin\left(t + n\tfrac{\pi}{2}\right), K\right]$$

So $\vec{c}(t)$ moves in a circle of radius 1, in a plane parallel to xy-plane, through point $(0,0,K)$.

(c) (c) How can we resolve the fact that the trajectories of **F** and **G** are both the same (circular about the z axis) yet **F** is rotational and **G** is not? [HINT: The property of being rotational is a local condition, that is, a property of the fluid in the neighborhood of a point.]

The trajectory has to do with revolving around an axis, not rotating about a

point. $\nabla \times \vec{F}$ and $\nabla \times \vec{G}$ have to do with rotating about a point in space. Thus, for $\vec{G}$, a point revolves around the z-axis, but does not rotate about its own axis (parallel to z-axis).

29. **29.** Let $\mathbf{F} = -(GmM\mathbf{r}/r^3)$ be the gravitational force field defined on $\mathbb{R}^3 \setminus \{\mathbf{0}\}$.

(a) Show that div $\mathbf{F} = 0$.

(b) Show that $\mathbf{F} \neq$ curl $\mathbf{G}$ for any C^1 vector field $\mathbf{G}$ on $\mathbb{R}^3 \setminus \{\mathbf{0}\}$.

(a) Let $\vec{r} = (x,y,z)$ $\therefore \vec{F}(x,y,z) = \dfrac{-GMm}{(x^2+y^2+z^2)^{3/2}}(x,y,z)$

$$\therefore \frac{\partial F_1}{\partial x} = -GMm\left[\frac{(x^2+y^2+z^2)^{3/2}(1) - (x)\left(\frac{3}{2}\right)(x^2+y^2+z^2)^{\frac{1}{2}}(2x)}{(x^2+y^2+z^2)^3}\right]$$

$$= -GMm\,\frac{(x^2+y^2+z^2)^{\frac{1}{2}}\left[(x^2+y^2+z^2) - 3x^2\right]}{(x^2+y^2+z^2)^3}$$

$$= \frac{-GMm\,(-2x^2+y^2+z^2)}{(x^2+y^2+z^2)^{5/2}}$$

Similarly, $\dfrac{\partial F_2}{\partial y} = \dfrac{-GMm\,(x^2-2y^2+z^2)}{(x^2+y^2+z^2)^{5/2}}$

$$\frac{\partial F_3}{\partial z} = -\frac{GMm\left(x^2+y^2-2z^2\right)}{\left(x^2+y^2+z^2\right)^{5/2}}$$

$$\therefore \operatorname{div}\vec{F} = -\frac{GMm\left[\left(-2x^2+y^2+z^2\right)+\left(x^2-2y^2+z^2\right)+\left(x^2+y^2-2z^2\right)\right]}{\left(x^2+y^2+z^2\right)^{5/2}}$$

$$= -\frac{GMm\,(0)}{\left(x^2+y^2+z^2\right)^{5/2}} = \underline{0}$$

(b) $$\nabla \times \vec{F} = -GMm\left(\nabla \times \frac{\vec{r}}{r^3}\right)$$

$$\nabla \times \frac{\vec{r}}{r^3} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \dfrac{x}{\left(x^2+y^2+z^2\right)^{3/2}} & \dfrac{y}{\left(x^2+y^2+z^2\right)^{3/2}} & \dfrac{z}{\left(x^2+y^2+z^2\right)^{3/2}} \end{vmatrix}$$

Note $$\frac{\partial}{\partial x}\frac{1}{\left(x^2+y^2+z^2\right)^{3/2}} = -\frac{(3/2)\left(x^2+y^2+z^2\right)^{\frac{1}{2}}(2x)}{\left(x^2+y^2+z^2\right)^{3}}$$

$$= \frac{-3x}{\left(x^2+y^2+z^2\right)^{5/2}}$$

Similarly for $\partial/\partial y$ and $\partial/\partial z$

$$\therefore \nabla \times \frac{\vec{r}}{r^3} = \left[\frac{-3yz}{\left(x^2+y^2+z^2\right)^{5/2}} - \frac{-3yz}{\left(x^2+y^2+z^2\right)^{5/2}},\right.$$

$$\frac{-3xz}{(x^2+y^2+z^2)^{5/2}} - \frac{-3xz}{(x^2+y^2+z^2)^{5/2}},$$

$$\left.\frac{-3xy}{(x^2+y^2+z^2)^{5/2}} - \frac{-3xy}{(x^2+y^2+z^2)^{5/2}}\right]$$

$$= [0,0,0]$$

But this excludes the origin $(0,0,0)$, and Theorem 8, p. 459, requires all of R^3.

(Note gravitational force is irrotational in region excluding the origin).

So try a different approach. Assume there is a $\vec{G}$ s.t. $\nabla \times \vec{G} = \vec{F}$.

Consider a sphere S of radius $r=1$ centered at the origin. Let C be a unit circle, such as the equator of S. Let S_1 be the upper hemisphere, S_2 be the lower hemisphere.

By Stokes' Theorem,

$$\iint_{S_1} (\nabla \times \vec{G}) \cdot d\vec{S} = \int_{\partial S} \vec{G} \cdot d\vec{s} = \int_{C^+} \vec{G} \cdot d\vec{s},$$

where C^+ is assigned a positive orientation for C.

$$\therefore \iint_{S_2} (\nabla \times \vec{G}) \cdot d\vec{S} = -\int_{\partial S} \vec{G} \cdot d\vec{s} = \int_{C^-} \vec{G} \cdot d\vec{s}$$

$$\therefore \iint_{S} (\nabla \times \vec{G}) \cdot d\vec{S} = \iint_{S_1} (\nabla \times \vec{G}) \cdot d\vec{S} + \iint_{S_2} (\nabla \times \vec{G}) \cdot d\vec{S}$$

$$= \int_{C^+} \vec{G} \cdot d\vec{s} + \int_{C^-} \vec{G} \cdot d\vec{s} = 0$$

$\therefore$ Since $\nabla \times \vec{G} = \vec{F}$, $\iint_S \vec{F} \cdot d\vec{S} = 0$ [1]

But $\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \vec{n}\, dS$

$$= -GMm \iint_S \frac{\vec{r} \cdot \vec{n}}{\|\vec{r}\|^3}\, dS$$

But $\|\vec{r}\| = 1$, and $\vec{r} \cdot \vec{n} = \|\vec{r}\|\|\vec{n}\| \cos\theta = \cos\theta = 1$

since $\theta = 0$ ($\vec{r}$ and $\vec{n}$ are parallel on S).

$\|\vec{r}\| = 1$ because S is a unit sphere.

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = -GMm \iint_S dS = -GMm(4\pi) \quad [2]$$

$\therefore$ [1] says $\iint_S \vec{F} \cdot d\vec{S} = 0$

and [2] says $\iint_S \vec{F} \cdot d\vec{S} = -GMm(4\pi)$

$\therefore$ There can't be a $\vec{G}$ s.t. $\nabla \times \vec{G} = \vec{F}$

8.4 Gauss' Theorem

Note Title 7/11/2017

In Exercises 1 to 4, verify the divergence theorem for the given region W, boundary ∂W oriented outward, and vector field $\mathbf{F}$.

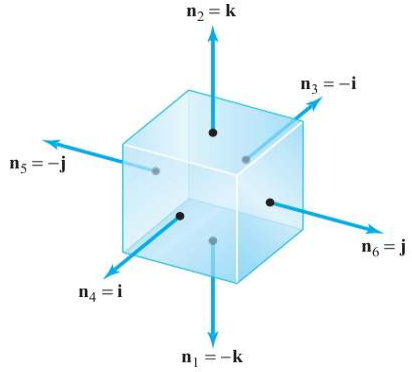
1.

1. $W = [0, 1] \times [0, 1] \times [0, 1]$
$\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$$

$$\therefore \iiint_W \nabla \cdot \vec{F}\, dV = \iiint_W 3\, dV = 3\iiint_W dV = 3(1) = \underline{3}$$

$$\iint_{\partial W} \vec{F} \cdot d\vec{S} = \iint_{\partial W} \vec{F} \cdot \vec{n}\, dS = \sum_{i=1}^{6} \iint_{S_i} \vec{F} \cdot \hat{n}_i\, dS_i$$



$$= \iint_{S_1} (x\hat{i} + y\hat{j} + 0\hat{k}) \cdot (-\hat{k})\, dS_1$$

$$+ \iint_{S_2} (x\hat{i} + y\hat{j} + (1)\hat{k}) \cdot (\hat{k})\, dS_2$$

$$+ \iint_{S_3} (0\hat{i} + y\hat{j} + z\hat{k}) \cdot (-\hat{i})\, dS_3 + \iint_{S_4} ((1)\hat{i} + y\hat{j} + z\hat{k}) \cdot (\hat{i})\, dS_4$$

$$+ \iint_{S_5} (x\hat{i} + 0\hat{j} + z\hat{k}) \cdot (-\hat{j})\, dS_5 + \iint_{S_6} (x\hat{i} + (1)\hat{j} + z\hat{k}) \cdot (\hat{j})\, dS_6$$

$$= 0 + \iint_{S_2} dS_2 + 0 + \iint_{S_4} dS_4 + 0 + \iint_{S_6} dS_6$$

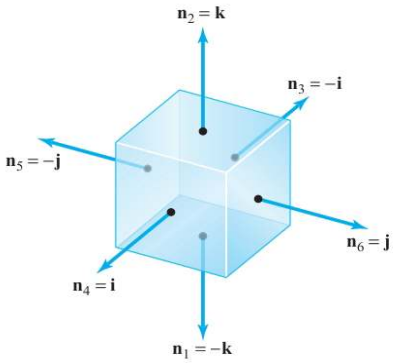
$$= 1+1+1 = \underline{3} \quad \text{as} \quad \iint_{S_i} dS_i = 1, \text{ The area of the face of a cube.}$$

$$\therefore \iiint_W \nabla \cdot \vec{F}\, dV = \underline{\iint_{\partial W} \vec{F} \cdot d\vec{S}} = 3$$

2. **2.** *W* as in Exercise 1, and $\mathbf{F} = zy\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(zy) + \frac{\partial}{\partial y}(xz) + \frac{\partial}{\partial z}(xy) = 0+0+0 = 0$$

$$\therefore \iiint_W \nabla \cdot \vec{F}\, dV = \underline{0}$$



$$\iint_{\partial W} \vec{F} \cdot d\vec{S} = \sum_{i=1}^{6} \iint_{S_i} \vec{F} \cdot \hat{n}_i\, dS_i$$

$$= \iint_{S_1} (0\hat{i} + 0\hat{j} + xy\hat{k}) \cdot (-\hat{k})\, dS_1 + \iint_{S_2} (y\hat{i} + x\hat{j} + xy\hat{k}) \cdot \hat{k}\, dS_2$$

($z = 0$ for S_1; $z = 1$ for S_2)

$$+ \iint_{S_3} (zy\hat{i} + 0\hat{j} + 0\hat{k}) \cdot (-\hat{i})\, dS_3 + \iint_{S_4} (zy\hat{i} + z\hat{j} + y\hat{k}) \cdot (\hat{i})\, dS_4$$

($x = 0$ for S_3; $x = 1$ for S_4)

$$+\iint_{S_5} (0\hat{i} + xz\hat{j} + 0\hat{k})\cdot(-\hat{j})\,dS_5 + \iint_{S_6} (z\hat{i} + xz\hat{j} + x\hat{k})\cdot(\hat{j})\,dS_6$$

($y=0$ for S_5; $y=1$ for S_6)

$$= \iint_{S_1} -xy\,dS_1 + \iint_{S_2} xy\,dS_2 + \iint_{S_3} -zy\,dS_3 + \iint_{S_4} zy\,dS_4$$

$$+ \iint_{S_5} -xz\,dS_5 + \iint_{S_6} xz\,dS_6$$

A parametrization for S_1 would be $T^1(u,v) = (u,v,0)$
S_2 would be $T^2(u,v) = (u,v,1)$.

$$\therefore T^1_u \times T^1_v = (1,0,0)\times(0,1,0) = (0,0,1)$$

$$T^2_u \times T^2_v = (1,0,0)\times(0,1,0) = (0,0,1)$$

$$\therefore \|T^1_u \times T^1_v\| = \|T^2_u \times T^2_v\| = 1$$

$$\therefore \iint_{S_1} -xy\,dS_1 = \int_0^1\int_0^1 -uv(1)\,du\,dv$$

and $$\iint_{S_2} xy\,dS_2 = \int_0^1\int_0^1 uv(1)\,du\,dv$$

$$\therefore \iint_{S_1} -xy\,dS_1 + \iint_{S_2} xy\,dS_2 = 0$$

Similarly, $\iint_{S_3} -zy\,dS_3 + \iint_{S_4} zy\,dS_4 = 0$

and $\iint_{S_5} -xz\,dS_5 + \iint_{S_6} xz\,dS_6 = 0$

$\therefore \iint_{\partial W} \vec{F} \cdot d\vec{S} = 0 + 0 + 0 = 0$

$\therefore \iiint_W \nabla \cdot \vec{F}\,dV = \iint_{\partial W} \vec{F} \cdot d\vec{S} = 0$

3. **3.** $W = \{(x, y, z) : x^2 + y^2 + z^2 \leq 1\}$ (the unit ball),
$\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

$\nabla \cdot \vec{F} = 1 + 1 + 1 = 3$

$\therefore \iiint_W (\nabla \cdot \vec{F})\,dV = \iiint_W (3)\,dV = 3\left(\frac{4}{3}\pi(1)^3\right) = 4\pi$

For ∂W, let $T(\phi, \theta) = [\cos\theta\sin\phi, \sin\theta\sin\phi, \cos\phi]$,
$0 \leq \theta \leq 2\pi$, $0 \leq \phi \leq \pi$

$\therefore \vec{F}(T(\phi, \theta)) = [\cos\theta\sin\phi, \sin\theta\sin\phi, \cos\phi]$

$T_\phi = (\cos\theta\cos\phi, \sin\theta\cos\phi, -\sin\phi)$

$$T_\theta = (-\sin\theta\sin\phi, \cos\theta\sin\phi, 0)$$

$$T_\phi \times T_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos\theta\cos\phi & \sin\theta\cos\phi & -\sin\phi \\ -\sin\theta\sin\phi & \cos\theta\sin\phi & 0 \end{vmatrix}$$

$$= (\cos\theta\sin^2\phi, \sin\theta\sin^2\phi, \cos\phi\sin\phi)$$

Note: $\cos\phi\sin\phi\,\hat{k}$ means $T_\phi \times T_\theta$ points "out".

$$\therefore \vec{F}(T(\phi,\theta)) \cdot (T_\phi \times T_\theta) =$$

$$(\cos^2\theta\sin^3\phi) + (\sin^2\theta\sin^3\phi) + (\cos^2\phi\sin\phi)$$

$$= \sin^3\phi + \cos^2\phi\sin\phi = (\sin^2\phi + \cos^2\phi)\sin\phi = \sin\phi$$

$$\therefore \iint_{\partial W} \vec{F} \cdot d\vec{S} = \int_0^{2\pi}\int_0^{\pi} \vec{F} \cdot (T_\phi \times T_\theta)\, d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \sin\phi\, d\phi\, d\theta = 2\pi \int_0^{\pi} \sin\phi\, d\phi = 2\pi\left[-\cos\phi\right]_0^{\pi}$$

$$= 2\pi\left[(1) - (-1)\right] = \underline{4\pi}$$

$$\therefore \iiint_W (\nabla \cdot \vec{F})\, dV = \iint_{\partial W} \vec{F} \cdot d\vec{S} = 4\pi$$

$\Longleftarrow$

4. **4.** *W* as in Exercise 3, and $\mathbf{F} = -y\mathbf{i} + x\mathbf{j} + z\mathbf{k}$

$\nabla \cdot \vec{F} = 0 + 0 + 1 = 1.$

$$\therefore \iiint_W (\nabla \cdot \vec{F})\, dV = \iiint_W dV = \frac{4}{3}\pi$$

From #3, $T_\phi \times T_\theta = (\cos\theta \sin^2\phi, \sin\theta \sin^2\phi, \cos\phi \sin\phi)$

$$\vec{F}(T(\phi,\theta)) = (-\sin\theta\sin\phi, \cos\theta\sin\phi, \cos\phi)$$

$$\therefore \vec{F} \cdot (T_\phi \times T_\theta) = (-\cos\theta\sin\theta\sin^3\phi) + (\cos\theta\sin\theta\sin^3\phi) + (\cos^2\phi\sin\phi)$$

$$= \cos^2\phi\sin\phi$$

$$\therefore \iint_{\partial W} \vec{F} \cdot d\vec{S} = \int_0^{2\pi}\int_0^{\pi} \vec{F} \cdot (T_\phi \times T_\theta)\, d\phi\, d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \cos^2\phi\sin\phi\, d\phi\, d\theta = 2\pi\left[-\frac{\cos^3\phi}{3}\right]_{\phi=0}^{\phi=\pi}$$

$$= 2\pi\left[\left(\frac{1}{3}\right) - \left(-\frac{1}{3}\right)\right] = \frac{4}{3}\pi$$

$$\therefore \iiint_W (\nabla \cdot \vec{F})\, dV = \iint_{\partial W} \vec{F} \cdot d\vec{S} = \frac{4}{3}\pi$$

5. **5.** Use the divergence theorem to calculate the flux of $\mathbf{F} = (x - y)\mathbf{i} + (y - z)\mathbf{j} + (z - x)\mathbf{k}$ out of the unit sphere.

$$\text{Flux} = \iint_S \vec{F} \cdot d\vec{S} = \iiint_W (\nabla \cdot \vec{F})\, dV$$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x-y) + \frac{\partial}{\partial y}(y-z) + \frac{\partial}{\partial z}(z-x) = 1 + 1 + 1 = 3$$

$$\therefore \iiint_W (3)\, dV = 3\left(\tfrac{4}{3}\pi\right) = \underline{\underline{4\pi}}$$

as volume of sphere of radius $R = \frac{4}{3}\pi R^3$

6. **6.** Let $\mathbf{F} = x^3\mathbf{i} + y^3\mathbf{j} + z^3\mathbf{k}$. Evaluate the surface integral of $\mathbf{F}$ over the unit sphere.

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_W (\nabla \cdot \vec{F})\, dV = \iiint_W (3x^2 + 3y^2 + 3z^2)\, dV$$

$x^2 + y^2 + z^2 = 1$ for unit sphere

$$= \iiint_W 3\, dV = 3\left(\tfrac{4}{3}\pi\right) = \underline{\underline{4\pi}}$$

7. **7.** Evaluate $\iint_{\partial W} \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and W is the unit cube (in the first octant). Perform the calculation directly and check by using the divergence theorem.

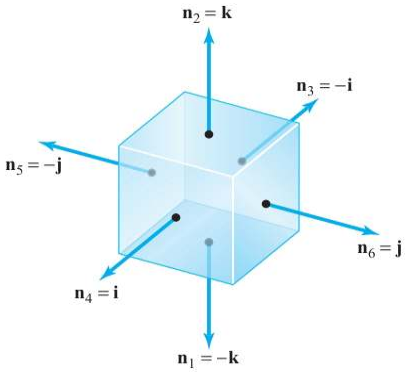
This was done in #1 above. $\iiint_W \nabla \cdot \vec{F}\, dV = \underline{\underline{3}}$

8.

8. Repeat Exercise 7 for

(a) $\mathbf{F} = \mathbf{i} + \mathbf{j} + \mathbf{k}$

(b) $\mathbf{F} = x^2\mathbf{i} + x^2\mathbf{j} + z^2\mathbf{k}$



(a) $\nabla \cdot \vec{F} = 0+0+0 = 0 \Rightarrow \iiint_W (\nabla \cdot \vec{F})\, dV = 0$

$$\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \hat{n}\, dS = \sum_{i=1}^{6} \iint_{S_i} \vec{F} \cdot \hat{n}_i\, dS_i$$

$$= \iint_{S_1} (\hat{i}+\hat{j}+\hat{k}) \cdot (-\hat{k})\, dS_1 + \iint_{S_2} (\hat{i}+\hat{j}+\hat{k}) \cdot (\hat{k})\, dS_2$$

$$+ \iint_{S_3} (\hat{i}+\hat{j}+\hat{k}) \cdot (-\hat{i})\, dS_3 + \iint_{S_4} (\hat{i}+\hat{j}+\hat{k}) \cdot (\hat{i})\, dS_4$$

$$+ \iint_{S_5} (\hat{i}+\hat{j}+\hat{k}) \cdot (-\hat{j})\, dS_5 + \iint_{S_6} (\hat{i}+\hat{j}+\hat{k}) \cdot (\hat{j})\, dS_6$$

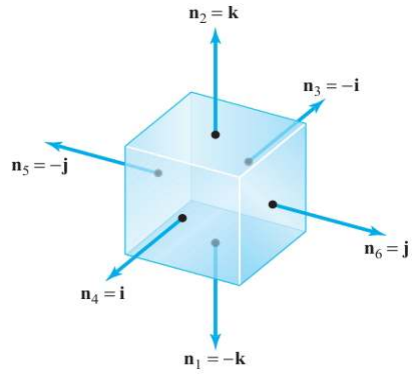
$$= -\iint_{S_1} dS_1 + \iint_{S_2} dS_2 - \iint_{S_3} dS_3 + \iint_{S_4} dS_4 - \iint_{S_5} dS_5 + \iint_{S_6} dS_6$$

$$= -1 + 1 - 1 + 1 - 1 + 1 = \underline{0}$$

(5) $\nabla \cdot \vec{F} = 2x + 2z \quad \therefore \iiint_W (2x+2z)\,dV$

$$= \int_0^1 \int_0^1 \int_0^1 2x + 2z \, dx\,dy\,dz = \int_0^1 \int_0^1 x^2 + 2xz \Big|_0^1 \, dy\,dz$$

$$= \int_0^1 \int_0^1 (1+2z)\,dy\,dz = \int_0^1 1 + 2z \, dz = 2 + z^2 \Big|_0^1 = \underline{2}$$



$$\iint_S \vec{F} \cdot \hat{n}\, dS =$$

$$\iint_{S_1} (x^2\hat{i} + x^2\hat{j} + z^2\hat{k}) \cdot (-\hat{k})\, dS_1 + \iint_{S_2} (x^2\hat{i} + x^2\hat{j} + z^2\hat{k}) \cdot \hat{k}\, dS_2$$

$$+ \iint_{S_3} (x^2\hat{i} + x^2\hat{j} + z^2\hat{k})(-\hat{i})\, dS_3 + \iint_{S_4} (x^2\hat{i} + x^2\hat{j} + z^2\hat{k}) \cdot (\hat{i})\, dS_4$$

$$+ \iint_{S_5} (x^2\hat{i} + x^2\hat{j} + z^2\hat{k})(-\hat{j})\, dS_5 + \iint_{S_6} (x^2\hat{i} + x^2\hat{j} + z^2\hat{k}) \cdot (\hat{j})\, dS_6$$

$$= \iint_{S_1} -z^2\, dS_1 + \iint_{S_2} z^2\, dS_2 + \iint_{S_3} -x^2\, dS_3 + \iint_{S_4} x^2\, dS_4 + \iint_{S_5} -x^2\, dS_5 + \iint_{S_6} x^2\, dS_6$$

$S_1 : T(u,v) = (u,v,0)$ $S_2 : T(u,v) = (u,v,1)$

$S_3 : T(u,v) = (0,u,v)$ $S_4 : T(u,v) = (1,u,v)$

$S_5 : T(u,v) = (u,0,v)$ $S_6 : T(u,v) = (u,1,v)$

for all, $\|T_u \times T_v\| = 1$

$$\therefore = \iint_{D_1}^{z=0} 0 \, du\,dv + \iint_{D_2}^{z=1} 1 \, du\,dv + \iint_{D_3}^{x=0} 0 \, du\,dv + \iint_{D_4}^{x=1} 1 \, du\,dv$$

$$+ \iint_{D_5}^{x=u} -u^2 \, du\,dv + \iint_{D_6}^{x=u} u^2 \, du\,dv$$

$$= \overset{D_1}{0} + \overset{D_2}{1} + \overset{D_3}{0} + \overset{D_4}{1} + \int_0^1 \int_0^1 \overset{\text{cancel}}{\leftarrow} -u^2 \, du\,dv + \int_0^1 \int_0^1 u^2 \, du\,dv = \underline{2}$$

$$\therefore \iiint_W (\nabla \cdot \vec{F}) \, dV = \iint_S \vec{F} \cdot d\vec{S} = \underline{2}$$

9.

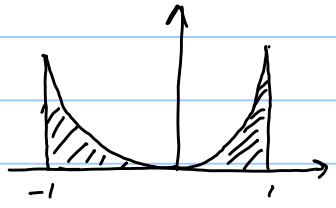
9. Let $\mathbf{F} = y\mathbf{i} + z\mathbf{j} + xz\mathbf{k}$. Evaluate $\iint_{\partial W} \mathbf{F} \cdot d\mathbf{S}$ for each of the following regions W:

(a) $x^2 + y^2 \le z \le 1$

(b) $x^2 + y^2 \le z \le 1$ and $x \ge 0$

(c) $x^2 + y^2 \le z \le 1$ and $x \le 0$

(a) This is the outside of a paraboloid



Easier to compute $\iiint_W (\nabla \cdot \vec{F})\, dV$

$\therefore \nabla \cdot \vec{F} = (0,0,x)$

$\therefore \iiint_W x\, dV$ Use cylindrical coordinates

$x = r\cos\theta \quad y = r\sin\theta \quad z = z$

$0 \le \theta \le 2\pi, \quad 0 \le r \le 1, \quad r^2 \le z \le 1$

$$= \int_0^{2\pi}\int_0^1\int_{r^2}^1 (r\cos\theta)\, r\, dz\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^1 r^2\cos\theta\, z\Big|_{z=r^2}^{z=1} dr\, d\theta = \int_0^{2\pi}\int_0^1 r^2\cos\theta - r^4\cos\theta\, dr\, d\theta$$

$$= \int_0^{2\pi} \frac{r^3}{3}\cos\theta - \frac{r^5}{5}\cos\theta\Big|_{r=0}^{r=1} d\theta = \int_0^{2\pi} \frac{\cos\theta}{3} - \frac{\cos\theta}{5}\, d\theta$$

$$= \frac{\sin\theta}{3} - \frac{\sin\theta}{5}\Big|_0^{2\pi} = 0 \quad \therefore \iint_{\partial W} \vec{F}\cdot d\vec{S} = \underline{0}$$

(5) same as (a), only $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

$$\therefore \iiint_W x\, dV = \frac{\sin\theta}{3} - \frac{\sin\theta}{5}\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$$

$$= \left(\frac{1}{3} - \frac{1}{5}\right) - \left[\left(-\frac{1}{3}\right) - \left(-\frac{1}{5}\right)\right] = \frac{2}{3} - \frac{2}{5} = \underline{\underline{\frac{4}{15}}}$$

(c) Same as (G), only $\frac{\pi}{2} \le \theta \le \frac{3}{2}\pi$

$$\therefore \iint_{\partial W} \vec{F} \cdot d\vec{S} = \iiint_W x\,dV = \frac{\sin\theta}{3} - \frac{\sin\theta}{5}\Bigg|_{\frac{\pi}{2}}^{\frac{3}{2}\pi}$$

$$= \left[\left(-\frac{1}{3}\right) - \left(-\frac{1}{5}\right)\right] - \left[\frac{1}{3} - \frac{1}{5}\right] = -\frac{2}{3} + \frac{2}{5} = \underline{\underline{-\frac{4}{15}}}$$

10. **10.** Repeat Exercise 9 for $\mathbf{F} = (x - y)\mathbf{i} + (y - z)\mathbf{j} + (z - x)\mathbf{k}$. [The solution to part (b) only is in the Study Guide to this text.]

Compute $\iiint_W (\nabla \cdot \vec{F})\,dV$ $\quad \nabla \cdot \vec{F} = 1 + 1 + 1 = 3$

$$\therefore \iint_{\partial W} \vec{F} \cdot d\vec{S} = 3 \iiint_W dV = 3 \iiint_W r\,dr\,d\theta\,dz$$

using cylindrical coordinates

(a) $0 \le \theta \le 2\pi$, $0 \le r \le 1$, $r^2 \le z \le 1$

$$\therefore 3 \int_0^{2\pi} \int_0^1 \int_{r^2}^1 r\,dz\,dr\,d\theta = 3 \int_0^{2\pi} \int_0^1 rz\Big|_{z=r^2}^{z=1} dr\,d\theta$$

$$= 3\int_0^{2\pi}\int_0^1 r - r^3\, dr\, d\theta = 3\int_0^{2\pi} \frac{r^2}{2} - \frac{r^4}{4}\Big|_0^1 d\theta$$

$$= 3\int_0^{2\pi} \frac{1}{4}\, d\theta = \frac{3}{4}(2\pi) = \underline{\frac{3}{2}\pi}$$

(b) Same as (a), only $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$

$$\therefore 3\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{4}\, d\theta = \underline{\frac{3}{4}\pi}$$

(c) Same as (a), only $\frac{\pi}{2} \le \theta \le \frac{3}{2}\pi$

$$\therefore 3\int_{\frac{\pi}{2}}^{\frac{3}{2}\pi} \frac{1}{4}\, d\theta = \underline{\frac{3}{4}\pi}$$

11. **11.** Find the flux of the vector field $\mathbf{F} = (x - y^2)\mathbf{i} + y\mathbf{j} + x^3\mathbf{k}$ out of the rectangular solid $[0, 1] \times [1, 2] \times [1, 4]$.

$$\text{Flux} = \iint_S \vec{F} \cdot d\vec{S} = \iiint_W (\nabla \cdot \vec{F})\, dV$$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x - y^2) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(x^3) = 1 + 1 + 0 = 2$$

Volume of rectangular solid $= 1 \times 1 \times 3 = 3$

$$\therefore \iiint_W (\nabla \cdot \vec{F})\, dV = \iiint_W (2)\, dV = 2\iiint_W dV = 2(3)$$

$$= \underline{6}$$

12. **12.** Evaluate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F} = 3xy^2\mathbf{i} + 3x^2y\mathbf{j} + z^3\mathbf{k}$ and S is the surface of the unit sphere.

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_W (\nabla \cdot \vec{F})\, dV = \iiint_W \frac{\partial}{\partial x}(3xy^2) + \frac{\partial}{\partial y}(3x^2y) + \frac{\partial}{\partial z}(z^3)\, dV$$

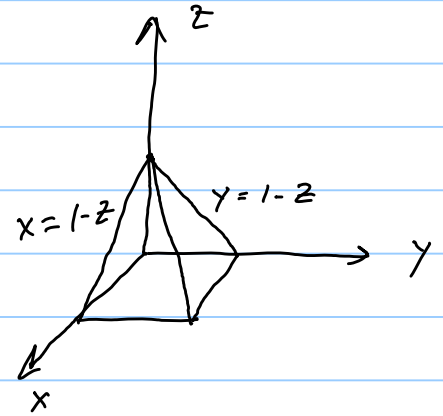
$$= \iiint_W (3y^2 + 3x^2 + 3z^2)\, dV = 3\iiint dV = 3\left(\frac{4}{3}\pi\right)$$

$x^2 + y^2 + z^2 = 1$ for unit sphere

$$= \underline{4\pi}$$ as unit sphere has volume $\frac{4}{3}\pi(1)^3$

13. **13.** Let W be the pyramid with top vertex $(0, 0, 1)$, and base vertices at $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, and $(1, 1, 0)$. Let S be the two-dimensional closed surface bounding W, oriented outward from W. Use Gauss' theorem to calculate $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where:

$$\mathbf{F}(x, y, z) = (x^2y, 3y^2z, 9z^2x).$$



$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_W (\nabla \cdot \vec{F})\, dV,$$

and $\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(3y^2 z) + \frac{\partial}{\partial z}(9z^2 x)$

$$= 2xy + 6yz + 18zx$$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \iiint_W (2xy + 6yz + 18zx)\, dV$$

$$= \int_0^1 \int_0^{1-z} \int_0^{1-z} 2xy + 6yz + 18xz \; dx\,dy\,dz$$

$$= \int_0^1 \int_0^{1-z} x^2 y + 6xyz + 9x^2 z \Big|_{x=0}^{x=1-z} dy\,dz$$

$$= \int_0^1 \int_0^{1-z} y(1-z)^2 + 6yz(1-z) + 9(1-z)^2 z \; dy\,dz$$

$$= \int_0^1 \frac{y^2}{2}(1-z)^2 + 3y^2 z(1-z) + 9y(1-z)^2 z \Big|_{y=0}^{y=1-z} dz$$

$$= \int_0^1 \frac{(1-z)^4}{2} + 3z(1-z)^3 + 9(1-z)^3 z \; dz$$

$(1-z)^3 \left[\frac{1-z}{2} + 3z + 9z \right] = (1-z)^3 \left[\frac{1}{2} + \frac{23}{2} z \right]$

$(1-z)^3 = 1 - 3z + 3z^2 - z^3$

$$= \int_0^1 \frac{1}{2} - \frac{3}{2}z + \frac{3}{2}z^2 - \frac{1}{2}z^3 + \frac{23}{2}z - \frac{69}{2}z^2 + \frac{69}{2}z^3 - \frac{23}{2}z^4 \, dz$$

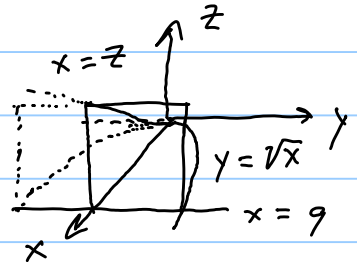
$$= \int_0^1 \frac{1}{2} + 10z - 33z^2 + 34z^3 - \frac{23}{2}z^4 \, dz$$

$$= \frac{z}{2} + 5z^2 - 11z^3 + \frac{17}{2}z^4 - \frac{23}{10}z^5 \Big|_0^1$$

$$= \frac{1}{2} + 5 - 11 + \frac{17}{2} - \frac{23}{10} = \frac{5 + 50 - 110 + 85 - 23}{10} = \frac{7}{10}$$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \frac{7}{10}$$

14. **14.** Let W be the three-dimensional solid enclosed by the surfaces $x = y^2$, $x = 9$, $z = 0$, and $x = z$. Let S be the boundary of W. Use Gauss' theorem to find the flux of $\mathbf{F}(x, y, z) = (3x - 5y)\,i + (4z - 2y)\,j + (8yz)\,k$ across S: $\iint_S \mathbf{F} \cdot d\mathbf{S}$.



$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(3x - 5y) + \frac{\partial}{\partial y}(4z - 2y) + \frac{\partial}{\partial z}(8yz) = 3 - 2 + 8y$$

$$\therefore \iint_S \vec{F} \cdot d\vec{S} = \iiint_W 1 + 8y \, dV$$

$$\therefore 0 \le x \le 9, \; -\sqrt{x} \le y \le \sqrt{x}, \; 0 \le z \le x$$

$$\therefore \iiint_W (1+8y)\,dV = \int_0^9 \int_0^x \int_{-\sqrt{x}}^{\sqrt{x}} (1+8y)\,dy\,dz\,dx$$

$$= \int_0^9 \int_0^x y + 4y^2 \Big|_{y=-\sqrt{x}}^{y=\sqrt{x}} dz\,dx$$

$\sqrt{x} + 4x - (-\sqrt{x} + 4x)$

$$= \int_0^9 \int_0^x 2\sqrt{x}\,dz\,dx = \int_0^9 2\sqrt{x}\,z \Big|_{z=0}^{z=x} dx = \int_0^9 2x\sqrt{x}\,dx$$

$$= \int_0^9 2x^{\frac{3}{2}}\,dx = \frac{4}{5}x^{\frac{5}{2}} \Big|_0^9 = \frac{4}{5}(3)^5 = \frac{4}{5}(729)$$

$$= \frac{2916}{5}$$

15. **15.** Evaluate $\iint_{\partial W} \mathbf{F}\cdot\mathbf{n}\,dA$, where $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} - z\mathbf{k}$ and W is the unit cube in the first octant. Perform the calculation directly and check by using the divergence theorem.

$$\nabla\cdot\vec{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(-z) = 1 + 1 - 1 = 1$$

$$\therefore \iiint_W (\nabla\cdot\vec{F})\,dV = \iiint_W dV = 1 \text{ for unit cube.}$$

Note area of face of unit cube = 1.

$$\therefore \iint_{\partial W} \vec{F}\cdot\hat{n}\,dA = \iint_{\partial W} \underbrace{\vec{F}(1,y,z)}_{=(1,y,-z)}\cdot\hat{i}\,dA + \iint_{\partial W} \underbrace{\vec{F}(0,y,z)}_{=(0,y,-z)}\cdot(-\hat{i})\,dA$$

$$+ \iint_{\partial W} \underbrace{\vec{F}(x,1,z)}_{=(x,1,-z)}\cdot\hat{j}\,dA + \iint_{\partial W} \underbrace{\vec{F}(x,0,z)}_{=(x,0,-z)}(-\hat{j})\,dA$$

$$+ \iint_{\partial W} \underbrace{\vec{F}(x,y,1)}_{=(x,y,-1)}\cdot\hat{k}\,dA + \iint_{\partial W} \underbrace{\vec{F}(x,y,0)}_{=(x,y,0)}\cdot(-\hat{k})\,dA$$

$$= \iint_{\partial W}(1)\,dA + \iint_{\partial W}(0)\,dA + \iint_{\partial W}(1)\,dA + \iint_{\partial W}(0)\,dA + \iint_{\partial W}(-1)\,dA + \iint_{\partial W}(0)\,dA$$

$$= 1+0+1+0-1+0 = \underline{1}$$

16. **16.** Evaluate the surface integral $\iint_{\partial S}\mathbf{F}\cdot\mathbf{n}\,dA$, where $\mathbf{F}(x, y, z) = \mathbf{i}+\mathbf{j}+z(x^2+y^2)^2\mathbf{k}$ and ∂S is the surface of the cylinder $x^2+y^2 \le 1,\ 0 \le z \le 1$.

$$\iint_{\partial S}\vec{F}\cdot\hat{n}\,dA = \iint_{\partial S}\vec{F}\cdot d\vec{S} = \iiint_{W}(\nabla\cdot\vec{F})\,dV$$

$$\nabla\cdot\vec{F} = \frac{\partial}{\partial x}(1) + \frac{\partial}{\partial y}(1) + \frac{\partial}{\partial z}\left(z(x^2+y^2)^2\right) = 0+0+(x^2+y^2)^2$$

Using cylindrical coordinates, $x = r\cos\theta,\ y = r\sin\theta,\ z = z,$

$x^2+y^2=r^2$, $0 \le \theta \le 2\pi$, $0 \le r \le 1$, $0 \le z \le 1$

$dx\,dy\,dz = r\,dr\,d\theta\,dz$

$$\therefore \iiint_W (x^2+y^2)^2\,dV = \int_0^{2\pi}\int_0^1\int_0^1 r^4 \cdot r\,dr\,dz\,d\theta$$

$$= 2\pi\left[\frac{r^6}{6}\right]_{r=0}^{r=1} = \frac{\pi}{3}$$

17. **17.** Prove that

$$\iiint_W (\nabla f)\cdot\mathbf{F}\,dx\,dy\,dz = \iint_{\partial W} f\mathbf{F}\cdot\mathbf{n}\,dS - \iiint_W f\nabla\cdot\mathbf{F}\,dx\,dy\,dz.$$

Note that $div(f\vec{F}) = f(div\vec{F}) + \vec{F}\cdot(grad f)$

or, $\nabla\cdot(f\vec{F}) = f\,\nabla\cdot\vec{F} + \vec{F}\cdot(\nabla f)$ (text, p. 255)

From Gauss' Divergence Theorem,

$$\iiint_W \nabla\cdot(f\vec{F})\,dV = \iint_{\partial W}(f\vec{F})\cdot d\vec{S} = \iint_{\partial W}(f\vec{F})\cdot\hat{n}\,dS$$

$$\therefore \iiint_W \nabla\cdot(f\vec{F})\,dV = \iiint_W \left[f\,\nabla\cdot\vec{F} + \vec{F}\cdot(\nabla f)\right]dV = \iint_{\partial W}(f\vec{F})\cdot\hat{n}\,dS$$

So, $\iiint_W [f \nabla \cdot \vec{F} + \vec{F} \cdot (\nabla f)] \, dV =$

$$\iiint_W f \nabla \cdot \vec{F} \, dV + \iiint_W \vec{F} \cdot (\nabla f) \, dV = \iint_{\partial W} (f \vec{F}) \cdot \hat{n} \, dS$$

and since $\vec{F} \cdot (\nabla f) = \nabla f \cdot \vec{F}$,

$$\iiint_W \nabla f \cdot \vec{F} \, dV = \iint_{\partial W} (f \vec{F}) \cdot \hat{n} \, dS - \iiint_W f \nabla \cdot \vec{F} \, dV$$

18.

18. Prove the identity

$$\nabla \cdot (\mathbf{F} \times \mathbf{G}) = \mathbf{G} \cdot (\nabla \times \mathbf{F}) - \mathbf{F} \cdot (\nabla \times \mathbf{G}).$$

This is #8 in text, p. 255

Let $\vec{F} = (F_1, F_2, F_3)$, $\vec{G} = (G_1, G_2, G_3)$

$$\vec{F} \times \vec{G} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ F_1 & F_2 & F_3 \\ G_1 & G_2 & G_3 \end{vmatrix} = \left[F_2 G_3 - F_3 G_2, F_3 G_1 - F_1 G_3, F_1 G_2 - F_2 G_1 \right]$$

$$\therefore \nabla \cdot (\vec{F} \times \vec{G}) = \frac{\partial}{\partial x}(F_2 G_3 - F_3 G_2) + \frac{\partial}{\partial y}(F_3 G_1 - F_1 G_3) + \frac{\partial}{\partial z}(F_1 G_2 - F_2 G_1)$$

$$= F_2 \frac{\partial G_3}{\partial x} + G_3 \frac{\partial F_2}{\partial x} - F_3 \frac{\partial G_2}{\partial x} - G_2 \frac{\partial F_3}{\partial x}$$

$$+ F_3 \frac{\partial G_1}{\partial y} + G_1 \frac{\partial F_3}{\partial y} - F_1 \frac{\partial G_3}{\partial y} - G_3 \frac{\partial F_1}{\partial y}$$

$$+ F_1 \frac{\partial G_2}{\partial z} + G_2 \frac{\partial F_1}{\partial z} - F_2 \frac{\partial G_1}{\partial z} - G_1 \frac{\partial F_2}{\partial z}$$

$$= G_1\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right) + G_2\left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}\right) + G_3\left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)$$

$$- F_1\left(\frac{\partial G_3}{\partial y} - \frac{\partial G_2}{\partial z}\right) - F_2\left(\frac{\partial G_1}{\partial z} - \frac{\partial G_3}{\partial x}\right) - F_3\left(\frac{G_2}{\partial x} - \frac{\partial G_1}{\partial y}\right)$$

$$= (G_1, G_2, G_3) \cdot \left[\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}, \frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x}, \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right]$$

$$- (F_1, F_2, F_3) \cdot \left[\frac{\partial G_3}{\partial y} - \frac{\partial G_2}{\partial z}, \frac{\partial G_1}{\partial z} - \frac{\partial G_3}{\partial x}, \frac{\partial G_2}{\partial x} - \frac{\partial G_1}{\partial y}\right]$$

$$= \vec{G} \cdot (\nabla \times \vec{F}) - \vec{F} \cdot (\nabla \times \vec{G})$$

19. **19.** Show that $\iiint_W (1/r^2)\, dx\, dy\, dz = \iint_{\partial W} (\mathbf{r} \cdot \mathbf{n}/r^2)\, dS$, where $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

(a) Assume $(0,0,0) \notin W$, so $r \neq 0$. Using Gauss' Divergence Th.,

$$\iint_{\partial W} \frac{\vec{r}}{r^2} \cdot \hat{n}\, dS = \iint_{\partial W} \frac{\vec{r}}{r^2} \cdot d\vec{S} = \iiint_W \left(\nabla \cdot \frac{\vec{r}}{r^2}\right) dV$$

Note $r^2 = \left(\sqrt{x^2+y^2+z^2}\right)^2 = x^2+y^2+z^2$

$$\nabla \cdot \left(\frac{\vec{r}}{r^2} \right) = \nabla \cdot \left(\frac{x}{x^2+y^2+z^2}, \frac{y}{x^2+y^2+z^2}, \frac{z}{x^2+y^2+z^2} \right)$$

$$\frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2+z^2} \right) = \frac{(x^2+y^2+z^2)(1) - (x)(2x)}{(x^2+y^2+z^2)^2}$$

$$= \frac{-x^2+y^2+z^2}{r^4}$$

Similarly, $\frac{\partial}{\partial y}\left(\frac{y}{x^2+y^2+z^2} \right) = \frac{x^2-y^2+z^2}{r^4}$

$$\frac{\partial}{\partial z}\left(\frac{z}{x^2+y^2+z^2} \right) = \frac{x^2+y^2-z^2}{r^4}$$

$$\therefore \nabla \cdot \left(\frac{\vec{r}}{r^2} \right) = \frac{(-x^2+y^2+z^2) + (x^2-y^2+z^2) + (x^2+y^2-z^2)}{r^4}$$

$$= \frac{x^2+y^2+z^2}{r^4} = \frac{r^2}{r^4} = \underline{\frac{1}{r^2}}$$

$$\therefore \iint_{\partial W} \frac{\vec{r}}{r^2} \cdot \hat{n} \, dS = \iiint_W \left(\nabla \cdot \frac{\vec{r}}{r^2} \right) dV = \iiint_W \left(\frac{1}{r^2} \right) dV$$

(5) Now assume $(0,0,0) \in W$, and assume $(0,0,0) \notin \partial W$ so that a small ball B_ϵ of radius $\epsilon > 0$ is completely with W: $B_\epsilon \subset W$.

$\therefore B_\epsilon = \{(x,y,z) : (x^2+y^2+z^2)^{\frac{1}{2}} < \epsilon\}$

Let $S = W - B_\epsilon$, or $S = W \setminus B_\epsilon$

$\therefore$ From (a) $\iiint_S \left(\frac{1}{r^2}\right) dV = \iint_{\partial S} \left(\frac{\vec{r}\cdot\hat{n}}{r^2}\right) dS$

$= \iint_{\partial W} \left(\frac{\vec{r}\cdot\hat{n}}{r^2}\right) dS - \iint_{\partial B_\epsilon} \left(\frac{\vec{r}\cdot\hat{n}}{r^2}\right) dS$

as the normal for ∂B_ϵ points in the opposite direction as the normal for S at the interface for B_ϵ.

But for B_ϵ, $\hat{n} = \frac{\vec{r}}{r}$, so that $\frac{\vec{r}\cdot\hat{n}}{r^2} = \frac{\vec{r}\cdot\vec{r}}{r^3}$

$= \frac{r^2}{r^3} = \frac{1}{r} = \frac{1}{\epsilon}$ since $r = \epsilon$.

$\therefore \iint_{B_\epsilon} \left(\frac{\vec{r}\cdot\hat{n}}{r^2}\right) dS = \iint_{B_\epsilon} \frac{1}{\epsilon} dS = \frac{1}{\epsilon} \iint_{B_\epsilon} dS$

$= \frac{1}{\epsilon}\left(4\pi\epsilon^2\right) = 4\pi\epsilon$

$$\therefore \iiint_S \left(\frac{1}{r^2}\right) dV = \iint_{\partial W} \left(\frac{\vec{r}\cdot\hat{n}}{r^2}\right) dS - 4\pi\epsilon$$

$$\therefore \iiint_W \left(\frac{1}{r^2}\right) dx\,dy\,dz = \lim_{\epsilon\to 0} \iiint_S \left(\frac{1}{r^2}\right) dV$$

$$= \lim_{\epsilon\to 0} \iint_{\partial W} \left(\frac{\vec{r}\cdot\hat{n}}{r^2}\right) dS - \lim_{\epsilon\to 0} 4\pi\epsilon$$

$$= \iint_{\partial W} \left(\frac{\vec{r}\cdot\hat{n}}{r^2}\right) dS - 0 = \iint_{\partial W} \left(\frac{\vec{r}\cdot\hat{n}}{r^2}\right) dS$$

$\therefore$ (a) & (b) $\Rightarrow$

$$\iiint_W \left(\frac{1}{r^2}\right) dV = \iint_{\partial W} \left(\frac{\vec{r}\cdot\hat{n}}{r^2}\right) dS$$

whether or not $(0,0,0) \in W$, and $(0,0,0) \notin \partial W$.

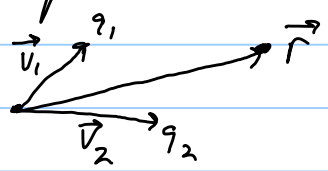
20.

20. Fix vectors $\mathbf{v}_1, \ldots, \mathbf{v}_k \in \mathbb{R}^3$ and numbers ("charges") $q_1, \ldots, q_k$. Define the function ϕ by $\phi(x, y, z) = \sum_{i=1}^{k} q_i/(4\pi \|\mathbf{r} - \mathbf{v}_i\|)$, where $\mathbf{r} = (x, y, z)$. Show that for a closed surface S and $\mathbf{E} = -\nabla\phi$,

$$\iint_S \mathbf{E} \cdot d\mathbf{S} = Q,$$

where Q is the total charge inside S. (Assume that Gauss' law from Theorem 10 applies and that none of the charges are on S.)

Assume $\vec{v}_i$ are vectors from origin to charge q_i, and $\vec{r}$ is some other point in space $\mathbb{R}^3$.



From Exercise #38 of section 4.4 (text, p. 260), which showed $\nabla\left(\frac{1}{r}\right) = \nabla\left(\frac{1}{\|\vec{r}\|}\right) = -\frac{\vec{r}}{r^3}$, $r \neq 0$ [1]

and Example 6 of section 8.4 (text, p. 470),

$\vec{E} = -\nabla\phi = \frac{Q}{4\pi}\left(\frac{\vec{r}}{r^3}\right)$, in this problem,

$$\vec{E} = -\nabla\phi = -\nabla\left[\sum_{i=1}^{k} \frac{q_i}{4\pi}\left(\frac{1}{\|\vec{r} - \vec{v}_i\|}\right)\right]$$

$$= \sum_{i=1}^{k} -\nabla\left[\frac{q_i}{4\pi}\left(\frac{1}{\|\vec{r} - \vec{v}_i\|}\right)\right]$$

$$= \sum_{i=1}^{k} \frac{q_i}{4\pi}\left(\frac{\vec{r} - \vec{v}_i}{\|\vec{r} - \vec{v}_i\|^3}\right), \quad \vec{r} \neq \vec{v}_i \quad \text{from [1]}$$

$$\therefore \iint_S \vec{E} \cdot d\vec{S} = \iint_S \left[\sum_{i=1}^{k} \frac{q_i}{4\pi}\left(\frac{\vec{r} - \vec{v}_i}{\|\vec{r} - \vec{v}_i\|^3}\right)\right] \cdot d\vec{S}$$

$$= \iint_S \left[\sum_{i=1}^{K} \frac{q_i}{4\pi} \left(\frac{\vec{r} - \vec{v}_i}{\|\vec{r} - \vec{v}_i\|^3} \right) \right] \cdot \hat{n} \, dS$$ definition of surface integral

$$= \iint_S \left[\sum_{i=1}^{K} \frac{q_i}{4\pi} \frac{(\vec{r} - \vec{v}_i) \cdot \hat{n}}{\|\vec{r} - \vec{v}_i\|^3} \right] dS$$ linearity of dot product

$$= \sum_{i=1}^{K} \left[\frac{q_i}{4\pi} \iint_S \frac{(\vec{r} - \vec{v}_i) \cdot \vec{n}}{\|\vec{r} - \vec{v}_i\|^3} \, dS \right] \quad [2]$$ linearity of integrals

In Theorem 10, text p. 469, $(0,0,0) \in M$ means the origin of $\vec{r}$ in the Theorem is in region M.

To use Theorem 10, the origin of $\vec{r} - \vec{v}_i$ is where the charge q_i is located, so the origin of $\vec{r} - \vec{v}_i$ inside S means q_i is inside S, or $q_i \in S$.

$$\therefore \iint_S \frac{(\vec{r} - \vec{v}_i) \cdot \hat{n}}{\|\vec{r} - \vec{v}_i\|^3} \, dS = \begin{cases} 4\pi & \text{for } q_i \text{ inside } S \\ 0 & \text{for } q_i \text{ outside } S \end{cases}$$

$\therefore$ [2] becomes $$\sum_{i=1}^{K} \left[\frac{q_i}{4\pi} \begin{pmatrix} 4\pi \text{ if } q_i \in S \\ 0 \text{ if } q_i \notin S \end{pmatrix} \right] = Q$$

where Q = sum of charges q_i inside S.

$$\therefore \iint_S \vec{E} \cdot d\vec{S} = Q$$

21.

21. Prove ***Green's identities***

$$\iint_{\partial W} f\nabla g \cdot \mathbf{n}\, dS = \iiint_W (f\nabla^2 g + \nabla f \cdot \nabla g)\, dV$$

and

$$\iint_{\partial W} (f\nabla g - g\nabla f) \cdot \mathbf{n}\, dS = \iiint_W (f\nabla^2 g - g\nabla^2 f)\, dV.$$

(a) $$\iint_{\partial w} (f\nabla g)\cdot \hat{n}\, dS = \iiint_w \nabla\cdot(f\nabla g)\, dV$$ by Gauss' Divergence Th.

But $\nabla\cdot(f\nabla g) = f(\operatorname{div}\nabla g) + \nabla g \cdot \nabla f$ #7, Extra p. 255

$= f\nabla^2 g + \nabla f \cdot \nabla g$ since $\vec{x}\cdot\vec{y} = \vec{y}\cdot\vec{x}$

and since $\operatorname{div}\nabla g = \operatorname{div}(\partial g/\partial x, \partial g/\partial y, \partial g/\partial z)$

$$= \frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} + \frac{\partial^2 g}{\partial z^2} = \nabla^2 g$$

$$\therefore \iint_{\partial w} (f\nabla g)\cdot \hat{n}\, dS = \iiint_w (f\nabla^2 g + \nabla f\cdot\nabla g)\, dV$$

(b) $$\iint_{\partial w} (f\nabla g - g\nabla f)\cdot \hat{n}\, dS = \iiint_w \nabla\cdot(f\nabla g - g\nabla f)\, dV$$

by Gauss' Divergence Theorem

But $\operatorname{div}(\vec{F}+\vec{G}) = \operatorname{div}\vec{F} + \operatorname{div}\vec{G}$ #5, 4.7.4 p. 255

$\therefore \nabla\cdot(f\nabla g - g\nabla f) = \nabla\cdot(f\nabla g) - \nabla\cdot(g\nabla f)$

and $\operatorname{div}(f\vec{F}) = f\operatorname{div}\vec{F} + \vec{F}\cdot\nabla f$ #7, p. 255

$\therefore \nabla\cdot(f\nabla g) = f\operatorname{div}(\nabla g) + \nabla g\cdot\nabla f$

$= f\nabla^2 g + \nabla g\cdot\nabla f$

$\nabla\cdot(g\nabla f) = g\operatorname{div}(\nabla f) + \nabla f\cdot\nabla g$

$= g\nabla^2 f + \nabla f\cdot\nabla g$

$\therefore \nabla\cdot(f\nabla g) - \nabla\cdot(g\nabla f) = f\nabla^2 g + \nabla g\cdot\nabla f - g\nabla^2 f - \nabla f\cdot\nabla g$

$= f\nabla^2 g - g\nabla^2 f$

$\therefore \nabla\cdot(f\nabla g - g\nabla f) = f\nabla^2 g - g\nabla^2 f$

$\therefore \iiint_W \nabla\cdot(f\nabla g - g\nabla f)\,dV = \iiint_W (f\nabla^2 g - g\nabla^2 f)\,dV$

$\therefore \iint_{\partial W} (f\nabla g - g\nabla f)\cdot\hat{n}\,dS = \iiint_W (f\nabla^2 g - g\nabla^2 f)\,dV$

22. **22.** Suppose $\mathbf{F}$ satisfies div $\mathbf{F} = 0$ and curl $\mathbf{F} = \mathbf{0}$ on all of $\mathbb{R}^3$. Show that we can write $\mathbf{F} = \nabla f$, where $\nabla^2 f = 0$.

curl $\vec{F} = \vec{0} \Rightarrow \exists f$ s.t. $\nabla f = \vec{F}$

$\therefore$ div $(\nabla f) =$ div $\vec{F} = 0$, and div $(\nabla f) = \nabla^2 f$

$\therefore \nabla^2 f = 0$

23. **23.** Let ρ be a continuous function on $\mathbb{R}^3$ such that $\rho(\mathbf{q}) = 0$ except for $\mathbf{q}$ in some region W. Let $\mathbf{q} \in W$ be denoted by $\mathbf{q} = (x, y, z)$. The ***potential*** of ρ is defined to be the function

$$\phi(\mathbf{p}) = \iiint_W \frac{\rho(\mathbf{q})}{4\pi \|\mathbf{p} - \mathbf{q}\|} \, dV(\mathbf{q}),$$

where $\|\mathbf{p} - \mathbf{q}\|$ is the distance between $\mathbf{p}$ and $\mathbf{q}$.

(a) Using the method of Theorem 10, show that

$$\iint_{\partial W} \nabla\phi \cdot \mathbf{n} \, dS = -\iiint_W \rho \, dV$$

for those regions W that can be partitioned into a finite union of symmetric elementary regions.

(a) Detailed answer given in back of text.

Not sure why the notation $dV(\vec{q})$ is used other than to indicate W contains $\vec{q}$.

Note $\nabla\phi(\vec{p}) = \left[\frac{\partial\phi(\vec{p})}{\partial x}, \frac{\partial\phi(\vec{p})}{\partial y}, \frac{\partial\phi(\vec{p})}{\partial z} \right]$

and $\frac{\partial\phi(\vec{p})}{\partial x} = \frac{\partial}{\partial x} \iiint_W \frac{\rho(\vec{q})}{4\pi \|\vec{p} - \vec{q}\|} dV(\vec{q})$

using differentiation under integral

$$= \iiint_W \frac{\partial}{\partial x} \frac{\rho(\vec{q})}{4\pi \|\vec{p}-\vec{q}\|} \, dV(\vec{q})$$

$$= \iiint_W \frac{\rho(\vec{q})}{4\pi} \frac{\partial}{\partial x}\left(\frac{1}{\|\vec{p}-\vec{q}\|}\right) dV(\vec{q}) \quad [1]$$

Similarly for $\partial/\partial y$ and $\partial/\partial z$

$\therefore \nabla\phi(\vec{p})$ is a vector with triple integrals for each of its 3 components.

Note from #38 of text, p. 35, $\nabla\left(\frac{1}{r}\right) = \frac{-\vec{r}}{\|\vec{r}\|^3}$

where $r = \sqrt{x^2+y^2+z^2}$, $r \neq 0$, so

$$\nabla\left(\frac{1}{\|\vec{p}-\vec{q}\|}\right) = -\frac{\vec{p}-\vec{q}}{\|\vec{p}-\vec{q}\|^3}, \quad \vec{p} \neq \vec{q}$$

$\therefore$ [1] along with similar values for $\partial/\partial y$, $\partial/\partial y$,

$$\nabla\phi(\vec{p}) = -\iiint_W \frac{\rho(\vec{q})}{4\pi}\left(\frac{\vec{p}-\vec{q}}{\|\vec{p}-\vec{q}\|^3}\right) dV(\vec{q}) \quad [2]$$

in which the notation:

$$\iiint_W \vec{r}\, dV \text{ means } \left[\iiint_W r_x\, dV, \iiint_W r_y\, dV, \iiint_W r_z\, dV\right] \quad [3]$$

For a given volume V around $\vec{q}$, let $\hat{n}$ be the unit outward normal, and take the dot product of $\hat{n}$ with the 3 component in [2]:

$$\nabla\phi(\vec{p})\cdot\hat{n} = \left[-\iiint_W \frac{\rho(\vec{q})}{4\pi}\left(\frac{\vec{p}-\vec{q}}{\|\vec{p}-\vec{q}\|^3}\right)dV(\vec{q})\right]\cdot\hat{n}$$

$$= -\iiint_W \frac{\rho(\vec{q})}{4\pi}\,\frac{(\vec{p}-\vec{q})\cdot\hat{n}}{\|\vec{p}-\vec{q}\|^3}\,dV(\vec{q}) \quad [4]$$

using the same notation indicated in [3]

If $\vec{p}$ varies in $V \cup \partial V$, then take surface integral of the scalar valued function in [4],

$$\iint_{\partial V} \nabla\phi(\vec{p})\cdot\hat{n}\,dS = -\iint_{\partial V}\left[\iiint_W \frac{\rho(\vec{q})}{4\pi}\,\frac{(\vec{p}-\vec{q})\cdot\hat{n}}{\|\vec{p}-\vec{q}\|^3}\,dV(\vec{q})\right]dS \quad [5]$$

where the above is evaluated with $\vec{p}$ on ∂V surface.

A simplified expression for the integral on the right side of the equation is:

$$\iint_{\partial V}\left[\iiint_W \rho_x\,dV+\iiint_W \rho_y\,dV+\iiint_W \rho_z\,dV\right]dS$$

$$=\iint_{\partial V}\left[\iiint_W \rho_x\,dV\right]dS+\iint_{\partial V}\left[\iiint_W \rho_y\,dV\right]dS$$

$$+\iint_{\partial V}\left[\iiint_W \rho_z\,dV\right]dS \qquad \text{using linearity of integrals}$$

Each term, using Fubini's Theorem can be written as a 5-integral iterated integral, e.g.,

$$\iint_{\partial V}\left[\iiint_W \rho_x\,dV\right]dS=\int_{u_0}^{u_1}\int_{v_0}^{v_1}\left[\int_{x_0}^{x_1}\int_{y_0}^{y_1}\int_{z_0}^{z_1}\rho_x\,dz\,dy\,dx\right]dv\,du$$

and again by Fubini's Theorem, changing the order,

$$=\int_x\int_y\int_z\int_u\int_v \rho_x\,dv\,du\,dz\,dy\,dx$$

$$=\iiint_W\left[\iint_{\partial V}\rho_x\,dS\right]dV$$

$\therefore$ [5] can be rewritten as:

$$\iint_{\partial V} \nabla\phi(\vec{\rho})\cdot\hat{n}\,dS = -\iiint_W \left[\iint_{\partial V} \frac{\rho(\vec{q})}{4\pi}\frac{(\vec{\rho}-\vec{q})\cdot\hat{n}}{\|\vec{\rho}-\vec{q}\|^3}\,dS\right] dV(\vec{q})$$

In Theorem 10, text p. 469, $(0,0,0)\in M$ means the origin of $\vec{r}$ in the Theorem is in region M.

To use Theorem 10, the origin of $\vec{\rho}-\vec{q}$ inside ∂V (i.e., within V) means $\vec{q}$ is in V. By Theorem 10,

$$\vec{q}\in V \Rightarrow \iint_{\partial V} \frac{(\vec{\rho}-\vec{q})\cdot\hat{n}}{\|\vec{\rho}-\vec{q}\|^3}\,dS = 4\pi$$

assuming V is a symmetric elementary region.

$$\therefore \iint_{\partial V} \nabla\phi(\vec{\rho})\cdot\hat{n}\,dS = -\iiint_W \rho(\vec{q})\,dV(\vec{q})$$

if $q\notin V$, then $\iint_{\partial V} \frac{(\vec{\rho}-\vec{q})\cdot\hat{n}}{\|\vec{\rho}-\vec{q}\|^3}\,dS = 0$

$\therefore$ For $\vec{q}\in V\cap W$, $\iint_{\partial V} \nabla\phi(\vec{\rho})\cdot\hat{n}\,dS = -\iiint_{V\cap W} \rho(\vec{q})\,dV(\vec{q})$

$\rho = 0$ outside W takes care of the case of $\vec{q}\in V$

and $\vec{q} \notin W$. $\therefore$ The restriction $V \cap W$ can be dropped, so $\iint_{\partial V} \nabla\phi(\vec{p}) \cdot \hat{n}\, dS = -\iiint_V \rho(\vec{q})\, dV(\vec{q})$

The answer in the first addresses if V is not a symmetric elementary region.

(6) (b) Show that ϕ satisfies ***Poisson's equation***

$$\nabla^2\phi = -\rho.$$

[HINT: Use part (a).] (Notice that if ρ is a charge density, then the integral defining ϕ may be thought of as the sum of the potential at $\mathbf{p}$ caused by point charges distributed over W according to the density ρ.)

From (a), $\iint_{\partial W} \nabla\phi \cdot \hat{n}\, dS = -\iiint_W \rho\, dV$

and by Gauss' Divergence Theorem,

$$\iint_{\partial W} \nabla\phi \cdot \hat{n}\, dS = \iiint_W \nabla \cdot \nabla\phi\, dV = \iiint_W \nabla^2\phi\, dV$$

$$\therefore \iiint_W \nabla^2\phi\, dV = -\iiint_W \rho\, dV$$

Since $\nabla^2\phi$ and ρ are both continuous, and the above equation holds for arbitrarily small regions, Then $\nabla^2\phi = -\rho$.

Suppose they differed by a small amount.

Choose a small enough volume of W so that

$\nabla^2\phi$ is >0 for all W or <0 for all W.

Continuity ensures this is possible. Similarly for ρ. $\therefore$ Over the smaller of the two volumes, $\nabla^2\phi + \rho \neq 0$ since they differ by a small amount.

$$\therefore \iiint_W \nabla^2\phi + \rho \, dV \neq 0 \Rightarrow \iiint_W \nabla^2\phi \neq -\iiint_W \rho$$

a contradiction.

24. **24.** Suppose $\mathbf{F}$ is tangent to the closed surface $S = \partial W$ of a region W. Prove that

$$\iiint_W (\operatorname{div} \mathbf{F})\, dV = 0.$$

By Gauss' Divergence Theorem,

$$\iiint_W (\operatorname{div} \vec{F})\, dV = \iint_{\partial W} \vec{F} \cdot \hat{n}\, dS,$$ where $\hat{n}$ is the unit normal to the surface ∂W at any point of ∂W. Since $\vec{F}$ is tangential to ∂W at each point, $\vec{F} \cdot \hat{n} = 0$, so $\iint_{\partial W} \vec{F} \cdot \hat{n}\, dS = 0$.

$\therefore \iiint_W (\operatorname{div} \vec{F})\, dV = 0$

25. **25.** Use Gauss' law and symmetry to prove that the electric field due to a charge Q evenly spread over the surface of a sphere is the same outside the surface as the field from a point charge Q located at the center of the sphere. What is the field inside the sphere?

Let S be a sphere centered at the origin, with charge evenly distributed. Let M be a surface that

either completely encloses S, or is completely inside S. Take any point $\vec{\rho}$ on the surface of M, whose origin is at (0,0,0).

Note that Q is composed of a large finite number of charges, q_i, so that for every q_i, draw a vector from q_i to the head of $\vec{\rho}$. This vector can be decomposed of a component perpendicular to $\vec{\rho}$, and a component along $\vec{\rho}$. By symmetry of the sphere, this is an "opposite" charge q_j whose similar decomposed components are such that the component perpendicular to $\vec{\rho}$ cancels the q_i component perpendicular to $\vec{\rho}$.

Thus, whether M is inside or outside, the effective summation of $\vec{E}$ from all the q_i to $\vec{\rho}$ is along the vector $\vec{\rho}$.

As discussed in Example 6, text p. 470, and using Theorem 10, text p. 469, the flux $\int_{\partial m} \vec{E} \cdot d\vec{S}$ from $\vec{E}$ generated by q_i is q_i if q_i is inside m, and 0 is q_i is outside m. Thus, from all the q_i, the flux is Q if m is outside S, and 0 if m is inside S.

By the symmetry of the sphere S, and noting $\vec{p}$ can be anywhere around S, $\vec{E}$ is therefore uniform outside S and inside S. As stated above, $\vec{E} = \vec{0}$ inside S. Since $\vec{E}$ is uniform outside S, and $\iint_{\partial m} \vec{E} \cdot d\vec{S} = Q$ whether Q is distributed evenly over S, or is at a point at the origin.

26. **26.** Reformulate Exercise 25 in terms of gravitational fields.

The argument and result is the same as

Theorem 10, text p. 469, still applies, and the potential function is of the same form:

$$\phi(x,y,z) = -\frac{kM}{4\pi r}, \text{ so } \vec{g} = -\nabla\phi = -\frac{kM}{4\pi}\left(\frac{\vec{r}}{r^3}\right)$$

so that $\vec{F} = -\frac{kMm}{4\pi}\left(\frac{\vec{r}}{r^3}\right)$

where $\frac{k}{4\pi} = G$, universal gravitational constant.

27. **27.** Show how Gauss' law can be used to solve part (b) of Exercise 29 in Section 8.3.

29. Let $\mathbf{F} = -(GmM\mathbf{r}/r^3)$ be the gravitational force field defined on $\mathbb{R}^3\backslash\{\mathbf{0}\}$.

(b) Show that $\mathbf{F} \neq \text{curl } \mathbf{G}$ for any C^1 vector field $\mathbf{G}$ on $\mathbb{R}^3\backslash\{\mathbf{0}\}$.

Since $\frac{\vec{F}}{m} = -GM\left(\frac{\vec{r}}{r^3}\right)$, by Theorem 10, text p. 469,

$$\iint_{\partial S} -GM\left(\frac{\vec{r}}{r^3}\right)\cdot\hat{n}\, dS = 4\pi \text{ for any surface } S$$

around the mass M, or $\iint_{\partial S} \frac{\vec{E}}{m}\cdot d\vec{S} = 4\pi$

By Gauss' Divergence Theorem,

$$\iint_{\partial S} \frac{\vec{F}}{m} \cdot d\vec{S} = \iiint_W \left(\operatorname{div} \frac{\vec{F}}{m}\right) dV = \frac{1}{m} \iiint_W \nabla \cdot \vec{F}\, dV,$$

so $\iiint_W \nabla \cdot \vec{F}\, dV = 4\pi m$, where W is the region enclosed by S. But if $\vec{F} = \nabla \times \vec{G}$ for some G, then $\nabla \cdot (\nabla \times \vec{G}) = 0$, so that

$$\iiint_W \nabla \cdot \vec{F}\, dV = \iiint_W \nabla \cdot (\nabla \times \vec{G})\, dV = 0,$$ contradicting the value of $4\pi m$, which is not 0.

$\therefore$ There can be no $\vec{G}$ s.t. $\vec{F} = \nabla \times \vec{G}$.

28. **28.** Let S be a closed surface. Use Gauss' theorem to show that if $\mathbf{F}$ is a C^2 vector field, then we have $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = 0$.

$$\iint_{\partial S} (\nabla \times \vec{F}) \cdot d\vec{S} = \iiint_W \nabla \cdot (\nabla \times \vec{F})\, dV,$$ where W is the region enclosed by S. But $\nabla \cdot (\nabla \times \vec{F}) = 0$ (see text, p. 253).

$$\therefore 0 = \iiint_W \nabla \cdot (\nabla \times \vec{F})\, dV = \iint_{\partial S} (\nabla \times \vec{F}) \cdot d\vec{S}$$

29.

29. Let S be the surface of region W. Show that

$$\iint_S \mathbf{r} \cdot \mathbf{n}\, dS = 3 \text{ volume } (W).$$

Explain this geometrically.

Let $\vec{r} = (x, y, z)$, a radial vector from the origin.

$$\therefore \nabla \cdot \vec{r} = \frac{\partial (x)}{\partial x} + \frac{\partial (y)}{\partial y} + \frac{\partial (z)}{\partial z} = 1 + 1 + 1 = 3.$$

$$\therefore \iint_{\partial S} \vec{r} \cdot \hat{n}\, dS = \iiint_W \nabla \cdot \vec{r}\, dV = \iiint_W 3\, dV =$$

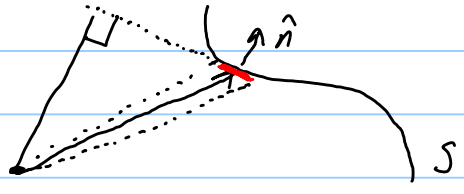
$$3 \iiint_W dV = 3 \,(\text{volume of } W)$$

Geometric Interpretation.

$\vec{r} \cdot \hat{n}\, dS = \|\vec{r}\| \cos\theta\, dS$ since $\|\hat{n}\| = 1$

Consider dS a small patch of area of surface S

$\|\vec{r}\| \cos\theta$ is effectively the height



of the cone or pyramid with base dS on surface S_1 and so $\vec{r} \cdot \hat{n}\, dS = (\text{height})(\text{base})$ for a pyramid P_i with base dS.

From above, $\iint_{\partial S} \vec{r} \cdot \hat{n}\, dS = 3\,(\text{volume of } W)$,

so, $\iint_{\partial S} \frac{1}{3} \vec{r} \cdot \hat{n}\, dS = \text{volume of } W.$

$\therefore$ Volume of a pyramid generated from a patch of S, call it P_i with apex at $(0,0,0)$ and base dS, must be $\frac{1}{3}(\text{base})(\text{height})$ so that

$$\sum_{i=1}^{n} P_i = \text{volume of } W.$$

$\therefore$ Formula for volume of pyramid with base dS and height r is $\frac{1}{3} r\, dS$.

30.

30. For a steady-state charge distribution and divergence-free current distribution, the electric and magnetic fields $\mathbf{E}(x, y, z)$ and $\mathbf{H}(x, y, z)$ satisfy

$$\nabla \times \mathbf{E} = \mathbf{0}, \quad \nabla \cdot \mathbf{H} = 0, \quad \nabla \cdot \mathbf{J} = 0, \quad \nabla \cdot \mathbf{E} = \rho, \quad \text{and} \quad \nabla \times \mathbf{H} = \mathbf{J}.$$

Here $\rho = \rho(x, y, z)$ and $\mathbf{J}(x, y, z)$ are assumed to be known. The radiation that the fields produce through a surface S is determined by a radiation flux density vector field, called the ***Poynting*** vector field,

$$\mathbf{P} = \mathbf{E} \times \mathbf{H}.$$

(a)

(a) If S is a *closed* surface, show that the radiation flux—that is, the flux of $\mathbf{P}$ through S—is given by

$$\iint_S \mathbf{P} \cdot d\mathbf{S} = -\iiint_V \mathbf{E} \cdot \mathbf{J}\, dV,$$

where V is the region enclosed by S.

$$\iint_S \vec{P} \cdot d\vec{S} = \iiint_V \nabla \cdot (\vec{E} \times \vec{H})\, dV$$

But $\nabla \cdot (\vec{E} \times \vec{H}) = H \cdot (\nabla \times \vec{E}) - \vec{E} \cdot (\nabla \times \vec{H})$

using # 8, text p. 255

It's given that $\nabla \times \vec{E} = 0$ and $\nabla \times \vec{H} = \vec{J}$

$\therefore \nabla \cdot (\vec{E} \times \vec{H}) = 0 - \vec{E} \cdot (\vec{J}) = -\vec{E} \cdot \vec{J}$

$\therefore \iint_S \vec{P} \cdot d\vec{S} = -\iiint_V \vec{E} \cdot \vec{J}\, dV$

(6) (b) Examples of such fields are

$$\mathbf{E}(x, y, z) = z\mathbf{j} + y\mathbf{k} \quad \text{and} \quad \mathbf{H}(x, y, z) = -xy\mathbf{i} + x\mathbf{j} + yz\mathbf{k}.$$

In this case, find the flux of the Poynting vector through the hemispherical shell shown in Figure 8.4.9. (Notice that it is an *open* surface.)

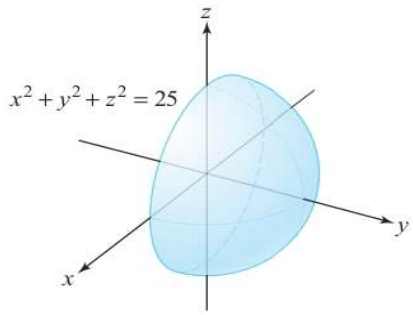


figure **8.4.9** The surface for Exercise 30.

$$\text{Flux} = \iint_{\partial S} \vec{\rho} \cdot d\vec{S} = \iint_{\partial S} \vec{\rho} \cdot \hat{n}\, dS$$

$$\vec{\rho} = \vec{E} \times \vec{H} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & z & y \\ -xy & x & yz \end{vmatrix} = (yz^2 - xy,\ -xy^2,\ xyz)$$

$$\hat{n} = \frac{1}{\sqrt{x^2+y^2+z^2}}(x, y, z) = \frac{1}{5}(x, y, z)$$

$$\therefore \vec{\rho} \cdot \vec{n} = \frac{1}{5}\left\{ xyz^2 - x^2y - xy^3 + xyz^2 \right\}$$

$$= \frac{2}{5} xyz^2 - x^2y - xy^3$$

Let $T(\phi, \theta) = (5\cos\theta\sin\phi,\ 5\sin\theta\sin\phi,\ 5\cos\phi)$

$$T_\phi \times T_\theta = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5\cos\theta\cos\phi & 5\sin\theta\cos\phi & -5\sin\phi \\ -5\sin\theta\sin\phi & 5\cos\theta\sin\phi & 0 \end{vmatrix}$$

$$= \left\{ 25\cos\theta\sin^2\phi,\ 25\sin\theta\sin^2\phi,\ 25\cos\phi\sin\phi \right\}$$

(note: points outward)

$$\therefore \|T_\phi \times T_\theta\| = \sqrt{25^2 \sin^4\phi + 25^2 \cos^2\phi \sin^2\phi} = 25\sin\phi$$

$$\left(\vec{F}\cdot\vec{n}\right)\left(T(\phi,\theta)\right) = \frac{2}{5}\left(5\cos\theta\sin\phi\right)\left(5\sin\theta\sin\phi\right)\left(25\cos^2\phi\right)$$

$$- \left(25\cos^2\theta\sin^2\phi\right)\left(5\sin\theta\sin\phi\right)$$

$$- \left(5\cos\theta\sin\phi\right)\left(5^3\sin^3\theta\sin^3\phi\right)$$

$$= 2(5^3)\cos\theta\sin\theta\sin^2\phi\cos^2\phi$$

$$- 5^3\cos^2\theta\sin\theta\sin^3\phi$$

$$- 5^4\cos\theta\sin^3\theta\sin^4\phi$$

$$\therefore \iint_{\partial S} \vec{F}\cdot\vec{n}\, dS = \int_0^{\pi}\int_0^{\pi} (2)(5^5)\cos\theta\sin\theta\sin^3\phi\cos\phi\, d\theta d\phi$$

$$- 5^5 \int_0^{\pi}\int_0^{\pi} \cos^2\theta\sin\theta\sin^4\phi\, d\theta d\phi$$

$\sin^2\phi = \frac{1-\cos 2\phi}{2}$ $\cos^2\phi = \frac{1+\cos 2\phi}{2}$

$\sin^4\phi = \left(\frac{1-\cos 2\phi}{2}\right)^2 = \frac{1}{4}\left(1-2\cos 2\phi+\cos^2 2\phi\right)$

$= \frac{1}{4} - \frac{\cos 2\phi}{2} + \frac{1}{4}\left(\frac{1+\cos 4\phi}{2}\right)$

$$- 5^6 \int_0^{\pi}\int_0^{\pi} \cos\theta\sin^3\theta\sin^5\phi\, d\theta d\phi$$

$$= 5^5 \left[\sin^2\theta\right]_0^{\pi} \left[\frac{\sin^4\phi}{4}\right]_0^{\pi}$$

($\sin^2\theta$: $=0$ at both limits; $\frac{\sin^4\phi}{4}$: $=0$ at both limits)

$$+ 5^5 \left[\frac{\cos^3\theta}{3} \right]_0^{\pi} \left[\frac{3}{8}\phi - \frac{\sin 2\phi}{4} + \frac{\sin 4\phi}{32} \right]_0^{\pi}$$

$$- 5^6 \left[\frac{\sin^4\theta}{4} \right]_0^{\pi} \int_0^{\pi} \sin^5\phi \, d\phi \qquad (\sin^4\theta \text{ term} = 0)$$

$$= 5^5 \left[-\frac{1}{3} - \frac{1}{3} \right] \left[\frac{3}{8}\pi \right] = 5^5 \left(-\frac{2}{3}\right)\left(\frac{3}{8}\right)\pi = -\frac{5^5}{4}\pi$$

$$= -\frac{3125}{4}\pi$$

If S_1 is the flat base, $\iiint_W \nabla \cdot \vec{F}\, dV - \iint_{\partial S_1} \vec{F} \cdot d\vec{S}$ gives the same calculation.

(c) The fields of part (b) produce a Poynting vector field passing through the toroidal surface shown in Figure 8.4.10. What is the flux through this torus?

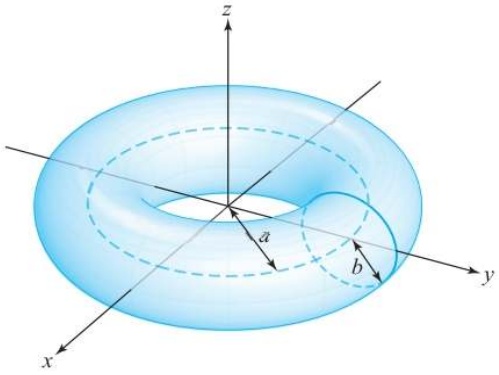


figure **8.4.10** The surface for Exercise 30(c).

Note the torus is a closed surface.

From (b), $\vec{P} = (yz^2 - xy, -xy^2, xyz)$

$$\nabla \cdot \vec{P} = (-y) + (-2xy) + (xy) = -y - xy$$

$$\iint_{\partial S} \vec{P} \cdot d\vec{S} = \iiint_W \nabla \cdot \vec{P}\, dV = \iiint_W -y - xy \, dV$$

The region of the torus can be parametrized as:

$$T(r,\theta,\phi) = \left[(a + r\cos\phi)\cos\theta, (a + r\cos\phi)\sin\theta, r\sin\phi\right]$$

where $a > b$, $0 \le r \le b$, $0 \le \theta \le 2\pi$, $0 \le \phi \le 2\pi$

where θ measures the angle in the xy-plane from the x-axis, and ϕ measures the angle from the xy-plane of the cross section of the torus.

$$\therefore T_r = \left[\cos\theta\cos\phi, \sin\theta\cos\phi, \sin\phi\right]$$

$$T_\theta = \left[-(a + r\cos\phi)\sin\theta, (a + r\cos\phi)\cos\theta, 0\right]$$

$$T_\phi = \left[-r\cos\theta\sin\phi, -r\sin\theta\sin\phi, r\cos\phi\right]$$

$$\therefore \frac{\partial(x,y,z)}{\partial(r,\theta,\phi)} = \cos\theta\cos\phi\left(ar\cos\theta\cos\phi + r^2\cos\theta\cos^2\phi\right)$$

combine ↗ ↘ combine ↙ ↖ combine

$$- \sin\theta\cos\phi\left(-ar\sin\theta\cos\phi - r^2\sin\theta\cos^2\phi\right)$$

$$+ \sin\phi\left(ar\sin^2\theta\sin\phi + r^2\sin^2\theta\cos\phi\sin\phi\right.$$

combine ↗ ↘ combine ↙ ↖ combine

$$\left. + ar\cos^2\theta\sin\phi + r^2\cos^2\theta\cos\phi\sin\phi\right)$$

$$= ar\cos^2\phi + r^2\cos^3\phi + ar\sin^2\phi + r^2\cos\phi\sin^2\phi$$

$$= ar + r^2\cos\phi\left(\cos^2\theta + \sin^2\theta\right)$$

$$= ar + r^2\cos\phi = r(a + r\cos\phi)$$

Note that for $0 \le r \le b$, $a + r\cos\phi > 0$

$$\therefore \left|\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}\right| = r(a + r\cos\phi) = \text{Jacobian}$$

If $f(x,y,z) = -y - xy = -y(1+x)$

$$f(T(r,\theta,\phi)) = f\left[(a + r\cos\phi)\cos\theta, (a + r\cos\phi)\sin\theta, r\sin\phi\right]$$

$$= -(a + r\cos\phi)\sin\theta\left[1 + (a + r\cos\phi)\cos\theta\right]$$

$$= -(a + r\cos\phi)\sin\theta - (a + r\cos\phi)^2\cos\theta\sin\theta$$

$$\therefore f(T(r,\theta,\phi))\left|\frac{\partial(x,y,z)}{\partial(r,\theta,\phi)}\right| =$$

$$-r(a + r\cos\phi)^2\sin\theta - r(a + r\cos\phi)^3\cos\theta\sin\theta$$

$$\therefore \iint_{\partial S} \vec{F}\cdot d\vec{S} = \iiint_W -y - xy\, dx\,dy\,dz$$

$$= \int_0^{2\pi}\int_0^{2\pi}\int_0^{b} -r(a + r\cos\phi)^2\sin\theta - r(a + r\cos\phi)^3\cos\theta\sin\theta\, dr\,d\theta\,d\phi$$

$$= \int_0^{2\pi}\int_0^{b} r(a+r\cos\phi)^2\, dr\, d\phi \Big[\cos\theta\Big]_0^{2\pi}$$

($\cos\theta$ evaluated from 0 to 2π $= 0$)

$$- \int_0^{2\pi}\int_0^{b} r(a+r\cos\phi)^3\, dr\, d\phi \left[\frac{\sin^2\theta}{2}\right]_0^{2\pi}$$

($\frac{\sin^2\theta}{2}$ evaluated from 0 to 2π $= 0$)

$$= 0 - 0 = \underline{0}$$

$$\therefore \iint_{\partial S} \vec{F} \cdot d\vec{S} = 0 = \text{flux through torus}$$

8.5 Differential Forms

Note Title 7/21/2017

1. **1.** Evaluate $\omega \wedge \eta$ if

(a) $\omega = 2x\,dx + y\,dy$
$\eta = x^3\,dx + y^2\,dy$

(a) $\omega \wedge \eta = (2x\,dx + y\,dy) \wedge (x^3\,dx + y^2\,dy)$

$= (2x\,dx + y\,dy) \wedge (x^3\,dx)$

$+ (2x\,dx + y\,dy) \wedge (y^2\,dy)$

$= 2x^4 (dx \wedge dx) + x^3 y (dy \wedge dx)$ ($dx \wedge dx = 0$)

$+ 2xy^2 (dx \wedge dy) + y^3 (dy \wedge dy)$ ($dy \wedge dy = 0$)

$= x^3 y (dy \wedge dx) + 2xy^2 (dx \wedge dy)$

$= -x^3 y (dx \wedge dy) + 2xy^2 (dx \wedge dy)$

$= (2xy^2 - x^3 y)\,dx\,dy$

(6) (b) $\omega = x\,dx - y\,dy$
$\eta = y\,dx + x\,dy$

$\omega \wedge \eta = (x\,dx - y\,dy) \wedge (y\,dx + x\,dy)$

$= (x\,dx) \wedge (y\,dx) - (y\,dy) \wedge (y\,dx)$

$+ (x\,dx) \wedge (x\,dy) - (y\,dy) \wedge (x\,dy)$

$$= xy\,(dx \wedge dx) - y^2\,(dy \wedge dx) + x^2\,(dx \wedge dy) - xy\,(dy \wedge dy)$$

($dx \wedge dx = 0$, $dy \wedge dy = 0$)

$$= -y^2\,(dy \wedge dx) + x^2\,(dx \wedge dy)$$

$$= y^2\,(dx \wedge dy) + x^2\,(dx \wedge dy)$$

$$= (x^2 + y^2)\,dx\,dy$$

(c) (c) $\omega = x\,dx + y\,dy + z\,dz$
$\eta = z\,dx\,dy + x\,dy\,dz + y\,dz\,dx$

$$\omega \wedge \eta = (x\,dx + y\,dy + z\,dz) \wedge (z\,dx\,dy + x\,dy\,dz + y\,dz\,dx)$$

$$= xz\,dx \wedge (dx \wedge dy) + yz\,dy \wedge (dx \wedge dy) + z^2\,dz \wedge (dx \wedge dy)$$
$$+ x^2\,dx \wedge (dy \wedge dz) + xy\,dy \wedge (dy \wedge dz) + xz\,dz \wedge (dy \wedge dz)$$
$$+ xy\,dx \wedge (dz \wedge dx) + y^2\,dy \wedge (dz \wedge dx) + yz\,dz \wedge (dz \wedge dx)$$

($dx \wedge dx = 0$, $dy \wedge dy = 0$, $dz \wedge dz = 0$)

$$= z^2\,(dx \wedge dy) \wedge dz + x^2\,dx \wedge (dy \wedge dz) + y^2\,dy \wedge (dz \wedge dx)$$

$$= (x^2 + z^2)\,dx\,dy\,dz - y^2\,dy \wedge (dx \wedge dz)$$

$$= (x^2 + z^2)\,dx\,dy\,dz - y^2\,(dy \wedge dx) \wedge dz$$

$$= (x^2 + z^2)\,dx\,dy\,dz + y^2\,(dx \wedge dy) \wedge dz$$

$$= (x^2 + y^2 + z^2)\,dx\,dy\,dz$$

(d) (d) $\omega = xy\,dy\,dz + x^2\,dx\,dy$
$\eta = dx + dz$

$$\omega \wedge \eta = (xy\,dy\,dz + x^2\,dx\,dy) \wedge (dx + dz)$$

$$= xy\,(dy \wedge dz) \wedge dx + x^2 (dx \wedge dy) \wedge dx \quad (= 0)$$

$$+ xy\,(dy \wedge dz) \wedge dz + x^2 (dx \wedge dy) \wedge dz \quad (= 0)$$

$$= xy\,(dy \wedge dz) \wedge dx + x^2\,dx\,dy\,dz$$

$$= xy\,dx \wedge (dy \wedge dz) + x^2\,dx\,dy\,dz$$

$$= (xy + x^2)\,dx\,dy\,dz$$

(e) (e) $\omega = e^{xyz}\,dx\,dy$
$\eta = e^{-xyz}\,dz$

$$\omega \wedge \eta = (e^{xyz}\,dx\,dy) \wedge (e^{-xyz}\,dz)$$

$$= (dx \wedge dy) \wedge dz = dx\,dy\,dz$$

as $e^{xyz} \cdot e^{-xyz} = e^0 = 1$

2. **2.** Prove that

$$(a_1\,dx + a_2\,dy + a_3\,dz) \wedge (b_1\,dy\,dz + b_2\,dz\,dx + b_3\,dx\,dy) = \left(\sum_{i=1}^{3} a_i b_i\right) dx\,dy\,dz.$$

$$(a_1\,dx + a_2\,dy + a_3\,dz) \wedge (b_1\,dy\,dz + b_2\,dz\,dx + b_3\,dx\,dy)$$

$$= a_1 b_1\,dx \wedge (dy \wedge dz) + a_2 b_1\,dy \wedge (dy \wedge dz) + a_3 b_1\,dz \wedge (dy \wedge dz)$$
$(= 0)$ $(= 0)$

$$+ a_2 b_1\, dy \wedge (dy \wedge dz) + a_2 b_2\, dy \wedge (dz \wedge dx) + a_2 b_3\, dy \wedge (dx \wedge dy)$$
($dy \wedge (dy \wedge dz) = 0$, $dy \wedge (dz \wedge dx) = 0$)

$$+ a_3 b_1\, dz \wedge (dy \wedge dz) + a_3 b_2\, dz \wedge (dz \wedge dx) + a_3 b_3\, dz \wedge (dx \wedge dy)$$
($dz \wedge (dy \wedge dz) = 0$, $dz \wedge (dz \wedge dx) = 0$)

$$= a_1 b_1\, dx\, dy\, dz + a_2 b_2\, dy \wedge (dz \wedge dx) + a_3 b_3\, dz \wedge (dx \wedge dy)$$

$$= a_1 b_1\, dx\, dy\, dz - a_2 b_2\, dy \wedge (dx \wedge dz) + a_3 b_3\, (dx \wedge dy) \wedge dz$$

$$= a_1 b_1\, dx\, dy\, dz - a_2 b_2\, (dy \wedge dx) \wedge dz + a_3 b_3\, dx\, dy\, dz$$

$$= a_1 b_1\, dx\, dy\, dz + a_2 b_2\, (dx \wedge dy) \wedge dz + a_3 b_3\, dx\, dy\, dz$$

$$= (a_1 b_1 + a_2 b_2 + a_3 b_3)\, dx\, dy\, dz$$

$$= \left(\sum_{i=1}^{3} a_i b_i\right) dx\, dy\, dz$$

3.

3. Find $d\omega$ in the following examples:

(a) $\omega = x^2 y + y^3$

(a) $$d\omega = \frac{\partial (x^2 y + y^3)}{\partial x}\, dx + \frac{\partial (x^2 y + y^3)}{\partial y}\, dy + \frac{\partial (x^2 y + y^3)}{\partial z}\, dz$$

$$= 2xy\, dx + (x^2 + 3y^2)\, dy$$

(b)

(b) $\omega = y^2 \cos x\, dy + xy\, dx + dz$

$$d\omega = d(y^2 \cos x\, dy) + d(xy\, dx) + d(dz)$$

$$= d(y^2\cos x \wedge dy) + d(xy \wedge dx)$$

$$= d(y^2\cos x)\wedge dy + (y^2\cos x)\wedge \underbrace{d(dy)}_{=0} + d(xy)\wedge dx + xy \wedge \underbrace{d(dx)}_{=0}$$

$$= (-y^2\sin x\, dx + 2y\cos x\, dy)\wedge dy + (y\,dx + x\,dy)\wedge dx$$

$$= -y^2\sin x\, dx\,dy + x\,dy\,dx$$

$$= \underline{(-y^2\sin x - x)\,dx\,dy}$$

(c) $\omega = xy\,dy + (x+y)^2\,dx$

$$d\omega = d(xy\wedge dy) + d((x+y)^2\wedge dx)$$

$$= \left[\frac{\partial(xy)}{\partial x}dx + \frac{\partial(xy)}{\partial y}dy\right]\wedge dy + (-1)^0 xy\wedge \underbrace{d(dy)}_{=0}$$

$$+ \left[\frac{\partial(x+y)^2}{\partial x}dx + \frac{\partial(x+y)^2}{\partial y}dy\right]\wedge dx + (-1)^0(x+y)^2\wedge \underbrace{d(dx)}_{=0}$$

$$= (y\,dx + x\,dy)\wedge dy + (2(x+y)dx + 2(x+y)dy)\,dx$$

$$= y\,dx\,dy + 2(x+y)\,dy\,dx = y\,dx\,dy - (2x+2y)\,dx\,dy$$

$$= \underline{-(2x+y)\,dx\,dy}$$

(d) (d) $\omega = x\,dx\,dy + z\,dy\,dz + y\,dz\,dx$

$$d\omega = d(x \wedge dx \wedge dy) + d(z \wedge dy \wedge dz) + d(y \wedge dz \wedge dx)$$

$$= d(x) \wedge (dx \wedge dy) + (-1)^0 x \wedge d(dx \wedge dy)$$

$$+ d(z) \wedge (dy \wedge dz) + (-1)^0 z \wedge d(dy \wedge dz)$$

$$+ d(y) \wedge (dz \wedge dx) + (-1)^0 y \wedge d(dz \wedge dx)$$

$$= dx \wedge \overset{=0}{(dx \wedge dy)} + x \wedge \left[\overset{=0}{d(dx)} \wedge dy + (-1)^1 dx \wedge \overset{=0}{d(dy)} \right]$$

$$+ dz \wedge \overset{=0}{(dy \wedge dz)} + z \wedge \left[\overset{=0}{d(dy)} \wedge dz + (-1)^1 dy \wedge \overset{=0}{d(dz)} \right]$$

$$+ dy \wedge (dz \wedge dx) + y \wedge \left[\overset{=0}{d(dz)} \wedge dx + (-1)^1 dz \wedge \overset{=0}{d(dx)} \right]$$

$$= dy \wedge (dz \wedge dx) = -dy \wedge (dx \wedge dz)$$

$$= -(dy \wedge dx) \wedge dz = (dx \wedge dy) \wedge dz$$

$$= \underline{dx\,dy\,dz}$$

(e) (e) $\omega = (x^2 + y^2)\,dy\,dz$

$$d\omega = d\left[(x^2+y^2) \wedge dy \wedge dz \right]$$

$$= d(x^2+y^2) \wedge (dy \wedge dz) + (-1)^0 (x^2+y^2) \wedge \overset{=0}{d(dy \wedge dz)}$$

$$= \left(\frac{\partial (x^2+y^2)}{\partial x} dx + \frac{\partial (x^2+y^2)}{\partial y} dy \right) \wedge (dy \wedge dz)$$

$$= (2x\,dx + 2y\,dy) \wedge (dy \wedge dz)$$

$$= 2x\,dx \wedge (dy \wedge dz) + 2y\,dy \wedge (dy \wedge dz) \qquad dy \wedge dy = 0$$

$$= 2x\,dx\,dy\,dz$$

(f) (f) $\omega = (x^2 + y^2 + z^2)\,dz$

$$d\omega = d\left[(x^2+y^2+z^2) \wedge dz\right]$$

$$= (2x\,dx + 2y\,dy + 2z\,dz) \wedge dz$$

$$= 2x\,dx\,dz + 2y\,dy\,dz \qquad 2z\,dz \wedge dz = 0$$

$$= -2x\,dz\,dx + 2y\,dy\,dz \qquad (dz\,dx \text{ is standard cyclic notation})$$

(g) (g) $\omega = \dfrac{-x}{x^2+y^2}\,dx + \dfrac{y}{x^2+y^2}\,dy$

$$d\omega = d\left(\frac{-x}{x^2+y^2} \wedge dx\right) + d\left(\frac{y}{x^2+y^2} \wedge dy\right) \qquad d(dx) = 0,\ d(dy) = 0$$

$$= \left[\frac{(x^2+y^2)(-1) - (-x)(2x)}{(x^2+y^2)^2}\,dx + \frac{(x^2+y^2)(0) - (-x)(2y)}{(x^2+y^2)^2}\,dy\right] \wedge dx \qquad dy \wedge dx = 0$$

$$+ \left[\frac{(x^2+y^2)(0) - (y)(2x)}{(x^2+y^2)^2}\,dx + \frac{(x^2+y^2)(1) - (y)(2y)}{(x^2+y^2)^2}\,dy\right] \wedge dy \qquad dx \wedge dy = 0$$

$$= \frac{2xy}{(x^2+y^2)^2}\,dy\,dx + \frac{-2xy}{(x^2+y^2)^2}\,dx\,dy$$

$$= \frac{-2xy}{(x^2+y^2)^2}\,dx\,dy \quad \frac{-2xy}{(x^2+y^2)^2}\,dx\,dy$$

$$= \frac{-4xy}{(x^2+y^2)^2}\,dx\,dy$$

(4) (h) $\omega = x^2 y\,dy\,dz$

$$d\omega = d[x^2 y \wedge (dy \wedge dz)]$$

$$= (2xy\,dx + x^2\,dy) \wedge (dy \wedge dz) + (-1)^0 x^2 y \wedge \underbrace{d(dy \wedge dz)}_{=0}$$

$$= 2xy\,dx \wedge (dy \wedge dz) + \underbrace{x^2\,dy \wedge (dy \wedge dz)}_{=0}$$

$$= 2xy\,dx\,dy\,dz$$

4. **4.** Let C be the line segment from the point $(-2, 0, 1)$ to $(3, 6, 9)$. Let $\omega_1 = y\,dx + x\,dy + xy\,dz$, $\omega_2 = z\,dx + y\,dy + 2x\,dz$, and $f(x, y, z) = xy$. Calculate the following:

(a) $\int_C f\omega_1$ (b) $\int_C f\omega_2$ (c) $\int_C \omega_1 + \omega_2$

For C: let $\vec{c}(t) = (-2,0,1) + t[(3,6,9) - (-2,0,1)]$, $0 \le t \le 1$

$$= (-2,0,1) + t(5,6,8) = (-2+5t, 6t, 1+8t)$$

$$\therefore \vec{c}(t) = (-2+5t, 6t, 1+8t), \quad \vec{c}'(t) = (5, 6, 8)$$

(6) $$\int_C f\,\omega_1 = \int_C xy^2\,dx + x^2y\,dy + x^2y^2\,dz$$

$$= \int_0^1 (-2+5t)(6t)^2(5) + (-2+5t)^2(6t)(6) + (-2+5t)^2(6t)^2(8)\,dt$$

$$= \int_0^1 (-2+5t)(6t)\left[30t + (-12+30t) + (-96t+240t^2)\right] dt$$

$$= \int_0^1 (-12t+30t^2)(-12-36t+240t^2)\,dt$$

$$144t - 360t^2$$
$$432t^2 - 1080t^3$$
$$-2880t^3 + 7200t^4$$

$$= \int_0^1 144t + 72t^2 - 3960t^3 + 7200t^4\,dt$$

$$= 72t^2 + 24t^3 - 990t^4 + 1440t^5 \Big|_0^1$$

$$= 72 + 24 - 990 + 1440 = \underline{546}$$

(6) $\int_C f\omega_2 = \int_C xyz\,dx + xy^2\,dy + 2x^2y\,dz$

$= \int_0^1 (-2+5t)(6t)(1+8t)(5) + (-2+5t)(6t)^2(6) + 2(-2+5t)^2(6t)(8)\,dt$

$(-72t^2+180t^3)(6)$ $\quad (4-20t+25t^2)96t$

$(-12t+30t^2)(5+40t)$

$(-60t+150t^2-480t^2+1200t^3)$

$= \int_0^1 (-60t-330t^2+1200t^3) + (-432t^2+1080t^3) + (384t-1920t^2+2400t^3)\,dt$

$= \int_0^1 (324t-2682t^2+4680t^3)\,dt$

$= 162t^2 - 894t^3 + 1170t^4 \Big|_0^1 = 162-894+1170$

$= \underline{438}$

(c) $\int_C \omega_1+\omega_2 = \int_C (y+z)dx + (x+y)dy + (2x+xy)dz$

$= \int_0^1 (6t+1+8t)(5) + (-2+5t+6t)(6) + (-4+10t-12t+30t^2)(8)\,dt$

$$= \int_0^1 (5 + 70t) + (-12 + 66t) + (-32 - 16t + 240t^2)\, dt$$

$$= \int_0^1 (-39 + 120t + 240t^2)\, dt$$

$$= -39t + 60t^2 + 80t^3 \Big|_0^1 = -39 + 60 + 80$$

$$= \underline{101}$$

5.

5. Let C be parameterized by $c(t) = (t^2 + 4t,\ t + 1)$, $t \in [0, \pi]$. Let $\omega_1 = y\,dx + x\,dy$, $\omega_2 = y^2\,dx + x^2\,dy$, and $f(x, y) = x$. Calculate the following:

(a) $\int_C f\omega_1$ (b) $\int_C f\omega_2$ (c) $\int_C \omega_1 + \omega_2$

$$\vec{c}\,'(t) = (2t + 4,\ 1)$$

(a)
$$\int_C f\omega_1 = \int_C xy\,dx + x^2\,dy$$

$$= \int_0^\pi (t^2 + 4t)(t + 1)(2t + 4) + (t^2 + 4t)^2(1)\, dt$$

$(t^3 + 5t^2 + 4t)(2t + 4)$ $\qquad$ $t^4 + 8t^3 + 16t^2$

$$= \int_0^\pi (2t^4 + 10t^3 + 8t^2 + 4t^3 + 20t^2 + 16t) + (t^4 + 8t^3 + 16t^2)\, dt$$

$$= \int_0^{\pi} (16t + 44t^2 + 22t^3 + 3t^4)\,dt$$

$$= 8t^2 + \frac{44}{3}t^3 + \frac{22}{4}t^4 + \frac{3}{5}t^5 \Big|_0^{\pi}$$

$$= 8\pi^2 + \frac{44}{3}\pi^3 + \frac{11}{2}\pi^4 + \frac{3}{5}\pi^5$$

(5) $$\int_C f\omega_2 = \int_C xy^2\,dx + x^3\,dy$$

$$= \int_0^{\pi} (t^2+4t)(t+1)^2(2t+4) + (t^2+4t)^3(1)\,dt$$

$(2t^3+12t^2+16t)(t^2+2t+1)$ $\quad t^6 + 3(t^2)^2(4t) + 3(t^2)(4t)^2 + 64t^3$

$2t^5 + 12t^4 + 16t^3$

$4t^4 + 24t^3 + 32t^2$

$2t^3 + 12t^2 + 16t$

$$= \int_0^{\pi} (2t^5 + 16t^4 + 42t^3 + 44t^2 + 16t) + (t^6 + 12t^5 + 48t^4 + 64t^3)\,dt$$

$$= \int_0^{\pi} (t^6 + 14t^5 + 64t^4 + 106t^3 + 44t^2 + 16t)\,dt$$

$$= \frac{1}{7}t^7 + \frac{14}{6}t^6 + \frac{64}{5}t^5 + \frac{106}{4}t^4 + \frac{44}{3}t^3 + 8t^2 \Big|_0^{\pi}$$

$$= \frac{1}{7}\pi^7 + \frac{7}{3}\pi^6 + \frac{64}{5}\pi^5 + \frac{53}{2}\pi^4 + \frac{44}{3}\pi^3 + 8\pi^2$$

(c) $\int_C \omega_1 + \omega_2 = \int_C (y+y^2)\,dx + (x+x^2)\,dy$

$= \int_C y(y+1)\,dx + x(x+1)\,dy$

$= \int_0^{\pi} (t+1)(t+2)(2t+4) + (t^2+4t)(t^2+4t+1)(1)\,dt$

$(t^2+3t+2)(2t+4)$

$2t^3+6t^2+4t$

$4t^2+12t+8$

$t^4+4t^3+t^2$

$4t^3+16t^2+4t$

$= \int_0^{\pi} (2t^3+10t^2+16t+8) + (t^4+8t^3+17t^2+4t)\,dt$

$= \int_0^{\pi} (t^4+10t^3+27t^2+20t+8)\,dt$

$= \frac{t^5}{5} + \frac{10}{4}t^4 + 9t^3 + 10t^2 + 8t \Big|_0^{\pi}$

$= \frac{\pi^5}{5} + \frac{5}{2}\pi^4 + 9\pi^3 + 10\pi^2 + 8\pi$

C.

6. Let $\mathbf{V}\colon K \to \mathbb{R}^3$ be a vector field defined by $\mathbf{V}(x, y, z) = G(x, y, z)\mathbf{i} + H(x, y, z)\mathbf{j} + F(x, y, z)\mathbf{k}$, and let η be the 2-form on K given by

$$\eta = F\,dx\,dy + G\,dy\,dz + H\,dz\,dx.$$

Show that $d\eta = (\operatorname{div}\mathbf{V})\,dx\,dy\,dz$.

Example 15, p. 486 of text shows that

$$d\eta = \left(\frac{\partial F}{\partial z} + \frac{\partial G}{\partial x} + \frac{\partial H}{\partial y}\right) dx\,dy\,dz$$

But $\operatorname{div}\vec{V} = \frac{\partial G}{\partial x} + \frac{\partial H}{\partial y} + \frac{\partial F}{\partial z}$

$\therefore\ d\eta = (\operatorname{div}\vec{V})\,dx\,dy\,dz$

7.

7. If $\mathbf{V} = A(x, y, z)\mathbf{i} + B(x, y, z)\mathbf{j} + C(x, y, z)\mathbf{k}$ is a vector field on $K \subset \mathbb{R}^3$, define the operation Form_2: Vector Fields $\to$ 2-forms by

$$\text{Form}_2(\mathbf{V}) = A\,dy\,dz + B\,dz\,dx + C\,dx\,dy.$$

(G)

(a) Show that $\text{Form}_2(\alpha\mathbf{V}_1 + \mathbf{V}_2) = \alpha\ \text{Form}_2(\mathbf{V}_1) + \text{Form}_2(\mathbf{V}_2)$, where α is a real number.

Let $\vec{V}_1 = [A_1(x,y,z),\ B_1(x,y,z),\ C_1(x,y,z)]$

$\vec{V}_2 = [A_2(x,y,z),\ B_2(x,y,z),\ C_2(x,y,z)]$

$\therefore\ \alpha\vec{V}_1 + \vec{V}_2 = [\alpha A_1 + A_2,\ \alpha B_1 + B_2,\ \alpha C_1 + C_2]$

$\therefore$ By definition, and noting α is a 0-form,

$$\text{Form}_2(\alpha\vec{V}_1 + \vec{V}_2) = (\alpha A_1 + A_2)\,dy\,dz + (\alpha B_1 + B_2)\,dz\,dx + (\alpha C_1 + C_2)\,dx\,dy$$

$$= \alpha A_1\,dy\,dz + A_2\,dy\,dz + \alpha B_1\,dz\,dx + B_2\,dz\,dx + \alpha C_1\,dx\,dy + C_2\,dx\,dy$$

By definition of addition of 2-forms (p. 479 of text)

$$= \alpha(A_1\,dy\,dz + B_1\,dz\,dx + C_1\,dx\,dy) + A_2\,dy\,dz + B_2\,dz\,dx + C_2\,dx\,dy$$

Def. of product on 2-forms (p. 479)

$$= \alpha\,\text{Form}_2(\vec{V}_1) + \text{Form}_2(\vec{V}_2)$$

(6) (b) Show that $\text{Form}_2(\text{curl } \mathbf{V}) = d\omega$, where $\omega = A\,dx + B\,dy + C\,dz$.

$$\text{Curl } \vec{V} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ A & B & C \end{vmatrix} = (C_y - B_z,\ A_z - C_x,\ B_x - A_y)$$

$$\therefore \text{Form}_2(\text{curl } \vec{V}) =$$

$$(C_y - B_z)\,dy\,dz + (A_z - C_x)\,dz\,dx + (B_x - A_y)\,dx\,dy \quad [1]$$

$$d\omega = d(A \wedge dx) + d(B \wedge dy) + d(C \wedge dz)$$

$$= (A_x\,dx + A_y\,dy + A_z\,dz) \wedge dx + (-1)^0 A \wedge \underbrace{d(dx)}_{=0}$$

$$+ (B_x\,dx + B_y\,dy + B_z\,dz) \wedge dy + (-1)^0 B \wedge \underbrace{d(dy)}_{=0}$$

$$+ (C_x\,dx + C_y\,dy + C_z\,dz) \wedge dz + (-1)^0 C \wedge \overset{=0}{d(dz)}$$

$$= \overset{=0}{A_x(dx \wedge dx)} + A_y(dy \wedge dx) + A_z(dz \wedge dx)$$

$$+ B_x(dx \wedge dy) + \overset{=0}{B_y(dy \wedge dy)} + B_z(dz \wedge dy)$$

$$+ C_x(dx \wedge dz) + C_y(dy \wedge dz) + \overset{=0}{C_z(dz \wedge dz)}$$

$$= -A_y\,dx\,dy + A_z\,dz\,dx$$

$$+ B_x\,dx\,dy - B_z\,dy\,dz$$

$$- C_x\,dz\,dx + C_y\,dy\,dz$$

put in cyclic order

$$= (C_y - B_z)\,dy\,dz + (A_z - C_x)\,dz\,dx + (B_x - A_y)\,dx\,dy \quad [2]$$

Since [1] = [2], $\text{Form}_2(\text{curl}\,\vec{V}) = d\omega$

8.

8. Using the differential form version of Stokes' theorem, prove the vector field version in Section 8.2. Repeat for Gauss' theorem.

(a) Stokes' Theorem

Let ω be a 1-form on an open set K that contains an oriented surface S in $\mathbb{R}^3$ with boundary ∂S, a simple closed curve oriented the same as S.

Let $\omega = P\,dx + Q\,dy + R\,dz$, $\vec{F} = (P, Q, R)$

As shown in #7(b) above,

$$d\omega = (R_y - Q_z)\,dy\,dz + (P_z - R_x)\,dz\,dx + (Q_x - P_y)\,dx\,dy \quad [1]$$

And note $\nabla \times \vec{F} = (R_y - Q_z, P_z - R_x, Q_x - P_y)$ [2]

Let $\vec{\phi}(u,v) = [x(u,v), y(u,v), z(u,v)]$ be a parametrization of S. Note that

$$\vec{\phi}_u \times \vec{\phi}_v = (y_u z_v - y_v z_u, z_u x_v - z_v x_u, x_u y_v - x_v y_u)$$

$$= \left[\frac{\partial(y,z)}{\partial(u,v)}, \frac{\partial(z,x)}{\partial(u,v)}, \frac{\partial(x,y)}{\partial(u,v)} \right] \quad [3]$$

Let $\vec{c}(t) = [x(t), y(t), z(t)]$ be an orientation preserving parametrization of ∂S, $\vec{c}(t): [a,b] \to \mathbb{R}^3$

We are given $$\iint_S d\omega = \int_{\partial S} \omega \quad [4]$$

By definition, p. 481 of text, and from [1],

$$\iint_S d\omega = \iint_S (Q_x - P_y)\,dx\,dy + (R_y - Q_z)\,dy\,dz + (P_z - R_x)\,dz\,dx$$

$$= \iint_D \left[(Q_x - P_y) \cdot \frac{\partial(x,y)}{\partial(u,v)} + (R_y - Q_z) \cdot \frac{\partial(y,z)}{\partial(u,v)} + (P_z - R_x) \frac{\partial(z,x)}{\partial(u,v)} \right] du\,dv$$

$$= \iint_D (Q_x - P_y, R_y - Q_z, P_z - R_x) \cdot \left[\frac{\partial(x,y)}{\partial(u,v)}, \frac{\partial(y,z)}{\partial(u,v)}, \frac{\partial(z,x)}{\partial(u,v)} \right] du\,dv$$

dot product

$$= \iint_D (R_y - Q_z, P_z - R_x, Q_x - P_y) \cdot \left[\frac{\partial(y,z)}{\partial(u,v)}, \frac{\partial(z,x)}{\partial(u,v)}, \frac{\partial(x,y)}{\partial(u,v)} \right] du\,dv$$

$$= \iint_D (\nabla \times \vec{F}) \cdot (\vec{\phi}_u \times \vec{\phi}_v)\, du\,dv$$ From [2], [3]

$$= \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$ From definition, p. 400 of text

$$\therefore \iint_S dw = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S}$$ [5]

Now, by definition on p. 480 of text,

$$\int_{\partial S} \omega = \int_{\partial S} P\,dx + Q\,dy + R\,dz$$

$$= \int_a^b \left[P \cdot \frac{dx}{dt} + Q \frac{dy}{dt} + R \frac{dz}{dt} \right] dt$$

$$= \int_a^b (P, Q, R) \cdot \vec{c}\,'(t)\, dt$$

$$= \int_a^b \vec{F} \cdot \vec{c}\,'(t)\, dt$$

$$= \int_{\vec{c}} \vec{F} \cdot d\vec{s}$$

definition of line integral, p. 359 of text

$$\therefore \int_{\partial S} \omega = \int_{\vec{c}} \vec{F} \cdot d\vec{s} \qquad [6]$$

$\therefore$ From [4], [5], [6],

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \iint_S d\omega = \int_{\partial S} \omega = \int_{\vec{c}} \vec{F} \cdot d\vec{s}$$

or,

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\vec{c}} \vec{F} \cdot d\vec{s}$$

(6) Gauss' Theorem

Let W be an elementary region in $\mathbb{R}^3$ with ∂W the boundary and given outward orientation. Let η be a 2-form on an open set K containing W, so that $\eta = F\,dx\,dy + G\,dy\,dz + H\,dz\,dx$

and $\vec{F} = (G, H, F)$, $F, G, H : \mathbb{R}^3 \to \mathbb{R}$

Note the order of components for $\vec{F}$.

As shown by Example 15, p. 486-7 of text,

$$d\eta = (F_z + G_x + H_y)\,dx\,dy\,dz$$

$$= (G_x + H_y + F_z)\,dx\,dy\,dz$$

$$\therefore d\eta = (\nabla \cdot \vec{F})\,dx\,dy\,dz \qquad [1]$$

We are given $$\iiint_W d\eta = \iint_{\partial W} \eta \qquad [2]$$

$$\therefore \iiint_W d\eta = \iiint_W (\nabla \cdot \vec{F})\,dx\,dy\,dz \qquad [3]$$

Let $\vec{\phi}(u,v) = [x(u,v), y(u,v), z(u,v)]$ be a parametrization of ∂W. Note that

$$\vec{\phi}_u \times \vec{\phi}_v = \left(y_u z_v - y_v z_u,\ z_u x_v - z_v x_u,\ x_u y_v - x_v y_u\right)$$

$$= \left[\frac{\partial(y,z)}{\partial(u,v)},\ \frac{\partial(z,x)}{\partial(u,v)},\ \frac{\partial(x,y)}{\partial(u,v)}\right]$$

Now, from definition on p. 481 of text,

$$\iint_{\partial W} n = \iint_{\partial W} F\,dx\,dy + G\,dy\,dz + H\,dz\,dx$$

$$= \iint_D \left[(F)\frac{\partial(x,y)}{\partial(u,v)} + (G)\frac{\partial(y,z)}{\partial(u,v)} + (H)\frac{\partial(z,x)}{\partial(u,v)}\right] du\,dv$$

$$= \iint_D (F,G,H) \cdot \left[\frac{\partial(x,y)}{\partial(u,v)}, \frac{\partial(y,z)}{\partial(u,v)}, \frac{\partial(z,x)}{\partial(u,v)}\right] du\,dv$$

dot product

$$= \iint_D (G,H,F) \cdot \left[\frac{\partial(y,z)}{\partial(u,v)}, \frac{\partial(z,x)}{\partial(u,v)}, \frac{\partial(x,y)}{\partial(u,v)}\right] du\,dv$$

$$= \iint_D \vec{F} \cdot (\vec{\phi}_u \times \vec{\phi}_v)\,du\,dv = \iint_{\partial W} \vec{F} \cdot d\vec{S}$$

$$\therefore \iint_{\partial W} n = \iint_{\partial W} \vec{F} \cdot d\vec{S} \qquad [4]$$

$\therefore$ From [2], [3], and [4],

$$\iiint_W (\nabla \cdot \vec{F})\, dx\,dy\,dz = \iiint_W dn = \iint_{\partial W} n = \iint_{\partial W} \vec{F} \cdot d\vec{S}$$

i.e., $\iiint_W dn = \iint_{\partial W} n$ implies

$$\iiint_W (\nabla \cdot \vec{F})\, dx\,dy\,dz = \iint_{\partial W} \vec{F} \cdot d\vec{S}$$

9. **9.** Interpret Theorem 16 in the case $k = 1$.

Note: "Theorem 16" is from the 5th edition.

It is Theorem 14 on p. 488 for the 6th edition.

> **Theorem** 14 General Stokes' Theorem Let M be an oriented k-manifold in $\mathbb{R}^3$ ($k = 2$ or 3) contained in some open set K. Suppose ω is $(k-1)$-form on K. Then
>
> $$\int_{\partial M} \omega = \int_M d\omega.$$

So, M is an oriented 1-manifold in $\mathbb{R}^3$, which would be a line, C, perhaps curved, and ω

would be a 0-form on K; i.e., a function $f: \mathbb{R}^3 \to 1$. The boundary for M would be the endpoints of the line.

From p. 485 of the text, $dw = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$

$$\therefore \int_M dw = \int_C \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz$$

which is a line integral.

If $\vec{c}(t): [a,b] \to \mathbb{R}^3$ is an orientation preserving parametrization, $\vec{c}(t) = [x(t), y(t), z(t)]$, then

$$\int_M dw = \int_a^b \left[f_x\, x'(t) + f_y\, y'(t) + f_z\, z'(t)\right] dt$$

$$= \int_a^b \nabla f(\vec{c}(t)) \cdot \vec{c}\,'(t)\, dt = \int_{\vec{c}} \nabla f \cdot d\vec{s}$$

and $$\int_a^b \nabla f(\vec{c}(t)) \cdot \vec{c}\,'(t)\, dt = f(\vec{c}(b)) - f(\vec{c}(a))$$

since $\frac{d}{dt}\left[f(\vec{c}(t))\right] = \nabla f \cdot \vec{c}\,'(t)$

$\therefore$ Using ω for f, and $\vec{B} = \vec{c}(b)$, $\vec{A} = \vec{c}(a)$

$$\int_{\partial m} \omega = \int_m d\omega = \omega(\vec{B}) - \omega(\vec{A})$$

10.

10. Let $\omega = (x+y)\,dz + (y+z)\,dx + (x+z)\,dy$, and let S be the upper part of the unit sphere; that is, S is the set of (x, y, z) with $x^2 + y^2 + z^2 = 1$ and $z \geq 0$. ∂S is the unit circle in the xy plane. Evaluate $\int_{\partial S} \omega$ both directly and by Stokes' theorem.

Let $\vec{c}(\theta) = (\cos\theta, \sin\theta, 0)$, $0 \leq \theta \leq 2\pi$, for a parametrization for ∂S.

$$\therefore \int_{\partial S} \omega = \int_{\vec{c}} (y+z)\,dx + (x+z)\,dy + (x+y)\,dz$$

$$= \int_0^{2\pi} (\sin\theta + 0)(-\sin\theta) + (\cos\theta + 0)\cos\theta + (\sin\theta + \cos\theta)0\,d\theta$$

$$= \int_0^{2\pi} \cos^2\theta - \sin^2\theta\,d\theta = \int_0^{2\pi} \cos(2\theta)\,d\theta = \frac{\sin 2\theta}{2}\Big|_0^{2\pi}$$

$= 0 - 0 = 0$

$\therefore \int_{\partial S} \omega = 0$

From Stoke's Theorem, $\int_{\partial S} \omega = \iint_S d\omega$

$\therefore d\omega = d[(x+y)dz] + d[(y+z)dx] + d[(x+z)dy]$

$= d[(x+y) \wedge dz] + d[(y+z) \wedge dx] + d[(x+z) \wedge dy]$

$= \left[\frac{\partial(x+y)}{\partial x} dx + \frac{\partial(x+y)}{\partial y} dy + \frac{\partial(x+y)}{\partial z} dz\right] \wedge dz$

$+ \left[\frac{\partial(y+z)}{\partial x} dx + \frac{\partial(y+z)}{\partial y} dy + \frac{\partial(y+z)}{\partial z} dz\right] \wedge dx$

$+ \left[\frac{\partial(x+z)}{\partial x} dx + \frac{\partial(x+z)}{\partial y} dy + \frac{\partial(x+z)}{\partial z} dz\right] \wedge dy$

$= [dx + dy] \wedge dz$

$+ [dy + dz] \wedge dx$

$+ [dx + dz] \wedge dy$

$= -dz\,dx + dy\,dz + (-dx\,dy + dz\,dx) + (dx\,dy - dy\,dz)$

$= 0$

$\therefore \; d\omega = 0 \Rightarrow \iint_S d\omega = 0$

11.

11. Let T be the triangular solid bounded by the xy plane, the xz plane, the yz plane, and the plane $2x + 3y + 6z = 12$. Compute

$$\iint_{\partial T} F_1 \, dx\,dy + F_2 \, dy\,dz + F_3 \, dz\,dx$$

directly and by Gauss' theorem, if

(a) $F_1 = 3y$, $F_2 = 18z$, $F_3 = -12$; and

(b) $F_1 = z$, $F_2 = x^2$, $F_3 = y$.

Need a parametrization for the various surfaces, each with outward orientation.

Note $2x + 3y + 6z = 12$ intersects axes at: $(6,0,0)$, $(0,4,0)$, $(0,0,2)$

xy-plane ($z = 0$), $\therefore \; 2x + 3y = 12$, $3y = 12 - 2x$, $y = 4 - \frac{2}{3}x$

$\therefore \; \phi(u,v) = (u, v, 0)$, $0 \le u \le 6$, $0 \le v \le 4 - \frac{2}{3}u$

$\therefore \; \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \quad \frac{\partial(y,z)}{\partial(u,v)} = \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix} = 0 \quad \frac{\partial(z,x)}{\partial(u,v)} = 0$

But note $\phi_u \times \phi_v = (0, 0, 1)$, which is upward pointing ($\hat{k}$ points up), and we want downward, or outward pointing. $\therefore$ Our computation for

The xy-plane component will be the negative of the computed value.

yz-plane ($x=0$), $\therefore 3y+6z=12$, $z=2-\frac{y}{2}$

$\therefore \phi(u,v)=(0,u,v)$, $\quad 0\le u\le 4$, $0\le v\le 2-\frac{u}{2}$

$\therefore \frac{\partial(x,y)}{\partial(u,v)}=0$, $\frac{\partial(y,z)}{\partial(u,v)}=\begin{vmatrix}1&0\\0&1\end{vmatrix}=1$ $\frac{\partial(z,x)}{\partial(u,v)}=0$

Note $\phi_u \times \phi_v = (1,0,0)=\hat{i}$, which points inward to the triangular solid. $\therefore$ Take negative of computed value for yz-plane component.

xz-plane ($y=0$), $\therefore 2x+6z=12$, $6z=12-2x$, $z=2-\frac{x}{3}$

$\therefore \phi(u,v)=(u,0,v)$, $\quad 0\le u\le 6$, $0\le v\le 2-\frac{u}{3}$

$\therefore \frac{\partial(x,y)}{\partial(u,v)}=0$, $\frac{\partial(y,z)}{\partial(u,v)}=0$, $\frac{\partial(z,x)}{\partial(u,v)}=\begin{vmatrix}0&1\\1&0\end{vmatrix}=-1$

Note $\phi_u \times \phi_v = (0,-1,0)=-\hat{j}$ which points outward to triangular solid.

$2x+3y+6z=12$ plane, $6z=12-2x-3y$, $z=2-\frac{x}{3}-\frac{y}{2}$

$\phi(u,v)=\left(u, v, 2-\frac{u}{3}-\frac{v}{2}\right)$, $0\le u\le 6$, $0\le v\le 4-\frac{2}{3}u$

$\therefore \frac{\partial(x,y)}{\partial(u,v)}=\begin{vmatrix}1&0\\0&1\end{vmatrix}=1$ $\frac{\partial(y,z)}{\partial(u,v)}=\begin{vmatrix}0&1\\-\frac{1}{3}&-\frac{1}{2}\end{vmatrix}=\frac{1}{3}$ $\frac{\partial(z,x)}{\partial(u,v)}=\begin{vmatrix}-\frac{1}{3}&-\frac{1}{2}\\1&0\end{vmatrix}=\frac{1}{2}$

Note $\phi_u\times\phi_v=\left(\frac{1}{3},\frac{1}{2},1\right)$, which points outward to the triangular solid.

$\therefore$ If $\partial T=\sum_{i=1}^{4} T_i$, T_1 = xy-plane component
T_2 = yz-plane component
T_3 = xz-plane component
and T_4 = $2x+3y+6z$ plane component

Then $\iint_{\partial T} F_1\,dx\,dy+F_2\,dy\,dz+F_3\,dz\,dx$

$$=-\iint_{D_1} F_1(1)\,du\,dv-\iint_{D_2} F_2(1)\,du\,dv+\iint_{D_3} F_3(-1)\,du\,dv$$

xy part, yz part, xz part

$$+\iint_{D_4} F_1(1)+F_2\left(\frac{1}{3}\right)+F_3\left(\frac{1}{2}\right)du\,dv$$

$2x+3y+6z=12$ part

where the Jacobian is in parenthesis within the integral, and the sign before the integral is from the orientation.

$$= -\int_0^6 \int_0^{4-\frac{2}{3}u} F_1 \, dv\,du - \int_0^4 \int_0^{2-\frac{u}{2}} F_2 \, dv\,du - \int_0^6 \int_0^{2-\frac{u}{3}} F_3 \, dv\,du$$

xy $\qquad$ yz $\qquad$ zx

$$+ \int_0^6 \int_0^{4-\frac{2}{3}u} \left[F_1(1) + F_2\left(\tfrac{1}{3}\right) + F_3\left(\tfrac{1}{2}\right) \right] dv\,du$$

$2x + 3y + 6z = 12$

(4) (a) $F_1 = 3y,\ F_2 = 18z,\ F_3 = -12;$

From above, $\iint_{\partial T} (3y)\,dx\,dy + (18z)\,dy\,dz + (-12)\,dz\,dx$

$$= -\int_0^6 \int_0^{4-\frac{2}{3}u} (3v)\,dv\,du - \int_0^4 \int_0^{2-\frac{u}{2}} 18v\,dv\,du - \int_0^6 \int_0^{2-\frac{u}{3}} (-12)\,dv\,du$$

$\phi(u,v) = (u,v,0)$ $\qquad$ $\phi(u,v) = (0,u,v)$ $\qquad$ $\phi(u,v) = (u,0,v)$

$$+ \int_0^6 \int_0^{4-\frac{2}{3}u} (3v) + 18\left(2 - \tfrac{u}{3} - \tfrac{v}{2}\right)\left(\tfrac{1}{3}\right) + (-12)\left(\tfrac{1}{2}\right) dv\,du$$

$3v$ $\qquad$ $12 - 2u - 3v$ $\qquad$ $-6 \quad = 6 - 2u$

$\phi(u,v) = \left(u, v, 2 - \frac{u}{3} - \frac{v}{2}\right)$

$$= -\int_0^6 \frac{3v^2}{2}\Big|_0^{4-\frac{2}{3}u} du - \int_0^4 9v^2\Big|_0^{2-\frac{u}{2}} du - \int_0^6 -12v\Big|_0^{2-\frac{u}{3}} du$$

xy part $\qquad$ yz part $\qquad$ zx part

$$+ \int_0^6 (6-2u)v\Big|_{v=0}^{v=4-\frac{2}{3}u}$$

$2x + 3y + 6z = 12$ part

$$= -\int_0^6 \frac{3}{2}\left(16 - \frac{16}{3}u + \frac{4}{9}u^2\right)du - \int_0^4 9\left(4 - 2u + \frac{u^2}{4}\right)du - \int_0^6 -12\left(2 - \frac{u}{3}\right)du$$

$$+ \int_0^6 (6-2u)\left(4 - \frac{2}{3}u\right)du$$

$24 - 8u - 4u + \frac{4}{3}u^2$

$$= -\left[24u - 4u^2 + \frac{2}{9}u^3\Big|_0^6\right] - \left[36u - 9u^2 + \frac{3}{4}u^3\Big|_0^4\right] - \left[-24u + 2u^2\Big|_0^6\right]$$

$$+ \left[24u - 6u^2 + \frac{4}{9}u^3\Big|_0^6\right]$$

$$= -\left[24(6) - 4(36) + \frac{2}{9}(36\cdot 6)\right] - \left[36(4) - 9(16) + 48\right] - \left[-24(6) + 72\right]$$

$$+ \left[24(6) - 216 + \frac{4}{9}(36\cdot 6)\right]$$

$$= -[48] - [48] - [-72] + [24] = -96 + 96$$

$$= \underline{0}$$

By Gauss' Divergence Theorem,

$$d(3y\,dx\,dy + 18z\,dy\,dz - 12\,dz\,dx)$$

$$= d(3y \wedge dx \wedge dy) + d(18z \wedge dy \wedge dz) - d(12 \wedge dz \wedge dx)$$

$$= d(3y) \wedge (dx \wedge dy) + 3y \wedge d(dx \wedge dy)$$ ($d(dx \wedge dy) = 0$)

$$+ d(18z) \wedge (dy \wedge dz) + 18z \wedge d(dy$$

$$- d(12) \wedge (dz \wedge dx) - 12 \wedge d(dz \wedge dx)$$ ($d(dz \wedge dx) = 0$)

$$= (0\,dx + 3\,dy + 0\,dz) \wedge (dx \wedge dy)$$

$$+ (0\,dx + 0\,dy + 18\,dz) \wedge (dy \wedge dz)$$

$$- (0) \wedge (dz \wedge dx)$$

$$= 3[dy \wedge (dx \wedge dy)] + 18[dz \wedge (dy \wedge dz)]$$

$$= -3[(dy \wedge dy) \wedge dx] - 18[(dz \wedge dz) \wedge dy] = 0$$ ($dy \wedge dy = 0$, $dz \wedge dz = 0$)

$$\therefore \iiint_T d(F_1\,dx\,dy + F_2\,dy\,dz + F_3\,dz\,dx) = \iiint_T 0 = \underline{0}$$

(6) (b) $F_1 = z,\ F_2 = x^2,\ F_3 = y.$

Using the developed formula above,

$$\therefore \iint_{\partial T} (z)\,dx\,dy + (x^2)\,dy\,dz + (y)\,dz\,dx$$

$$= -\int_0^6 \int_0^{4-\frac{2}{3}u} F_1\,dv\,du - \int_0^4 \int_0^{2-\frac{u}{2}} F_2\,dv\,du - \int_0^6 \int_0^{2-\frac{u}{3}} F_3\,dv\,du$$

F_1: xy, F_2: yz, F_3: zx

$$+\int_0^6\int_0^{4-\frac{2}{3}u}\left[F_1(1)+F_2\left(\tfrac{1}{3}\right)+F_3\left(\tfrac{1}{2}\right)\right]dv\,du$$

$2x+3y+6z=12$

$$=-\int_0^6\int_0^{4-\frac{2}{3}u}0\,du\,dv-\int_0^4\int_0^{2-\frac{u}{2}}0\,du\,dv-\int_0^6\int_0^{2-\frac{u}{3}}0\,du\,dv$$

$\phi(u,v)=(u,v,0)$ $\therefore F_1(u,v,0)=0$

$\phi(u,v)=(0,u,v)$ $\therefore F_2(0,u,v)=0$

$\phi(u,v)=(u,0,v)$ $\therefore F_3(u,0,v)=0$

$$+\int_0^6\int_0^{4-\frac{2}{3}u}\left(2-\tfrac{u}{3}-\tfrac{v}{2}\right)+\left(u^2\right)\left(\tfrac{1}{3}\right)+\left(v\right)\left(\tfrac{1}{2}\right)dv\,du$$

$\phi(u,v)=\left(u,v,2-\frac{u}{3}-\frac{v}{2}\right)$

$F_1(\phi(u,v))=2-\frac{u}{3}-\frac{v}{2}$ $\quad F_2(\phi(u,v))=u^2$ $\quad F_3(\phi(u,v))=v$

$$=0-0-0$$

$$+\int_0^6\int_0^{4-\frac{2}{3}u}2-\frac{u}{3}+\frac{u^2}{3}\,dv\,du$$

$$=\int_0^6\left(2-\tfrac{u}{3}+\tfrac{u^2}{3}\right)\left(4-\tfrac{2}{3}u\right)du=\int_0^6 8-\tfrac{8}{3}u+\tfrac{14}{9}u^2-\tfrac{2}{9}u^3\,du$$

$8-\frac{4}{3}u+\frac{4}{3}u^2-\frac{4}{3}u+\frac{2}{9}u^2-\frac{2}{9}u^3$

$$=8u-\tfrac{4}{3}u^2+\tfrac{14}{27}u^3-\tfrac{1}{18}u^4\Big|_0^6$$

$$=48-48+\frac{14\cdot6\cdot6\cdot6}{3\cdot3\cdot3}-\frac{6\cdot6\cdot6\cdot6}{2\cdot3\cdot3}=112-72=\underline{40}$$

By Gauss' Theorem

$$d(z\,dx\,dy + x^2\,dy\,dz + y\,dz\,dx)$$

$$= d(z)\wedge(dx\wedge dy) + z\wedge \underbrace{d(dx\wedge dy)}_{=0}$$

$$+ d(x^2)\wedge(dy\wedge dz) + x^2\wedge \underbrace{d(dy\wedge dz)}_{=0}$$

$$+ d(y)\wedge(dz\wedge dx) + y\wedge \underbrace{d(dz\wedge dx)}_{=0}$$

$$= (0\,dx + 0\,dy + dz)\wedge(dx\wedge dy)$$

$$+ (2x\,dx + 0\,dy + 0\,dz)\wedge(dy\wedge dz)$$

$$+ (0\,dx + dy + 0\,dz)\wedge(dz\wedge dx)$$

$$= dx\,dy\,dz + 2x\,dx\,dy\,dz + dx\,dy\,dz$$

$$= (2 + 2x)\,dx\,dy\,dz$$

$$\therefore \iiint_T d(z\,dx\,dy + x^2\,dy\,dz + y\,dz\,dx) = \iiint_T (2+2x)\,dx\,dy\,dz$$

$$= \int_0^6 \int_0^{4-\frac{2}{3}x} \int_0^{2-\frac{x}{3}-\frac{y}{2}} (2+2x)\,dz\,dy\,dx$$

$$= \int_0^6 \int_0^{4-\frac{2}{3}x} (2+2x)\left(2-\frac{x}{3}-\frac{y}{2}\right) dy\, dx$$

$4 - \frac{2x}{3} - y + 4x - \frac{2}{3}x^2 - xy$

$$= \int_0^6 \int_0^{4-\frac{2}{3}x} \left(4 + \frac{10}{3}x - y - \frac{2}{3}x^2 - xy\right) dy\, dx$$

$$= \int_0^6 4y + \frac{10}{3}xy - \frac{y^2}{2} - \frac{2}{3}x^2 y - x\frac{y^2}{2}\Big|_{y=0}^{y=4-\frac{2}{3}x} dx$$

$$= \int_0^6 4(4-\tfrac{2}{3}x) + \frac{10}{3}x(4-\tfrac{2}{3}x) - \frac{(4-\frac{2}{3}x)^2}{2} - \frac{2}{3}x^2(4-\tfrac{2}{3}x) - \frac{x}{2}(4-\tfrac{2}{3}x)^2 \, dx$$

$16 - \frac{16}{3}x + \frac{4}{9}x^2$

$16 - \frac{8}{3}x + \frac{40}{3}x - \frac{20}{9}x^2 - \left(8 - \frac{8}{3}x + \frac{2}{9}x^2\right) - \frac{8}{3}x^2 + \frac{4}{9}x^3 - 8x + \frac{8}{3}x^2 - \frac{2}{9}x^3$

$$= \int_0^6 8 + \frac{16}{3}x - \frac{22}{9}x^2 + \frac{2}{9}x^3 \, dx$$

$$= 8x + \frac{8}{3}x^2 - \frac{22}{27}x^3 + \frac{2}{36}x^4\Big|_0^6$$

$$= 48 + 96 - \frac{22 \cdot 6 \cdot 6 \cdot 6}{3 \cdot 3 \cdot 3} + \frac{2}{36} \cdot 36 \cdot 36$$

$$= 144 - 176 + 72$$

$$= \underline{40}$$

12. **12.** Evaluate $\iint_S \omega$, where $\omega = z\,dx\,dy + x\,dy\,dz + y\,dz\,dx$ and S is the unit sphere, directly and by Gauss' theorem.

(a) Directly

Let $\vec{T}(\phi,\theta) = (\cos\theta\sin\phi, \sin\theta\sin\phi, \cos\phi)$, $0 \le \theta \le 2\pi$, $0 \le \phi \le \pi$

be a parametrization of the unit sphere.

$\vec{T}_\phi \times \vec{T}_\theta = (\cos\theta\sin^2\phi, \sin\theta\sin^2\phi, \cos\phi\sin\phi)$ p. 401 of text

and this points outward.

$$\frac{\partial(x,y)}{\partial(\phi,\theta)} = \begin{vmatrix} \cos\theta\cos\phi & -\sin\theta\sin\phi \\ \sin\theta\cos\phi & \cos\theta\sin\phi \end{vmatrix} = \cos\phi\sin\phi$$

$$\frac{\partial(y,z)}{\partial(\phi,\theta)} = \begin{vmatrix} \sin\theta\cos\phi & \cos\theta\sin\phi \\ -\sin\phi & 0 \end{vmatrix} = \cos\theta\sin^2\phi$$

$$\frac{\partial(z,x)}{\partial(\phi,\theta)} = \begin{vmatrix} -\sin\phi & 0 \\ \cos\theta\cos\phi & -\sin\theta\sin\phi \end{vmatrix} = \sin\theta\sin^2\phi$$

By definition,

$$\iint_S \omega = \iint_S \left[z\frac{\partial(x,y)}{\partial(\phi,\theta)} + x\frac{\partial(y,z)}{\partial(\phi,\theta)} + y\frac{\partial(z,x)}{\partial(\phi,\theta)} \right] d\phi\,d\theta$$

$$= \int_0^{2\pi}\int_0^{\pi} \Big[(\cos\phi)(\cos\phi\sin\phi) + (\cos\theta\sin\phi)(\cos\theta\sin^2\phi) + (\sin\theta\sin\phi)(\sin\theta\sin^2\phi) \Big] d\phi\,d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi} \cos^2\phi \sin\phi + \cos^2\theta \sin^3\phi + \sin^2\theta \sin^3\phi \; d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi} \cos^2\phi \sin\phi + \sin^3\phi \; d\phi \, d\theta$$

$$= \int_0^{2\pi} \int_0^{\pi} \sin\phi \; d\phi \, d\theta = 2\pi \left[-\cos\phi \right]_{\phi=0}^{\phi=\pi}$$

$$= 2\pi \left[1 - (-1) \right] = \underline{4\pi}$$

(5) Gauss' Theorem

$$\iint_S \omega = \iiint_B d\omega \quad , \text{ where } B = \text{solid unit ball.}$$

$$\omega = z \, dx dy + x \, dy dz + y \, dz dx$$

By Example #15 p. 486 of text,

$$d\omega = \left(\frac{\partial(z)}{\partial z} + \frac{\partial(x)}{\partial x} + \frac{\partial(y)}{\partial y} \right) dx dy dz = 3 \, dx dy dz$$

$$\therefore \iint_S \omega = \iiint_B 3 \, dx dy dz$$

$= 3$ (volume of unit ball)

$= 3\left(\frac{4}{3}\pi\right) = \underline{4\pi}$

13. **13.** Let R be an elementary region in $\mathbb{R}^3$. Show that the volume of R is given by the formula

$$v(R) = \frac{1}{3}\iint_{\partial R} x\,dy\,dz + y\,dz\,dx + z\,dx\,dy.$$

Let $\omega = x\,dy\,dz + y\,dz\,dx + z\,dx\,dy$

From Example # 15 p. 486 of text,

$$d\omega = \left(\frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z)\right) dx\,dy\,dz = 3\,dx\,dy\,dz$$

By Gauss' Divergence Theorem,

$$\iint_{\partial R} \omega = \iiint_R d\omega = \iiint_R 3\,dx\,dy\,dz$$

$$= 3\iiint_R dx\,dy\,dz = 3\,(\text{volume of } R)$$

$$\therefore \text{vol}(R) = \frac{1}{3}\iint_{\partial R} \omega = \frac{1}{3}\iint_{\partial R} x\,dy\,dz + y\,dz\,dx + z\,dx\,dy$$

14.

14. In Section 4.2, we saw that the length $l(\mathbf{c})$ of a curve $\mathbf{c}(t) = (x(t), y(t), z(t))$, $a \le t \le b$, was given by the formula

$$l(\mathbf{c}) = \int d\mathbf{s} = \int_a^b \left(\frac{ds}{dt}\right) dt$$

where, loosely speaking, $(ds)^2 = (dx)^2 + (dy)^2 + (dz)^2$, that is,

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}.$$

Now suppose a surface S is given in parametrized form by $\Phi(u, v) = (x(u, v), y(u, v), z(u, v))$, where $(u, v) \in D$. Show that the area of S can be expressed as

$$A(S) = \iint_D dS,$$

where formally $(dS)^2 = (dx \wedge dy)^2 + (dy \wedge dz)^2 + (dz \wedge dx)^2$, a formula requiring interpretation. [HINT:

$$dx = \frac{\partial x}{\partial u}\, du + \frac{\partial x}{\partial v}\, dv,$$

and similarly for dy and dz. Use the law of forms for the basic 1-forms du and dv. Then dS turns out to be a function times the basic 2-form $du\, dv$, which we can integrate over D.]

From definition #(1) p. 485 of text,

$$dx = x_u\, du + x_v\, dv$$

$$dy = y_u\, du + y_v\, dv$$

$$dz = z_u\, du + z_v\, dv$$

$$\therefore\ dx \wedge dy = (x_u\, du + x_v\, dv) \wedge (y_u\, du + y_v\, dv)$$

$$= x_u y_u \underbrace{(du \wedge du)}_{=0} + x_v y_u (dv \wedge du)$$

$$+ x_u y_v (du \wedge dv) + \underbrace{x_v y_v (dv \wedge dv)}_{=0}$$

$$= (x_u y_v - x_v y_u)\, du\, dv = \frac{\partial(x, y)}{\partial(u, v)}\, du\, dv$$

Similarly,

$$dy \wedge dz = \frac{\partial(y, z)}{\partial(u, v)}\, du\, dv \qquad dz \wedge dx = \frac{\partial(z, x)}{\partial(u, v)}\, du\, dv$$

If ω is a 2-form, the product $(\omega)^2$ hasn't been defined. So if you define $(f\,du\,dv)^2 = f^2(du\,dv)^2$ for a 0-form f, and allow distributivity, then

$$(dS)^2 = (dx\wedge dy)^2 + (dy\wedge dz)^2 + (dz\wedge dx)^2$$

$$= \left(\left[\frac{\partial(x,y)}{\partial(u,v)}\right]^2 + \left[\frac{\partial(y,z)}{\partial(u,v)}\right]^2 + \left[\frac{\partial(z,x)}{\partial(u,v)}\right]^2 \right) (du\,dv)^2$$

$$\therefore dS = \sqrt{\left[\frac{\partial(x,y)}{\partial(u,v)}\right]^2 + \left[\frac{\partial(y,z)}{\partial(u,v)}\right]^2 + \left[\frac{\partial(z,x)}{\partial(u,v)}\right]^2}\; du\,dv$$

$$\therefore A(S) = \iint_D dS$$

$$= \iint_D \sqrt{\left[\frac{\partial(x,y)}{\partial(u,v)}\right]^2 + \left[\frac{\partial(y,z)}{\partial(u,v)}\right]^2 + \left[\frac{\partial(z,x)}{\partial(u,v)}\right]^2}\; du\,dv$$

which agrees with the formula (3), p. 384 of text.

Note $\dfrac{\partial(z,x)}{\partial(u,v)} = -\dfrac{\partial(x,z)}{\partial(u,v)}$ so that $\left[\dfrac{\partial(z,x)}{\partial(u,v)}\right]^2 = \left[\dfrac{\partial(x,z)}{\partial(u,v)}\right]^2$

Review Exercises for Chapter 8

Note Title 8/15/2017

1. **1.** Let $\mathbf{F} = 2yz\mathbf{i} + (-x + 3y + 2)\mathbf{j} + (x^2 + z)\mathbf{k}$. Evaluate $\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$, where S is the cylinder $x^2 + y^2 = a^2$, $0 \le z \le 1$ (without the top and bottom). What if the top and bottom are included?

(a) Without top + bottom

Parametrization for S: $T(\theta, z) = (a\cos\theta, a\sin\theta, z)$, $0 \le \theta \le 2\pi$, $0 \le z \le 1$.

$\therefore T_\theta = (-a\sin\theta, a\cos\theta, 0)$ $T_z = (0, 0, 1)$

$\therefore T_\theta \times T_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -a\sin\theta & a\cos\theta & 0 \\ 0 & 0 & 1 \end{vmatrix} = (a\cos\theta, a\sin\theta, 0)$ which points "out"

$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2yz & -x+3y+2 & x^2+z \end{vmatrix} = (0, 2y-2x, -1-2z)$

$\therefore (\nabla \times \vec{F}) \circ (T(\theta, z)) = (0, 2a\sin\theta - 2a\cos\theta, -1-2z)$

$\therefore \left[(\nabla \times \vec{F}) \circ (T(\theta, z)) \right] \cdot (T_\theta \times T_z) =$

$(0, 2a\sin\theta - 2a\cos\theta, -1-2z) \cdot (a\cos\theta, a\sin\theta, 0) =$

$2a^2\sin^2\theta - 2a^2\cos\theta\sin\theta$

$$\therefore \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_0^{2\pi} \int_0^1 2a^2 \sin^2\theta - 2a^2 \cos\theta \sin\theta \, dz \, d\theta$$

$$= \int_0^{2\pi} 2a^2 \sin^2\theta - 2a^2 \cos\theta \sin\theta \, d\theta$$

$$= 2a^2 \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{2\pi} - 2a^2 \left[\frac{\sin^2\theta}{2} \right]_0^{2\pi}$$

($\frac{\sin 2\theta}{4} = 0$, $\frac{\sin^2\theta}{2} = 0$)

$$= 2\pi a^2$$

(5) With top & bottom

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = 0$$ since S is a closed surface (see #25, p. 452 of text).

2. Let W be a region in $\mathbb{R}^3$ with boundary ∂W. Prove the identity

$$\iint_{\partial W} [\mathbf{F} \times (\nabla \times \mathbf{G})] \cdot dS = \iiint_W (\nabla \times \mathbf{F}) \cdot (\nabla \times \mathbf{G}) \, dV - \iiint_W \mathbf{F} \cdot (\nabla \times \nabla \times \mathbf{G}) \, dV.$$

Should be $\iint_{\partial W} [\vec{F} \times (\nabla \times \vec{G})] \cdot d\vec{S}$

From Gauss' Divergence Theorem,

$$\iint_{\partial W} [\vec{F} \times (\nabla \times \vec{G})] \cdot d\vec{S} = \iiint_W \nabla \cdot [\vec{F} \times (\nabla \times \vec{G})] \, dV$$

But from table p. 255 of text, #8,

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$$

$$\therefore \nabla \cdot [\vec{F} \times (\nabla \times \vec{G})] = (\nabla \times \vec{G}) \cdot (\nabla \times \vec{F}) - \vec{F} \cdot (\nabla \times (\nabla \times \vec{G}))$$

$$\therefore \iint_{\partial W} [\vec{F} \times (\nabla \times \vec{G})] \cdot d\vec{S} = \iiint_W \nabla \cdot [\vec{F} \times (\nabla \times \vec{G})] \, dV$$

$$= \iiint_W [(\nabla \times \vec{G}) \cdot (\nabla \times \vec{F}) - \vec{F} \cdot (\nabla \times (\nabla \times \vec{G}))] \, dV$$

$$= \iiint_W (\nabla \times \vec{F}) \cdot (\nabla \times \vec{G}) \, dV - \iiint_W \vec{F} \cdot (\nabla \times (\nabla \times \vec{G})) \, dV$$

using linearity of integrals and commutivity of dot products.

3.

3. Let $\mathbf{F} = x^2 y\mathbf{i} + z^8\mathbf{j} - 2xyz\mathbf{k}$. Evaluate the integral of $\mathbf{F}$ over the surface of the unit cube.

Need to evaluate $\iint_{\partial W} \vec{F} \cdot d\vec{S}$

W = cube,
∂W = boundary or surface of cube

By Gauss' Divergence Theorem, $\iint_{\partial W} \vec{F} \cdot d\vec{S} = \iiint_W \nabla \cdot \vec{F}\, dV$

Here, $\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(z^8) + \frac{\partial}{\partial z}(-2xyz)$

$= 2xy - 2xy = 0.$

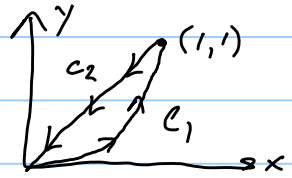
$\therefore \iiint_W \nabla \cdot \vec{F}\, dV = 0 \Rightarrow \iint_{\partial W} \vec{F} \cdot d\vec{S} = \underline{0}$

4.

4. Verify Green's theorem for the line integral

$$\int_C x^2 y\, dx + y\, dy,$$

when C is the boundary of the region between the curves $y = x$ and $y = x^3$, $0 \le x \le 1$.



(a) For $0 \le x \le 1$, $x^3 \le x$.

A parametrization for $y = x^3$ is $\vec{c}_1(t) = (t, t^3)$, $0 \le t \le 1$

and for $y = x$, $\vec{c}_2(t) = (2-t, 2-t)$, $1 \le t \le 2$

$\vec{c}_1'(t) = (1, 3t^2)$ $\quad$ $\vec{c}_2'(t) = (-1, -1)$

$\therefore \int_C x^2 y\, dx + y\, dy = \int_{\vec{c}_1} x^2 y\, dx + y\, dy + \int_{\vec{c}_2} x^2 y\, dx + y\, dy$

$$= \int_0^1 t^2(t^3)(1) + t^3(3t^2)\,dt$$

$$+ \int_1^2 (2-t)^2(2-t)(-1) + (2-t)(-1)\,dt$$

$$= \int_0^1 4t^5\,dt + \int_1^2 -(2-t)^3 + t - 2\,dt$$

$$= \frac{4}{6}t^6\Big|_0^1 + \frac{(2-t)^4}{4} + \frac{t^2}{2} - 2t\Big|_1^2$$

$$= \frac{2}{3} + \left[-2 - \left(\frac{1}{4} + \frac{1}{2} - 2\right)\right] = \frac{2}{3} - \frac{3}{4} = \underline{-\frac{1}{12}}$$

(5) From Green's Theorem,

$$\int_C x^2y\,dx + y\,dy = \iint_D \frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x^2y)\,dx\,dy$$

$$= \iint_D -x^2\,dx\,dy = \int_0^1 \int_{x^3}^{x} (-x^2)\,dy\,dx$$

$$= \int_0^1 (-x^2)(x - x^3)\,dx = \int_0^1 -x^3 + x^5\,dx$$

$$= \frac{-x^4}{4} + \frac{x^6}{6}\Big|_0^1 = -\frac{1}{4} + \frac{1}{6} = -\frac{1}{12}$$

$$\therefore (a) + (b) \Rightarrow \int_C x^2 y\,dx + y\,dy = \iint_D \frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x^2 y)\,dx\,dy$$

$$= -\frac{1}{12}$$

5.

5. (a) Show that $\mathbf{F} = (x^3 - 2xy^3)\mathbf{i} - 3x^2y^2\mathbf{j}$ is a gradient vector field.

(b) Evaluate the integral of $\mathbf{F}$ along the path $x = \cos^3\theta$, $y = \sin^3\theta$, $0 \le \theta \le \pi/2$.

(a) $\vec{F} = \nabla f$ for some function f.

$\therefore f_x = x^3 - 2xy^3 \Rightarrow f = \frac{x^4}{4} - x^2y^3 + g(y)$,

$g(y)$ some function of y.

$\therefore f_y = -3x^2y^2 + g'(y) = -3x^2y^2 \Rightarrow g'(y) = 0$,

$\therefore g(y) = C$, a constant.

$\therefore f(x,y) = \frac{x^4}{4} - x^2y^3 + C$, C a constant

and $\therefore \nabla f = \vec{F}$

(5) $\int_{\vec{C}} \vec{F} \cdot d\vec{s} = \int_0^{\frac{\pi}{2}} \vec{F}(\vec{c}(t)) \cdot \vec{c}'(t)\, dt =$

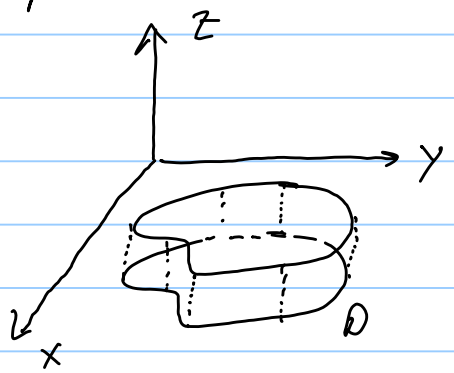
$\int_0^{\frac{\pi}{2}} \nabla f(\vec{c}(t))\, dt = f\left[\vec{c}\left(\frac{\pi}{2}\right)\right] - f[\vec{c}(0)]$

$= f[0,1] - f[1,0]$

$= (0 + c) - \left(\frac{1}{4} + c\right) = \underline{-\frac{1}{4}}$

6. **6.** Can you derive Green's theorem in the plane from Gauss' theorem?

Let D be a simple region in $\mathbb{R}^2$ and C be its boundary. Let $P, Q : D \to \mathbb{R}$ be C^1 functions. Let $\vec{F} = (Q, -P, 0)$, and define W to be the volume defined as having height 1, so D is the base and the top is D with z-coordinate 1.



$$\therefore \iiint_W (\nabla \cdot \vec{F})\, dV = \iiint_W (Q_x - P_y)\, dV$$

$$= \iint_D \left[\int_0^1 (Q_x - P_y)\, dz \right] dx\, dy = \iint_D (Q_x - P_y)\, dx\, dy$$

$$= \iint_{\partial W} \vec{F} \cdot d\vec{S} = \iint_{D_{Top}} \vec{F} \cdot d\vec{S} + \iint_{D_{Bottom}} \vec{F} \cdot d\vec{S} +$$

$$\iint_{W_{side}} \vec{F} \cdot d\vec{S}$$

since $\partial W = D_{Top} \cup D_{Bottom} \cup W_{side}$

$$= \iint_{D_{Top}} \vec{F} \cdot \hat{n}_{top}\, dS + \iint_{D_{Bottom}} \vec{F} \cdot \hat{n}_{Bottom}\, dS + \iint_{W_{side}} \vec{F} \cdot \hat{n}_{side}\, dS$$

$$= \iint_{W_{side}} \vec{F} \cdot \hat{n}_{side}\, dS$$ since $\vec{F} \cdot \hat{n}_{Top} = 0$ and $\vec{F} \cdot \hat{n}_{Bottom} = 0$

since $\vec{F}$ has no z-component

$\therefore$ So far, $\iint_D (Q_x - P_y)\,dx\,dy = \iint_{W_{side}} \vec{F} \cdot \hat{n}_{side}\,dS$ [1]

For a parametrization of W_{side}, let $\vec{c}(t) = (x(t), y(t))$ be a parametrization of D, and $\phi(t,z) = [x(t), y(t), z]$, $0 \le z \le 1$, be a parametrization of W_{side}.

$\therefore \phi_t = [x'(t), y'(t), 0] \quad \phi_z = [0, 0, 1]$

$$\therefore \phi_t \times \phi_z = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x' & y' & 0 \\ 0 & 0 & 1 \end{vmatrix} = (y'(t), -x'(t), 0)$$

$$\therefore \hat{n}_{side} = \frac{(y'(t), -x'(t), 0)}{\sqrt{(x')^2 + (y')^2}}$$

$$\therefore \iint_{W_{side}} \vec{F} \cdot \hat{n}_{side}\,dS = \iint_{W_{side}} (Q, -P, 0) \cdot \frac{(y', -x', 0)}{\sqrt{(x')^2 + (y')^2}}\,dS$$

$$= \iint_{W_{side}} \frac{P \cdot x'(t) + Q \cdot y'(t)}{\sqrt{(x')^2 + (y')^2}}\,dt\,dz$$

$$= \int_{\partial D} \left[\int_0^1 \frac{P \cdot x'(t) + Q \cdot y'(t)}{\sqrt{(x')^2 + (y')^2}} \, dz \right] dt$$

$$= \int_{\partial D} \frac{P \cdot x'(t) + Q \cdot y'(t)}{\sqrt{(x')^2 + (y')^2}} \, dt$$

$$= \int_{\partial D} [P, Q] \cdot \frac{\vec{c}\,'(t)}{\|\vec{c}\,'(t)\|} \, dt = \int_{\partial D} [P, Q] \cdot \hat{n} \, ds$$

where $\hat{n}$ = positively oriented unit normal to ∂D.

$$= \int_{\partial D} P\,dx + Q\,dy \qquad [2]$$

$\therefore$ [1], [2] $\Rightarrow$ $$\iint_D Q_x - P_y \, dx\,dy = \int_{\partial D} P\,dx + Q\,dy$$

7.

7. (a) Show that $\mathbf{F} = 6xy(\cos z)\mathbf{i} + 3x^2(\cos z)\mathbf{j} - 3x^2 y(\sin z)\mathbf{k}$ is conservative (see Section 8.3).

(b) Find f such that $\mathbf{F} = \nabla f$.

(c) Evaluate the integral of $\mathbf{F}$ along the curve $x = \cos^3\theta$, $y = \sin^3\theta$, $z = 0$, $0 \le \theta \le \pi/2$.

(a) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 6xy\cos z & 3x^2\cos z & -3x^2 y\sin z \end{vmatrix}$

$= \{ -3x^2\sin z + 3x^2\sin z,$

$-6xy\sin z - (-6xy\sin z),$

$6x\cos z - 6x\cos z \}$

$= \{0, 0, 0\} = \vec{0}$

$\therefore \vec{F}$ is conservative (Theorem 7, p. 453 text).

(b) $f_x = 6xy\cos z \Rightarrow f(x,y,z) = 3x^2 y\cos z + g(y,z)$

$g(y,z)$ some function of y, z

$\therefore f_y = 3x^2\cos z + g_y(y,z) = 3x^2\cos z$

$\Rightarrow g_y = 0 \Rightarrow g(y,z) = h(z)$

$h(z)$ some function of z.

$\therefore f(x,y,z) = 3x^2 y\cos z + h(z)$

$f_z = -3x^2 y\sin z + h'(z) = -3x^2 y\sin z$

$\Rightarrow h'(z) = 0 \Rightarrow h(z) = K$, some constant

$\therefore f(x,y,z) = 3x^2 y\cos z + K$, K some constant

(c) Endpoints: $\theta = 0 : [\cos^3(0), \sin^3(0), 0] = [1,0,0]$

$\theta = \frac{\pi}{2} : [\cos^3(\frac{\pi}{2}), \sin^3(\frac{\pi}{2}), 0] = [0,1,0]$

$\therefore \int_{\vec{C}} \vec{F}\cdot d\vec{s} = f(\vec{c}(\frac{\pi}{2})) - f(\vec{c}(0))$

$= f(1,0,0) - f(0,1,0)$

$= 3x^2 y\cos z + K \Big|_{(1,0,0)}^{(0,1,0)}$

$= (0+K) - (0+K) = \underline{0}$

8. **8.** Let $\mathbf{r}(x, y, z) = (x, y, z)$, $r = \|\mathbf{r}\|$. Show that $\nabla^2(\log r) = 1/r^2$ and $\nabla^2(r^n) = n(n+1)r^{n-2}$.

Assume $r \neq 0$

(a) $\nabla^2(\log r) = \nabla\cdot\nabla(\log r)$

$\nabla \log r = \frac{1}{r}(r_x, r_y, r_z)$

$$= \frac{1}{r} \cdot \frac{1}{2}(x^2+y^2+z^2)^{-\frac{1}{2}}(2x, 2y, 2z)$$

$$= \frac{1}{r^2}(x, y, z) = \left(\frac{x}{r^2}, \frac{y}{r^2}, \frac{z}{r^2}\right)$$

$$\therefore \nabla \cdot \left(\frac{x}{r^2}, \frac{y}{r^2}, \frac{z}{r^2}\right) = \left[\frac{\partial}{\partial x}\left(\frac{x}{r^2}\right), \frac{\partial}{\partial y}\left(\frac{y}{r^2}\right), \frac{\partial}{\partial z}\left(\frac{z}{r^2}\right)\right]$$

$$\frac{\partial}{\partial x}\left(\frac{x}{r^2}\right) = \frac{r^2(1) - x(2r)\left(\frac{1}{2}\right)(x^2+y^2+z^2)^{-\frac{1}{2}}(2x)}{r^4}$$

$$= \frac{r^2 - x\,r\left(\frac{1}{r}\right)2x}{r^4} = \frac{r^2 - 2x^2}{r^4}$$

Similarly, $\frac{\partial}{\partial y}\left(\frac{y}{r^2}\right) = \frac{r^2-2y^2}{r^4}$ $\frac{\partial}{\partial z}\left(\frac{z}{r^2}\right) = \frac{r^2-2z^2}{r^4}$

$$\therefore \nabla \cdot \left(\frac{x}{r^2}, \frac{y}{r^2}, \frac{z}{r^2}\right) = \frac{r^2-2x^2}{r^4} + \frac{r^2-2y^2}{r^4} + \frac{r^2-2z^2}{r^4}$$

$$= \frac{3r^2 - 2(x^2+y^2+z^2)}{r^4} = \frac{3r^2-2r^2}{r^4}$$

$$= \frac{r^2}{r^4} = \frac{1}{r^2}$$

$$\therefore \nabla^2(\log r) = \frac{1}{r^2}$$

(5) $\nabla^2(r^n) = n(n+1)r^{n-2}$

$n=0$: $\nabla^2(1) = 0 = 0(0+1)r^{0-2} = 0$

$n \neq 0 : \nabla^2(r^n) = \nabla \cdot \nabla(r^n)$

$$\nabla(r^n) = \nabla (x^2+y^2+z^2)^{\frac{n}{2}}$$

$$\frac{\partial}{\partial x}(x^2+y^2+z^2)^{\frac{n}{2}} = \frac{n}{2}(x^2+y^2+z^2)^{\frac{n-2}{2}}(2x)$$

$$= n(x)(x^2+y^2+z^2)^{\frac{n-2}{2}} = n(x)\, r^{n-2}$$

Similarly, $\frac{\partial}{\partial y}(r^n) = n(y)(x^2+y^2+z^2)^{\frac{n-2}{2}} = n(y) r^{n-2}$

$$\frac{\partial}{\partial z}(r^n) = n(z)(x^2+y^2+z^2)^{\frac{n-2}{2}} = n(z) r^{n-2}$$

$$\therefore \frac{\partial^2(r^n)}{\partial x^2} = n r^{n-2} + n x \frac{\partial}{\partial x}(r^{n-2})$$

$$= n r^{n-2} + n x (n-2)(r^{n-3})\left(\tfrac{1}{2}\right)(x^2+y^2+z^2)^{-\frac{1}{2}}(2x)$$

$$= n r^{n-2} + x^2 (n)(n-2)(r^{n-3})\left(\tfrac{1}{r}\right)$$

$$= n r^{n-2} + x^2 (n)(n-2)(r^{n-4})$$

$$= n r^{n-2} + (n)(n-2)(r^{n-2})\left(\frac{x^2}{r^2}\right)$$

$$= n r^{n-2}\left[1 + (n-2)\frac{x^2}{r^2}\right]$$

Similarly, $\frac{\partial^2}{\partial y^2}(r^n) = n r^{n-2}\left[1 + (n-2)\frac{y^2}{r^2}\right]$

$$\frac{\partial^2}{\partial z^2}(r^n) = n r^{n-2}\left[1 + (n-2)\frac{z^2}{r^2}\right]$$

$$\therefore \nabla^2(r^n) = n r^{n-2}\left[1 + 1 + 1 + \frac{(n-2)}{r^2}(x^2+y^2+z^2)\right]$$

$$= n r^{n-2}\left[3 + \frac{(n-2)}{r^2} r^2\right]$$

$$= n r^{n-2}\left[3 + n - 2\right]$$

$$= \underline{n(n+1) r^{n-2}}$$

9.

9. Let the velocity of a fluid be described by $\mathbf{F} = 6xz\mathbf{i} + x^2y\mathbf{j} + yz\mathbf{k}$. Compute the rate at which fluid is leaving the unit cube.

This is the flux through the cube $= \iint_S \vec{F} \cdot d\vec{S}$

i.e., the surface integral of the vector field $\vec{F}$ over S.

By Gauss' Divergence Theorem, $\iiint_W \nabla \cdot \vec{F}\, dV = \iint_S \vec{F} \cdot d\vec{S}$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(6xz) + \frac{\partial}{\partial y}(x^2y) + \frac{\partial}{\partial z}(yz)$$

$$= 6z + x^2 + y$$

The unit cube is presumably from $(0,0,0)$ to $(1,1,1)$.

$$\therefore \iiint_W \nabla \cdot \vec{F}\, dV = \int_0^1 \int_0^1 \int_0^1 (6z + x^2 + y)\, dx\, dy\, dz$$

$$= \int_0^1 \int_0^1 6zx + \frac{x^3}{3} + yx \Big|_{x=0}^{x=1} dy\, dz$$

$$= \int_0^1 \int_0^1 \left(6z + \frac{1}{3} + y\right) dy\, dz = \int_0^1 6zy + \frac{y}{3} + \frac{y^2}{2} \Big|_{y=0}^{y=1} dz$$

$$= \int_0^1 6z + \frac{5}{6}\, dz = 3z^2 + \frac{5}{6}z \Big|_0^1 = 3 + \frac{5}{6}$$

$$= \frac{23}{6}$$

10. **10.** Let $\mathbf{F} = x^2\mathbf{i} + (x^2 y - 2xy)\mathbf{j} - x^2 z\mathbf{k}$. Does there exist a $\mathbf{G}$ such that $\mathbf{F} = \nabla \times \mathbf{G}$?

If such a $\vec{G}$ existed, then $\operatorname{div} \vec{F} = \operatorname{div}(\nabla \times \vec{G}) = 0$

Here, $\nabla \cdot \vec{F} = (2x) + (x^2 - 2x) + (-x^2) = 0$

By Theorem 8, p. 459 of text, yes there

is a $\vec{G}$ s.t. $\vec{F} = \nabla \times \vec{G}$.

To find such a $\vec{G}$, use formula from #20, p. 461 of text.

11. **11.** Let **a** be a constant vector and $\mathbf{F} = \mathbf{a} \times \mathbf{r}$ [as usual, $\mathbf{r}(x, y, z) = (x, y, z)$]. Is **F** conservative? If so, find a potential for it.

$\vec{F}$ is conservative $\Leftrightarrow \nabla \times \vec{F} = \vec{0}$

$\therefore$ Look at $\nabla \times (\vec{a} \times \vec{r})$

Let $\vec{a} = (a_1, a_2, a_3)$

$$\therefore \vec{a} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ x & y & z \end{vmatrix} = (a_2 z - a_3 y,\ a_3 x - a_1 z,\ a_1 y - a_2 x)$$

$$\therefore \nabla \times (\vec{a} \times \vec{r}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ a_2 z - a_3 y & a_3 x - a_1 z & a_1 y - a_2 x \end{vmatrix}$$

$$= \left[a_1 - (-a_1),\ a_2 - (-a_2),\ a_3 - (-a_3) \right] = \left[2a_1, 2a_2, 2a_3 \right]$$

$$= 2\vec{a}$$

Assuming $\vec{a} \neq \vec{0}$, then no, $\vec{F}$ is not conservative.

12. **12.** Show that the fields **F** in (a) and (b) are conservative and find a function f such that $\mathbf{F} = \nabla f$.

(a) $\mathbf{F} = (y^2 e^{xy^2})\mathbf{i} + (2y\, e^{xy^2})\mathbf{j}$

(b) $\mathbf{F} = (\sin y)\mathbf{i} + (x \cos y)\mathbf{i} + (e^z)\mathbf{k}$

$\vec{F}$ conservative $\Leftrightarrow \nabla \times \vec{F} = \vec{0}$

(a) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 e^{xy^2} & 2y\, e^{xy^2} & 0 \end{vmatrix}$

$= [0, 0, 2y\, e^{xy^2}(y^2) - (2y e^{xy^2} + y^2 e^{xy^2} 2xy)]$

$= [0, 0, 2y^3 e^{xy^2} - 2y e^{xy^2} + 2xy^3 e^{xy^2}]$

$\neq \vec{0}$

$\therefore \vec{F}$ is <u>not</u> conservative

(b) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \sin y & x \cos y & e^z \end{vmatrix}$

$= (0, 0, \cos y - \cos y) = \vec{0}$

$\therefore f_x = \sin y$, $\therefore f(x, y, z) = x \sin y + g(y, z)$

$g(y, z)$ some function of y, z

$\therefore f_y = x \cos y + g_y(y, z) = x \cos y$

$\Rightarrow g_y(y,z) = 0 \Rightarrow g(y,z) = h(z)$,

$h(z)$ some function of z.

$\therefore f(x,y,z) = x \sin y + h(z)$

$\therefore f_z = e^z = h'(z) \Rightarrow h(z) = e^z + K$, K a constant

$\therefore f(x,y,z) = x \sin y + e^z + K$, K some constant

$\therefore \nabla f = \vec{F}$

13.

13. (a) Let $f(x, y, z) = 3xye^{z^2}$. Compute ∇f.

(b) Let $\mathbf{c}(t) = (3\cos^3 t, \sin^2 t, e^t)$, $0 \le t \le \pi$. Evaluate

$$\int_{\mathbf{c}} \nabla f \cdot d\mathbf{s}.$$

(c) Verify directly Stokes' theorem for gradient vector fields $\mathbf{F} = \nabla f$.

(a) $f_x = 3ye^{z^2}$ $f_y = 3xe^{z^2}$ $f_z = 3xye^{z^2}(2z)$

$\therefore \nabla f = (3ye^{z^2}, 3xe^{z^2}, 6xyze^{z^2})$

(b) $\int_{\vec{c}} \nabla f \cdot d\vec{s} = \int_0^{\pi} \nabla f(\vec{c}(t)) \cdot \vec{c}'(t)\, dt$

$= f(\vec{c}(\pi)) - f(\vec{c}(0)) = f(-3, 0, e^{\pi}) - f(3, 0, 1)$

$= 3(-3)(0)\, e^{(e^{\pi})^2} - 3(3)(0)e^{1} = 0 - 0 = \underline{0}$

(c) Stokes' Theorem says $\iint_S \nabla \times (\nabla f) \cdot d\vec{S} = \int_{\partial S} \nabla f \cdot d\vec{s}$

From (b), $\int_{\partial S} \nabla f \cdot d\vec{s} = \int_{\vec{c}} \nabla f \cdot d\vec{s} = 0$

Look at $\nabla \times (\nabla f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3ye^{z^2} & 3xe^{z^2} & 6xyze^{z^2} \end{vmatrix}$

$= \left[6xze^{z^2} - 6xze^{z^2},\ 6yze^{z^2} - 6yze^{z^2},\ 3e^{z^2} - 3e^{z^2} \right]$

$= \{0, 0, 0\} = \vec{0}$

$\therefore \iint_S \nabla \times (\nabla f) \cdot d\vec{S} = \iint_S \vec{0} \cdot d\vec{S} = \underline{0}$

$\therefore \iint_S \nabla \times (\nabla f) \cdot d\vec{S} = \int_{\partial S} (\nabla f) \cdot d\vec{s} = 0$

14. **14.** Using Green's theorem, or otherwise, evaluate $\int_C x^3\,dy - y^3\,dx$, where C is the unit circle $(x^2 + y^2 = 1)$.

(a) $$\int_C -y^3\,dx + x^3\,dy = \iint_D \frac{\partial}{\partial x}(x^3) - \frac{\partial}{\partial y}(-y^3)\,dx\,dy$$

$$= \iint_D 3x^2 + 3y^2\,dx\,dy$$ Note: $x^2+y^2=1$, not $x^2+y^2=1$ for D.

Let $x = r\cos\theta$ $y = r\sin\theta$ Jacobian $= r$

$0 \le \theta \le 2\pi$, $0 \le r \le 1$

$$\therefore = \int_0^{2\pi}\int_0^1 3r^2(r)\,dr\,d\theta$$

$$= 2\pi\left(\frac{3}{4}r^4\right)\Big|_{r=0}^{r=1} = \underline{\frac{3}{2}\pi}$$

(b) Computing directly the line integral,

Let $\vec{c}(\theta) = (\cos\theta, \sin\theta)$, $0 \le \theta \le 2\pi$.

$$\therefore \int_C x^3\,dy - y^3\,dx = \int_0^{2\pi} \cos^3\theta(\cos\theta) - \sin^3\theta(-\sin\theta)\,d\theta$$

$$= \int_0^{2\pi} \cos^4\theta + \sin^4\theta \, d\theta$$

$$= \left(\frac{3}{8}\theta + \frac{\sin 2\theta}{4} + \frac{\sin 4\theta}{32}\right)\Big|_0^{2\pi} + \left(\frac{3\theta}{8} - \frac{\sin 2\theta}{4} + \frac{\sin 4\theta}{32}\right)\Big|_0^{2\pi}$$

($\cos^4\theta$ $\quad$ $\sin^4\theta$)

$$= \frac{3}{8}(2\pi) + \frac{3(2\pi)}{8} = \underline{\frac{3}{2}\pi}$$

15. **15.** Evaluate the integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + 3\mathbf{k}$ and where S is the surface of the unit sphere $x^2 + y^2 + z^2 = 1$.

By Gauss' Theorem, $\iint_S \vec{F} \cdot d\vec{S} = \iiint_W \nabla \cdot \vec{F} \, dV$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(3) = 1 + 1 + 0 = 2$$

$$\therefore \iiint_W \nabla \cdot \vec{F} \, dV = 2 \iiint_W dV = 2\left(\frac{4}{3}\pi(1)^3\right) = \underline{\frac{8}{3}\pi}$$

16. **16.** (a) State Stokes' theorem for surfaces in $\mathbb{R}^3$.

(b) Let $\mathbf{F}$ be a vector field on $\mathbb{R}^3$ satisfying $\nabla \times \mathbf{F} = \mathbf{0}$. Use Stokes' theorem to show that $\int_C \mathbf{F} \cdot d\mathbf{s} = 0$ where C is a closed curve.

(a) This is a restatement of Theorem 6, p. 444 of the text.

Let S be an oriented surface in $\mathbb{R}^3$ defined by a 1-to-1 parametrization $\phi: D \subset \mathbb{R}^2 \to S$, where D is a simple region, or region that is the union of simple regions. Let ∂S denote the oriented boundary of S (and so is a closed curve), and let $\vec{F}$ be a C^1 vector field on S. Then

$$\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \int_{\partial S} \vec{F} \cdot d\vec{s}$$

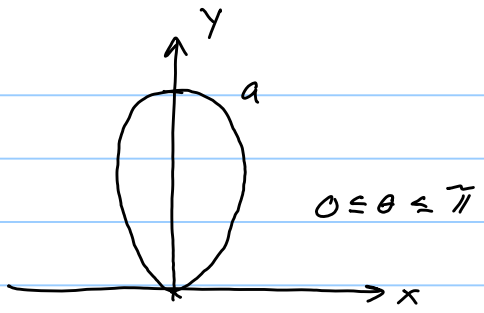
If S has no boundary, then the integral on the left is zero.

(b) Since $\nabla \times \vec{F} = 0$, Then $\iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = \iint_S \vec{0} \cdot d\vec{S} = 0$

$\therefore$ From (a), $\int_{\partial S} \vec{F} \cdot d\vec{s} = \int_C \vec{F} \cdot d\vec{s} = \iint_S (\nabla \times \vec{F}) \cdot d\vec{S} = 0$

17.

17. Use Green's theorem to find the area of the loop of the curve $x = a\sin\theta\cos\theta$, $y = a\sin^2\theta$, for $a > 0$ and $0 \le \theta \le \pi$.



$\theta = 0 : (x,y) = (0,0)$

$\theta = \pi : (x,y) = (0,0)$

Using Th. 2, p. 433 of text

$$A = \frac{1}{2}\int_{\partial D} x\,dy - y\,dx = \frac{1}{2}\iint_D \left(\frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(-y)\right) dx\,dy$$

$$= \iint_D dx\,dy = \text{area of loop}$$

$$x = a\sin\theta\cos\theta = \frac{a}{2}\sin 2\theta \qquad \frac{dx}{d\theta} = a\cos 2\theta$$

$$y = a\sin^2\theta = \frac{a}{2}(1-\cos 2\theta) \qquad \frac{dy}{d\theta} = a\sin 2\theta$$

$$\therefore A = \frac{1}{2}\int_0^{\pi} \left(\frac{a}{2}\sin 2\theta\right)(a\sin 2\theta) - \frac{a}{2}(1-\cos 2\theta)(a\cos 2\theta)\,d\theta$$

$$= \frac{1}{2}\int_0^{\pi} \frac{a^2}{2}\sin^2 2\theta - \frac{a^2}{2}\cos 2\theta + \frac{a^2}{2}\cos^2 2\theta\,d\theta$$

$$= \frac{1}{2}\int_0^{\pi} \frac{a^2}{2} - \frac{a^2}{2}\cos 2\theta\,d\theta$$

$$= \frac{1}{2}\left[\frac{a^2}{2}\theta - \frac{a^2}{4}\sin 2\theta\right]_{\theta=0}^{\theta=\pi}$$

$$= \frac{1}{2}\left(\frac{a^2}{2}\pi\right) = \underline{\frac{a^2}{4}\pi}$$

18.

18. Evaluate $\int_C yz\,dx + xz\,dy + xy\,dz$, where C is the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the surface $z = y^2$.

Note for $f(x,y,z) = xyz$, $\nabla f = (yz, xz, xy)$

The curve of intersection C is a closed curve.

$$\therefore \int_C yz\,dx + xz\,dy + xy\,dz = \int_C (\nabla f)\cdot d\vec{s} = \underline{0}$$

as $f(\text{endpoint}) = f(\text{startpoint})$, so

$$\int_C (\nabla f)\cdot d\vec{s} = f(\text{endpoint}) - f(\text{startpoint}) = 0$$

19.

19. Evaluate $\int_C (x + y)\,dx + (2x - z)\,dy + (y + z)\,dz$, where C is the perimeter of the triangle connecting $(2, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 6)$, in that order.

Parametrize C:

$(2,0,0)$ to $(0,3,0)$: $(2,0,0) + t[(0,3,0) - (2,0,0)]$, $0 \le t \le 1$

$$= (2,0,0) + t(-2,3,0) = (2-2t, 3t, 0)$$

$(0,3,0)$ to $(0,0,6)$: $(0,3,0) + (t-1)[(0,0,6)-(0,3,0)]$, $1 \le t \le 2$

$$= (0,3,0) + (t-1)(0,-3,6)$$

$$= (0, 6-3t, 6t-6)$$

$(0,0,6)$ to $(2,0,0)$: $(0,0,6) + (t-2)[(2,0,0)-(0,0,6)]$, $2 \le t \le 3$

$$= (0,0,6) + (t-2)(2,0,-6)$$

$$= (2t-4, 0, 18-6t)$$

$$\therefore \vec{c}(t) = \begin{cases} (2-2t, 3t, 0) & 0 \le t \le 1 \\ (0, 6-3t, 6t-6) & 1 \le t \le 2 \\ (2t-4, 0, 18-6t) & 2 \le t \le 3 \end{cases}$$

$$\vec{c}\,'(t) = \begin{cases} (-2,3,0) & 0 \le t \le 1 \\ (0,-3,6) & 1 \le t \le 2 \\ (2,0,-6) & 2 \le t \le 3 \end{cases}$$

$4t - 8 - 18 + 6t$

$$\therefore \int_C (x+y)\,dx + (2x-z)\,dy + (y+z)\,dz$$

$$= \int_0^1 \underbrace{(2-2t+3t)}_{x+y}\underbrace{(-2)}_{dx} + \underbrace{(4-4t)}_{2x-z}\underbrace{(3)}_{dy} + \underbrace{(3t)}_{y+z}\underbrace{(0)}_{dz}\,dt$$

$-4-2t$ $\qquad$ $12-12t$

$$+\int_1^2 \underbrace{(6-3t)}_{x+y}\,\underbrace{dx}_{}\,(0) + \underbrace{(6-6t)}_{2x-z}\,\underbrace{dy}_{}\,(-3) + \underbrace{(3t)}_{y+z}\,\underbrace{dz}_{}\,(6)\,dt$$

$$-18+18t \qquad 18t$$

$$+\int_2^3 \underbrace{(2t-4)}_{x+y}\,\underbrace{dx}_{}\,(2) + \underbrace{(-26+10t)}_{2x-z}\,\underbrace{dy}_{}\,(0) + \underbrace{(18-6t)}_{y+z}\,\underbrace{dz}_{}\,(-6)\,dt$$

$$4t-8 \qquad -108+36t$$

$$= \int_0^1 (8-14t)\,dt + \int_1^2 (-18+36t)\,dt + \int_2^3 (-116+40t)\,dt$$

$$= 8t-7t^2\Big|_0^1 + (-18t+18t^2)\Big|_1^2 + (-116t+20t^2)\Big|_2^3$$

$$= (8-7) + [(-36+72)-(-18+18)] + [(-348+180)-(-232+80)]$$

$$= 1+36+(-16) = \underline{21}$$

20.

20. Which of the following are conservative fields on $\mathbb{R}^3$? For those that are, find a function f such that $\mathbf{F} = \nabla f$.

(a) $\mathbf{F}(x, y, z) = 3x^2y\mathbf{i} + x^3\mathbf{j} + 5\mathbf{k}$

(b) $\mathbf{F}(x, y, z) = (x+z)\mathbf{i} - (y+z)\mathbf{j} + (x-y)\mathbf{k}$

(c) $\mathbf{F}(x, y, z) = 2xy^3\mathbf{i} + x^2z^3\mathbf{j} + 3x^2yz^2\mathbf{k}$

$\vec{F}$ is conservative $\Leftrightarrow \nabla \times \vec{F} = \vec{0}$

(a) $$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3x^2y & x^3 & 5 \end{vmatrix} = (0-0,\ 0-0,\ 3x^2-3x^2) = \vec{0}$$

$\therefore$ yes, $\vec{F}$ is conservative

Let $f(x,y,z)$ be s.t. $\nabla f = \vec{F}$

$\therefore f_x = 3x^2y \Rightarrow f = x^3y + g(y,z)$,

g some function of y, z.

$\therefore f_y = x^3 + g_y(y,z) = x^3 \Rightarrow g_y = 0$

$\Rightarrow g = h(z)$, h some function of z.

$\therefore f(x,y,z) = x^3y + h(z)$.

$f_z = h'(z) = 5 \Rightarrow h(z) = 5z + K$, K some constant

$\therefore f(x,y,z) = x^3y + 5z + K$, K some constant

(6) (b) $\mathbf{F}(x, y, z) = (x+z)\mathbf{i} - (y+z)\mathbf{j} + (x-y)\mathbf{k}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x+z & -(y+z) & x-y \end{vmatrix} = [-1-(-1), 1-1, 0-0]$$

$$= \vec{0}$$

$\therefore$ yes, $\vec{F}$ is conservative

Let $f(x,y,z)$ be s.t. $\nabla f = \vec{F}$

$\therefore f_x = x + z \Rightarrow f = \frac{x^2}{2} + xz + g(y,z)$,

g some function of y, z

$\therefore f_y = g_y(y,z) = -y - z \Rightarrow g = -\frac{y^2}{2} - yz + c$

$\therefore f = \frac{x^2}{2} + xz - \frac{y^2}{2} - yz + c$, c a constant

$\therefore f_z = x - y =$ $\hat{k}$ component of $\vec{F}$

$\therefore f(x,y,z) = \frac{x^2}{2} + xz - \frac{y^2}{2} - yz + c$, c a constant

(c) (c) $\mathbf{F}(x, y, z) = 2xy^3\mathbf{i} + x^2z^3\mathbf{j} + 3x^2yz^2\mathbf{k}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xy^3 & x^2z^3 & 3x^2yz^2 \end{vmatrix} = \left[3x^2z^2 - 3x^2z^2,\ 0 - 6xyz^2,\ 2xz^3 - 6xy^2\right]$$

$$= \left[0, -6xyz^2, 2xz^3 - 6xy^2\right] \neq \vec{0} \text{ for } (x,y,z) \neq \vec{0}$$

$\therefore$ No, $\vec{F}$ is not conservative.

21.

21. Consider the following two vector fields in $\mathbb{R}^3$:

(i) $\mathbf{F}(x, y, z) = y^2\mathbf{i} - z^2\mathbf{j} + x^2\mathbf{k}$
(ii) $\mathbf{G}(x, y, z) = (x^3 - 3xy^2)\mathbf{i} + (y^3 - 3x^2y)\mathbf{j} + z\mathbf{k}$

(a) Which of these fields (if any) are conservative on $\mathbb{R}^3$? (That is, which are gradient fields?) Give reasons for your answer.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^2 & -z^2 & x^2 \end{vmatrix} = (0-(-2z),\ 0-2x,\ 0-2y)$$

$$\neq \vec{0}$$

$\therefore \nabla \times \vec{F} \neq \vec{0} \Rightarrow \vec{F}$ not conservative

$$\nabla \times \vec{G} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^3 - 3xy^2 & y^3 - 3x^2y & z \end{vmatrix}$$

$$= [0-0,\ 0-0,\ -6xy-(-6xy)] = \vec{0}$$

$\therefore \vec{G}$ is conservative

(b) Find potential for the fields that are conservative.

Let $f(x,y,z)$ be s.t. $\nabla f = \vec{G}$

$$\therefore f_x = x^3 - 3xy^2 \Rightarrow f = \frac{x^4}{4} - \frac{3}{2}x^2y^2 + g(y,z)$$

g some function of y, z

$\therefore f_y = -3x^2y + g_y(y,z) = y^3 - 3x^2y$

$\Rightarrow g_y = y^3 \Rightarrow g = \frac{y^4}{4} + h(z)$,

h some function of z

$\therefore f = \frac{x^4}{4} - \frac{3}{2}x^2y^2 + \frac{y^4}{4} + h(z)$

$\therefore f_z = h'(z) = z \Rightarrow h(z) = \frac{z^2}{2} + c$, c a constant

$\therefore f(x,y,z) = \frac{x^4}{4} - \frac{3}{2}x^2y^2 + \frac{y^4}{4} + \frac{z^2}{2} + C$, C a constant

(c)

(c) Let α be the path that goes from (0, 0, 0) to (1, 1, 1) by following edges of the cube $0 \le x \le 1$, $0 \le y \le 1$, $0 \le z \le 1$ from (0, 0, 0) to (0, 0, 1) to (0, 1, 1) to (1, 1, 1). Let β be the path from (0, 0, 0) to (1, 1, 1) directly along the diagonal of the cube. Find the values of the line integrals

$$\int_\alpha \mathbf{F} \cdot d\mathbf{s}, \quad \int_\alpha \mathbf{G} \cdot d\mathbf{s}, \quad \int_\beta \mathbf{F} \cdot d\mathbf{s}, \quad \int_\beta \mathbf{G} \cdot d\mathbf{s}.$$

(1) $\int_\alpha \vec{F} \cdot d\vec{s}$

Parametrization for α:

(0,0,0) to (0,0,1): $(0,0,t)$, $0 \le t \le 1$

$(0,0,1)$ to $(0,1,1)$: $(0, t-1, 1)$, $1 \le t \le 2$

$(0,1,1)$ to $(1,1,1)$: $(t-2, 1, 1)$, $2 \le t \le 3$

$$\therefore \alpha(t) = \begin{cases} (0,0,t) & 0 \le t \le 1 \\ (0, t-1, 1) & 1 \le t \le 2 \\ (t-2, 1, 1) & 2 \le t \le 3 \end{cases}$$

$$\alpha'(t) = \begin{cases} (0,0,1) & 0 \le t \le 1 \\ (0,1,0) & 1 \le t \le 2 \\ (1,0,0) & 2 \le t \le 3 \end{cases}$$

$$\vec{F}(\alpha(t)) = \begin{cases} (0, -t^2, 0) & 0 \le t \le 1 \\ ((t-1)^2, 0, 0) & 1 \le t \le 2 \\ (0, 0, (t-2)^2) & 2 \le t \le 3 \end{cases}$$

$$\therefore \vec{F}(\alpha(t)) \cdot \alpha'(t) = \begin{cases} 0 & 0 \le t \le 1 \\ 0 & 1 \le t \le 2 \\ 0 & 2 \le t \le 3 \end{cases}$$

$$\therefore \int_{\alpha} \vec{F} \cdot d\vec{s} = \int_0^1 0 \, dt + \int_1^2 0 \, dt + \int_2^3 0 \, dt = \underline{0}$$

(2) $\displaystyle\int_{\alpha} \vec{G} \cdot d\vec{s}$

Since $\vec{G}$ is conservative and $\nabla f = \vec{G}$ from

(6), where $f = \frac{x^4}{4} - \frac{3}{2}x^2y^2 + \frac{y^4}{4} + \frac{z^2}{2} + C$,

$$\int_\alpha \vec{G} \cdot d\vec{s} = f(\text{endpoint}) - f(\text{startpoint})$$

$$= f(1,1,1) - f(0,0,0)$$

$$= \left[\frac{1}{4} - \frac{3}{2} + \frac{1}{4} + \frac{1}{2} + c\right] - [c] = \underline{-\frac{1}{2}}$$

(3) $\int_\beta \vec{F} \cdot d\vec{s}$

Parametrization for $\beta(t) = (t,t,t)$, $0 \le t \le 1$

$$\therefore \beta'(t) = (1,1,1)$$

$$\vec{F}(\beta(t)) = [t^2, -t^2, t^2]$$

$$\therefore \vec{F}(\beta(t)) \cdot \beta'(t) = t^2 - t^2 + t^2 = t^2$$

$$\therefore \int_\beta \vec{F} \cdot d\vec{s} = \int_0^1 t^2\,dt = \frac{t^3}{3}\Big|_0^1 = \underline{\frac{1}{3}}$$

(4) Since $\vec{G}$ is conservative, path doesn't matter.

$\therefore$ Answer is the same as in (2)

$$\int_\beta \vec{G} \cdot d\vec{s} = -\frac{1}{2}$$

22.

22. Consider the *constant* vector field $\mathbf{F}(x, y, z) = \mathbf{i} + 2\mathbf{j} - \mathbf{k}$ in $\mathbb{R}^3$.

(a)

(a) Find a scalar field $\phi(x, y, z)$ in $\mathbb{R}^3$ such that $\nabla\phi = \mathbf{F}$ in $\mathbb{R}^3$ and $\phi(0, 0, 0) = 0$.

$\phi_x = 1 \Rightarrow \phi = x + g(y,z)$,

g a function of y, z

$\phi_y = 2 \Rightarrow g_y = 2 \Rightarrow g = 2y + h(z)$,

h a function of z.

$\therefore \phi = x + 2y + h(z)$, $\phi_z = -1 \Rightarrow h'(z) = -1$

$\Rightarrow h(z) = -z + C$

$\therefore \phi(x,y,z) = x + 2y - z + C$, C a constant

(6) (b) On the sphere Σ of radius 2 about the origin, find all the points at which
(i) ϕ is a maximum, and
(ii) ϕ is a minimum.

We have $\phi(x,y,z) = x + 2y - z$

Subject to constraint

$g(x,y,z) = x^2 + y^2 + z^2 = 4$

Note $\nabla g = (2x, 2y, 2z) \neq \vec{0}$ for (x,y,z) on the sphere.

Since $\phi(x,y,z)$ is continuous on Σ, a closed compact set, ϕ has a max and min on Σ.

$\therefore$ By Theorem 8, p. 186 of text, by method of Lagrange Multipliers, if $\vec{x}_{max} \in \Sigma$ and $\vec{x}_{min} \in \Sigma$ is where ϕ has its max and min, respectively, Then $\exists\, \lambda, \delta$ s.t.

$$\nabla \phi(\vec{x}_{max}) = \lambda \nabla g(\vec{x}_{max})$$

$$\nabla \phi(\vec{x}_{min}) = \delta \nabla g(\vec{x}_{min})$$

Or, since $\nabla\phi = (1, 2, -1)$ and $\nabla g = (2x, 2y, 2z)$,

$(1, 2, -1) = \lambda(2a, 2b, 2c)$, $\vec{x}_{max} = (a, b, c)$

and, $(1, 2, -1) = \delta(2r, 2s, 2t)$, $\vec{x}_{min} = (r, s, t)$

$\therefore \lambda \neq 0, \delta \neq 0$

$\therefore 1 = 2a\lambda \qquad \frac{1}{2\lambda} = a$

$2 = 2b\lambda \qquad \frac{1}{\lambda} = b$

$-1 = 2c\lambda \qquad -\frac{1}{2\lambda} = c$

$a^2 + b^2 + c^2 = 4$

$\therefore \left(\frac{1}{2\lambda}\right)^2 + \left(\frac{1}{\lambda}\right)^2 + \left(-\frac{1}{2\lambda}\right)^2 = 4$

$\frac{1}{4\lambda^2} + \frac{4}{4\lambda^2} + \frac{1}{4\lambda^2} = 4, \quad 6 = 16\lambda^2$

$\lambda^2 = \frac{3}{8}, \quad \lambda = \pm\sqrt{\frac{3}{8}}, \quad \frac{1}{\lambda} = \pm\sqrt{\frac{8}{3}}$

$\therefore (a, b, c) = \pm\left(\frac{1}{2}\sqrt{\frac{8}{3}}, \sqrt{\frac{8}{3}}, -\frac{1}{2}\sqrt{\frac{8}{3}}\right)$

Equations for δ are identical.

Note The max and min of ϕ are also local max and min.

$\therefore \phi\left(\frac{1}{2}\sqrt{\frac{8}{3}}, \sqrt{\frac{8}{3}}, -\frac{1}{2}\sqrt{\frac{8}{3}}\right) = \left(\frac{1}{2}\sqrt{\frac{8}{3}}\right) + 2\sqrt{\frac{8}{3}} - \left(-\frac{1}{2}\sqrt{\frac{8}{3}}\right) + c$

$= 3\sqrt{\frac{8}{3}} + c = \sqrt{24} + c$

$$\phi\left(-\tfrac{1}{2}\sqrt{\tfrac{8}{3}}, -\sqrt{\tfrac{8}{3}}, \tfrac{1}{2}\sqrt{\tfrac{8}{3}}\right) = \left(-\tfrac{1}{2}\sqrt{\tfrac{8}{3}}\right) + \left(-2\sqrt{\tfrac{8}{3}}\right) - \left(\tfrac{1}{2}\sqrt{\tfrac{8}{3}}\right) + c$$

$$= -3\sqrt{\tfrac{8}{3}} + c = -\sqrt{24} + c$$

$\therefore$ (i) ϕ is a maximum at $(x,y,z) = \left(\frac{1}{2}\sqrt{\frac{8}{3}}, \sqrt{\frac{8}{3}}, -\frac{1}{2}\sqrt{\frac{8}{3}}\right)$

(ii) ϕ is a minimum at $(x,y,z) = \left(-\frac{1}{2}\sqrt{\frac{8}{3}}, -\sqrt{\frac{8}{3}}, \frac{1}{2}\sqrt{\frac{8}{3}}\right)$

(c) (c) Compute the maximum and minimum values of ϕ on Σ.

As calculated above,

$\phi_{max} = \sqrt{24} + c$

$\phi_{min} = -\sqrt{24} + c$

c a constant

23. **23.** Let $\mathbf{F}$ be a C^1 vector field and suppose $\nabla \cdot \mathbf{F}(x_0, y_0, z_0) > 0$. Show that for a sufficiently small sphere S centered at (x_0, y_0, z_0), the flux of $\mathbf{F}$ out of S is positive.

Since $\vec{F}$ is C^1, $\nabla \cdot \vec{F}$ is a continuous function.

Since $\nabla \cdot \vec{F}(x_0, y_0, z_0) > 0$, then there is a ball (sphere) of radius $\delta > 0$, B_δ, s.t. $\nabla \cdot \vec{F}(x,y,z) > 0$

for all $(x,y,z) \in B_\delta$.

$\therefore \iiint_{B_\delta} (\nabla \cdot \vec{F})\, dV > 0$ since this integral is a sum of positive terms.

By Gauss' Divergence Theorem,

$$\iint_{\partial B_\delta} \vec{F} \cdot d\vec{S} = \iiint_{B_\delta} (\nabla \cdot \vec{F})\, dV$$

$\therefore \iint_{\partial B_\delta} \vec{F} \cdot d\vec{S} > 0$, which means the flux of $\vec{F}$ out of B_δ is positive.

24.

24. Let $B \subset \mathbb{R}^3$ be a planar region, and let $O \in \mathbb{R}^3$ be a point. If we connect all points in B to O, we get a cone, say C, with vertex O and base B. Show that

$$\text{Volume } (C) = \frac{1}{3} \text{ area } (B)\ h,$$

where h is the distance of O from the plane of B, using the following steps.

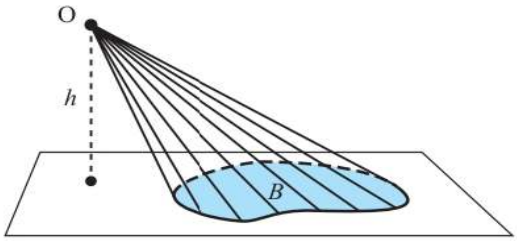


figure **8.R.1**

1. Let O be the origin of the coordinate system. Define $\mathbf{r}(x, y, z) := (x, y, z)$. Evaluate the flux of $\mathbf{r}$ through the boundary of C, that is, $\iint_{\partial C} \mathbf{r} \cdot \mathbf{n}\, dA$, where $\mathbf{n}$ is the outward unit normal to ∂C.

$\vec{r}$ never "exits" the side of C, as $\vec{r}$ always points toward the base B. The flux through B is the perpendicular component of each $\vec{r}$ relative to B. ∴ For each $\vec{r}$, this component is $\|\vec{r}\|\cos\theta$, where θ is the angle $\vec{r}$ makes with the vertical. For each $\vec{r}$, $\|\vec{r}\|\cos\theta = h$.

∴ The flux of $\vec{r}$ through B is (Area B)h

$$\therefore \iint_{\partial C} \vec{r} \cdot \hat{n}\, dA = (\text{Area of } B)h$$

2. Evaluate the total divergence $\iiint_C \nabla \cdot \mathbf{r}\, dV$.

$$\nabla \cdot \vec{r} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$$

$$\therefore \iiint_C (\nabla \cdot \vec{r})\, dV = \iiint_C 3\, dV$$

$$= 3\,(\text{Volume of } C)$$

3. Use Gauss' theorem, which states that the total divergence of a vector field within a region enclosed by a surface is equal to the flux of that vector field through the surface:

$$\iiint_C \nabla \cdot \mathbf{r}\, dV = \iint_{\partial C} \mathbf{r} \cdot \mathbf{n}\, dA.$$

$$3\,(\text{Volume } C) = \iiint_C (\nabla \cdot \vec{r})\, dV = \iint_{\partial C} \vec{r} \cdot \hat{n}\, dA = (\text{Area } B)h$$

Gauss' Divergence Theorem

$$\therefore \text{Volume } C = \frac{1}{3}(\text{Area of } B)h$$